

Velocity Vault Code Source

Generated 2026-05-25 20:03:38 for the v92 weather-disaster and road-condition build.

This PDF/HTML companion contains the current standalone Velocity Vault source for the v92 weather, disaster, road-condition, train, and phone-cleanup build. This update adds route-specific sun, clouds, rain, snow, night, storm, tornado, quake, flood, tidal-surge, and passing-train moments; paints wet roads with clear blue glossy sheen and puddles; paints oil and black ice as dark shiny patches; adds slush and cracked pavement cues; lightly reduces grip/speed on slick or damaged surfaces; removes oversized phone side-road rails/grid/arrow clutter; and documents that Apex Vault Command Cloud needs a fresh static zip upload after local changes unless an automatic deploy hook is configured. The build remains standalone, uses local profiles/saves, and does not depend on remote game assets or licensed vehicle models.

index.html

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
  <meta http-equiv="Content-Security-Policy" content="default-src 'self'; script-src 'self'; style-src 'self'; img-src 'self' data:;
connect-src 'self'; manifest-src 'self'; worker-src 'self'; object-src 'none'; base-uri 'none'; form-action 'none'; frame-ancestors
'none'; upgrade-insecure-requests">
  <meta name="referrer" content="no-referrer">
  <meta name="theme-color" content="#09100f">
  <meta name="mobile-web-app-capable" content="yes">
  <meta name="apple-mobile-web-app-capable" content="yes">
  <meta name="apple-mobile-web-app-title" content="Velocity Vault">
  <title>Velocity Vault</title>
  <link rel="manifest" href="manifest.json">
  <link rel="icon" href="app-icon.svg" type="image/svg+xml">
  <link rel="apple-touch-icon" href="app-icon.svg">
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <main class="shell">
    <section class="stage" aria-label="Velocity Vault race game">
      <canvas id="glCanvas" width="1280" height="720" aria-label="WebGL 3D racing renderer"></canvas>
      <canvas id="gameCanvas" width="1280" height="720" aria-label="Racing game canvas"></canvas>
      <div class="scanline" aria-hidden="true"></div>
      <div id="hud" class="hud">
        <div class="hud-left">
          <div class="brand-mark">Velocity Vault</div>
          <div class="chips">
            <span id="profileChip" class="chip">No Driver</span>
            <span id="modeChip" class="chip">Select Profile</span>
            <span id="missionChip" class="chip">Mission Ready</span>
            <span id="controllerChip" class="chip">Controller: Off</span>
          </div>
        </div>
        <div class="hud-stats" aria-live="polite">
          <div><span>Speed</span><strong id="speedStat">0</strong></div>
          <div><span>Focus</span><strong id="focusStat">100</strong></div>
          <div><span>Damage</span><strong id="damageStat">0%</strong></div>
          <div><span>Coins</span><strong id="coinStat">0</strong></div>
          <div><span id="repLabel">Rep</span><strong id="repStat">0</strong></div>
        </div>
      </div>
      <div id="toast" class="toast" role="status" aria-live="polite"></div>
      <div class="mobile-drive-controls" aria-label="Mobile driving controls">
        <button class="mobile-control control-size" data-control="size" type="button" aria-label="Switch compact controls" aria-
pressed="true">
          <span>Size</span>
          <strong id="touchSizeLabel">Full</strong>
        </button>
        <button class="mobile-control control-mode" data-control="mode" type="button" aria-label="Switch touch control mode" aria-
pressed="false">
          <span>Controls</span>
          <strong id="touchModeLabel">Hold</strong>
        </button>
        <div class="drive-stick" aria-label="Compact drive control">
          <button class="mobile-control stick-up" data-control="gas" type="button" aria-label="Accelerate">&uarr;</button>
          <button class="mobile-control stick-left" data-control="left" type="button" aria-label="Steer left">&larr;</button>
          <button class="mobile-control stick-center" data-control="neutral" type="button" aria-label="Center
steering">&bull;</button>
          <button class="mobile-control stick-right" data-control="right" type="button" aria-label="Steer right">&rarr;</button>
          <button class="mobile-control stick-down" data-control="brake" type="button" aria-label="Brake or reverse">&darr;</button>
        </div>
        <div class="direction-pad" aria-label="Directional steering pad">
          <button class="mobile-control steer-left" data-control="left" type="button" aria-label="Steer left">&larr;</button>
          <button class="mobile-control steer-neutral" data-control="neutral" type="button" aria-label="Center
steering">Center</button>
          <button class="mobile-control steer-right" data-control="right" type="button" aria-label="Steer right">&rarr;</button>
        </div>
        <div class="pedal-pad" aria-label="Throttle and brake controls">
          <button class="mobile-c
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ontrol brake" data-control="brake" type="button" aria-label="Brake or reverse">Brake</button>
  <button class="mobile-control gas" data-control="gas" type="button" aria-label="Accelerate">Gas</button>
  <button class="mobile-control boost" data-control="boost" type="button" aria-label="Boost">Boost</button>
</div>
</div>
</section>

<aside class="panel" aria-label="Game controls">
  <div id="loginView" class="view active">
    <h1>Driver Gate</h1>
    <p class="subtle">Profiles stay on this device. Choose a driver band, create a short callsign, and race.</p>
    <label class="field">
      <span>Driver callsign</span>
      <input id="driverName" type="text" maxlength="16" autocomplete="off" spellcheck="false" placeholder="TurboAce">
    </label>
    <label class="field">
      <span>Local PIN</span>
      <input id="driverPin" type="password" inputmode="numeric" maxlength="6" autocomplete="off" placeholder="Optional">
    </label>
    <div class="age-grid" role="radiogroup" aria-label="Age group">
      <button class="age-card selected" data-age="rookie" type="button">
        <strong>Rookie</strong>
        <span>5-8</span>
      </button>
      <button class="age-card" data-age="prodigy" type="button">
        <strong>Prodigy</strong>
        <span>9-12</span>
      </button>
      <button class="age-card" data-age="elite" type="button">
        <strong>Elite</strong>
        <span>13+</span>
      </button>
      <button class="age-card" data-age="family" type="button">
        <strong>Family</strong>
        <span>All</span>
      </button>
    </div>
    <button id="quickPlay" class="primary" type="button">Play</button>
    <button id="startProfile" class="ghost" type="button">Enter Garage</button>
    <button id="installApp" class="ghost" type="button">Install / Add to Phone</button>
    <button id="resetProfiles" class="ghost danger" type="button">Clear Local Profiles</button>
  </div>

  <div id="garageView" class="view">
    <div class="section-head">
      <div>
        <h2>Race Hub</h2>
        <p id="driverSummary" class="subtle"></p>
      </div>
      <button id="logoutBtn" class="icon-btn" type="button" aria-label="Switch driver" title="Switch driver">Back</button>
    </div>

    <div class="tabs" role="tablist" aria-label="Garage tabs">
      <button class="tab active" data-tab="races" type="button">Races</button>
      <button class="tab" data-tab="vehicles" type="button">Vehicles</button>
      <button class="tab" data-tab="upgrades" type="button">Upgrades</button>
      <button class="tab" data-tab="missions" type="button">Missions</button>
      <button class="tab" data-tab="ai" type="button">AI</button>
    </div>

    <div id="racesTab" class="tab-panel active">
      <div id="raceList" class="race-list"></div>
      <div id="scenarioList" class="scenario-list" aria-label="Multiplayer scenarios"></div>
      <button id="launchRace" class="primary" type="button">Launch Race</button>
      <button id="resumeRace" class="ghost" type="button">Resume Saved Race</button>
    </div>

    <div id="upgradesTab" class="tab-panel">
      <div id="upgradeList" class="upgrade-list"></div>
    </div>

    <div id="vehiclesTab" class="tab-panel">
      <div id="vehicleList" class="vehicle-list"></div>
    </div>

    <div id="missionsTab" class="tab-panel">
      <div id="missionList" class="mission-list"></div>
    </div>

    <div id="aiTab" class="tab-panel">
      <div id="directorCard" class="director-card"></div>
      <button id="refreshDirector" class="ghost" type="button">Refresh AI Tuning</button>
    </div>
  </div>

  <div id="raceView" class="view race-controls">
    <div class="section-head">
      <div>
        <h2 id="raceTitle">Race Live</h2>
        <p id="raceBrief" class="subtle"></p>
      </div>
      <button id="pauseBtn" class="icon-btn" type="button" ari
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a-label="Pause race" title="Pause race">Pause</button>
</div>
<button id="saveRace" class="ghost save-race" type="button">Save Race</button>
<button id="resetCar" class="ghost reset-car" type="button">Reset Car</button>
<div class="control-grid">
  <button id="leftBtn" type="button" aria-label="Steer left">Left</button>
  <button id="gasBtn" type="button" aria-label="Accelerate">Gas</button>
  <button id="brakeBtn" type="button" aria-label="Brake or reverse">Brake / Reverse</button>
  <button id="boostBtn" type="button" aria-label="Boost">Boost</button>
  <button id="rightBtn" type="button" aria-label="Steer right">Right</button>
</div>
<div class="camera-grid" role="group" aria-label="Driver view">
  <button class="camera-btn active" data-camera="chase" type="button">Full Car</button>
  <button class="camera-btn" data-camera="hood" type="button">Half Car</button>
  <button class="camera-btn" data-camera="cockpit" type="button">Windshield</button>
</div>
<div class="renderer-grid" role="group" aria-label="Graphics renderer">
  <button class="renderer-btn active" data-renderer="webgl" type="button">WebGL 3D</button>
  <button class="renderer-btn" data-renderer="canvas" type="button">Classic 2D</button>
</div>
<div class="controller-card">
  <strong id="controllerName">Bluetooth controller ready</strong>
  <span>Pair a controller with your device, press any controller button, then steer with the left stick or D-pad. RT / A
accelerates, LT brakes or reverses, B boosts. Y resets the vehicle. Start pauses.</span>
</div>
<div class="progress-card distance-progress">
  <span>Distance</span>
  <div class="bar"><i id="distanceBar"></i></div>
</div>
<div class="progress-card mission-progress">
  <span>Mission</span>
  <div class="bar"><i id="missionBar"></i></div>
</div>
<div class="progress-card damage-progress">
  <span>Damage</span>
  <div class="bar damage-bar"><i id="damageBar"></i></div>
</div>
<button id="endRace" class="ghost" type="button">Return to Hub</button>
</div>

<div id="finishView" class="view">
  <h2 id="finishTitle">Race Complete</h2>
  <div id="finishStats" class="finish-stats"></div>
  <button id="nextRace" class="primary" type="button">Next Challenge</button>
  <button id="toGarage" class="ghost" type="button">Garage</button>
</div>
</aside>
</main>
<script src="phone-asset-pipeline.js" defer></script>
<script src="webgl-pipeline.js" defer></script>
<script src="game.js" defer></script>
</body>
</html>
```

styles.css

```
:root {
  color-scheme: dark;
  --bg: #09100f;
  --panel: #111817;
  --panel-2: #172220;
  --line: rgba(255, 255, 255, 0.14);
  --text: #f4fbf8;
  --muted: #a9bbb5;
  --lime: #bbf24a;
  --cyan: #46d9ff;
  --red: #ff5b6b;
  --gold: #ffd166;
  --road: #252927;
  --ink: #050807;
  --shadow: 0 24px 80px rgba(0, 0, 0, 0.42);
}

* {
  box-sizing: border-box;
}

html,
body {
  height: 100%;
}

html {
  min-height: 100%;
}

body {
  margin: 0;
  min-width: 320px;
  overflow: hidden;
  background:
    radial-gradient(circle at 18% 20%, rgba(70, 217, 255, 0.16), transparent 30%),
    linear-gradient(130deg, #070a09 0%, #0a1714 48%, #17140a 100%);
  color: var(--text);
  font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
}

button,
input {
  font: inherit;
}

button {
  border: 0;
  color: inherit;
  cursor: pointer;
}

.shell {
  display: grid;
  grid-template-columns: minmax(0, 1fr) 390px;
  gap: 18px;
  height: 100vh;
  padding: 18px;
}

.stage {
  position: relative;
  min-width: 0;
  overflow: hidden;
  border: 1px solid var(--line);
  border-radius: 8px;
  background: #050807;
  box-shadow: var(--shadow);
  touch-action: none;
}

.stage canvas {
  position: absolute;
  inset: 0;
  display: block;
  width: 100%;
  height: 100%;
}

#glCanvas {
  z-index: 1;
  opacity: 0;
}

#gameCanvas {
  z-index: 2;
  background: #050807;
}

.stage.webgl #glCanvas {
  opacity: 1;
}

.stage.webgl #gameCanvas {
  background: transparent;
}

.scanline {
  position: absolute;
  inset: 0;
  z-index: 3;
  pointer-events: none;
  background: linear-gradient(180deg, rgba(255,255,255,0.035), transparent 22%, transparent 78%, rgba(0,0,0,0.22));
  mix-blend-mode: soft-light;
}

.hud {
  position: absolute;
  z-index: 4;
  top: 16px;
  left: 16px;
  right: 16px;
```

```

display: flex;
justify-content: space-between;
gap: 16px;
pointer-events: none;
}

.brand-mark {
width: fit-content;
padding: 8px 10px;
border: 1px solid rgba(187, 242, 74, 0.38);
border-radius: 6px;
background: rgba(5, 8, 7, 0.72);
color: var(--lime);
font-weight: 900;
letter-spacing: 0;
text-transform: uppercase;
}

.chips {
display: flex;
flex-wrap: wrap;
gap: 8px;
margin-top: 10px;
}

.chip,
.hud-stats div,
.toast {
border: 1px solid var(--line);
border-radius: 6px;
background: rgba(5, 8, 7, 0.72);
backdrop-filter: blur(10px);
}

.chip {
padding: 6px 9px;
color: var(--muted);
font-size: 12px;
font-weight: 800;
}

.hud-stats {
display: grid;
grid-template-columns: repeat(5, minmax(70px, 1fr));
gap: 8px;
}

.hud-stats div {
padding: 8px 10px;
text-align: right;
}

.hud-stats span {
display: block;
color: var(--muted);
font-size: 11px;
text-transform: uppercase;
}

.hud-stats strong {
color: var(--text);
font-size: 20px;
}

.toast {
position: absolute;
z-index: 5;
left: 50%;
bottom: 24px;
max-width: min(520px, calc(100% - 48px));
padding: 12px 14px;
opacity: 0;
transform: translateX(-50%) translateY(16px);
transition: opacity 180ms ease, transform 180ms ease;
color: var(--text);
font-weight: 800;
}

.toast.show {
opacity: 1;
transform: translateX(-50%) translateY(0);
}

.mobile-drive-controls {
display: none;
touch-action: none;
}

.mobile-control {
touch-action: none;
user-select: none;
-webkit-user-select: none;
-webkit-touch-callout: none;
}

.drive-stick,
.direction-pad,
.pedal-pad {
display: none;
touch-action: none;
}

.panel {
overflow: hidden auto;
border: 1px solid var(--line);
border-radius: 8px;
background: linear-gradient(180deg, rgba(17, 24, 23, 0.98), rgba(11, 16, 15, 0.98));
box-shadow: var(--shadow);
}

.view {
display: none;
padding: 22px;
}

.view.active {
display: block;
}

```

```
}

h1,
h2 {
  margin: 0;
  line-height: 1.05;
  letter-spacing: 0;
}

h1 {
  font-size: 36px;
}

h2 {
  font-size: 26px;
}

.subtle {
  margin: 8px 0 18px;
  color: var(--muted);
  line-height: 1.45;
}

.field {
  display: grid;
  gap: 7px;
  margin-bottom: 14px;
}

.field span {
  color: var(--muted);
  font-size: 13px;
  font-weight: 800;
}

input {
  width: 100%;
  min-height: 46px;
  border: 1px solid var(--line);
  border-radi
```

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```
us: 6px;
outline: none;
background: #0b1110;
color: var(--text);
padding: 0 12px;
}

input:focus {
border-color: var(--cyan);
box-shadow: 0 0 3px rgba(70, 217, 255, 0.16);
}

.age-grid {
display: grid;
grid-template-columns: repeat(2, minmax(0, 1fr));
gap: 10px;
margin: 18px 0;
}

.age-card,
.race-card,
.upgrade-card,
.mission-card,
.progress-card {
border: 1px solid var(--line);
border-radius: 8px;
background: var(--panel-2);
}

.age-card {
min-height: 78px;
padding: 12px;
text-align: left;
}

.age-card strong,
.age-card span {
display: block;
}

.age-card span {
margin-top: 5px;
color: var(--muted);
}

.age-card.selected {
border-color: var(--lime);
background: linear-gradient(135deg, rgba(187, 242, 74, 0.18), rgba(70, 217, 255, 0.08));
}

.primary,
.ghost {
width: 100%;
min-height: 48px;
border-radius: 6px;
font-weight: 900;
}

.primary {
background: linear-gradient(135deg, var(--lime), var(--cyan));
color: #06100e;
}

.ghost {
margin-top: 10px;
border: 1px solid var(--line);
background: transparent;
color: var(--text);
}

.ghost.disabled {
cursor: not-allowed;
opacity: 0.45;
}

.danger {
color: #ffc5cc;
}

.section-head {
display: flex;
align-items: flex-start;
justify-content: space-between;
gap: 12px;
margin-bottom: 14px;
}

.icon-btn {
flex: 0 0 44px;
width: 44px;
height: 44px;
border: 1px solid var(--line);
border-radius: 6px;
background: #0b1110;
font-size: 18px;
font-weight: 900;
}

.tabs {
display: grid;
grid-template-columns: repeat(5, 1fr);
gap: 6px;
margin: 14px 0 16px;
padding: 4px;
border: 1px solid var(--line);
border-radius: 8px;
background: #0b1110;
}

.tab {
min-height: 38px;
border-radius: 6px;
background: transparent;
}
```



```

    color: var(--muted);
    font-weight: 900;
}

.tab.active {
    background: var(--panel-2);
    color: var(--text);
}

.tab-panel {
    display: none;
}

.tab-panel.active {
    display: block;
}

.race-list,
.scenario-list,
.upgrade-list,
.vehicle-list,
.mission-list {
    display: grid;
    gap: 10px;
    margin-bottom: 14px;
}

.race-card,
.scenario-card,
.upgrade-card,
.vehicle-card,
.mission-card {
    padding: 13px;
}

.race-card,
.scenario-card,
.vehicle-card {
    cursor: pointer;
}

.race-card.selected,
.scenario-card.selected,
.vehicle-card.selected {
    border-color: var(--cyan);
    box-shadow: inset 0 0 0 1px rgba(70, 217, 255, 0.4);
}

.race-card.disabled,
.vehicle-card.locked {
    cursor: not-allowed;
    opacity: 0.58;
}

.scenario-list {
    margin-top: 2px;
}

.scenario-card {
    border: 1px solid var(--line);
    border-radius: 8px;
    background: linear-gradient(135deg, rgba(70, 217, 255, 0.08), rgba(187, 242, 74, 0.06));
    text-align: left;
}

.vehicle-card {
    display: grid;
    grid-template-columns: 76px 1fr;
    gap: 12px;
    width: 100%;
    text-align: left;
    align-items: center;
}

.vehicle-card .card-top,
.vehicle-card .tiny {
    grid-column: 2;
}

.vehicle-preview {
    --vehicle-color: #46d9ff;
    grid-row: 1 / span 2;
    width: 74px;
    height: 52px;
    position: relative;
    border-radius: 7px;
    border: 1px solid rgba(244, 251, 248, 0.14);
    background:
        linear-gradient(135deg, rgba(255,255,255,0.1), rgba(5,8,7,0.34)),
        #101817;
    overflow: hidden;
}

.vehicle-preview-canvas {
    position: absolute;
    inset: 0;
    z-index: 1;
    display: block;
    width: 100%;
    height: 100%;
}

.vehicle-preview.asset-preview::before,
.vehicle-preview.asset-preview::after {
    display: none;
}

.vehicle-preview::before {
    content: "";
    position: absolute;
    left: 15px;
    right: 15px;
    top: 11px;
    bottom: 10px;
    border-radius: 10px 10px 5px 5px;
    background: linear-gradient(135deg, color-mix(in srgb, var(--vehicle-color), white 24%), var(--vehicle-color) 52%, color-mix(in srgb, var(--vehicle-color), black 42%));
}

```

```
    box-shadow: 0 10px 18px rgba(0,0,0,0.36);
}

.vehicle-preview::after {
  content: "";
  position: absolute;
  left: 9px;
  right: 9px;
  bottom: 8px;
  height: 8px;
  border-radius: 99px;
  background: #050807;
}

.vehicle-preview[data-type="f1"]::before,
.vehicle-preview[data-type="prototype"]::before {
  clip-path: polygon(50% 0, 84% 22%, 72% 92%, 28% 92%, 16% 22%);
}

.vehicle-pr
```

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```
eview[data-type="truck"]::before,
.vehicle-preview[data-type="semi"]::before,
.vehicle-preview[data-type="tractor"]::before,
.vehicle-preview[data-type="monster"]::before,
.vehicle-preview[data-type="tank"]::before {
  left: 11px;
  right: 11px;
  border-radius: 5px;
}

.vehicle-preview[data-type="semi"]::before {
  left: 7px;
  right: 7px;
  top: 9px;
  bottom: 8px;
}

.vehicle-preview[data-type="tractor"]::before {
  left: 18px;
  right: 18px;
  top: 8px;
  bottom: 14px;
}

.vehicle-preview[data-type="tractor"]::after {
  left: 12px;
  right: 12px;
  height: 14px;
}

.vehicle-preview[data-type="monster"]::after {
  height: 13px;
}

.vehicle-preview[data-type="boat"]::before,
.vehicle-preview[data-type="snowmobile"]::before,
.vehicle-preview[data-type="airplane"]::before {
  clip-path: polygon(50% 0, 91% 70%, 70% 100%, 30% 100%, 9% 70%);
}

.vehicle-preview[data-type="helicopter"]::after {
  left: 12px;
  right: 12px;
  top: 8px;
  bottom: auto;
  height: 3px;
  background: rgba(244,251,248,0.75);
}

.card-top {
  display: flex;
  justify-content: space-between;
  gap: 10px;
  margin-bottom: 8px;
}

.card-top strong {
  font-size: 16px;
}

.badge {
  align-self: flex-start;
  white-space: nowrap;
  border-radius: 999px;
  background: rgba(255, 209, 102, 0.18);
  color: var(--gold);
  padding: 4px 8px;
  font-size: 12px;
  font-weight: 900;
}

.tiny {
  color: var(--muted);
  font-size: 12px;
  line-height: 1.35;
}

.upgrade-action {
  display: grid;
  grid-template-columns: 1fr 96px;
  gap: 8px;
  align-items: center;
  margin-top: 10px;
}

.upgrade-action button {
  min-height: 36px;
  border-radius: 6px;
  background: var(--lime);
  color: var(--ink);
  font-weight: 900;
}

.upgrade-action button:disabled {
  cursor: not-allowed;
  opacity: 0.45;
}

.control-grid {
  display: grid;
  grid-template-columns: repeat(5, minmax(0, 1fr));
  gap: 8px;
  margin: 18px 0;
}

.control-grid button,
.camera-grid button,
.renderer-grid button {
  min-height: 72px;
  border: 1px solid var(--line);
  border-radius: 8px;
  background: var(--panel-2);
  font-size: 18px;
}
```

```

font-weight: 900;
touch-action: manipulation;
user-select: none;
}

.camera-grid {
display: grid;
grid-template-columns: repeat(3, 1fr);
gap: 8px;
margin-bottom: 10px;
}

.renderer-grid {
display: grid;
grid-template-columns: repeat(2, 1fr);
gap: 8px;
margin-bottom: 10px;
}

.camera-grid button,
.renderer-grid button {
min-height: 46px;
color: var(--muted);
font-size: 13px;
}

.camera-grid button.active,
.renderer-grid button.active {
border-color: var(--cyan);
color: var(--text);
background: linear-gradient(135deg, rgba(70, 217, 255, 0.18), rgba(187, 242, 74, 0.1));
}

#boostBtn {
background: linear-gradient(135deg, rgba(255, 209, 102, 0.96), rgba(255, 91, 107, 0.88));
color: #160a05;
}

#gasBtn {
background: linear-gradient(135deg, rgba(187, 242, 74, 0.92), rgba(70, 217, 255, 0.74));
color: #06100e;
}

#brakeBtn {
background: linear-gradient(135deg, rgba(244, 251, 248, 0.14), rgba(255, 91, 107, 0.28));
}

.control-grid button.pressed {
transform: translateY(1px);
filter: brightness(1.18);
box-shadow: inset 0 0 0 2px rgba(244, 251, 248, 0.22);
}

.progress-card {
padding: 12px;
margin-bottom: 10px;
}

.controller-card,
.director-card {
border: 1px solid var(--line);
border-radius: 8px;
background: #0b1110;
padding: 12px;
margin-bottom: 10px;
}

.controller-card strong,
.controller-card span,
.director-card strong,
.director-card span {
display: block;
}

.controller-card strong,
.director-card strong {
margin-bottom: 6px;
color: var(--cyan);
}

.controller-card span,
.director-card span {
color: var(--muted);
font-size: 12px;
line-height: 1.42;
}

.director-card ul {
margin: 10px 0 0;
padding-left: 18px;
color: var(--muted);
font-size: 12px;
line-height: 1.45;
}

.director-card li + li {
margin-top: 6px;
}

.progress-card span {
display: block;
margin-bottom: 8px;
color: var(--muted);
font-weight: 800;
}

.bar {
height: 12px;
overflow: hidden;
border-radius: 999px;
background: #080d0c;
}

.bar i {
display: block;
width: 0%;
height: 100%;

```

```
border-radius: inherit;
```

styles.css continued 4/10

```
background: linear-gradient(90deg, var(--lime), var(--cyan));
}

.damage-bar i {
background: linear-gradient(90deg, var(--lime), var(--amber) 48%, var(--red));
}

.finish-stats {
display: grid;
gap: 10px;
margin: 18px 0;
}

.finish-stats div {
display: flex;
justify-content: space-between;
border-bottom: 1px solid var(--line);
padding-bottom: 9px;
}

@media (min-width: 981px) {
body.race-live .shell {
grid-template-columns: minmax(0, 1fr) clamp(280px, 28vw, 360px);
gap: 12px;
padding: 12px;
}

body.race-live .panel {
min-width: 0;
max-height: calc(100dvh - 24px);
}

body.race-live .view {
padding: clamp(12px, 1.4vw, 18px);
}

body.race-live .section-head {
gap: 8px;
margin-bottom: 10px;
}

body.race-live .section-head h2 {
font-size: clamp(18px, 1.9vw, 24px);
}

body.race-live .subtle {
margin-bottom: 10px;
font-size: 12px;
}

body.race-live .control-grid {
grid-template-columns: repeat(5, minmax(42px, 1fr));
gap: 6px;
margin: 10px 0;
}

body.race-live .control-grid button {
min-height: clamp(46px, 8vh, 64px);
padding: 0 4px;
overflow: hidden;
font-size: clamp(12px, 1.2vw, 16px);
text-overflow: ellipsis;
}

body.race-live .camera-grid,
body.race-live .renderer-grid {
gap: 6px;
}

body.race-live .camera-grid button,
body.race-live .renderer-grid button {
min-height: 38px;
padding: 0 4px;
overflow: hidden;
text-overflow: ellipsis;
}

body.race-live #pauseBtn,
body.race-live #saveRace,
body.race-live #resetCar,
body.race-live #endRace {
min-height: 42px;
}

body.race-live .mobile-drive-controls,
body.race-live .drive-stick,
body.race-live .direction-pad,
body.race-live .pedal-pad,
body.race-live .mobile-control.control-mode,
body.race-live .mobile-control.control-size {
display: none !important;
pointer-events: none;
}
}

@media (max-width: 980px) {
body {
overflow: auto;
}

.shell {
grid-template-columns: 1fr;
height: auto;
min-height: 100vh;
}

.stage {
min-height: 58vh;
}

.hud {
position: absolute;
display: grid;
}
```

```

}

.hud-stats {
  grid-template-columns: repeat(2, minmax(82px, 1fr));
}
}

@media (max-width: 560px) {
  .shell {
    padding: 10px;
    gap: 10px;
  }

  .stage {
    min-height: 50vh;
  }

  .hud {
    inset: 10px 10px auto;
  }

  .brand-mark,
  .chip {
    font-size: 11px;
  }

  .hud-stats strong {
    font-size: 17px;
  }

  .view {
    padding: 16px;
  }
}

@media (max-width: 980px), (pointer: coarse) {
  body.race-live {
    position: fixed;
    inset: 0;
    width: var(--vw, 100dvw);
    height: var(--vh, 100dvh);
    overflow: hidden;
    overscroll-behavior: none;
  }

  body.race-live .shell {
    position: fixed;
    inset: 0;
    display: block;
    width: var(--vw, 100dvw);
    height: var(--vh, 100dvh);
    min-height: 0;
    padding: 0;
    overflow: hidden;
  }

  body.race-live .stage {
    width: 100%;
    height: 100%;
    min-height: 0;
    border: 0;
    border-radius: 0;
    touch-action: none;
  }

  body.race-live .panel {
    position: fixed;
    inset: 0;
    z-index: 10;
    overflow: visible;
    pointer-events: none;
    border: 0;
    border-radius: 0;
    background: transparent;
    box-shadow: none;
  }

  body.race-live #raceView {
    display: block;
    padding: 0;
    pointer-events: none;
  }

  body.race-live .control-grid,
  body.race-live .controller-card,
  body.race-live .progress-card,
  body.race-live .section-head div {
    display: none;
  }

  body.race-live #pauseBtn,
  body.race-live #saveRace,
  body.race-live #resetCar,
  body.race-live #endRace,
  body.race-live .mobile-control {
    pointer-events: auto;
    backdrop-filter: blur(10px);
  }

  body.race-live #pauseBtn,
  body.race-live #saveRace,
  body.race-live #resetCar,
  body.race-live #endRace {
    position: fixed;
    right: max(8px, env(safe-area-inset-right));
    z-index: 32;
    width: 76px;
    min-height: 34px;
    margin: 0;
    font-size: 11px;
    background: rgba(5, 8, 7, 0.68);
  }

  body.race-live #pauseBtn {
    top: max(8px, env(safe-area-i

```

styles.css continued 5/10

```
nset-top));
}

body.race-live #saveRace {
  top: calc(max(8px, env(safe-area-inset-top)) + 40px);
}

body.race-live #resetCar {
  top: calc(max(8px, env(safe-area-inset-top)) + 80px);
  background: rgba(255, 91, 107, 0.34);
}

body.race-live #endRace {
  top: calc(max(8px, env(safe-area-inset-top)) + 120px);
}

body.race-live .mobile-drive-controls {
  position: fixed;
  inset: 0;
  z-index: 30;
  display: block;
  width: var(--vw, 100dvw);
  height: var(--vh, 100dvh);
  pointer-events: auto;
}

body.race-live .direction-pad,
body.race-live .pedal-pad {
  position: fixed;
  z-index: 31;
  display: grid;
  pointer-events: auto;
}

body.race-live .direction-pad {
  left: max(8px, env(safe-area-inset-left));
  bottom: max(12px, env(safe-area-inset-bottom));
  grid-template-columns: repeat(3, clamp(42px, 12vw, 62px));
  gap: 6px;
}

body.race-live .pedal-pad {
  right: max(8px, env(safe-area-inset-right));
  bottom: max(12px, env(safe-area-inset-bottom));
  grid-template-columns: clamp(54px, 15vw, 76px) clamp(62px, 17vw, 88px);
  grid-template-rows: clamp(42px, 11vw, 56px) clamp(50px, 13vw, 68px);
  gap: 6px;
}

body.race-live .mobile-control {
  min-width: 0;
  min-height: 0;
  border: 1px solid rgba(244, 251, 248, 0.28);
  border-radius: 13px;
  background:
    linear-gradient(145deg, rgba(255,255,255,0.14), rgba(5,8,7,0.36)),
    rgba(5, 8, 7, 0.58);
  color: #f4fbf8;
  font-size: clamp(10px, 3vw, 13px);
  font-weight: 900;
  opacity: 0.94;
  box-shadow: 0 10px 30px rgba(0,0,0,0.26);
}

body.race-live .mobile-control.steer-left,
body.race-live .mobile-control.steer-neutral,
body.race-live .mobile-control.steer-right {
  width: 100%;
  height: clamp(42px, 12vw, 62px);
}

body.race-live .mobile-control.gas {
  grid-column: 2;
  grid-row: 1 / span 2;
  background: linear-gradient(135deg, rgba(187, 242, 74, 0.74), rgba(70, 217, 255, 0.6));
  color: #06100e;
}

body.race-live .mobile-control.brake {
  grid-column: 1;
  grid-row: 2;
}

body.race-live .mobile-control.boost {
  grid-column: 1;
  grid-row: 1;
  background: linear-gradient(135deg, rgba(255, 209, 102, 0.78), rgba(255, 91, 107, 0.72));
  color: #160a05;
}

body.race-live .mobile-control.control-mode {
  position: fixed;
  left: 50%;
  bottom: max(12px, env(safe-area-inset-bottom));
  z-index: 31;
  width: clamp(74px, 18vw, 112px);
  height: clamp(32px, 8vw, 44px);
  transform: translateX(-50%);
  pointer-events: auto;
  background: rgba(5, 8, 7, 0.62);
}

body.race-live .mobile-control.control-mode span,
body.race-live .mobile-control.control-mode strong {
  display: block;
  line-height: 1.05;
}

body.race-live .mobile-control.control-mode span {
  color: rgba(244, 251, 248, 0.72);
  font-size: 9px;
  text-transform: uppercase;
}
```



```

body.race-live .mobile-control.control-mode strong {
  font-size: 12px;
}

body.race-live.touch-toggle .mobile-control.control-mode {
  border-color: rgba(187, 242, 74, 0.72);
  background: linear-gradient(135deg, rgba(187, 242, 74, 0.72), rgba(70, 217, 255, 0.5));
  color: #06100e;
}

}

@media (orientation: portrait) and (max-width: 980px) {
  body.race-live .hud {
    top: max(8px, env(safe-area-inset-top));
    left: 8px;
    right: 90px;
    display: block;
  }

  body.race-live .brand-mark,
  body.race-live .chips {
    display: none;
  }

  body.race-live .hud-stats {
    grid-template-columns: repeat(3, minmax(48px, 1fr));
    gap: 5px;
  }

  body.race-live .hud-stats div:nth-child(n+4),
  body.race-live #saveRace,
  body.race-live #endRace,
  body.race-live .camera-grid,
  body.race-live .renderer-grid {
    display: none;
  }

  body.race-live #resetCar {
    top: calc(max(8px, env(safe-area-inset-top)) + 42px);
  }

  body.race-live .direction-pad {
    bottom: calc(max(12px, env(safe-area-inset-bottom)) + 2px);
    grid-template-columns: repeat(3, clamp(40px, 13vw, 54px));
  }

  body.race-live .pedal-pad {
    bottom: calc(max(12px, env(safe-area-inset-bottom)) + 2px);
    grid-template-columns: clamp(50px, 14vw, 62px) c
  }

```

styles.css continued 6/10

```
lamp(58px, 16vw, 74px);
  grid-template-rows: clamp(40px, 11vw, 52px) clamp(46px, 13vw, 60px);
}

body.race-live .mobile-control.steer-left,
body.race-live .mobile-control.steer-neutral,
body.race-live .mobile-control.steer-right {
  height: clamp(40px, 13vw, 54px);
}

body.race-live .mobile-control.control-mode {
  bottom: calc(max(12px, env(safe-area-inset-bottom)) + 68px);
}
}

@media (orientation: landscape) and (max-width: 980px) and (max-height: 760px), (orientation: landscape) and (pointer: coarse) {
  body.race-live {
    overflow: hidden;
  }

  body:not(.race-live) {
    overflow: auto;
  }

  body:not(.race-live) .stage {
    touch-action: pan-y;
  }

  body:not(.race-live) .shell {
    height: auto;
    min-height: 100dvh;
  }

  body.race-live .shell {
    position: fixed;
    inset: 0;
    display: block;
    width: var(--vw, 100dvw);
    height: var(--vh, 100dvh);
    min-height: 0;
    padding: 0;
    overflow: hidden;
  }

  body.race-live .stage {
    width: 100%;
    height: 100%;
    min-height: 0;
    border-radius: 0;
    border: 0;
  }

  body.race-live .hud {
    top: 6px;
    left: 8px;
    right: 8px;
    display: flex;
    align-items: flex-start;
    gap: 8px;
  }

  body.race-live .brand-mark {
    padding: 6px 8px;
    font-size: 12px;
  }

  body.race-live .chips {
    display: none;
  }

  body.race-live .hud-stats {
    grid-template-columns: repeat(5, minmax(44px, 54px));
    gap: 5px;
    margin-left: auto;
    margin-right: 78px;
  }

  body.race-live .hud-stats div {
    padding: 5px 6px;
  }

  body.race-live .hud-stats span {
    font-size: 8px;
  }

  body.race-live .hud-stats strong {
    font-size: 14px;
  }

  body.race-live .panel {
    position: fixed;
    inset: 0;
    z-index: 10;
    overflow: visible;
    pointer-events: none;
    border: 0;
    border-radius: 0;
    background: transparent;
    box-shadow: none;
  }

  body.race-live #raceView {
    display: block;
    padding: 0;
    pointer-events: none;
  }

  body.race-live .section-head {
    position: fixed;
    top: 8px;
    right: 8px;
    z-index: 6;
    display: block;
  }
}
```

```

    margin: 0;
    pointer-events: none;
}

body.race-live .section-head div {
    display: none;
}

body.race-live #pauseBtn,
body.race-live #saveRace,
body.race-live #resetCar,
body.race-live #endRace,
body.race-live .camera-grid button,
body.race-live .renderer-grid button,
body.race-live .mobile-control {
    pointer-events: auto;
    backdrop-filter: blur(10px);
}

body.race-live #pauseBtn {
    position: fixed;
    top: 8px;
    right: 8px;
    z-index: 12;
    width: 68px;
    height: 38px;
    font-size: 12px;
    background: rgba(5, 8, 7, 0.72);
}

body.race-live #saveRace {
    position: fixed;
    top: 52px;
    right: 8px;
    z-index: 12;
    width: 88px;
    min-height: 36px;
    margin: 0;
    font-size: 12px;
    background: rgba(5, 8, 7, 0.58);
}

body.race-live #endRace {
    position: fixed;
    top: 140px;
    right: 8px;
    z-index: 12;
    width: 88px;
    min-height: 36px;
    margin: 0;
    font-size: 12px;
    background: rgba(5, 8, 7, 0.58);
}

body.race-live #resetCar {
    position: fixed;
    top: 96px;
    right: 8px;
    z-index: 12;
    width: 88px;
    min-height: 36px;
    margin: 0;
    font-size: 12px;
    background: rgba(255, 91, 107, 0.28);
}

body.race-live .control-grid {
    display: none;
}

body.race-live .mobile-drive-controls {
    position: absolute;
    inset: 0;
    z-index: 8;
    display: block;
    pointer-events: auto;
}

body.race-live .direction-pad,
body.race-live .pedal-pad {
    position: absolute;
    z-index: 9;
    display: grid;
    pointer-events: auto;
}

body.race-live .direction-pad {
    left: max(10px, env(safe-area-inset-left));
    bottom: max(12px, env(safe-area-inset-bottom));
    grid-template-columns: repeat(3, clamp(44px, 11dvh, 66px));
    gap: 6px;
}

body.race-live .pedal-pad {
    right: max(10px, env(safe-area-inset-right));
    bottom: max(12px, env(safe-area-inset-bottom));
    grid-template-columns: clamp(58px, 14dvh, 86px) clamp(66px, 16dvh, 100px);
    grid-template-rows: clamp(42px, 9d

```

styles.css continued 7/10

```
vh, 58px) clamp(48px, 11dvh, 68px);
gap: 6px;
}

body.race-live .mobile-control {
  min-width: 0;
  min-height: 0;
  border: 1px solid rgba(244, 251, 248, 0.26);
  border-radius: 14px;
  background:
    linear-gradient(145deg, rgba(255,255,255,0.12), rgba(5,8,7,0.34)),
    rgba(5, 8, 7, 0.54);
  color: #f4fbf8;
  font-size: clamp(10px, 2.8dvh, 13px);
  font-weight: 900;
  opacity: 0.92;
  box-shadow: 0 10px 30px rgba(0,0,0,0.24);
}

body.race-live .mobile-control.pressed {
  filter: brightness(1.25);
  box-shadow: inset 0 0 2px rgba(244, 251, 248, 0.28), 0 10px 30px rgba(0,0,0,0.24);
}

body.race-live .mobile-control.steer-left,
body.race-live .mobile-control.steer-neutral,
body.race-live .mobile-control.steer-right {
  width: 100%;
  height: clamp(44px, 11dvh, 66px);
}

body.race-live .mobile-control.gas {
  grid-column: 2;
  grid-row: 1 / span 2;
  background: linear-gradient(135deg, rgba(187, 242, 74, 0.72), rgba(70, 217, 255, 0.58));
  color: #06100e;
}

body.race-live .mobile-control.brake {
  grid-column: 1;
  grid-row: 2;
}

body.race-live .mobile-control.boost {
  grid-column: 1;
  grid-row: 1;
  background: linear-gradient(135deg, rgba(255, 209, 102, 0.78), rgba(255, 91, 107, 0.72));
  color: #160a05;
}

body.race-live .mobile-control.control-mode {
  position: absolute;
  left: 50%;
  bottom: max(10px, env(safe-area-inset-bottom));
  z-index: 9;
  width: clamp(70px, 17dvh, 104px);
  height: clamp(32px, 7dvh, 42px);
  transform: translateX(-50%);
  pointer-events: auto;
  background: rgba(5, 8, 7, 0.58);
}

body.race-live .mobile-control.control-mode span,
body.race-live .mobile-control.control-mode strong {
  display: block;
  line-height: 1.05;
}

body.race-live .mobile-control.control-mode span {
  color: rgba(244, 251, 248, 0.72);
  font-size: 9px;
  text-transform: uppercase;
}

body.race-live .mobile-control.control-mode strong {
  font-size: 12px;
}

body.race-live.touch-toggle .mobile-control.control-mode {
  border-color: rgba(187, 242, 74, 0.72);
  background: linear-gradient(135deg, rgba(187, 242, 74, 0.72), rgba(70, 217, 255, 0.5));
  color: #06100e;
}

body.race-live .camera-grid {
  position: fixed;
  left: 50%;
  top: 8px;
  bottom: auto;
  z-index: 4;
  width: min(280px, 38vw);
  transform: translateX(-50%);
  gap: 5px;
  margin: 0;
  pointer-events: none;
}

body.race-live .camera-grid button {
  min-height: 32px;
  padding: 0 5px;
  overflow: hidden;
  background: rgba(5, 8, 7, 0.48);
  font-size: 10px;
  text-overflow: ellipsis;
}

body.race-live .renderer-grid {
  position: fixed;
  left: 50%;
  top: 46px;
  bottom: auto;
  z-index: 4;
  width: min(210px, 30vw);
```

```

        transform: translateX(-50%);
        gap: 5px;
        margin: 0;
        pointer-events: none;
    }

    body.race-live .renderer-grid button {
        min-height: 30px;
        padding: 0 5px;
        overflow: hidden;
        background: rgba(5, 8, 7, 0.48);
        font-size: 10px;
        text-overflow: ellipsis;
    }

    body.race-live .controller-card {
        display: none;
    }

    body.race-live .progress-card {
        display: none;
    }
}

@media (orientation: landscape) and (max-width: 980px) and (max-height: 430px) {
    body.race-live .hud-stats {
        grid-template-columns: repeat(3, 46px);
        margin-right: 64px;
    }

    body.race-live .brand-mark,
    body.race-live .hud-stats div:nth-child(n+4) {
        display: none;
    }

    body.race-live #saveRace,
    body.race-live #endRace {
        display: none;
    }

    body.race-live #pauseBtn,
    body.race-live #resetCar {
        width: 62px;
        min-height: 32px;
        height: 32px;
        font-size: 11px;
    }

    body.race-live #resetCar {
        top: 46px;
    }

    body.race-live .camera-grid,
    body.race-live .renderer-grid {
        display: none;
    }

    body.race-live .direction-pad {
        grid-template-columns: repeat(3, clamp(40px, 11dvh, 52px));
    }

    body.race-live .pedal-pad {
        grid-template-columns: clamp(52px, 14dvh, 68px) clamp(58px, 16dvh, 76px);
        grid-template-rows: clamp(38px, 9dvh, 48px) clamp(44px, 11dvh, 56px);
    }
}

body.race-live

```

styles.css continued 8/10

```
.mobile-control.steer-left,
body.race-live .mobile-control.steer-neutral,
body.race-live .mobile-control.steer-right {
  height: clamp(40px, 11dvh, 52px);
}

@media (max-width: 980px), (pointer: coarse) {
  body.race-live .control-size {
    position: fixed;
    left: 50%;
    bottom: max(12px, env(safe-area-inset-bottom));
    z-index: 34;
    width: 58px;
    height: 32px;
    transform: translateX(calc(-50% - 72px));
    pointer-events: auto;
    border-radius: 999px;
    background: rgba(5, 8, 7, 0.72);
  }

  body.race-live .control-size span,
  body.race-live .control-size strong {
    display: block;
    line-height: 1;
  }

  body.race-live .control-size span {
    color: rgba(244, 251, 248, 0.68);
    font-size: 8px;
    text-transform: uppercase;
  }

  body.race-live .control-size strong {
    font-size: 11px;
  }

  body.race-live.control-mini .control-mode {
    display: none;
  }

  body.race-live.control-mini .drive-stick {
    position: fixed;
    left: max(10px, env(safe-area-inset-left));
    bottom: max(10px, env(safe-area-inset-bottom));
    z-index: 34;
    display: grid;
    grid-template-columns: repeat(3, 42px);
    grid-template-rows: repeat(3, 42px);
    gap: 4px;
    width: 138px;
    height: 138px;
    padding: 4px;
    pointer-events: auto;
    border: 1px solid rgba(244, 251, 248, 0.22);
    border-radius: 999px;
    background:
      radial-gradient(circle at 50% 50%, rgba(244, 251, 248, 0.12), rgba(5, 8, 7, 0.52) 66%),
      rgba(5, 8, 7, 0.34);
    box-shadow: 0 14px 34px rgba(0, 0, 0, 0.34);
    backdrop-filter: blur(10px);
    -webkit-backdrop-filter: blur(10px);
  }

  body.race-live.control-mini .drive-stick .mobile-control {
    width: 42px;
    height: 42px;
    border-radius: 999px;
    font-size: 18px;
    line-height: 1;
  }

  body.race-live.control-mini .stick-up {
    grid-column: 2;
    grid-row: 1;
  }

  body.race-live.control-mini .stick-left {
    grid-column: 1;
    grid-row: 2;
  }

  body.race-live.control-mini .stick-center {
    grid-column: 2;
    grid-row: 2;
    opacity: 0.58;
  }

  body.race-live.control-mini .stick-right {
    grid-column: 3;
    grid-row: 2;
  }

  body.race-live.control-mini .stick-down {
    grid-column: 2;
    grid-row: 3;
  }

  body.race-live.control-mini .direction-pad,
  body.race-live.control-mini .pedal-pad {
    display: none;
  }

  body.race-live.control-mini .direction-pad {
    left: max(8px, env(safe-area-inset-left));
    bottom: max(10px, env(safe-area-inset-bottom));
    grid-template-columns: repeat(2, clamp(34px, 9vw, 46px));
    gap: 5px;
  }

  body.race-live.control-mini .mobile-control.steer-neutral {
    display: none;
  }
}
```

```

body.race-live.control-mini .mobile-control.steer-left,
body.race-live.control-mini .mobile-control.steer-right {
  width: clamp(34px, 9vw, 46px);
  height: clamp(34px, 9vw, 46px);
  border-radius: 999px;
  font-size: 0;
}

body.race-live.control-mini .mobile-control.steer-left::after,
body.race-live.control-mini .mobile-control.steer-right::after {
  font-size: clamp(17px, 4.5vw, 22px);
  line-height: 1;
}

body.race-live.control-mini .mobile-control.steer-left::after {
  content: "<";
}

body.race-live.control-mini .mobile-control.steer-right::after {
  content: ">";
}

body.race-live.control-mini .pedal-pad {
  right: max(8px, env(safe-area-inset-right));
  bottom: max(10px, env(safe-area-inset-bottom));
  grid-template-columns: clamp(36px, 10vw, 48px) clamp(42px, 12vw, 58px);
  grid-template-rows: clamp(30px, 8vw, 40px) clamp(34px, 9vw, 46px);
  gap: 5px;
}

body.race-live.control-mini .mobile-control.gas {
  width: 100%;
  height: 100%;
  border-radius: 999px;
  font-size: clamp(10px, 2.7vw, 12px);
}

body.race-live.control-mini .mobile-control.brake,
body.race-live.control-mini .mobile-control.boost {
  border-radius: 999px;
  font-size: 0;
}

body.race-live.control-mini .mobile-control.brake::after,
body.race-live.control-mini .mobile-control.boost::after {
  font-size: clamp(9px, 2.7vw, 11px);
  line-height: 1;
}

body.race-live.control-mini .mobile-control.brake::after {
  content: "Brake";
}

```

styles.css continued 9/10

```
body.race-live.control-mini .mobile-control.boost::after {
  content: "Boost";
}

body.race-live.control-mini .control-size {
  left: max(13px, env(safe-area-inset-left));
  bottom: calc(max(10px, env(safe-area-inset-bottom)) + 146px);
  width: 52px;
  height: 30px;
  transform: none;
  opacity: 0.88;
}

@media (orientation: landscape) and (max-width: 980px) and (max-height: 760px), (orientation: landscape) and (pointer: coarse) {
  body.race-live.control-mini .drive-stick {
    grid-template-columns: repeat(3, 36px);
    grid-template-rows: repeat(3, 36px);
    width: 120px;
    height: 120px;
  }

  body.race-live.control-mini .drive-stick .mobile-control {
    width: 36px;
    height: 36px;
    font-size: 16px;
  }

  body.race-live.control-mini .control-size {
    bottom: calc(max(10px, env(safe-area-inset-bottom)) + 128px);
  }

  body.race-live.control-mini .direction-pad {
    grid-template-columns: repeat(2, clamp(34px, 10dvh, 44px));
  }

  body.race-live.control-mini .mobile-control.steer-left,
  body.race-live.control-mini .mobile-control.steer-right {
    width: clamp(34px, 10dvh, 44px);
    height: clamp(34px, 10dvh, 44px);
  }

  body.race-live.control-mini .pedal-pad {
    grid-template-columns: clamp(36px, 10dvh, 48px) clamp(42px, 12dvh, 56px);
    grid-template-rows: clamp(30px, 8dvh, 38px) clamp(34px, 10dvh, 46px);
  }
}

@media (orientation: portrait) and (max-width: 980px) {
  body.race-live.control-mini .direction-pad {
    bottom: max(10px, env(safe-area-inset-bottom));
  }

  body.race-live.control-mini .pedal-pad {
    bottom: max(10px, env(safe-area-inset-bottom));
  }

  body.race-live.control-mini .control-size {
    bottom: calc(max(10px, env(safe-area-inset-bottom)) + 146px);
  }
}

@media (max-width: 980px), (pointer: coarse) {
  body.race-live.control-mini .drive-stick {
    --stick-origin-x: 68px;
    --stick-origin-y: calc(var(--vvh, 100dvh) - 68px);
    --stick-knob-x: 0px;
    --stick-knob-y: 0px;
    display: block;
    width: 112px;
    height: 112px;
    padding: 0;
    left: max(12px, env(safe-area-inset-left));
    bottom: max(12px, env(safe-area-inset-bottom));
    grid-template-columns: none;
    grid-template-rows: none;
    overflow: visible;
    opacity: 0.78;
    transition: opacity 140ms ease, filter 140ms ease;
  }

  body.race-live.control-mini .drive-stick.floating-stick-active {
    left: var(--stick-origin-x);
    top: var(--stick-origin-y);
    bottom: auto;
    transform: translate(-50%, -50%);
    opacity: 0.96;
    filter: brightness(1.12);
  }

  body.race-live.control-mini .drive-stick .mobile-control {
    position: absolute;
    inset: 0;
    width: 100%;
    height: 100%;
    opacity: 0;
    pointer-events: none;
  }

  body.race-live.control-mini .drive-stick::before,
  body.race-live.control-mini .drive-stick::after {
    position: absolute;
    content: "";
    pointer-events: none;
    border-radius: 999px;
  }

  body.race-live.control-mini .drive-stick::before {
    inset: 0;
    border: 1px solid rgba(244, 251, 248, 0.28);
    background:
      radial-gradient(circle at 50% 50%, rgba(187, 242, 74, 0.18), rgba(70, 217, 255, 0.1) 28%, rgba(5, 8, 7, 0.5) 66%),
      linear-gradient(135deg, rgba(255, 255, 255, 0.12), rgba(5, 8, 7, 0.18));
  }
}
```



```

    box-shadow: 0 16px 34px rgba(0, 0, 0, 0.36), inset 0 0 26px rgba(70, 217, 255, 0.12);
  }

body.race-live.control-mini .drive-stick::after {
  left: 50%;
  top: 50%;
  width: 46px;
  height: 46px;
  transform: translate(calc(-50% + var(--stick-knob-x)), calc(-50% + var(--stick-knob-y)));
  border: 1px solid rgba(244, 251, 248, 0.42);
  background:
    radial-gradient(circle at 36% 28%, rgba(255,255,255,0.72), rgba(255,255,255,0.14) 28%, rgba(187,242,74,0.72) 54%,
    rgba(70,217,255,0.72));
  box-shadow: 0 8px 22px rgba(0, 0, 0, 0.36), 0 0 18px rgba(70, 217, 255, 0.24);
}

body.race-live.control-mini .control-size {
  bottom: calc(max(10px, env(safe-area-inset-bottom)) + 120px);
}
}

@media (orientation: landscape) and (max-width: 990px) and (max-height: 760px), (orientation: landscape) and (pointer: coarse) {
  body.race-live.control-mini .drive-stick {
    width: 96px;
    height:

```

styles.css continued 10/10

```
96px;
  left: max(10px, env(safe-area-inset-left));
  bottom: max(10px, env(safe-area-inset-bottom));
}

body.race-live.control-mini .drive-stick::after {
  width: 38px;
  height: 38px;
}

body.race-live.control-mini .control-size {
  bottom: calc(max(10px, env(safe-area-inset-bottom)) + 104px);
}
}
```

game.js

```
"use strict";

const $ = (selector) => document.querySelector(selector);
const $$ = (selector) => Array.from(document.querySelectorAll(selector));

const canvas = $("#gameCanvas");
const ctx = canvas.getContext("2d");
const glCanvas = $("#glCanvas");

const storeKey = "velocityVaultProfilesV1";
const saveKey = "velocityVaultSavedRaceV1";
const starterGarageVersion = 76;
const raceDistanceMultiplier = 7.4;
const minimumRaceSeconds = 72;
const ageBands = {
  rookie: { label: "Rookie 5-8", help: "Wide lanes, bigger coin streaks, cheerful missions.", speed: 0.86, traffic: 0.72, rewards: 1.18 },
  prodigy: { label: "Prodigy 9-12", help: "Sharper goals, more traffic, better rewards.", speed: 1, traffic: 1, rewards: 1 },
  elite: { label: "Elite 13+", help: "Fast pace, tighter windows, premium rep bonuses.", speed: 1.17, traffic: 1.25, rewards: 1.15 },
  family: { label: "Family All", help: "Balanced settings made for pass-and-play.", speed: 0.96, traffic: 0.9, rewards: 1.05 }
};

const races = [
  { id: "neon", name: "Pacific Coast Highway Sprint", length: 3600, target: "Collect 18 coins", type: "coins", goal: 18, reward: 150, rep: 16, theme: ["#0c1a1b", "#46d9ff", "#bbf24a"], place: "coast", sign: "Pacific Coast", mood: "golden coast", vehicleTypes: ["car", "f1", "prototype", "truck"] },
  { id: "skyline", name: "Chicago Skyline Night Run", length: 4300, target: "Keep focus above 60", type: "focus", goal: 60, reward: 180, rep: 20, theme: ["#12151d", "#ffd166", "#46d9ff"], place: "city", sign: "Lakefront Loop", mood: "wet downtown", vehicleTypes: ["car", "f1", "prototype", "truck"] },
  { id: "canyon", name: "Sedona Red Rock Clash", length: 5000, target: "Dodge 30 rivals", type: "dodges", goal: 30, reward: 220, rep: 28, theme: ["#15110c", "#ff5b6b", "#ffd166"], place: "canyon", sign: "Sedona Route", mood: "desert chase", vehicleTypes: ["car", "truck", "monster"] },
  { id: "vault", name: "Rocky Mountain Grand Prix", length: 6200, target: "Score 5000 points", type: "score", goal: 5000, reward: 320, rep: 42, theme: ["#09100f", "#bbf24a", "#46d9ff"], place: "alpine", sign: "Mountain Pass", mood: "storm pass", vehicleTypes: ["car", "f1", "prototype", "truck"] },
  { id: "harbor", name: "Miami Harbor Boat Rush", length: 4200, target: "Score 3500 points", type: "score", goal: 3500, reward: 240, rep: 30, theme: ["#07161b", "#46d9ff", "#ffd166"], place: "harbor", sign: "Harbor Run", mood: "water sprint", vehicleTypes: ["boat"] },
  { id: "snowfield", name: "Aspen Snowmobile Dash", length: 3900, target: "Keep focus above 55", type: "focus", goal: 55, reward: 210, rep: 26, theme: ["#0b1519", "#f4fbf8", "#46d9ff"], place: "snow", sign: "Snow Pass", mood: "ice grip", vehicleTypes: ["snowmobile"] },
  { id: "airstrip", name: "Nevada Airfield Scramble", length: 5200, target: "Dodge 28 rivals", type: "dodges", goal: 28, reward: 280, rep: 36, theme: ["#11131a", "#ffd166", "#ff5b6b"], place: "airfield", sign: "Runway 4", mood: "air chase", vehicleTypes: ["airplane", "helicopter", "fighterjet", "drone"] },
  { id: "pursuit", name: "Metro Police Interceptor Run", length: 4700, target: "Keep focus above 64", type: "focus", goal: 64, reward: 260, rep: 34, theme: ["#0b1218", "#46d9ff", "#ff3348"], place: "pursuit", sign: "Interceptor Route", mood: "lights and sirens", unlock: 24, vehicleTypes: ["policecar"], hotPursuit: true },
  { id: "freight", name: "Texas Big Rig Freight Run", length: 5600, target: "Score 4200 points", type: "score", goal: 4200, reward: 270, rep: 34, theme: ["#10140f", "#ffd166", "#46d9ff"], place: "freight", sign: "Texas Freightway", mood: "big rig draft", unlock: 0, vehicleTypes: ["semi", "truck"] },
  { id: "interstate", name: "Great Plains Semi Convoy Trial", length: 6500, target: "Keep focus above 58", type: "focus", goal: 58, reward: 340, rep: 42, theme: ["#101712", "#ffd166", "#46d9ff"], place: "interstate", sign: "Convoy Trial", mood: "diesel endurance", unlock: 32, vehicleTypes: ["semi"] },
  { id: "farmrally", name: "Iowa Tractor Rally", length: 3300, target: "Collect 16 route markers", type: "coins", goal: 16, reward: 190, rep: 22, theme: ["#10180e", "#bbf24a", "#ffd166"], place: "farm", sign: "Iowa Backroads", mood: " " }
```

game.js continued 2/162

```
field sprint", unlock: 0, vehicleTypes: ["tractor", "truck"] },
  { id: "monsterjam", name: "Utah Monster Truck Dirt Arena", length: 4500, target: "Dodge 26 rivals", type: "dodges", goal: 26,
    reward: 300, rep: 36, theme: ["#140f0b", "#ff8c42", "#bbf24a"], place: "monsterpark", sign: "Dirt Arena", mood: "mud and stadium
    lights", unlock: 36, vehicleTypes: ["monster"] },
  { id: "armory", name: "Nevada Armor Proving Ground", length: 5200, target: "Score 4300 points", type: "score", goal: 4300, reward:
    350, rep: 46, theme: ["#10130e", "#bbf24a", "#ffd166"], place: "military", sign: "Proving Ground", mood: "armored range", unlock:
    46, vehicleTypes: ["tank"] },
  { id: "skycommand", name: "High Desert Sky Command", length: 6700, target: "Dodge 36 rivals", type: "dodges", goal: 36, reward:
    390, rep: 54, theme: ["#07111a", "#46d9ff", "#ffd166"], place: "skybase", sign: "Sky Command", mood: "jets drones and bomber lanes",
    unlock: 74, vehicleTypes: ["fighterjet", "bomber", "drone", "airplane"] },
  { id: "tokyo", name: "Tokyo Neon Expressway", length: 5700, target: "Keep focus above 62", type: "focus", goal: 62, reward: 310,
    rep: 40, theme: ["#100d1c", "#ff4fd8", "#46d9ff"], place: "tokyo", sign: "Tokyo Express", mood: "neon tunnel", unlock: 44,
    vehicleTypes: ["car", "f1", "prototype"] },
  { id: "sahara", name: "Sahara Desert Rally", length: 6100, target: "Dodge 34 rivals", type: "dodges", goal: 34, reward: 330, rep:
    44, theme: ["#160f0a", "#ffb74a", "#f4fbf8"], place: "desert", sign: "Sahara Rally", mood: "sand storm", unlock: 66, vehicleTypes:
    ["car", "truck", "monster", "tank"] },
  { id: "rainforest", name: "Amazon Rainforest Rush", length: 5400, target: "Score 4600 points", type: "score", goal: 4600, reward:
    320, rep: 42, theme: ["#07130d", "#36d98a", "#ffd166"], place: "rainforest", sign: "Amazon Route", mood: "jungle rain", unlock: 88,
    vehicleTypes: ["truck", "monster", "tractor", "car"] },
  { id: "eurotour", name: "Swiss Alps Grand Tour", length: 6400, target: "Score 5600 points", type: "score", goal: 5600, reward:
    360, rep: 48, theme: ["#091119", "#dce8ef", "#46d9ff"], place: "europe", sign: "Swiss Alps", mood: "euro pass", unlock: 110,
    vehicleTypes: ["car", "f1", "prototype", "airplane", "helicopter"] }
];

const vehicleDefs = [
  { id: "street", name: "Street Supercar", type: "car", desc: "Fast all-around modern racing feel.", speed: 1, handling: 1, mass: 1,
    color: "#1bb7e8", price: 0, unlock: 0, class: "road" },
  { id: "rally", name: "Rally Coupe", type: "car", desc: "Widebody dirt-road racer with fast lane changes.", speed: 0.96, handling:
    1.16, mass: 0.96, color: "#ff8c42", price: 1800, unlock: 28, class: "road", model: "Rally widebody" },
  { id: "drift", name: "Drift Street Coupe", type: "car", desc: "Slide-focused city build with extra drift control.", speed: 0.98,
    handling: 1.24, mass: 0.92, color: "#ff4fd8", price: 2200, unlock: 38, class: "road", model: "Drift kit" },
  { id: "interceptor", name: "Police Interceptor", type: "policecar", desc: "Pursuit-class cruiser with lightbar heat control and
    upgradeable chase grip.", speed: 1.02, handling: 1.06, mass: 1.16, color: "#dce8ef", price: 2800, unlock: 34, class: "pursuit",
    model: "Interceptor cruiser" },
  { id: "pursuitsuv", name: "Pursuit SUV", type: "policecar", desc: "Heavier pursuit build with stronger contact resistance and
    upgradeable torque.", speed: 0.9, handling: 0.84, mass: 1.48, color: "#101820", price: 3800, unlock: 56, class: "pursuit", model:
    "Pursuit utility" },
  { id: "f1", name: "F1 Open-wheel", type: "f1", desc: "Sharp steering and high top speed.", speed: 1.18, handling: 1.18, mass:
    0.72, color: "#ff3348", price: 2600, unlock: 45, class: "road" },
  { id: "grandprix", name: "Grand Prix Prototype", type: "prototype", desc: "Stable, fast, low race body.", speed: 1.12, handling:
    1.08, mass: 0.82, color: "#f4fbf8", price: 3200, unlock: 58, class: "road" },
  { id: "truck", name: "Performance Truck", type: "truck", desc: "Heavier, stable, strong contact resistance.", speed: 0.88,
    handling: 0.82, mass: 1.35, color: "#ffd166", price: 1200, unlock: 20, class: "road" },
  { id: "buggy", name: "Desert Buggy", type: "truck", desc: "Light off-road machi
```

game.js continued 3/162

```
ne built for dunes and shortcut branches.", speed: 0.92, handling: 1.04, mass: 0.88, color: "#ffb74a", price: 2400, unlock: 50,
class: "rally", model: "Open desert buggy" },
{ id: "semi", name: "Semi Truck Racer", type: "semi", desc: "Huge highway pull, heavy drafting, upgrade into a freight rocket.",
speed: 0.72, handling: 0.58, mass: 2.1, color: "#dce8ef", price: 3400, unlock: 60, class: "freight" },
{ id: "aerosemi", name: "Aero Cab Semi", type: "semi", desc: "Streamlined convoy racer with stronger highway speed and upgrade
headroom.", speed: 0.84, handling: 0.62, mass: 1.95, color: "#46d9ff", price: 5200, unlock: 86, class: "freight", model: "Aero cab"
},
{ id: "hauler", name: "Heavy Hauler Semi", type: "semi", desc: "Big torque freight build for long interstate races and heavy
contact.", speed: 0.66, handling: 0.52, mass: 2.45, color: "#ffdi66", price: 6400, unlock: 112, class: "freight", model: "Long-haul
sleeper" },
{ id: "tractor", name: "Racing Tractor", type: "tractor", desc: "Farm rally machine with tough grip and upgradeable agility.",
speed: 0.62, handling: 0.7, mass: 1.65, color: "#36d98a", price: 1400, unlock: 22, class: "farm" },
{ id: "monster", name: "Monster Truck", type: "monster", desc: "Huge stance, slower but tough.", speed: 0.76, handling: 0.7, mass:
1.75, color: "#bbf24a", price: 4200, unlock: 72, class: "monster" },
{ id: "stadiummonster", name: "Stadium Monster Truck", type: "monster", desc: "Lighter arena monster with quicker steering for
dirt stadium races.", speed: 0.84, handling: 0.82, mass: 1.55, color: "#ff8c42", price: 5600, unlock: 96, class: "monster", model:
"Stadium lift" },
{ id: "crawler", name: "Rock Crawler Monster", type: "monster", desc: "Heavy lifted build for rough desert and crush-car
sections.", speed: 0.68, handling: 0.9, mass: 1.95, color: "#bbf24a", price: 6800, unlock: 122, class: "monster", model: "Crawler
cage" },
{ id: "tank", name: "Armored Tank", type: "tank", desc: "Slow, heavy, almost unstoppable.", speed: 0.56, handling: 0.52, mass:
2.35, color: "#6d7667", price: 5200, unlock: 92, class: "military" },
{ id: "scouttank", name: "Scout Tank", type: "tank", desc: "Quicker armored racer with lighter plating and sharper range
handling.", speed: 0.64, handling: 0.62, mass: 2.02, color: "#8d986f", price: 6200, unlock: 116, class: "military", model: "Recon
armor" },
{ id: "heavytank", name: "Heavy Battle Tank", type: "tank", desc: "Bulkier armor, slower turn-in, stronger route-clearing hits.",
speed: 0.48, handling: 0.46, mass: 2.75, color: "#3f4a38", price: 7600, unlock: 138, class: "military", model: "Heavy armor" },
{ id: "snowmobile", name: "Snowmobile", type: "snowmobile", desc: "Light and quick on snow routes.", speed: 0.94, handling: 1.15,
mass: 0.58, color: "#f4fbf8", price: 1600, unlock: 26, class: "snow" },
{ id: "boat", name: "Race Boat", type: "boat", desc: "Best fit for harbor water sprints.", speed: 1.05, handling: 0.86, mass:
0.92, color: "#46d9ff", price: 1900, unlock: 30, class: "water" },
{ id: "hydro", name: "Hydroplane Racer", type: "boat", desc: "Higher-speed water rocket for harbor and marina cuts.", speed: 1.18,
handling: 0.78, mass: 0.78, color: "#6fffe9", price: 3600, unlock: 64, class: "water", model: "Hydroplane" },
{ id: "patrolboat", name: "Armored Patrol Boat", type: "boat", desc: "Heavy water build for chase scenarios and rough wakes.",
speed: 0.88, handling: 0.74, mass: 1.35, color: "#dce8ef", price: 4600, unlock: 82, class: "water", model: "Patrol hull" },
{ id: "offshorecat", name: "Offshore Catamaran", type: "boat", desc: "Twin-hull speed boat for rough wake racing and long harbor
sprints.", speed: 1.25, handling: 0.72, mass: 0.86, color: "#ff8c42", price: 5600, unlock: 104, class: "water", model: "Twin hull"
},
{ id: "helicopter", name: "Pursuit Helicopter", type: "helicopter", desc: "Air-style handling with wide steering.", speed: 0.92,
handling: 0.98, mass: 0.9, color: "#dce8ef", price: 6000, unlock: 110, class: "air" },
{ id: "attackheli", name: "Attack Helicopter", type: "helicopter", desc: "Faster air mission build with stronger pursuit
handling.", speed: 1.02, handling: 0.9, mass: 1.05, color: "#6d
```

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```
7667", price: 7000, unlock: 145, class: "air", model: "Attack rotor" },
  { id: "racingheli", name: "Racing Helicopter", type: "helicopter", desc: "Light rotorcraft tuned for faster sky gates and sharper air slaloms.", speed: 1.1, handling: 1.02, mass: 0.76, color: "#fffd166", price: 8200, unlock: 152, class: "air", model: "Racing rotor" },
  { id: "airplane", name: "Sport Airplane", type: "airplane", desc: "Fast runway and airfield races.", speed: 1.22, handling: 0.78, mass: 0.74, color: "#ff5b6b", price: 7000, unlock: 128, class: "air" },
  { id: "jetplane", name: "Jet Stunt Plane", type: "airplane", desc: "High-speed sky racer for long airfield and mountain routes.", speed: 1.34, handling: 0.68, mass: 0.82, color: "#f4fbf8", price: 8800, unlock: 158, class: "air", model: "Jet stunt wing" },
  { id: "falconfighter", name: "Falcon Fighter Jet", type: "fighterjet", desc: "Fast tactical jet with boost-heavy air racing and upgradeable turn response.", speed: 1.42, handling: 0.74, mass: 0.9, color: "#dce8ef", price: 11200, unlock: 182, class: "air", model: "Fighter jet" },
  { id: "nightfighter", name: "Night Fighter Jet", type: "fighterjet", desc: "Sleek high-speed fighter build with stronger stealth-style handling upgrades.", speed: 1.48, handling: 0.68, mass: 0.98, color: "#3f4a60", price: 13800, unlock: 214, class: "air", model: "Night fighter" },
  { id: "bomber", name: "Heavy Bomber", type: "bomber", desc: "Large air racer with slower turns, huge stability, and upgradeable thrust.", speed: 1.08, handling: 0.52, mass: 1.85, color: "#6d7667", price: 12600, unlock: 198, class: "air", model: "Heavy bomber" },
},
{ id: "surveydrone", name: "Survey Drone", type: "drone", desc: "Small quad drone for tight sky gates and precision upgrade builds.", speed: 0.96, handling: 1.36, mass: 0.34, color: "#46d9ff", price: 4200, unlock: 72, class: "air", model: "Quad drone" },
{ id: "racingdrone", name: "Racing Drone", type: "drone", desc: "Ultra-agile drone tuned for rapid lane changes and gate runs.", speed: 1.16, handling: 1.48, mass: 0.28, color: "#bbf24a", price: 6600, unlock: 118, class: "air", model: "Racing quad" }
];

const opponentNames = ["Vega", "Knox", "Ryder", "Nova", "Sable", "Mako", "Jett", "Blitz"];

const upgradeDefs = [
  { id: "engine", name: "Ion Engine", desc: "Higher top speed and faster score flow.", base: 320 },
  { id: "tires", name: "Grip Tires", desc: "Quicker steering and smoother dodges.", base: 280 },
  { id: "shield", name: "Pulse Shield", desc: "Protects focus during collisions.", base: 360 },
  { id: "magnet", name: "Coin Magnet", desc: "Pulls coins from wider lanes.", base: 300 },
  { id: "boost", name: "Clean Boost", desc: "Longer boost bursts with less focus drain.", base: 340 }
];

const paintPalette = ["#1bb7e8", "#ff3348", "#ff8c42", "#fffd166", "#bbf24a", "#f4fbf8", "#ff4fd8", "#36d98a", "#6fffe9", "#dce8ef", "#6d7667", "#3f4a38"];
const pickupLaneSlots = [-1.42, -0.72, 0, 0.72, 1.42];
const airVehicleTypes = ["airplane", "helicopter", "fighterjet", "bomber", "drone"];
const waterVehicleTypes = ["boat"];

function vehicleTypeOf(vehicleOrType) {
  return typeof vehicleOrType === "string" ? vehicleOrType : vehicleOrType && vehicleOrType.type ? vehicleOrType.type : "";
}

function isAirVehicleType(vehicleOrType) {
  return airVehicleTypes.includes(vehicleTypeOf(vehicleOrType));
}

function isWaterVehicleType(vehicleOrType) {
  return waterVehicleTypes.includes(vehicleTypeOf(vehicleOrType));
}

const vehicleRaceRules = {
  car: { places: ["coast", "city", "canyon", "alpine", "tokyo", "desert", "rainforest", "europe"], label: "road circuits" },
  policecar: { places: ["pursuit", "city", "tokyo"], label: "police pursuit routes" },
  f1: { places: ["coast", "city", "alpine", "tokyo", "europe"], label: "road and grand prix circuits" },
  prototype: { places: ["coast", "city", "alpine", "tokyo", "europe"], label: "road and grand prix circuits" },
  truck: { places: ["coast", "city", "canyon", "alpine", "freight", "farm", "desert", "rainforest"], label: "road, freight, and rally routes" },
  semi: { places: ["freight", "interstate"], label: "semi and freight routes" },
  tractor: { places: ["farm", "rainfores"]
}
```

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```
t"], label: "farm rally routes" },
monster: { places: ["monsterpark", "canyon", "desert", "rainforest"], label: "monster truck arenas" },
tank: { places: ["military", "desert"], label: "armored proving routes" },
snowmobile: { places: ["snow"], label: "snow routes" },
boat: { places: ["harbor"], label: "water routes" },
helicopter: { places: ["airfield", "skybase", "desert", "rainforest", "europe", "military"], label: "sky routes" },
airplane: { places: ["airfield", "skybase", "desert", "europe"], label: "sky routes" },
fighterjet: { places: ["airfield", "skybase", "military", "desert"], label: "fighter jet sky routes" },
bomber: { places: ["skybase", "military", "desert"], label: "bomber sky routes" },
drone: { places: ["airfield", "skybase", "military", "rainforest", "europe"], label: "drone sky routes" }
};

const multiplayerScenarios = [
  { id: "solo", name: "Solo Race", desc: "Standard race against the route pack.", allies: 0, extraOpponents: 0, heatBoost: 0,
    reward: 1 },
  { id: "ghostduel", name: "Local Ghost Duel", desc: "Adds a Player 2 ghost and one extra rival for pass-and-play style races.",
    allies: 1, extraOpponents: 1, heatBoost: 0.02, reward: 1.05 },
  { id: "crewrelay", name: "Crew Relay", desc: "Adds two local ghost teammates and a bigger rival field.", allies: 2,
    extraOpponents: 2, heatBoost: 0.04, reward: 1.1 },
  { id: "pursuitcrew", name: "Hot Pursuit Crew", desc: "Crew race with rising police pressure over longer routes.", allies: 1,
    extraOpponents: 2, hotPursuit: true, heatBoost: 0.22, reward: 1.18 }
];

const missionDefs = [
  { id: "first", text: "Finish your first race", reward: 90, test: (p) => p.stats.races >= 1 },
  { id: "collector", text: "Collect 75 total coins", reward: 130, test: (p) => p.stats.totalCoins >= 75 },
  { id: "steady", text: "Complete 3 races with focus above 50", reward: 190, test: (p) => p.stats.steadyRuns >= 3 },
  { id: "vault", text: "Earn 120 reputation", reward: 260, test: (p) => p.rep >= 120 }
];

const directorEvents = [
  { id: "coinrush", name: "AI Coin Rush", focus: "more coin gates", coinRate: 1.42, traffic: 0.92, reward: 1.12 },
  { id: "draftline", name: "AI Draft Line", focus: "speed and clean dodges", coinRate: 0.95, traffic: 1.1, reward: 1.18 },
  { id: "shieldlab", name: "AI Shield Lab", focus: "recovery after impacts", coinRate: 1.06, traffic: 0.86, reward: 1.08 },
  { id: "neonmix", name: "AI Neon Mix", focus: "balanced variety", coinRate: 1.14, traffic: 1, reward: 1.1 }
];

let profiles = loadProfiles();
let selectedAge = "rookie";
let activeProfile = null;
let selectedRace = races[0];
let selectedScenario = scenarioById(localStorage.getItem("velocityVaultScenarioMode") || "solo");
let view = "login";
let tab = "races";
let toastTimer = 0;
let deferredInstallPrompt = null;
let cameraMode = localStorage.getItem("velocityVaultCameraMode") || "chase";
let rendererMode = localStorage.getItem("velocityVaultRendererMode") || "webgl";
let touchDriveMode = localStorage.getItem("velocityVaultTouchDriveMode") === "toggle" ? "toggle" : "hold";
let touchControlSize = localStorage.getItem("velocityVaultTouchControlSize") === "full" ? "full" : "mini";
let webglRenderer = null;
let audioSystem = null;

const input = {
  left: false,
  right: false,
  gas: false,
  brake: false,
  boost: false,
  paused: false,
  touchSteer: 0,
  gamepadSteer: 0,
  gamepadGas: false,
  gamepadBrake: false,
  gamepadBoost: false,
  gamepadPauseHeld: false,
  gamepadResetHeld: false
};

const controllerState = {
  connected: false,
  name: "",
  index: null
};

const mobileTouchState = {
  active: new Map(),
  usingTouchEvents: false,
  stickSteer: 0,
  driveStickActive: false,
  latchedStickControl: "",
  modeToggleAt: 0,
  floatingStickId: null,
  floatingStickKey: "",
  floatingStickOriginX: 0,
  floatingStickOriginY: 0,
  floatingStickKnobX: 0,
  floatingStickKnobY: 0
};

const raceState = {
  active: false,
  distance: 0,
  speed: 0,
  focus: 100,
  score: 0,
  coins: 0,
  dodges: 0,
  overtakes: 0,
  passesAgainst: 0,
  civilianHits: 0,
  penaltyCoins: 0,
  penaltyRep: 0,
  combo: 1,
  lane: 0,
  x: 0,
  visualLane: 0,
  lateralVelocity: 0,
  steerAngle: 0,
  throt
```

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```
    tleLoad: 0,
    brakeHeat: 0,
    slip: 0,
    damage: 0,
    damageAlertTimer: 0,
    damageAlertLabel: "",
    hazardWarningTimer: 0,
    hazardWarningLabel: "",
    resetTimer: 0,
    resetReason: "",
    crashCooldown: 0,
    roadCurve: 0,
    roadTurn: 0,
    roadOffset: 0,
    spawnClock: 0,
    coinClock: 0,
    civilianClock: 0,
    oncomingClock: 0,
    cannonCooldown: 0,
    routeFeatureClock: 0,
    hideCooldown: 0,
    scenarioId: "solo",
    scenarioLabel: "Solo Race",
    teamScore: 0,
    rivals: [],
    police: [],
    opponents: [],
    coinsOnRoad: [],
    civilians: [],
    oncoming: [],
    routeFeatures: [],
    particles: [],
    elapsed: 0,
    heat: 0,
    heatClock: 0,
    autoSaveClock: 0,
    chaseActive: false,
    cameraShake: 0,
    countdown: 0,
    goalIntroTimer: 0,
    finished: false,
    director: null
};

function loadProfiles() {
  try {
    const raw = localStorage.getItem(storeKey);
    return raw ? JSON.parse(raw) : {};
  } catch {
    return {};
  }
}

function saveProfiles() {
  localStorage.setItem(storeKey, JSON.stringify(profiles));
}

function selectedVehicle() {
  const id = activeProfile && activeProfile.selectedVehicle ? activeProfile.selectedVehicle : "street";
  return vehicleDefs.find((vehicle) => vehicle.id === id) || vehicleDefs[0];
}

function vehicleById(id) {
  return vehicleDefs.find((vehicle) => vehicle.id === id) || vehicleDefs[0];
}

function scenarioById(id) {
  return multiplayerScenarios.find((scenario) => scenario.id === id) || multiplayerScenarios[0];
}

function activeScenario() {
  selectedScenario = scenarioById(selectedScenario && selectedScenario.id);
  return selectedScenario;
}

function defaultUpgrades() {
  return { engine: 0, tires: 0, shield: 0, magnet: 0, boost: 0 };
}

function defaultBuild(vehicle = vehicleDefs[0]) {
  return {
    color: vehicle.color,
    trim: "#050807",
    upgrades: defaultUpgrades(),
    plate: ""
  };
};

function normalizeBuild(vehicle, build, legacyUpgrades = null) {
  const base = defaultBuild(vehicle);
  const source = build && typeof build === "object" ? build : {};
  const upgrades = Object.assign(
    base.upgrades,
    legacyUpgrades && typeof legacyUpgrades === "object" ? legacyUpgrades : {},
    source.upgrades || {}
  );
  Object.keys(upgrades).forEach((key) => {
    upgrades[key] = Math.max(0, Math.min(5, Number(upgrades[key]) || 0));
  });
  return {
    color: source.color || base.color,
    trim: source.trim || base.trim,
    upgrades,
    plate: sanitizePlate(source.plate || "")
  };
};

function sanitizePlate(value) {
  const source = String(value || "").toUpperCase().replace(/[^A-Z0-9_-]/g, "");
  return source.slice(0, 8);
}

function driverPlate(profile = activeProfile) {
```



```

    if (!profile) return "RACER";
    const build = vehicleBuild(profile, profile.selectedVehicle || "street");
    return sanitizePlate(build.plate || profile.name || "RACER");
}

function vehicleBuild(profile = activeProfile, vehicleId = null) {
    const vehicle = vehicleById(vehicleId || (profile && profile.selectedVehicle) || "street");
    if (!profile) return defaultBuild(vehicle);
    profile.vehicleBuilds = profile.vehicleBuilds && typeof profile.vehicleBuilds === "object" ? profile.vehicleBuilds : {};
    const legacy = vehicleOwned(profile, vehicle.id) ? profile.upgrades : null;
    profile.vehicleBuilds[vehicle.id] = normalizeBuild(vehicle, profile.vehicleBuilds[vehicle.id], legacy);
    return profile.vehicleBuilds[vehicle.id];
}

function activeVehicleUpgrades(profile = activeProfile) {
    return vehicleBuild(profile, profile && profile.selectedVehicle).upgrades;
}

function selectedVehicleColor(vehicle = selectedVehicle()) {
    return vehicleBuild(activeProfile, vehicle.id).color || vehicle.color;
}

function vehicleOwned(profile = activeProfile, vehicleId = "street") {
    if (!profile) return vehicleId === "street";
    const owned = Array.isArray(profile.ownedVehicles) ? profile.ownedVehicles : ["street"];
    return owned.includes(vehicleId);
}

function vehicleUnlocked(profile = activeProfile, vehicle = vehicleDefs[0]) {
    return !vehicle.unlock || (profile && profile.rep >= vehicle.unlock);
}

function raceCompatibleWithVehicle(race = selectedRace, vehicle = selectedVehicle()) {
    if (!race || !vehicle) return false;

```

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```
if (Array.isArray(race.vehicleTypes) && !race.vehicleTypes.includes(vehicle.type)) return false;
if (Array.isArray(race.vehicleClasses) && !race.vehicleClasses.includes(vehicle.class)) return false;
const rule = vehicleRaceRules[vehicle.type] || vehicleRaceRules.car;
return rule.places.includes(race.place);
}

function raceVehicleLabel(race) {
  if (!race || !Array.isArray(race.vehicleTypes) || !race.vehicleTypes.length) return "open class";
  const names = race.vehicleTypes.map((type) => {
    if (type === "semi") return "semis";
    if (type === "monster") return "monster trucks";
    if (type === "tank") return "tanks";
    if (type === "boat") return "boats";
    if (type === "airplane") return "planes";
    if (type === "helicopter") return "helicopters";
    if (type === "fighterjet") return "fighter jets";
    if (type === "bomber") return "bombers";
    if (type === "drone") return "drones";
    if (type === "policecar") return "police cars";
    if (type === "tractor") return "tractors";
    if (type === "snowmobile") return "snowmobiles";
    if (type === "f1") return "F1";
    return `${type}s`;
  });
  return names.length > 3 ? `${names.slice(0, 3).join(", ")} + ` : names.join(", ");
}

function compatibilityLabel(vehicle = selectedVehicle()) {
  const rule = vehicleRaceRules[vehicle.type] || vehicleRaceRules.car;
  return rule.label;
}

function firstCompatibleRace(vehicle = selectedVehicle(), profile = activeProfile) {
  return races.find((race, index) => {
    const requiredRep = race.unlock ?? index * 22;
    return (!profile || profile.rep >= requiredRep) && raceCompatibleWithVehicle(race, vehicle);
  }) || races[0];
}

function buyOrSelectVehicle(vehicle) {
  if (!activeProfile) return;
  if (!vehicleUnlocked(activeProfile, vehicle)) {
    showToast(`${vehicle.name} needs ${vehicle.unlock} reputation.`);
    return;
  }
  activeProfile.ownedVehicles = Array.isArray(activeProfile.ownedVehicles) ? activeProfile.ownedVehicles : ["street"];
  if (!vehicleOwned(activeProfile, vehicle.id)) {
    if (activeProfile.coins < vehicle.price) {
      showToast(`${vehicle.name} costs ${vehicle.price} coins.`);
      return;
    }
    activeProfile.coins -= vehicle.price;
    activeProfile.ownedVehicles.push(vehicle.id);
    activeProfile.vehicleBuilds = activeProfile.vehicleBuilds && typeof activeProfile.vehicleBuilds === "object" ?
    activeProfile.vehicleBuilds : {};
    activeProfile.vehicleBuilds[vehicle.id] = normalizeBuild(vehicle, null, null);
    activeProfile.vehicleBuilds[vehicle.id].plate = sanitizePlate(activeProfile.name);
    showToast(`${vehicle.name} added to your garage.`);
  } else if (activeProfile.selectedVehicle === vehicle.id) {
    cycleVehiclePaint(vehicle);
    return;
  }
  activeProfile.selectedVehicle = vehicle.id;
  const build = vehicleBuild(activeProfile, vehicle.id);
  activeProfile.upgrades = Object.assign(defaultUpgrades(), build.upgrades);
  if (!raceCompatibleWithVehicle(selectedRace, vehicle)) selectedRace = firstCompatibleRace(vehicle, activeProfile);
  profiles[activeProfile.id] = activeProfile;
  saveProfiles();
  renderHub();
  updateHud();
  showToast(`${vehicle.name} selected for ${compatibilityLabel(vehicle)}.`);
}

function cycleVehiclePaint(vehicle = selectedVehicle()) {
  if (!activeProfile) return;
  const build = vehicleBuild(activeProfile, vehicle.id);
  const current = build.color || vehicle.color;
  const currentIndex = paintPalette.indexOf(current);
  build.color = paintPalette[(currentIndex + 1 + paintPalette.length) % paintPalette.length];
  profiles[activeProfile.id] = activeProfile;
  saveProfiles();
  renderHub();
  showToast(`${vehicle.name} paint changed.`);
}

function raceLength(race = selectedRace) {
  const baseLength = race && race.length ? race.length : races[0].length;
  return Math.round(baseLength * raceDistanceMultiplier);
}

const routeWorlds = {
  coast: { country: "USA", scene: "Pacific Coast", cue: "ocean cliffs", turn: 1.05, seed: 1.2 },
  city: { country: "USA", scene: "Chicago Loop", cue: "wet downtown", turn: 0.95, seed: 2.4 },
  canyon: { country: "USA", scene: "Sedona Canyon", cue: "red rock sweepers", turn: 1.24, seed: 3.6 },
  alpine: {
```

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```
country: "USA", scene: "Rocky Mountain Pass", cue: "storm pass", turn: 1.32, seed: 4.8 },
harbor: { country: "USA", scene: "Miami Harbor", cue: "bridge run", turn: 0.92, seed: 5.7 },
snow: { country: "USA", scene: "Aspen Snowfields", cue: "ice bends", turn: 1.08, seed: 6.9 },
airfield: { country: "USA", scene: "Nevada Airfield", cue: "runway chicanes", turn: 0.72, seed: 7.4 },
pursuit: { country: "USA", scene: "Metro Pursuit Grid", cue: "lights and sirens", turn: 1.08, seed: 7.82 },
freight: { country: "USA", scene: "Texas Freightway", cue: "wide interstate", turn: 0.82, seed: 8.1 },
interstate: { country: "USA", scene: "Great Plains Interstate", cue: "wide convoy lanes", turn: 0.62, seed: 8.45 },
farm: { country: "USA", scene: "Iowa Backroads", cue: "rolling farm roads", turn: 0.88, seed: 8.9 },
monsterpark: { country: "USA", scene: "Utah Dirt Arena", cue: "mud lanes and stadium banks", turn: 0.98, seed: 9.15 },
military: { country: "USA", scene: "Nevada Proving Ground", cue: "concrete range roads", turn: 0.74, seed: 9.35 },
skybase: { country: "USA", scene: "High Desert Sky Command", cue: "high-altitude gates", turn: 0.86, seed: 9.48 },
tokyo: { country: "Japan", scene: "Tokyo Expressway", cue: "neon overpasses", turn: 1.18, seed: 9.6 },
desert: { country: "Morocco", scene: "Sahara Rally", cue: "sand ridges", turn: 1.12, seed: 10.5 },
rainforest: { country: "Brazil", scene: "Amazon Rainforest", cue: "jungle tunnels", turn: 1.02, seed: 11.4 },
europe: { country: "Switzerland", scene: "Swiss Alps Tour", cue: "village switchbacks", turn: 1.36, seed: 12.2 }
};

const genAiSceneDesigns = {
  coast: { style: "coastal grand tour", surface: "#1b2424", accent: "#46d9ff", light: "#ffd166", props: ["surf", "cliff", "camera", "crowd", "brake"] },
  city: { style: "downtown street circuit", surface: "#151b1d", accent: "#46d9ff", light: "#f4fbf8", props: ["tower", "barrier", "crowd", "crosswalk", "brake"] },
  canyon: { style: "red-rock drift course", surface: "#2a1c14", accent: "#ff5b6b", light: "#ffd166", props: ["rock", "dust", "camera", "barrier", "brake"] },
  alpine: { style: "mountain pass race", surface: "#1b2727", accent: "#bbf24a", light: "#dce8ef", props: ["pine", "snowcap", "crowd", "guard", "brake"] },
  harbor: { style: "waterfront tunnel sprint", surface: "#14242a", accent: "#46d9ff", light: "#ffd166", props: ["crane", "boat", "barrier", "camera", "brake"] },
  snow: { style: "winter rally stage", surface: "#344141", accent: "#f4fbf8", light: "#46d9ff", props: ["snowbank", "pine", "flag", "crowd", "brake"] },
  airfield: { style: "runway speed trial", surface: "#252723", accent: "#ffd166", light: "#ff5b6b", props: ["hangar", "beacon", "plane", "barrier", "brake"] },
  pursuit: { style: "police interceptor chase grid", surface: "#121b20", accent: "#46d9ff", light: "#ff3348", props: ["trafficSignal", "roadblock", "policeHQ", "lightbarSign", "billboard"] },
  freight: { style: "freightway endurance race", surface: "#22241d", accent: "#ffd166", light: "#46d9ff", props: ["trailer", "gantry", "crowd", "barrier", "brake"] },
  interstate: { style: "semi convoy endurance trial", surface: "#20231d", accent: "#ffd166", light: "#46d9ff", props: ["trailer", "gantry", "truckStop", "weighStation", "dieselSign"] },
  farm: { style: "rural rally festival", surface: "#3d3424", accent: "#bbf24a", light: "#ffd166", props: ["barn", "field", "tractor", "crowd", "brake"] },
  monsterpark: { style: "monster truck dirt arena", surface: "#3c2414", accent: "#ff8c42", light: "#bbf24a", props: ["grandstand", "dirtBerm", "crushCarStack", "stadiumRamp", "scoreboard"] },
  military: { style: "armored proving ground", surface: "#2b3024", accent: "#bbf24a", light: "#ffd166", props: ["bunker", "watchTower", "rangeTarget", "blastWall", "checkpoint"] },
  skybase: { style: "high desert sky command", surface: "#172532", accent: "#46d9ff", light: "#ffd166", props: ["controlTower", "runwayLight", "hangar", "rangeTarget", "checkpoint"] },
  tokyo: { style: "neon expressway battle", surface: "#151424", accent: "#ff4fd8", light: "#46d9ff", props: ["neon", "tunnel", "tower", "crowd", "brake"] },
  desert: { style: "open desert rally"
```

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```
, surface: "#4b3321", accent: "#ffb74a", light: "#f4fbf8", props: ["dune", "dust", "camera", "barrier", "brake"] },
rainforest: { style: "jungle wet rally", surface: "#17241e", accent: "#36d98a", light: "#fffd16", props: ["canopy", "rain",
"bridge", "crowd", "brake"] },
europe: { style: "euro alpine tour", surface: "#263235", accent: "#dce8ef", light: "#46d9ff", props: ["village", "snowcap",
"guard", "crowd", "brake"] }
};

function routeWorldInfo(place = "city") {
  return routeWorlds[place] || routeWorlds.city;
}

function genAiSceneDesign(place = "city") {
  return genAiSceneDesigns[place] || genAiSceneDesigns.city;
}

function roadTurnAt(distance, race = selectedRace) {
  const info = routeWorldInfo(race && race.place ? race.place : "city");
  const d = Number(distance) || 0;
  const longSweep = Math.sin(d * 0.00034 + info.seed) * 0.82;
  const countryBend = Math.sin(d * 0.00072 + info.seed * 1.9) * 0.42;
  const shortSet = Math.sin(d * 0.00118 + info.seed * 0.58) * 0.18;
  const scenicSettle = Math.sin(d * 0.00009 + info.seed * 3.4) * 0.22;
  return Math.max(-1.32, Math.min(1.32, (longSweep + countryBend + shortSet + scenicSettle) * info.turn));
}

function routeTurnName(turn) {
  if (turn > 0.58) return "Right sweep";
  if (turn < -0.58) return "Left sweep";
  if (turn > 0.24) return "Right bend";
  if (turn < -0.24) return "Left bend";
  return "Fast straight";
}

function loadSavedRace() {
  try {
    const raw = localStorage.getItem(saveKey);
    return raw ? JSON.parse(raw) : null;
  } catch {
    return null;
  }
}

function clearSavedRace() {
  localStorage.removeItem(saveKey);
}

function makeProfile(name, pin, age) {
  return {
    id: `${age}:${name.toLowerCase()}`,
    name,
    pinHash: pin ? hashPin(pin) : "",
    age,
    coins: 1250,
    rep: 44,
    selectedVehicle: "street",
    upgrades: defaultUpgrades(),
    ownedVehicles: ["street"],
    vehicleBuilds: {
      street: Object.assign(defaultBuild(vehicleById("street")), { plate: sanitizePlate(name) })
    },
    completedMissions: [],
    stats: { races: 0, wins: 0, totalCoins: 0, steadyRuns: 0, bestScore: 0 },
    garageStarterVersion: starterGarageVersion
  };
}

function normalizeProfile(profile) {
  profile.coins = Number(profile.coins) || 0;
  profile.rep = Number(profile.rep) || 0;
  profile.selectedVehicle = profile.selectedVehicle || "street";
  profile.upgrades = Object.assign(defaultUpgrades(), profile.upgrades || {});
  profile.ownedVehicles = Array.isArray(profile.ownedVehicles) ? profile.ownedVehicles : ["street"];
  if (!profile.ownedVehicles.includes("street")) profile.ownedVehicles.unshift("street");
  profile.vehicleBuilds = profile.vehicleBuilds && typeof profile.vehicleBuilds === "object" ? profile.vehicleBuilds : {};
  vehicleDefs.forEach((vehicle) => {
    if (profile.ownedVehicles.includes(vehicle.id) || vehicle.id === profile.selectedVehicle) {
      profile.vehicleBuilds[vehicle.id] = normalizeBuild(vehicle, profile.vehicleBuilds[vehicle.id], profile.upgrades);
      if (!profile.vehicleBuilds[vehicle.id].plate) profile.vehicleBuilds[vehicle.id].plate = sanitizePlate(profile.name);
    }
  });
  if (!vehicleOwned(profile, profile.selectedVehicle)) profile.selectedVehicle = "street";
  profile.upgrades = Object.assign(defaultUpgrades(), vehicleBuild(profile, profile.selectedVehicle).upgrades);
  profile.completedMissions = Array.isArray(profile.completedMissions) ? profile.completedMissions : [];
  profile.stats = Object.assign({ races: 0, wins: 0, totalCoins: 0, steadyRuns: 0, bestScore: 0 }, profile.stats || {});
  if (profile.garageStarterVersion !== starterGarageVersion) {
    profile.coins = Math.max(profile.coins, 1250);
    profile.rep = Math.max(profile.rep, 44);
    profile.garageStarterVersion = starterGarageVersion;
  }
  return profile;
}

function hashPin(pin) {
  let hash = 2166136261;
  for (let i = 0; i < pin.length; i += 1) {
    hash ^= pin.charCodeAt(i);
    hash = Math.imul(hash, 16777619);
  }
  return (hash >>> 0).toString(36);
}

function sanitizeName(value) {
  return value.replace(/^[^a-zA-Z0-9_-]/g, "").slice(0, 16) || "TurboAce";
}

function showView(next) {
  view = next;
  $$(".view").forEach((e
```

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```
1) => el.classList.remove("active"));
  $('#${next}View').classList.add("active");
  document.body.classList.toggle("race-live", next === "race");
  if (next !== "race") clearTouchDriveInputs();
  setTouchDriveMode(touchDriveMode, true);
  syncViewportSize();
  fitCanvas();
}

function showToast(message) {
  const toast = $('#toast');
  toast.textContent = message;
  toast.classList.add("show");
  clearTimeout(toastTimer);
  toastTimer = setTimeout(() => toast.classList.remove("show"), 2300);
}

function startAudio() {
  if (audioSystem || !window.AudioContext && !window.webkitAudioContext) return;
  const AudioCtor = window.AudioContext || window.webkitAudioContext;
  const ctxAudio = new AudioCtor();
  const master = ctxAudio.createGain();
  master.gain.value = 0.045;
  master.connect(ctxAudio.destination);

  const engine = ctxAudio.createOscillator();
  const engineGain = ctxAudio.createGain();
  engine.type = "sawtooth";
  engine.frequency.value = 72;
  engineGain.gain.value = 0.18;
  engine.connect(engineGain);
  engineGain.connect(master);
  engine.start();

  const siren = ctxAudio.createOscillator();
  const sirenGain = ctxAudio.createGain();
  siren.type = "triangle";
  siren.frequency.value = 620;
  sirenGain.gain.value = 0;
  siren.connect(sirenGain);
  sirenGain.connect(master);
  siren.start();

  audioSystem = { ctx: ctxAudio, master, engine, engineGain, siren, sirenGain };
}

function updateAudio() {
  if (!audioSystem) return;
  const now = audioSystem.ctx.currentTime;
  const speed = raceState.active ? raceState.speed : 0;
  const boostInput = input.boost || input.gamepadBoost;
  audioSystem.engine.frequency.setTargetAtTime(64 + speed * 0.62 + (boostInput ? 70 : 0), now, 0.045);
  audioSystem.engineGain.gain.setTargetAtTime(raceState.active ? 0.12 + Math.min(0.18, speed / 1200) : 0.035, now, 0.08);
  const sirenLevel = raceState.active && raceState.chaseActive ? 0.14 : 0;
  const sirenSweep = 620 + Math.sin(raceState.elapsed * 7.2) * 210;
  audioSystem.siren.frequency.setTargetAtTime(sirenSweep, now, 0.03);
  audioSystem.sirenGain.gain.setTargetAtTime(sirenLevel, now, 0.06);
}

function playHitSound(kind = "impact") {
  if (!audioSystem) return;
  const now = audioSystem.ctx.currentTime;
  const osc = audioSystem.ctx.createOscillator();
  const gain = audioSystem.ctx.createGain();
  osc.type = kind === "coin" ? "sine" : "square";
  osc.frequency.value = kind === "coin" ? 880 : kind === "police" ? 150 : 210;
  gain.gain.setValueAtTime(kind === "coin" ? 0.07 : 0.11, now);
  gain.gain.exponentialRampToValueAtTime(0.001, now + (kind === "coin" ? 0.14 : 0.22));
  osc.connect(gain);
  gain.connect(audioSystem.master);
  osc.start(now);
  osc.stop(now + 0.24);
}

function profilePower(profile) {
  const build = vehicleBuild(profile, profile && profile.selectedVehicle);
  return Object.values(build.upgrades).reduce((sum, level) => sum + level, 0);
}

function directorDay() {
  return Math.floor(Date.now() / 86400000);
}

function getDirector(profile) {
  const day = directorDay();
  const seed = profile.name.length * 17 + profile.rep * 3 + profile.stats.races * 11 + day;
  const event = directorEvents[Math.abs(seed) % directorEvents.length];
  const upgrades = activeVehicleUpgrades(profile);
  let weakest = upgradeDefs[0];
  upgradeDefs.forEach((def) => {
    if (upgrades[def.id] < upgrades[weakest.id]) weakest = def;
  });
  const winRate = profile.stats.races ? profile.stats.wins / profile.stats.races : 0;
  const assist = winRate < 0.35 ? "extra reward support" : winRate > 0.7 ? "faster rival pacing" : "balanced variety";
  return {
    day,
    event,
    recommended: weakest,
    assist,
    reward: event.reward + (winRate < 0.35 ? 0.08 : 0),
    traffic: event.traffic + (winRate > 0.7 ? 0.08 : 0),
    coinRate: event.coinRate + (profile.stats.totalCoins < 40 ? 0.12 : 0)
  };
}

function renderDirector() {
  if (!activeProfile) return;
  const director = getDirector(activeProfile);
  const card = $('#directorCard');
  card.innerHTML = `
    <strong>${director.event.name}</strong>
    <span>Local agentic tuning is active for this driver. It studies only this profile's on-device race stats, then rotates events
  `;
}
```

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```
and upgrade nudges to keep the game fresh.</span>
<ul>
  <li>Today's focus: ${director.event.focus}</li>
  <li>Recommended upgrade: ${director.recommended.name}</li>
  <li>Adaptive assist: ${director.assist}</li>
  <li>Bonus rewards: ${Math.round((director.reward - 1) * 100)}%</li>
</ul>
`;
}

function upgradeCost(profile, def) {
  const level = activeVehicleUpgrades(profile)[def.id];
  return Math.round(def.base * Math.pow(1.92, level));
}

function setActiveProfile(profile) {
  activeProfile = profile;
  selectedRace = races[Math.min(profile.stats.races, races.length - 1)];
  if (!raceCompatibleWithVehicle(selectedRace, selectedVehicle())) selectedRace = firstCompatibleRace(selectedVehicle(), profile);
  renderHub();
  updateHud();
  showView("garage");
  showToast("Welcome, ${profile.name}. Garage unlocked.");
}

function enterProfile(openRace = false) {
  const name = sanitizeName($("#driverName").value);
  const pin = ($("#driverPin").value.replace(/D/g, "").slice(0, 6));
  const id = `${selectedAge}:${name.toLowerCase()}`;
  const existing = profiles[id];
  if (existing && existing.pinHash && existing.pinHash !== hashPin(pin)) {
    showToast("PIN did not match this local profile.");
    return null;
  }
  const profile = normalizeProfile(existing || makeProfile(name, pin, selectedAge));
  profiles[profile.id] = profile;
  saveProfiles();
  setActiveProfile(profile);
  if (openRace) {
    selectedRace = races[Math.min(profile.stats.races, races.length - 1)];
    if (!raceCompatibleWithVehicle(selectedRace, selectedVehicle())) selectedRace = firstCompatibleRace(selectedVehicle(), profile);
    launchRace();
  }
  return profile;
}

function renderHub() {
  if (!activeProfile) return;
  $("#driversSummary").textContent = `${ageBands[activeProfile.age].label} | ${activeProfile.coins} coins | ${activeProfile.rep} rep | ${selectedVehicle().name} | ${activeScenario().name} | Plate ${driverPlate(activeProfile)} | Power ${profilePower(activeProfile)}`;
  renderRaces();
  renderScenarios();
  renderVehicles();
  renderUpgrades();
  renderMissions();
  renderDirector();
  renderSavedRaceButton();
}

function renderSavedRaceButton() {
  const btn = $("#resumeRace");
  const saved = loadSavedRace();
  const canResume = saved && activeProfile && saved.profileId === activeProfile.id;
  btn.disabled = !canResume;
  if (canResume) {
    const race = races.find((item) => item.id === saved.selectedRaceId);
    btn.textContent = `Resume ${race ? race.name : "Saved Race"}`;
  } else {
    btn.textContent = saved ? "Saved Race For Another Driver" : "No Saved Race";
  }
}

function renderRaces() {
  const list = $("#raceList");
  list.innerHTML = "";
  const director = getDirector(activeProfile);
  const vehicle = selectedVehicle();
  races.forEach((race, index) => {
    const requiredRep = race.unlock ?? index * 22;
    const locked = activeProfile.rep < requiredRep;
    const incompatible = !raceCompatibleWithVehicle(race, vehicle);
    const card = document.createElement("button");
    card.type = "button";
    card.className = `race-card ${selectedRace.id === race.id ? "selected" : ""}`;
    card.disabled = locked || incompatible;
    card.innerHTML = `
      <div class="card-top"><strong>${race.name}</strong><span class="badge">${locked ? `${requiredRep} rep` : incompatible ? `Needs ${raceVehicleLabel(race)}` : `${Math.round(race.reward * director.reward)} coins`}</span></div>
      <div class="tiny">${race.target} | ${Math.round(raceLength(race) / 100)} sectors | ${routeWorldInfo(race.place).scene} | ${raceVehicleLabel(race)} | ${director.event.name}</div>
    `;
    card.addEventListener("click", () => {
      if (locked || incompatible) return;
      selectedRace = race;
      renderRaces();
    });
    list.appendChild(card);
  });
}

function renderScenarios() {
  const list = $("#scenarioList");
  if (!list) return;
  list.innerHTML = "";
  const current = activeScenario();
  multiplayerScenarios.forEach((scenario) => {
    const card = document.createElement("button");
    card.type = "button";
    card.className = `scenario-card ${current.id === scenario.id ? "selected" : ""}`;
    card.innerHTML = `

```

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```
= `
  <div class="card-top"><strong>${scenario.name}</strong><span class="badge">${scenario.hotPursuit ? "Pursuit" : scenario.allies
? "Local MP" : "Solo"}</span></div>
  <div class="tiny">${scenario.desc}</div>
  `;
  card.addEventListener("click", () => {
    selectedScenario = scenario;
    localStorage.setItem("velocityVaultScenarioMode", scenario.id);
    renderScenarios();
    renderHub();
    showToast(` ${scenario.name} selected.`);
  });
  list.appendChild(card);
});
}

function renderVehicles() {
  const list = $("#vehicleList");
  if (!list) return;
  list.innerHTML = "";
  const current = selectedVehicle();
  vehicleDefs.forEach((vehicle) => {
    const owned = vehicleOwned(activeProfile, vehicle.id);
    const unlocked = vehicleUnlocked(activeProfile, vehicle.id);
    const build = vehicleBuild(activeProfile, vehicle.id);
    const status = current.id === vehicle.id ? "Active" : owned ? "Select" : !unlocked ? `${vehicle.unlock} rep` : `${vehicle.price}
coins`;
    const card = document.createElement("button");
    card.type = "button";
    card.className = `vehicle-card ${current.id === vehicle.id ? "selected" : ""} ${!unlocked ? "locked" : ""}`;
    card.innerHTML = `
      <div class="vehicle-preview asset-preview" data-type="${vehicle.type}">
        <canvas class="vehicle-preview-canvas" width="180" height="116" aria-hidden="true"></canvas>
      </div>
      <div class="card-top"><strong>${vehicle.name}</strong><span class="badge">${status}</span></div>
      <div class="tiny">${vehicle.desc} | ${vehicle.model ? `Model ${vehicle.model}` : ""} | ${compatibilityLabel(vehicle)} | Plate
      ${build.plate} || sanitizePlate(activeProfile.name)} | Tap active vehicle to repaint</div>
    `;
    drawVehiclePreview(card.querySelector(".vehicle-preview-canvas"), Object.assign({}, vehicle, { color: build.color ||
vehicle.color }));
    card.addEventListener("click", () => buyOrSelectVehicle(vehicle));
    list.appendChild(card);
  });
}

function drawVehiclePreview(canvasEl, vehicle) {
  if (!canvasEl || !vehicle) return;
  const previewCtx = canvasEl.getContext("2d");
  const w = canvasEl.width;
  const h = canvasEl.height;
  previewCtx.clearRect(0, 0, w, h);
  const road = previewCtx.createLinearGradient(0, 0, 0, h);
  road.addColorStop(0, "rgba(21,31,30,0.96)");
  road.addColorStop(0.58, "rgba(10,16,15,0.98)");
  road.addColorStop(1, "rgba(5,8,7,0.98)");
  previewCtx.fillStyle = road;
  previewCtx.fillRect(0, 0, w, h);
  previewCtx.strokeStyle = "rgba(244,251,248,0.12)";
  previewCtx.lineWidth = 2;
  previewCtx.beginPath();
  previewCtx.moveTo(w * 0.18, 0);
  previewCtx.lineTo(w * 0.88, h);
  previewCtx.moveTo(w * 0.82, 0);
  previewCtx.lineTo(w * 0.92, h);
  previewCtx.stroke();
  previewCtx.strokeStyle = "rgba(255,209,102,0.32)";
  previewCtx.setLineDash([12, 10]);
  previewCtx.beginPath();
  previewCtx.moveTo(w * 0.5, h * 0.08);
  previewCtx.lineTo(w * 0.5, h * 0.96);
  previewCtx.stroke();
  previewCtx.setLineDash([]);

  const assets = window.VelocityPhoneAssets;
  const sprite = assets && assets.ready && typeof assets.getVehicleSprite === "function"
    ? assets.getVehicleSprite(vehicle.type, vehicle.color, { damage: 0 })
    : null;
  const baseWidth = ["semi", "truck", "monster", "tank", "tractor"].includes(vehicle.type) ? 100 : 88;
  const spriteW = vehicle.type === "semi" ? 122 : baseWidth;
  const spriteH = vehicle.type === "semi" ? 104 : isAirVehicleType(vehicle.type) ? 98 : 102;
  previewCtx.save();
  previewCtx.translate(w * 0.5, h * 0.56);
  drawPreviewGroundContact(previewCtx, spriteW, spriteH, vehicle.type);
  if (sprite) {
    previewCtx.imageSmoothingEnabled = true;
    previewCtx.drawImage(sprite, -spriteW / 2, -spriteH * 0.54, spriteW, spriteH);
  } else {
    previewCtx.fillStyle = vehicle.color;
    roundRect(-spriteW * 0.28, -spriteH * 0.42, spriteW * 0.56, spriteH * 0.8, 8);
    previewCtx.fill();
  }
  previewCtx.restore();
}

function drawPreviewGroundContact(previewCtx, w, h, vehicleType = "car") {
  previewCtx.save();
  const air = isAirVehicleType(vehicleType);
  const water = vehicleType === "boat";
  const contactY = h * (air ? 0.46 : water ? 0.5 : 0.56);
  const
```

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```
shadow = previewCtx.createRadialGradient(0, contactY, w * 0.08, 0, contactY + h * 0.06, w * 0.64);
shadow.addColorStop(0, air ? "rgba(0,0,0,0.28)" : "rgba(0,0,0,0.62)");
shadow.addColorStop(1, "rgba(0,0,0,0)");
previewCtx.fillStyle = shadow;
previewCtx.beginPath();
previewCtx.ellipse(0, contactY + h * 0.08, w * 0.62, h * 0.13, 0, 0, Math.PI * 2);
previewCtx.fill();
if (!air && !water) {
  previewCtx.fillStyle = "rgba(3,5,5,0.78)";
  previewCtx.beginPath();
  previewCtx.ellipse(-w * 0.28, h * 0.38, w * 0.14, h * 0.055, 0, 0, Math.PI * 2);
  previewCtx.ellipse(w * 0.28, h * 0.38, w * 0.14, h * 0.055, 0, 0, Math.PI * 2);
  previewCtx.ellipse(-w * 0.28, h * 0.65, w * 0.16, h * 0.07, 0, 0, Math.PI * 2);
  previewCtx.ellipse(w * 0.28, h * 0.65, w * 0.16, h * 0.07, 0, 0, Math.PI * 2);
  previewCtx.fill();
}
previewCtx.restore();
}

function renderUpgrades() {
  const list = $("#upgradeList");
  list.innerHTML = "";
  const director = getDirector(activeProfile);
  const build = vehicleBuild(activeProfile, activeProfile.selectedVehicle);
  upgradeDefs.forEach((def) => {
    const level = build.upgrades[def.id];
    const cost = upgradeCost(activeProfile, def);
    const card = document.createElement("div");
    card.className = "upgrade-card";
    card.innerHTML = `
      <div class="card-top"><strong>${def.name}</strong><span class="badge">${director.recommended.id === def.id ? "AI Pick" : `Lv
${level}/5`}</span></div>
      <div class="tiny">${def.desc}</div>
      <div class="upgrade-action">
        <div class="bar"><i></i></div>
        <button type="button" ${level >= 5 || activeProfile.coins < cost ? "disabled" : ""}>${level >= 5 ? "MAX" : `Buy
${cost}`}</button>
      </div>
    `;
    card.querySelector(".bar i").style.width = `${(level / 5) * 100}%`;
    card.querySelector("button").addEventListener("click", () => buyUpgrade(def));
    list.appendChild(card);
  });
}

function renderMissions() {
  const list = $("#missionList");
  list.innerHTML = "";
  missionDefs.forEach((mission) => {
    const done = activeProfile.completedMissions.includes(mission.id);
    const ready = !done && mission.test(activeProfile);
    const card = document.createElement("div");
    card.className = "mission-card";
    card.innerHTML = `
      <div class="card-top"><strong>${mission.text}</strong><span class="badge">${done ? "Claimed" : `${mission.reward}
coins`}</span></div>
      <div class="tiny">${ready ? "Ready to claim." : done ? "Nice work." : "Keep racing to unlock this reward."}</div>
      <div>
        <button class="${ready ? "primary" : ""}" type="button">Claim Reward</button>
      </div>
    `;
    const claim = card.querySelector("button");
    if (claim) claim.addEventListener("click", () => claimMission(mission));
    list.appendChild(card);
  });
}

function buyUpgrade(def) {
  const build = vehicleBuild(activeProfile, activeProfile.selectedVehicle);
  const level = build.upgrades[def.id];
  if (level >= 5) return;
  const cost = upgradeCost(activeProfile, def);
  if (activeProfile.coins < cost) return showToast("More coins needed.");
  activeProfile.coins -= cost;
  build.upgrades[def.id] += 1;
  activeProfile.upgrades = Object.assign(defaultUpgrades(), build.upgrades);
  profiles[activeProfile.id] = activeProfile;
  saveProfiles();
  renderHub();
  updateHud();
  showToast(`${selectedVehicle().name} ${def.name} upgraded.`);
}

function claimMission(mission) {
  activeProfile.completedMissions.push(mission.id);
  activeProfile.coins += mission.reward;
  activeProfile.rep += 8;
  profiles[activeProfile.id] = activeProfile;
  saveProfiles();
  renderHub();
  updateHud();
  showToast("Mission reward claimed.");
}

function launchRace() {
  startAudio();
  if (audioSystem && audioSystem.ctx.state === "suspended") audioSystem.ctx.resume();
  const age = ageBands[activeProfile.age];
  const director = getDirector(activeProfile);
  const vehicle = selectedVehicle();
  const scenario = activeScenario();
  if (!vehicleOwned(activeProfile, vehicle.id)) {
    showToast("Buy or select an owned vehicle first.");
    renderVehicles();
    return;
  }
  if (!raceCompatibleWithVehicle(selectedRace, vehicle)) {
    selectedRace =
  }
```


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```
firstCompatibleRace(vehicle, activeProfile);
renderRaces();
showToast(`${vehicle.name} switched to a ${compatibilityLabel(vehicle)} route.`);
}
Object.assign(raceState, {
  active: true,
  distance: 0,
  speed: 0,
  focus: 100,
  score: 0,
  coins: 0,
  dodges: 0,
  overtakes: 0,
  passesAgainst: 0,
  civilianHits: 0,
  penaltyCoins: 0,
  penaltyRep: 0,
  combo: 1,
  lane: 0,
  x: 0,
  visualLane: 0,
  lateralVelocity: 0,
  steerAngle: 0,
  throttleLoad: 0,
  brakeHeat: 0,
  slip: 0,
  damage: 0,
  damageAlertTimer: 0,
  damageAlertLabel: "",
  hazardWarningTimer: 0,
  hazardWarningLabel: "",
  resetTimer: 0,
  resetReason: "",
  crashCooldown: 0,
  roadCurve: 0,
  roadTurn: 0,
  roadOffset: 0,
  spawnClock: 0,
  coinClock: 0,
  civilianClock: 5.5,
  oncomingClock: 8,
  cannonCooldown: 0,
  routeFeatureClock: 4.5,
  hideCooldown: 0,
  scenarioId: scenario.id,
  scenarioLabel: scenario.name,
  teamScore: 0,
  rivals: [],
  police: [],
  opponents: makeOpponents(vehicle, age, director),
  coinsOnRoad: [],
  civilians: [],
  oncoming: [],
  routeFeatures: [],
  particles: [],
  elapsed: 0,
  heat: 0,
  heatClock: 1.7,
  autoSaveClock: 0,
  chaseActive: false,
  cameraShake: 0,
  countdown: 0,
  goalIntroTimer: 5.8,
  finished: false,
  director
});
$("#raceTitle").textContent = selectedRace.name;
const route = routeWorldInfo(selectedRace.place);
$("#raceBrief").textContent = `Goal: ${raceGoalText()} | ${route.country}: ${route.scene} | ${compatibilityLabel(vehicle)} | ${scenario.name}`;
$("#modeChip").textContent = `${route.country} | ${route.scene}`;
$("#missionChip").textContent = `${vehicle.name} | ${scenario.name} | ${selectedRace.target}`;
showView("race");
saveProfiles();
saveRace(true);
showToast(`${vehicle.name} on grid. ${raceGoalText()}`);
}

function saveRace(silent = false) {
  const quiet = silent === true;
  if (!activeProfile) return;
  if (!raceState.active) {
    saveProfiles();
    if (!quiet) showToast("Garage progress saved on this device.");
    renderSavedRaceButton();
    return;
  }
  const snapshot = {
    version: 1,
    savedAt: Date.now(),
    profileId: activeProfile.id,
    selectedRaceId: selectedRace.id,
    cameraMode,
    inputPaused: input.paused,
    state: {
      active: true,
      distance: raceState.distance,
      speed: raceState.speed,
      focus: raceState.focus,
      score: raceState.score,
      coins: raceState.coins,
      dodges: raceState.dodges,
      overtakes: raceState.overtakes,
      passesAgainst: raceState.passesAgainst,
      civilianHits: raceState.civilianHits,
      penaltyCoins: raceState.penaltyCoins,
      penaltyRep: raceState.penaltyRep,
      combo: raceState.combo,
      lane: raceState.lane,
      x: raceState.x,
      visualLane: raceState.visualLane,
      lateralVelocity: raceState.lateralVelocity,
      steerAngle: raceState.steerAngle,
```

```
throttleLoad: raceState.throttleLoad,  
brakeHeat: raceState.brakeHeat,  
slip: raceState.slip,  
damage: raceState.damage,  
resetTimer: raceState.resetTimer,  
resetReason: raceState.resetReason,  
crashCooldown: raceState.crashCooldown,  
roadCurve: raceState.roadCurve,  
roadTurn: raceState.roadTurn,  
roadOffset: raceState.roadOffset,  
spawnClock: raceState.spawnClock,  
coinClock: raceState.coinClock,  
civilianClock: raceState.civilianClock,  
oncomingClock: raceState.oncomingClock,  
cannonCooldown: raceState.cannonCooldown,  
routeFeatureClock: raceState.routeFeatureClock,  
hideCooldown: raceState.hideCooldown,  
scenarioId: raceState.scenarioId,  
scenarioLabel: raceState.scenarioLabel,  
teamScore: raceState.teamScore,  
rivals: raceState.rivals,  
police: raceState.police,  
opponents: raceState.opponents,  
coinsOnRoad: raceState.coinsOnRoad,  
civilians: raceState.civilians,  
oncoming: raceState.oncoming,  
routeFeatures: raceState.routeFeatures,  
elapsed: raceState.elapsed,  
heat: raceState.heat,  
heatClock: raceState.heatClock
```

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```
,
    autoSaveClock: raceState.autoSaveClock,
    chaseActive: raceState.chaseActive,
    cameraShake: 0,
    countdown: 0,
    goalIntroTimer: 0,
    finished: false
  }
};
localStorage.setItem(saveKey, JSON.stringify(snapshot));
saveProfiles();
renderSaveRaceButton();
raceState.autoSaveClock = 0;
if (!quiet) showToast("Race saved on this device.");
}

function resumeSavedRace() {
  const saved = loadSavedRace();
  if (!saved || !activeProfile || saved.profileId !== activeProfile.id) {
    showToast("No saved race for this driver.");
    renderSaveRaceButton();
    return;
  }
  const race = races.find((item) => item.id === saved.selectedRaceId) || races[0];
  selectedRace = race;
  selectedScenario = scenarioById(saved.state.scenarioId || selectedScenario.id);
  setCameraMode(saved.cameraMode || cameraMode, true);
  const director = getDirector(activeProfile);
  Object.assign(raceState, saved.state, {
    active: true,
    particles: [],
    opponents: saved.state.opponents || makeOpponents(selectedVehicle(), ageBands[activeProfile.age], director),
    civilians: saved.state.civilians || [],
    oncoming: saved.state.oncoming || [],
    routeFeatures: saved.state.routeFeatures || [],
    director,
    cameraShake: 0
  });
  raceState.visuallane = Number(saved.state.visuallane);
  if (!Number.isFinite(raceState.visuallane)) raceState.visuallane = raceState.lane || 0;
  raceState.lateralVelocity = Number(saved.state.lateralVelocity) || 0;
  raceState.steerAngle = Number(saved.state.steerAngle) || 0;
  raceState.throttleLoad = Number(saved.state.throttleLoad) || 0;
  raceState.brakeHeat = Number(saved.state.brakeHeat) || 0;
  raceState.slip = Number(saved.state.slip) || 0;
  raceState.damage = Math.max(0, Math.min(100, Number(saved.state.damage) || 0));
  raceState.resetTimer = Math.max(0, Number(saved.state.resetTimer) || 0);
  raceState.resetReason = saved.state.resetReason || "";
  raceState.crashCooldown = Math.max(0, Number(saved.state.crashCooldown) || 0);
  raceState.goalIntroTimer = Math.max(0, Number(saved.state.goalIntroTimer) || 0);
  raceState.roadCurve = Number(saved.state.roadCurve) || 0;
  raceState.roadTurn = Number(saved.state.roadTurn) || raceState.roadCurve || 0;
  raceState.overtakes = Number(saved.state.overtakes) || 0;
  raceState.passesAgainst = Number(saved.state.passesAgainst) || 0;
  raceState.civilianHits = Number(saved.state.civilianHits) || 0;
  raceState.penaltyCoins = Number(saved.state.penaltyCoins) || 0;
  raceState.penaltyRep = Number(saved.state.penaltyRep) || 0;
  raceState.cannonCooldown = Number(saved.state.cannonCooldown) || 0;
  raceState.routeFeatureClock = Number(saved.state.routeFeatureClock) || 0;
  raceState.hideCooldown = Number(saved.state.hideCooldown) || 0;
  raceState.autoSaveClock = Number(saved.state.autoSaveClock) || 0;
  raceState.scenarioId = selectedScenario.id;
  raceState.scenarioLabel = selectedScenario.name;
  raceState.teamScore = Number(saved.state.teamScore) || 0;
  raceState.opponents.forEach((opponent) => {
    const gap = Number(opponent.distance) - raceState.distance;
    opponent.wasAhead = gap > 4;
    opponent.passCooldown = Math.max(0, Number(opponent.passCooldown) || 0);
    opponent.surgePhase = Number(opponent.surgePhase) || Math.random() * Math.PI * 2;
  });
  input.paused = Boolean(saved.inputPaused);
  $("#pauseBtn").textContent = input.paused ? "Play" : "Pause";
  $("#raceTitle").textContent = selectedRace.name;
  const route = routeWorldInfo(selectedRace.place);
  $("#raceBrief").textContent = `Goal: ${raceGoalText()} | ${route.country}: ${route.scene} |
${compatibilityLabel(selectedVehicle())} | ${selectedScenario.name}`;
  $("#modeChip").textContent = `${route.country} | ${route.scene}`;
  $("#missionChip").textContent = raceState.chaseActive ? `Police heat ${Math.round(raceState.heat)}% | Escape clean` :
`${selectedRace.target} | ${director.event.name}`;
  startAudio();
  if (audioSystem && audioSystem.ctx.state === "suspended") audioSystem.ctx.resume();
  showView("race");
  updateRaceUi();
  showToast("Saved race resumed.");
}

function endRace(manual = false) {
  if (!raceState.active) return;
  if (manual) {
```

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```
saveRace();
  raceState.active = false;
  updateHud();
  showView("garage");
  renderHub();
  return;
}
raceState.active = false;
const success = missionProgress() >= 1 && raceState.focus > 0;
const finishPosition = playerPosition();
const age = ageBands[activeProfile.age];
const directorReward = raceState.director ? raceState.director.reward : 1;
const scenarioReward = scenarioById(raceState.scenarioId).reward || 1;
const podiumBonus = finishPosition === 1 ? 120 : finishPosition <= 3 ? 60 : 0;
const grossReward = Math.round((selectedRace.reward + raceState.coins * 5 + (success ? 70 : 18) + podiumBonus +
(raceState.teamScore || 0) * 0.08) * age.rewards * directorReward * scenarioReward);
const reward = Math.max(0, grossReward - Math.round(raceState.penaltyCoins || 0));
const grossRep = success ? selectedRace.rep + (finishPosition === 1 ? 8 : 0) : Math.ceil(selectedRace.rep / 3);
const rep = Math.max(0, grossRep - Math.round(raceState.penaltyRep || 0));
activeProfile.coins += reward;
activeProfile.rep += rep;
activeProfile.upgrades = Object.assign(defaultUpgrades(), activeVehicleUpgrades(activeProfile));
activeProfile.stats.races += 1;
activeProfile.stats.wins += success && finishPosition === 1 ? 1 : 0;
activeProfile.stats.totalCoins += raceState.coins;
activeProfile.stats.bestScore = Math.max(activeProfile.stats.bestScore, Math.round(raceState.score));
if (raceState.focus >= 50) activeProfile.stats.steadyRuns += 1;
profiles[activeProfile.id] = activeProfile;
saveProfiles();
clearSavedRace();
$("#finishTitle").textContent = success && finishPosition === 1 ? "Race Won" : success ? "Challenge Cleared" : "Race Complete";
$("#finishStats").innerHTML = `
  <div><span>Position</span><strong>${finishPosition}/${raceRankings().length}</strong></div>
  <div><span>Coins earned</span><strong>${reward}</strong></div>
  <div><span>Reputation</span><strong>${rep}</strong></div>
  <div><span>Penalties</span><strong>${raceState.civilianHits || 0}</strong></div>
  <div><span>Overtakes</span><strong>${raceState.overtakes}</strong></div>
  <div><span>Crew score</span><strong>${Math.round(raceState.teamScore || 0)}</strong></div>
  <div><span>Score</span><strong>${Math.round(raceState.score)}</strong></div>
  <div><span>Damage</span><strong>${Math.round(raceState.damage)}%</strong></div>
  <div><span>Focus left</span><strong>${Math.max(0, Math.round(raceState.focus))}</strong></div>
  `;
  updateHud();
  renderHub();
  showView("finish");
}

function missionProgress() {
  if (selectedRace.type === "coins") return raceState.coins / selectedRace.goal;
  if (selectedRace.type === "focus") return raceState.focus / selectedRace.goal;
  if (selectedRace.type === "dodges") return raceState.dodges / selectedRace.goal;
  return raceState.score / selectedRace.goal;
}

function raceGoalText(race = selectedRace, vehicle = selectedVehicle()) {
  const route = routeWorldInfo(race.place);
  const scenario = activeScenario();
  const clean = vehicle.type === "tank" ? "clear the route" : isWaterVehicleType(vehicle.type) ? "hold the water line" :
  isAirVehicleType(vehicle.type) ? "fly the gates" : vehicle.type === "policecar" ? "run the pursuit line" : "race the pack";
  const pursuit = scenario.hotPursuit ? "Police pressure climbs the longer you stay visible. " : "";
  return `${race.target}; ${clean} through ${route.scene}. ${pursuit}Use hideouts and shortcut branches when they appear.`;
}

function updateHud() {
  $("#profileChip").textContent = activeProfile ? `${activeProfile.name} | ${ageBands[activeProfile.age].label}` : "No Driver";
  $("#coinStat").textContent = activeProfile ? activeProfile.coins : 0;
  $("#repLabel").textContent = raceState.active ? "Heat" : "Rep";
  $("#repStat").textContent = raceState.active ? Math.round(raceState.heat) : activeProfile ? activeProfile.rep : 0;
  $("#speedStat").textContent = Math.max(0, Math.round(raceState.speed));
  $("#focusStat").textContent = Math.max(0, Math.round(raceState.focus));
  const damageStat = $("#damageStat");
  if (damageStat) damageStat.textContent = raceState.active ? `${Math.round(raceState.damage)}%` : "0%";
  if (
```

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```
raceState.active) {
  const pos = playerPosition();
  const leader = raceRankings()[0];
  const fieldSize = raceRankings().length;
  const turnName = routeTurnName(raceState.roadTurn || raceState.roadCurve || 0);
  $("#missionChip").textContent = raceState.chaseActive
    ? `P${pos}/${fieldSize} | ${turnName} | Heat ${Math.round(raceState.heat)}%`
    : `P${pos}/${fieldSize} | ${turnName} | Leader ${leader.name}`;
  if ((raceState.teamScore || 0) > 0) $("#missionChip").textContent = `P${pos}/${fieldSize} | Crew
${Math.round(raceState.teamScore)} | ${turnName}`;
  if (raceState.speed < 0) $("#missionChip").textContent = "Hold Gas to accelerate | Brake to stop/reverse";
  if ((raceState.hazardWarningTimer || 0) > 0) $("#missionChip").textContent = raceState.hazardWarningLabel || "Traffic ahead";
  if ((raceState.damageAlertTimer || 0) > 0) $("#missionChip").textContent = raceState.damageAlertLabel || "Damage taken";
  if (raceState.resetTimer > 0) $("#missionChip").textContent = `Vehicle reset in ${Math.ceil(raceState.resetTimer)} |
${raceState.resetReason}`;
}
}

function setCameraMode(mode, quiet = false) {
  cameraMode = mode;
  localStorage.setItem("velocityVaultCameraMode", mode);
  $(".camera-btn").forEach(btn => btn.classList.toggle("active", btn.dataset.camera === mode));
  if (!quiet) {
    const labels = { chase: "Third-person chase camera", hood: "Low hood camera", cockpit: "Windshield driver view" };
    showToast(labels[mode]);
  }
}

function applyPhoneGraphicsDefaults() {
  const key = "velocityVaultPhoneGraphicsDefaultsV56";
  if (!phoneGraphicsActive() || localStorage.getItem(key)) return;
  rendererMode = "canvas";
  if (cameraMode === "chase") cameraMode = "hood";
  touchDriveMode = "hold";
  touchControlSize = "mini";
  localStorage.setItem("velocityVaultRendererMode", rendererMode);
  localStorage.setItem("velocityVaultCameraMode", cameraMode);
  localStorage.setItem("velocityVaultTouchDriveMode", touchDriveMode);
  localStorage.setItem("velocityVaultTouchControlSize", touchControlSize);
  localStorage.setItem(key, "applied");
}

function initWebGLRenderer() {
  if (!glCanvas || webglRenderer) return;
  try {
    webglRenderer = window.VelocityWebGLPipeline ? window.VelocityWebGLPipeline(glCanvas) : null;
  } catch {
    webglRenderer = null;
    rendererMode = "canvas";
    localStorage.setItem("velocityVaultRendererMode", rendererMode);
  }
}

function useWebGLRenderer() {
  if (phoneGraphicsActive()) return false;
  return rendererMode === "webgl" && webglRenderer && webglRenderer.ready;
}

function phoneGraphicsActive() {
  const assets = window.VelocityPhoneAssets;
  if (assets && typeof assets.isPhoneViewport === "function" && assets.isPhoneViewport()) return true;
  return window.matchMedia && window.matchMedia("(pointer: coarse)").matches;
}

function phoneCleanRoadActive() {
  return phoneGraphicsActive();
}

function setRendererMode(mode, quiet = false) {
  if (mode === "webgl" && phoneGraphicsActive()) mode = "canvas";
  rendererMode = mode;
  if (mode === "webgl") initWebGLRenderer();
  if (mode === "webgl" && !webglRenderer) rendererMode = "canvas";
  localStorage.setItem("velocityVaultRendererMode", rendererMode);
  $(".renderer-btn").forEach(btn => btn.classList.toggle("active", btn.dataset.renderer === rendererMode));
  $(".stage").classList.toggle("webgl", useWebGLRenderer());
  if (!quiet) showToast(useWebGLRenderer() ? "WebGL 3D renderer enabled." : "Classic 2D renderer enabled.");
}

function clearTouchDriveInputs() {
  input.left = false;
  input.right = false;
  input.touchSteer = 0;
  input.gas = false;
  input.brake = false;
  input.boost = false;
  mobileTouchState.active.clear();
  mobileTouchState.stickSteer = 0;
  mobileTouchState.driveStickActive = false;
  mobileTouchState.latchedStickControl = "";
  mobileTouchState.floatingStickId = null;
  mobileTouchState.floatingStickKey = "";
  mobileTouchState.floatingStickOriginX = 0;
  mobileTouchState.floatingStickOriginY = 0;
  mobileTouchState.floatingStickKnobX = 0;
  mobileTouchState.floatingStickKnobY = 0;
  setFloatingStickVisual(false);
}

function useFloatingStickControls() {
  return document
```

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```
t.body.classList.contains("race-live") && touchControlSize === "mini";
}

function floatingStickRadius() {
  const stick = $(".drive-stick");
  if (!stick) return 58;
  const rect = stick.getBoundingClientRect();
  return Math.max(42, Math.min(72, Math.min(rect.width || 116, rect.height || 116) * 0.44));
}

function setFloatingStickVisual(active, originX = mobileTouchState.floatingStickOriginX, originY =
mobileTouchState.floatingStickOriginY, knobX = 0, knobY = 0) {
  const stick = $(".drive-stick");
  if (!stick) return;
  stick.classList.toggle("floating-stick-active", Boolean(active));
  stick.style.setProperty("--stick-origin-x", `${Math.round(originX)}px`);
  stick.style.setProperty("--stick-origin-y", `${Math.round(originY)}px`);
  stick.style.setProperty("--stick-knob-x", `${Math.round(knobX)}px`);
  stick.style.setProperty("--stick-knob-y", `${Math.round(knobY)}px`);
}

function floatingStickControlAt(clientX, clientY) {
  const radius = floatingStickRadius();
  const dx = clientX - mobileTouchState.floatingStickOriginX;
  const dy = clientY - mobileTouchState.floatingStickOriginY;
  const distance = Math.hypot(dx, dy);
  const limit = distance > radius ? radius / distance : 1;
  const knobX = dx * limit;
  const knobY = dy * limit;
  mobileTouchState.floatingStickKnobX = knobX;
  mobileTouchState.floatingStickKnobY = knobY;
  setFloatingStickVisual(true, mobileTouchState.floatingStickOriginX, mobileTouchState.floatingStickOriginY, knobX, knobY);
  const steer = Math.max(-1, Math.min(1, dx / radius));
  const controls = ["stick"];
  if (dy > radius * 0.72) {
    controls.push("brake");
  } else {
    controls.push("gas");
  }
  if (steer < -0.07) controls.push("left");
  if (steer > 0.07) controls.push("right");
  if (Math.abs(steer) > 0.035) controls.push(`steer=${steer.toFixed(2)}`);
  return controls.join(":");
}

function beginFloatingStick(id, clientX, clientY) {
  if (!useFloatingStickControls()) return false;
  if (mobileTouchState.floatingStickId !== null && mobileTouchState.floatingStickId !== id) return false;
  const key = `floating-stick-${id}`;
  mobileTouchState.floatingStickId = id;
  mobileTouchState.floatingStickKey = key;
  mobileTouchState.floatingStickOriginX = clientX;
  mobileTouchState.floatingStickOriginY = clientY;
  mobileTouchState.latchedStickControl = "";
  mobileTouchState.active.set(key, floatingStickControlAt(clientX, clientY));
  applyMobileTouchSnapshot();
  return true;
}

function moveFloatingStick(id, clientX, clientY) {
  if (mobileTouchState.floatingStickId !== id) return false;
  mobileTouchState.active.set(mobileTouchState.floatingStickKey, floatingStickControlAt(clientX, clientY));
  applyMobileTouchSnapshot();
  return true;
}

function endFloatingStick(id) {
  if (mobileTouchState.floatingStickId !== id) return false;
  mobileTouchState.active.delete(mobileTouchState.floatingStickKey);
  mobileTouchState.floatingStickId = null;
  mobileTouchState.floatingStickKey = "";
  mobileTouchState.floatingStickKnobX = 0;
  mobileTouchState.floatingStickKnobY = 0;
  setFloatingStickVisual(false);
  applyMobileTouchSnapshot();
  return true;
}

function setTouchDriveMode(mode, quiet = false) {
  touchDriveMode = mode === "toggle" ? "toggle" : "hold";
  localStorage.setItem("velocityVaultTouchDriveMode", touchDriveMode);
  document.body.classList.toggle("touch-toggle", touchDriveMode === "toggle");
  const label = $("#touchModeLabel");
  const modeButton = $(".control-mode");
  if (label) label.textContent = touchDriveMode === "toggle" ? "Toggle" : "Hold";
  if (modeButton) modeButton.setAttribute("aria-pressed", touchDriveMode === "toggle" ? "true" : "false");
  if (!quiet) {
    clearTouchDriveInputs();
    updateRaceUi();
    showToast(touchDriveMode === "toggle" ? "Touch controls set to toggle. Tap Gas or Brake to hold it on." : "Touch controls set to hold. Press and hold to drive.");
  }
}

function setTouchControlSize(size, quiet = false) {
  touchControlSize = size === "full" ? "full" : "mini";
  localStorage.setItem("velocityVaultTouchControlSize", touchControlSize);
  document.body.classList.toggle("control-mini", touchControlSize === "mini");
  const label = $("#touchSizeLab
```

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```
el");
const sizeButton = $(".control-size");
if (label) label.textContent = touchControlSize === "mini" ? "Full" : "Mini";
if (sizeButton) {
  sizeButton.setAttribute("aria-pressed", touchControlSize === "mini" ? "true" : "false");
  sizeButton.setAttribute("aria-label", touchControlSize === "mini" ? "Switch to full controls" : "Switch to mini controls");
}
if (!quiet) {
  clearTouchDriveInputs();
  updateRaceUI();
  showToast(touchControlSize === "mini" ? "Floating one-thumb joystick on. Drag anywhere on the driving screen." : "Full controls on. Tap Mini for more screen space.");
}
}

function setDriveInput(control, pressed) {
  if (control === "left") {
    input.left = pressed;
    if (pressed) input.right = false;
  } else if (control === "right") {
    input.right = pressed;
    if (pressed) input.left = false;
  } else if (control === "neutral") {
    input.left = false;
    input.right = false;
  } else if (control === "gas") {
    input.gas = pressed;
    if (pressed) input.brake = false;
  } else if (control === "brake") {
    input.brake = pressed;
    if (pressed) input.gas = false;
  } else if (control === "boost") {
    input.boost = pressed;
  }
}

function toggleDriveInput(control) {
  if (control === "left") {
    setDriveInput("left", !input.left);
  } else if (control === "right") {
    setDriveInput("right", !input.right);
  } else if (control === "neutral") {
    setDriveInput("neutral", true);
  } else if (control === "gas") {
    setDriveInput("gas", !input.gas);
  } else if (control === "brake") {
    setDriveInput("brake", !input.brake);
  } else if (control === "boost") {
    setDriveInput("boost", !input.boost);
  }
}

function isDriveStickControl(control) {
  return typeof control === "string" && control.startsWith("stick:");
}

function driveStickControlAt(stick, clientX, clientY) {
  const rect = stick.getBoundingClientRect();
  if (!rect.width || !rect.height) return "";
  const clampedX = Math.max(rect.left, Math.min(rect.right, clientX));
  const clampedY = Math.max(rect.top, Math.min(rect.bottom, clientY));
  const nx = (clampedX - rect.left) / rect.width - 0.5;
  const ny = (clampedY - rect.top) / rect.height - 0.5;
  const dead = 0.08;
  const controls = ["stick"];
  if (touchControlSize === "mini") {
    if (ny > 0.34) controls.push("brake");
    else controls.push("gas");
  } else {
    if (ny < -dead) controls.push("gas");
    if (ny > dead) controls.push("brake");
  }
  if (nx < -dead) controls.push("left");
  if (nx > dead) controls.push("right");
  if (controls.length === 1) controls.push("neutral");
  const steer = Math.max(-1, Math.min(1, nx / 0.42));
  if (Math.abs(steer) > dead) controls.push(`steer:${steer.toFixed(2)}`);
  return controls.join(":");
}

function addTouchControls(control, controls) {
  if (!control) return;
  if (!isDriveStickControl(control)) {
    controls.add(control);
    return;
  }
  control.split(":").forEach((part) => {
    if (part && part !== "stick") controls.add(part);
  });
}

function driveStickSteerValue(control) {
  if (!isDriveStickControl(control)) return 0;
  const analog = control.split(":").find((part) => part.startsWith("steer="));
  if (analog) {
    const value = Number(analog.slice(6));
    if (Number.isFinite(value)) return Math.max(-1, Math.min(1, value));
  }
  const parts = control.split(":");
  const left = parts.includes("left");
  const right = parts.includes("right");
  return right && !left ? 1 : left && !right ? -1 : 0;
}

function mobileControlAt(clientX, clientY) {
  if (document.body.classList.contains("control-mini")) {
    const miniStick = $(".drive-stick");
    if (miniStick) {
      const rect = miniStick.getBoundingClientRect();
      const margin = Math.max(24, Math.min(rect.width, rect.height) * 0.22);
      const insideStick =
        clientX >= rect.left - margin &&
        clientX <= rect.right + margin &&
```

```
    clientY >= rect.top - margin &&  
    clientY <= rect.bottom + margin;  
    if (insideStick) return driveStickControlAt(miniStick, clientX, clientY);  
  }  
}  
const el = document.elementFromPoint(clientX, clientY);  
const stick = el && el.closest ? el.closest(".drive-stick") :
```


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```
    null;
    if (stick && document.body.classList.contains("control-mini")) return driveStickControlAt(stick, clientX, clientY);
    const button = el && el.closest ? el.closest(".mobile-control") : null;
    return button && button.dataset ? button.dataset.control : "";
}

function applyMobileTouchSnapshot() {
    const hadDriveStick = mobileTouchState.driveStickActive;
    const controls = new Set();
    let hasDriveStick = false;
    let stickSteer = 0;
    if (touchDriveMode === "toggle" && mobileTouchState.latchedStickControl) {
        hasDriveStick = true;
        stickSteer = driveStickSteerValue(mobileTouchState.latchedStickControl);
        addTouchControls(mobileTouchState.latchedStickControl, controls);
    }
    mobileTouchState.active.forEach((control) => {
        if (isDriveStickControl(control)) {
            hasDriveStick = true;
            const value = driveStickSteerValue(control);
            if (Math.abs(value) > Math.abs(stickSteer)) stickSteer = value;
        }
        addTouchControls(control, controls);
    });
    if (touchDriveMode === "toggle" && !hasDriveStick && !hadDriveStick) return;
    mobileTouchState.driveStickActive = hasDriveStick;
    mobileTouchState.stickSteer = hasDriveStick ? stickSteer : 0;
    const neutral = controls.has("neutral");
    const left = controls.has("left");
    const right = controls.has("right");
    const gas = controls.has("gas");
    const brake = controls.has("brake");
    input.left = neutral ? false : left && !right;
    input.right = neutral ? false : right && !left;
    input.touchSteer = neutral ? 0 : mobileTouchState.stickSteer;
    input.gas = gas && !brake;
    input.brake = brake && !gas;
    input.boost = controls.has("boost");
    if (touchDriveMode === "toggle" && neutral) mobileTouchState.latchedStickControl = "";
    updateRaceUi();
}

function handleMobileTouchStart(event) {
    mobileTouchState.usingTouchEvent = true;
    event.preventDefault();
    startAudio();
    if (audioSystem && audioSystem.ctx.state === "suspended") audioSystem.ctx.resume();
    Array.from(event.changedTouches).forEach((touch) => {
        const el = document.elementFromPoint(touch.clientX, touch.clientY);
        if (useFloatingStickControls() && !(el && el.closest && el.closest(".mobile-control:not(.stick-up):not(.stick-left):not(.stick-center):not(.stick-right):not(.stick-down)"))) {
            beginFloatingStick(touch.identifier, touch.clientX, touch.clientY);
            return;
        }
        if (useFloatingStickControls() && el && el.closest && el.closest(".drive-stick")) {
            beginFloatingStick(touch.identifier, touch.clientX, touch.clientY);
            return;
        }
        const control = mobileControlAt(touch.clientX, touch.clientY);
        if (!control) return;
        if (control === "size") {
            setTouchControlSize(touchControlSize === "mini" ? "full" : "mini");
            return;
        }
        if (control === "mode") {
            const now = performance.now();
            if (now - mobileTouchState.modeToggleAt > 260) {
                mobileTouchState.modeToggleAt = now;
                setTouchDriveMode(touchDriveMode === "toggle" ? "hold" : "toggle");
            }
            return;
        }
        if (touchDriveMode === "toggle") {
            if (isDriveStickControl(control)) {
                mobileTouchState.latchedStickControl = control.includes("neutral") ? "" : control;
                mobileTouchState.active.set(touch.identifier, control);
            } else {
                toggleDriveInput(control);
                updateRaceUi();
            }
            return;
        }
        mobileTouchState.active.set(touch.identifier, control);
    });
    applyMobileTouchSnapshot();
}

function handleMobileTouchMove(event) {
    event.preventDefault();
    Array.from(event.changedTouches).forEach((touch) => {
        if (moveFloatingStick(touch.identifier, touch.clientX, touch.clientY)) return;
        const previous = mobileTouchState.active.get(touch.identifier);
        const control = mobileControlAt(touch.clientX, touch.clientY);
        if (touchDriveMode === "toggle") {
            if (isDriveStickControl(control)) {
                mobileTouchState.latchedStickControl = control.includes("neutral") ? "" : control;
                mobileTouchState.active.set(touch.identifier, control);
            } else if (isDriveStickControl(previous)) {
                mobileTouchState.active.delete(touch.identifier);
            }
            return;
        }
        if (control
```

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```
&& control !== "mode" && control !== "size") {
  mobileTouchState.active.set(touch.identifier, control);
} else {
  mobileTouchState.active.delete(touch.identifier);
}
});
applyMobileTouchSnapshot();
}

function handleMobileTouchEnd(event) {
  event.preventDefault();
  Array.from(event.changedTouches).forEach((touch) => {
    if (endFloatingStick(touch.identifier)) return;
    mobileTouchState.active.delete(touch.identifier);
  });
  applyMobileTouchSnapshot();
}

function bindDriveStickPointerControl() {
  const stick = $(".drive-stick");
  if (!stick) return;
  const pointerKey = "drive-stick-pointer";
  const applyStick = (event) => {
    if (event.pointerType === "touch" && mobileTouchState.usingTouchEvents) return;
    if (useFloatingStickControls()) return;
    if (touchControlSize !== "mini") return;
    event.preventDefault();
    const control = driveStickControlAt(stick, event.clientX, event.clientY);
    if (touchDriveMode === "toggle") mobileTouchState.latchedStickControl = control.includes("neutral") ? "" : control;
    mobileTouchState.active.set(pointerKey, control);
    applyMobileTouchSnapshot();
  };
  const stopStick = (event) => {
    if (event && event.pointerType === "touch" && mobileTouchState.usingTouchEvents) return;
    if (event) event.preventDefault();
    mobileTouchState.active.delete(pointerKey);
    applyMobileTouchSnapshot();
    if (stick.releasePointerCapture && event && event.pointerId !== undefined) {
      try {
        stick.releasePointerCapture(event.pointerId);
      } catch {
        // Capture may already be released by the browser.
      }
    }
  };
  stick.addEventListener("pointerdown", (event) => {
    if (event.pointerType === "touch" && mobileTouchState.usingTouchEvents) return;
    startAudio();
    if (audioSystem && audioSystem.ctx.state === "suspended") audioSystem.ctx.resume();
    if (stick.setPointerCapture && event.pointerId !== undefined) {
      try {
        stick.setPointerCapture(event.pointerId);
      } catch {
        // Some browsers reject capture after synthetic pointer events.
      }
    }
    applyStick(event);
  });
  stick.addEventListener("pointermove", applyStick);
  stick.addEventListener("pointerup", stopStick);
  stick.addEventListener("pointercancel", stopStick);
  stick.addEventListener("lostpointercapture", stopStick);
}

function floatingStickPointerAllowed(event) {
  if (!useFloatingStickControls()) return false;
  if (event.pointerType === "touch" || event.pointerType === "pen") return true;
  const compactViewport = window.matchMedia && window.matchMedia("(max-width: 980px)").matches;
  return phoneGraphicsActive() || compactViewport;
}

function bindFloatingPhoneJoystick() {
  const stick = $(".drive-stick");
  const layer = $(".mobile-drive-controls");
  const targets = [canvas, stick, layer].filter(Boolean);
  const start = (event) => {
    if (!floatingStickPointerAllowed(event)) return;
    const target = event.target;
    if (target && target.closest && target.closest(".mobile-control:not(.stick-up):not(.stick-left):not(.stick-center):not(.stick-right):not(.stick-down)") return;
    event.preventDefault();
    startAudio();
    if (audioSystem && audioSystem.ctx.state === "suspended") audioSystem.ctx.resume();
    const id = `pointer-${event.pointerId}`;
    if (!beginFloatingStick(id, event.clientX, event.clientY)) return;
    if (event.currentTarget.setPointerCapture && event.pointerId !== undefined) {
      try {
        event.currentTarget.setPointerCapture(event.pointerId);
      } catch {
        // Synthetic pointer events may not allow capture.
      }
    }
  };
  const move = (event) => {
    const id = `pointer-${event.pointerId}`;
    if (!moveFloatingStick(id, event.clientX, event.clientY)) return;
    event.preventDefault();
  };
  const stop = (event) => {
    const id = `pointer-${event.pointerId}`;
    if (!endFloatingStick(id)) return;
    event.preventDefault();
    if (event.currentTarget.releasePointerCapture && event.pointerId !== undefined) {
      try {
        event.currentTarget.releasePointerCapture(event.pointerId);
      } catch {
        // Capture may already be gone.
      }
    }
  };
}
```

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```
    }
  };
  targets.forEach((target) => {
    target.addEventListener("pointerdown", start);
    target.addEventListener("pointermove", move);
    target.addEventListener("pointerup", stop);
    target.addEventListener("pointercancel", stop);
    target.addEventListener("lostpointercapture", stop);
  });
}

function registerOfflineApp() {
  if (!("serviceWorker" in navigator)) return;
  if (!location.protocol.startsWith("http")) return;
  navigator.serviceWorker.register("./sw.js").then(() => {
    showToast("Offline app mode ready.");
  }).catch(() => {
    showToast("Offline app mode needs HTTPS hosting.");
  });
}

function installApp() {
  const isStandalone = window.matchMedia("(display-mode: standalone)").matches || navigator.standalone;
  if (isStandalone) {
    showToast("Velocity Vault is already running as an app.");
    return;
  }
  if (deferredInstallPrompt) {
    deferredInstallPrompt.prompt();
    deferredInstallPrompt.userChoice.finally(() => {
      deferredInstallPrompt = null;
    });
    return;
  }
  if (!location.protocol.startsWith("http")) {
    showToast("Install needs the secure GitHub Pages link. Then use Share > Add to Home Screen on iPhone, or Install on Android.");
    return;
  }
  showToast("iPhone: Safari Share > Add to Home Screen. Android/desktop: browser menu > Install / Add to Phone.");
}

function makeOpponents(playerVehicle, age, director) {
  const routeTypes = raceTrafficTypes(selectedRace, playerVehicle);
  const compatiblePool = vehicleDefs.filter((vehicle) => vehicle.id !== playerVehicle.id && (raceCompatibleWithVehicle(selectedRace, vehicle) || routeTypes.includes(vehicle.type)));
  const pool = compatiblePool.length ? compatiblePool : vehicleDefs.filter((vehicle) => vehicle.id !== playerVehicle.id);
  const scenario = activeScenario();
  const grid = [
    { gap: 1000, lane: -1.95, bias: 0.58, speed: 28 },
    { gap: 5200, lane: 0.86, bias: 0.66, speed: 34 },
    { gap: -1800, lane: 1.34, bias: 0.74, speed: 32 },
    { gap: 9800, lane: 1.95, bias: 0.74, speed: 42 },
    { gap: -4500, lane: -1.28, bias: 0.84, speed: 38 },
    { gap: 14400, lane: -0.76, bias: 0.78, speed: 44 },
    { gap: -7200, lane: 1.95, bias: 0.86, speed: 40 },
    { gap: 18800, lane: 1.18, bias: 0.82, speed: 48 }
  ];
  const allyGrid = [
    { gap: -720, lane: 1.72, bias: 0.72, speed: 31 },
    { gap: -2380, lane: -1.72, bias: 0.82, speed: 36 }
  ];
  const roster = [
    ...opponentNames.slice(0, 5 + (scenario.extraOpponents || 0)).map((name) => ({ name, ally: false })),
    ...Array.from({ length: scenario.allies || 0 }, (_, index) => ({ name: `P${index + 2} Ghost`, ally: true }))
  ];
  let rivalIndex = 0;
  let allyIndex = 0;
  return roster.map((entry, index) => {
    const { name, ally } = entry;
    const vehicle = ally ? playerVehicle : pool[(rivalIndex * 2 + selectedRace.id.length) % pool.length];
    const slot = ally ? allyGrid[allyIndex % allyGrid.length] : grid[rivalIndex % grid.length];
    if (ally) allyIndex += 1;
    else rivalIndex += 1;
    const startGap = slot.gap + (Math.random() - 0.5) * 18;
    return {
      name,
      vehicleId: vehicle.id,
      lane: slot.lane,
      homeLane: slot.lane,
      distance: startGap,
      speed: slot.speed + index * 2,
      targetBias: slot.bias + Math.random() * 0.035 + (ally ? 0.035 : 0),
      focus: 100,
      damage: 0,
      wrecked: false,
      color: ally ? selectedVehicleColor(playerVehicle) : vehicle.color,
      multiplayerRole: ally ? "ally" : "rival",
      wobble: Math.random() * Math.PI * 2,
      surgePhase: Math.random() * Math.PI * 2,
      packPhase: Math.random() * Math.PI * 2,
      spreadGap: slot.gap,
      bumpCooldown: 0,
      passCooldown: 0,
      wasAhead: startGap > 0,
      laneVelocity: 0,
      spin: 0,
      finished: false,
      ageSpeed: age.speed,
      aiTune: director.traffic
    };
  });
}

function updateOpponents(dt, maxSpeed) {
  const length = raceLength();
  const playerVehicle = selectedVehicle();
  const fieldSpeed = maxSpeed / Math.max(0.68, playerVehicle.speed);
  raceState.opponents.forEach((opponent, index) => {
    if (opponent.finished) return;
    opponent.damage = Math.max(0, Math.min(100, Num
```

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```
ber(opponent.damage) || 0));
opponent.laneVelocity = Number(opponent.laneVelocity) || 0;
opponent.spin = Number(opponent.spin) || 0;
opponent.passCooldown = Math.max(0, (Number(opponent.passCooldown) || 0) - dt);
opponent.surgePhase = Number(opponent.surgePhase) || index * 1.3;
opponent.packPhase = Number(opponent.packPhase) || index * 0.9;
const vehicle = vehicleById(opponent.vehicleId);
const routePush = routeVehicleBoost(selectedRace.place, vehicle.type, true);
const gapToPlayer = opponent.distance - raceState.distance;
let racePressure = gapToPlayer > 0
  ? Math.max(-0.07, -gapToPlayer / 2400)
  : Math.min(0.2, Math.abs(gapToPlayer) / 980);
if (gapToPlayer < -20 && raceState.elapsed < 26) racePressure = Math.min(racePressure, 0.045);
const duelSurge = Math.abs(gapToPlayer) < 260 ? Math.sin(raceState.elapsed * (1.05 + index * 0.08) + opponent.surgePhase) *
0.085 : 0;
let spreadPressure = 0;
raceState.opponents.forEach((other) => {
  if (other === opponent || other.finished) return;
  const otherGap = other.distance - opponent.distance;
  const minGap = phoneGraphicsActive() ? 420 : 320;
  if (Math.abs(otherGap) < minGap) spreadPressure += otherGap >= 0 ? -0.075 : 0.075;
});
const damageDrag = Math.max(0.28, 1 - ((opponent.damage || 0) / 100) * 0.58);
const vehiclePace = fieldSpeed * (0.78 + vehicle.speed * 0.24);
const packPace = 0.92 + opponent.aiTune * 0.045 + racePressure + duelSurge + Math.max(-0.08, Math.min(0.08, spreadPressure));
const target = opponent.wrecked
  ? Math.max(18, maxSpeed * 0.16 * damageDrag)
  : vehiclePace * opponent.targetBias * routePush * packPace * damageDrag;
opponent.speed += (target - opponent.speed) * Math.min(1, dt * (opponent.wrecked ? 0.45 : 0.85 + vehicle.handling * 0.35));
opponent.distance = Math.min(length + 80, opponent.distance + Math.max(0, opponent.speed) * dt);
opponent.wobble += dt * (0.85 + index * 0.08);
const homeLane = Number.isFinite(Number(opponent.homeLane)) ? Number(opponent.homeLane) : ((index % 5) - 2);
const flowLane = homeLane * 0.82 + Math.sin(opponent.wobble + opponent.packPhase) * 0.42 + Math.sin(opponent.wobble * 0.47 +
index) * 0.18;
const closePassWindow = gapToPlayer > -180 && gapToPlayer < 260;
const sideFromPlayer = Math.sign(opponent.lane - raceState.lane) || (index % 2 ? -1 : 1);
const passSide = closePassWindow ? sideFromPlayer : (index % 2 ? -1 : 1);
const clearance = closePassWindow ? 1.86 : 1.08;
const duellLane = Math.max(-2.08, Math.min(2.08, raceState.lane + passSide * clearance + Math.sin(opponent.wobble * 1.6) *
0.14));
const laneTarget = closePassWindow && !opponent.wrecked ? duellLane : flowLane;
if (!opponent.wrecked) {
  opponent.lane += (laneTarget - opponent.lane) * dt * (closePassWindow ? 1.35 : 0.42) * vehicle.handling * Math.max(0.42,
damageDrag);
}
opponent.lane += (opponent.laneVelocity || 0) * dt;
opponent.laneVelocity = (opponent.laneVelocity || 0) * Math.max(0, 1 - dt * (opponent.wrecked ? 0.75 : 1.35));
opponent.lane = Math.max(-2.15, Math.min(2.15, opponent.lane));
opponent.spin += (opponent.wrecked ? 1.8 : 0.2) * dt * Math.sign(opponent.laneVelocity || 1);
opponent.bumpCooldown = Math.max(0, (opponent.bumpCooldown || 0) - dt);
const phoneMode = phoneGraphicsActive();
const laneOverlap = Math.abs(opponent.lane - raceState.lane);
const wheelToWheel = Math.abs(opponent.distance - raceState.distance) < (phoneMode ? 24 : 32) && laneOverlap < (phoneMode ? 0.56
: 0.64);
if (wheelToWheel && opponent.bumpCooldown <= 0 && raceState.resetTimer <= 0) {
  opponent.bumpCooldown = phoneMode ? 1.45 : 1.15;
  const side = Math.sign(opponent.lane - raceState.lane) || (index % 2 ? -1 : 1);
  const shove = 0.82 - Math.min(0.82, laneOverlap);
  opponent.speed *= phoneMode ? 0.9 : 0.84;
  opponent.laneVelocity += side * (phoneMode ? 2.25 : 1.75) * (1 + shove);
  opponent.focus = Math.max(0, opponent.focus - (phoneMode ? 10 : 16));
  raceState.speed *= phoneMode ? 0.93 : 0.87;
  raceState.lateralVelocity -= side * (phoneMode ? 0
```

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```
.78 : 1.12) * (1 + shove * 0.65);
    raceState.lane = Math.max(-2.18, Math.min(2.18, raceState.lane - side * (phoneMode ? 0.055 : 0.07) * (1 + shove)));
    raceState.cameraShake = Math.max(raceState.cameraShake, phoneMode ? 4 : 7);
    applyVehicleDamage(phoneMode ? 3.2 : 8) / Math.max(0.72, selectedVehicle().mass), `${opponent.name} side contact`);
    applyOpponentDamage(opponent, (phoneMode ? 12 : 17) / Math.max(0.74, vehicle.mass), `${opponent.name} damaged`, canvas.width /
2 + raceState.lane * laneWidth(), canvas.height * 0.76, opponent.color || "#ffd166");
    burst(canvas.width / 2 + raceState.lane * laneWidth(), canvas.height * 0.76, opponent.color || "#ffd166");
    playHitSound("Impact");
  }
  if (opponent.multiplayerRole === "ally" && Math.abs(gapToPlayer) < 560 && !opponent.wrecked) {
    raceState.teamScore = (raceState.teamScore || 0) + dt * (10 + Math.max(0, raceState.speed) * 0.035);
  }
  const gapAfter = opponent.distance - raceState.distance;
  const previouslyAhead = opponent.wasAhead !== false;
  if (previouslyAhead && gapAfter < -2 && raceState.elapsed > 1.4) {
    raceState.overtakes = (raceState.overtakes || 0) + 1;
    raceState.score += 240 * raceState.combo;
    raceState.teamScore = (raceState.teamScore || 0) + (opponent.multiplayerRole === "ally" ? 80 : 120);
    raceState.combo = Math.min(5, raceState.combo + 0.22);
    opponent.passCooldown = 1.8;
    showToast(`Overtake: ${opponent.name}`);
    burst(canvas.width / 2 + raceState.lane * laneWidth(), canvas.height * 0.72, opponent.color || "#ffd166");
    playHitSound("coin");
  } else if (!previouslyAhead && gapAfter > 6 && opponent.passCooldown <= 0 && raceState.elapsed > 3) {
    raceState.passesAgainst = (raceState.passesAgainst || 0) + 1;
    raceState.combo = Math.max(1, raceState.combo * 0.94);
    opponent.passCooldown = 2.1;
    showToast(`${opponent.name} passed you. Draft back.`);
  }
  if (gapAfter > 3) opponent.wasAhead = true;
  if (gapAfter < -2) opponent.wasAhead = false;
  if (opponent.distance >= length) opponent.finished = true;
});
enforceOpponentSpacing(length, dt);
}

function phonePassLaneSlots(playerLane = raceState.lane) {
  const slots = [-1.95, -0.72, 0.72, 1.95];
  return slots.sort((a, b) => Math.abs(b - playerLane) - Math.abs(a - playerLane));
}

function enforceOpponentSpacing(length, dt) {
  const active = raceState.opponents
    .filter((opponent) => opponent && !opponent.finished)
    .sort((a, b) => a.distance - b.distance);
  const phoneMode = phoneGraphicsActive();
  const minGap = phoneMode ? 1500 : 1250;
  const laneSlots = phonePassLaneSlots();
  active.forEach((opponent, index) => {
    if (index > 0) {
      const previous = active[index - 1];
      const gap = opponent.distance - previous.distance;
      if (gap < minGap) {
        opponent.distance = Math.min(length + 80, previous.distance + minGap);
        opponent.speed = Math.max(opponent.speed, previous.speed * 0.98);
      }
    }
  });
  const playerGap = opponent.distance - raceState.distance;
  if (playerGap > -420 && playerGap < 2200 && !opponent.wrecked) {
    const nearPlayerLane = Math.abs(opponent.lane - raceState.lane) < 0.9 && playerGap > -110 && playerGap < 520;
    const laneConflict = active.some((other) => {
      if (other === opponent || other.finished) return false;
      const otherGap = other.distance - opponent.distance;
      return Math.abs(otherGap) < (phoneMode ? 420 : 360) && Math.abs(other.lane - opponent.lane) < 0.72;
    });
    const openSlot = laneSlots.find((slot) => Math.abs(slot - raceState.lane) > 1.05 && active.every((other) => {
      if (other === opponent) return true;
      const otherGap = other.distance - raceState.distance;
      return Math.abs(otherGap - playerGap) > (phoneMode ? 420 : 360) || Math.abs(other.lane - slot) > 0.72;
    }));
    const currentHome = Number.isFinite(Number(opponent.homeLane)) ? Number(opponent.homeLane) : opponent.lane;
    const laneTarget = nearPlayerLane || laneConflict
      ? (openSlot ?? (opponent.lane < raceState.lane ? -1.95 : 1.95))
      : currentHome;
    opponent.ho
```

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```
meLane = laneTarget;
const laneRate = nearPlayerLane ? 2.4 : laneConflict ? 1.25 : 0.24;
opponent.lane += (laneTarget - opponent.lane) * Math.min(1, dt * laneRate);
if (nearPlayerLane) opponent.speed *= Math.max(0.96, 1 - dt * 0.16);
opponent.lane = Math.max(-2.15, Math.min(2.15, opponent.lane));
}
});
}

function visiblePhoneTrafficCount() {
  if (!phoneGraphicsActive()) return 0;
  let count = 0;
  const inRange = (object) => {
    const gap = ensureRoadDistance(object) - raceState.distance;
    return gap > -180 && gap < 1150;
  };
  raceState.rivals.forEach((rival) => {
    if (!rival.wrecked && inRange(rival)) count += 1;
  });
  raceState.police.forEach((unit) => {
    if (!unit.wrecked && inRange(unit)) count += 1;
  });
  raceState.opponents.forEach((opponent) => {
    const gap = opponent.distance - raceState.distance;
    if (!opponent.wrecked && gap > -180 && gap < 1150) count += 1;
  });
  (raceState.oncoming || []).forEach((unit) => {
    if (!unit.wrecked && inRange(unit)) count += 1;
  });
  return count;
}

function choosePhoneSpawnLane() {
  const slots = phonePassLaneSlots();
  const busy = (lane) => {
    const closeToPlayer = Math.abs(lane - raceState.lane) < 1.05;
    if (closeToPlayer) return true;
    return raceState.opponents.some((opponent) => {
      const gap = opponent.distance - raceState.distance;
      return gap > -160 && gap < 1200 && Math.abs(opponent.lane - lane) < 0.76;
    }) || raceState.rivals.some((rival) => {
      const gap = ensureRoadDistance(rival) - raceState.distance;
      return gap > -160 && gap < 1200 && Math.abs(rival.lane - lane) < 0.76;
    }) || raceState.police.some((unit) => {
      const gap = ensureRoadDistance(unit) - raceState.distance;
      return gap > -160 && gap < 1200 && Math.abs(unit.lane - lane) < 0.76;
    }) || (raceState.oncoming || []).some((unit) => {
      const gap = ensureRoadDistance(unit) - raceState.distance;
      return gap > -160 && gap < 1200 && Math.abs(unit.lane - lane) < 0.76;
    });
  };
  return slots.find((lane) => !busy(lane)) ?? slots.find((lane) => Math.abs(lane - raceState.lane) > 1.05) ?? -1.95;
}

function collisionScreenYForObject(object) {
  const distance = ensureRoadDistance(object);
  const lane = Number(object && object.lane) || 0;
  const projected = roadObjectPos(lane, distance);
  if (projected && Number.isFinite(projected.y)) return projected.y;
  return roadObjectY(object);
}

function playerContactWindow(kind = "traffic", phoneMode = phoneGraphicsActive()) {
  if (phoneMode) {
    return kind === "civilian"
      ? { lane: 0.2, ahead: 8, behind: 9, scrapeLane: 0.24, scrapeAhead: 10, scrapeBehind: -8 }
      : kind === "police"
      ? { lane: 0.26, ahead: 12, behind: 12, scrapeLane: 0.46, scrapeAhead: 16, scrapeBehind: -12 }
      : { lane: 0.24, ahead: 10, behind: 10, scrapeLane: 0.42, scrapeAhead: 14, scrapeBehind: -10 };
  }
  return kind === "civilian"
    ? { lane: 0.24, ahead: 10, behind: 7, scrapeLane: 0.28, scrapeAhead: 12, scrapeBehind: -7 }
    : kind === "police"
    ? { lane: 0.34, ahead: 22, behind: 18, scrapeLane: 0.58, scrapeAhead: 26, scrapeBehind: -18 }
    : { lane: 0.32, ahead: 18, behind: 16, scrapeLane: 0.54, scrapeAhead: 22, scrapeBehind: -16 };
}

function playerVehicleContact(object, kind = "traffic") {
  const objectLane = Number(object && object.lane) || 0;
  const laneGap = Math.abs(objectLane - raceState.lane);
  const gap = ensureRoadDistance(object) - raceState.distance;
  const phoneMode = phoneGraphicsActive();
  const contactWindow = playerContactWindow(kind, phoneMode);
  if (phoneMode) {
    return {
      laneHit: laneGap < contactWindow.lane,
      yHit: gap > -contactWindow.behind && gap < contactWindow.ahead,
      gap,
      laneGap
    };
  }
  const screenY = collisionScreenYForObject(object);
  const screenAligned = screenY > canvas.height * 0.7 && screenY < canvas.height * 1.02;
  return {
    laneHit: laneGap < contactWindow.lane,
    yHit: gap > -contactWindow.behind && gap < contactWindow.ahead && screenAligned,
    gap,
    screenY,
    laneGap
  };
}

function visiblePlayerVehicleContact(object, kind = "traffic") {
  const contac
```

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```
t = playerVehicleContact(object, kind);
const gap = Number.isFinite(contact.gap) ? contact.gap : ensureRoadDistance(object) - raceState.distance;
const screenY = Number.isFinite(contact.screenY) ? contact.screenY : collisionScreenYForObject(object);
const phoneMode = phoneGraphicsActive();
const visibleTop = phoneMode ? canvas.height * 0.58 : canvas.height * 0.66;
const visibleBottom = phoneMode ? canvas.height * 1.02 : canvas.height * 0.98;
const contactWindow = playerContactWindow(kind, phoneMode);
const aheadLimit = contactWindow.ahead;
const rearForgiveness = -contactWindow.behind;
const visible = screenY >= visibleTop && screenY <= visibleBottom;
const avoidableOverlap = gap >= rearForgiveness && gap <= aheadLimit;
const sideScrapeLane = contactWindow.scrapeLane;
const sideScrapeGap = gap >= contactWindow.scrapeBehind && gap <= contactWindow.scrapeAhead;
const sideScrape = kind !== "civilian" && contact.laneGap < sideScrapeLane && visible && sideScrapeGap;
return {
  ...contact,
  gap,
  screenY,
  visible,
  hit: Boolean((contact.laneHit && contact.yHit && visible && avoidableOverlap) || sideScrape)
};
}

function updateHazardWarning(object, kind = "traffic") {
  if (!raceState.active || raceState.resetTimer > 0) return;
  const gap = ensureRoadDistance(object) - raceState.distance;
  const laneGap = Math.abs((Number(object) && object.lane) || 0) - raceState.lane;
  const screenY = collisionScreenYForObject(object);
  const phoneMode = phoneGraphicsActive();
  const warningRange = phoneMode ? 260 : 190;
  if (gap > 38 && gap < warningRange && laneGap < 0.96 && screenY > canvas.height * 0.42 && screenY < canvas.height * 0.88) {
    raceState.hazardWarningTimer = Math.max(raceState.hazardWarningTimer || 0, 0.35);
    raceState.hazardWarningLabel = kind === "civilian"
      ? ((object && object.kind === "animal") ? "ANIMAL AHEAD - SLOW DOWN" : "CIVILIAN AHEAD - CHANGE LANES")
      : kind === "police"
        ? "POLICE AHEAD - CHANGE LANES"
        : "CAR AHEAD - CHANGE LANES";
  }
}

function routeVehicleBoost(place, vehicleType, rival = false) {
  if (place === "snow" && vehicleType === "snowmobile") return rival ? 1.12 : 1.1;
  if (place === "harbor" && vehicleType === "boat") return rival ? 1.14 : 1.12;
  if ((place === "airfield" || place === "skybase") && isAirVehicleType(vehicleType)) return rival ? 1.12 : 1.1;
  if (place === "pursuit" && vehicleType === "policecar") return rival ? 1.14 : 1.12;
  if (place === "freight" && (vehicleType === "semi" || vehicleType === "truck")) return rival ? 1.16 : 1.14;
  if (place === "interstate" && vehicleType === "semi") return rival ? 1.2 : 1.18;
  if (place === "farm" && vehicleType === "tractor") return rival ? 1.18 : 1.16;
  if (place === "monsterpark" && vehicleType === "monster") return rival ? 1.18 : 1.16;
  if (place === "military" && vehicleType === "tank") return rival ? 1.18 : 1.16;
  if (place === "tokyo" && (vehicleType === "car" || vehicleType === "f1" || vehicleType === "prototype")) return rival ? 1.1 :
1.08;
  if (place === "desert" && (vehicleType === "monster" || vehicleType === "truck")) return rival ? 1.12 : 1.1;
  if (place === "rainforest" && (vehicleType === "truck" || vehicleType === "monster" || vehicleType === "tractor")) return rival ?
1.1 : 1.08;
  if (place === "europe" && (vehicleType === "f1" || vehicleType === "prototype" || vehicleType === "car" ||
isAirVehicleType(vehicleType))) return rival ? 1.1 : 1.08;
  return 1;
}

function raceTrafficTypes(race = selectedRace, vehicle = selectedVehicle()) {
  if (vehicle.type === "boat" || race.place === "harbor") return ["boat", "boat", "boat", "boat"];
  if (isAirVehicleType(vehicle.type)) return ["airplane", "helicopter", "fighterjet", "drone", "bomber"];
  if (vehicle.type === "policecar" || race.place === "pursuit") return ["policecar", "car", "truck", "policecar"];
  if (vehicle.type === "tank") return ["tank", "truck", "semi", "monster"];
  if (vehicle.type === "monster") return ["monster", "truck", "monster", "semi"];
  if (vehicle.type === "tractor") return ["tractor", "tractor", "truck"];
  if (vehicle.type === "semi") return ["semi", "semi", "truck"];
```

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```
if (vehicle.type === "snowmobile" || race.place === "snow") return ["snowmobile", "snowmobile", "truck"];
if (race.place === "freight") return ["semi", "semi", "truck", "monster"];
if (race.place === "interstate") return ["semi", "semi", "truck", "semi"];
if (race.place === "farm") return ["tractor", "tractor", "truck", "monster"];
if (race.place === "monsterpark") return ["monster", "monster", "truck", "monster"];
if (race.place === "military") return ["tank", "tank", "truck", "semi"];
if (race.place === "skybase") return ["fighterjet", "drone", "airplane", "bomber"];
if (race.place === "pursuit") return ["policecar", "car", "truck", "policecar"];
if (race.place === "tokyo") return ["car", "f1", "prototype", "semi"];
if (race.place === "desert") return ["monster", "truck", "semi", "car"];
if (race.place === "rainforest") return ["truck", "monster", "tractor", "car"];
return ["car", "f1", "prototype", "truck", "monster"];
}

function raceAllowsPolice(race = selectedRace, vehicle = selectedVehicle()) {
  if (activeScenario().hotPursuit) return true;
  if (vehicle.type === "policecar") return false;
  if (race.place === "military" && vehicle.type === "tank") return true;
  return Boolean(race.hotPursuit || /hot pursuit/i.test(`${race.name} ${race.mood}`));
}

function pursuitIntensity() {
  const scenario = activeScenario();
  const routeProgress = raceLength() ? Math.max(0, Math.min(1, raceState.distance / raceLength())) : 0;
  const longRacePressure = Math.max(0, Math.min(1, raceState.elapsed / 160));
  const heatPressure = Math.max(0, Math.min(1, raceState.heat / 100));
  const scenarioPressure = scenario.hotPursuit ? 0.34 : scenario.heatBoost || 0;
  return Math.max(0, Math.min(1, heatPressure * 0.52 + longRacePressure * 0.24 + routeProgress * 0.18 + scenarioPressure));
}

function raceHasOncomingTraffic(race = selectedRace, vehicle = selectedVehicle()) {
  if (isWaterVehicleType(vehicle.type) || isAirVehicleType(vehicle.type) || vehicle.type === "tank") return false;
  return ["city", "tokyo", "coast", "desert", "europe", "pursuit"].includes(race.place);
}

function raceAllowsCivilians(race = selectedRace, vehicle = selectedVehicle()) {
  if (isAirVehicleType(vehicle.type) || isWaterVehicleType(vehicle.type)) return false;
  return !["airfield", "snow", "military", "monsterpark", "skybase"].includes(race.place);
}

function raceIsNight(race = selectedRace) {
  return race.place === "city" || race.place === "tokyo" || /night|neon|tunnel/i.test(race.name + " " + race.mood);
}

function raceRankings() {
  const racers = [{ name: "You", distance: raceState.distance, player: true }];
  raceState.opponents.forEach((opponent) => racers.push({
    name: opponent.name,
    distance: opponent.distance,
    player: false
  }));
  racers.sort((a, b) => b.distance - a.distance);
  return racers;
}

function playerPosition() {
  return raceRankings().findIndex((racer) => racer.player) + 1;
}

function spawnRival() {
  const age = ageBands[activeProfile.age];
  const director = raceState.director || getDirector(activeProfile);
  const laneOptions = [-2, -1, 0, 1, 2].filter((candidate) => Math.abs(candidate - raceState.lane) > 0.72);
  const lane = phoneGraphicsActive() ? choosePhoneSpawnLane() : (laneOptions[Math.floor(Math.random() * laneOptions.length)] ?? (Math.floor(Math.random() * 5) - 2));
  const routeTypes = raceTrafficTypes(selectedRace, selectedVehicle());
  const type = routeTypes[Math.floor(Math.random() * routeTypes.length)];
  const def = vehicleDefs.find((vehicle) => vehicle.type === type) || vehicleDefs[0];
  raceState.rivals.push({
    lane,
    distance: roadSpawnDistance(0.34, 0.43),
    w: type === "semi" ? 82 : type === "tractor" ? 64 : type === "monster" || type === "truck" ? 68 : 54,
    h: isAirVehicleType(type) ? 104 : 92,
    speed: (38 + Math.random() * 58) * age.traffic * director.traffic,
    color: Math.random() > 0.45 ? def.color : (Math.random() > 0.5 ? "#ff5b6b" : "#ffd166"),
    type,
    passed: false,
    contactCooldown: 0,
    damage: 0,
    wrecked: false,
    laneVelocity: 0,
    spin: 0
  });
}

function spawnPoliceUnit() {
  const lane = phoneGraphicsActive() ? choosePh
```


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```
oneSpawnLane() : Math.floor(Math.random() * 5) - 2;
const intensity = pursuitIntensity();
const heavy = intensity > 0.68 && Math.random() < 0.38;
raceState.police.push({
  lane,
  distance: roadSpawnDistance(0.34, 0.42),
  w: heavy ? 74 : 62,
  h: heavy ? 118 : 112,
  speed: (heavy ? 48 : 58) + Math.random() * 62 + raceState.heat * (heavy ? 0.2 : 0.26),
  type: heavy ? "truck" : "car",
  label: heavy ? "POLICE SUV" : "POLICE",
  passed: false,
  contactCooldown: 0,
  damage: 0,
  wrecked: false,
  laneVelocity: 0,
  spin: 0
});
}

function spawnCivilian() {
  const side = Math.random() > 0.5 ? 1 : -1;
  const species = routeAnimalSpecies(selectedRace && selectedRace.place ? selectedRace.place : "city", Math.random() * 9999);
  const animal = Boolean(species) && Math.random() < routeAnimalChance(selectedRace && selectedRace.place ? selectedRace.place :
"city");
  const crossing = animal ? Math.random() < 0.18 : Math.random() < 0.26;
  raceState.civilians.push({
    lane: crossing ? side * (animal ? 1.7 + Math.random() * 0.24 : 1.9 + Math.random() * 0.22) : side * (animal ? 2.42 +
Math.random() * 0.34 : 2.34 + Math.random() * 0.28),
    distance: roadSpawnDistance(0.24, 0.34),
    side,
    crossing,
    speed: crossing ? (animal ? 0.42 + Math.random() * 0.34 : 0.34 + Math.random() * 0.26) : 0,
    step: Math.random() * Math.PI * 2,
    hit: false,
    warned: false,
    kind: animal ? "animal" : "person",
    species: animal ? species : "",
    jacket: animal ? "" : (Math.random() > 0.5 ? "#ffd166" : "#46d9ff")
  });
}

function routeAnimalChance(place = "city") {
  const chances = {
    coast: 0.32,
    city: 0.14,
    canyon: 0.34,
    alpine: 0.34,
    harbor: 0.2,
    freight: 0.12,
    interstate: 0.16,
    farm: 0.48,
    tokyo: 0.1,
    desert: 0.32,
    rainforest: 0.42,
    europe: 0.3,
    pursuit: 0.12
  };
  return chances[place] || 0.18;
}

function routeAnimalSet(place = "city") {
  const sets = {
    coast: ["dog", "shoreBird", "horse"],
    city: ["dog", "bird"],
    canyon: ["deer", "coyote", "bird"],
    alpine: ["deer", "goat", "bird"],
    harbor: ["dog", "shoreBird"],
    freight: ["dog", "bird"],
    interstate: ["deer", "dog"],
    farm: ["cow", "horse", "sheep", "dog"],
    tokyo: ["dog", "bird"],
    desert: ["camel", "coyote", "bird"],
    rainforest: ["monkey", "tapir", "bird"],
    europe: ["deer", "sheep", "dog"],
    pursuit: ["dog", "bird"]
  };
  return sets[place] || sets.city;
}

function routeAnimalSpecies(place = "city", seed = 0) {
  const list = routeAnimalSet(place);
  if (!list || !list.length) return "";
  return list[Math.abs(Math.floor(seed)) % list.length];
}

function spawnOncomingTraffic() {
  const laneChoices = [-1.55, -0.48, 0.48, 1.55].filter((lane) => Math.abs(lane - raceState.lane) > 0.48);
  const routeTypes = raceTrafficTypes(selectedRace, selectedVehicle()).filter((type) => !isWaterVehicleType(type) &&
!isAirVehicleType(type));
  const type = routeTypes[Math.floor(Math.random() * routeTypes.length)] || "car";
  const def = vehicleDefs.find((vehicle) => vehicle.type === type) || vehicleDefs[0];
  raceState.oncoming.push({
    lane: laneChoices[Math.floor(Math.random() * laneChoices.length)] || -1.55,
    distance: roadSpawnDistance(0.18, 0.3) + 360,
    speed: 54 + Math.random() * 44,
    w: type === "semi" ? 82 : type === "monster" || type === "truck" ? 68 : 56,
    h: type === "semi" ? 112 : 94,
    color: def.color,
    type,
    damage: 0,
    wrecked: false,
    contactCooldown: 0,
    laneVelocity: 0,
    spin: Math.PI
  });
}

function routeFeatureCatalog(race = selectedRace, vehicle = selectedVehicle()) {
  if (vehicle.type === "boat") {
    return [
      { type: "hide", label: "Marina Cover", detail: "duck behind docks", icon: "dock" },
      { type: "shortcut", label: "Canal Cut", detail: "side-water sprint", icon: "water" }
    ]
  }
}
```

```
    ];  
  }  
  if (isAirVehicleType(vehicle.type)) {  
    return [  
      { type: "hide", label: "Cloud Cover", detail: "drop heat in cloud banks", icon: "cloud" },  
      { type: "shortcut", label: "Sky Gate", detail: "faster air corridor", icon: "gate" }  
    ];  
  }  
}  
const byPlace = {  
  city: [  
    { type
```

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```
: "hide", label: "Parking Garage", detail: "building hideout", icon: "building" },
  { type: "hide", label: "Pole Cutoff", detail: "telephone pole shadow", icon: "pole" },
  { type: "shortcut", label: "Alley Branch", detail: "extra city road", icon: "road" }
],
tokyo: [
  { type: "hide", label: "Neon Garage", detail: "building hideout", icon: "building" },
  { type: "shortcut", label: "Tunnel Branch", detail: "expressway slip road", icon: "road" }
],
canyon: [
  { type: "hide", label: "Cave Pull-Off", detail: "cave hideout", icon: "cave" },
  { type: "shortcut", label: "Rock Cut", detail: "side canyon track", icon: "mountain" }
],
alpine: [
  { type: "hide", label: "Mountain Tunnel", detail: "mountain cover", icon: "mountain" },
  { type: "shortcut", label: "Service Road", detail: "extra mountain road", icon: "road" }
],
desert: [
  { type: "hide", label: "Dune Cave", detail: "desert cave cover", icon: "cave" },
  { type: "shortcut", label: "Wadi Track", detail: "extra desert track", icon: "road" }
],
rainforest: [
  { type: "hide", label: "Canopy Cover", detail: "jungle hideout", icon: "building" },
  { type: "shortcut", label: "Bridge Bypass", detail: "extra jungle track", icon: "road" }
],
freight: [
  { type: "hide", label: "Truck Stop", detail: "building hideout", icon: "building" },
  { type: "shortcut", label: "Service Lane", detail: "extra freight road", icon: "road" }
],
interstate: [
  { type: "hide", label: "Weigh Station", detail: "pull behind the scale house", icon: "building" },
  { type: "shortcut", label: "Convoy Bypass", detail: "express freight lane", icon: "road" }
],
farm: [
  { type: "hide", label: "Barn Cover", detail: "building hideout", icon: "building" },
  { type: "shortcut", label: "Field Track", detail: "extra dirt track", icon: "road" }
],
monsterpark: [
  { type: "hide", label: "Stadium Tunnel", detail: "duck under the stands", icon: "building" },
  { type: "shortcut", label: "Berm Cut", detail: "wide dirt shortcut", icon: "road" }
],
military: [
  { type: "hide", label: "Bunker Cover", detail: "armored hideout", icon: "building" },
  { type: "shortcut", label: "Range Cut", detail: "concrete proving lane", icon: "road" }
],
pursuit: [
  { type: "hide", label: "Station Garage", detail: "police garage cover", icon: "building" },
  { type: "hide", label: "Underpass Cover", detail: "drop heat below the overpass", icon: "building" },
  { type: "shortcut", label: "Interceptor Lane", detail: "lights-and-sirens bypass", icon: "road" }
],
airfield: [
  { type: "hide", label: "Hangar Cover", detail: "building hideout", icon: "building" },
  { type: "shortcut", label: "Taxiway Cut", detail: "extra runway lane", icon: "road" }
],
snow: [
  { type: "hide", label: "Pine Shelter", detail: "snow cover", icon: "mountain" },
  { type: "shortcut", label: "Ice Cut", detail: "extra snow track", icon: "road" }
],
coast: [
  { type: "hide", label: "Cliff Pull-Off", detail: "coastal cover", icon: "mountain" },
  { type: "shortcut", label: "Beach Road", detail: "extra coast road", icon: "road" }
],
europe: [
  { type: "hide", label: "Village Arch", detail: "building hideout", icon: "building" },
  { type: "shortcut", label: "Switchback Cut", detail: "extra alpine road", icon: "road" }
]
];
return byPlace[race.place] || byPlace.city;
}

function spawnRouteFeature() {
  const catalog = routeFeatureCatalog();
  const wantsHide = (raceState.chaseActive || raceState.heat > 42 || activeScenario().hotPursuit) && Math.random() < 0.62;
  const candidates = catalog.filter((feature) => feature.type === (wantsHide ? "hide" : "shortcut"));
  const feature = (candidates.length ? candidates : catalog[Math.floor(Math.random() * (candidates.length ? candidates.length : catalog.length))]);
  const side = Math.random() > 0.5 ? 1 : -1;
  raceState.routeFeatures.push({
    id: `${feature.type}:${Date.now()}:${Math.random()}`,
    type: feature.type,
    label: feature.label,
    deta
```

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```
    il: feature.detail,
    icon: feature.icon,
    lane: side * (feature.type === "hide" ? 2.1 : 1.82),
    distance: roadSpawnDistance(0.16, 0.28) + 260,
    side,
    pulse: Math.random() * Math.PI * 2,
    used: false,
    warned: false
  });
}

function safePickupLane(lane) {
  const value = Number(lane);
  if (!Number.isFinite(value)) return 0;
  return pickupLaneSlots.reduce((best, candidate) => Math.abs(candidate - value) < Math.abs(best - value) ? candidate : best,
  pickupLaneSlots[0]);
}

function choosePickupLane() {
  const sorted = pickupLaneSlots
  .map((lane) => ({ lane, distanceFromPlayer: Math.abs(lane - (raceState.lane || 0)), jitter: Math.random() })))
  .sort((a, b) => b.distanceFromPlayer + b.jitter * 0.18 - (a.distanceFromPlayer + a.jitter * 0.18));
  const preferAway = Math.random() < 0.58 ? sorted.slice(0, 3) : sorted;
  return preferAway[Math.floor(Math.random() * preferAway.length)].lane;
}

function updateRouteFeatures(dt) {
  raceState.routeFeatures = Array.isArray(raceState.routeFeatures) ? raceState.routeFeatures : [];
  raceState.routeFeatures.forEach((feature) => {
    feature.pulse = (feature.pulse || 0) + dt * 4;
    const gap = ensureRoadDistance(feature) - raceState.distance;
    const laneGap = Math.abs((feature.lane || 0) - raceState.lane);
    if (!feature.warned && gap > 36 && gap < 260 && laneGap < 1.25) {
      feature.warned = true;
      raceState.hazardWarningTimer = Math.max(raceState.hazardWarningTimer || 0, 0.75);
      raceState.hazardWarningLabel = feature.type === "hide" ? `${feature.label.toUpperCase()} - SLOW AND MOVE OVER` :
`${feature.label.toUpperCase()} - CUT IN`;
    }
    if (feature.used || gap < -18 || gap > 44 || laneGap > 0.46) return;
    if (feature.type === "hide") {
      if (raceState.speed > 118) {
        raceState.hazardWarningTimer = Math.max(raceState.hazardWarningTimer || 0, 0.6);
        raceState.hazardWarningLabel = "TOO FAST TO HIDE - BRAKE";
        return;
      }
      feature.used = true;
      raceState.hideCooldown = 2.4;
      raceState.heat = Math.max(0, raceState.heat - (activeScenario().hotPursuit ? 34 : 26));
      raceState.chaseActive = raceState.heat > 20 && raceState.police.length > 0;
      raceState.focus = Math.min(100, raceState.focus + 5);
      raceState.score += 420;
      raceState.teamScore = (raceState.teamScore || 0) + 80;
      showToast(`${feature.label}: heat dropped.`);
      burst(canvas.width / 2 + raceState.lane * laneWidth(), canvas.height * 0.72, "#46d9ff");
    } else {
      feature.used = true;
      const gain = Math.min(460, Math.max(180, raceLength() * 0.018));
      raceState.distance = Math.max(raceState.distance, Math.min(raceLength() - 120, raceState.distance + gain));
      raceState.speed = Math.min(raceState.speed + 24, raceState.speed * 1.08 + 12);
      raceState.score += 520 * raceState.combo;
      raceState.combo = Math.min(5, raceState.combo + 0.28);
      raceState.teamScore = (raceState.teamScore || 0) + 90;
      showToast(`${feature.label}: shortcut gained.`);
      burst(canvas.width / 2 + raceState.lane * laneWidth(), canvas.height * 0.72, "#bbf24a");
    }
  });
  raceState.routeFeatures = raceState.routeFeatures.filter((feature) => roadObjectY(feature) < canvas.height + 220 &&
  feature.distance > raceState.distance - 360 && !feature.used);
}

function routeFeatureSize(feature) {
  if (feature.type === "hide") return feature.icon === "pole" ? { w: 42, h: 108 } : { w: 92, h: 86 };
  return { w: 104, h: 42 };
}

function fireTankCannon() {
  raceState.cannonCooldown = 1.25;
  const targets = [
    ...raceState.rivals.map((target) => ({ target, group: "traffic", gap: ensureRoadDistance(target) - raceState.distance })),
    ...raceState.police.map((target) => ({ target, group: "police", gap: ensureRoadDistance(target) - raceState.distance })),
    ...raceState.oncoming.map((target) => ({ target, group: "traffic", gap: ensureRoadDistance(target) - raceState.distance })),
    ...raceState.opponents.map((target) => ({ target, group: "opponent", gap: target.distance - raceState.distance }))]
  ].filter((item) => item.target && !item.target.wrecked && item.gap > 18 && item.gap < 520 && Math.abs((ite
```

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```
m.target.lane || 0) - raceState.lane) < 0.86)
    .sort((a, b) => a.gap - b.gap);
const hit = targets[0];
const x = canvas.width / 2 + raceState.lane * laneWidth();
const y = canvas.height * 0.68;
burst(x, y, "#ff1666");
playHitSound("police");
raceState.focus = Math.max(0, raceState.focus - 2.5);
if (!hit) {
    showToast("Tank cannon fired. No target in lane.");
    return;
}
if (hit.group === "opponent") {
    applyOpponentDamage(hit.target, 58, `${hit.target.name} disabled`, x, y, hit.target.color || "#ff1666");
} else {
    applyTrafficDamage(hit.target, 72, hit.group === "police" ? "Interceptor disabled" : "Vehicle cleared", x, y, hit.target.color || "#ff1666");
}
raceState.score += 260;
showToast("Route cleared.");
}

function spawnCoin() {
    const lane = choosePickupLane();
    raceState.coinsOnRoad.push({ lane, distance: roadSpawnDistance(0.35, 0.45), r: 13, pulse: Math.random() * 10 });
}

function tick(dt) {
    if (!raceState.active || input.paused) return;
    const upgrades = activeVehicleUpgrades(activeProfile);
    const age = ageBands[activeProfile.age];
    const director = raceState.director || getDirector(activeProfile);
    const vehicle = selectedVehicle();
    const scenario = activeScenario();
    const gasInput = input.gas || input.gamepadGas;
    const brakeInput = input.brake || input.gamepadBrake;
    const boostInput = input.boost || input.gamepadBoost;
    raceState.crashCooldown = Math.max(0, raceState.crashCooldown - dt);
    raceState.damageAlertTimer = Math.max(0, (raceState.damageAlertTimer || 0) - dt);
    raceState.hazardWarningTimer = Math.max(0, (raceState.hazardWarningTimer || 0) - dt);
    raceState.goalIntroTimer = Math.max(0, (raceState.goalIntroTimer || 0) - dt);
    if (raceState.resetTimer > 0) {
        raceState.resetTimer = Math.max(0, raceState.resetTimer - dt);
        raceState.speed *= Math.max(0, 1 - dt * 5.6);
        raceState.lateralVelocity *= Math.max(0, 1 - dt * 6.8);
        raceState.steerAngle *= Math.max(0, 1 - dt * 6.4);
        raceState.slip = Math.max(raceState.slip * 0.9, 0.55);
        raceState.cameraShake = Math.max(raceState.cameraShake, 2 + raceState.resetTimer * 1.8);
        emitDrivingEffects(dt, false, true, false, 0);
        if (raceState.resetTimer <= 0) completeVehicleReset();
        updateRaceUi();
        updateAudio();
        return;
    }
    const routeProgress = raceLength() ? Math.max(0, Math.min(1, raceState.distance / raceLength())) : 0;
    const roadCondition = routeWeatherCondition(selectedRace.place, routeProgress, raceState.elapsed || 0);
    raceState.roadCondition = roadCondition;
    const roadGrip = weatherRoadGrip(roadCondition.surface);
    const roadSpeed = weatherRoadSpeed(roadCondition.surface);
    const surfaceBoost = routeVehicleBoost(selectedRace.place, vehicle.type);
    const damageRatio = Math.min(0.86, (raceState.damage || 0) / 120);
    const performanceHealth = Math.max(0.42, 1 - damageRatio * 0.44);
    const handlingHealth = Math.max(0.48, 1 - damageRatio * 0.34);
    const maxSpeed = (245 + upgrades.engine * 26) * age.speed * vehicle.speed * surfaceBoost * performanceHealth * roadSpeed;
    const keySteer = Number(input.right) - Number(input.left);
    const touchSteer = Number(input.touchSteer) || 0;
    const manualSteer = Math.abs(touchSteer) > Math.abs(keySteer) ? touchSteer : keySteer;
    const steerInput = Math.abs(input.gamepadSteer) > Math.abs(manualSteer) ? input.gamepadSteer : manualSteer;
    const boostPower = boostInput && gasInput && raceState.focus > 2 ? (60 + upgrades.boost * 17) * performanceHealth : 0;
    const speedLimit = maxSpeed + boostPower;
    const forwardSpeed = Math.max(0, raceState.speed);
    const reverseSpeed = Math.min(0, raceState.speed);
    const throttleAccel = (42 + upgrades.engine * 4.8) * age.speed * vehicle.speed * surfaceBoost * performanceHealth * roadSpeed;
    const boostAccel = boostInput && gasInput && raceState.focus > 2 ? (54 + upgrades.boost * 9) * performanceHealth : 0;
    const brakeForce = ((88 + upgrades.tires * 5) / Math.max(0.72, vehicle.mass)) * Math.max(0.64, 1 - damageRatio * 0.24);
    const reverseAccel = 34 / Math.max(0.8, vehicle.mass);
    const rollingDrag = raceState.speed > 0 ? (10 + forwardSpeed * 0.035 + forwardSpeed * forwardSp
```

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```
eed * 0.00045) : (14 + Math.abs(reverseSpeed) * 0.12);

if (gasInput) {
  raceState.speed += (throttleAccel + boostAccel) * dt;
}
if (brakeInput) {
  raceState.speed -= (raceState.speed > 4 ? brakeForce : reverseAccel) * dt;
}
if (!gasInput && !brakeInput && Math.abs(raceState.speed) > 0.05) {
  const drag = Math.min(Math.abs(raceState.speed), rollingDrag * dt);
  raceState.speed += raceState.speed > 0 ? -drag : drag;
}
raceState.speed = Math.max(-38, Math.min(speedLimit, raceState.speed));
raceState.throttleLoad += ((gasInput ? 1 : 0) - raceState.throttleLoad) * Math.min(1, dt * 4.4);
raceState.brakeHeat += ((brakeInput && Math.abs(raceState.speed) > 12 ? 1 : 0) - raceState.brakeHeat) * Math.min(1, dt * 5.2);
if (boostInput && gasInput && raceState.focus > 2) raceState.focus -= dt * Math.max(4, 13 - upgrades.boost * 1.4);

const speedGrip = Math.max(0.54, Math.min(1.18, Math.abs(raceState.speed) / 112));
const steerTarget = steerInput * (0.92 + upgrades.tires * 0.045) * handlingHealth * roadGrip * (raceState.speed < -1 ? -0.62 : 1);
raceState.steerAngle += (steerTarget - raceState.steerAngle) * Math.min(1, dt * (5.4 + vehicle.handling * 2.2) * handlingHealth);
const lateralAccel = raceState.steerAngle * speedGrip * (4.2 + vehicle.handling * 1.35 + upgrades.tires * 0.28) * handlingHealth *
roadGrip;
raceState.lateralVelocity += lateralAccel * dt;
const grip = (2.15 + vehicle.handling * 1.25 + upgrades.tires * 0.32 + (brakeInput ? 0.75 : 0)) * handlingHealth * roadGrip;
raceState.lateralVelocity *= Math.max(0, 1 - dt * grip);
if (Math.abs(steerInput) > 0.05 && Math.abs(raceState.speed) < 46) {
  raceState.lateralVelocity += steerInput * dt * (2.1 + vehicle.handling * 0.48) * handlingHealth * roadGrip;
}
raceState.lane += raceState.lateralVelocity * dt;
raceState.x = raceState.lane;
const targetVisuallane = raceState.lane;
raceState.visuallane = Number.isFinite(raceState.visuallane) ? raceState.visuallane : targetVisuallane;
raceState.visuallane += (targetVisuallane - raceState.visuallane) * Math.min(1, dt * 8.5);
const offRoad = Math.max(0, Math.abs(raceState.lane) - 2.04);
if (offRoad > 0) {
  raceState.lane = Math.max(-2.28, Math.min(2.28, raceState.lane));
  raceState.lateralVelocity -= Math.sign(raceState.lane) * offRoad * 6.5 * dt;
  raceState.speed *= Math.max(0.985, 1 - dt * (0.22 + offRoad * 0.24));
  raceState.focus -= dt * offRoad * 1.8;
  if (Math.abs(raceState.speed) > 90) {
    raceState.hazardWarningTimer = Math.max(raceState.hazardWarningTimer || 0, 0.45);
    raceState.hazardWarningLabel = "ROAD EDGE - STEER BACK";
  }
}
}
raceState.slip = Math.min(1, Math.abs(raceState.lateralVelocity) * 0.42 + Math.abs(raceState.steerAngle) * speedGrip * 0.18 +
raceState.brakeHeat * 0.22 + offRoad * 0.5 + (1 - roadGrip) * 0.34);
raceState.distance = Math.max(0, raceState.distance + Math.max(0, raceState.speed) * dt);
raceState.roadOffset += raceState.speed * dt;
raceState.elapsed += dt;
const roadShake = roadCondition.hazard === "earthquake" ? Math.sin((raceState.elapsed || 0) * 9.5) * 0.08 : 0;
const turnTarget = roadTurnAt(raceState.distance) + roadShake;
raceState.roadTurn += (turnTarget - (raceState.roadTurn || 0)) * Math.min(1, dt * 0.72);
raceState.roadCurve += ((raceState.roadTurn || 0) - (raceState.roadCurve || 0)) * Math.min(1, dt * 0.65);
raceState.score += dt * Math.max(0, raceState.speed) * raceState.combo * 0.32;
raceState.heat = Math.min(100, raceState.heat + dt * (gasInput ? 0.8 + Math.max(0, raceState.speed) / 185 + scenario.heatBoost :
0.1) + (boostInput && gasInput ? dt * 2.4 : 0));
raceState.heatClock -= dt;
raceState.cameraShake = Math.max(0, raceState.cameraShake - dt * 18);
if (roadCondition.hazard === "earthquake") raceState.cameraShake = Math.max(raceState.cameraShake, 1.4 + Math.abs(roadShake) *
18);
emitDrivingEffects(dt, gasInput, brakeInput, boostInput, steerInput);
updateOpponents(dt, maxSpeed);
raceState.spawnClock -= dt;
raceState.coinClock -= dt;
raceState.civilianClock = Math.max(0, (raceState.civilianClock || 0) - dt);
raceState.oncomingClock = Math.max(0, (raceState.oncoming
```

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```
clock || 0) - dt);
raceState.cannonCooldown = Math.max(0, (raceState.cannonCooldown || 0) - dt);
raceState.routeFeatureClock = Math.max(0, (raceState.routeFeatureClock || 0) - dt);
raceState.hideCooldown = Math.max(0, (raceState.hideCooldown || 0) - dt);
const phoneMode = phoneGraphicsActive();
const trafficRoom = !phoneMode || visiblePhoneTrafficCount() < 3;
if (raceState.spawnClock <= 0 && raceState.speed > 45 && raceState.elapsed > 5 && trafficRoom) {
  spawnRival();
  raceState.spawnClock = phoneMode
    ? Math.max(3.4, 4.8 - age.traffic * director.traffic * 0.18)
    : Math.max(1.15, 2.1 - age.traffic * director.traffic * 0.16 - raceState.elapsed * 0.002);
} else if (phoneMode && raceState.spawnClock <= 0) {
  raceState.spawnClock = 1.4;
}
if (raceState.coinClock <= 0 && raceState.speed > 30) {
  spawnCoin();
  raceState.coinClock = Math.max(0.78, (1.15 - upgrades.magnet * 0.045) / Math.max(0.8, director.coinRate * 0.82));
}
if (raceAllowsCivilians(selectedRace, vehicle) && raceState.civilianClock <= 0 && raceState.speed > 34) {
  spawnCivilian();
  raceState.civilianClock = 8.5 + Math.random() * 10 + (phoneMode ? 3 : 0);
}
if (raceHasOncomingTraffic(selectedRace, vehicle) && raceState.oncomingClock <= 0 && raceState.speed > 55 && (!phoneMode || visiblePhoneTrafficCount() < 3)) {
  spawnOncomingTraffic();
  raceState.oncomingClock = 7.5 + Math.random() * 8;
}
if (vehicle.type === "tank" && boostInput && raceState.cannonCooldown <= 0) fireTankCannon();
if (raceState.routeFeatureClock <= 0 && raceState.speed > 38 && raceState.elapsed > 7) {
  spawnRouteFeature();
  raceState.routeFeatureClock = 8 + Math.random() * 8 + (phoneMode ? 3 : 0);
}
updateRouteFeatures(dt);
const policeEligible = raceAllowsPolice(selectedRace, vehicle) || raceState.heat > 82;
const intensity = pursuitIntensity();
const policeLimit = phoneMode ? (intensity > 0.75 ? 3 : 2) : (intensity > 0.72 ? 5 : intensity > 0.48 ? 3 : 2);
if (policeEligible && raceState.heat > 38 && raceState.heatClock <= 0 && raceState.speed > 45 && raceState.police.length < policeLimit && (!phoneMode || visiblePhoneTrafficCount() < 4)) {
  raceState.chaseActive = true;
  const spawnCount = !phoneMode && intensity > 0.68 && raceState.police.length < policeLimit - 1 ? 2 : 1;
  for (let i = 0; i < spawnCount; i += 1) spawnPoliceUnit();
  raceState.heatClock = Math.max(1.9, 9.2 - intensity * 6.1 - upgrades.engine * 0.08);
  showToast(intensity > 0.72 ? "Hot pursuit escalating. More units inbound." : "Pursuit unit entering the route.");
}
moveObjects(dt);
if (raceState.focus <= 0) {
  raceState.focus = 18;
  raceState.damage = Math.max(raceState.damage || 0, 82);
  triggerVehicleReset("Focus recovered");
}
if (raceState.distance >= raceLength() && raceState.elapsed >= minimumRaceSeconds) endRace(false);
raceState.autoSaveClock = (raceState.autoSaveClock || 0) + dt;
if (raceState.autoSaveClock >= 7.5) saveRace(true);
updateRaceUi();
updateAudio();
}

function moveObjects(dt) {
  const carLane = raceState.lane;
  const vehicle = selectedVehicle();
  raceState.rivals.forEach((rival) => {
    ensureRoadDistance(rival, -120);
    rival.damage = Math.max(0, Math.min(100, Number(rival.damage) || 0));
    rival.laneVelocity = Number(rival.laneVelocity) || 0;
    rival.spin = Number(rival.spin) || 0;
    rival.contactCooldown = Math.max(0, (rival.contactCooldown || 0) - dt);
    const damageDrag = Math.max(0.24, 1 - ((rival.damage || 0) / 100) * 0.66);
    const beforeY = roadObjectY(rival);
    if (rival.wrecked) {
      rival.speed *= Math.max(0, 1 - dt * 1.65);
      rival.spin += dt * (2.2 + Math.abs(rival.laneVelocity || 0)) * Math.sign(rival.laneVelocity || 1);
    } else {
      rival.speed *= Math.max(0, 1 - dt * (1 - damageDrag) * 0.18);
      const avoid = beforeY > canvas.height * 0.34 && beforeY < canvas.height * 0.7 && Math.abs(rival.lane - carLane) < 0.85
        ? Math.sign(rival.lane - carLane || 1) * 0.55
        : 0;
      rival.laneVelocity += avoid * dt;
    }
    rival.lane += (rival.laneVelocity || 0) * dt;
    rival.laneVelocity = (rival.laneVelocity || 0) * Math.m
```

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```
ax(0, 1 - dt * 1.2);
rival.lane = Math.max(-2.24, Math.min(2.24, rival.lane));
rival.distance += Math.max(8, rival.speed * damageDrag) * dt;
const screenY = roadObjectY(rival);
if (!rival.passed && !rival.wrecked && screenY > canvas.height * 0.72) {
  rival.passed = true;
  raceState.dodges += 1;
  raceState.combo = Math.min(5, raceState.combo + 0.12);
}
updateHazardWarning(rival, "traffic");
const contact = visiblePlayerVehicleContact(rival, "traffic");
if (contact.hit && rival.contactCooldown <= 0 && raceState.resetTimer <= 0) {
  const phoneMode = phoneGraphicsActive();
  rival.contactCooldown = phoneMode ? 1.35 : 0.55;
  const shield = activeVehicleUpgrades(activeProfile).shield;
  const impact = Math.max(phoneMode ? 3.2 : 8, ((28 - shield * 2.7) * (phoneMode ? 0.28 : 1)) / Math.max(0.72, vehicle.mass));
  const side = Math.sign(rival.lane - carLane) || (Math.random() > 0.5 ? 1 : -1);
  applyVehicleDamage(impact, "Visible traffic contact");
  applyTrafficDamage(rival, impact * 1.65, "Traffic disabled", canvas.width / 2 + carLane * laneWidth(), canvas.height * 0.76,
    rival.color || "#ff5b6b");
  rival.color || "#ff5b6b";
  raceState.combo = 1;
  raceState.cameraShake = Math.max(raceState.cameraShake, phoneMode ? 4 : 7);
  raceState.speed *= Math.max(phoneMode ? 0.68 : 0.42, (phoneMode ? 0.88 : 0.78) - impact * 0.007);
  raceState.lateralVelocity -= side * (phoneMode ? 0.62 : 1.05);
  rival.laneVelocity += side * (phoneMode ? 1.75 : 1.35);
  rival.speed *= Math.max(phoneMode ? 0.42 : 0.24, (phoneMode ? 0.72 : 0.62) - impact * 0.006);
  burst(canvas.width / 2 + carLane * laneWidth(), canvas.height * 0.76, "#ff5b6b");
  playHitSound("impact");
}
});
raceState.police.forEach((unit) => {
  ensureRoadDistance(unit, -170);
  unit.damage = Math.max(0, Math.min(100, Number(unit.damage) || 0));
  unit.laneVelocity = Number(unit.laneVelocity) || 0;
  unit.spin = Number(unit.spin) || 0;
  unit.contactCooldown = Math.max(0, (unit.contactCooldown || 0) - dt);
  const damageDrag = Math.max(0.22, 1 - ((unit.damage || 0) / 100) * 0.62);
  if (unit.wrecked) {
    unit.speed *= Math.max(0, 1 - dt * 1.45);
    unit.spin += dt * (2.4 + Math.abs(unit.laneVelocity) || 0) * Math.sign(unit.laneVelocity || 1);
  } else {
    const pursuitPull = Math.sign(carLane - unit.lane) * dt * (0.18 + raceState.heat / 260) * damageDrag;
    unit.lane += pursuitPull;
  }
  unit.lane += (unit.laneVelocity || 0) * dt;
  unit.laneVelocity = (unit.laneVelocity || 0) * Math.max(0, 1 - dt * 1.05);
  unit.lane = Math.max(-2.2, Math.min(2.2, unit.lane));
  unit.distance += Math.max(8, unit.speed * damageDrag) * dt;
  const screenY = roadObjectY(unit);
  if (!unit.passed && !unit.wrecked && screenY > canvas.height * 0.72) {
    unit.passed = true;
    raceState.dodges += 1;
    raceState.combo = Math.min(5, raceState.combo + 0.2);
    raceState.score += 180;
    raceState.heat = Math.max(10, raceState.heat - 4);
  }
  updateHazardWarning(unit, "police");
  const contact = visiblePlayerVehicleContact(unit, "police");
  if (contact.hit && unit.contactCooldown <= 0 && raceState.resetTimer <= 0) {
    const phoneMode = phoneGraphicsActive();
    unit.contactCooldown = phoneMode ? 1.45 : 0.65;
    const shield = activeVehicleUpgrades(activeProfile).shield;
    const impact = Math.max(phoneMode ? 4.5 : 12, ((38 - shield * 2.5) * (phoneMode ? 0.26 : 1)) / Math.max(0.72, vehicle.mass));
    const side = Math.sign(unit.lane - carLane) || (Math.random() > 0.5 ? 1 : -1);
    applyVehicleDamage(impact, "Visible police contact");
    applyTrafficDamage(unit, impact * 1.25, "Interceptor damaged", canvas.width / 2 + carLane * laneWidth(), canvas.height * 0.78,
      "#46d9ff");
    raceState.combo = 1;
    raceState.cameraShake = Math.max(raceState.cameraShake, phoneMode ? 6 : 12);
    raceState.heat = Math.min(100, raceState.heat + (phoneMode ? 7 : 12));
    raceState.speed *= Math.max(phoneMode ? 0.62 : 0.36, (phoneMode ? 0.84 : 0.72) - impact * 0.006);
    raceState.lateralVelocity -= side * (phoneMode ? 0.72 : 1.3);
  }
  u
```


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```
nit.laneVelocity += side * (phoneMode ? 1.8 : 1.5);
unit.speed *= Math.max(phoneMode ? 0.38 : 0.22, (phoneMode ? 0.7 : 0.58) - impact * 0.004);
burst(canvas.width / 2 + carLane * laneWidth(), canvas.height * 0.78, "#46d9ff");
burst(canvas.width / 2 + carLane * laneWidth(), canvas.height * 0.78, "#ff3348");
playHitSound("police");
}
});
raceState.oncoming = Array.isArray(raceState.oncoming) ? raceState.oncoming : [];
raceState.oncoming.forEach((unit) => {
  ensureRoadDistance(unit, -180);
  unit.damage = Math.max(0, Math.min(100, Number(unit.damage) || 0));
  unit.contactCooldown = Math.max(0, (unit.contactCooldown || 0) - dt);
  unit.laneVelocity = Number(unit.laneVelocity) || 0;
  unit.spin = Number(unit.spin) || Math.PI;
  const damageDrag = Math.max(0.26, 1 - ((unit.damage || 0) / 100) * 0.64);
  if (unit.wrecked) {
    unit.speed *= Math.max(0, 1 - dt * 1.5);
    unit.spin += dt * 2.1;
  } else {
    unit.distance -= Math.max(10, unit.speed * damageDrag) * dt;
    unit.lane += (unit.laneVelocity || 0) * dt;
    unit.laneVelocity *= Math.max(0, 1 - dt * 1.2);
  }
  updateHazardWarning(unit, "traffic");
  const contact = visiblePlayerVehicleContact(unit, "traffic");
  if (contact.hit && unit.contactCooldown <= 0 && raceState.resetTimer <= 0) {
    const phoneMode = phoneGraphicsActive();
    unit.contactCooldown = phoneMode ? 1.4 : 0.6;
    const shield = activeVehicleUpgrades(activeProfile).shield;
    const impact = Math.max(phoneMode ? 4 : 11, ((34 - shield * 2.4) * (phoneMode ? 0.3 : 1)) / Math.max(0.72, vehicle.mass));
    const side = Math.sign(unit.lane - carLane) || (Math.random() > 0.5 ? 1 : -1);
    applyVehicleDamage(impact, "Oncoming traffic contact");
    applyTrafficDamage(unit, impact * 1.45, "Oncoming vehicle disabled", canvas.width / 2 + carLane * laneWidth(), canvas.height *
0.76, unit.color || "#ffd166");
    raceState.combo = 1;
    raceState.cameraShake = Math.max(raceState.cameraShake, phoneMode ? 5 : 10);
    raceState.speed *= Math.max(phoneMode ? 0.6 : 0.32, (phoneMode ? 0.82 : 0.68) - impact * 0.006);
    raceState.lateralVelocity -= side * (phoneMode ? 0.7 : 1.25);
    unit.laneVelocity += side * 1.45;
    burst(canvas.width / 2 + carLane * laneWidth(), canvas.height * 0.76, unit.color || "#ffd166");
    playHitSound("impact");
  }
});
raceState.civilians = Array.isArray(raceState.civilians) ? raceState.civilians : [];
raceState.civilians.forEach((person) => {
  ensureRoadDistance(person, -120);
  const animal = person.kind === "animal";
  person.step = (person.step || 0) + dt * (animal ? 8 : 6);
  if (person.crossing && !person.hit) {
    person.lane -= person.side * (person.speed || 0.42) * dt;
    if (Math.abs(person.lane) < 0.55 && !person.warned) {
      person.warned = true;
      raceState.hazardWarningTimer = Math.max(raceState.hazardWarningTimer || 0, animal ? 1 : 0.8);
      raceState.hazardWarningLabel = animal ? "ANIMAL CROSSING - SLOW DOWN" : "CIVILIAN CROSSING - AVOID";
    }
  }
  updateHazardWarning(person, "civilian");
  const contact = visiblePlayerVehicleContact(person, "civilian");
  if (contact.hit && !person.hit && raceState.resetTimer <= 0) {
    person.hit = true;
    raceState.civilianHits = (raceState.civilianHits || 0) + 1;
    raceState.penaltyCoins = (raceState.penaltyCoins || 0) + (animal ? 120 : 160);
    raceState.penaltyRep = (raceState.penaltyRep || 0) + (animal ? 4 : 5);
    raceState.coins = Math.max(0, raceState.coins - (animal ? 1 : 2));
    raceState.score = Math.max(0, raceState.score - (animal ? 420 : 520));
    raceState.focus = Math.max(0, raceState.focus - (animal ? 9 : 12));
    raceState.combo = 1;
    raceState.cameraShake = Math.max(raceState.cameraShake, animal ? 3 : 6);
    raceState.speed *= animal ? 0.72 : 0.82;
    burst(canvas.width / 2 + raceState.lane * laneWidth(), canvas.height * 0.78, animal ? "#bbf24a" : "#f4fbf8");
    showToast(animal ? "Animal hazard penalty. Slow down and steer clear." : "Civilian hit. Reputation and coins penalized.");
    playHitSound(animal ? "coin" : "impa
```

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```
ct");
    }
  });
  const magnet = activeVehicleUpgrades(activeProfile).magnet;
  raceState.coinsOnRoad.forEach((coin) => {
    coin.lane = safePickupLane(coin.lane);
    ensureRoadDistance(coin, -60);
    const screenY = roadObjectY(coin);
    coin.pulse += dt * 8;
    const catchRange = 0.25 + magnet * 0.065;
    const laneHit = Math.abs(coin.lane - carLane) < catchRange;
    const yHit = screenY > canvas.height * 0.62 && screenY < canvas.height * 0.9;
    if (laneHit && yHit && !coin.hit) {
      coin.hit = true;
      raceState.coins += 1;
      raceState.score += 120 * raceState.combo;
      raceState.combo = Math.min(5, raceState.combo + 0.18);
      burst(canvas.width / 2 + carLane * laneWidth(), canvas.height * 0.76, "#ffd166");
      playHitSound("coin");
    }
  });
  raceState.rivals = raceState.rivals.filter((r) => roadObjectY(r) < canvas.height + 220 && r.distance > raceState.distance - 360);
  raceState.police = raceState.police.filter((p) => roadObjectY(p) < canvas.height + 240 && p.distance > raceState.distance - 390);
  raceState.oncoming = raceState.oncoming.filter((p) => roadObjectY(p) < canvas.height + 260 && p.distance > raceState.distance - 420);
  raceState.civilians = raceState.civilians.filter((p) => roadObjectY(p) < canvas.height + 180 && p.distance > raceState.distance - 300 && !p.hit);
  if (!raceState.police.length && raceState.heat < 16) raceState.chaseActive = false;
  raceState.coinsOnRoad = raceState.coinsOnRoad.filter((c) => roadObjectY(c) < canvas.height + 80 && !c.hit);
  raceState.particles.forEach((p) => {
    p.x += p.vx * dt;
    p.y += p.vy * dt;
    p.vx *= Math.max(0, 1 - dt * 1.8);
    p.vy *= Math.max(0, 1 - dt * 1.2);
    p.life -= dt;
  });
  raceState.particles = raceState.particles.filter((p) => p.life > 0);
}

function applyOpponentDamage(opponent, amount, reason, x, y, color) {
  if (!opponent || amount <= 0) return;
  opponent.damage = Math.max(0, Math.min(100, (opponent.damage || 0) + amount));
  opponent.bumpCooldown = Math.max(opponent.bumpCooldown || 0, 0.7);
  opponent.focus = Math.max(0, (opponent.focus || 100) - amount * 0.85);
  if (opponent.damage >= 100 && !opponent.wrecked) {
    opponent.wrecked = true;
    opponent.speed *= 0.22;
    opponent.laneVelocity += (Math.random() > 0.5 ? 1 : -1) * 1.4;
    opponent.spin += 0.8;
    showToast(`${reason}. Rival limping.`);
  }
  burst(x, y, color || "#ffd166");
  if (opponent.damage > 48) emitVehicleSmoke(x, y, opponent.damage, color || "#888888");
}

function applyTrafficDamage(target, amount, reason, x, y, color) {
  if (!target || amount <= 0) return;
  target.damage = Math.max(0, Math.min(100, (target.damage || 0) + amount));
  if (target.damage >= 100 && !target.wrecked) {
    target.wrecked = true;
    target.speed *= 0.18;
    target.spin += 1.2;
    target.laneVelocity += (Math.random() > 0.5 ? 1 : -1) * 1.2;
    showToast(reason);
  }
  if (target.damage > 35) emitVehicleSmoke(x, y, target.damage, color || "#777777");
}

function emitVehicleSmoke(x, y, damage, color = "#777777") {
  const count = Math.min(10, 3 + Math.floor(damage / 18));
  for (let i = 0; i < count; i += 1) {
    const life = 0.45 + Math.random() * 0.5;
    raceState.particles.push({
      x: x + (Math.random() - 0.5) * 32,
      y: y + (Math.random() - 0.5) * 22,
      vx: (Math.random() - 0.5) * 80,
      vy: -40 - Math.random() * 80,
      life,
      maxLife: life,
      radius: 10 + Math.random() * 18,
      color: damage > 72 ? "rgba(22,22,20,0.48)" : "rgba(120,126,120,0.34)"
    });
  }
}

function applyVehicleDamage(amount, reason = "Impact", quiet = false) {
  if (!raceState.active || raceState.resetTimer > 0 || amount <= 0) return;
  const shield = activeProfile ? activeVehicleUpgrades(activeProfile).shield : 0;
  const vehicle = selectedVehicle();
  const massResist = Math.max(0.55, Math.min(1.15, 1.18 / Math.max(0.72, vehicle.mass)));
  const shieldResist = Math.max(0.58, 1 - shield * 0.055);
  const finalAmount = amount * massResist * shieldResist;
  raceState.damage = Math.max(0, Math.min(100, (raceState.damage || 0) + finalAmount));
  raceState.focus = Math.max(0, raceState.focus - Math.max(3, finalAmount * 0.64));
  raceState
```

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```
.crashCooldown = Math.max(raceState.crashCooldown, quiet ? 0.18 : 0.75);
raceState.damageAlertTimer = Math.max(raceState.damageAlertTimer || 0, quiet ? 0.65 : 1.45);
raceState.damageAlertLabel = `${reason}: +${Math.max(1, Math.round(finalAmount))}%`;
if (!quiet) {
  const label = raceState.damage >= 100 ? "Critical damage" : `${reason}: ${Math.round(raceState.damage)}% damage`;
  showRoast(label);
}
if (raceState.damage >= 100) triggerVehicleReset(reason);
}

function triggerVehicleReset(reason = "Critical damage") {
  if (!raceState.active || raceState.resetTimer > 0) return;
  raceState.resetTimer = 3.2;
  raceState.resetReason = reason;
  raceState.speed = Math.min(18, Math.max(-4, raceState.speed * 0.18));
  raceState.lateralVelocity = 0;
  raceState.steerAngle = 0;
  raceState.combo = 1;
  raceState.cameraShake = Math.max(raceState.cameraShake, 18);
  raceState.focus = Math.max(8, raceState.focus - 7);
  clearTouchDriveInputs();
  burst(canvas.width / 2 + raceState.lane * laneWidth(), canvas.height * 0.78, "#ff5b6b");
  burst(canvas.width / 2 + raceState.lane * laneWidth(), canvas.height * 0.78, "#ffd166");
  playHitSound("impact");
  showRoast(`${reason}. Resetting vehicle.`);
}

function completeVehicleReset() {
  raceState.resetTimer = 0;
  raceState.resetReason = "";
  raceState.damage = Math.min(42, raceState.damage * 0.42);
  raceState.lane = 0;
  raceState.x = 0;
  raceState.visualLane = 0;
  raceState.speed = 0;
  raceState.lateralVelocity = 0;
  raceState.steerAngle = 0;
  raceState.slip = 0;
  raceState.crashCooldown = 1.2;
  showRoast("Vehicle reset. Damage stabilized.");
}

function manualResetVehicle() {
  if (!raceState.active) return;
  if (raceState.resetTimer > 0) return;
  raceState.damage = Math.min(100, raceState.damage + 6);
  triggerVehicleReset("Manual reset");
}

function burst(x, y, color) {
  for (let i = 0; i < 16; i += 1) {
    const life = 0.35 + Math.random() * 0.35;
    raceState.particles.push({
      x,
      y,
      vx: (Math.random() - 0.5) * 240,
      vy: (Math.random() - 0.8) * 220,
      life,
      maxLife: life,
      radius: 4 + Math.random() * 4,
      color
    });
  }
}

function emitDrivingEffects(dt, gasInput, brakeInput, boostInput, steerInput) {
  if (!raceState.active || canvas.width <= 0 || canvas.height <= 0) return;
  const speed = Math.abs(raceState.speed);
  const x = canvas.width / 2 + (raceState.lane - cameraLaneOffset(1)) * laneWidth();
  const baseY = cameraMode === "cockpit" ? canvas.height * 0.78 : cameraMode === "hood" ? canvas.height * 0.93 : canvas.height * 0.87;
  const slipChance = Math.min(0.75, raceState.slip * 0.34 + (brakeInput && speed > 24 ? 0.12 : 0) + (Math.abs(steerInput) > 0 && speed > 90 ? 0.06 : 0));
  if (Math.random() < slipChance * dt * 18) {
    const side = Math.random() > 0.5 ? -1 : 1;
    const life = 0.35 + Math.random() * 0.28;
    raceState.particles.push({
      x: x + side * laneWidth() * 0.24 + (Math.random() - 0.5) * 18,
      y: baseY + Math.random() * 24,
      vx: -raceState.lateralVelocity * 34 + (Math.random() - 0.5) * 42,
      vy: -32 - Math.random() * 34,
      life,
      maxLife: life,
      radius: 10 + Math.random() * 18,
      color: brakeInput ? "rgba(190,205,205,0.42)" : "rgba(120,134,128,0.32)"
    });
  }
  const driftPower = Math.max(0, Math.min(1, ((raceState.slip || 0) - 0.16) * 1.95 + Math.abs(raceState.lateralVelocity || 0) * 0.07));
  if (driftPower > 0.08 && speed > 36 && Math.random() < dt * (18 + driftPower * 36)) {
    const side = Math.sign(raceState.lateralVelocity || raceState.steerAngle || (Math.random() - 0.5)) || 1;
    const life = 0.42 + Math.random() * 0.38;
    const dusty = selectedRace.place === "desert" || selectedRace.place === "canyon" || selectedRace.place === "farm";
    raceState.particles.push({
      x: x - side * laneWidth() * (0.18 + Math.random() * 0.18),
      y: baseY + 8 + Math.random() * 18,
      vx: -side * (34 + Math.random() * 72) - raceState.lateralVelocity * 18,
      vy: -18 - Math.random() * 44,
      life,
      maxLife: life,
      radius: 16 + Math.random() * 34,
      color: dusty ? "rgba(255,183,74,0.28)" : "rgba(205,218,214,0.34)"
    });
  }
}
1
```

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```
f (boostInput && gasInput && raceState.focus > 2 && Math.random() < dt * 22) {
  const life = 0.25 + Math.random() * 0.18;
  raceState.particles.push({
    x: x + (Math.random() - 0.5) * laneWidth() * 0.5,
    y: baseY + 18 + Math.random() * 22,
    vx: (Math.random() - 0.5) * 36,
    vy: 80 + Math.random() * 90,
    life,
    maxLife: life,
    radius: 12 + Math.random() * 16,
    color: "rgba(255,209,102,0.58)"
  });
}
if (gasInput && speed < 20 && Math.random() < dt * 8) {
  const life = 0.28 + Math.random() * 0.22;
  raceState.particles.push({
    x: x + (Math.random() - 0.5) * laneWidth() * 0.42,
    y: baseY + 12,
    vx: (Math.random() - 0.5) * 24,
    vy: 34 + Math.random() * 36,
    life,
    maxLife: life,
    radius: 8 + Math.random() * 12,
    color: "rgba(244,251,248,0.26)"
  });
}
const damage = raceState.damage || 0;
if (damage > 24 && Math.random() < dt * (4 + damage * 0.12)) {
  const life = 0.48 + Math.random() * 0.5;
  raceState.particles.push({
    x: x + (Math.random() - 0.5) * laneWidth() * 0.38,
    y: baseY + 8 + Math.random() * 18,
    vx: (Math.random() - 0.5) * 36 - raceState.lateralVelocity * 18,
    vy: 38 + Math.random() * 80,
    life,
    maxLife: life,
    radius: 12 + Math.random() * 24,
    color: damage > 70 ? "rgba(25,25,24,0.42)" : "rgba(135,138,132,0.28)"
  });
}
}

function updateRaceUi() {
  $("#distanceBar").style.width = `${Math.min(100, (raceState.distance / raceLength()) * 100)}%`;
  $("#missionBar").style.width = `${Math.min(100, missionProgress() * 100)}%`;
  const damageBar = $("#damageBar");
  if (damageBar) damageBar.style.width = `${Math.min(100, raceState.damage || 0)}%`;
  const resetCar = $("#resetCar");
  if (resetCar) resetCar.textContent = raceState.resetTimer > 0 ? `Reset ${Math.ceil(raceState.resetTimer)}` : "Reset Car";
  $("#gasBtn").classList.toggle("pressed", input.gas || input.gamepadGas);
  $("#brakeBtn").classList.toggle("pressed", input.brake || input.gamepadBrake);
  $("#boostBtn").classList.toggle("pressed", input.boost || input.gamepadBoost);
  $(".mobile-control").forEach(button => {
    const control = button.dataset.control;
    const steer = Number(input.touchSteer) || 0;
    const pressed = control === "left" ? input.left || steer < -0.08
      : control === "right" ? input.right || steer > 0.08
      : control === "neutral" ? !input.left && !input.right && Math.abs(steer) <= 0.08
      : control === "gas" ? input.gas || input.gamepadGas
      : control === "brake" ? input.brake || input.gamepadBrake
      : control === "boost" ? input.boost || input.gamepadBoost
      : control === "mode" ? touchDriveMode === "toggle"
      : control === "size" ? touchControlSize === "mini"
      : false;
    button.classList.toggle("pressed", pressed);
  });
  const touchLabel = $("#touchModeLabel");
  if (touchLabel) touchLabel.textContent = touchDriveMode === "toggle" ? "Toggle" : "Hold";
  const sizeLabel = $("#touchSizeLabel");
  if (sizeLabel) sizeLabel.textContent = touchControlSize === "mini" ? "Full" : "Mini";
  updateHud();
}

function laneWidth() {
  if (cameraMode === "cockpit") return Math.min(canvas.width * 0.145, 118);
  if (cameraMode === "hood") return Math.min(canvas.width * 0.13, 104);
  return Math.min(canvas.width * 0.115, 92);
}

function cameraLaneOffset(depth = 1) {
  if (!raceState.active) return 0;
  const lane = Number.isFinite(raceState.visualLane) ? raceState.visualLane : raceState.lane || 0;
  const follow = Math.max(0.18, Math.min(1.08, depth));
  return lane * follow;
}

let last = performance.now();
function loop(now = performance.now()) {
  const dt = Math.min(0.04, (now - last) / 1000);
  last = now;
  pollGamepad();
  tick(dt);
  drawFrame();
}

function drawFrame() {
  const w = canvas.width;
  const h = canvas.height;
  const shake = raceState.cameraShake;
  const theme = selectedRace ? selectedRace.theme : races[0].theme;
  if (useWebGLRenderer()) {
    $(".stage").classList.add("webgl");
    ctx.clearRect(0, 0, w, h);
    webglRenderer.resize(w, h);
    webglRenderer.render({
      width: w,

```

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```
        height: h,
        raceState,
        selectedRace: selectedRace || races[0],
        selectedVehicle: Object.assign({}, selectedVehicle(), { color: selectedVehicleColor(selectedVehicle()) }),
        vehicleDefs,
        cameraMode
    });
    ctx.save();
    if (shake > 0) ctx.translate((Math.random() - 0.5) * shake, (Math.random() - 0.5) * shake * 0.55);
    drawWebGLRouteAtmosphere(w, h, theme);
    drawPhoneConsoleChasePass(w, h, theme);
    drawRealisticDrivingPass(w, h, theme);
    drawWebGLFlatPavementPass(w, h, theme);
    drawGenAI RacingScenePass(w, h, theme);
    drawRouteStageLandmarks(w, h, theme);
    drawRealWorldDetailPass(w, h, theme);
    drawRouteLightingPass(w, h, theme);
    if (raceState.active) {
        drawObjects();
        drawCar(w, h);
    } else {
        drawDemoPursuitTraffic(w, h);
    }
    drawDriftStylePass(w, h, theme);
    drawCameraOverlay(w, h, theme);
    drawDamageOverlay(w, h, theme);
    if (raceState.active) drawRaceStandings(w, h);
    drawParticles();
    drawPhoneFilmicPost(w, h, theme);
    drawCinematicGrade(w, h, theme);
    drawPhoneConsolePostPass(w, h, theme);
    if (!raceState.active) drawAttract(w, h);
    drawRaceGoalIntro(w, h, theme);
    if (input.paused && raceState.active) drawPause(w, h);
    ctx.restore();
    requestAnimationFrame(loop);
    return;
}
$(".stage").classList.remove("webgl");
if (phoneGraphicsActive()) {
    drawPhoneCanvasFrame(w, h, theme, shake);
    requestAnimationFrame(loop);
    return;
}
ctx.save();
if (shake > 0) ctx.translate((Math.random() - 0.5) * shake, (Math.random() - 0.5) * shake * 0.55);
const sky = ctx.createLinearGradient(0, 0, 0, h);
sky.addColorStop(0, theme[0]);
sky.addColorStop(0.6, "#08100f");
sky.addColorStop(1, "#050807");
ctx.fillStyle = sky;
ctx.fillRect(0, 0, w, h);
drawScenery(w, h, theme);
drawRoad(w, h, theme);
drawPhoneConsoleChasePass(w, h, theme);
drawRealisticDrivingPass(w, h, theme);
drawPhoneAssetTexturePass(w, h, theme);
drawRoadWeightPass(w, h, theme);
drawRoadMotionPass(w, h, theme);
drawPhoneUltraGraphicsPass(w, h, theme);
drawGenAI RacingScenePass(w, h, theme);
drawRouteStageLandmarks(w, h, theme);
drawRealWorldDetailPass(w, h, theme);
drawRouteLightingPass(w, h, theme);
if (!raceState.active) drawDemoPursuitTraffic(w, h);
drawObjects();
drawCar(w, h);
drawDriftStylePass(w, h, theme);
drawCameraOverlay(w, h, theme);
drawDamageOverlay(w, h, theme);
if (raceState.active) drawRaceStandings(w, h);
if (raceState.active) drawPhoneLaneRadar(w, h, theme);
drawParticles();
drawPhoneFilmicPost(w, h, theme);
drawCinematicGrade(w, h, theme);
drawPhoneConsolePostPass(w, h, theme);
if (!raceState.active) drawAttract(w, h);
drawRaceGoalIntro(w, h, theme);
if (input.paused && raceState.active) drawPause(w, h);
ctx.restore();
requestAnimationFrame(loop);
}

function drawPhoneCanvasFrame(w, h, theme, shake) {
    ctx.save();
    if (shake > 0) ctx.translate((Math.random() - 0.5) * shake, (Math.random() - 0.5) * shake * 0.55);
    const sky = ctx.createLinearGradient(0, 0, 0, h);
    sky.addColorStop(0, theme[0]);
    sky.addColorStop(0.62, "#08100f");
    sky.addColorStop(1, "#050807");
    ctx.fillStyle = sky;
    ctx.fillRect(0, 0, w, h);
    drawPhoneConsoleChasePass(w, h, theme);
    drawRealisticDrivingPass(w, h, theme);
    drawPhoneAssetTexturePass(w, h, theme);
    drawPhoneRoadContrastPass(w, h, theme);
    drawGenAI RacingScenePass(w, h, theme);
    drawRouteStageLandmarks(w, h, theme);
    drawRealWorldDetailPass(w, h, theme);
    drawRouteLightingPass(w, h, theme);
    if (!raceState.active) drawDemoPursuitTraffic(w, h);
    drawObjects();
    drawCar(w, h);
    drawDriftStylePass(w, h, theme);
    drawCameraOverlay(w, h, theme);
    drawDamageOverlay(w, h, theme);
    if (raceState.active) drawRaceStandings(w, h);
    if (raceState.active) drawPhoneLaneRadar(w, h, theme);
    drawParticles();
    drawPhoneFilmicPost(w, h, theme);
    drawCinematicGrade(w, h, theme);
    drawPhoneConsolePostPass(w, h, theme);
    if (!raceState.active) drawAttract(w, h);
}
```

```
drawRaceGoalIntro(w, h, theme);
if (input.paused && raceState.active) drawPause(w, h);
ctx.restore();
}

function drawWebGLRouteAtmospher
```

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```
e(w, h, theme) {
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const phoneMode = phoneGraphicsActive();
  const cleanRoad = phoneCleanRoadActive();
  const skyLimit = phoneMode ? h * 0.34 : h * 0.62;
  ctx.save();
  const sky = ctx.createLinearGradient(0, 0, 0, skyLimit);
  sky.addColorStop(0, `${theme[0]}99`);
  sky.addColorStop(0.6, "rgba(5,8,7,0.08)");
  sky.addColorStop(1, "rgba(5,8,7,0)");
  ctx.fillStyle = sky;
  ctx.fillRect(0, 0, w, skyLimit);

  const glow = ctx.createRadialGradient(w * 0.5, h * 0.42, w * 0.06, w * 0.5, h * 0.55, w * 0.62);
  glow.addColorStop(0, phoneMode ? "rgba(244,251,248,0.025)" : "rgba(244,251,248,0.08)");
  glow.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = glow;
  ctx.fillRect(0, 0, w, phoneMode ? h * 0.42 : h);

  if (place === "tokyo" || place === "city") {
    ctx.globalAlpha = phoneMode ? (place === "tokyo" ? 0.12 : 0.06) : (place === "tokyo" ? 0.18 : 0.08);
    for (let i = 0; i < 14; i += 1) {
      const x = (i * 83 + raceState.roadOffset * 0.07) % (w + 140) - 70;
      const y = phoneMode ? h * (0.19 + (i % 4) * 0.032) : h * (0.25 + (i % 5) * 0.058);
      const dashW = 24 + (i % 4) * 18;
      const neon = ctx.createLinearGradient(x, y, x + dashW, y);
      neon.addColorStop(0, "rgba(255,255,255,0)");
      neon.addColorStop(0.5, place === "tokyo" ? "rgba(255,79,216,0.44)" : "rgba(244,251,248,0.18)");
      neon.addColorStop(1, "rgba(255,255,255,0)");
      ctx.strokeStyle = neon;
      ctx.lineWidth = 1.4;
      ctx.beginPath();
      ctx.moveTo(x, y);
      ctx.lineTo(x + dashW, y + 1);
      ctx.stroke();
    }
    ctx.globalAlpha = 1;
  }

  if (place === "desert" || place === "canyon" || place === "freight") {
    const dust = ctx.createLinearGradient(0, h * 0.38, 0, h);
    dust.addColorStop(0, "rgba(255,183,74,0)");
    dust.addColorStop(0.72, "rgba(255,183,74,0.12)");
    dust.addColorStop(1, "rgba(255,183,74,0.18)");
    ctx.fillStyle = dust;
    ctx.fillRect(0, h * 0.34, w, h * 0.66);
  }

  if (!cleanRoad && (place === "rainforest" || place === "snow")) {
    ctx.strokeStyle = place === "snow" ? "rgba(244,251,248,0.28)" : "rgba(185,255,220,0.16)";
    ctx.lineWidth = place === "snow" ? 2 : 1;
    for (let i = 0; i < 58; i += 1) {
      const x = (i * 53 + raceState.elapsed * (place === "snow" ? 24 : 70)) % w;
      const y = (i * 37 + raceState.elapsed * (place === "snow" ? 18 : 110)) % h;
      ctx.beginPath();
      ctx.moveTo(x, y);
      ctx.lineTo(x - (place === "snow" ? 4 : 12), y + (place === "snow" ? 6 : 28));
      ctx.stroke();
    }
  }

  ctx.restore();
}

function drawPhoneConsoleChasePass(w, h, theme) {
  if (!phoneGraphicsActive()) return;
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const speed = Math.max(0, raceState.speed || 0);
  const cleanRoad = phoneCleanRoadActive();
  const horizon = cameraMode === "cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const roadTop = cameraMode === "cockpit" ? w * 0.11 : cameraMode === "hood" ? w * 0.14 : w * 0.13;
  const roadBottom = cameraMode === "cockpit" ? w * 1.02 : cameraMode === "hood" ? w * 0.98 : w * 1.08;
  const wet = ["city", "tokyo", "coast", "harbor", "rainforest"].includes(place);
  const warm = ["desert", "canyon", "farm", "freight"].includes(place);
  const webglActive = useWebGLRenderer();
  const turn = raceState.roadTurn || raceState.roadCurve || 0;
  const roadCenter = (t) => {
    const clamped = Math.max(0, Math.min(1.08, t));
    const farPull = (1 - clamped) * (1 - clamped);
    const roadWave = Math.sin(clamped * 4.2 + (raceState.roadOffset || 0) * 0.004) * Math.abs(turn) * w * 0.018;
    const laneShift = cameraLaneOffset(0.25 + clamped * 0.85) * laneWidth() * (0.18 + clamped * 0.95);
    return w * 0.5 + turn * farPull * w * 0.26 + roadWave - laneShift;
  };

  ctx.save();
  const skyBottom = webglActive ? horizon + h * 0.1 : horizon + h * 0.24;
  const skyWash = ctx.createLinearGradient(0, 0, 0, skyBottom);
  skyWash.addColorStop(0, `${theme[0]}d9`);
  skyWash.addColorStop(0.54, warm ? "rgba(178,118,56,0.18)" : "rgba(70,217,255,0.11)");
  skyWash.addColorStop(1, "rgba(5,8,7,0)");
  ctx
```

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```
.fillStyle = skyWash;
ctx.fillRect(0, 0, w, skyBottom);

drawPhoneSceneryBackdrop(w, h, horizon, place, theme);
drawPhoneConsoleLandmarks(w, h, horizon, place, theme);
if (webgLActive) {
  ctx.restore();
  return;
}

ctx.beginPath();
ctx.moveTo(roadCenter(0) - roadTop, horizon);
ctx.lineTo(roadCenter(0) + roadTop, horizon);
ctx.lineTo(roadCenter(1) + roadBottom * 0.62, h + 8);
ctx.lineTo(roadCenter(1) - roadBottom * 0.62, h + 8);
ctx.closePath();
ctx.clip();

const asphalt = ctx.createLinearGradient(0, horizon, 0, h);
asphalt.addColorStop(0, wet ? "#2c3a3a" : "#292f2c");
asphalt.addColorStop(0.36, wet ? "#182223" : "#171c1a");
asphalt.addColorStop(1, "#030505");
ctx.fillStyle = asphalt;
ctx.fillRect(0, horizon, w, h - horizon);

if (!cleanRoad) {
  const edgeAlpha = 0.92;
  for (let side = -1; side <= 1; side += 2) {
    const shoulder = ctx.createLinearGradient(0, horizon, 0, h);
    shoulder.addColorStop(0, "rgba(244,251,248,0.36)");
    shoulder.addColorStop(0.42, "rgba(244,251,248,0.2)");
    shoulder.addColorStop(1, "rgba(244,251,248,0.1)");
    ctx.globalAlpha = edgeAlpha;
    ctx.strokeStyle = shoulder;
    ctx.lineWidth = Math.max(3, w * 0.005);
    ctx.beginPath();
    ctx.moveTo(roadCenter(0) + side * roadTop * 0.96, horizon + 1);
    ctx.lineTo(roadCenter(1) + side * roadBottom * 0.62, h + 8);
    ctx.stroke();
    ctx.strokeStyle = side < 0 ? "rgba(255,91,107,0.5)" : "rgba(70,217,255,0.46)";
    ctx.lineWidth = Math.max(5, w * 0.008);
    ctx.globalAlpha = 0.46;
    ctx.beginPath();
    ctx.moveTo(roadCenter(0) + side * roadTop * 0.76, horizon + 8);
    ctx.lineTo(roadCenter(1) + side * roadBottom * 0.48, h + 8);
    ctx.stroke();
  }
}
ctx.globalAlpha = 1;

if (wet && !cleanRoad) {
  ctx.globalCompositeOperation = "screen";
  for (let i = 0; i < 12; i += 1) {
    const y = horizon + (((i * 62 + raceState.roadOffset * (0.72 + speed / 380)) % (h - horizon + 90)) - 45);
    if (y < horizon || y > h) continue;
    const t = (y - horizon) / Math.max(1, h - horizon);
    const ribbonW = w * (0.08 + t * 0.44);
    const x = roadCenter(t) + Math.sin(i * 1.9 + raceState.elapsed * 0.7) * w * 0.06;
    const shine = ctx.createLinearGradient(x - ribbonW, y, x + ribbonW, y);
    shine.addColorStop(0, "rgba(255,255,255,0)");
    shine.addColorStop(0.5, "rgba(210,245,255,${0.08 + t * 0.18})");
    shine.addColorStop(1, "rgba(255,255,255,0)");
    ctx.strokeStyle = shine;
    ctx.lineWidth = 1 + t * 6;
    ctx.beginPath();
    ctx.moveTo(x - ribbonW, y);
    ctx.quadraticCurveTo(x, y + 3 + t * 12, x + ribbonW, y + 2);
    ctx.stroke();
  }
  ctx.globalCompositeOperation = "source-over";
}

const roadX = (lane, t) => roadCenter(t) + lane * laneWidth() * (0.26 + t * 1.24);
const paint = (lane, y, length, width, alpha) => {
  const t1 = Math.max(0, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
  const t2 = Math.max(0, Math.min(1.12, (y + length - horizon) / Math.max(1, h - horizon)));
  if (t2 <= 0 || t1 >= 1.12) return;
  const x1 = roadX(lane, t1);
  const x2 = roadX(lane, t2);
  ctx.globalAlpha = alpha;
  ctx.fillStyle = "rgba(230,238,234,0.82)";
  ctx.beginPath();
  ctx.moveTo(x1 - width * (0.25 + t1), y);
  ctx.lineTo(x1 + width * (0.25 + t1), y);
  ctx.lineTo(x2 + width * (0.28 + t2 * 1.25), y + length);
  ctx.lineTo(x2 - width * (0.28 + t2 * 1.25), y + length);
  ctx.closePath();
  ctx.fill();
};

const motion = raceState.roadOffset * (0.96 + Math.min(1.8, speed / 130));
const paintCount = cleanRoad ? 8 : 17;
for (let i = 0; i < paintCount; i += 1) {
  const y = horizon + (((i * 78 + motion) % (h - horizon + 120)) - 52);
  const t = Math.max(0, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
  paint(-0.5, y, 14 + t * 42, 1.5 + t * 5.2, 0.24 + t * 0.34);
  paint(0.5, y + 12, 14 + t * 38, 1.5 + t * 5.2, 0.2 + t * 0.3);
}
ctx.globalAlpha = 1;

if (!cleanRoad) {
  for (let side = -1; side <= 1; side += 2) {
    for (let i = 0; i < 12; i += 1) {
      const y = horizon + (((i * 66 + motion * 0.74) % (h - horizon + 100)) -
```


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```
28);
    if (y < horizon || y > h) continue;
    const t = Math.max(0, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    const x = roadX(side * (1.72 + t * 0.24), t);
    const dashW = 10 + t * 46;
    const dashH = 2 + t * 6;
    ctx.globalAlpha = 0.1 + t * 0.34;
    ctx.fillStyle = side < 0 ? "rgba(255,91,107,0.32)" : "rgba(70,217,255,0.28)";
    ctx.save();
    ctx.translate(x, y);
    ctx.rotate(side * turn * -0.08);
    roundRect(-dashW / 2, -dashH / 2, dashW, dashH, 2);
    ctx.fill();
    ctx.restore();
  }
}
ctx.globalAlpha = 1;

if (!cleanRoad && Math.abs(turn) > 0.22) {
  const side = turn > 0 ? 1 : -1;
  ctx.save();
  ctx.globalCompositeOperation = "screen";
  for (let i = 0; i < 8; i += 1) {
    const y = horizon + ((i * 74 + motion * 0.82) % (h - horizon + 120)) - 36;
    const t = Math.max(0.04, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    const x = roadX(side * 2.72, t);
    const boardW = 16 + t * 42;
    const boardH = 5 + t * 11;
    ctx.globalAlpha = 0.18 + t * 0.42;
    ctx.fillStyle = side > 0 ? `${theme[2]}aa` : `${theme[1]}aa`;
    ctx.save();
    ctx.translate(x, y);
    ctx.rotate(side * -0.34);
    roundRect(-boardW / 2, -boardH / 2, boardW, boardH, 2);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.76)";
    for (let dot = -1; dot <= 1; dot += 1) {
      ctx.beginPath();
      ctx.arc(dot * boardW * 0.18, 0, Math.max(1.4, boardH * 0.18), 0, Math.PI * 2);
      ctx.fill();
    }
    ctx.restore();
  }
  ctx.restore();
  ctx.globalAlpha = 1;
}

if (!cleanRoad && speed > 90) {
  ctx.globalCompositeOperation = "screen";
  ctx.globalAlpha = Math.min(0.2, (speed - 80) / 700);
  for (let side = -1; side <= 1; side += 2) {
    for (let i = 0; i < 7; i += 1) {
      const y = horizon + h * (0.12 + i * 0.11);
      ctx.strokeStyle = i % 2 ? "rgba(255,255,255,0.22)" : `${theme[1]}66`;
      ctx.lineWidth = 1 + i * 0.35;
      ctx.beginPath();
      ctx.moveTo(side < 0 ? 0 : w, y);
      ctx.lineTo(side < 0 ? w * 0.14 : w * 0.86, y + h * 0.015);
      ctx.stroke();
    }
  }
  ctx.restore();
}

function drawPhoneRoadContrastPass(w, h, theme) {
  if (!phoneGraphicsActive()) return;
  if (phoneCleanRoadActive()) return;
  const horizon = cameraMode === "cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const roadTop = cameraMode === "cockpit" ? w * 0.11 : cameraMode === "hood" ? w * 0.14 : w * 0.13;
  const roadBottom = cameraMode === "cockpit" ? w * 1.02 : cameraMode === "hood" ? w * 0.98 : w * 1.08;
  const turn = raceState.roadTurn || raceState.roadCurve || 0;
  const roadCenter = (t) => {
    const clamped = Math.max(0, Math.min(1.08, t));
    const farPull = (1 - clamped) * (1 - clamped);
    return w * 0.5 + turn * farPull * w * 0.26;
  };
  ctx.save();
  ctx.beginPath();
  ctx.moveTo(roadCenter(0) - roadTop, horizon);
  ctx.lineTo(roadCenter(0) + roadTop, horizon);
  ctx.lineTo(roadCenter(1) + roadBottom * 0.62, h + 8);
  ctx.lineTo(roadCenter(1) - roadBottom * 0.62, h + 8);
  ctx.closePath();
  ctx.clip();
  const shade = ctx.createLinearGradient(0, horizon, 0, h);
  shade.addColorStop(0, "rgba(0,0,0,0.16)");
  shade.addColorStop(0.48, "rgba(0,0,0,0.28)");
  shade.addColorStop(1, "rgba(0,0,0,0.48)");
  ctx.fillStyle = shade;
  ctx.fillRect(0, horizon, w, h - horizon);
  const motion = raceState.roadOffset * (1.08 + Math.min(1.8, Math.max(0, raceState.speed || 0) / 130));
  const laneX = (lane, t) => roadCenter(t) + lane * laneWidth() * (0.34 + t * 1.08);
  for (let lane = -1.5; lane <= 1.5; lane += 1) {
    for (let i = 0; i < 9; i += 1) {
      const y = horizon + ((i * 118 + motion) % (h - horizon + 160)) - 60;
      if (y < horizon || y > h + 50) continue;
      const t = Math.max(0, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
      const x = laneX(lane, t);
      ctx.globalAlpha = 0.38 + t * 0.28;
      ctx.fillStyle = "rgba(244,251,248,0.88)";
      roundRect(x - (2 + t * 4), y, 4 + t * 8, 16 + t * 36, 2 + t * 3);
      ctx.fill();
    }
  }
  ctx.globalAlpha = 1;
  for (let side = -1; side <= 1; side +
```

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```
= 2) {
  ctx.strokeStyle = side < 0 ? "rgba(255,91,107,0.72)" : "rgba(70,217,255,0.68)";
  ctx.lineWidth = Math.max(4, w * 0.006);
  ctx.beginPath();
  ctx.moveTo(roadCenter(0) + side * roadTop * 0.82, horizon + 4);
  ctx.lineTo(roadCenter(1) + side * roadBottom * 0.5, h + 8);
  ctx.stroke();
}
ctx.restore();
}

function drawPhoneSceneryBackdrop(w, h, horizon, place, theme) {
  if (!phoneGraphicsActive()) return;
  const offset = (raceState.roadOffset || 0) * 0.018;
  ctx.save();
  ctx.globalAlpha = 0.9;
  const farY = horizon - h * 0.02;
  if (place === "city" || place === "tokyo") {
    for (let i = 0; i < 22; i += 1) {
      const x = ((i * 94 - offset * (1.2 + (i % 3) * 0.18)) % (w + 180)) - 90;
      const bw = 34 + (i % 5) * 13;
      const bh = h * (0.2 + (i % 6) * 0.045);
      const glow = place === "tokyo" ? "rgba(255,79,216,0.26)" : "rgba(70,217,255,0.2)";
      const tower = ctx.createLinearGradient(x, farY - bh, x + bw, farY);
      tower.addColorStop(0, glow);
      tower.addColorStop(0.45, "rgba(16,25,28,0.9)");
      tower.addColorStop(1, "rgba(5,8,7,0.82)");
      ctx.fillStyle = tower;
      roundRect(x, farY - bh, bw, bh, 3);
      ctx.fill();
      ctx.fillStyle = i % 2 ? `${theme[1]}88` : `${theme[2]}77`;
      for (let yy = 14; yy < bh - 10; yy += 24) {
        ctx.globalAlpha = 0.42;
        ctx.fillRect(x + 8 + (yy % 3) * 7, farY - bh + yy, 5, 12);
        ctx.fillRect(x + bw - 15, farY - bh + yy + 5, 5, 10);
      }
      ctx.globalAlpha = 0.9;
    }
  }
  else if (place === "coast" || place === "harbor") {
    const water = ctx.createLinearGradient(0, horizon - h * 0.01, 0, horizon + h * 0.2);
    water.addColorStop(0, "rgba(70,217,255,0.16)");
    water.addColorStop(1, "rgba(4,21,24,0.42)");
    ctx.fillStyle = water;
    ctx.fillRect(0, horizon - h * 0.02, w, h * 0.24);
    for (let i = 0; i < 12; i += 1) {
      const x = ((i * 128 - offset * 1.9) % (w + 180)) - 90;
      ctx.fillStyle = "rgba(8,38,43,0.86)";
      ctx.beginPath();
      ctx.ellipse(x, horizon + h * 0.06, 54 + (i % 4) * 12, 12 + (i % 3) * 4, 0, 0, Math.PI * 2);
      ctx.fill();
      ctx.fillStyle = "rgba(244,251,248,0.28)";
      roundRect(x - 26, horizon + h * 0.025, 52, 6, 2);
      ctx.fill();
    }
  }
  else if (place === "alpine" || place === "snow" || place === "europe") {
    for (let i = 0; i < 26; i += 1) {
      const x = ((i * 180 - offset) % (w + 260)) - 130;
      const ridgeW = 180 + (i % 3) * 45;
      const ridgeH = h * (0.13 + (i % 4) * 0.025);
      ctx.fillStyle = place === "snow" ? "rgba(210,228,230,0.42)" : "rgba(102,130,126,0.38)";
      ctx.beginPath();
      ctx.ellipse(x, horizon + h * 0.025, ridgeW * 0.5, ridgeH, -0.08, 0, Math.PI * 2);
      ctx.fill();
      ctx.fillStyle = "rgba(244,251,248,0.36)";
      roundRect(x - ridgeW * 0.22, horizon - ridgeH * 0.28, ridgeW * 0.44, 7, 4);
      ctx.fill();
    }
  }
  else {
    for (let i = 0; i < 26; i += 1) {
      const x = ((i * 82 - offset * 2.2) % (w + 160)) - 80;
      const treeH = h * (0.1 + (i % 5) * 0.018);
      ctx.fillStyle = place === "desert" || place === "canyon" ? "rgba(132,72,38,0.64)" : "rgba(24,78,52,0.66)";
      ctx.beginPath();
      ctx.ellipse(x, horizon + h * 0.04, 34 + (i % 4) * 8, treeH * 0.42, 0, 0, Math.PI * 2);
      ctx.fill();
      ctx.fillStyle = place === "farm" ? "rgba(187,242,74,0.24)" : "rgba(54,217,138,0.2)";
      ctx.fillRect(x - 2, horizon - treeH * 0.42, 4, treeH * 0.72);
    }
  }
  const roadside = ctx.createLinearGradient(0, horizon + h * 0.06, 0, h);
  roadside.addColorStop(0, "rgba(7,20,18,0)");
  roadside.addColorStop(1, "rgba(7,20,18,0.45)");
  ctx.fillStyle = roadside;
  ctx.fillRect(0, horizon + h * 0.05, w, h * 0.42);
  ctx.restore();
}

function drawPhoneConsoleLandmarks(w, h, horizon, place, theme) {
  ctx.save();
  ctx.globalAlpha = 0.52;
  const baseY = horizon + h * 0.05;
  const offset = (raceState.roadOffset * 0.025) % 180;
  const route = routeWorldInfo(place);
  for (let i = 0; i < 18; i += 1) {
    const side = i % 2 ? -1 : 1;
    const x = side < 0 ? (i * 74 - offset) % (w * 0.48) : w - ((i * 78 + offset) % (w * 0.48));
    if (place === "city" || place === "tokyo") {
      const b
```

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```
w = 20 + (i % 4) * 9;
const bh = h * (0.14 + (i % 5) * 0.035);
ctx.fillStyle = place === "tokyo" ? "rgba(18,14,36,0.82)" : "rgba(14,20,23,0.86)";
roundRect(x, baseY - bh, bw, bh, 2);
ctx.fill();
ctx.fillStyle = i % 2 ? `${theme[1]}66` : `${theme[2]}55`;
ctx.fillRect(x + bw * 0.22, baseY - bh + 8, bw * 0.12, bh * 0.62);
ctx.fillRect(x + bw * 0.62, baseY - bh + 18, bw * 0.12, bh * 0.48);
} else if (place === "coast" || place === "harbor") {
  ctx.fillStyle = "rgba(8,38,43,0.72)";
  ctx.beginPath();
  ctx.ellipse(x, baseY + 8, 48 + (i % 3) * 12, 12 + (i % 4) * 3, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.16)";
  roundRect(x - 24, baseY - 5, 48, 5, 2);
  ctx.fill();
  ctx.fillStyle = `${theme[1]}33`;
  roundRect(x - 8, baseY - 24 - (i % 3) * 4, 16, 20 + (i % 3) * 4, 2);
  ctx.fill();
} else if (place === "desert" || place === "canyon") {
  ctx.fillStyle = "rgba(132,72,38,0.58)";
  ctx.beginPath();
  ctx.moveTo(x - 68, baseY + 14);
  ctx.quadraticCurveTo(x, baseY - 26 - (i % 3) * 12, x + 74, baseY + 16);
  ctx.closePath();
  ctx.fill();
} else if (place === "snow" || place === "alpine" || place === "europe") {
  ctx.fillStyle = "rgba(150,174,174,0.26)";
  ctx.beginPath();
  ctx.ellipse(x, baseY + 8, 68 + (i % 4) * 12, 20 + (i % 3) * 5, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.42)";
  roundRect(x - 26, baseY - 3 - (i % 2) * 4, 52, 5, 3);
  ctx.fill();
} else {
  ctx.fillStyle = place === "farm" ? "rgba(60,92,42,0.58)" : "rgba(24,78,52,0.55)";
  ctx.beginPath();
  ctx.ellipse(x, baseY + 8, 44 + (i % 4) * 12, 20 + (i % 3) * 6, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(10,28,18,0.64)";
  ctx.fillRect(x - 2, baseY - 40, 4, 50);
}
}
if (phoneGraphicsActive()) {
  ctx.restore();
  return;
}
const signSide = Math.sin((raceState.roadOffset || 0) * 0.002 + route.seed) > 0 ? 1 : -1;
const signX = signSide > 0 ? w * 0.82 : w * 0.18;
const signY = horizon + h * 0.08;
ctx.globalAlpha = 0.74;
ctx.fillStyle = "rgba(4,9,9,0.68)";
roundRect(signX - w * 0.095, signY - h * 0.035, w * 0.19, h * 0.07, 4);
ctx.fill();
ctx.strokeStyle = `${theme[1]}88`;
ctx.lineWidth = 1.4;
ctx.stroke();
ctx.fillStyle = "rgba(244,251,248,0.9)";
ctx.font = `700 ${Math.max(8, Math.min(13, w * 0.014))}px Inter, system-ui, sans-serif`;
ctx.textAlign = "center";
ctx.fillText(route.country.toUpperCase(), signX, signY - h * 0.005);
ctx.font = `600 ${Math.max(7, Math.min(11, w * 0.012))}px Inter, system-ui, sans-serif`;
ctx.fillStyle = `${theme[2]}dd`;
ctx.fillText(route.scene.toUpperCase(), signX, signY + h * 0.018);
ctx.restore();
}

function drawPhoneConsolePostPass(w, h, theme) {
  if (!phoneGraphicsActive()) return;
  const speed = Math.max(0, raceState.speed || 0);
  const turn = raceState.roadTurn || raceState.roadCurve || 0;
  ctx.save();
  ctx.globalCompositeOperation = "screen";
  const headlight = ctx.createRadialGradient(w * 0.5, h * 0.76, w * 0.04, w * 0.5, h * 0.76, w * 0.55);
  headlight.addColorStop(0, "rgba(244,251,248,0.12)");
  headlight.addColorStop(0.55, `${theme[1]}12`);
  headlight.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = headlight;
  ctx.fillRect(0, h * 0.35, w, h * 0.65);
  ctx.restore();

  if (Math.abs(turn) > 0.08) {
    ctx.save();
    const leanSide = turn > 0 ? 1 : -1;
    const pull = Math.min(0.26, Math.abs(turn) * 0.11 + speed / 1800);
    const turnGlow = ctx.createLinearGradient(leanSide > 0 ? w : 0, 0, leanSide > 0 ? w * 0.45 : w * 0.55, 0);
    turnGlow.addColorStop(0, `${theme[1]}33`);
    turnGlow.addColorStop(0.42, "rgba(244,251,248,0.04)");
    turnGlow.addColorStop(1, "rgba(0,0,0,0)");
    ctx.globalCompositeOperation = "screen";
    ctx.globalAlpha = pull;
    ctx.fillStyle = turnGlow;
    ctx.fillRect(0, 0, w, h);
    ctx.restore();
  }

  if (speed > 110) {
    ctx.save();
    ctx.globalAlpha = Math.min(0.24, (speed - 100) / 520);
    const blur = ctx.createLinearGradient(0, 0, w, 0);
    blur.addColorStop(0, "rgba(244,251,2
```

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48,0.18)");
    blur.addColorStop(0.18, "rgba(244,251,248,0)");
    blur.addColorStop(0.82, "rgba(244,251,248,0)");
    blur.addColorStop(1, "rgba(244,251,248,0.18)");
    ctx.fillStyle = blur;
    ctx.fillRect(0, 0, w, h);
    ctx.restore();
}

ctx.save();
const cinematic = ctx.createLinearGradient(0, 0, 0, h);
cinematic.addColorStop(0, "rgba(0,0,0,0.24)");
cinematic.addColorStop(0.5, "rgba(0,0,0,0)");
cinematic.addColorStop(1, "rgba(0,0,0,0.36)");
ctx.fillStyle = cinematic;
ctx.fillRect(0, 0, w, h);
ctx.restore();
}

function hashText(value = "") {
    let hash = 2166136261;
    for (let i = 0; i < value.length; i += 1) {
        hash ^= value.charCodeAt(i);
        hash = Math.imul(hash, 16777619);
    }
    return hash >>> 0;
}

function seededUnit(seed, salt = 0) {
    const n = Math.sin(seed * 12.9898 + salt * 78.233) * 43758.5453;
    return n - Math.floor(n);
}

function perspectiveRoadCenter(w, t) {
    const turn = raceState.roadTurn || raceState.roadCurve || 0;
    const clamped = Math.max(0, Math.min(1.08, t));
    const farPull = (1 - clamped) * (1 - clamped);
    const laneShift = cameraLaneOffset(0.24 + clamped * 0.86) * laneWidth() * (0.18 + clamped * 0.95);
    return w * 0.5 + turn * farPull * w * 0.27 + Math.sin(clamped * 4.1 + (raceState.roadOffset || 0) * 0.003) * Math.abs(turn) * w *
    0.015 - laneShift;
}

function drawGenAiRacingScenePass(w, h, theme) {
    if (!raceState.active && view !== "race") return;
    const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
    const design = genAiSceneDesign(place);
    const route = routeWorldInfo(place);
    const phoneMode = phoneGraphicsActive();
    const horizon = cameraMode === "cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
    const roadTop = cameraMode === "cockpit" ? w * 0.11 : cameraMode === "hood" ? w * 0.14 : w * 0.13;
    const roadBottom = cameraMode === "cockpit" ? w * 1.02 : cameraMode === "hood" ? w * 0.98 : w * 1.08;
    const seed = hashText(`${selectedRace ? selectedRace.id : "race"}:${design.style}`);
    drawGenAiHeroBackdrop(w, h, horizon, design, theme, place, route, seed);
    drawGenAiSurfaceDetails(w, h, theme, horizon, roadTop, roadBottom, design, seed);
    ctx.save();
    const count = phoneMode ? 10 : 18;
    const offset = (raceState.roadOffset || 0) * (phoneMode ? 1.08 : 0.84);
    for (let i = 0; i < count; i += 1) {
        const side = i % 2 ? -1 : 1;
        const loopSpan = h - horizon + 240;
        const raw = ((i * (phoneMode ? 148 : 126) + offset * (0.46 + seededUnit(seed, i) * 0.26)) % loopSpan) - 92;
        const y = horizon + raw;
        if (y < horizon - 18 || y > h + 70) continue;
        const t = Math.max(0.02, Math.min(1.05, (y - horizon) / Math.max(1, h - horizon)));
        const center = perspectiveRoadCenter(w, t);
        const roadHalf = roadTop * 0.82 + (roadBottom * 0.58 - roadTop * 0.82) * t;
        const x = center + side * (roadHalf + w * (0.05 + seededUnit(seed, i + 4) * 0.11) * (0.42 + t));
        const scale = 0.36 + t * 1.06;
        const prop = worldSceneryProp(design.props[i % design.props.length], place, i, phoneMode);
        drawGenAiTrackProp(prop, x, y, scale, side, design, theme, route, seed + i * 13);
    }
    drawGenAiCrowdAndCameras(w, h, horizon, roadTop, roadBottom, design);
    ctx.restore();
}

function routeStageInfo(place = "city", progress = 0) {
    const stageIndex = Math.max(0, Math.min(3, Math.floor(Math.max(0, Math.min(0.999, progress)) * 4)));
    const stages = {
        coast: ["Ocean Cliff", "cliff"], ["Beach Town", "palm"], ["Coastal Bridge", "bridge"], ["Finish Pier", "checkpoint"]],
        city: ["Lakefront", "tower"], ["Downtown", "crowd"], ["Underpass", "tunnel"], ["Loop Finish", "checkpoint"]],
        canyon: ["Red Rocks", "cliff"], ["Cave Road", "tunnel"], ["Mesa Sweep", "dune"], ["Canyon Finish", "checkpoint"]],
        alpine: ["Pine Pass", "pine"], ["Mountain Tunnel", "tunnel"], ["Snow Peaks", "mountain"], ["Summit Finish", "checkpoint"]],
        harbor: ["Marina", "pier"], ["Container Docks", "crane"], ["Bridge Cut", "bridge"], ["Harbor Finish", "checkpoint"]],
        snow: ["Aspen Forest", "pine"], ["Ice Village", "village"], ["Snow Tunnel", "tunnel"], ["Winter Finish", "checkpoint"]],
        airfield: ["Runway", "beacon"]
    };
}
```

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```
n"], ["Hangars", "hangar"], ["Taxiway", "plane"], ["Airfield Finish", "checkpoint"]],
  pursuit: [{"Station Exit", "policeHQ"}, {"Roadblock", "roadblock"}, {"Siren Alley", "lightbarSign"}, {"Pursuit Finish",
"checkpoint"}],
  freight: [{"Truck Stop", "trailer"}, {"Overpass", "bridge"}, {"Depot", "hangar"}, {"Freight Finish", "checkpoint"}],
  interstate: [{"Weigh Station", "weighStation"}, {"Convoy Lanes", "trailer"}, {"Diesel Stop", "truckStop"}, {"Freight Finish",
"checkpoint"}],
  farm: [{"Cornfields", "field"}, {"Barn Bend", "barn"}, {"Dirt Cut", "tractor"}, {"Farm Finish", "checkpoint"}],
  monsterpark: [{"Dirt Bowl", "dirtBerm"}, {"Crush Lane", "crushCarStack"}, {"Stadium Ramp", "stadiumRamp"}, {"Arena Finish",
"checkpoint"}],
  military: [{"Range Road", "rangeTarget"}, {"Bunker Cut", "bunker"}, {"Blast Wall", "blastWall"}, {"Armor Finish",
"checkpoint"}],
  skybase: [{"Launch Corridor", "beacon"}, {"Drone Gates", "plane"}, {"High Range", "rangeTarget"}, {"Sky Finish", "checkpoint"}],
  tokyo: [{"Neon Entry", "neon"}, {"Expressway", "tower"}, {"Tunnel Run", "tunnel"}, {"Tokyo Finish", "checkpoint"}],
  desert: [{"Dune Sea", "dune"}, {"Oasis Road", "palm"}, {"Rock Gate", "cliff"}, {"Rally Finish", "checkpoint"}],
  rainforest: [{"Jungle Entry", "canopy"}, {"Bridge Bypass", "bridge"}, {"Canopy Tunnel", "canopy"}, {"Rain Finish",
"checkpoint"}],
  europe: [{"Village Road", "village"}, {"Switchback", "mountain"}, {"Stone Tunnel", "tunnel"}, {"Alps Finish", "checkpoint"}]
  };
  const selected = (stages[place] || stages.city)[stageIndex];
  return { index: stageIndex, label: selected[0], prop: selected[1] };
}

function drawRouteStageLandmarks(w, h, theme) {
  if (!raceState.active && view !== "race") return;
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const length = raceLength();
  const progress = length ? Math.max(0, Math.min(1, raceState.distance / length)) : 0;
  const stage = routeStageInfo(place, progress);
  const design = genAiSceneDesign(place);
  const horizon = cameraMode === "cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const phoneMode = phoneGraphicsActive();
  const seed = hashText(`${place}:${stage.label}:${stage.index}`);
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.72 : 0.82;
  drawRouteHorizonSetPiece(w, h, horizon, stage, design, theme, seed);
  const loopSpan = h - horizon + 360;
  const count = phoneMode ? 7 : 11;
  for (let i = 0; i < count; i += 1) {
    const raw = ((i * (phoneMode ? 245 : 210) + (raceState.roadOffset || 0) * (0.58 + seededUnit(seed, i) * 0.16) + stage.index *
63) % loopSpan) - 104;
    const y = horizon + raw;
    if (y < horizon - 12 || y > h + 80) continue;
    const t = Math.max(0.04, Math.min(1.1, (y - horizon) / Math.max(1, h - horizon)));
    const side = i % 2 === 0 ? -1 : 1;
    const center = perspectiveRoadCenter(w, t);
    const sideOffset = w * (0.18 + t * 0.48 + seededUnit(seed, i + 10) * 0.07);
    const x = center + side * sideOffset;
    const scale = 0.38 + t * (phoneMode ? 0.92 : 1.22);
    const prop = worldSceneryProp(i % 6 === 0 ? stage.prop : design.props[(i + stage.index) % design.props.length], place, i +
stage.index, phoneMode);
    drawRouteStageProp(prop, x, y, scale, side, design, theme, seed + i * 17);
  }
  if (!phoneMode) drawRouteStageSign(w, h, horizon, stage, design, theme);
  drawFinishGate(w, h, theme);
  ctx.restore();
}

function drawRouteHorizonSetPiece(w, h, horizon, stage, design, theme, seed) {
  const x = w * (0.5 + Math.sin((raceState.distance || 0) * 0.00008 + seed) * 0.05);
  const y = horizon + h * 0.035;
  ctx.save();
  ctx.globalAlpha = 0.62;
  if (stage.prop === "tunnel") {
    ctx.fillStyle = "rgba(5,8,7,0.82)";
    roundRect(x - w * 0.16, y - h * 0.06, w * 0.32, h * 0.14, 12);
    ctx.fill();
    ctx.strokeStyle = `${design.accent}77`;
    ctx.lineWidth = Math.max(2, w * 0.004);
    ctx.stroke();
    ctx.fillStyle = "rgba(0,0,0,0.55)";
    ctx.beginPath();
    ctx.ellipse(x, y + h * 0.045, w * 0.09, h * 0.06, 0, 0, Math.PI * 2);
    ctx.fill();
  } else if (stage.prop === "bridge") {
    ctx.strokeStyle = `${design.light}88`;
    ctx.lineWidth
```

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```
dth = Math.max(2, w * 0.004);
ctx.beginPath();
ctx.moveTo(x - w * 0.27, y + h * 0.04);
ctx.quadraticCurveTo(x, y - h * 0.11, x + w * 0.27, y + h * 0.04);
ctx.stroke();
for (let i = -3; i <= 3; i += 1) {
  ctx.beginPath();
  ctx.moveTo(x + i * w * 0.07, y - h * 0.045);
  ctx.lineTo(x + i * w * 0.055, y + h * 0.055);
  ctx.stroke();
}
} else if (stage.prop === "checkpoint") {
  ctx.fillStyle = "rgba(5,8,7,0.72)";
  roundRect(x - w * 0.13, y - h * 0.055, w * 0.26, h * 0.05, 6);
  ctx.fill();
  ctx.strokeStyle = theme[1] || "#46d9ff";
  ctx.lineWidth = 2;
  ctx.stroke();
} else {
  ctx.fillStyle = `${design.accent}22`;
  ctx.beginPath();
  ctx.ellipse(x, y, w * 0.22, h * 0.045, 0, 0, Math.PI * 2);
  ctx.fill();
}
ctx.restore();
}

function drawRouteStageProp(prop, x, y, scale, side, design, theme, seed) {
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.globalAlpha = Math.max(0.42, Math.min(0.96, 0.42 + scale * 0.22));
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  if (prop === "tower" || prop === "neon" || prop === "village") {
    const bw = prop === "village" ? 52 : 34;
    const bh = prop === "village" ? 48 : 118;
    ctx.fillStyle = prop === "neon" ? "rgba(18,10,32,0.9)" : "rgba(10,18,20,0.88)";
    roundRect(-bw / 2, -bh, bw, bh, 4);
    ctx.fill();
    ctx.fillStyle = prop === "neon" ? `${accent}cc` : `${light}99`;
    for (let i = 0; i < 5; i += 1) ctx.fillRect(-bw * 0.3, -bh + 12 + i * 18, bw * 0.6, 4);
    if (prop === "village") {
      ctx.fillStyle = `${light}88`;
      ctx.beginPath();
      ctx.moveTo(-bw * 0.6, -bh);
      ctx.lineTo(0, -bh - 28);
      ctx.lineTo(bw * 0.6, -bh);
      ctx.closePath();
      ctx.fill();
    }
  } else if (prop === "palm" || prop === "pine" || prop === "canopy") {
    ctx.strokeStyle = prop === "palm" ? "rgba(140,100,55,0.9)" : "rgba(30,78,52,0.92)";
    ctx.lineWidth = 5;
    ctx.beginPath();
    ctx.moveTo(0, 12);
    ctx.lineTo(side * 6, -70);
    ctx.stroke();
    ctx.fillStyle = prop === "pine" ? "rgba(34,93,59,0.86)" : "rgba(54,217,138,0.62)";
    for (let i = 0; i < 3; i += 1) {
      ctx.beginPath();
      ctx.ellipse(side * (4 + i * 3), -70 + i * 12, 36 - i * 7, 12, side * 0.4, 0, Math.PI * 2);
      ctx.fill();
    }
  } else if (prop === "bridge" || prop === "crane" || prop === "beacon") {
    ctx.strokeStyle = `${light}99`;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(0, 16);
    ctx.lineTo(0, -86);
    ctx.moveTo(-42 * side, -62);
    ctx.lineTo(52 * side, -76);
    ctx.stroke();
    ctx.fillStyle = `${accent}aa`;
    ctx.beginPath();
    ctx.arc(0, -90, 7, 0, Math.PI * 2);
    ctx.fill();
  } else if (prop === "hangar" || prop === "barn" || prop === "trailer" || prop === "field" || prop === "tractor") {
    ctx.fillStyle = prop === "barn" ? "rgba(148,42,34,0.84)" : "rgba(22,29,29,0.84)";
    roundRect(-54, -42, 108, 54, 5);
    ctx.fill();
    ctx.fillStyle = prop === "barn" ? "rgba(255,209,102,0.7)" : `${light}77`;
    ctx.beginPath();
    ctx.moveTo(-62, -42);
    ctx.lineTo(0, -76);
    ctx.lineTo(62, -42);
    ctx.closePath();
    ctx.fill();
    if (prop === "trailer" || prop === "tractor") {
      ctx.fillStyle = "rgba(5,8,7,0.9)";
      ctx.beginPath();
      ctx.arc(-34, 14, 8, 0, Math.PI * 2);
      ctx.arc(34, 14, 8, 0, Math.PI * 2);
      ctx.fill();
    }
  } else {
    ctx.fillStyle = prop === "dune" ? "rgba(255,183,74,0.58)" : prop === "mountain" ? "rgba(130,160,160,0.5)" :
    "rgba(132,72,38,0.58)";
    ctx.beginPath();
    ctx.moveTo(-58, 16);
    ctx.quadraticCurveTo(-8, -68 - seededUnit(seed, 1) * 28, 64, 18);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.18)";
    ctx.fillRect(-18, -18, 42, 5);
  }
  ctx.restore();
}

function drawRouteStageSign(w, h, horizon, stage, design, theme) {
  const x = w * 0.2;
  const y = horizon + h * 0.08;
  ctx.save();
```

```
ctx.globalAlpha *= 0.88;
ctx.fillStyle = "rgba(5,8,7,0.78)";
roundRect(x - w * 0.095, y - 18, w * 0.19, 38, 6);
ctx.fill();
ctx.strokeStyle = design.accent || theme[1] || "#46d9ff";
ctx.lineWidth
```

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```
th = 2;
ctx.stroke();
ctx.fillStyle = "#f4fbf8";
ctx.font = `900 ${Math.max(9, Math.min(15, w * 0.013))}px system-ui`;
ctx.textAlign = "center";
ctx.textBaseline = "middle";
ctx.fillText(stage.label.toUpperCase(), x, y + 1);
ctx.restore();
}

function drawFinishGate(w, h, theme) {
  if (!raceState.active) return;
  const length = raceLength();
  const gap = length - raceState.distance;
  if (gap > 2600 || gap < -420) return;
  const horizon = cameraMode === "cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const p = roadObjectPos(0, length);
  const y = Math.max(horizon + h * 0.05, Math.min(h * 0.94, p.y));
  const scale = Math.max(0.34, Math.min(1.24, p.scale || 1));
  const half = laneWidth() * (2.15 + scale * 0.65);
  const x = p.x;
  const lineH = Math.max(8, 20 * scale);
  ctx.save();
  ctx.globalAlpha = Math.max(0.48, Math.min(0.96, 1 - Math.max(0, gap - 1800) / 1600));
  for (let i = 0; i < 12; i += 1) {
    ctx.fillStyle = i % 2 ? "#050807" : "#f4fbf8";
    const tileW = (half * 2) / 12;
    ctx.fillRect(x - half + i * tileW, y + lineH * 0.25, tileW + 1, lineH);
  }
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(x - half * 0.92, y - 86 * scale, 12 * scale, 108 * scale, 4 * scale);
  ctx.fill();
  roundRect(x + half * 0.92 - 12 * scale, y - 86 * scale, 12 * scale, 108 * scale, 4 * scale);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.88)";
  roundRect(x - half * 0.72, y - 104 * scale, half * 1.44, 34 * scale, 5 * scale);
  ctx.fill();
  ctx.strokeStyle = theme[1] || "#46d9ff";
  ctx.lineWidth = Math.max(1.5, 2.5 * scale);
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `900 ${Math.max(9, 17 * scale)}px system-ui`;
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText("FINISH", x, y - 87 * scale);
  ctx.restore();
}

function drawRealWorldDetailPass(w, h, theme) {
  if (!raceState.active && view !== "race") return;
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const design = genAiSceneDesign(place);
  const route = routeWorldInfo(place);
  const horizon = cameraMode === "cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const roadTop = cameraMode === "cockpit" ? w * 0.11 : cameraMode === "hood" ? w * 0.14 : w * 0.13;
  const roadBottom = cameraMode === "cockpit" ? w * 1.02 : cameraMode === "hood" ? w * 0.98 : w * 1.08;
  const seed = hashText(`v83:${place}:${route.scene}:${selectedVehicle().type}`);
  const worldMode = raceWorldViewMode(selectedVehicle(), place);
  ctx.save();
  drawRealWorldHorizonDetails(w, h, horizon, place, design, theme, seed);
  if (worldMode === "air" || worldMode === "water") {
    drawRealWorldWeatherDisasterPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed);
    drawGroundLockedWorldVisibilityPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
    ctx.restore();
    return;
  }
  drawRealWorldDepthInfrastructurePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldRouteAtmospherePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldRouteSectorPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldLivingRoutePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldPlaceIdentityPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldAmbientLifePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldDistrictMotionPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldRouteSignaturePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldRoadsideLife(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldRoadSurface(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldSurfaceLightingPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
  drawRealWorldWeatherDisasterPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed);
  drawRealWorldPavementStoryPass(w, h, horizon, roadTop, roadBottom, place, design, theme, se
```


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```
ed);
drawRealWorldRoadRulePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
drawRealWorldLightShadowPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
drawRealWorldCurveCues(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
drawRealWorldForegroundDepthPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
drawRealWorldNearSideInfrastructurePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
drawRealWorldEdgeMaterialPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
drawGroundLockedWorldVisibilityPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
drawRealWorldRouteDramaPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed);
drawHighVisibilityRealWorldDetailPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed);
ctx.restore();
}

function drawGroundLockedWorldVisibilityPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const length = raceLength();
  const progress = length ? Math.max(0, Math.min(1, raceState.distance / length)) : 0;
  const stage = routeStageInfo(place, progress);
  const route = routeWorldInfo(place);
  const motion = raceState.roadOffset || 0;
  const worldMode = raceWorldViewMode(selectedVehicle(), place);
  ctx.save();
  if (worldMode === "air") {
    drawAirRaceWorldViewPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode);
    ctx.restore();
    return;
  }
  if (worldMode === "water") {
    drawWaterRaceWorldViewPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode);
    ctx.restore();
    return;
  }
  drawReferenceInspiredDriveByScenePass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode);
  drawRouteAnchoredDriveByWorldPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode);
  drawGroundLockedSideDetailPass(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion, phoneMode);
  drawVisibleGroundRoadPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode);
  drawSpecialtySurfaceAndWorldPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode);
  drawGroundLockedNearDetailPass(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion, phoneMode);
  ctx.restore();
}

function drawHighVisibilityRealWorldDetailPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const route = routeWorldInfo(place);
  const length = Math.max(1, raceLength());
  const progress = Math.max(0, Math.min(1, raceState.distance / length));
  const stage = routeStageInfo(place, progress);
  const detailSet = highVisibilityDetailSet(place, stage);
  const rows = phoneMode ? 7 : 10;
  const motion = (raceState.roadOffset || 0) * (0.56 + Math.min(1.1, Math.max(0, raceState.speed || 0) / 260));
  ctx.save();
  clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 0 : 4);
  for (let i = 0; i < rows; i += 1) {
    const band = i / Math.max(1, rows - 1);
    const t = Math.max(0.42, Math.min(1.08, 0.44 + band * 0.58));
    const y = horizon + (h - horizon) * t + (((motion + i * 59 + seed * 0.003) % 62) - 31);
    if (y < horizon + 10 || y > h + 70) continue;
    const side = (i + Math.floor(seed % 3)) % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const projectedX = perspectiveRoadCenter(w, t) + side * (half * (phoneMode ? 1.01 : 1.06) + w * (phoneMode ? 0.02 : 0.032));
    const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, y, side, projectedX, phoneMode, phoneMode ? 4 : 10);
    const near = Math.max(0.12, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    const scale = Math.max(phoneMode ? 0.82 : 0.76, Math.min(phoneMode ? 2.08 : 2.32, (0.76 + near * (phoneMode ? 1.08 : 1.22)) * (0.94 + seededUnit(seed, i + 712) * 0.16)));
    const kind = detailSet[(i + Math.floor(seed % detailSet.length)) % detailSet.length];
    ctx.g
```

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```
    globalAlpha = Math.min(1, phoneMode ? 0.84 + near * 0.16 : 0.78 + near * 0.18);
    drawHighVisibilityGroundCue(x, y, scale, side, place, design, theme, seed + i * 43);
    drawGroundLockedOrLiveByShadow(x, y, scale * 0.76, side, theme);
    if (kind.startsWith("life:")) {
        drawLivingWorldActivityShape(kind.slice(5), x, y - scale * 3, scale * 0.92, side, place, route, design, theme, seed + i * 59);
    } else {
        drawGroundLockedPlaceProp(kind, x, y, scale, side, place, route, design, theme, seed + i * 59);
    }
}
ctx.restore();
ctx.globalAlpha = 1;
}

function highVisibilityDetailSet(place, stage) {
    if (stage && stage.prop === "bridge") return ["coastRail", "life:pierPeople", "lamp", "waterEdge", "parkedCar",
"life:sidewalkQueue"];
    if (stage && stage.prop === "tunnel") return ["tunnelPortal", "wallLamp", "lowBarrier", "life:cameraCrew", "roadsideHedge"];
    const sets = {
        coast: ["palm", "beachHouse", "life:beachJoggers", "life:cyclistLane", "coastRail", "life:pierPeople", "lighthouse"],
        city: ["storefront", "life:sidewalkQueue", "parkedCar", "streetLight", "busStop", "life:cyclistLane", "life:patioCrowd"],
        canyon: ["rockFace", "guardPost", "life:wildlifeLookout", "tourVan", "curveBoard", "life:hikerLookout"],
        alpine: ["pine", "stoneWall", "chalet", "life:wildlifeLookout", "snowPost", "life:hikerLookout"],
        harbor: ["container", "warehouse", "life:dockWorkers", "craneBase", "dockLight", "life:boatWake", "marinaDock"],
        snow: ["pine", "lodge", "snowFence", "life:skiQueue", "life:snowPlowCrew", "life:wildlifeLookout"],
        airfield: ["hangar", "runwayLight", "controlTower", "life:airportBeacon", "life:serviceCartMotion", "life:planeSpotters"],
        pursuit: ["policeHQ", "parkedCar", "streetLight", "life:policeCheckpoint", "lowBarrier", "life:cameraCrew"],
        freight: ["truckStop", "container", "serviceTruck", "life:forkliftSweep", "life:loadingCrew", "dieselSign"],
        interstate: ["truckStop", "weighStation", "serviceTruck", "life:dieselStopLife", "life:restStopLife", "streetLight"],
        farm: ["barn", "roadsideHedge", "tractorParked", "life:pastureAnimals", "hayBales", "life:fieldWorkers", "windmill"],
        monsterpark: ["grandstand", "dirtBerm", "crushCarStack", "life:stadiumCrowd", "life:pitCrew", "stadiumRamp"],
        military: ["bunker", "watchTower", "blastWall", "life:rangeCrew", "life:militaryPatrol", "rangeTarget"],
        skybase: ["controlTower", "hangar", "runwayLight", "life:airportBeacon", "life:radarCrew", "life:serviceCartMotion"],
        tokyo: ["neonShop", "life:neonCrowd", "streetLight", "parkedCar", "life:cyclistLane", "pedestrianBridge", "life:tramStopLife"],
        desert: ["dune", "supportTent", "rockFace", "life:rallyCrew", "life:wildlifeLookout", "rallyMarker"],
        rainforest: ["canopyTree", "marketStand", "roadsideHedge", "life:marketLife", "life:canopyAnimals", "waterfallSign"],
        europe: ["villageHouse", "cafe", "stoneWall", "life:patioCrowd", "life:cyclistLane", "life:pastureAnimals", "tramPost"]
    };
    return sets[place] || sets.city;
}

function drawHighVisibilityGroundCue(x, y, scale, side, place, design, theme, seed) {
    const accent = design.accent || theme[1] || "#46d9ff";
    const ground = place === "city" || place === "tokyo" || place === "europe" ? "rgba(205,214,208,0.32)"
        : place === "harbor" || place === "coast" ? "rgba(70,217,255,0.26)"
        : place === "farm" || place === "rainforest" ? "rgba(54,217,138,0.25)"
        : place === "snow" || place === "alpine" ? "rgba(244,251,248,0.3)"
        : place === "desert" || place === "canyon" ? "rgba(255,183,74,0.26)"
        : "rgba(180,192,188,0.24)";
    ctx.save();
    ctx.translate(x, y);
    ctx.scale(scale, scale);
    ctx.fillStyle = ground;
    ctx.beginPath();
    ctx.ellipse(side * 6, 16, 120, 22, -0.04 * side, 0, Math.PI * 2);
    ctx.fill();
    if (phoneGraphicsActive()) {
        ctx.fillStyle = "rgba(0,0,0,0.24)";
        ctx.beginPath();
        ctx.ellipse(side * 8, 18, 92, 12, -0.04 * side, 0, Math.PI * 2);
        ctx.fill();
    } else {
        ctx.strokeStyle = `${accent}88`;
        ctx.lineWidth = 3;
        ctx.beginPath();
        ctx.moveTo(-90 * side, 5);
        ctx.lineTo(96 * side, -2);
        ctx.stroke();
    }
    ctx.restore();
}
```

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```
});
}

function drawRealWorldRouteDramaPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const length = Math.max(1, raceLength());
  const progress = Math.max(0, Math.min(1, raceState.distance / length));
  const stage = routeStageInfo(place, progress);
  const turn = raceState.roadTurn || raceState.roadCurve || 0;
  const routeKind = realWorldRouteDramaKind(place, stage, turn);
  ctx.save();
  if (routeKind === "tunnel") {
    drawVisibleTunnelDriveThrough(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, phoneMode);
  } else if (routeKind === "mountain" || routeKind === "hill") {
    drawVisibleMountainHillDrive(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, turn, phoneMode);
  } else if (routeKind === "cityTunnel") {
    drawVisibleCityUnderpass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, turn, phoneMode);
  }
  drawVisibleRoadTurnCues(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, turn, phoneMode);
  ctx.restore();
  ctx.globalAlpha = 1;
}

function realWorldRouteDramaKind(place, stage, turn) {
  if (stage && stage.prop === "tunnel") return place === "city" || place === "tokyo" || place === "pursuit" ? "cityTunnel" : "tunnel";
  if (stage && (stage.prop === "mountain" || stage.prop === "cliff")) return "mountain";
  if (place === "alpine" || place === "europe" || place === "canyon") return Math.abs(turn) > 0.4 ? "mountain" : "hill";
  if (place === "desert" || place === "rainforest") return Math.abs(turn) > 0.5 ? "hill" : "";
  return Math.abs(turn) > 0.72 ? "hill" : "";
}

function drawVisibleTunnelDriveThrough(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, phoneMode) {
  const topY = horizon + h * (phoneMode ? 0.02 : 0.01);
  const mouthW = w * (phoneMode ? 0.72 : 0.62);
  const mouthH = h * (phoneMode ? 0.31 : 0.36);
  const x = w * 0.5 + Math.sin(seed + raceState.distance * 0.0002) * w * 0.035;
  ctx.globalAlpha = phoneMode ? 0.7 : 0.68;
  ctx.fillStyle = "rgba(4,6,6,0.72)";
  ctx.beginPath();
  ctx.ellipse(x, topY + mouthH * 0.5, mouthW * 0.5, mouthH * 0.58, 0, Math.PI, 0);
  ctx.lineTo(x + mouthW * 0.5, topY + mouthH * 0.58);
  ctx.lineTo(x - mouthW * 0.5, topY + mouthH * 0.58);
  ctx.closePath();
  ctx.fill();
  ctx.strokeStyle = "rgba(205,218,214,0.52)";
  ctx.lineWidth = Math.max(4, w * 0.01);
  ctx.stroke();

  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  const tunnelShade = ctx.createLinearGradient(0, horizon, 0, h);
  tunnelShade.addColorStop(0, "rgba(0,0,0,0.34)");
  tunnelShade.addColorStop(0.5, "rgba(0,0,0,0.18)");
  tunnelShade.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = tunnelShade;
  ctx.fillRect(0, horizon, w, h - horizon);
  ctx.globalCompositeOperation = "screen";
  for (let i = 0; i < (phoneMode ? 8 : 12); i += 1) {
    const y = horizon + ((i * 86 + raceState.roadOffset * 0.82 + seed * 0.01) % (h - horizon + 160)) - 48;
    if (y < horizon || y > h) continue;
    const t = Math.max(0.02, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    const center = perspectiveRoadCenter(w, t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t) * 0.82;
    ctx.strokeStyle = `${design.light || theme[2]}55`;
    ctx.lineWidth = Math.max(2, 2 + t * 8);
    ctx.beginPath();
    ctx.moveTo(center - half * 0.62, y);
    ctx.lineTo(center + half * 0.62, y + t * 8);
    ctx.stroke();
  }
  ctx.globalCompositeOperation = "source-over";
}

function drawVisibleCityUnderpass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, turn, phoneMode) {
  const y = horizon + h * 0.05;
  const x = w * 0.5 + turn * w * 0.05;
  ctx.globalAlpha = phoneMode ? 0.64 : 0.7;
  ctx.fillStyle = "rgba(5,8,7,0.68)";
  roundRect(x - w * 0.34, y - h * 0.055, w * 0.68, h * 0.09, 8);
  ctx.fill();
  ctx.strokeStyle = `${design.accent || theme[1]}88`;
  ctx.lineWidth = Math.max(2, w * 0.005);
  ctx.stroke();
  ctx.fillStyle = "rgba(244,251,248,0.22)";
  for (let i = -3; i <= 3; i += 1) {
    roundRect(x + i * w * 0.08 - w * 0.018, y - h * 0.028, w * 0.036, h * 0.012, 4);
    ctx.fill();
  }
}

function drawVisibleMountainHillDrive(w, h, horizon
```

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```
, roadTop, roadBottom, place, design, theme, seed, turn, phoneMode) {
  const color = place === "desert" || place === "canyon" ? "rgba(178,92,48,0.5)" : place === "snow" || place === "alpine" || place
=== "europe" ? "rgba(198,214,210,0.42)" : "rgba(35,118,68,0.38)";
  ctx.globalAlpha = phoneMode ? 0.58 : 0.62;
  ctx.fillStyle = color;
  for (let side = -1; side <= 1; side += 2) {
    ctx.beginPath();
    ctx.moveTo(side < 0 ? -w * 0.08 : w * 1.08, horizon + h * 0.26);
    ctx.quadraticCurveTo(w * (0.22 + side * 0.12) + turn * w * 0.1, horizon - h * 0.12, w * (0.48 + side * 0.16), horizon + h *
0.16);
    ctx.quadraticCurveTo(w * (0.34 + side * 0.12), horizon + h * 0.26, side < 0 ? -w * 0.08 : w * 1.08, horizon + h * 0.42);
    ctx.closePath();
    ctx.fill();
  }
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  const hill = ctx.createLinearGradient(0, horizon, 0, h);
  hill.addColorStop(0, "rgba(244,251,248,0.1)");
  hill.addColorStop(0.38, turn > 0 ? "rgba(0,0,0,0.04)" : "rgba(244,251,248,0.08)");
  hill.addColorStop(0.72, "rgba(0,0,0,0.16)");
  hill.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = hill;
  ctx.fillRect(0, horizon, w, h - horizon);
}

function drawVisibleRoadTurnCues(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, turn, phoneMode) {
  if (Math.abs(turn) < 0.22) return;
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  const side = turn > 0 ? 1 : -1;
  const count = phoneMode ? 3 : 7;
  for (let i = 0; i < count; i += 1) {
    const y = horizon + h * (0.22 + i * (phoneMode ? 0.12 : 0.105)) + (((raceState.roadOffset || 0) * 0.28) % 34);
    if (y > h + 30) continue;
    const t = Math.max(0.05, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    const center = perspectiveRoadCenter(w, t) + turn * w * 0.08 * (1 - t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t) * (0.24 + t * 0.2);
    ctx.globalAlpha = phoneMode ? Math.min(0.22, 0.08 + t * 0.14) : Math.min(0.48, 0.18 + t * 0.28);
    ctx.strokeStyle = phoneMode ? "rgba(244,251,248,0.72)" : `${design.light || theme[2]}cc`;
    ctx.lineWidth = Math.max(2, 2 + t * (phoneMode ? 3 : 6));
    ctx.lineCap = "round";
    ctx.beginPath();
    ctx.moveTo(center - side * half, y);
    ctx.quadraticCurveTo(center + side * half * 0.25, y + 18 * t, center + side * half, y + 6 + t * 14);
    ctx.stroke();
    if (!phoneMode) {
      ctx.fillStyle = `${design.light || theme[2]}aa`;
      ctx.beginPath();
      ctx.moveTo(center + side * (half + 14 * t), y + 6 + t * 14);
      ctx.lineTo(center + side * (half - 16 * t), y - 5 + t * 8);
      ctx.lineTo(center + side * (half - 8 * t), y + 22 + t * 12);
      ctx.closePath();
      ctx.fill();
    }
  }
  ctx.restore();
}

function raceWorldViewMode(vehicle = selectedVehicle(), place = selectedRace && selectedRace.place) {
  if (vehicle && isWaterVehicleType(vehicle.type)) return "water";
  if (vehicle && isAirVehicleType(vehicle.type)) return "air";
  return place === "harbor" && vehicle && vehicle.class === "water" ? "water" : "road";
}

function drawWaterRaceWorldViewPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode) {
  const motion = raceState.roadOffset || 0;
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  const water = ctx.createLinearGradient(0, horizon, 0, h);
  water.addColorStop(0, "rgba(75,216,255,0.44)");
  water.addColorStop(0.48, "rgba(12,72,86,0.72)");
  water.addColorStop(1, "rgba(2,24,32,0.94)");
  ctx.fillStyle = water;
  ctx.fillRect(0, horizon, w, h - horizon);
  ctx.globalCompositeOperation = "screen";
  for (let i = 0; i < (phoneMode ? 24 : 34); i += 1) {
    const y = horizon + (((i * 82 + motion * 1.55 + seed * 0.01) % (h - horizon + 180)) - 70);
    if (y < horizon || y > h + 40) continue;
    const t = Math.max(0.02, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
    const center = perspectiveRoadCenter(w, t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t) * (0.42 + seededUnit(seed, i) * 0.18);
    ctx.strokeStyle = i % 3 === 0 ? "rgba(244,251,248,0.46)" : "rgba(70,217,255,0.38)";
    ctx.lineWidth = 1.2 + t * (phoneMode ? 5.2 : 4.4);
    ctx.gl
```

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```
    obalAlpha = 0.2 + t * 0.42;
    ctx.beginPath();
    ctx.moveTo(center - half, y + Math.sin(i) * 6);
    ctx.quadraticCurveTo(center, y + 8 + t * 18, center + half, y + Math.cos(i) * 6);
    ctx.stroke();
  }
  ctx.globalCompositeOperation = "source-over";
  for (let i = 0; i < (phoneMode ? 10 : 14); i += 1) {
    const distance = raceState.distance + i * 230 + seededUnit(seed, i + 80) * 120;
    const side = i % 2 === 0 ? -1 : 1;
    const pos = roadObjectPos(side * (0.96 + seededUnit(seed, i + 90) * 0.18), distance);
    if (!pos || pos.y < horizon || pos.y > h + 50) continue;
    drawWaterBuoy(pos.x, pos.y, Math.max(0.22, pos.scale * 0.82), side, theme, seed + i * 19);
  }
  for (let i = 0; i < (phoneMode ? 5 : 7); i += 1) {
    const distance = raceState.distance + 340 + i * 420 + seededUnit(seed, i + 170) * 160;
    const side = i % 2 === 0 ? -1 : 1;
    const pos = roadObjectPos(side * (1.28 + seededUnit(seed, i + 180) * 0.18), distance);
    if (!pos || pos.y < horizon || pos.y > h + 60) continue;
    drawWaterSceneBoat(pos.x, pos.y, Math.max(0.28, pos.scale * 0.76), side, theme, seed + i * 31);
  }
  ctx.restore();

  ctx.save();
  clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 8 : 12);
  const bank = ctx.createLinearGradient(0, horizon, 0, h);
  bank.addColorStop(0, "rgba(180,150,96,0.24)");
  bank.addColorStop(0.48, "rgba(38,72,58,0.5)");
  bank.addColorStop(1, "rgba(16,34,28,0.86)");
  ctx.fillStyle = bank;
  ctx.fillRect(0, horizon, w, h - horizon);
  const shorelineProps = ["marinaDock", "lighthouse", "dockLight", "warehouse", "container", "serviceTruck", "routeSign"];
  for (let i = -1; i < (phoneMode ? 10 : 14); i += 1) {
    const distance = raceState.distance + i * 250 + seededUnit(seed, i + 240) * 120;
    const side = seededUnit(seed, i + 250) > 0.48 ? 1 : -1;
    const pos = roadObjectPos(side * 2.05, distance);
    if (!pos || pos.y < horizon - 24 || pos.y > h + 90) continue;
    const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, pos.y, side, pos.x, phoneMode, phoneMode ? 18 : 28);
    const scale = Math.max(0.32, Math.min(phoneMode ? 1.48 : 1.68, pos.scale * (0.88 + (pos.depth || 0.5) * 0.56)));
    ctx.globalAlpha = Math.min(0.96, 0.72 + (pos.depth || 0.4) * 0.24);
    drawGroundLockedDriveByShadow(x, pos.y, scale * 0.9, side, theme);
    drawGroundLockedPlaceProp(shorelineProps[Math.abs(i) % shorelineProps.length], x, pos.y, scale, side, place, route, design,
  theme, seed + i * 61);
    if (i % 3 === 0) {
      const liveKind = i % 2 === 0 ? "pierPeople" : "dockWorkers";
      drawLivingWorldActivityShape(liveKind, x + side * scale * 42, pos.y + scale * 6, scale * 0.72, side, place, route, design,
  theme, seed + i * 67);
    }
  }
  ctx.restore();
  ctx.globalAlpha = 1;
}

function drawAirRaceWorldViewPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode) {
  const motion = raceState.roadOffset || 0;
  ctx.save();
  const sky = ctx.createLinearGradient(0, 0, 0, h);
  sky.addColorStop(0, `${theme[0]}ee`);
  sky.addColorStop(0.42, "rgba(30,76,100,0.86)");
  sky.addColorStop(0.42, "rgba(8,18,28,0.94)");
  ctx.fillStyle = sky;
  ctx.fillRect(0, 0, w, h);

  const groundTop = h * (phoneMode ? 0.54 : 0.58);
  const ground = ctx.createLinearGradient(0, groundTop, 0, h);
  ground.addColorStop(0, "rgba(34,86,68,0.1)");
  ground.addColorStop(0.48, "rgba(42,72,54,0.28)");
  ground.addColorStop(1, "rgba(13,28,24,0.62)");
  ctx.fillStyle = ground;
  ctx.fillRect(0, groundTop, w, h - groundTop);
  for (let i = 0; i < (phoneMode ? 24 : 34); i += 1) {
    const x = ((i * 137 - motion * 0.18 + seed * 0.02) % (w + 220)) - 110;
    const y = groundTop + ((i * 47 + motion * 0.08) % Math.max(1, h - groundTop + 80)) - 40;
    const patchW = 54 + seededUnit(seed, i + 10) * (phoneMode ? 96 : 150);
    const patchH = 18 + seededUnit(seed, i + 20) * 42;
    ctx.fillStyle = i % 4 === 0 ? "rgba(255,209,102,0.18)" : i % 3 === 0 ? "rgba(70,217,255,0.12)" : "rgba(36,217,138,0.16)";
    roundRect(x, y, patchW, patchH, 3);
    ctx.fill();
  }

  ctx.globalCompositeOperation = "screen";
  for (let i = 0; i < (phoneMode ? 12 : 18); i += 1) {
    const x = ((i * 113 - mot
```

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```
ion * 0.28 + seed * 0.01) % (w + 180)) - 90;
const y = h * (0.16 + seededUnit(seed, i + 50) * 0.48);
const cloudW = 60 + seededUnit(seed, i + 60) * (phoneMode ? 88 : 130);
ctx.fillStyle = "rgba(244,251,248,0.22)";
ctx.beginPath();
ctx.ellipse(x, y, cloudW * 0.42, cloudW * 0.12, 0, 0, Math.PI * 2);
ctx.ellipse(x + cloudW * 0.28, y + 3, cloudW * 0.34, cloudW * 0.11, 0, 0, Math.PI * 2);
ctx.ellipse(x - cloudW * 0.22, y + 5, cloudW * 0.3, cloudW * 0.1, 0, 0, Math.PI * 2);
ctx.fill();
}
ctx.globalCompositeOperation = "source-over";

for (let i = 0; i < (phoneMode ? 8 : 10); i += 1) {
  const distance = raceState.distance + 240 + i * 330 + seededUnit(seed, i + 110) * 130;
  const center = roadObjectPos(0, distance);
  if (!center || center.y < horizon - 40 || center.y > h + 60) continue;
  const gateScale = Math.max(0.34, center.scale * (phoneMode ? 0.86 : 0.96));
  drawAirGate(center.x, center.y - gateScale * 36, gateScale, theme, seed + i * 17);
}

for (let i = 0; i < (phoneMode ? 5 : 7); i += 1) {
  const distance = raceState.distance + 400 + i * 520 + seededUnit(seed, i + 150) * 160;
  const side = i % 2 === 0 ? -1 : 1;
  const pos = roadObjectPos(side * (1.2 + seededUnit(seed, i + 160) * 0.28), distance);
  if (!pos || pos.y < horizon || pos.y > h + 80) continue;
  drawAirTrafficSilhouette(pos.x, pos.y - pos.scale * 30, Math.max(0.24, pos.scale * 0.78), side, theme, seed + i * 29);
}

for (let i = 0; i < (phoneMode ? 4 : 6); i += 1) {
  const side = i % 2 === 0 ? -1 : 1;
  const y = groundTop + h * (0.08 + seededUnit(seed, i + 210) * 0.26);
  const x = side < 0 ? w * (0.12 + seededUnit(seed, i + 220) * 0.12) : w * (0.88 - seededUnit(seed, i + 230) * 0.12);
  const s = phoneMode ? 0.42 + seededUnit(seed, i + 240) * 0.12 : 0.56 + seededUnit(seed, i + 240) * 0.18;
  const kind = i % 3 === 0 ? "airportBeacon" : i % 3 === 1 ? "serviceCartMotion" : "planeSpotters";
  ctx.globalAlpha = phoneMode ? 0.5 : 0.62;
  drawLivingWorldActivityShape(kind, x, y, s, side, place, route, design, theme, seed + i * 71);
}
ctx.restore();
ctx.globalAlpha = 1;
}

function drawWaterBuoy(x, y, scale, side, theme, seed) {
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.fillStyle = "rgba(0,0,0,0.38)";
  ctx.beginPath();
  ctx.ellipse(0, 12, 26, 8, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = side < 0 ? "#ff5b6b" : "#ffd166";
  ctx.beginPath();
  ctx.moveTo(0, -34);
  ctx.lineTo(18, 8);
  ctx.lineTo(-18, 8);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.86)";
  ctx.fillRect(-14, -8, 28, 8);
  ctx.restore();
}

function drawWaterSceneBoat(x, y, scale, side, theme, seed) {
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.fillStyle = "rgba(70,217,255,0.22)";
  ctx.beginPath();
  ctx.ellipse(side * -10, 16, 62, 13, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = seededUnit(seed, 1) > 0.5 ? "#f4fbf8" : theme[1] || "#46d9ff";
  ctx.beginPath();
  ctx.moveTo(-54, -10);
  ctx.lineTo(44, -18);
  ctx.lineTo(62, -2);
  ctx.lineTo(42, 14);
  ctx.lineTo(-42, 16);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.78)";
  roundRect(-12, -34, 42, 20, 5);
  ctx.fill();
  ctx.restore();
}

function drawAirGate(x, y, scale, theme, seed) {
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.strokeStyle = theme[1] || "#46d9ff";
  ctx.lineWidth = 5;
  ctx.globalAlpha = 0.68;
  ctx.beginPath();
  ctx.ellipse(0, 0, 78, 34, 0, 0, Math.PI * 2);
  ctx.stroke();
  ctx.strokeStyle = "rgba(244,251,248,0.54)";
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(-78, 0);
  ctx.lineTo(-110, 0);
  ctx.moveTo(78, 0);
  ctx.lineTo(110, 0);
  ctx.stroke();
  ctx.restore();
}

function drawAirTrafficSilhouette(x, y, scale, side, theme, seed) {
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale * side, scale);
  ctx.fillStyle = "rgba(244,251,248,0.78)";
  ctx.beginPath();
  ctx.moveTo(-58, 0);
  ctx.lineTo(48, -9);
  ctx.lineTo(66, 0);
  ctx.lineTo(48, 9);
```

```
ctx.closePath();
ctx.fill();
ctx.fillStyle = theme[2] || "#ffd166";
ctx.beginPath();
ctx.moveTo(-8, -4);
ctx.lineTo(24, -44);
c
```

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```
tx.lineTo(34, -38);
ctx.lineTo(16, -3);
ctx.closePath();
ctx.fill();
ctx.restore();
}

function drawVisibleGroundRoadPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode) {
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  const roadWash = ctx.createLinearGradient(0, horizon, 0, h);
  roadWash.addColorStop(0, phoneMode ? "rgba(26,36,34,0.18)" : "rgba(26,36,34,0.12)");
  roadWash.addColorStop(0.52, phoneMode ? "rgba(8,12,12,0.32)" : "rgba(8,12,12,0.24)");
  roadWash.addColorStop(1, phoneMode ? "rgba(0,0,0,0.48)" : "rgba(0,0,0,0.34)");
  ctx.fillStyle = roadWash;
  ctx.fillRect(0, horizon, w, h - horizon);

  const edgeColor = place === "desert" || place === "farm" || place === "canyon" ? "rgba(255,209,102,0.74)" :
  "rgba(244,251,248,0.78)";
  for (let side = -1; side <= 1; side += 2) {
    if (phoneMode) {
      ctx.globalAlpha = 0.18;
      ctx.strokeStyle = edgeColor;
      ctx.lineWidth = Math.max(2, w * 0.004);
      for (let i = 0; i < 3; i += 1) {
        const t = 0.58 + i * 0.13;
        const y = horizon + (h - horizon) * t;
        const half = realWorldRoadHalf(roadTop, roadBottom, t);
        const x = perspectiveRoadCenter(w, t) + side * half * 0.72;
        ctx.beginPath();
        ctx.moveTo(x - side * (8 + t * 12), y);
        ctx.lineTo(x + side * (34 + t * 22), y + 2 + t * 3);
        ctx.stroke();
      }
      continue;
    }
    ctx.strokeStyle = edgeColor;
    ctx.lineWidth = Math.max(3, w * (phoneMode ? 0.008 : 0.006));
    ctx.globalAlpha = phoneMode ? 0.72 : 0.62;
    ctx.beginPath();
    ctx.moveTo(perspectiveRoadCenter(w, 0.04) + side * roadTop * 0.78, horizon + 3);
    ctx.lineTo(perspectiveRoadCenter(w, 1) + side * roadBottom * 0.54, h + 8);
    ctx.stroke();

    ctx.strokeStyle = side < 0 ? "rgba(255,91,107,0.58)" : "rgba(70,217,255,0.52)";
    ctx.lineWidth = Math.max(5, w * (phoneMode ? 0.012 : 0.008));
    ctx.globalAlpha = phoneMode ? 0.5 : 0.38;
    ctx.beginPath();
    ctx.moveTo(perspectiveRoadCenter(w, 0.08) + side * roadTop * 0.98, horizon + 7);
    ctx.lineTo(perspectiveRoadCenter(w, 1) + side * roadBottom * 0.66, h + 14);
    ctx.stroke();
  }

  const span = h - horizon + 240;
  const laneAlpha = phoneMode ? 0.42 : 0.58;
  ctx.globalCompositeOperation = "screen";
  const visibleLanes = phoneMode ? [-0.5, 0.5] : [-1.5, -0.5, 0.5, 1.5];
  visibleLanes.forEach((lane) => {
    for (let i = 0; i < (phoneMode ? 8 : 16); i += 1) {
      const y = horizon + (((i * 118 + motion * 1.5 + lane * 37) % span) - 76);
      if (y < horizon || y > h + 55) continue;
      const t = Math.max(0.03, Math.min(1.06, (y - horizon) / Math.max(1, h - horizon)));
      const dashH = 14 + t * (phoneMode ? 44 : 74);
      const dashW = 5 + t * (phoneMode ? 7 : 12);
      ctx.globalAlpha = laneAlpha * (0.42 + t * 0.58);
      drawPerspectivePavementQuad(w, horizon, lane, y, dashH, dashW, "rgba(244,251,248,0.86)");
    }
  });
  ctx.globalCompositeOperation = "source-over";

  for (let i = 0; i < (phoneMode ? 10 : 14); i += 1) {
    const y = horizon + (((i * 94 + motion * 0.9 + seed * 0.002) % span) - 42);
    if (y < horizon || y > h + 30) continue;
    const t = Math.max(0.05, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    const center = perspectiveRoadCenter(w, t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t) * 0.6;
    ctx.globalAlpha = phoneMode ? 0.16 + t * 0.12 : 0.12 + t * 0.1;
    ctx.fillStyle = i % 2 ? "rgba(244,251,248,0.3)" : "rgba(0,0,0,0.68)";
    roundRect(center - half, y, half * 2, 3 + t * 9, Math.max(2, 2 + t * 3));
    ctx.fill();
  }
  ctx.restore();
  ctx.globalAlpha = 1;
}

function drawSpecialtySurfaceAndWorldPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode) {
  const vehicle = selectedVehicle();
  const specialty = place === "interstate" || place === "monsterpark" || place === "military" || vehicle.type === "semi" ||
  vehicle.type === "monster" || vehicle.type === "tank";
  if (!specialty) return;
  const motion = raceState.roadOffset || 0;
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  if (place === "monsterpark" || vehicle.type === "m
```


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```
onster") {
  const dirt = ctx.createLinearGradient(0, horizon, 0, h);
  dirt.addColorStop(0, "rgba(118,68,34,0.28)");
  dirt.addColorStop(0.58, "rgba(82,44,22,0.42)");
  dirt.addColorStop(1, "rgba(42,24,14,0.64)");
  ctx.fillStyle = dirt;
  ctx.fillRect(0, horizon, w, h - horizon);
  for (let i = 0; i < (phoneMode ? 16 : 22); i += 1) {
    const y = horizon + (((1 * 92 + motion * 1.22 + seed * 0.004) % (h - horizon + 170)) - 60);
    if (y < horizon || y > h + 44) continue;
    const t = Math.max(0.04, Math.min(1.1, (y - horizon) / Math.max(1, h - horizon)));
    const center = perspectiveRoadCenter(w, t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t) * (0.24 + seededUnit(seed, i) * 0.34);
    ctx.strokeStyle = i % 3 === 0 ? "rgba(255,209,102,0.3)" : "rgba(0,0,0,0.36)";
    ctx.lineWidth = 5 + t * 13;
    ctx.globalAlpha = 0.18 + t * 0.3;
    ctx.beginPath();
    ctx.moveTo(center - half, y);
    ctx.quadraticCurveTo(center, y + 8 + t * 12, center + half, y + seededUnit(seed, i + 3) * 16);
    ctx.stroke();
  }
} else if (place === "military" || vehicle.type === "tank") {
  for (let i = 0; i < (phoneMode ? 10 : 14); i += 1) {
    const y = horizon + (((i * 118 + motion * 0.86 + seed * 0.003) % (h - horizon + 160)) - 42);
    if (y < horizon || y > h + 30) continue;
    const t = Math.max(0.04, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
    const center = perspectiveRoadCenter(w, t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t) * 0.56;
    ctx.fillStyle = i % 2 ? "rgba(180,190,170,0.16)" : "rgba(0,0,0,0.22)";
    roundRect(center - half, y, half * 2, 5 + t * 12, 3);
    ctx.fill();
    ctx.strokeStyle = "rgba(187,242,74,0.24)";
    ctx.lineWidth = 2 + t * 4;
    ctx.beginPath();
    ctx.moveTo(center - half * 0.76, y + 8);
    ctx.lineTo(center + half * 0.76, y + 6);
    ctx.stroke();
  }
} else {
  for (let i = 0; i < (phoneMode ? 9 : 13); i += 1) {
    const y = horizon + (((i * 128 + motion * 1.05 + seed * 0.002) % (h - horizon + 180)) - 50);
    if (y < horizon || y > h + 48) continue;
    const t = Math.max(0.04, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
    const center = perspectiveRoadCenter(w, t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    ctx.fillStyle = "rgba(255,209,102,0.18)";
    roundRect(center - half * 0.82, y, half * 1.64, 4 + t * 8, 3);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.34)";
    ctx.font = `${Math.max(8, 9 + t * 10)}px system-ui`;
    ctx.textAlign = "center";
    ctx.fillText(i % 2 ? "TRUCK LANE" : "WEIGH", center, y + 12 + t * 14, half * 1.1);
  }
}
ctx.restore();

const propSets = {
  interstate: ["weighStation", "overheadSign", "dieselSign", "truckStop", "serviceTruck", "billboard"],
  monsterpark: ["grandstand", "scoreboard", "crushCarStack", "stadiumRamp", "dirtBerm", "spectators"],
  military: ["watchTower", "bunker", "rangeTarget", "blastWall", "checkpointArch", "serviceTruck"]
};
const props = propSets[place] || (vehicle.type === "semi" ? propSets.interstate : vehicle.type === "monster" ? propSets.monsterpark : vehicle.type === "tank" ? propSets.military : null);
if (!props) return;
ctx.save();
clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 12 : 18);
for (let i = -1; i < (phoneMode ? 11 : 16); i += 1) {
  const distance = raceState.distance + 180 + i * (phoneMode ? 230 : 250) + seededUnit(seed, i + 400) * 130;
  const side = seededUnit(seed, i + 410) > 0.5 ? 1 : -1;
  const pos = roadObjectPos(side * (2.15 + seededUnit(seed, i + 420) * 0.26), distance);
  if (!pos || pos.y < horizon - 34 || pos.y > h + 96) continue;
  const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, pos.y, side, pos.x, phoneMode, phoneMode ? 20 : 32);
  const near = Math.max(0, Math.min(1, pos.depth || 0));
  const scale = Math.max(0.34, Math.min(phoneMode ? 1.68 : 1.92, pos.scale * (0.92 + near * 0.58)));
  ctx.globalAlpha = Math.min(0.98, 0.74 + near * 0.24);
  drawGroundLockedDriveByShadow(x, pos.y, scale, side, theme);
  drawGroundLockedPla
```

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```
ceProp(props[Math.abs(i) % props.length], x, pos.y, scale, side, place, route, design, theme, seed + i * 83);
}
ctx.restore();
ctx.globalAlpha = 1;
}

function referenceInspiredSceneSet(place, stage) {
  if (stage && stage.prop === "tunnel") return ["tunnelMouthScene", "maintenanceBay", "concreteShoulder", "tunnelLightRun"];
  if (stage && (stage.prop === "bridge" || stage.prop === "pier")) return ["bridgeGuardrailScene", "waterSpanScene",
"inspectionTruckScene", "pierDeckScene"];
  if (stage && stage.prop === "checkpoint") return ["marshalPullout", "cameraCrewSetback", "crowdBehindBarrier",
"serviceVanPullout"];
  const sets = {
    coast: ["shorelineHomes", "beachPullout", "palmShoulder", "marinaShore", "seaCliffRail", "beachBystanders"],
    city: ["storefrontBlock", "parkedStreet", "glassOfficeSetback", "sidewalkCafe", "crosswalkCorner", "bystanderCorner"],
    canyon: ["rockCutGuardrail", "photoPullout", "desertScrubBasin", "mesaWall", "tourVanPullout", "wildlifePullout"],
    alpine: ["pineSlope", "stoneGuardwall", "chaletCluster", "snowbankCurve", "mountainPullout", "wildlifeSlope"],
    harbor: ["containerYardRow", "dockWarehouse", "craneSilhouette", "marinaPier", "serviceQuay", "dockBystanders"],
    snow: ["pineSnowfield", "snowFenceRun", "skiLodgeCluster", "plowPullout", "stoneGuardwall"],
    airfield: ["hangarApron", "runwayServiceRow", "controlTowerSet", "beaconFence", "fuelTruckPad"],
    pursuit: ["cityPoliceBlock", "parkedStreet", "streetLightRun", "crowdBehindBarrier", "serviceVanPullout"],
    freight: ["truckStopRow", "semiLot", "soundWallTraffic", "dieselCanopy", "containerYardRow"],
    interstate: ["truckStopRow", "soundWallTraffic", "dieselCanopy", "semiLot", "restAreaPullout"],
    farm: ["fieldFenceRows", "barnYard", "grainSiloSet", "tractorField", "countyRoadShoulder", "pastureAnimals"],
    monsterpark: ["grandstandBerm", "dirtArenaEdge", "crushCarYard", "servicePitRow", "crowdBehindBarrier"],
    military: ["bunkerSetback", "watchTowerRidge", "blastWallRun", "serviceConvoyPullout", "rangeMound"],
    skybase: ["hangarApron", "controlTowerSet", "beaconFence", "serviceConvoyPullout", "radarPad"],
    tokyo: ["neonStreetRow", "railOverpassScene", "taxiCurb", "towerAlley", "wetSidewalkGlow", "bystanderCorner"],
    desert: ["openDesertHighway", "telephonePoleLine", "rallyCampSparse", "distantButtes", "desertScrubBasin", "wildlifePullout"],
    rainforest: ["canopyVillage", "riverBridgeScene", "marketRoadside", "mistWaterfall", "woodRailRun", "canopyAnimals"],
    europe: ["stoneVillageStreet", "terraceVineyard", "switchbackWall", "tramStreet", "cafeCurve", "pastureAnimals"]
  };
  return sets[place] || sets.city;
}

function drawReferenceInspiredDriveByScenePass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode) {
  const length = Math.max(1, raceLength());
  const spacing = phoneMode ? 760 : 620;
  const lookBehind = phoneMode ? 260 : 340;
  const lookAhead = phoneMode ? 2300 : 3000;
  const first = Math.floor((raceState.distance - lookBehind) / spacing) - 1;
  const last = Math.ceil((raceState.distance + lookAhead) / spacing) + 1;
  const scenes = [];
  for (let index = first; index <= last; index += 1) {
    const baseDistance = index * spacing + seededUnit(seed, index + 510) * spacing * 0.38;
    if (baseDistance < 0 || baseDistance > length + lookAhead * 0.24) continue;
    const gap = baseDistance - raceState.distance;
    if (gap < -lookBehind || gap > lookAhead) continue;
    const progress = Math.max(0, Math.min(1, baseDistance / length));
    const stage = routeStageInfo(place, progress);
    const set = referenceInspiredSceneSet(place, stage);
    const side = seededUnit(seed, index + 520) > 0.5 ? 1 : -1;
    const lane = side * (2.56 + seededUnit(seed, index + 530) * (phoneMode ? 0.5 : 0.72));
    const projected = roadObjectPos(lane, baseDistance);
    if (!projected || projected.y < horizon - 52 || projected.y > h + 130) continue;
    scenes.push({
      x: projected.x,
      y: projected.y,
      scale: projected.scale,
      depth: projected.depth || Math.max(0, Math.min(1, (projected.y - horizon) / Math.max(1, h - horizon))),
      side,
      k
    });
  }
}
```

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```
ind: set[Math.abs(index + stage.index) % set.length],
  seed: seed + index * 173,
  distance: baseDistance
});
if (!phoneMode && index % 3 === 0) {
  const otherDistance = baseDistance + spacing * (0.42 + seededUnit(seed, index + 540) * 0.16);
  const otherProgress = Math.max(0, Math.min(1, otherDistance / length));
  const otherStage = routeStageInfo(place, otherProgress);
  const otherSet = referenceInspiredSceneSet(place, otherStage);
  const otherSide = -side;
  const otherProjected = roadObjectPos(otherSide * (2.46 + seededUnit(seed, index + 550) * 0.58), otherDistance);
  if (otherProjected && otherProjected.y >= horizon - 52 && otherProjected.y <= h + 130) {
    scenes.push({
      x: otherProjected.x,
      y: otherProjected.y,
      scale: otherProjected.scale,
      depth: otherProjected.depth || Math.max(0, Math.min(1, (otherProjected.y - horizon) / Math.max(1, h - horizon))),
      side: otherSide,
      kind: otherSet[Math.abs(index) + otherStage.index + 2] % otherSet.length,
      seed: seed + index * 191,
      distance: otherDistance
    });
  }
}
}
scenes.sort((a, b) => a.y - b.y);
ctx.save();
clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 18 : 24);
const lastBySide = { "-1": -9999, "1": -9999 };
scenes.forEach((scene) => {
  const sideKey = String(scene.side < 0 ? -1 : 1);
  const minGap = phoneMode ? 220 : 150;
  if (scene.y - lastBySide[sideKey] < minGap) return;
  lastBySide[sideKey] = scene.y;
  const near = Math.max(0, Math.min(1, scene.depth || 0));
  const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, scene.y, scene.side, scene.x, phoneMode, phoneMode ? 42 : 58);
  const scale = Math.max(0.36, Math.min(phoneMode ? 1.74 : 2.02, scene.scale * (0.92 + near * 0.72)));
  ctx.globalAlpha = Math.min(0.98, (phoneMode ? 0.72 : 0.56) + near * 0.28);
  drawReferenceInspiredSceneCluster(scene.kind, x, scene.y, scale, scene.side, place, route, design, theme, scene.seed);
});
ctx.restore();
ctx.globalAlpha = 1;
}

function drawReferenceInspiredSceneCluster(kind, x, y, scale, side, place, route, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  drawReferenceSceneGroundPatch(kind, side, place, design, theme);
  if (/storefront|parkedStreet|glassOffice|sidewalkCafe|crosswalk|neonStreet|taxiCurb|towerAlley|wetSidewalk|stoneVillage|tramStreet|cafeCurve|cityPolice/i.test(kind)) {
    drawReferenceBuildingRow(side, kind, place, accent, light, seed);
    if (!/glassOffice|towerAlley/i.test(kind)) drawReferenceParkedVehicleLine(side, accent, light, seed);
    if (/sidewalk|cafe|crosswalk|taxi|stoneVillage|neon|police/i.test(kind)) drawReferencePeopleLine(side, accent, light, seed);
    drawReferenceStreetFurniture(side, kind, accent, light);
  } else if (/shoreline|beach|palm|marina|seaCliff|waterSpan|pierDeck/i.test(kind)) {
    drawReferenceWaterEdge(side, kind, accent, light, seed);
    drawReferenceTreeLine("palm", side, seed);
    if (/homes|beach|marina/i.test(kind)) drawReferenceSmallHouseRow(side, accent, light, seed);
    if (/seaCliff|bridge|waterSpan/i.test(kind)) drawReferenceGuardrail(side, "rgba(214,226,221,0.62)");
  } else if (/rock|mesa|desert|telephone|rallyCamp|distantButtes|photoPullout|tourVan|wildlifePullout/i.test(kind)) {
    drawReferenceRockAndDesertScene(side, kind, accent, light, seed);
    drawReferenceGuardrail(side, "rgba(210,190,156,0.42)");
    if (/telephone/i.test(kind)) drawReferenceUtilityPoleLine(side, accent, light);
    if (/rallyCamp|photoPullout|tourVan/i.test(kind)) drawReferenceParkedVehicleLine(side, accent, light, seed);
  } else if (/pine|stoneGuard|chalet|snowbank|mountainPullout|skiLodge|snowFence|plow|wildlifeSlope/i.test(kind)) {
    drawReferenceTreeLine("pine", side, seed);
    drawReferenceMountainVillage(side, kind, accent, light, seed);
    drawReferenceGuardrail(side, /snow|pine/i.test(kind) ? "rgba(244,251,248,0.58)" : "rgba(170,178,170,0.58)");
  } else
```

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```
e if (/container|dockWarehouse|crane|serviceQuay|truckStop|semiLot|soundWall|diesel|restArea/i.test(kind)) {
  drawReferenceIndustrialScene(side, kind, accent, light, seed);
  if (/truckStop|semiLot|diesel|restArea/i.test(kind)) drawReferenceParkedVehicleLine(side, light, accent, seed);
} else if (/field|barn|grainSilo|tractor|countyRoad|pastureAnimals/i.test(kind)) {
  drawReferenceFarmScene(side, kind, accent, light, seed);
} else if (/hangar|runway|controlTower|beacon|fuelTruck|radar/i.test(kind)) {
  drawReferenceAirfieldScene(side, kind, accent, light, seed);
} else if (/bunker|watchTower|blastWall|serviceConvoy|rangeMound|concreteShoulder/i.test(kind)) {
  drawReferenceMilitaryScene(side, kind, accent, light, seed);
} else if (/grandstand|dirtArena|crushCar|servicePit|Berm/i.test(kind)) {
  drawReferenceMonsterScene(side, kind, accent, light, seed);
} else if (/canopy|riverBridge|market|mistWaterfall|woodRail|canopyAnimals/i.test(kind)) {
  drawReferenceRainforestScene(side, kind, accent, light, seed);
} else {
  drawReferenceBuildingRow(side, kind, place, accent, light, seed);
  drawReferenceParkedVehicleLine(side, accent, light, seed);
}
if (/bystander|crowd|cafe|crosswalk|market|dock|beach|station|service|village|corner/i.test(kind) && seededUnit(seed, 730) > 0.18)
{
  drawReferenceBystanderBarrier(side, accent, light, seed + 17);
}
if (routeAnimalSet(place).length &&
(/animal|wildlife|pasture|farm|field|pine|forest|beach|shore|desert|rainforest|village|slope|pullout/i.test(kind) ||
seededUnit(seed, 740) > 0.62)) {
  drawReferenceAnimalGroup(routeAnimalSpecies(place, seed + 741), side, accent, light, seed + 23);
}
ctx.restore();
}

function drawReferenceSceneGroundPatch(kind, side, place, design, theme) {
  const ground = place === "desert" || place === "canyon" ? "rgba(178,92,48,0.26)"
  : place === "coast" || place === "harbor" ? "rgba(70,217,255,0.18)"
  : place === "farm" ? "rgba(115,146,56,0.2)"
  : place === "snow" || place === "alpine" ? "rgba(244,251,248,0.2)"
  : "rgba(16,24,23,0.26)";
  ctx.fillStyle = "rgba(0,0,0,0.36)";
  ctx.beginPath();
  ctx.ellipse(side * -10, 18, 126, 18, -0.04 * side, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = ground;
  ctx.beginPath();
  ctx.moveTo(-136 * side, 14);
  ctx.lineTo(118 * side, 2);
  ctx.lineTo(146 * side, 34);
  ctx.lineTo(-118 * side, 40);
  ctx.closePath();
  ctx.fill();
}

function drawReferenceBuildingRow(side, kind, place, accent, light, seed) {
  const night = place === "tokyo" || /neon|wetSidewalk|towerAlley/i.test(kind);
  const village = /stoneVillage|tramStreet|cafeCurve/i.test(kind);
  const baseColor = night ? "rgba(18,10,34,0.9)" : village ? "rgba(72,54,42,0.88)" : "rgba(12,22,24,0.88)";
  for (let i = -1; i <= 2; i += 1) {
    const bx = side * (i * 42 + seededUnit(seed, i + 3) * 10);
    const bw = village ? 44 + seededUnit(seed, i + 11) * 14 : 34 + seededUnit(seed, i + 11) * 26;
    const bh = village ? 46 + seededUnit(seed, i + 22) * 24 : 64 + seededUnit(seed, i + 22) * 94;
    ctx.fillStyle = baseColor;
    roundRect(bx - bw / 2, 12 - bh, bw, bh, 4);
    ctx.fill();
    if (village) {
      ctx.fillStyle = i % 2 ? "rgba(156,72,48,0.9)" : "rgba(104,74,54,0.9)";
      ctx.beginPath();
      ctx.moveTo(bx - bw * 0.62, 12 - bh);
      ctx.lineTo(bx, -bh - 10);
      ctx.lineTo(bx + bw * 0.62, 12 - bh);
      ctx.closePath();
      ctx.fill();
    }
    ctx.fillStyle = night ? (i % 2 ? `${accent}92` : `${light}88`) : "rgba(244,251,248,0.36)";
    for (let row = 0; row < Math.min(6, Math.floor(bh / 18)); row += 1) {
      ctx.fillRect(bx - bw * 0.28, 23 - bh + row * 15, bw * 0.16, 4);
      ctx.fillRect(bx + bw * 0.1, 24 - bh + row * 15, bw * 0.18, 4);
    }
  }
}

function drawReferenceSmallHouseRow(side, accent, light, seed) {
  for (let i = -1; i <= 1; i += 1) {
    const bx = side * (i * 48 + 8);
    ctx.fillStyle = i % 2 ? "rgba(228,208,170,0.82)" : "rgba(28,46,48,0.82)";
    roundRect(bx - 28, -42, 56, 52, 5);
    ctx.fill();
    ctx.fillStyle = i % 2 ? `${accent}99` : `${light}88`;
    ctx.beginPath();
    ctx.moveTo(bx - 34, -42);
    ctx.lineTo(bx, -66);
    ctx.lineTo(bx +
```

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```
34, -42);
    ctx.closePath();
    ctx.fill();
}
}

function drawReferenceWaterEdge(side, kind, accent, light, seed) {
    const phoneMode = phoneGraphicsActive();
    ctx.fillStyle = "rgba(70,217,255,0.22)";
    ctx.beginPath();
    ctx.ellipse(side * -14, 18, 130, 22, -0.05 * side, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.lineWidth = 2;
    for (let i = 0; i < (phoneMode ? 1 : 3); i += 1) {
        ctx.beginPath();
        ctx.moveTo(-108 * side, i * 8 + 10);
        ctx.quadraticCurveTo(-20 * side, i * 8 + 4 + Math.sin(seed + i) * 5, 104 * side, i * 8 + 12);
        ctx.stroke();
    }
    if (/marina|pier|waterSpan/i.test(kind)) {
        if (phoneMode) {
            ctx.fillStyle = "rgba(126,86,48,0.68)";
            roundRect(-94, -19, 188, 24, 6);
            ctx.fill();
            ctx.strokeStyle = "rgba(244,251,248,0.24)";
            ctx.lineWidth = 3;
            ctx.beginPath();
            ctx.moveTo(-82 * side, -13);
            ctx.lineTo(82 * side, -23);
            ctx.stroke();
        } else {
            ctx.strokeStyle = "rgba(138,102,62,0.78)";
            ctx.lineWidth = 5;
            ctx.beginPath();
            ctx.moveTo(-94 * side, -4);
            ctx.lineTo(92 * side, -18);
            ctx.moveTo(-94 * side, 12);
            ctx.lineTo(92 * side, -2);
            for (let p = -3; p <= 3; p += 1) {
                ctx.moveTo(p * 28 * side, -20);
                ctx.lineTo(p * 28 * side, 20);
            }
            ctx.stroke();
        }
    }
}

function drawReferenceRockAndDesertScene(side, kind, accent, light, seed) {
    const main = /canyon|rock|mesa|Buttes/i.test(kind) ? "rgba(172,82,42,0.72)" : "rgba(216,142,66,0.5)";
    for (let i = -1; i <= 1; i += 1) {
        const bx = side * (i * 52 + seededUnit(seed, i + 4) * 16);
        const peak = -42 - seededUnit(seed, i + 14) * 52;
        ctx.fillStyle = main;
        ctx.beginPath();
        ctx.moveTo(bx - 58, 18);
        ctx.lineTo(bx - 24, peak * 0.72);
        ctx.lineTo(bx + 8, -18);
        ctx.lineTo(bx + 44, peak);
        ctx.lineTo(bx + 72, 18);
        ctx.closePath();
        ctx.fill();
    }
    ctx.fillStyle = "rgba(96,118,52,0.72)";
    for (let i = -3; i <= 3; i += 1) {
        ctx.beginPath();
        ctx.ellipse(side * (i * 24 + 6), 4 + (i % 2) * 7, 14, 9, 0.2 * i, 0, Math.PI * 2);
        ctx.fill();
    }
}

function drawReferenceMountainVillage(side, kind, accent, light, seed) {
    if (/chalet|Lodge|mountainPullout/i.test(kind)) {
        drawReferenceSmallHouseRow(side, accent, light, seed);
    }
    ctx.fillStyle = /snow|pine/i.test(kind) ? "rgba(244,251,248,0.42)" : "rgba(150,168,160,0.34)";
    ctx.beginPath();
    ctx.moveTo(-128, 18);
    ctx.lineTo(-58, -58);
    ctx.lineTo(-16, -20);
    ctx.lineTo(46, -86);
    ctx.lineTo(128, 20);
    ctx.closePath();
    ctx.fill();
}

function drawReferenceIndustrialScene(side, kind, accent, light, seed) {
    ctx.fillStyle = /dock|container|crane/i.test(kind) ? "rgba(18,34,36,0.88)" : "rgba(28,30,28,0.88)";
    roundRect(side > 0 ? -86 : -70, -42, 156, 54, 5);
    ctx.fill();
    const colors = ["#ff5b6b", "#46d9ff", "#ffd166", "#dce8ef"];
    for (let i = -2; i <= 2; i += 1) {
        ctx.fillStyle = colors[(Math.abs(i) + Math.floor(seed)) % colors.length];
        roundRect(side * (i * 30) - 14, -26 + (i % 2) * 12, 28, 18, 2);
        ctx.fill();
    }
    if (/crane|container|dock/i.test(kind)) {
        ctx.strokeStyle = `${light}99`;
        ctx.lineWidth = 4;
        ctx.beginPath();
        ctx.moveTo(52 * side, 12);
        ctx.lineTo(34 * side, -86);
        ctx.lineTo(-72 * side, -72);
        ctx.moveTo(34 * side, -86);
        ctx.lineTo(4 * side, -18);
        ctx.stroke();
    }
}

function drawReferenceFarmScene(side, kind, accent, light, seed) {
    ctx.strokeStyle = "rgba(187,242,74,0.35)";
    ctx.lineWidth = 2;
    for (let i = -4; i <= 4; i += 1) {
```

```
    ctx.beginPath();
    ctx.moveTo(-126 * side, i * 8 + 16);
    ctx.lineTo(126 * side, i * 2 + 2);
    ctx.stroke();
}
if (/barn|county|field/i.test(kind)) {
    ctx.fillStyle = "rgba(148,42,34,0.88)";
    roundRect(side > 0 ? -44 : -32, -52, 76, 62, 5);
    ctx.fill();
    ctx.fillStyle = "rgba(96,28,24,0.9)";
    ctx.beginPath();
    ctx.moveTo(-52 * side, -52);
    ctx.lineTo(-6 * side, -86);
    ctx.lineTo(44 * side, -52);
    ctx.closePath();
    ctx.fill();
}
if (/silo|grain/i.test(kind)) {
    ctx.fillStyle = "rgba(220,224,210,0.72)";
    roundRect(side > 0 ? 48 : -80, -
```

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```
82, 32, 92, 8);
    ctx.fill();
  }
  drawReferenceGuardrail(side, "rgba(150,117,80,0.58)");
}

function drawReferenceAirfieldScene(side, kind, accent, light, seed) {
  ctx.fillStyle = "rgba(22,26,26,0.88)";
  roundRect(side > 0 ? -104 : -76, -38, 180, 52, 8);
  ctx.fill();
  ctx.strokeStyle = "rgba(216,224,213,0.58)";
  ctx.lineWidth = 3;
  for (let i = -3; i <= 3; i += 1) {
    ctx.beginPath();
    ctx.moveTo(i * 30 * side, 16);
    ctx.lineTo(i * 34 * side, -28);
    ctx.stroke();
  }
  if (/control|beacon|radar/i.test(kind)) {
    ctx.fillStyle = "rgba(214,226,221,0.72)";
    roundRect(side > 0 ? 62 : -96, -86, 34, 94, 5);
    ctx.fill();
    ctx.fillStyle = accent;
    ctx.beginPath();
    ctx.arc(78 * side, -96, 8, 0, Math.PI * 2);
    ctx.fill();
  }
}

function drawReferenceMilitaryScene(side, kind, accent, light, seed) {
  ctx.fillStyle = "rgba(58,68,54,0.82)";
  roundRect(side > 0 ? -86 : -46, -30, 132, 42, 4);
  ctx.fill();
  ctx.fillStyle = "rgba(26,32,28,0.9)";
  roundRect(side > 0 ? -46 : -26, -58, 72, 42, 5);
  ctx.fill();
  ctx.strokeStyle = "rgba(187,242,74,0.44)";
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-110 * side, -4);
  ctx.lineTo(110 * side, -14);
  ctx.stroke();
  if (/watchTower/i.test(kind)) {
    ctx.strokeStyle = "rgba(170,182,160,0.72)";
    ctx.beginPath();
    ctx.moveTo(60 * side, 12);
    ctx.lineTo(48 * side, -82);
    ctx.moveTo(96 * side, 12);
    ctx.lineTo(84 * side, -82);
    ctx.stroke();
    ctx.fillStyle = "rgba(20,26,22,0.86)";
    roundRect(side > 0 ? 38 : -96, -102, 58, 26, 3);
    ctx.fill();
  }
}

function drawReferenceMonsterScene(side, kind, accent, light, seed) {
  ctx.fillStyle = "rgba(110,62,30,0.58)";
  ctx.beginPath();
  ctx.ellipse(0, 10, 122, 30, -0.08 * side, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(8,12,12,0.82)";
  roundRect(side > 0 ? -94 : -48, -70, 142, 42, 5);
  ctx.fill();
  ctx.fillStyle = `${accent}88`;
  for (let i = -2; i <= 2; i += 1) {
    const carX = i * 28 * side;
    roundRect(carX - 17, -14 + (i % 2) * 5, 34, 14, 2);
    ctx.fill();
  }
}

function drawReferenceRainforestScene(side, kind, accent, light, seed) {
  drawReferenceTreeLine("canopy", side, seed);
  if (/market|Village/i.test(kind)) {
    ctx.fillStyle = "rgba(62,42,26,0.86)";
    roundRect(side > 0 ? -62 : -42, -42, 104, 52, 5);
    ctx.fill();
    ctx.fillStyle = `${light}88`;
    ctx.beginPath();
    ctx.moveTo(-70 * side, -42);
    ctx.lineTo(-12 * side, -74);
    ctx.lineTo(54 * side, -42);
    ctx.closePath();
    ctx.fill();
  }
  if (/river|Waterfall/i.test(kind)) {
    ctx.fillStyle = "rgba(70,217,255,0.24)";
    ctx.beginPath();
    ctx.ellipse(0, 20, 116, 18, -0.05 * side, 0, Math.PI * 2);
    ctx.fill();
    if (/Waterfall/i.test(kind)) {
      ctx.fillStyle = "rgba(222,249,255,0.45)";
      roundRect(side > 0 ? 28 : -50, -86, 22, 94, 6);
      ctx.fill();
    }
  }
  drawReferenceGuardrail(side, "rgba(124,86,48,0.6)");
}

function drawReferenceTreeLine(kind, side, seed) {
  for (let i = -2; i <= 2; i += 1) {
    const x = side * (i * 34 + seededUnit(seed, i + 12) * 9);
    const trunk = kind === "palm" ? "rgba(136,88,46,0.86)" : "rgba(70,54,34,0.86)";
    ctx.strokeStyle = trunk;
    ctx.lineWidth = kind === "canopy" ? 5 : 4;
    ctx.beginPath();
    ctx.moveTo(x, 14);
    ctx.lineTo(x + side * 4, -48 - seededUnit(seed, i + 18) * 26);
    ctx.stroke();
    ctx.fillStyle = kind === "palm" ? "rgba(54,164,88,0.8)" : kind === "canopy" ? "rgba(32,118,58,0.82)" : "rgba(24,92,54,0.84)";
    if (kind === "pine") {
      for (let tier = 0; tier < 3; tier += 1) {
```

```

        ctx.beginPath();
        ctx.moveTo(x, -82 + tier * 22);
        ctx.lineTo(x - 28, -38 + tier * 18);
        ctx.lineTo(x + 30, -38 + tier * 18);
        ctx.closePath();
        ctx.fill();
    }
} else {
    ctx.beginPath();
    ctx.ellipse(x + side * 4, -64, kind === "canopy" ? 42 : 30, kind === "palm" ? 10 : 22, 0.15 * side, 0, Math.PI * 2);
    ctx.fill();
    ctx.beginPath();
    ctx.ellipse(x - side * 14, -52, kind === "canopy" ? 34 : 26, kind === "palm" ? 9 : 20, -0.35 * side, 0, Math.PI * 2);
    ctx.fill();
}
}
}
function drawReferenceParkedVehicleLine(side,

```


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```
    accent, light, seed) {
    for (let i = -1; i <= 1; i += 1) {
        const x = side * (i * 42 + 8);
        const paint = i % 2 ? accent : light;
        ctx.fillStyle = "rgba(0,0,0,0.42)";
        ctx.beginPath();
        ctx.ellipse(x, 17, 34, 6, 0, 0, Math.PI * 2);
        ctx.fill();
        ctx.fillStyle = paint;
        roundRect(x - 30, -12, 60, 27, 6);
        ctx.fill();
        ctx.fillStyle = "rgba(5,8,7,0.8)";
        roundRect(x - 12, -22, 24, 13, 4);
        ctx.fill();
        ctx.fillStyle = "#050807";
        ctx.beginPath();
        ctx.arc(x - 20, 15, 6, 0, Math.PI * 2);
        ctx.arc(x + 20, 15, 6, 0, Math.PI * 2);
        ctx.fill();
    }
}

function drawReferencePeopleLine(side, accent, light, seed) {
    const colors = [accent, light, "#f4fbf8", "#ff5b6b"];
    for (let i = -2; i <= 2; i += 1) {
        const x = side * (i * 13 + 2);
        ctx.fillStyle = colors[(Math.abs(i) + Math.floor(seed)) % colors.length];
        ctx.beginPath();
        ctx.arc(x, -18 + (i % 2) * 3, 4, 0, Math.PI * 2);
        ctx.fill();
        roundRect(x - 4, -14 + (i % 2) * 3, 8, 18, 3);
        ctx.fill();
    }
}

function drawReferenceBystanderBarrier(side, accent, light, seed) {
    ctx.save();
    ctx.strokeStyle = "rgba(244,251,248,0.48)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-74 * side, 6);
    ctx.lineTo(82 * side, -3);
    ctx.moveTo(-74 * side, -8);
    ctx.lineTo(82 * side, -17);
    for (let i = -2; i <= 2; i += 1) {
        ctx.moveTo(i * 32 * side, -22);
        ctx.lineTo(i * 32 * side, 10);
    }
    ctx.stroke();
    ctx.translate(side * 4, -24);
    drawReferencePeopleLine(side, accent, light, seed);
    ctx.restore();
}

function drawReferenceAnimalGroup(species, side, accent, light, seed) {
    if (!species) return;
    const count = species === "bird" || species === "shoreBird" ? 3 : species === "sheep" ? 3 : 2;
    ctx.save();
    for (let i = 0; i < count; i += 1) {
        const x = side * (-34 + i * 32 + seededUnit(seed, i + 2) * 10);
        const y = 15 + (i % 2) * 5;
        const s = species === "camel" || species === "horse" || species === "cow" ? 1.08 : species === "bird" || species === "shoreBird" ? 0.54 : 0.82;
        drawReferenceAnimalShape(species, x, y, s, side, accent, light, seed + i * 19);
    }
    ctx.restore();
}

function drawReferenceAnimalShape(species, x, y, scale, side, accent, light, seed) {
    const body = species === "cow" ? "rgba(244,251,248,0.86)" :
        species === "horse" || species === "deer" ? "rgba(132,86,46,0.86)" :
        species === "camel" ? "rgba(196,142,78,0.86)" :
        species === "sheep" ? "rgba(230,230,210,0.88)" :
        species === "monkey" ? "rgba(108,72,46,0.86)" :
        species === "tapir" ? "rgba(74,64,58,0.86)" :
        species === "bird" || species === "shoreBird" ? "rgba(244,251,248,0.82)" :
        "rgba(90,70,52,0.86)";
    ctx.save();
    ctx.translate(x, y);
    ctx.scale(scale * side, scale);
    ctx.fillStyle = "rgba(0,0,0,0.32)";
    ctx.beginPath();
    ctx.ellipse(0, 8, species === "bird" || species === "shoreBird" ? 16 : 30, 5, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = body;
    if (species === "bird" || species === "shoreBird") {
        ctx.beginPath();
        ctx.ellipse(0, -4, 18, 7, -0.1, 0, Math.PI * 2);
        ctx.fill();
        ctx.beginPath();
        ctx.arc(14, -9, 5, 0, Math.PI * 2);
        ctx.fill();
        ctx.strokeStyle = "rgba(40,40,34,0.72)";
        ctx.lineWidth = 2;
        ctx.beginPath();
        ctx.moveTo(-4, 2);
        ctx.lineTo(-9, 10);
        ctx.moveTo(7, 2);
        ctx.lineTo(2, 10);
        ctx.stroke();
    } else {
        ctx.beginPath();
        ctx.ellipse(0, -10, species === "camel" ? 34 : 30, species === "sheep" ? 16 : 13, 0, 0, Math.PI * 2);
        ctx.fill();
        if (species === "camel") {
            ctx.beginPath();
            ctx.ellipse(-10, -24, 11, 16, 0, 0, Math.PI * 2);
            ctx.fill();
        }
        ctx.beginPath();
        ctx.arc(28, -18, species === "horse" || species === "deer" ? 9 : 8, 0, Math.PI * 2);
        ctx.fill();
        if (species === "deer" || species === "goat") {
```

```
ctx.strokeStyle = light;
ctx.lineWidth = 2;
ctx.beginPath();
ctx.moveTo(31, -26);
ctx.lineTo(38, -42);
ctx.moveTo(25, -26);
ctx.lineTo(18, -39);
ctx.stroke();
}
ctx.strokeStyle = "rgba(36,28,20,0.78)";
ctx.lineWidth = 3;
ctx.beginPath();
for (le
```

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```
t leg = -1; leg <= 1; leg += 2) {
  ctx.moveTo(leg * 15, 0);
  ctx.lineTo(leg * 17, 13);
  ctx.moveTo(leg * 2, 1);
  ctx.lineTo(leg * 4, 13);
}
ctx.stroke();
if (species === "cow") {
  ctx.fillStyle = "rgba(30,25,20,0.76)";
  ctx.beginPath();
  ctx.ellipse(-8, -13, 8, 5, 0.2, 0, Math.PI * 2);
  ctx.ellipse(10, -8, 7, 4, -0.1, 0, Math.PI * 2);
  ctx.fill();
}
}
ctx.restore();
}

function drawReferenceStreetFurniture(side, kind, accent, light) {
  ctx.strokeStyle = "rgba(205,218,214,0.64)";
  ctx.lineWidth = 3;
  for (let i = -1; i <= 1; i += 2) {
    const x = side * (80 + i * 12);
    ctx.beginPath();
    ctx.moveTo(x, 12);
    ctx.lineTo(x, -74);
    ctx.stroke();
    ctx.globalCompositeOperation = "screen";
    ctx.fillStyle = /neon|wet/i.test(kind) ? accent : light;
    ctx.beginPath();
    ctx.arc(x, -80, 7, 0, Math.PI * 2);
    ctx.fill();
    ctx.globalCompositeOperation = "source-over";
  }
}

function drawReferenceUtilityPoleLine(side, accent, light) {
  ctx.strokeStyle = "rgba(120,86,54,0.72)";
  ctx.lineWidth = 4;
  for (let i = -2; i <= 2; i += 1) {
    const x = side * (i * 42);
    ctx.beginPath();
    ctx.moveTo(x, 16);
    ctx.lineTo(x, -74);
    ctx.moveTo(x - 20 * side, -54);
    ctx.lineTo(x + 20 * side, -54);
    ctx.stroke();
  }
  ctx.strokeStyle = "rgba(216,226,221,0.26)";
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(-106 * side, -58);
  ctx.bezierCurveTo(-42 * side, -66, 30 * side, -42, 108 * side, -56);
  ctx.stroke();
}

function drawReferenceGuardrail(side, color) {
  if (phoneGraphicsActive()) {
    ctx.strokeStyle = "rgba(214,226,221,0.32)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-58 * side, -4);
    ctx.quadraticCurveTo(0, -10, 60 * side, -6);
    ctx.moveTo(-52 * side, 8);
    ctx.quadraticCurveTo(0, 3, 54 * side, 5);
    ctx.stroke();
    ctx.fillStyle = "rgba(0,0,0,0.2)";
    ctx.beginPath();
    ctx.ellipse(0, 12, 68, 10, -0.03 * side, 0, Math.PI * 2);
    ctx.fill();
    return;
  }
  ctx.strokeStyle = color;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-112 * side, -10);
  ctx.lineTo(112 * side, -20);
  ctx.moveTo(-112 * side, 8);
  ctx.lineTo(112 * side, -2);
  if (!phoneGraphicsActive()) {
    for (let i = -4; i <= 4; i += 1) {
      ctx.moveTo(i * 28 * side, -26);
      ctx.lineTo(i * 28 * side, 18);
    }
  }
  } else {
    ctx.moveTo(-84 * side, -24);
    ctx.quadraticCurveTo(0, -34, 88 * side, -28);
  }
  ctx.stroke();
  if (phoneGraphicsActive()) {
    ctx.fillStyle = "rgba(0,0,0,0.22)";
    ctx.beginPath();
    ctx.ellipse(0, 9, 106, 12, -0.04 * side, 0, Math.PI * 2);
    ctx.fill();
  }
}

function groundLockedPlaceProps(place, stage) {
  if (stage && stage.prop === "tunnel") return ["barrier", "wallLamp", "tunnelFan", "routeSign"];
  if (stage && (stage.prop === "bridge" || stage.prop === "pier")) return ["bridgeRail", "lamp", "waterEdge", "routeSign"];
  const props = {
    coast: ["palm", "beachHouse", "seawall", "spectators", "routeSign"],
    city: ["storefront", "streetLight", "parkedCar", "busStop", "crowd"],
    canyon: ["rockFace", "guardPost", "desertScrub", "curveBoard", "tourVan"],
    alpine: ["pine", "stonewall", "chalet", "snowPost", "spectators"],
    harbor: ["container", "dockLight", "warehouse", "craneBase", "serviceTruck"],
    snow: ["pine", "snowFence", "lodge", "skiPost", "spectators"],
    airfield: ["runwayLight", "hangar", "serviceCart", "towerSign", "barrier"],
    pursuit: ["policeHQ", "parkedCar", "streetLight", "crowd", "lowBarrier"],
    freight: ["container", "railSignal", "truckStop", "fence", "serviceTruck"],
    interstate: ["weighStation", "dieselSign", "truckStop", "serviceTruck", "overheadSign"],
    farm: ["fence", "barn", "tractorParked", "hayBales", "windmill"],
    monsterpark: ["grandstand", "dirtBerm", "crushCarStack", "stadiumRamp", "scoreboard"],
  }
```

```
military: ["bunker", "watchTower", "rangeTarget", "blastWall", "barrier"],
skybase: ["controlTower", "runwayLight", "hangar", "rangeTarget", "serviceCart"],
tokyo: ["neonShop", "streetLight", "parkedCar", "crowd", "expressSign"],
desert: ["dune", "desertScrub", "supportTent", "rallyMarker"],
```

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```
"rockFace"],
rainforest: ["canopyTree", "marketStand", "woodRail", "waterfallSign", "crowd"],
europe: ["stoneWall", "cafe", "tramPost", "villageHouse", "spectators"]
};
return props[place] || props.city;
}

function routeDriveByProps(place, stage) {
const stageProps = {
  tunnel: ["tunnelPortal", "wallLamp", "barrier", "trafficSignal", "routeSign"],
  bridge: ["overpassPier", "bridgeRail", "billboard", "lamp", "routeSign"],
  pier: ["marinaDock", "dockLight", "lighthouse", "bridgeRail", "routeSign"],
  checkpoint: ["grandstand", "checkpointArch", "crowd", "billboard", "routeSign"],
  beacon: ["runwayLight", "controlTower", "hangar", "serviceCart", "billboard"],
  hangar: ["hangar", "serviceTruck", "controlTower", "runwayLight", "routeSign"],
  plane: ["serviceCart", "controlTower", "hangar", "runwayLight", "billboard"],
  policeHQ: ["policeHQ", "parkedCar", "streetLight", "crowd", "lowBarrier"],
  roadblock: ["roadblock", "parkedCar", "lowBarrier", "streetLight", "crowd"],
  lightbarSign: ["lightbarSign", "parkedCar", "streetLight", "policeHQ", "lowBarrier"],
  barn: ["barn", "fence", "hayBales", "tractorParked", "windmill"],
  tractor: ["tractorParked", "hayBales", "fence", "barn", "routeSign"],
  field: ["fence", "barn", "windmill", "hayBales", "serviceStation"],
  weighStation: ["weighStation", "dieselSign", "truckStop", "serviceTruck", "overheadSign"],
  crane: ["container", "craneBase", "warehouse", "serviceTruck", "billboard"],
  trailer: ["truckStop", "serviceTruck", "container", "billboard", "trafficSignal"],
  dirtBerm: ["dirtBerm", "grandstand", "scoreboard", "stadiumRamp", "crushCarStack"],
  crushCarStack: ["crushCarStack", "stadiumRamp", "grandstand", "dirtBerm", "scoreboard"],
  stadiumRamp: ["stadiumRamp", "crushCarStack", "grandstand", "dirtBerm", "scoreboard"],
  bunker: ["bunker", "watchTower", "rangeTarget", "blastWall", "checkpointArch"],
  rangeTarget: ["rangeTarget", "blastWall", "bunker", "watchTower", "routeSign"],
  blastWall: ["blastWall", "bunker", "rangeTarget", "watchTower", "checkpointArch"],
  village: ["villageHouse", "cafe", "stoneWall", "tramPost", "crowd"],
  mountain: ["pine", "chalet", "stoneWall", "billboard", "routeSign"],
  pine: ["pine", "lodge", "snowFence", "spectators", "routeSign"],
  palm: ["palm", "beachHouse", "serviceStation", "spectators", "routeSign"],
  tower: ["storefront", "billboard", "trafficSignal", "pedestrianBridge", "crowd"],
  neon: ["neonShop", "billboard", "trafficSignal", "pedestrianBridge", "crowd"],
  cliff: ["rockFace", "curveBoard", "guardPost", "tourVan", "routeSign"],
  dune: ["dune", "supportTent", "rallyMarker", "desertScrub", "serviceTruck"],
  canopy: ["canopyTree", "marketStand", "woodRail", "waterfallSign", "crowd"]
};
if (stage && stageProps[stage.prop]) return stageProps[stage.prop];
const props = {
  coast: ["palm", "beachHouse", "lighthouse", "billboard", "spectators", "routeSign"],
  city: ["storefront", "trafficSignal", "billboard", "pedestrianBridge", "parkedCar", "crowd"],
  canyon: ["rockFace", "curveBoard", "tourVan", "supportTent", "routeSign", "guardPost"],
  alpine: ["pine", "chalet", "stoneWall", "skiPost", "spectators", "routeSign"],
  harbor: ["container", "craneBase", "warehouse", "marinaDock", "serviceTruck", "dockLight"],
  snow: ["pine", "lodge", "snowFence", "skiPost", "spectators", "routeSign"],
  airfield: ["runwayLight", "hangar", "controlTower", "serviceCart", "billboard", "barrier"],
  pursuit: ["policeHQ", "parkedCar", "streetLight", "crowd", "lowBarrier", "trafficSignal"],
  freight: ["truckStop", "container", "railSignal", "serviceTruck", "billboard", "trafficSignal"],
  interstate: ["weighStation", "overheadSign", "dieselSign", "truckStop", "serviceTruck", "billboard"],
  farm: ["fence", "barn", "tractorParked", "hayBales", "windmill", "serviceStation"],
  monsterpark: ["grandstand", "dirtBerm", "crushCarStack", "stadiumRamp", "scoreboard", "spectators"],
  military: ["bunker", "watchTower", "rangeTarget", "blastWall", "checkpointArch", "serviceTruck"],
  skybase: ["controlTower", "runwayLight", "hangar", "ran
```

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```
geTarget", "checkpointArch", "serviceCart"],
  tokyo: ["neonShop", "billboard", "trafficSignal", "pedestrianBridge", "parkedCar", "crowd"],
  desert: ["dune", "supportTent", "rallyMarker", "desertScrub", "rockFace", "serviceTruck"],
  rainforest: ["canopyTree", "marketStand", "woodRail", "waterfallSign", "crowd", "routeSign"],
  europe: ["stoneWall", "cafe", "tramPost", "villageHouse", "billboard", "spectators"]
};
return props[place] || props.city;
}

function isSignLikeWorldProp(kind = "") {
  return /sign|billboard|marker|scoreboard|roadblock|lightbar|trafficSignal|overhead|diesel|weigh|checkpoint/i.test(kind);
}

function realWorldGroundFallback(place = "city", index = 0) {
  const sets = {
    coast: ["palm", "beachHouse", "seawall", "spectators", "parkedCar"],
    city: ["storefront", "streetLight", "parkedCar", "crowd", "busStop"],
    canyon: ["rockFace", "desertScrub", "guardPost", "tourVan", "fence"],
    alpine: ["pine", "stoneWall", "chalet", "snowPost", "spectators"],
    harbor: ["warehouse", "container", "dockLight", "serviceTruck", "marinaDock"],
    snow: ["pine", "snowFence", "lodge", "skiPost", "spectators"],
    airfield: ["hangar", "runwayLight", "serviceCart", "controlTower", "barrier"],
    pursuit: ["policeHQ", "parkedCar", "streetLight", "crowd", "lowBarrier"],
    freight: ["truckStop", "container", "serviceTruck", "fence", "streetLight"],
    interstate: ["truckStop", "serviceTruck", "streetLight", "fence", "parkedCar"],
    farm: ["fence", "barn", "tractorParked", "hayBales", "windmill"],
    monsterpark: ["grandstand", "dirtBerm", "crushCarStack", "stadiumRamp", "spectators"],
    military: ["bunker", "watchTower", "blastWall", "serviceTruck", "barrier"],
    skybase: ["controlTower", "runwayLight", "hangar", "serviceCart", "barrier"],
    tokyo: ["neonShop", "streetLight", "parkedCar", "crowd", "pedestrianBridge"],
    desert: ["dune", "desertScrub", "supportTent", "rockFace", "serviceTruck"],
    rainforest: ["canopyTree", "marketStand", "woodRail", "crowd", "parkedCar"],
    europe: ["stoneWall", "cafe", "tramPost", "villageHouse", "spectators"]
  };
  const set = sets[place] || sets.city;
  return set[Math.abs(index) % set.length];
}

function worldSceneryProp(kind, place = "city", index = 0, phoneMode = phoneGraphicsActive()) {
  if (!isSignLikeWorldProp(kind)) return kind;
  if (phoneMode) return realWorldGroundFallback(place, index);
  return Math.abs(index) % 3 === 0 ? kind : realWorldGroundFallback(place, index);
}

function drawRouteAnchoredDriveByWorldPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed, phoneMode) {
  const length = Math.max(1, raceLength());
  const spacing = phoneMode ? 430 : 340;
  const lookBehind = phoneMode ? 160 : 220;
  const lookAhead = phoneMode ? 1850 : 2400;
  const first = Math.floor((raceState.distance - lookBehind) / spacing) - 1;
  const last = Math.ceil((raceState.distance + lookAhead) / spacing) + 1;
  const drawList = [];
  for (let index = first; index <= last; index += 1) {
    const baseDistance = index * spacing + seededUnit(seed, index + 11) * spacing * 0.44;
    if (baseDistance < 0 || baseDistance > length + lookAhead * 0.25) continue;
    const gap = baseDistance - raceState.distance;
    if (gap < -lookBehind || gap > lookAhead) continue;
    const progress = Math.max(0, Math.min(1, baseDistance / length));
    const stage = routeStageInfo(place, progress);
    const props = routeDriveByProps(place, stage);
    const side = seededUnit(seed, index + 22) > 0.5 ? 1 : -1;
    const lane = side * (2.04 + seededUnit(seed, index + 33) * 0.5);
    const projected = roadObjectPos(lane, baseDistance);
    if (!projected || projected.y < horizon - 36 || projected.y > h + 110) continue;
    drawList.push({
      x: projected.x,
      y: projected.y,
      scale: projected.scale,
      depth: projected.depth || Math.max(0, Math.min(1, (projected.y - horizon) / Math.max(1, h - horizon))),
      side,
      kind: worldSceneryProp(props[Math.abs(index) % props.length], place, index, phoneMode),
      stage,
      seed: seed + index * 131,
      distance: baseDistance
    });
  }
  if (!phoneMode && index % 5 === 0) {
```

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```
const otherSide = -side;
const companionLane = otherSide * (1.94 + seededUnit(seed, index + 44) * 0.42);
const companionDistance = baseDistance + spacing * (0.22 + seededUnit(seed, index + 55) * 0.18);
const companionProgress = Math.max(0, Math.min(1, companionDistance / length));
const companionStage = routeStageInfo(place, companionProgress);
const companionProps = routeDriveByProps(place, companionStage);
const companion = roadObjectPos(companionLane, companionDistance);
if (companion && companion.y >= horizon - 36 && companion.y <= h + 110) {
  drawList.push({
    x: companion.x,
    y: companion.y,
    scale: companion.scale,
    depth: companion.depth || Math.max(0, Math.min(1, (companion.y - horizon) / Math.max(1, h - horizon))),
    side: otherSide,
    kind: worldSceneryProp(companionProps[Math.abs(index) + 2] % companionProps.length], place, index + 2, phoneMode),
    stage: companionStage,
    seed: seed + index * 151,
    distance: companionDistance
  });
}
}
}
drawList.sort((a, b) => a.y - b.y);
ctx.save();
clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 10 : 14);
const laneBands = { "-1": -9999, "1": -9999 };
let recentSignY = -9999;
drawList.forEach((item) => {
  const sideKey = String(item.side < 0 ? -1 : 1);
  const minGap = phoneMode ? (isSignLikeWorldProp(item.kind) ? 170 : 104) : (isSignLikeWorldProp(item.kind) ? 118 : 70);
  if (item.y - laneBands[sideKey] < minGap) return;
  if (isSignLikeWorldProp(item.kind) && item.y - recentSignY < (phoneMode ? 260 : 170)) return;
  laneBands[sideKey] = item.y;
  if (isSignLikeWorldProp(item.kind)) recentSignY = item.y;
  const near = Math.max(0, Math.min(1, item.depth || 0));
  const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, item.y, item.side, item.x, phoneMode, phoneMode ? 18 : 28);
  const scale = Math.max(0.32, Math.min(phoneMode ? 1.66 : 1.86, item.scale * (phoneMode ? 0.96 : 1.02) * (0.86 + near * 0.54)));
  if (near > 0.42 && seededUnit(item.seed, 8) > 0.4) {
    ctx.save();
    ctx.globalAlpha = phoneMode ? 0.46 + near * 0.22 : 0.36 + near * 0.2;
    drawGroundLockedDriveByShadow(x, item.y, scale, item.side, theme);
    ctx.restore();
  }
  ctx.globalAlpha = Math.min(1, (phoneMode ? 0.88 : 0.84) + near * 0.18);
  drawGroundLockedPlaceProp(item.kind, x, item.y, scale, item.side, place, Object.assign({}, route, { scene: item.stage.label ||
route.scene })), design, theme, item.seed);
});
ctx.restore();
ctx.globalAlpha = 1;
}

function drawGroundLockedDriveByShadow(x, y, scale, side, theme) {
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.fillStyle = "rgba(0,0,0,0.68)";
  ctx.beginPath();
  ctx.ellipse(side * -4, 10, 92, 15, -0.04 * side, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = `${theme[1] || "#46d9ff"}33`;
  roundRect(-84, 0, 168, 7, 4);
  ctx.fill();
  ctx.restore();
}

function drawGroundLockedSideDetailPass(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion,
phoneMode) {
  const props = groundLockedPlaceProps(place, stage);
  const count = phoneMode ? 7 : 14;
  const span = h - horizon + 420;
  ctx.save();
  clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 6 : 10);
  const lastYBySide = { "-1": -9999, "1": -9999 };
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * (phoneMode ? 190 : 158) + motion * (0.62 + seededUnit(seed, i) * 0.16) + seed * 0.003) % span) - 96);
    if (y < horizon + 4 || y > h + 70) continue;
    const t = Math.max(0.06, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
    const side = i % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const shoulder = half * (1.06 + seededUnit(seed, i + 40) * 0.2) + w * (0.018 + seededUnit(seed, i + 60) * 0.035);
    const projectedX = perspectiveRoadCenter(w, t) + side * shoulder;
    const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, y, side, projectedX, phoneMode, phoneMode ? 10 : 18);
    const scale = (0.38 + t * (phoneMode ? 1
```

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```
.02 : 1.18)) * (0.9 + seededUnit(seed, i + 80) * 0.24);
const kind = worldSceneryProp(props[i % props.length], place, i, phoneMode);
const sideKey = String(side);
const rowGap = phoneMode ? (isSignLikeWorldProp(kind) ? 155 : 92) : (isSignLikeWorldProp(kind) ? 108 : 64);
if (y - lastYBySide[sideKey] < rowGap) continue;
lastYBySide[sideKey] = y;
ctx.globalAlpha = Math.min(1, (phoneMode ? 0.7 : 0.74) + t * 0.25);
drawGroundLockedPlaceProp(kind, x, y, scale, side, place, route, design, theme, seed + i * 71);
}
ctx.restore();
ctx.globalAlpha = 1;
}

function drawGroundLockedNearDetailPass(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion,
phoneMode) {
  const rows = phoneMode ? 6 : 8;
  ctx.save();
  clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 2 : 4);
  for (let i = 0; i < rows; i += 1) {
    const t = 0.54 + i / Math.max(1, rows - 1) * 0.48;
    const y = horizon + (h - horizon) * t + ((motion * 0.45 + i * 29) % 52) - 26;
    if (y < horizon || y > h + 48) continue;
    const side = i % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const projectedX = perspectiveRoadCenter(w, t) + side * (half * 0.96 + w * 0.02);
    const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, y, side, projectedX, phoneMode, phoneMode ? 4 : 8);
    const scale = 0.42 + t * (phoneMode ? 0.78 : 0.92);
    ctx.globalAlpha = phoneMode ? 0.72 : 0.64;
    drawGroundLockedPlaceProp(i % 3 === 0 ? "curbBlock" : i % 3 === 1 ? "reflectorPost" : "lowBarrier", x, y, scale, side, place,
route, design, theme, seed + i * 97);
  }
  ctx.restore();
  ctx.globalAlpha = 1;
}

function drawGroundLockedPlaceProp(kind, x, y, scale, side, place, route, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const label = route && route.scene ? route.scene : "Route";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.fillStyle = "rgba(0,0,0,0.6)";
  ctx.beginPath();
  ctx.ellipse(0, 5, 58, 10, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.42)";
  roundRect(-46, -2, 92, 5, 3);
  ctx.fill();

  if (kind === "streetLight" || kind === "lamp" || kind === "dockLight" || kind === "runwayLight" || kind === "wallLamp") {
    ctx.strokeStyle = "rgba(214,226,221,0.84)";
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(0, 0);
    ctx.lineTo(0, -74);
    ctx.stroke();
    ctx.fillStyle = light;
    ctx.beginPath();
    ctx.arc(0, -82, 9, 0, Math.PI * 2);
    ctx.fill();
    ctx.globalCompositeOperation = "screen";
    ctx.fillStyle = `${light}44`;
    ctx.beginPath();
    ctx.ellipse(0, -40, 22, 52, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.globalCompositeOperation = "source-over";
  } else if (kind === "palm" || kind === "pine" || kind === "canopyTree" || kind === "desertScrub") {
    ctx.fillStyle = kind === "desertScrub" ? "rgba(92,118,48,0.86)" : kind === "palm" ? "rgba(39,124,78,0.9)" :
"rgba(24,88,56,0.92)";
    if (kind === "desertScrub") {
      for (let i = -2; i <= 2; i += 1) {
        ctx.beginPath();
        ctx.ellipse(i * 11, -8 - Math.abs(i) * 2, 15, 22, i * 0.24, 0, Math.PI * 2);
        ctx.fill();
      }
    } else {
      ctx.fillStyle = kind === "palm" ? "rgba(132,88,48,0.9)" : "rgba(72,54,34,0.9)";
      roundRect(-5, -72, 10, 74, 4);
      ctx.fill();
      ctx.fillStyle = kind === "palm" ? "rgba(44,146,82,0.94)" : "rgba(24,94,58,0.94)";
      for (let i = 0; i < 6; i += 1) {
        ctx.save();
        ctx.translate(0, -76);
        ctx.rotate(i * Math.PI / 3 + (kind === "palm" ? 0.18 : 0));
        ctx.beginPath();
        ctx.ellipse(24, 0, kind === "palm" ? 34 : 18, kind === "palm" ? 8 : 26, 0, 0, Math.PI * 2);
        ctx.fill();
        ctx.restore();
      }
    }
  } else if (kind === "storefront" || kind === "neonShop" || kind === "cafe" || kind === "villageHouse" || kind === "beachHouse" ||
kind === "chalet" || kind === "lodge" || kind === "barn" || kind === "warehouse" || kind === "hangar") {
    const buildingColor = kind === "barn" ? "rgba(154,44,36,0.92)" : kind === "neonShop"
```


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```
? "rgba(18,12,34,0.94)" : kind === "warehouse" || kind === "hangar" ? "rgba(36,46,46,0.92)" : "rgba(42,54,52,0.92)";
  ctx.fillStyle = buildingColor;
  roundRect(-58, -72, 116, 72, 6);
  ctx.fill();
  ctx.fillStyle = kind === "barn" || kind === "chalet" || kind === "lodge" || kind === "villageHouse" ? light : accent;
  ctx.beginPath();
  ctx.moveTo(-66, -72);
  ctx.lineTo(0, -112);
  ctx.lineTo(66, -72);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.62)";
  for (let i = -1; i <= 1; i += 1) {
    roundRect(i * 32 - 8, -52, 16, 18, 3);
    ctx.fill();
  }
  ctx.fillStyle = "rgba(5,8,7,0.86)";
  roundRect(-10, -30, 20, 30, 3);
  ctx.fill();
  if (kind === "neonShop" || kind === "storefront" || kind === "cafe") {
    ctx.fillStyle = `${accent}cc`;
    roundRect(-44, -84, 88, 16, 4);
    ctx.fill();
  }
} else if (kind === "parkedCar" || kind === "tourVan" || kind === "serviceTruck" || kind === "serviceCart" || kind ===
"tractorParked") {
  ctx.fillStyle = kind === "tractorParked" ? "#bbf24a" : kind === "serviceTruck" ? light : accent;
  roundRect(-48, -28, 96, 28, 7);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.78)";
  roundRect(-26, -42, 44, 20, 5);
  ctx.fill();
  ctx.fillStyle = "#020303";
  ctx.beginPath();
  ctx.arc(-28, 0, kind === "tractorParked" ? 13 : 9, 0, Math.PI * 2);
  ctx.arc(28, 0, kind === "tractorParked" ? 16 : 9, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.72)";
  ctx.fillRect(-44, -18, 10, 5);
  ctx.fillRect(34, -18, 10, 5);
} else if (kind === "crowd" || kind === "spectators") {
  for (let i = -4; i <= 4; i += 1) {
    const color = i % 3 === 0 ? accent : i % 3 === 1 ? light : "#f4fbf8";
    ctx.fillStyle = color;
    ctx.beginPath();
    ctx.arc(i * 12, -38 - (i % 2) * 4, 5, 0, Math.PI * 2);
    ctx.fill();
    roundRect(i * 12 - 4, -34, 8, 28, 3);
    ctx.fill();
  }
  ctx.strokeStyle = "rgba(244,251,248,0.58)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-66, -8);
  ctx.lineTo(66, -8);
  ctx.stroke();
} else if (kind === "billboard") {
  ctx.strokeStyle = "rgba(214,226,221,0.82)";
  ctx.lineWidth = 5;
  ctx.beginPath();
  ctx.moveTo(-48, 0);
  ctx.lineTo(-48, -84);
  ctx.moveTo(48, 0);
  ctx.lineTo(48, -84);
  ctx.stroke();
  ctx.fillStyle = "rgba(5,8,7,0.94)";
  roundRect(-96, -132, 192, 60, 7);
  ctx.fill();
  ctx.strokeStyle = accent;
  ctx.lineWidth = 3;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 16px system-ui";
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(label.toUpperCase(), 0, -108, 162);
  ctx.fillStyle = light;
  ctx.font = "900 10px system-ui";
  ctx.fillText("NEXT SECTOR", 0, -87, 142);
  ctx.fillStyle = `${accent}66`;
  ctx.fillRect(-84, -76, 168, 5);
} else if (kind === "roadblock" || kind === "lightbarSign" || kind === "policeHQ") {
  if (kind === "policeHQ") {
    ctx.fillStyle = "rgba(14,24,30,0.96)";
    roundRect(-82, -74, 164, 74, 6);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.7)";
    for (let i = -2; i <= 2; i += 1) {
      roundRect(i * 28 - 8, -54, 16, 18, 3);
      ctx.fill();
    }
    ctx.fillStyle = `${accent}dd`;
    roundRect(-58, -92, 116, 18, 4);
    ctx.fill();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = "900 11px system-ui";
    ctx.textAlign = "center";
    ctx.textBaseline = "middle";
    ctx.fillText("POLICE HQ", 0, -83, 96);
  } else if (kind === "roadblock") {
    ctx.strokeStyle = "rgba(214,226,221,0.82)";
    ctx.lineWidth = 5;
    ctx.beginPath();
    ctx.moveTo(-72, 0);
    ctx.lineTo(-72, -34);
    ctx.moveTo(72, 0);
    ctx.lineTo(72, -34);
    ctx.stroke();
    ctx.fillStyle = "rgba(5,8,7,0.88)";
    roundRect(-94, -42, 188, 18, 4);
    ctx.fill();
    for (let i = -3; i <= 3; i += 1) {
      ctx.fillStyle = i % 2 ? "#f4fbf8" : "#ff3348";
```

```
    ctx.fillRect(i * 28 - 10, -39, 20, 12);  
  }  
  ctx.fillStyle = "rgba(255,209,102,0.9)";  
  for (let i = -2; i <= 2; i += 1) {  
    ctx.beginPath();  
    ctx.mo
```

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```
veTo(i * 42 - 9, -2);
    ctx.lineTo(i * 42 + 9, -2);
    ctx.lineTo(i * 42 + 5, -24);
    ctx.lineTo(i * 42 - 5, -24);
    ctx.closePath();
    ctx.fill();
  }
} else {
  ctx.strokeStyle = "rgba(214,226,221,0.84)";
  ctx.lineWidth = 5;
  ctx.beginPath();
  ctx.moveTo(-54, 0);
  ctx.lineTo(-54, -44);
  ctx.moveTo(54, 0);
  ctx.lineTo(54, -44);
  ctx.stroke();
  ctx.fillStyle = "rgba(5,8,7,0.94)";
  roundRect(-72, -64, 144, 28, 5);
  ctx.fill();
  ctx.strokeStyle = kind === "lightbarSign" ? "#46d9ff" : "#ff3348";
  ctx.lineWidth = 3;
  ctx.stroke();
  ctx.fillStyle = "#ff3348";
  ctx.fillRect(-58, -54, 48, 7);
  ctx.fillStyle = "#46d9ff";
  ctx.fillRect(10, -54, 48, 7);
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 9px system-ui";
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText("INTERCEPTOR", 0, -43, 112);
}
} else if (kind === "trafficSignal") {
  ctx.strokeStyle = "rgba(214,226,221,0.84)";
  ctx.lineWidth = 5;
  ctx.beginPath();
  ctx.moveTo(0, 0);
  ctx.lineTo(0, -92);
  ctx.lineTo(52, -92);
  ctx.stroke();
  ctx.fillStyle = "rgba(5,8,7,0.94)";
  roundRect(48, -116, 28, 64, 8);
  ctx.fill();
  ["#ff3348", "#fffd166", "#36d98a"].forEach((color, index) => {
    ctx.fillStyle = color;
    ctx.beginPath();
    ctx.arc(62, -102 + index * 19, 6, 0, Math.PI * 2);
    ctx.fill();
  });
  ctx.fillStyle = `${accent}66`;
  roundRect(-62, -70, 74, 18, 4);
  ctx.fill();
} else if (kind === "pedestrianBridge" || kind === "overpassPier" || kind === "tunnelPortal" || kind === "checkpointArch") {
  ctx.strokeStyle = kind === "checkpointArch" ? light : "rgba(214,226,221,0.76)";
  ctx.lineWidth = kind === "tunnelPortal" ? 9 : 7;
  ctx.beginPath();
  ctx.moveTo(-94, 0);
  ctx.lineTo(-94, -104);
  ctx.moveTo(94, 0);
  ctx.lineTo(94, -104);
  ctx.moveTo(-108, -104);
  ctx.lineTo(108, -104);
  ctx.stroke();
  ctx.fillStyle = kind === "tunnelPortal" ? "rgba(5,8,7,0.9)" : "rgba(22,32,32,0.86)";
  if (kind === "tunnelPortal") {
    ctx.beginPath();
    ctx.arc(0, -4, 88, Math.PI, 0);
    ctx.lineTo(88, 0);
    ctx.lineTo(-88, 0);
    ctx.closePath();
    ctx.fill();
  } else {
    roundRect(-116, -126, 232, 34, 6);
    ctx.fill();
  }
  ctx.fillStyle = kind === "checkpointArch" ? accent : light;
  ctx.font = "900 14px system-ui";
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(kind === "checkpointArch" ? "CHECKPOINT" : label.toUpperCase(), 0, -109, 170);
} else if (kind === "serviceStation") {
  ctx.fillStyle = "rgba(28,38,36,0.94)";
  roundRect(-86, -54, 172, 54, 7);
  ctx.fill();
  ctx.fillStyle = light;
  roundRect(-98, -86, 196, 24, 5);
  ctx.fill();
  ctx.fillStyle = `${accent}dd`;
  ctx.fillRect(-84, -78, 168, 6);
  ctx.strokeStyle = "rgba(214,226,221,0.72)";
  ctx.lineWidth = 5;
  ctx.beginPath();
  ctx.moveTo(-62, 0);
  ctx.lineTo(-62, -62);
  ctx.moveTo(62, 0);
  ctx.lineTo(62, -62);
  ctx.stroke();
  ctx.fillStyle = "rgba(5,8,7,0.86)";
  roundRect(-32, -42, 24, 42, 4);
  ctx.fill();
  roundRect(12, -42, 24, 42, 4);
  ctx.fill();
} else if (kind === "lighthouse" || kind === "controlTower") {
  ctx.fillStyle = kind === "lighthouse" ? "rgba(238,242,232,0.92)" : "rgba(198,215,212,0.9)";
  roundRect(-24, -112, 48, 112, 8);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-38, -136, 76, 28, 6);
  ctx.fill();
  ctx.fillStyle = accent;
  ctx.beginPath();
  ctx.arc(0, -122, 9, 0, Math.PI * 2);
  ctx.fill();
}
```

```
ctx.globalCompositeOperation = "screen";
ctx.fillStyle = `${accent}33`;
ctx.beginPath();
ctx.ellipse(0, -122, 92, 18, 0, 0, Math.PI * 2);
ctx.fill();
ctx.globalCompositeOperation = "source-over";
} else if (kind === "marinaDock") {
  ctx.fillStyle = "rgba(70,217,255,0.18)";
  ctx.beginPath();
  ctx.ellipse(0, 5, 108, 20, 0, 0, Math.PI * 2);
  ctx.fill();
  if (phoneGraphicsActive()) {
    ctx.fillStyle = "rgba(132,92,48,0.72)";
    roundRect(-94, -26, 188,
```

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```
24, 7);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-82, -18);
    ctx.lineTo(84, -24);
    ctx.moveTo(-82, -8);
    ctx.lineTo(84, -14);
    ctx.stroke();
  } else {
    ctx.strokeStyle = "rgba(214,180,112,0.84)";
    ctx.lineWidth = 6;
    ctx.beginPath();
    ctx.moveTo(-96, -12);
    ctx.lineTo(96, -20);
    for (let i = -3; i <= 3; i += 1) {
      ctx.moveTo(i * 30, -18);
      ctx.lineTo(i * 30, 18);
    }
    ctx.stroke();
  }
  ctx.fillStyle = `${light}aa`;
  ctx.beginPath();
  ctx.arc(72, -46, 8, 0, Math.PI * 2);
  ctx.fill();
} else if (kind === "weighStation" || kind === "dieselSign" || kind === "overheadSign") {
  ctx.strokeStyle = "rgba(214,226,221,0.82)";
  ctx.lineWidth = 6;
  ctx.beginPath();
  ctx.moveTo(-76, 0);
  ctx.lineTo(-76, -96);
  ctx.moveTo(76, 0);
  ctx.lineTo(76, -96);
  if (kind === "overheadSign") {
    ctx.moveTo(-118, -96);
    ctx.lineTo(118, -96);
  }
  ctx.stroke();
  ctx.fillStyle = "rgba(18,24,20,0.95)";
  roundRect(-112, -132, 224, 48, 7);
  ctx.fill();
  ctx.strokeStyle = kind === "dieselSign" ? light : accent;
  ctx.lineWidth = 3;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 14px system-ui";
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(kind === "weighStation" ? "WEIGH STATION" : kind === "dieselSign" ? "DIESEL STOP" : label.toUpperCase(), 0, -111,
188);
  ctx.fillStyle = `${light}99`;
  ctx.fillRect(-92, -88, 184, 6);
} else if (kind === "grandstand" || kind === "scoreboard") {
  ctx.fillStyle = "rgba(18,24,22,0.92)";
  roundRect(-108, -74, 216, 74, 6);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.22)";
  for (let row = 0; row < 3; row += 1) {
    ctx.fillRect(-96, -62 + row * 18, 192, 5);
  }
  for (let i = -6; i <= 6; i += 1) {
    ctx.fillStyle = i % 2 ? accent : light;
    ctx.beginPath();
    ctx.arc(i * 14, -50 + (i % 3) * 12, 4, 0, Math.PI * 2);
    ctx.fill();
  }
  if (kind === "scoreboard") {
    ctx.fillStyle = "rgba(5,8,7,0.96)";
    roundRect(-82, -124, 164, 42, 5);
    ctx.fill();
    ctx.strokeStyle = accent;
    ctx.lineWidth = 2;
    ctx.stroke();
    ctx.fillStyle = light;
    ctx.font = "900 13px system-ui";
    ctx.textAlign = "center";
    ctx.textBaseline = "middle";
    ctx.fillText("ARENA SPLIT", 0, -103, 132);
  }
} else if (kind === "stadiumRamp" || kind === "dirtBerm" || kind === "crushCarStack") {
  if (kind === "crushCarStack") {
    for (let i = 0; i < 4; i += 1) {
      ctx.fillStyle = i % 2 ? accent : light;
      roundRect(-82 + i * 42, -20 - (i % 2) * 14, 52, 20, 4);
      ctx.fill();
      ctx.fillStyle = "rgba(5,8,7,0.8)";
      ctx.fillRect(-72 + i * 42, -15 - (i % 2) * 14, 22, 6);
    }
  } else {
    ctx.fillStyle = kind === "stadiumRamp" ? "rgba(196,106,42,0.92)" : "rgba(128,78,38,0.92)";
    ctx.beginPath();
    ctx.moveTo(-104, 0);
    ctx.lineTo(kind === "stadiumRamp" ? 96 : 54, kind === "stadiumRamp" ? -58 : -34);
    ctx.lineTo(112, 0);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(255,209,102,0.34)";
    roundRect(-72, -14, 142, 7, 4);
    ctx.fill();
  }
} else if (kind === "bunker" || kind === "watchTower" || kind === "rangeTarget" || kind === "blastWall") {
  if (kind === "watchTower") {
    ctx.strokeStyle = "rgba(214,226,221,0.74)";
    ctx.lineWidth = 5;
    ctx.beginPath();
    ctx.moveTo(-34, 0);
    ctx.lineTo(-18, -94);
    ctx.moveTo(34, 0);
    ctx.lineTo(18, -94);
    ctx.moveTo(-42, -42);
    ctx.lineTo(42, -42);
    ctx.stroke();
  }
}
```

```
ctx.fillStyle = "rgba(40,52,42,0.94)";
roundRect(-46, -124, 92, 36, 5);
ctx.fill();
} else if (kind === "rangeTarget") {
ctx.strokeStyle = "rgba(214,226,221,0.78)";
ctx.lineWidth = 4;
ctx.beginPath();
ctx.moveTo(0, 0);
ctx.lineTo(0, -92);
ctx.stroke();
ctx.fillStyle = "rgba(244,251,248,0.88)";
ctx.beginPath();
ctx.arc(0, -106, 38, 0, Math.PI * 2);
ctx.f
```

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```
ill();
  ctx.fillStyle = "#ff5b6b";
  ctx.beginPath();
  ctx.arc(0, -100, 24, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "#f4fbf8";
  ctx.beginPath();
  ctx.arc(0, -100, 10, 0, Math.PI * 2);
  ctx.fill();
} else {
  ctx.fillStyle = kind === "blastWall" ? "rgba(72,82,70,0.94)" : "rgba(48,60,48,0.94)";
  roundRect(-92, -50, 184, 50, 4);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.66)";
  for (let i = -2; i <= 2; i += 1) {
    roundRect(i * 34 - 10, -38, 20, 14, 2);
    ctx.fill();
  }
  ctx.strokeStyle = `${light}99`;
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-82, -10);
  ctx.lineTo(82, -18);
  ctx.stroke();
}
} else if (kind === "routeSign" || kind === "curveBoard" || kind === "towerSign" || kind === "expressSign" || kind ===
"waterfallSign" || kind === "busStop" || kind === "rallyMarker") {
  ctx.strokeStyle = "rgba(214,226,221,0.8)";
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-28, 0);
  ctx.lineTo(-28, -54);
  ctx.moveTo(28, 0);
  ctx.lineTo(28, -54);
  ctx.stroke();
  ctx.fillStyle = "rgba(5,8,7,0.9)";
  roundRect(-72, -92, 144, 44, 6);
  ctx.fill();
  ctx.strokeStyle = accent;
  ctx.lineWidth = 2;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 12px system-ui";
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(label.toUpperCase(), 0, -72, 126);
  ctx.fillStyle = light;
  ctx.fillRect(-54, -55, 108, 5);
} else if (kind === "rockFace" || kind === "dune" || kind === "hayBales" || kind === "container") {
  const fill = kind === "container" ? "rgba(36,69,82,0.92)" : kind === "hayBales" || kind === "dune" ? "rgba(205,148,62,0.88)" :
"rgba(144,79,49,0.88)";
  ctx.fillStyle = fill;
  if (kind === "container") {
    for (let i = 0; i < 3; i += 1) {
      roundRect(-62 + i * 42, -34 - i % 2 * 10, 38, 34, 4);
      ctx.fill();
    }
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.lineWidth = 2;
    for (let i = -2; i <= 2; i += 1) {
      ctx.beginPath();
      ctx.moveTo(i * 24, -34);
      ctx.lineTo(i * 24, 0);
      ctx.stroke();
    }
  } else {
    ctx.beginPath();
    ctx.moveTo(-72, 0);
    ctx.lineTo(-44, -52);
    ctx.lineTo(-8, -32);
    ctx.lineTo(24, -76);
    ctx.lineTo(72, 0);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.18)";
    roundRect(-40, -26, 72, 7, 4);
    ctx.fill();
  }
} else if (kind === "roadsideHedge" || kind === "coastRail" || kind === "fence" || kind === "snowFence" || kind === "woodRail" ||
kind === "stoneWall" || kind === "seawall" || kind === "bridgeRail" || kind === "guardPost" || kind === "lowBarrier" || kind ===
"curbBlock" || kind === "reflectorPost" || kind === "barrier") {
  ctx.strokeStyle = kind === "stoneWall" || kind === "seawall" ? "rgba(192,202,196,0.86)" : kind === "snowFence" ?
"rgba(244,251,248,0.88)" : `${light}cc`;
  ctx.lineWidth = kind === "reflectorPost" ? 5 : 6;
  if (kind === "roadsideHedge") {
    const hedge = place === "desert" || place === "canyon" ? "rgba(156,92,42,0.54)" : place === "snow" || place === "alpine" ?
"rgba(222,236,232,0.58)" : "rgba(44,132,74,0.58)";
    ctx.fillStyle = hedge;
    ctx.beginPath();
    ctx.ellipse(0, -12, 96, 24, -0.04, 0, Math.PI * 2);
    ctx.ellipse(-44, -23, 42, 18, 0.16, 0, Math.PI * 2);
    ctx.ellipse(44, -24, 44, 18, -0.12, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-86, -6);
    ctx.lineTo(90, -13);
    ctx.stroke();
  } else if (phoneGraphicsActive() && kind !== "reflectorPost") {
    ctx.fillStyle = kind === "stoneWall" || kind === "seawall" ? "rgba(174,186,180,0.62)" : kind === "snowFence" ?
"rgba(244,251,248,0.52)" : kind === "coastRail" || kind === "bridgeRail" ? "rgba(205,218,214,0.42)" : "rgba(78,96,76,0.42)";
    roundRect(-54, -24, 108, 20, 7);
    ctx.fill();
    ctx.strokeStyle = kind === "coastRail" || kind === "bridgeRail" ? "rgba(244,251,248,0.34)" : `${light}77`;
    ctx.linew
```

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```
idth = 3;
    ctx.beginPath();
    ctx.moveTo(-52, -24);
    ctx.quadraticCurveTo(0, -31, 54, -25);
    ctx.moveTo(-48, -8);
    ctx.quadraticCurveTo(0, -14, 50, -10);
    ctx.stroke();
    ctx.fillStyle = `${accent}55`;
    ctx.beginPath();
    ctx.ellipse(-30, -21, 8, 4, 0, 0, Math.PI * 2);
    ctx.ellipse(24, -23, 9, 4, 0, 0, Math.PI * 2);
    ctx.fill();
  } else {
    ctx.beginPath();
    if (kind === "reflectorPost") {
      ctx.moveTo(0, 0);
      ctx.lineTo(0, -46);
    } else {
      ctx.moveTo(-76, -20);
      ctx.lineTo(76, -28);
      ctx.moveTo(-76, -4);
      ctx.lineTo(76, -12);
      for (let i = -2; i <= 2; i += 1) {
        ctx.moveTo(i * 32, -42);
        ctx.lineTo(i * 32, 4);
      }
    }
    ctx.stroke();
  }
}
if (kind === "reflectorPost") {
  ctx.fillStyle = accent;
  ctx.beginPath();
  ctx.arc(0, -48, 8, 0, Math.PI * 2);
  ctx.fill();
}
} else if (kind === "craneBase" || kind === "windmill" || kind === "tramPost" || kind === "skiPost" || kind === "railSignal" ||
kind === "tunnelFan") {
  ctx.strokeStyle = `${light}cc`;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(0, 0);
  ctx.lineTo(0, -96);
  ctx.moveTo(-34, -72);
  ctx.lineTo(48, -88);
  ctx.stroke();
  ctx.fillStyle = accent;
  ctx.beginPath();
  ctx.arc(0, -98, 9, 0, Math.PI * 2);
  ctx.fill();
} else if (kind === "waterEdge" || kind === "marketStand" || kind === "supportTent" || kind === "truckStop") {
  ctx.fillStyle = kind === "waterEdge" ? "rgba(70,217,255,0.22)" : "rgba(36,46,44,0.9)";
  roundRect(-70, -38, 140, 38, 5);
  ctx.fill();
  ctx.fillStyle = light;
  ctx.beginPath();
  ctx.moveTo(-78, -38);
  ctx.lineTo(0, -72);
  ctx.lineTo(78, -38);
  ctx.closePath();
  ctx.fill();
} else {
  ctx.fillStyle = "rgba(42,54,52,0.86)";
  roundRect(-52, -48, 104, 48, 6);
  ctx.fill();
}
}
ctx.restore();
}

function drawRealWorldDepthInfrastructurePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  if (phoneCleanRoadActive()) return;
  const phoneMode = phoneGraphicsActive();
  const length = raceLength();
  const progress = length ? Math.max(0, Math.min(1, raceState.distance / length)) : 0;
  const stage = routeStageInfo(place, progress);
  const route = routeWorldInfo(place);
  const depthSeed = hashText(`v81-depth:${place}:${stage.label}:${route.scene}`);
  const motion = raceState.roadOffset || 0;
  ctx.save();
  drawDepthParallaxBands(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, depthSeed, motion, phoneMode);
  drawDepthSideInfrastructure(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, depthSeed, motion, phoneMode);
  drawDepthOverRoadInfrastructure(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, depthSeed, motion,
phoneMode);
  drawDepthFarTrafficFlow(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, depthSeed, motion, phoneMode);
  ctx.restore();
}

function depthInfrastructureProfile(place, stage) {
  if (stage.prop === "tunnel") return { band: "tunnelMouth", side: "concretewall", overhead: "tunnelLights", traffic:
"serviceLights" };
  if (stage.prop === "bridge" || stage.prop === "pier") return { band: "waterSpan", side: "bridgeRail", overhead: "cables", traffic:
"farCars" };
  const profiles = {
    coast: { band: "waterCliff", side: "coastRail", overhead: "powerLines", traffic: "farCars" },
    city: { band: "cityBlocks", side: "soundWall", overhead: "gantry", traffic: "farCars" },
    canyon: { band: "rockCuts", side: "guardRail", overhead: "utilityLines", traffic: "tourTraffic" },
    alpine: { band: "mountainLayers", side: "snowFence", overhead: "liftCable", traffic: "shuttle" },
    harbor: { band: "portCranes", side: "dockWall", overhead: "craneBoom", traffic: "yardTraffic" },
    snow: { band: "snowVillage", side: "snowFence", overhead: "liftCable", traffic: "shuttle" },
    airfield: { band: "runwayApron", side: "perimeterFence", overhead: "beaconRow", traffic: "serviceCarts" },
    freight: { band: "railYard", side: "soundWall", overhead: "gantry", traffic: "semiFlow" },
    farm: { band: "fiel
```


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```
dRows", side: "splitFence", overhead: "powerLines", traffic: "farmTraffic" },
  tokyo: { band: "neonTowers", side: "expressWall", overhead: "expressSigns", traffic: "farCars" },
  desert: { band: "duneLayers", side: "rallyFence", overhead: "utilityLines", traffic: "supportFlow" },
  rainforest: { band: "canopyLayers", side: "woodRail", overhead: "vineLines", traffic: "marketFlow" },
  europe: { band: "villageSlope", side: "stoneWall", overhead: "tramWire", traffic: "tourTraffic" }
};
return profiles[place] || profiles.city;
}

function drawDepthParallaxBands(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion, phoneMode) {
  const profile = depthInfrastructureProfile(place, stage);
  const bandTop = Math.max(0, horizon - h * 0.17);
  const bandHeight = h * (phoneMode ? 0.22 : 0.26);
  const offset = motion * 0.006;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.26 : 0.36;
  const skyFade = ctx.createLinearGradient(0, bandTop, 0, horizon + bandHeight);
  skyFade.addColorStop(0, "rgba(0,0,0,0)");
  skyFade.addColorStop(0.7, `${design.accent} || theme[1]10`);
  skyFade.addColorStop(1, "rgba(0,0,0,0.08)");
  ctx.fillStyle = skyFade;
  ctx.fillRect(0, bandTop, w, horizon + bandHeight - bandTop);
  const count = phoneMode ? 10 : 16;
  for (let i = 0; i < count; i += 1) {
    const sideDrift = ((i * w * 0.18 - offset * (10 + i * 3 * 4)) % (w + 180)) - 90;
    const y = horizon + h * (0.025 + seededUnit(seed, i + 4) * 0.12);
    const scale = 0.65 + seededUnit(seed, i + 10) * 0.75;
    drawDepthBandShape(profile.band, sideDrift, y, scale, place, design, theme, seed + i * 29);
  }
  ctx.restore();
}

function drawDepthBandShape(kind, x, y, scale, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const phoneMode = phoneGraphicsActive();
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  if (kind === "cityBlocks" || kind === "neonTowers") {
    for (let i = -2; i <= 2; i += 1) {
      const bw = 26 + seededUnit(seed, i + 1) * 28;
      const bh = 54 + seededUnit(seed, i + 8) * 96;
      ctx.fillStyle = kind === "neonTowers" ? "rgba(18,10,32,0.55)" : "rgba(8,18,20,0.5)";
      roundRect(i * 42 - bw / 2, 22 - bh, bw, bh, 4);
      ctx.fill();
      ctx.fillStyle = i % 2 ? `${accent}55` : `${light}44`;
      for (let row = 0; row < 4; row += 1) ctx.fillRect(i * 42 - bw * 0.26, 32 - bh + row * 18, bw * 0.52, 3);
    }
  } else if (kind === "mountainLayers" || kind === "rockCuts" || kind === "duneLayers" || kind === "villageSlope") {
    const color = kind === "duneLayers" ? "rgba(255,183,74,0.28)" : kind === "rockCuts" ? "rgba(178,92,48,0.28)" : kind === "villageSlope" ? "rgba(120,142,132,0.25)" : "rgba(160,172,164,0.26)";
    ctx.fillStyle = color;
    ctx.beginPath();
    ctx.moveTo(-110, 34);
    ctx.quadraticCurveTo(-58, -66 - seededUnit(seed, 1) * 42, -8, 8);
    ctx.quadraticCurveTo(44, -74 - seededUnit(seed, 2) * 48, 118, 34);
    ctx.closePath();
    ctx.fill();
    if (kind === "villageSlope") {
      ctx.fillStyle = "rgba(244,251,248,0.3)";
      for (let i = -2; i <= 2; i += 1) roundRect(i * 30 - 10, -8 + (i % 2) * 8, 20, 14, 2);
    }
  } else if (kind === "waterCliff" || kind === "waterSpan") {
    ctx.fillStyle = "rgba(70,217,255,0.16)";
    ctx.beginPath();
    ctx.ellipse(0, 18, 120, 20, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.22)";
    ctx.lineWidth = 2;
    for (let i = -2; i <= 2; i += 1) {
      ctx.beginPath();
      ctx.moveTo(-96, i * 8 + 14);
      ctx.quadraticCurveTo(-20, i * 8 + 4, 92, i * 8 + 14);
      ctx.stroke();
    }
  } else if (kind === "portCranes" || kind === "railYard" || kind === "runwayApron") {
    ctx.strokeStyle = `${light}55`;
    ctx.lineWidth = 4;
    for (let i = -2; i <= 2; i += 1) {
      ctx.beginPath();
      ctx.moveTo(i * 44, 26);
      ctx.lineTo(i * 44 + 8, -56);
      ctx.lineTo(i * 44 + 72, -42);
      ctx.stroke();
    }
    ctx.fillStyle = "rgba(5,8,7,0.36)";
    ctx.fillRect(-120, 18, 240, 18);
  } else if (kind === "fieldRows" || kind === "canopyLayers" || kind === "snowVillage") {
    c
```

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```
tx.fillStyle = kind === "canopyLayers" ? "rgba(30,104,56,0.26)" : kind === "snowVillage" ? "rgba(244,251,248,0.2)" :
"rgba(95,126,54,0.24)";
ctx.fillRect(-130, -12, 260, 48);
ctx.strokeStyle = kind === "snowVillage" ? "rgba(244,251,248,0.28)" : "rgba(187,242,74,0.2)";
ctx.lineWidth = 2;
const rows = phoneMode ? 2 : 7;
for (let i = -Math.floor(rows / 2); i <= Math.floor(rows / 2); i += 1) {
  ctx.beginPath();
  ctx.moveTo(-120, i * 9 + 16);
  ctx.lineTo(120, i * 4 + 18);
  ctx.stroke();
}
} else {
  ctx.fillStyle = `${accent}18`;
  ctx.beginPath();
  ctx.ellipse(0, 10, 116, 28, 0, 0, Math.PI * 2);
  ctx.fill();
}
}
ctx.restore();
}

function drawDepthSideInfrastructure(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion,
phoneMode) {
  const profile = depthInfrastructureProfile(place, stage);
  const count = phoneMode ? 8 : 13;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.34 : 0.48;
  for (let i = 0; i < count; i += 1) {
    const y = horizon + ((i * 104 + motion * 0.58 + seed * 0.004) % (h - horizon + 220)) - 74;
    if (y < horizon || y > h + 74) continue;
    const t = Math.max(0.05, Math.min(1.1, (y - horizon) / Math.max(1, h - horizon)));
    const side = i % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half * (0.92 + seededUnit(seed, i + 90) * 0.22));
    const scale = 0.28 + t * (phoneMode ? 0.62 : 0.84);
    drawDepthSidePiece(profile.side, x, y, scale, side, design, theme, seed + i * 37);
  }
  ctx.restore();
}

function drawDepthSidePiece(kind, x, y, scale, side, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const phoneMode = phoneGraphicsActive();
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  if (kind === "soundWall" || kind === "expressWall" || kind === "concretewall" || kind === "dockWall") {
    ctx.fillStyle = kind === "dockWall" ? "rgba(18,49,56,0.48)" : "rgba(16,22,22,0.48)";
    roundRect(-10 * side, -42, 86 * side, 56, 4);
    ctx.fill();
    ctx.strokeStyle = `${accent}44`;
    ctx.lineWidth = 3;
    if (!phoneMode) {
      for (let i = 0; i < 4; i += 1) {
        ctx.beginPath();
        ctx.moveTo((i * 22) * side, -38);
        ctx.lineTo((i * 22) * side, 8);
        ctx.stroke();
      }
    }
  } else if (kind === "snowFence" || kind === "splitFence" || kind === "rallyFence" || kind === "woodRail") {
    ctx.strokeStyle = kind === "snowFence" ? "rgba(244,251,248,0.55)" : `${light}66`;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-72 * side, -18);
    ctx.lineTo(72 * side, -26);
    ctx.moveTo(-72 * side, 3);
    ctx.lineTo(72 * side, -5);
    if (!phoneMode) {
      for (let i = -2; i <= 2; i += 1) {
        ctx.moveTo(i * 28 * side, -34);
        ctx.lineTo(i * 28 * side, 14);
      }
    }
    ctx.stroke();
  } else if (kind === "guardRail" || kind === "bridgeRail" || kind === "coastRail") {
    ctx.strokeStyle = "rgba(214,226,221,0.54)";
    ctx.lineWidth = 5;
    ctx.beginPath();
    ctx.moveTo(-82 * side, -14);
    ctx.lineTo(82 * side, -22);
    ctx.moveTo(-82 * side, 2);
    ctx.lineTo(82 * side, -6);
    ctx.stroke();
  } else if (kind === "perimeterFence" || kind === "stonewall") {
    ctx.strokeStyle = kind === "stonewall" ? "rgba(160,172,164,0.58)" : "rgba(216,226,221,0.38)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-84 * side, -22);
    ctx.lineTo(84 * side, -30);
    ctx.moveTo(-84 * side, 8);
    ctx.lineTo(84 * side, 0);
    if (!phoneMode) {
      for (let i = -3; i <= 3; i += 1) {
        ctx.moveTo(i * 24 * side, -34);
        ctx.lineTo(i * 24 * side, 14);
      }
    }
    ctx.stroke();
  } else {
    ctx.fillStyle = `${accent}28`;
    roundRect(-44, -28, 88, 36, 4);
    ctx.fill();
  }
  ctx.restore();
}

function drawDepthOverRoadInfrastructure(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion,
phoneMode) {
  const profile = depthInfrastructureProfile(place, stage);
  if (phoneMode) return;
  const count = phoneMode ? 3 : 5;
```


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```
ave();
ctx.globalAlpha = phoneMode ? 0.22 : 0.34;
for (let i = 0; i < count; i += 1) {
  const y = horizon + h * (0.035 + i * 0.085) + Math.sin(seed + i + motion * 0.002) * h * 0.012;
  if (y > h * 0.56) continue;
  drawDepthOverRoadPiece(profile.overhead, w, y, i, route, design, theme, seed + i * 43, phoneMode);
}
ctx.restore();
}

function drawDepthOverRoadPiece(kind, w, y, index, route, design, theme, seed, phoneMode) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  if (kind === "gantry" || kind === "expressSigns" || kind === "tunnelLights") {
    ctx.strokeStyle = "rgba(214,226,221,0.44)";
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(w * 0.14, y);
    ctx.lineTo(w * 0.86, y - 6);
    ctx.moveTo(w * 0.2, y);
    ctx.lineTo(w * 0.2, y + 44);
    ctx.moveTo(w * 0.8, y - 6);
    ctx.lineTo(w * 0.8, y + 38);
    ctx.stroke();
    if (!phoneMode) {
      ctx.fillStyle = "rgba(5,8,7,0.62)";
      roundRect(w * 0.39, y - 20, w * 0.22, 22, 4);
      ctx.fill();
      ctx.fillStyle = accent;
      ctx.font = `900 ${Math.max(7, Math.min(11, w * 0.011))}px system-ui`;
      ctx.textAlign = "center";
      ctx.textBaseline = "middle";
      ctx.fillText(route.scene.toUpperCase(), w * 0.5, y - 9, w * 0.18);
    }
  } else if (kind === "powerLines" || kind === "utilityLines" || kind === "tramWire" || kind === "vineLines" || kind === "liftCable" || kind === "cables") {
    ctx.strokeStyle = kind === "vineLines" ? "rgba(54,217,138,0.34)" : "rgba(216,226,221,0.32)";
    ctx.lineWidth = kind === "cables" ? 3 : 2;
    for (let i = 0; i < 3; i += 1) {
      ctx.beginPath();
      ctx.moveTo(-20, y + i * 12);
      ctx.bezierCurveTo(w * 0.22, y - 18 + i * 10, w * 0.62, y + 18 + i * 8, w + 20, y - 4 + i * 12);
      ctx.stroke();
    }
  } else if (kind === "craneBoom" || kind === "beaconRow") {
    ctx.strokeStyle = `${light}55`;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(w * 0.1, y + 22);
    ctx.lineTo(w * 0.9, y - 24);
    ctx.stroke();
    ctx.globalCompositeOperation = "screen";
    ctx.fillStyle = `${accent}66`;
    for (let i = 0; i < 6; i += 1) {
      ctx.beginPath();
      ctx.arc(w * (0.18 + i * 0.12), y - 8 + i % 2 * 9, 3.5, 0, Math.PI * 2);
      ctx.fill();
    }
    ctx.globalCompositeOperation = "source-over";
  } else {
    ctx.strokeStyle = `${accent}30`;
    ctx.lineWidth = 2;
    ctx.beginPath();
    ctx.moveTo(0, y);
    ctx.lineTo(w, y + Math.sin(seed) * 10);
    ctx.stroke();
  }
  ctx.restore();
}

function drawDepthFarTrafficFlow(w, h, horizon, roadTop, roadBottom, place, stage, route, design, theme, seed, motion, phoneMode) {
  const profile = depthInfrastructureProfile(place, stage);
  const count = phoneMode ? 6 : 10;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.24 : 0.36;
  for (let i = 0; i < count; i += 1) {
    const t = 0.12 + seededUnit(seed, i + 170) * 0.32;
    const y = horizon + (h - horizon) * t + Math.sin(motion * 0.004 + i) * 10;
    const side = i % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.13 + seededUnit(seed, i + 180) * 0.16));
    const scale = 0.25 + t * (phoneMode ? 0.34 : 0.48);
    drawDepthFarTrafficShape(profile.traffic, x, y, scale, side, design, theme, seed + i * 59);
  }
  ctx.restore();
}

function drawDepthFarTrafficShape(kind, x, y, scale, side, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const longBody = kind === "semiFlow" || kind === "yardTraffic" || kind === "supportFlow" || kind === "shuttle";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale * side, scale);
  ctx.fillStyle = "rgba(0,0,0,0.25)";
  ctx.beginPath();
  ctx.ellipse(0, 10, longBody ? 54 : 34, 5, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = kind === "serviceCarts" ? light : kind === "farmTraffic" || kind === "marketFlow" ? "#36d98a" : accent;
  roundRect(longBody ? -46 : -30, -14, longBody ? 92 : 60, 24, 4);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.72)";
  ctx.fillRect(longBody ? -32
```

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```
: -18, -9, longBody ? 24 : 16, 8);
ctx.fillRect(longBody ? 8 : 8, -9, longBody ? 24 : 16, 8);
ctx.fillStyle = "#050807";
ctx.beginPath();
ctx.arc(longBody ? -30 : -18, 11, 5, 0, Math.PI * 2);
ctx.arc(longBody ? 30 : 18, 11, 5, 0, Math.PI * 2);
ctx.fill();
ctx.restore();
}

function drawRealWorldRouteAtmospherePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  if (phoneCleanRoadActive()) return;
  const phoneMode = phoneGraphicsActive();
  const motion = raceState.roadOffset || 0;
  ctx.save();
  drawRouteShoulderTerrain(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode);
  drawRouteMicroLandmarks(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode);
  drawRouteWeatherAndAir(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode);
  ctx.restore();
}

function drawRealWorldRouteSectorPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  if (phoneCleanRoadActive()) return;
  const phoneMode = phoneGraphicsActive();
  const length = raceLength();
  const progress = length ? Math.max(0, Math.min(1, raceState.distance / length)) : 0;
  const stage = routeStageInfo(place, progress);
  const route = routeWorldInfo(place);
  const sectorSeed = hashText(`v73-sector:${place}:${stage.label}:${stage.index}`);
  const motion = raceState.roadOffset || 0;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.48 : 0.62;
  drawRouteSectorRoadEdgeDetail(w, h, horizon, roadTop, roadBottom, stage, place, design, theme, sectorSeed, motion, phoneMode);
  drawRouteSectorDistantSet(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, sectorSeed, motion, phoneMode);
  drawRouteSectorRoadsideStory(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, sectorSeed, motion, phoneMode);
  ctx.restore();
}

function drawRealWorldLivingRoutePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  if (phoneCleanRoadActive()) return;
  const phoneMode = phoneGraphicsActive();
  const length = raceLength();
  const progress = length ? Math.max(0, Math.min(1, raceState.distance / length)) : 0;
  const stage = routeStageInfo(place, progress);
  const route = routeWorldInfo(place);
  const livingSeed = hashText(`v74-living:${place}:${stage.label}:${route.scene}`);
  const motion = raceState.roadOffset || 0;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.42 : 0.58;
  drawLivingRouteActivityZones(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, livingSeed, motion, phoneMode);
  drawLivingRoutePeopleClusters(w, h, horizon, roadTop, roadBottom, stage, place, design, theme, livingSeed, motion, phoneMode);
  drawLivingRouteServiceVehicles(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, livingSeed, motion, phoneMode);
  ctx.restore();
}

function livingRouteActivitySet(place, stage) {
  if (stage.prop === "checkpoint") return ["cameraCrew", "marshalGroup", "finishBarrier", "sponsorTent", "timerBoard"];
  if (stage.prop === "tunnel") return ["maintenanceTruck", "tunnelCrew", "emergencyBay", "arrowBoard", "serviceDoor"];
  if (stage.prop === "bridge" || stage.prop === "pier") return ["bridgeCrew", "navLightCrew", "inspectionVan", "barrierCrew", "waterfrontCrowd"];
  const sets = {
    coast: ["lifeguardTruck", "beachSpectators", "surfShopStand", "cameraCrew", "roadCrew"],
    city: ["parkedSedan", "deliveryVan", "sidewalkCrowd", "foodKiosk", "trafficWorker"],
    canyon: ["rangerTruck", "photoPullout", "supportTent", "roadCrew", "lookoutCrowd"],
    alpine: ["rescueTruck", "skiSpectators", "chaletStand", "snowCrew", "shuttleVan"],
    harbor: ["forkliftCrew", "dockWorker", "containerLift", "serviceTruck", "marinaCrowd"],
    snow: ["plowTruck", "skiSpectators", "warmingTent", "snowCrew", "rescueTruck"],
    airfield: ["fuelTruck", "baggageCart", "groundCrew", "beaconCrew", "hangarVan"],
    freight: ["semiParked", "weighCrew", "yardForklift", "fuelTruck", "loadingCrew"],
    farm: ["farmPickup", "hayCrew", "produceStand", "tractorCrew", "countyCrowd"],
    tokyo: ["taxiQueue", "neonCrowd", "deliveryVan", "railWorker", "foodKiosk"],
  };
}
```

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```
desert: ["supportTruck", "rallyCrew", "checkpointTent", "shadeCrew", "waterTruck"],
rainforest: ["marketCrowd", "bridgeCrew", "riverGuide", "rainShelter", "supportTruck"],
europe: ["villageCrowd", "cafeStand", "marshalGroup", "tourVan", "stoneCrew"]
};
return sets[place] || sets.city;
}

function drawLivingRouteActivityZones(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, seed, motion,
phoneMode) {
  const count = phoneMode ? 6 : 10;
  const set = livingRouteActivitySet(place, stage);
  ctx.save();
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * 156 + motion * 0.54 + stage.index * 73 + seed * 0.003) % (h - horizon + 250)) - 78);
    if (y < horizon + 2 || y > h + 80) continue;
    const t = Math.max(0.06, Math.min(1.12, (y - horizon) / Math.max(1, h - horizon)));
    const side = i % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half * (0.98 + seededUnit(seed, i + 11) * 0.36) + w * 0.04);
    const scale = (0.36 + t * (phoneMode ? 0.82 : 1.08));
    const kind = set[(i + Math.floor(seed % set.length)) % set.length];
    drawLivingRouteActivityProp(kind, x, y, scale, side, route, place, design, theme, seed + i * 47);
  }
  ctx.restore();
}

function drawLivingRoutePeopleClusters(w, h, horizon, roadTop, roadBottom, stage, place, design, theme, seed, motion, phoneMode) {
  const clusters = phoneMode ? 5 : 9;
  ctx.save();
  ctx.globalAlpha *= phoneMode ? 0.72 : 0.84;
  for (let i = 0; i < clusters; i += 1) {
    const side = i % 2 === 0 ? -1 : 1;
    const y = horizon + (((i * 181 + motion * 0.46 + stage.index * 39) % (h - horizon + 240)) - 82);
    if (y < horizon || y > h + 72) continue;
    const t = Math.max(0.05, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half * (1.05 + seededUnit(seed, i + 70) * 0.42) + w * 0.02);
    const scale = 0.34 + t * (phoneMode ? 0.66 : 0.86);
    drawLivingPeopleCluster(x, y, scale, side, place, design, theme, seed + i * 61);
  }
  ctx.restore();
}

function drawLivingRouteServiceVehicles(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, seed, motion,
phoneMode) {
  const count = phoneMode ? 4 : 7;
  const set = livingRouteServiceVehicleSet(place, stage);
  ctx.save();
  ctx.globalAlpha *= phoneMode ? 0.68 : 0.82;
  for (let i = 0; i < count; i += 1) {
    const side = i % 2 === 0 ? -1 : 1;
    const y = horizon + (((i * 226 + motion * 0.36 + seed * 0.004) % (h - horizon + 260)) - 88);
    if (y < horizon + 8 || y > h + 86) continue;
    const t = Math.max(0.06, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half * (1.16 + seededUnit(seed, i + 120) * 0.34) + w * 0.035);
    const scale = 0.38 + t * (phoneMode ? 0.72 : 0.94);
    const kind = set[(i + stage.index) % set.length];
    drawLivingServiceVehicle(kind, x, y, scale, side, route, place, design, theme, seed + i * 71);
  }
  ctx.restore();
}

function livingRouteServiceVehicleSet(place, stage) {
  if (stage.prop === "checkpoint") return ["marshalCar", "cameraVan", "medicalCart"];
  if (stage.prop === "tunnel") return ["maintenanceTruck", "arrowTruck", "rescueVan"];
  if (stage.prop === "bridge" || stage.prop === "pier") return ["inspectionVan", "utilityTruck", "serviceCart"];
  const sets = {
    coast: ["lifeguardTruck", "cameraVan", "utilityTruck"],
    city: ["deliveryVan", "taxiQueue", "utilityTruck"],
    canyon: ["rangerTruck", "supportTruck", "cameraVan"],
    alpine: ["rescueVan", "shuttleVan", "snowService"],
    harbor: ["yardForklift", "utilityTruck", "dockCart"],
    snow: ["plowTruck", "rescueVan", "snowService"],
    airfield: ["fuelTruck", "baggageCart", "serviceCart"],
    freight: ["semiParked", "fuelTruck", "yardForklift"],
    farm: ["farmPickup", "tractorCrew", "utilityTruck"],
    tokyo: ["taxiQueue", "deliveryVan", "serviceCart"],
    desert: ["supportTruck", "waterTruck", "cameraVan"],
    rainforest: ["sup
```

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```
portTruck", "marketVan", "utilityTruck"],
  europe: ["tourVan", "marshalCar", "utilityTruck"]
};
return sets[place] || sets.city;
}

function drawLivingRouteActivityProp(kind, x, y, scale, side, route, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.fillStyle = "rgba(0,0,0,0.24)";
  ctx.beginPath();
  ctx.ellipse(0, 20, 56, 8, 0, 0, Math.PI * 2);
  ctx.fill();
  if (kind.includes("Tent") || kind.includes("Stand") || kind.includes("Kiosk") || kind.includes("Shelter") || kind.includes("Shop") || kind.includes("Awning")) {
    ctx.fillStyle = "rgba(5,8,7,0.72)";
    roundRect(-52, -38, 104, 56, 6);
    ctx.fill();
    ctx.fillStyle = `${accent}88`;
    ctx.beginPath();
    ctx.moveTo(-62, -38);
    ctx.lineTo(0, -70);
    ctx.lineTo(62, -38);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = `${light}88`;
    ctx.fillRect(-38, -18, 76, 6);
  } else if (kind.includes("Crew") || kind.includes("Worker") || kind.includes("Spectators") || kind.includes("Crowd") || kind.includes("Group")) {
    drawLivingPeopleCluster(0, 2, 1, side, place, design, theme, seed);
    ctx.strokeStyle = `${light}66`;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-58 * side, 12);
    ctx.lineTo(58 * side, 4);
    ctx.stroke();
  } else if (kind.includes("Barrier") || kind === "arrowBoard" || kind === "timerBoard") {
    ctx.strokeStyle = `${light}70`;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-64, 12);
    ctx.lineTo(64, 2);
    ctx.stroke();
    ctx.fillStyle = "rgba(5,8,7,0.78)";
    roundRect(-42, -42, 84, 30, 5);
    ctx.fill();
    ctx.fillStyle = accent;
    ctx.font = "900 9px system-ui";
    ctx.textAlign = "center";
    ctx.fillText(kind === "timerBoard" ? "TIME" : "KEEP OUT", 0, -23, 70);
  } else {
    drawLivingServiceVehicle(kind, 0, 0, 0.9, side, route, place, design, theme, seed);
  }
  ctx.restore();
}

function drawLivingPeopleCluster(x, y, scale, side, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const coats = [accent, light, "#f4fbf8", "#ff5b6b", "#36d98a"];
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  const count = 5 + Math.floor(seededUnit(seed, 2) * 5);
  for (let i = 0; i < count; i += 1) {
    const px = (i - count / 2) * 12 * side + Math.sin(seed + i) * 3;
    const py = -8 + (i % 3) * 5;
    ctx.fillStyle = coats[(i + Math.floor(seed)) % coats.length];
    ctx.beginPath();
    ctx.arc(px, py - 14, 4, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = `${coats[(i + 1) % coats.length]}cc`;
    roundRect(px - 4, py - 10, 8, 18, 3);
    ctx.fill();
    ctx.strokeStyle = "rgba(5,8,7,0.44)";
    ctx.lineWidth = 1.5;
    ctx.beginPath();
    ctx.moveTo(px - 5, py + 8);
    ctx.lineTo(px - 8, py + 16);
    ctx.moveTo(px + 5, py + 8);
    ctx.lineTo(px + 8, py + 16);
    ctx.stroke();
  }
  ctx.restore();
}

function drawLivingServiceVehicle(kind, x, y, scale, side, route, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const longBody = kind.includes("Truck") || kind.includes("semi") || kind.includes("fuel") || kind.includes("plow") || kind.includes("maintenance") || kind.includes("arrow");
  const compact = kind.includes("Cart") || kind.includes("Forklift") || kind.includes("baggage") || kind.includes("tractor");
  const paint = kind.includes("rescue") || kind.includes("medical") ? "#f4fbf8" : kind.includes("fuel") || kind.includes("taxi") ? "#ffd166" : kind.includes("farm") || kind.includes("tractor") ? "#36d98a" : kind.includes("plow") || kind.includes("snow") ? "#dce8ef" : accent;
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale * side, scale);
  ctx.fillStyle = "rgba(0,0,0,0.28)";
  ctx.beginPath();
  ctx.ellipse(0, 18, longBody ? 62 : 44, 8, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = paint;
  if (compact) {
    roundRect(-30, -18, 60, 32, 5);
  } else {
    roundRect(long
```

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```
Body ? -58 : -42, -22, longBody ? 116 : 84, 38, 6);
}
ctx.fill();
ctx.fillStyle = "rgba(5,8,7,0.78)";
ctx.fillRect(longBody ? -40 : -24, -14, longBody ? 28 : 20, 13);
ctx.fillRect(longBody ? 8 : 0, -14, longBody ? 30 : 20, 13);
ctx.fillStyle = "#050807";
ctx.beginPath();
ctx.arc(longBody ? -38 : -26, 18, 8, 0, Math.PI * 2);
ctx.arc(longBody ? 38 : 26, 18, 8, 0, Math.PI * 2);
ctx.fill();
if (kind.includes("arrow")) {
  ctx.fillStyle = "rgba(5,8,7,0.86)";
  roundRect(12, -48, 48, 24, 4);
  ctx.fill();
  ctx.fillStyle = light;
  ctx.beginPath();
  ctx.moveTo(24, -42);
  ctx.lineTo(48, -36);
  ctx.lineTo(24, -30);
  ctx.closePath();
  ctx.fill();
} else if (kind.includes("plow")) {
  ctx.fillStyle = `${light}88`;
  ctx.beginPath();
  ctx.moveTo(58, -10);
  ctx.lineTo(82, 8);
  ctx.lineTo(58, 24);
  ctx.closePath();
  ctx.fill();
} else if (kind.includes("fuel")) {
  ctx.strokeStyle = "rgba(5,8,7,0.54)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.arc(10, -2, 16, 0, Math.PI * 2);
  ctx.stroke();
} else if (kind.includes("marshal") || kind.includes("camera")) {
  ctx.fillStyle = `${light}aa`;
  ctx.beginPath();
  ctx.arc(0, -28, 5, 0, Math.PI * 2);
  ctx.fill();
}
ctx.restore();
}

function drawRouteSectorRoadEdgeDetail(w, h, horizon, roadTop, roadBottom, stage, place, design, theme, seed, motion, phoneMode) {
  const detail = routeSectorDetailKind(place, stage);
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  const count = phoneMode ? 5 : 8;
  for (let i = 0; i < count; i += 1) {
    const y = horizon + ((i * 184 + motion * 0.62 + stage.index * 81 + seed * 0.004) % (h - horizon + 260)) - 84);
    if (y < horizon || y > h * 0.97) continue;
    const t = Math.max(0.06, Math.min(1.05, (y - horizon) / Math.max(1, h - horizon)));
    const edgeLane = detail === "runway" ? 1.52 : detail === "crosswalk" ? 0 : i % 2 === 0 ? -1.68 : 1.68;
    const alpha = phoneMode ? 0.1 + t * 0.04 : 0.14 + t * 0.08;
    ctx.globalAlpha = alpha;
    if (detail === "tunnel") {
      drawPerspectivePavementQuad(w, horizon, 0, y, 18 + t * 48, laneWidth() * 4.8, "rgba(0,0,0,0.58)");
      drawPerspectivePavementQuad(w, horizon, -1.86, y + 8, 8 + t * 22, laneWidth() * 0.42, `${design.accent || theme[1]}55`);
      drawPerspectivePavementQuad(w, horizon, 1.86, y + 8, 8 + t * 22, laneWidth() * 0.42, `${design.accent || theme[1]}55`);
    } else if (detail === "bridge") {
      drawPerspectivePavementQuad(w, horizon, 0, y, 4 + t * 10, laneWidth() * 4.7, "rgba(244,251,248,0.34)");
      drawPerspectivePavementQuad(w, horizon, -1.95, y, 12 + t * 30, laneWidth() * 0.28, "rgba(180,192,188,0.5)");
      drawPerspectivePavementQuad(w, horizon, 1.95, y, 12 + t * 30, laneWidth() * 0.28, "rgba(180,192,188,0.5)");
    } else if (detail === "crosswalk") {
      for (let b = -2; b <= 2; b += 1) {
        drawPerspectivePavementQuad(w, horizon, 0, y + b * (10 + t * 5), 5 + t * 8, laneWidth() * 4.2, "rgba(244,251,248,0.54)");
      }
    } else if (detail === "runway") {
      for (let lane = -1.45; lane <= 1.45; lane += 0.72) {
        drawPerspectivePavementQuad(w, horizon, lane, y, 14 + t * 36, laneWidth() * 0.1, "rgba(255,209,102,0.42)");
      }
    } else if (detail === "yard") {
      drawPerspectivePavementQuad(w, horizon, edgeLane, y, 20 + t * 42, laneWidth() * 0.34, "rgba(255,209,102,0.36)");
      drawPerspectivePavementQuad(w, horizon, -edgeLane, y + 16, 12 + t * 30, laneWidth() * 0.24, "rgba(216,224,213,0.28)");
    } else if (detail === "snow") {
      drawPerspectivePavementQuad(w, horizon, edgeLane, y, 18 + t * 40, laneWidth() * 0.46, "rgba(244,251,248,0.38)");
      drawPerspectivePavementQuad(w, horizon, -edgeLane, y + 12, 16 + t * 36, laneWidth() * 0.42, "rgba(244,251,248,0.28)");
    } else if (detail === "dirt") {
      drawPerspectivePavementQuad(w, horizon, edgeLane, y, 16 + t * 44, laneWidth() * 0.5, "rgba(255,183,74,0.25)");
      drawPerspectivePavementQuad(w, horizon, -edgeLane, y + 18, 16 + t * 40, laneWidth() * 0.42, "rgba(117,86,47,0.24)");
    } else {
      drawPerspectivePavementQuad(w, horizon, edgeLane, y, 18 + t * 42, laneWidth()
```


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```
) * 0.34, `${design.accent || theme[1]}30`);
    drawPerspectivePavementQuad(w, horizon, -edgeLane, y + 14, 10 + t * 28, laneWidth() * 0.22, "rgba(244,251,248,0.24)");
  }
  ctx.restore();
}

function routeSectorDetailKind(place, stage) {
  if (stage.prop === "tunnel") return "tunnel";
  if (stage.prop === "bridge" || stage.prop === "pier") return "bridge";
  if (stage.prop === "checkpoint" || /finish/i.test(stage.label)) return "crosswalk";
  if (place === "airfield") return "runway";
  if (place === "freight" || place === "harbor") return "yard";
  if (place === "snow" || place === "alpine") return "snow";
  if (place === "desert" || place === "canyon" || place === "farm") return "dirt";
  if (place === "city" || place === "tokyo" || place === "europe") return "crosswalk";
  return "edge";
}

function drawRouteSectorDistantSet(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, seed, motion, phoneMode)
{
  const x = w * (0.5 + Math.sin(seed * 0.013 + motion * 0.0008) * 0.07);
  const y = horizon + h * (0.02 + seededUnit(seed, 1) * 0.05);
  const scale = phoneMode ? 0.74 : 0.92;
  ctx.save();
  ctx.globalAlpha *= phoneMode ? 0.42 : 0.56;
  drawRouteSectorSetPiece(stage.prop, x, y, scale, place, design, theme, seed);
  if (!phoneMode) {
    const signY = horizon + h * 0.145;
    const signT = 0.21;
    const signX = perspectiveRoadCenter(w, signT);
    const signW = Math.min(w * (phoneMode ? 0.38 : 0.5), phoneMode ? 150 : 260);
    ctx.fillStyle = "rgba(5,8,7,0.52)";
    roundRect(signX - signW / 2, signY - 15, signW, 30, 5);
    ctx.fill();
    ctx.strokeStyle = `${design.accent || theme[1]}55`;
    ctx.lineWidth = 1.5;
    ctx.stroke();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = `900 ${Math.max(8, Math.min(phoneMode ? 10 : 13, w * 0.011))}px system-ui`;
    ctx.textAlign = "center";
    ctx.textBaseline = "middle";
    ctx.fillText(`${route.country}: ${stage.label}`.toUpperCase(), signX, signY - 2, signW - 14);
    if (!phoneMode) {
      ctx.fillStyle = `${design.light || theme[2]}dd`;
      ctx.font = `800 ${Math.max(7, Math.min(10, w * 0.01))}px system-ui`;
      ctx.fillText(route.cue.toUpperCase(), signX, signY + 10, signW - 18);
    }
  }
  ctx.restore();
}

function drawRouteSectorSetPiece(prop, x, y, scale, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  if (prop === "tunnel") {
    ctx.fillStyle = "rgba(0,0,0,0.62)";
    roundRect(-132, -36, 264, 104, 14);
    ctx.fill();
    ctx.strokeStyle = `${accent}88`;
    ctx.lineWidth = 5;
    ctx.stroke();
    ctx.fillStyle = "rgba(244,251,248,0.16)";
    for (let i = -3; i <= 3; i += 1) ctx.fillRect(i * 32 - 8, -14, 16, 6);
  } else if (prop === "bridge" || prop === "pier") {
    ctx.strokeStyle = `${light}88`;
    ctx.lineWidth = 5;
    ctx.beginPath();
    ctx.moveTo(-168, 42);
    ctx.quadraticCurveTo(0, -86, 168, 42);
    ctx.stroke();
    for (let i = -4; i <= 4; i += 1) {
      ctx.beginPath();
      ctx.moveTo(i * 34, -22);
      ctx.lineTo(i * 25, 58);
      ctx.stroke();
    }
  } else if (prop === "checkpoint") {
    ctx.fillStyle = "rgba(5,8,7,0.76)";
    roundRect(-132, -34, 264, 48, 6);
    ctx.fill();
    ctx.strokeStyle = accent;
    ctx.lineWidth = 3;
    ctx.stroke();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = `900 18px system-ui`;
    ctx.textAlign = "center";
    ctx.fillText("SECTOR GATE", 0, -10, 180);
  } else if (prop === "tower" || prop === "neon" || prop === "village") {
    for (let i = -3; i <= 3; i += 1) {
      const bw = prop === "village" ? 42 : 28 + seededUnit(seed, i + 10) * 24;
      const bh = prop === "village" ? 52 : 84 + seededUnit(seed, i + 20) * 88;
      const bx = i * 36;
      ctx.fillStyle = prop === "neon" ? "rgba(18,10,32,0.78)" : prop === "village" ? "rgba(58,42,34,0.76)" : "rgba(8,18,20,0.78)";
      roundRect(bx - bw / 2, 22 - bh, bw, bh, 4);
      ctx.fill();
      ctx.fillStyle = i % 2 ? `${accent}77` : `${light}66`;
      ctx.fillRect(bx - bw * 0.26, 34 - bh, bw * 0.52, 4);
      if (prop === "village") {
        ctx.beginPath()
      }
    }
  }
}
```

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```
);
    ctx.moveTo(bx - bw * 0.62, 22 - bh);
    ctx.lineTo(bx, -bh - 12);
    ctx.lineTo(bx + bw * 0.62, 22 - bh);
    ctx.closePath();
    ctx.fill();
  }
} else if (prop === "pine" || prop === "canopy" || prop === "palm") {
  ctx.fillStyle = prop === "palm" ? "rgba(54,217,138,0.42)" : "rgba(33,102,62,0.46)";
  for (let i = -5; i <= 5; i += 1) {
    ctx.beginPath();
    ctx.ellipse(i * 28, -18 - seededUnit(seed, i + 30) * 36, 30, 42, 0, 0, Math.PI * 2);
    ctx.fill();
  }
} else if (prop === "field" || prop === "barn" || prop === "tractor") {
  ctx.fillStyle = "rgba(95,126,54,0.46)";
  ctx.fillRect(-180, -20, 360, 76);
  ctx.strokeStyle = "rgba(187,242,74,0.34)";
  ctx.lineWidth = 3;
  for (let i = -4; i <= 4; i += 1) {
    ctx.beginPath();
    ctx.moveTo(-180, i * 12);
    ctx.lineTo(180, i * 4 + 20);
    ctx.stroke();
  }
  ctx.fillStyle = "rgba(156,42,34,0.72)";
  roundRect(-44, -56, 88, 54, 4);
  ctx.fill();
} else {
  ctx.fillStyle = prop === "dune" ? "rgba(255,183,74,0.42)" : prop === "mountain" ? "rgba(160,172,164,0.42)" :
"rgba(178,92,48,0.44)";
  ctx.beginPath();
  ctx.moveTo(-180, 42);
  ctx.quadraticCurveTo(-80, -96, -16, 14);
  ctx.quadraticCurveTo(72, -120, 180, 42);
  ctx.closePath();
  ctx.fill();
}
ctx.restore();
}

function drawRouteSectorRoadsideStory(w, h, horizon, roadTop, roadBottom, stage, route, place, design, theme, seed, motion,
phoneMode) {
  const count = phoneMode ? 6 : 12;
  const set = routeSectorStorySet(place, stage);
  ctx.save();
  ctx.globalAlpha *= phoneMode ? 0.62 : 0.78;
  for (let i = 0; i < count; i += 1) {
    const side = i % 2 === 0 ? -1 : 1;
    const y = horizon + (((i * (phoneMode ? 198 : 150) + motion * (0.62 + seededUnit(seed, i) * 0.16) + stage.index * 71) % (h -
horizon + 320)) - 88);
    if (y < horizon + 4 || y > h + 72) continue;
    const t = Math.max(0.06, Math.min(1.1, (y - horizon) / Math.max(1, h - horizon)));
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half * (0.86 + seededUnit(seed, i + 10) * 0.28) + w * 0.03);
    const scale = (0.34 + t * (phoneMode ? 0.78 : 1.02));
    const kind = worldSceneryProp(set[(i + stage.index) % set.length], place, i + stage.index, phoneMode);
    drawRouteSectorStoryProp(kind, x, y, scale, side, route, place, design, theme, seed + i * 53);
  }
  ctx.restore();
}

function routeSectorStorySet(place, stage) {
  if (stage.prop === "tunnel") return ["tunnelLamp", "fanBox", "reflectorPost", "emergencyBay"];
  if (stage.prop === "bridge" || stage.prop === "pier") return ["bridgeCable", "navLight", "guardRail", "expansionPost"];
  if (stage.prop === "checkpoint") return ["marshalPost", "cameraCrew", "crowdFence", "sectorFlag"];
  const sets = {
    coast: ["lifeguardPost", "surfShop", "oceanFence", "mileMarker"],
    city: ["storefront", "trafficSignal", "taxiQueue", "newsStand"],
    canyon: ["rangerSign", "lookoutFence", "rockMarker", "touristCamera"],
    alpine: ["snowPole", "chaletSign", "skiBanner", "stoneWall"],
    harbor: ["dockBollard", "containerStack", "forkliftSign", "lifeRing"],
    snow: ["snowPole", "plowSign", "warmingHut", "skiBanner"],
    airfield: ["runwayLight", "windsock", "serviceCartSign", "fuelMarker"],
    freight: ["weighSign", "truckStopSign", "yardLamp", "palletLoad"],
    farm: ["countyMailbox", "hayStack", "cropSign", "tractorGate"],
    tokyo: ["neonStore", "railSignal", "taxiQueue", "expressSign"],
    desert: ["rallyFlag", "checkpointTent", "sandMarker", "supportTruckSign"],
    rainforest: ["marketAwning", "riverPost", "rainSign", "woodFence"],
    europe: ["villageLamp", "cafeAwning", "stoneWall", "alpsMarker"]
  };
  return sets[place] || sets.city;
}

function drawRouteSectorStoryProp(kind, x, y, scale, side, route, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.fillStyle = "rgba(0,0,0,0.28)";
  ctx.beginPath();
  ctx.ellipse(0, 18, 42, 7, 0, 0, Math.PI * 2);
  ctx.fill();
  if (ki
```

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```
nd.includes("Sign") || kind.includes("Marker") || kind === "mileMarker" || kind === "reflectorPost" || kind === "expressSign" ||
kind === "weighSign") {
  ctx.strokeStyle = "rgba(216,226,221,0.58)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(0, 18);
  ctx.lineTo(0, -42);
  ctx.stroke();
  ctx.fillStyle = `${accent}88`;
  roundRect(-36, -58, 72, 26, 4);
  ctx.fill();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 8px system-ui";
  ctx.textAlign = "center";
  const text = kind === "weighSign" ? "WEIGH" : kind === "expressSign" ? "EXP" : route.country.slice(0, 4).toUpperCase();
  ctx.fillText(text, 0, -42, 58);
} else if (kind.includes("Lamp") || kind === "trafficSignal" || kind === "runwayLight" || kind === "navLight" || kind ===
"railSignal") {
  ctx.strokeStyle = "rgba(216,226,221,0.58)";
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(0, 18);
  ctx.lineTo(0, -58);
  ctx.stroke();
  ctx.globalCompositeOperation = "screen";
  ctx.fillStyle = kind === "trafficSignal" ? "#ff5b6b" : light;
  ctx.beginPath();
  ctx.arc(0, -64, 9, 0, Math.PI * 2);
  ctx.fill();
  ctx.globalCompositeOperation = "source-over";
} else if (kind.includes("Fence") || kind === "guardRail" || kind === "stoneWall" || kind === "oceanFence" || kind ===
"lookoutFence" || kind === "woodFence") {
  ctx.strokeStyle = kind === "stoneWall" ? "rgba(160,172,164,0.62)" : `${light}66`;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-58 * side, -8);
  ctx.lineTo(58 * side, -16);
  ctx.moveTo(-58 * side, 10);
  ctx.lineTo(58 * side, 2);
  for (let i = -2; i <= 2; i += 1) {
    ctx.moveTo(i * 24 * side, -22);
    ctx.lineTo(i * 24 * side, 18);
  }
  ctx.stroke();
} else if (kind.includes("Awning") || kind.includes("Store") || kind === "storefront" || kind === "newsStand" || kind ===
"surfShop" || kind === "warmingHut") {
  ctx.fillStyle = "rgba(5,8,7,0.74)";
  roundRect(-48, -44, 96, 62, 5);
  ctx.fill();
  ctx.fillStyle = `${accent}88`;
  ctx.beginPath();
  ctx.moveTo(-56, -44);
  ctx.lineTo(0, -72);
  ctx.lineTo(56, -44);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = `${light}88`;
  ctx.fillRect(-34, -24, 68, 5);
} else if (kind.includes("Truck") || kind.includes("Cart") || kind === "taxiQueue" || kind === "serviceCartSign" || kind ===
"palletLoad" || kind === "hayStack") {
  ctx.fillStyle = kind === "hayStack" ? "rgba(255,209,102,0.72)" : `${accent}78`;
  roundRect(-42, -20, 84, 36, 5);
  ctx.fill();
  ctx.fillStyle = "#050807";
  ctx.beginPath();
  ctx.arc(-28, 18, 7, 0, Math.PI * 2);
  ctx.arc(28, 18, 7, 0, Math.PI * 2);
  ctx.fill();
} else if (kind === "bridgeCable" || kind === "expansionPost" || kind === "fanBox" || kind === "emergencyBay") {
  ctx.strokeStyle = `${light}7a`;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-52 * side, -38);
  ctx.lineTo(52 * side, 10);
  ctx.moveTo(-38 * side, -12);
  ctx.lineTo(38 * side, 16);
  ctx.stroke();
  ctx.fillStyle = `${accent}55`;
  roundRect(-18, -28, 36, 28, 4);
  ctx.fill();
} else {
  ctx.fillStyle = `${accent}50`;
  roundRect(-36, -34, 72, 52, 5);
  ctx.fill();
  ctx.fillStyle = `${light}66`;
  ctx.fillRect(-24, -14, 48, 5);
}
ctx.restore();
}

function routeTerrainPalette(place, design, theme) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const palettes = {
    coast: { near: "rgba(25,68,74,0.34)", far: "rgba(70,217,255,0.16)", line: "rgba(214,180,112,0.34)", detail: "wave" },
    city: { near: "rgba(16,20,21,0.46)", far: "rgba(42,52,54,0.34)", line: `${accent}28`, detail: "curb" },
    canyon: { near: "rgba(120,62,34,0.34)", far: "rgba(178,92,48,0.24)", line: "rgba(255,183,74,0.28)", detail: "rock" },
    alpine: { near: "rgba(42,71,61,0.34)", far: "rgba(210,232,239,0.18)", line: "rgba(244,251,248,0.38)", detail: "pine" },
    harbor: { near: "rgba(18,49,56,0.42)", far: "rgba(70,217,255,0.17)", line: "rgba(255,209,102,0.28)", detail: "dock" },
    snow: { near: "rgba(220,232,239,0.28)", far: "rgba(244,251,248,0.18)", line: "rgba(244,251,248,0.46)",
```

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```
detail: "snow" },
  airfield: { near: "rgba(45,56,44,0.34)", far: "rgba(255,209,102,0.12)", line: `${light}32`, detail: "runway" },
  freight: { near: "rgba(42,38,29,0.42)", far: "rgba(255,209,102,0.14)", line: "rgba(216,224,213,0.28)", detail: "yard" },
  farm: { near: "rgba(62,79,32,0.38)", far: "rgba(187,242,74,0.16)", line: "rgba(255,209,102,0.28)", detail: "field" },
  tokyo: { near: "rgba(18,13,32,0.48)", far: `${accent}18`, line: `${accent}42`, detail: "neon" },
  desert: { near: "rgba(123,78,35,0.36)", far: "rgba(255,183,74,0.18)", line: "rgba(244,251,248,0.24)", detail: "sand" },
  rainforest: { near: "rgba(21,72,42,0.38)", far: "rgba(54,217,138,0.16)", line: "rgba(255,209,102,0.24)", detail: "leaf" },
  europe: { near: "rgba(48,64,61,0.34)", far: "rgba(220,232,239,0.14)", line: `${light}28`, detail: "stone" }
};
return palettes[place] || palettes.city;
}

function drawRouteShoulderTerrain(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode) {
  const terrain = routeTerrainPalette(place, design, theme);
  const rows = phoneMode ? 8 : 12;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.58 : 0.72;
  for (let i = 0; i < rows; i += 1) {
    const t1 = Math.max(0.03, i / rows);
    const t2 = Math.min(1.08, (i + 1.15) / rows);
    const y1 = horizon + (h - horizon) * t1;
    const y2 = horizon + (h - horizon) * t2;
    const center1 = perspectiveRoadCenter(w, t1);
    const center2 = perspectiveRoadCenter(w, t2);
    const half1 = realWorldRoadHalf(roadTop, roadBottom, t1) * 0.72;
    const half2 = realWorldRoadHalf(roadTop, roadBottom, t2) * 0.72;
    const drift = Math.sin(motion * 0.003 + i * 0.7 + seed * 0.01) * w * 0.012;
    for (let side = -1; side <= 1; side += 2) {
      const edge1 = center1 + side * half1;
      const edge2 = center2 + side * half2;
      const outer1 = edge1 + side * (w * (0.22 + t1 * 0.18) + drift);
      const outer2 = edge2 + side * (w * (0.28 + t2 * 0.32) + drift * 1.4);
      const grad = ctx.createLinearGradient(edge1, y1, outer2, y2);
      grad.addColorStop(0, terrain.near);
      grad.addColorStop(1, terrain.far);
      ctx.fillStyle = grad;
      ctx.beginPath();
      ctx.moveTo(edge1, y1);
      ctx.lineTo(edge2, y2);
      ctx.lineTo(outer2, y2);
      ctx.lineTo(outer1, y1);
      ctx.closePath();
      ctx.fill();
      drawRouteTerrainTexture(terrain.detail, edge1, edge2, outer1, outer2, y1, y2, side, design, theme, seed + i * 31 + side * 11,
        phoneMode);
    }
  }
  ctx.restore();
}

function drawRouteTerrainTexture(detail, edge1, edge2, outer1, outer2, y1, y2, side, design, theme, seed, phoneMode) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  const midY = (y1 + y2) * 0.5;
  const t = Math.max(0.04, Math.min(1.1, (midY - y1) / Math.max(1, y2 - y1)));
  ctx.save();
  ctx.globalAlpha *= phoneMode ? 0.34 : 0.48;
  if (detail === "wave" || detail === "dock") {
    ctx.strokeStyle = detail === "dock" ? "rgba(214,180,112,0.34)" : "rgba(70,217,255,0.36)";
    ctx.lineWidth = 2;
    for (let i = 0; i < 3; i += 1) {
      const y = y1 + (i + 0.35) * (y2 - y1) / 3;
      const xA = edge1 + side * (18 + i * 16);
      const xB = outer1 - side * (20 + i * 8);
      ctx.beginPath();
      ctx.moveTo(xA, y);
      ctx.quadraticCurveTo((xA + xB) * 0.5, y + Math.sin(seed + i) * 7, xB, y + 2);
      ctx.stroke();
    }
  } else if (detail === "curb" || detail === "neon" || detail === "stone") {
    ctx.strokeStyle = detail === "neon" ? `${accent}6f` : detail === "stone" ? "rgba(190,202,196,0.34)" : "rgba(244,251,248,0.26)";
    ctx.lineWidth = 2 + Math.abs(y2 - y1) * 0.018;
    ctx.beginPath();
    ctx.moveTo(edge1 + side * 8, y1 + 4);
    ctx.lineTo(edge2 + side * 12, y2 - 2);
    ctx.stroke();
    for (let i = 0; i < 4; i += 1) {
      ctx.fillStyle = detail === "neon" ? `${accent}55` : "rgba(5,8,7,0.22)";
      const x = edge1 + side * (26 + seededUnit(seed, i) * 92);
      roundRect(x - 8, y1 + 8 + i * (y2 - y1) / 4, 18, 5, 2);
      ctx.fill();
    }
  } else if (detail === "field" || detail === "leaf" || detail === "pine") {
    ctx.strokeStyle = detail === "pine" ? "

```

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```
    rgba(34,93,59,0.48)" : "rgba(54,217,138,0.34)";
    ctx.lineWidth = 2;
    for (let i = 0; i < 5; i += 1) {
        const y = y1 + (i + 0.2) * (y2 - y1) / 5;
        ctx.beginPath();
        ctx.moveTo(edge1 + side * (20 + i * 5), y);
        ctx.lineTo(outer1 - side * (8 + seededUnit(seed, i) * 24), y + side * 3);
        ctx.stroke();
    }
} else if (detail === "snow" || detail === "sand" || detail === "rock") {
    ctx.fillStyle = detail === "snow" ? "rgba(244,251,248,0.34)" : detail === "sand" ? "rgba(255,183,74,0.23)" :
"rgba(178,92,48,0.26)";
    for (let i = 0; i < 5; i += 1) {
        const x = edge1 + side * (28 + seededUnit(seed, i) * Math.abs(outer1 - edge1) * 0.62);
        const y = y1 + seededUnit(seed, i + 9) * (y2 - y1);
        ctx.beginPath();
        ctx.ellipse(x, y, 10 + seededUnit(seed, i + 20) * 22, 2 + seededUnit(seed, i + 30) * 6, 0, 0, Math.PI * 2);
        ctx.fill();
    }
} else if (detail === "runway" || detail === "yard") {
    ctx.strokeStyle = detail === "runway" ? `${light}52` : "rgba(216,224,213,0.32)";
    ctx.lineWidth = 2;
    for (let i = 0; i < 3; i += 1) {
        const x = edge1 + side * (32 + i * 34);
        ctx.beginPath();
        ctx.moveTo(x, y1 + 6);
        ctx.lineTo(x + side * 24, y2 - 6);
        ctx.stroke();
    }
}
ctx.restore();
}

function drawRouteMicroLandmarks(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode) {
    const count = phoneMode ? 8 : 14;
    ctx.save();
    ctx.globalAlpha = phoneMode ? 0.42 : 0.58;
    for (let i = 0; i < count; i += 1) {
        const side = i % 2 === 0 ? -1 : 1;
        const t = 0.18 + seededUnit(seed, i + 820) * 0.74;
        const y = horizon + (h - horizon) * t + Math.sin(motion * 0.004 + i) * (4 + t * 12);
        if (y < horizon + 8 || y > h + 30) continue;
        const half = realWorldRoadHalf(roadTop, roadBottom, t);
        const x = perspectiveRoadCenter(w, t) + side * (half * (0.82 + seededUnit(seed, i + 830) * 0.3));
        const scale = (0.34 + t * 0.98) * (phoneMode ? 0.8 : 1);
        const kind = routeMicroLandmarkProp(routeMicroLandmarkKind(place, i, seed), place, i, phoneMode);
        drawRouteMicroLandmark(kind, x, y, scale, side, design, theme, seed + i * 43);
    }
    ctx.restore();
}

function routeMicroLandmarkKind(place, index, seed) {
    const sets = {
        coast: ["surfboardRack", "guardUmbrella", "beachFence", "shoreBirdSign"],
        city: ["taxiStand", "newsKiosk", "busShelter", "crosswalkPost"],
        canyon: ["milePost", "photoPullout", "rockStack", "desertGuard"],
        alpine: ["snowPole", "skiBanner", "stoneMarker", "chainSign"],
        harbor: ["bollard", "dockCart", "lifeRing", "containerCone"],
        snow: ["snowPole", "plowMarker", "skiBanner", "chainSign"],
        airfield: ["coneLight", "windSockMini", "serviceSign", "fuelCart"],
        freight: ["weighCone", "yardLight", "palletStack", "truckScale"],
        farm: ["mailbox", "hayBale", "farmGate", "produceSign"],
        tokyo: ["neonKiosk", "railPost", "taxiStand", "crosswalkPost"],
        desert: ["rallyFlag", "stoneMarker", "checkpointCone", "shadeTent"],
        rainforest: ["marketLantern", "woodPost", "riverSign", "leafBarrier"],
        europe: ["villageLamp", "stoneMarker", "cafeTable", "switchbackSign"]
    };
    const list = sets[place] || sets.city;
    return list[(index + Math.floor(seed % list.length)) % list.length];
}

function routeMicroLandmarkProp(kind, place, index, phoneMode) {
    if (!sign|marker|post|flag/i.test(kind)) return kind;
    if (!phoneMode && Math.abs(index) % 2 === 0) return kind;
    const sets = {
        coast: ["surfboardRack", "guardUmbrella", "beachFence", "cafeTable"],
        city: ["taxiStand", "newsKiosk", "busShelter", "cafeTable"],
        canyon: ["photoPullout", "rockStack", "desertGuard", "hayBale"],
        alpine: ["cafeTable", "hayBale", "busShelter", "snowPole"],
        harbor: ["bollard", "dockCart", "lifeRing", "containerCone"],
        snow: ["cafeTable", "hayBale", "busShelter", "snowPole"],
        airfield: ["fuelCart", "dockCart", "busShelter", "cafeTable"],
        freight: ["palletStack", "yardLight", "fuelCart", "dockCart"],
        farm: ["mailbox", "hayBale", "farmGate", "cafeTable"],
        tokyo: ["neonKiosk", "taxiStand", "busShelter", "cafeTable"],
        desert: ["shadeTent", "rockStack", ""]
    };
    const list = sets[place] || sets.city;
    return list[(index + Math.floor(seed % list.length)) % list.length];
}
```

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```
desertGuard", "hayBale"],
rainforest: ["marketLantern", "woodPost", "leafBarrier", "cafeTable"],
europe: ["villageLamp", "stoneMarker", "cafeTable", "busShelter"]
};
const list = sets[place] || sets.city;
return list[Math.abs(index) % list.length];
}

function drawRouteMicroLandmark(kind, x, y, scale, side, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.fillStyle = "rgba(0,0,0,0.24)";
  ctx.beginPath();
  ctx.ellipse(0, 18, 36, 6, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = "rgba(216,226,221,0.52)";
  ctx.lineWidth = 3;
  if (kind.includes("Sign") || kind === "milePost" || kind === "stoneMarker" || kind === "crosswalkPost" || kind === "switchbackSign") {
    ctx.beginPath();
    ctx.moveTo(0, 16);
    ctx.lineTo(0, -34);
    ctx.stroke();
    ctx.fillStyle = kind === "stoneMarker" ? "rgba(160,172,164,0.7)" : `${accent}88`;
    roundRect(-28, -48, 56, 22, 4);
    ctx.fill();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = "900 8px system-ui";
    ctx.textAlign = "center";
    ctx.fillText(kind === "milePost" ? "MILE" : kind === "switchbackSign" ? "TURN" : "SLOW", 0, -35, 46);
  } else if (kind.includes("Cone") || kind === "rallyFlag" || kind === "snowPole" || kind === "plowMarker" || kind === "coneLight") {
    ctx.fillStyle = kind === "snowPole" || kind === "plowMarker" ? "#f4fbf8" : "#ff5b6b";
    ctx.beginPath();
    ctx.moveTo(-12, 16);
    ctx.lineTo(0, -34);
    ctx.lineTo(12, 16);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = light;
    ctx.fillRect(-8, -8, 16, 5);
    if (kind === "rallyFlag") {
      ctx.fillStyle = accent;
      ctx.fillRect(2 * side, -36, 30 * side, 14);
    }
  } else if (kind.includes("Kiosk") || kind === "busShelter" || kind === "taxiStand" || kind === "guardUmbrella" || kind === "shadeTent" || kind === "cafeTable") {
    ctx.fillStyle = "rgba(5,8,7,0.72)";
    roundRect(-34, -34, 68, 50, 5);
    ctx.fill();
    ctx.fillStyle = `${accent}88`;
    ctx.beginPath();
    ctx.moveTo(-42, -34);
    ctx.lineTo(0, -58);
    ctx.lineTo(42, -34);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = `${light}88`;
    ctx.fillRect(-22, -18, 44, 5);
  } else if (kind.includes("Cart") || kind === "fuelCart" || kind === "dockCart" || kind === "palletStack" || kind === "hayBale") {
    ctx.fillStyle = kind === "hayBale" ? "rgba(255,209,102,0.72)" : `${accent}82`;
    roundRect(-28, -16, 56, 30, 4);
    ctx.fill();
    ctx.fillStyle = "#050807";
    ctx.beginPath();
    ctx.arc(-18, 17, 6, 0, Math.PI * 2);
    ctx.arc(18, 17, 6, 0, Math.PI * 2);
    ctx.fill();
  } else if (kind === "lifeRing" || kind === "bollard" || kind === "mailbox" || kind === "farmGate" || kind === "villageLamp") {
    ctx.strokeStyle = `${light}88`;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-24, 16);
    ctx.lineTo(-24, -26);
    ctx.moveTo(24, 16);
    ctx.lineTo(24, -26);
    ctx.moveTo(-28, -12);
    ctx.lineTo(28, -12);
    ctx.stroke();
    ctx.fillStyle = kind === "lifeRing" ? "#f4fbf8" : `${accent}77`;
    ctx.beginPath();
    ctx.arc(0, -28, 14, 0, Math.PI * 2);
    ctx.fill();
  } else {
    ctx.fillStyle = `${accent}44`;
    roundRect(-30, -24, 60, 40, 5);
    ctx.fill();
    ctx.fillStyle = `${light}66`;
    ctx.fillRect(-20, -10, 40, 5);
  }
  ctx.restore();
}

function drawRouteWeatherAndAir(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode) {
  const count = phoneMode ? 26 : 46;
  const mode = routeAirMode(place);
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.24 : 0.34;
  if (mode === "nightGlow") {
    ctx.globalCompositeOperation = "screen";
    for (let i = 0; i < 5; i += 1) {
      const x = w * (0.12 + seededUnit(seed, i + 900) * 0.76);
      const y = horizon + h * (0.02 + seededUnit(seed, i + 910) * 0.26);
      const r = w * (0.08 + seededUnit(seed, i + 920) * 0.08);
      const glow = ctx.createRadialGradient(x, y, 2, x, y, r);
      glow.addColorStop(0, `${design.accent || theme[1] || "#46d9ff"}44`);
      glow.addColorStop(1, "rgba(0,0,0,0)");
      ctx.fillStyle = glow;
    }
  }
}
```

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```
    ctx.fillRect(x - r, y - r, r * 2, r * 2);
  }
  ctx.globalCompositeOperation = "source-over";
}
for (let i = 0; i < count; i += 1) {
  const t = seededUnit(seed, i + 940);
  const laneAvoid = seededUnit(seed, i + 950) > 0.42 ? (seededUnit(seed, i + 951) > 0.5 ? -1 : 1) : 0;
  const y = horizon + ((1 * 67 + motion * (mode === "snow" ? 0.42 : 0.26) + seed * 0.003) % (h - horizon + 140)) - 40;
  if (y < horizon || y > h + 24) continue;
  const roadT = Math.max(0.05, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
  const half = realWorldRoadHalf(roadTop, roadBottom, roadT);
  const x = laneAvoid
    ? perspectiveRoadCenter(w, roadT) + laneAvoid * (half * (0.82 + seededUnit(seed, i + 960) * 0.45))
    : w * seededUnit(seed, i + 970);
  if (mode === "snow") {
    ctx.fillStyle = "rgba(244,251,248,0.62)";
    ctx.beginPath();
    ctx.arc(x, y, 1.2 + roadT * 2.1, 0, Math.PI * 2);
    ctx.fill();
  } else if (mode === "rain") {
    ctx.strokeStyle = "rgba(210,232,239,0.34)";
    ctx.lineWidth = 1.2;
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x - 8, y + 18);
    ctx.stroke();
  } else if (mode === "dust") {
    ctx.fillStyle = "rgba(255,183,74,0.18)";
    ctx.beginPath();
    ctx.ellipse(x, y, 8 + roadT * 18, 2 + roadT * 5, 0, 0, Math.PI * 2);
    ctx.fill();
  } else if (mode === "spray") {
    ctx.strokeStyle = "rgba(70,217,255,0.22)";
    ctx.lineWidth = 1.5;
    ctx.beginPath();
    ctx.moveTo(x - 12, y);
    ctx.quadraticCurveTo(x, y - 5, x + 14, y + 1);
    ctx.stroke();
  } else if (mode === "leaf") {
    ctx.fillStyle = t > 0.5 ? "rgba(54,217,138,0.28)" : "rgba(255,209,102,0.24)";
    ctx.beginPath();
    ctx.ellipse(x, y, 5 + roadT * 6, 2 + roadT * 3, Math.sin(seed + i), 0, Math.PI * 2);
    ctx.fill();
  }
}
ctx.restore();
}

function routeAirMode(place) {
  if (place === "snow" || place === "alpine") return "snow";
  if (place === "rainforest" || place === "city") return "rain";
  if (place === "desert" || place === "canyon" || place === "farm") return "dust";
  if (place === "coast" || place === "harbor") return "spray";
  if (place === "tokyo") return "nightGlow";
  return "leaf";
}

function routeWeatherCondition(place, progress = 0, elapsed = 0) {
  const phase = (Math.floor(progress * 8) + Math.floor(elapsed / 28)) % 4;
  const night = raceIsNight(selectedRace);
  const table = {
    coast: [
      { sky: "sun", surface: "wet", hazard: "spray", label: "sun glare on sea mist" },
      { sky: "cloud", surface: "standingWater", hazard: "coastalFlood", label: "water over low coast road" },
      { sky: "rain", surface: "wet", hazard: "train", label: "rain slick coast highway" },
      { sky: "storm", surface: "standingWater", hazard: "tidalSurge", label: "surging shoreline" }
    ],
    city: [
      { sky: "cloud", surface: "wet", hazard: "train", label: "wet downtown pavement" },
      { sky: "rain", surface: "standingWater", hazard: "train", label: "rain and light puddles" },
      { sky: "night" ? "night" : "sun", surface: "dry", hazard: "train", label: "clear city traffic" },
      { sky: "cloud", surface: "oilSlick", hazard: "train", label: "oil on city asphalt" }
    ],
    canyon: [
      { sky: "sun", surface: "dry", hazard: "rockfall", label: "dry canyon switchback" },
      { sky: "cloud", surface: "cracked", hazard: "earthquake", label: "quake-cracked canyon road" },
      { sky: "sun", surface: "dust", hazard: "train", label: "dusty canyon rail line" },
      { sky: "storm", surface: "wet", hazard: "rockfall", label: "storm runoff on rock road" }
    ],
    alpine: [
      { sky: "snow", surface: "snowSlush", hazard: "train", label: "snow slush mountain pass" },
      { sky: "cloud", surface: "blackIce", hazard: "train", label: "black ice warning" },
      { sky: "sun", surface: "wet", hazard: "meltwater", label: "sunny meltwater road" },
      { sky: "snow", surface: "snowSlush", hazard: "avalancheView", label: "snowy mountain road" }
    ],
    harbor: [
      { sky: "cloud", surface: "wet", hazard: "train", label: "wet port pavement" },
      { sky: "storm", surface: "standingWater",
```

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```
hazard: "tidalSurge", label: "harbor flood water" },
{ sky: "sun", surface: "oilSlick", hazard: "train", label: "diesel sheen by docks" },
{ sky: "rain", surface: "standingWater", hazard: "coastalFlood", label: "rain over dock road" }
],
snow: [
{ sky: "snow", surface: "snowSlush", hazard: "train", label: "snow-packed road" },
{ sky: "night", surface: "blackIce", hazard: "train", label: "black ice at night" },
{ sky: "cloud", surface: "snowSlush", hazard: "meltwater", label: "slush and meltwater" },
{ sky: "sun", surface: "wet", hazard: "train", label: "bright winter glare" }
],
airfield: [
{ sky: "sun", surface: "dry", hazard: "train", label: "clear runway heat shimmer" },
{ sky: "cloud", surface: "wet", hazard: "crosswind", label: "wet runway" },
{ sky: "storm", surface: "standingWater", hazard: "crosswind", label: "stormy runway spray" },
{ sky: "night", surface: "dry", hazard: "train", label: "night runway lights" }
],
freight: [
{ sky: "cloud", surface: "oilSlick", hazard: "train", label: "diesel-stained freight road" },
{ sky: "rain", surface: "wet", hazard: "train", label: "rainy rail corridor" },
{ sky: "sun", surface: "dry", hazard: "train", label: "sunlit freight bypass" },
{ sky: "cloud", surface: "cracked", hazard: "train", label: "patched industrial pavement" }
],
farm: [
{ sky: "sun", surface: "dry", hazard: "train", label: "clear county road" },
{ sky: "storm", surface: "wet", hazard: "tornado", label: "storm cell over fields" },
{ sky: "cloud", surface: "mud", hazard: "train", label: "mud at farm crossing" },
{ sky: "rain", surface: "standingWater", hazard: "train", label: "rain over low farm road" }
],
tokyo: [
{ sky: "night", surface: "wet", hazard: "train", label: "neon wet expressway" },
{ sky: "rain", surface: "standingWater", hazard: "train", label: "rain reflections and rail" },
{ sky: "night", surface: "oilSlick", hazard: "train", label: "dark glossy service lane" },
{ sky: "cloud", surface: "wet", hazard: "train", label: "cloudy expressway" }
],
desert: [
{ sky: "sun", surface: "dry", hazard: "heat", label: "dry desert highway" },
{ sky: "storm", surface: "dust", hazard: "tornado", label: "dust funnel in distance" },
{ sky: "cloud", surface: "cracked", hazard: "earthquake", label: "broken desert pavement" },
{ sky: "sun", surface: "dust", hazard: "train", label: "rail line through desert" }
],
rainforest: [
{ sky: "rain", surface: "wet", hazard: "train", label: "rainforest wet road" },
{ sky: "storm", surface: "standingWater", hazard: "floodedBridge", label: "flooded low bridge" },
{ sky: "cloud", surface: "mud", hazard: "train", label: "mud washed shoulder" },
{ sky: "sun", surface: "wet", hazard: "steam", label: "sun after rain" }
],
europe: [
{ sky: "cloud", surface: "wet", hazard: "train", label: "wet village road" },
{ sky: "sun", surface: "dry", hazard: "train", label: "clear alpine village" },
{ sky: "rain", surface: "standingWater", hazard: "train", label: "rain on stone road" },
{ sky: "snow", surface: "blackIce", hazard: "train", label: "icy switchback" }
]
};
const list = table[place] || table.city;
const condition = { ...list[phase % list.length] };
if (night && condition.sky !== "snow" && condition.sky !== "rain" && condition.sky !== "storm") condition.sky = "night";
return condition;
}

function weatherRoadGrip(surface) {
const grip = {
dry: 1,
dust: 0.94,
wet: 0.91,
mud: 0.86,
standingWater: 0.84,
oilSlick: 0.76,
blackIce: 0.68,
snowSlush: 0.79,
cracked: 0.88
};
return grip[surface] || 1;
}

function weatherRoadSpeed(surface) {
const speed = {
dry: 1,
dust: 0.98,
wet: 0.97,
mud: 0.92,
standingWater: 0.92,
oilSlick: 0.95,
blackIce: 0.9,
snowSlush: 0.88,
cracked: 0.94
};
return speed[surface] || 1;
}

function drawRealWorldPlaceIdentityPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
const route = rout
```


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```
eWorldInfo(place);
const phoneMode = phoneGraphicsActive();
const motion = (raceState.roadOffset || 0) * 0.018;
ctx.save();
const bandTop = Math.max(0, horizon - h * 0.18);
const bandBottom = horizon + h * 0.19;
const haze = ctx.createLinearGradient(0, bandTop, 0, bandBottom);
haze.addColorStop(0, "rgba(244,251,248,0)");
haze.addColorStop(0.6, `${design.accent}10`);
haze.addColorStop(1, "rgba(0,0,0,0.06)");
ctx.fillStyle = haze;
ctx.fillRect(0, bandTop, w, bandBottom - bandTop);

const skylineCount = phoneMode ? 8 : 13;
for (let i = 0; i < skylineCount; i += 1) {
  const side = i % 2 === 0 ? -1 : 1;
  const xBase = w * (side < 0 ? 0.08 : 0.92) + side * ((seededUnit(seed, i + 710) * 0.18 - 0.05) * w);
  const x = xBase + Math.sin(motion + i * 0.8) * w * 0.018;
  const y = horizon + h * (0.025 + seededUnit(seed, i + 720) * 0.12);
  const scale = (0.58 + seededUnit(seed, i + 730) * 0.62) * (phoneMode ? 0.82 : 1);
  const kind = realWorldPlaceIdentityKind(place, i, seed);
  drawPlaceIdentityObject(kind, x, y, scale, side, place, design, theme, seed + i * 89);
}

const boardCount = phoneMode ? 0 : 5;
for (let i = 0; i < boardCount; i += 1) {
  const side = i % 2 === 0 ? -1 : 1;
  const t = phoneMode ? 0.18 + seededUnit(seed, i + 780) * 0.16 : 0.28 + seededUnit(seed, i + 780) * 0.42;
  const half = realWorldRoadHalf(roadTop, roadBottom, t);
  const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.08 + seededUnit(seed, i + 790) * 0.08));
  const y = horizon + (h - horizon) * t + Math.sin(motion * 1.3 + i) * 8;
  drawPlaceDestinationBoard(x, y, 0.72 + t * 0.72, side, route, design, theme, seed + i * 97);
}
ctx.restore();
}

function realWorldPlaceIdentityKind(place, index, seed) {
  const sets = {
    coast: ["pierDeck", "beachHotel", "marinalLight", "cliffHome"],
    city: ["downtownTower", "elPlatform", "glassOffice", "cornerStore"],
    canyon: ["redRockWall", "lookoutRail", "rangerStation", "mesaHotel"],
    alpine: ["chalet", "skiLift", "stoneTunnel", "pineLodge"],
    harbor: ["containerStack", "shipCrane", "harborWarehouse", "drawBridge"],
    snow: ["snowLodge", "skiLift", "snowFence", "warmingHut"],
    airfield: ["hangarRow", "controlTower", "runwayBeacon", "serviceHangar"],
    freight: ["railYard", "grainSilo", "truckStop", "overheadGantry"],
    farm: ["grainSilo", "redBarn", "windmill", "farmStand"],
    tokyo: ["neonBlock", "expressPylon", "railPlatform", "towerSign"],
    desert: ["rallyCamp", "lowDune", "checkpointTent", "stoneOutpost"],
    rainforest: ["canopyWall", "riverDock", "woodMarket", "mistFalls"],
    europe: ["villageRoof", "stoneViaduct", "churchTower", "switchbackWall"]
  };
  const list = sets[place] || sets.city;
  return list[(index + Math.floor(seed % list.length)) % list.length];
}

function drawPlaceIdentityObject(kind, x, y, scale, side, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.globalAlpha = phoneGraphicsActive() ? 0.42 : 0.38;
  ctx.fillStyle = "rgba(0,0,0,0.28)";
  ctx.beginPath();
  ctx.ellipse(0, 20, 86, 10, 0, 0, Math.PI * 2);
  ctx.fill();
  if (kind === "downtownTower" || kind === "glassOffice" || kind === "neonBlock" || kind === "towerSign") {
    const widths = [34, 46, 28, 40];
    for (let i = 0; i < 4; i += 1) {
      const bw = widths[i];
      const bh = 72 + seededUnit(seed, i) * 82;
      const bx = (i - 1.6) * 34 * side;
      ctx.fillStyle = kind === "neonBlock" ? "rgba(18,12,35,0.82)" : "rgba(7,17,20,0.82)";
      roundRect(bx - bw / 2, 16 - bh, bw, bh, 4);
      ctx.fill();
      ctx.fillStyle = i % 2 ? `${accent}80` : `${light}6f`;
      for (let row = 0; row < 5; row += 1) ctx.fillRect(bx - bw * 0.28, 26 - bh + row * 16, bw * 0.56, 3);
    }
  } else if (kind === "elPlatform" || kind === "railPlatform" || kind === "stoneViaduct" || kind === "drawBridge" || kind === "expressPylon") {
    const phoneMode = phoneGraphicsActive();
    ctx.strokeStyle = kind === "stoneViaduct" ? "rgba(160,172,164,0.66)" : `${accent}70`;
    ctx.lineWidth = p
  }
```

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```
honeMode ? 8 : 5;
ctx.beginPath();
ctx.moveTo(-96 * side, -16);
ctx.lineTo(96 * side, -24);
ctx.moveTo(-92 * side, 2);
ctx.lineTo(92 * side, -6);
if (!phoneMode) {
  for (let i = -3; i <= 3; i += 1) {
    ctx.moveTo(i * 28 * side, -24);
    ctx.lineTo(i * 26 * side, 22);
  }
}
ctx.stroke();
} else if (kind === "shipCrane" || kind === "harborWarehouse" || kind === "containerStack" || kind === "railYard" || kind === "overheadGantry") {
  ctx.fillStyle = "rgba(16,25,26,0.76)";
  roundRect(-76, -30, 152, 48, 4);
  ctx.fill();
  ctx.strokeStyle = `${light}82`;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-54, -30);
  ctx.lineTo(-32, -96);
  ctx.lineTo(72 * side, -74);
  ctx.moveTo(-32, -96);
  ctx.lineTo(18 * side, -26);
  ctx.stroke();
  ctx.fillStyle = `${accent}62`;
  for (let i = -2; i <= 2; i += 1) ctx.fillRect(i * 28 - 12, -18 + (i % 2) * 11, 24, 10);
} else if (kind === "grainSilo" || kind === "redBarn" || kind === "windmill" || kind === "farmStand" || kind === "truckStop") {
  ctx.fillStyle = kind === "redBarn" ? "rgba(156,42,34,0.82)" : "rgba(216,224,213,0.66)";
  roundRect(-52, -58, 48, 76, 7);
  ctx.fill();
  ctx.fillStyle = kind === "redBarn" ? "rgba(104,30,26,0.86)" : "rgba(22,28,27,0.74)";
  roundRect(8, -40, 74, 58, 5);
  ctx.fill();
  ctx.fillStyle = `${light}77`;
  ctx.beginPath();
  ctx.moveTo(2, -40);
  ctx.lineTo(44, -74);
  ctx.lineTo(88, -40);
  ctx.closePath();
  ctx.fill();
  if (kind === "windmill") {
    ctx.strokeStyle = `${accent}80`;
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-28, -68);
    ctx.lineTo(-28, 16);
    for (let i = 0; i < 4; i += 1) {
      const a = i * Math.PI * 0.5 + seed * 0.03;
      ctx.moveTo(-28, -68);
      ctx.lineTo(-28 + Math.cos(a) * 34, -68 + Math.sin(a) * 34);
    }
    ctx.stroke();
  }
}
} else if (kind === "controlTower" || kind === "hangarRow" || kind === "serviceHangar" || kind === "runwayBeacon") {
  ctx.fillStyle = "rgba(20,25,25,0.78)";
  roundRect(-82, -26, 164, 44, 14);
  ctx.fill();
  ctx.fillStyle = "rgba(228,238,236,0.62)";
  roundRect(24 * side, -86, 42, 62, 5);
  ctx.fill();
  ctx.fillStyle = `${accent}99`;
  ctx.beginPath();
  ctx.arc(45 * side, -94, 8, 0, Math.PI * 2);
  ctx.fill();
} else if (kind === "skiLift" || kind === "chalet" || kind === "snowLodge" || kind === "pineLodge" || kind === "warmingHut") {
  ctx.strokeStyle = "rgba(244,251,248,0.58)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-96 * side, -86);
  ctx.lineTo(98 * side, -58);
  ctx.stroke();
  for (let i = -2; i <= 2; i += 1) {
    ctx.fillStyle = i % 2 ? "rgba(28,54,48,0.82)" : "rgba(96,62,34,0.82)";
    roundRect(i * 40 - 24, -42 + (i % 2) * 4, 48, 58, 5);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.72)";
    ctx.beginPath();
    ctx.moveTo(i * 40 - 31, -42 + (i % 2) * 4);
    ctx.lineTo(i * 40, -70 + (i % 2) * 4);
    ctx.lineTo(i * 40 + 31, -42 + (i % 2) * 4);
    ctx.closePath();
    ctx.fill();
  }
}
} else if (kind === "pierDeck" || kind === "marinaLight" || kind === "beachHotel" || kind === "riverDock") {
  const phoneMode = phoneGraphicsActive();
  ctx.fillStyle = "rgba(70,217,255,0.14)";
  ctx.beginPath();
  ctx.ellipse(0, 18, 112, 18, 0, 0, Math.PI * 2);
  ctx.fill();
  if (phoneMode) {
    ctx.fillStyle = "rgba(130,86,45,0.6)";
    roundRect(-92, -4, 184, 22, 7);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-78 * side, 2);
    ctx.lineTo(80 * side, -8);
    ctx.stroke();
  } else {
    ctx.strokeStyle = "rgba(214,180,112,0.68)";
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-92 * side, 2);
    ctx.lineTo(86 * side, -10);
    for (let i = -3; i <= 3; i += 1) {
      ctx.moveTo(i * 28 * side, -4);
      ctx.lineTo(i * 28 * side, 22);
    }
    ctx.stroke();
  }
}
```

```
    }  
    ctx.fillStyle = `${light}88`;  
    ctx.beginPath();  
    ctx.arc(74 * side, -38, 7, 0, Math.PI * 2);  
    ctx.fill();  
  } else if (kind === "canop
```

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```
yWall" || kind === "woodMarket" || kind === "mistFalls" || kind === "redRockWall" || kind === "lowDune" || kind === "stoneOutpost")
{
  const earth = place === "rainforest" ? "rgba(35,104,58,0.66)" : place === "desert" || place === "canyon" ?
"rgba(178,92,48,0.62)" : "rgba(126,135,126,0.58)";
  ctx.fillStyle = earth;
  ctx.beginPath();
  ctx.moveTo(-118, 18);
  ctx.bezierCurveTo(-72, -64, -18, -42, 18, -82);
  ctx.bezierCurveTo(54, -38, 98, -54, 124, 20);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.18)";
  roundRect(-28, -24, 72, 7, 3);
  ctx.fill();
} else {
  ctx.fillStyle = "rgba(32,42,42,0.66)";
  roundRect(-64, -56, 128, 74, 6);
  ctx.fill();
  ctx.fillStyle = `${light}66`;
  ctx.beginPath();
  ctx.moveTo(-72, -56);
  ctx.lineTo(0, -92);
  ctx.lineTo(72, -56);
  ctx.closePath();
  ctx.fill();
}
ctx.restore();
}

function drawPlaceDestinationBoard(x, y, scale, side, route, design, theme, seed) {
  const label = route.scene || "Race Route";
  const cue = route.cue || "next sector";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.globalAlpha = phoneGraphicsActive() ? 0.42 : 0.58;
  ctx.strokeStyle = "rgba(214,226,221,0.45)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-35, 18);
  ctx.lineTo(-35, 72);
  ctx.moveTo(35, 18);
  ctx.lineTo(35, 72);
  ctx.stroke();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-76, -28, 152, 48, 6);
  ctx.fill();
  ctx.strokeStyle = design.accent || theme[1] || "#46d9ff";
  ctx.lineWidth = 2;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 12px system-ui";
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(label.toUpperCase(), 0, -10, 132);
  ctx.fillStyle = `${design.light || theme[2] || "#ffd166"}dd`;
  ctx.font = "900 9px system-ui";
  ctx.fillText(`${side < 0 ? "<" : ">"} ${cue.toUpperCase()}`, 0, 10, 128);
  if (seededUnit(seed, 1) > 0.48) {
    ctx.fillStyle = `${design.accent || theme[1] || "#46d9ff"}55`;
    ctx.fillRect(-62, 20, 124, 4);
  }
  ctx.restore();
}

function realWorldRoadHalf(roadTop, roadBottom, t) {
  return roadTop * 0.84 + (roadBottom * 0.64 - roadTop * 0.84) * Math.max(0, Math.min(1.12, t));
}

function realWorldRoadX(w, lane, t) {
  return perspectiveRoadCenter(w, t) + lane * laneWidth() * (0.32 + t * 1.04);
}

function clipRealWorldRoad(w, h, horizon, roadTop, roadBottom) {
  ctx.beginPath();
  ctx.moveTo(perspectiveRoadCenter(w, 0) - roadTop * 0.86, horizon);
  ctx.lineTo(perspectiveRoadCenter(w, 0) + roadTop * 0.86, horizon);
  ctx.lineTo(perspectiveRoadCenter(w, 1) + roadBottom * 0.67, h + 18);
  ctx.lineTo(perspectiveRoadCenter(w, 1) - roadBottom * 0.67, h + 18);
  ctx.closePath();
  ctx.clip();
}

function clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, padding = 0) {
  ctx.beginPath();
  ctx.rect(-w * 0.2, horizon - h * 0.12, w * 1.4, h * 1.24);
  ctx.moveTo(perspectiveRoadCenter(w, 0) - roadTop * 0.86 - padding, horizon - 8);
  ctx.lineTo(perspectiveRoadCenter(w, 0) + roadTop * 0.86 + padding, horizon - 8);
  ctx.lineTo(perspectiveRoadCenter(w, 1) + roadBottom * 0.67 + padding, h + 24);
  ctx.lineTo(perspectiveRoadCenter(w, 1) - roadBottom * 0.67 - padding, h + 24);
  ctx.closePath();
  ctx.clip("evenodd");
}

function roadsideAnchorX(w, h, horizon, roadTop, roadBottom, y, side, projectedX, phoneMode, extra = 0) {
  const t = Math.max(0, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
  const edge = perspectiveRoadCenter(w, t) + side * realWorldRoadHalf(roadTop, roadBottom, t);
  const clearance = extra + w * (phoneMode ? 0.045 : 0.038) + t * w * (phoneMode ? 0.08 : 0.065);
  const safe = edge + side * clearance;
  if (side < 0) return Math.max(-w * 0.16, Math.min(projectedX, safe));
  return Math.min(w * 1.16, Math.max(projectedX, safe));
}

function drawPerspectivePavementQuad(w, horizon, lane, y, length, width, style) {
  const h = canvas.height;
  const y1 = Math.max(horizon, y);
  const y2 = Math.min(h + 60, y + length);
  const t1 = Math.max(0.02, Math.min(1.12, (y1 - horizon) / Math.max(1, h - horizon)));
  const t2 = Math.max(0.02, Math.min(1.12, (y2 - horizon) / Math.max(1, h - ho
```

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```
rizon));
const x1 = realWorldRoadX(w, lane, t1);
const x2 = realWorldRoadX(w, lane, t2);
const w1 = width * (0.28 + t1 * 0.78);
const w2 = width * (0.32 + t2 * 1.04);
ctx.fillStyle = style;
ctx.beginPath();
ctx.moveTo(x1 - w1 / 2, y1);
ctx.lineTo(x1 + w1 / 2, y1);
ctx.lineTo(x2 + w2 / 2, y2);
ctx.lineTo(x2 - w2 / 2, y2);
ctx.closePath();
ctx.fill();
}

function drawRealWorldRoadSurface(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const cleanPhone = phoneCleanRoadActive();
  const motion = (raceState.roadOffset || 0) * (0.7 + Math.min(1.4, Math.max(0, raceState.speed || 0) / 210));
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);

  const asphaltTone = ctx.createLinearGradient(0, horizon, 0, h);
  asphaltTone.addColorStop(0, `${design.surface}08`);
  asphaltTone.addColorStop(0.56, phoneMode ? "rgba(255,255,255,0.025)" : "rgba(255,255,255,0.045)");
  asphaltTone.addColorStop(1, "rgba(0,0,0,0.16)");
  ctx.fillStyle = asphaltTone;
  ctx.fillRect(0, horizon, w, h - horizon);

  const patchCount = phoneMode ? 18 : 32;
  for (let i = 0; i < patchCount; i += 1) {
    const span = h - horizon + 190;
    const y = horizon + (((i * 83 + motion * (0.58 + seededUnit(seed, i) * 0.28)) % span) - 64);
    if (y < horizon || y > h + 26) continue;
    const t = Math.max(0.02, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    const lane = -1.68 + seededUnit(seed, i + 20) * 3.36;
    const x = realWorldRoadX(w, lane, t);
    const patchW = (18 + seededUnit(seed, i + 30) * 56) * (0.5 + t * 1.18) * (phoneMode ? 0.72 : 1);
    const patchH = (5 + seededUnit(seed, i + 40) * 18) * (0.48 + t * 0.9);
    ctx.globalAlpha = phoneMode ? 0.08 + t * 0.06 : 0.11 + t * 0.12;
    ctx.fillStyle = seededUnit(seed, i + 50) > 0.52 ? "rgba(244,251,248,0.35)" : "rgba(0,0,0,0.62)";
    roundRect(x - patchW / 2, y, patchW, patchH, Math.max(2, patchH * 0.35));
    ctx.fill();
  }

  const reflectorCount = phoneMode ? 9 : 13;
  for (let lane = -1.5; lane <= 1.5; lane += 1) {
    for (let i = 0; i < reflectorCount; i += 1) {
      const y = horizon + (((i * 132 + motion * 1.05 + lane * 31) % (h - horizon + 170)) - 68);
      if (y < horizon || y > h + 28) continue;
      const t = Math.max(0.02, Math.min(1.05, (y - horizon) / Math.max(1, h - horizon)));
      const x = realWorldRoadX(w, lane, t);
      const dotW = 3 + t * (phoneMode ? 5 : 8);
      const dotH = 2 + t * 4;
      ctx.globalAlpha = 0.36 + t * 0.28;
      ctx.fillStyle = lane === -1.5 || lane === 1.5 ? "rgba(255,209,102,0.72)" : "rgba(244,251,248,0.74)";
      roundRect(x - dotW / 2, y, dotW, dotH, 2);
      ctx.fill();
    }
  }

  for (let side = -1; side <= 1; side += 2) {
    const blocks = phoneMode ? 12 : 18;
    for (let i = 0; i < blocks; i += 1) {
      const y = horizon + (((i * 72 + motion * 1.25 + side * 40) % (h - horizon + 130)) - 44);
      if (y < horizon || y > h + 20) continue;
      const t = Math.max(0.03, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
      const half = realWorldRoadHalf(roadTop, roadBottom, t);
      const x = perspectiveRoadCenter(w, t) + side * half * 0.95;
      ctx.globalAlpha = 0.2 + t * 0.34;
      ctx.fillStyle = i % 2 ? "rgba(244,251,248,0.5)" : "rgba(255,51,72,0.42)";
      roundRect(x - side * (8 + t * 16), y, side * (10 + t * 22), 5 + t * 13, 2);
      ctx.fill();
    }
  }

  if (!cleanPhone) {
    const crackCount = phoneMode ? 8 : 18;
    ctx.strokeStyle = "rgba(5,8,7,0.46)";
    ctx.lineCap = "round";
    for (let i = 0; i < crackCount; i += 1) {
      const y = horizon + (((i * 101 + motion * 0.5) % (h - horizon + 150)) - 40);
      if (y < horizon || y > h + 20) continue;
      const t = Math.max(0.03, Math.min(1.02, (y - horizon) / Math.max(1, h - horizon)));
      const x = realWorldRoadX(w, -1.6 + seededUnit(seed, i + 70) * 3.2, t);
      const len = (14 + seededUnit(seed, i + 80) * 54) * (0.4 + t);
      ctx.globalAlpha = 0.12 + t * 0.18;
      ctx.lineWidth = Math.max(1, 1 + t * 2);
      ctx.beginPath();
      ctx.moveTo(x - len * 0.46, y);
      ctx.lineTo(x - len * 0.12, y + 1 + t * 2);
      ctx.lineTo(x + 1
```

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```
en * 0.18, y - 1);
    ctx.lineTo(x + len * 0.5, y + 1 + t * 3);
    ctx.stroke();
  }
}

const paintLabel = place === "airfield" ? "RWY" : place === "freight" ? "TRUCK" : place === "farm" ? "RALLY" : place === "tokyo" ?
"EXP" : place === "city" ? "SLOW" : "";
if (paintLabel) {
  const labelY = horizon + ((h - horizon) * 0.62 + motion * 0.34) % ((h - horizon) * 0.32);
  const t = Math.max(0.2, Math.min(0.92, (labelY - horizon) / Math.max(1, h - horizon)));
  ctx.globalAlpha = phoneMode ? 0.12 : 0.2;
  ctx.fillStyle = "rgba(244,251,248,0.85)";
  ctx.font = `900 ${Math.max(12, Math.min(42, w * 0.04 * (0.64 + t)))}px system-ui`;
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(paintLabel, realWorldRoadX(w, 0, t), labelY);
}

if (place === "city" || place === "tokyo" || place === "europe") {
  const crossY = horizon + (((motion * 0.42 + seed) % (h - horizon + 280)) - 92);
  if (crossY > horizon && crossY < h * 0.92) {
    for (let i = -4; i <= 4; i += 1) {
      const y = crossY + i * 11;
      const t = Math.max(0.04, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
      ctx.globalAlpha = phoneMode ? 0.1 : 0.17;
      drawPerspectivePavementQuad(w, horizon, 0, y, 5 + t * 6, laneWidth() * 4.2, "rgba(244,251,248,0.76)");
    }
  }
}

drawRealWorldRoadHardware(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion);
ctx.globalAlpha = 1;
ctx.restore();
}

function drawRealWorldSurfaceLightingPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const cleanPhone = phoneCleanRoadActive();
  const profile = realWorldSurfaceLightingProfile(place);
  const speedFactor = Math.min(1.5, Math.max(0.18, (raceState.speed || 0) / 180));
  const motion = (raceState.roadOffset || 0) * (0.74 + speedFactor * 0.42);
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  drawSurfaceCrownAndShoulderLight(w, h, horizon, roadTop, roadBottom, profile, design, theme, phoneMode);
  if (!cleanPhone) {
    drawSurfaceMaterialStreaks(w, h, horizon, roadTop, roadBottom, profile, design, theme, seed, motion, phoneMode);
    drawSurfaceTireRubberAndSeams(w, h, horizon, roadTop, roadBottom, profile, design, theme, seed, motion, phoneMode);
  }
  drawSurfaceMovingReflectionPass(w, h, horizon, roadTop, roadBottom, profile, design, theme, seed, motion, phoneMode);
  ctx.restore();
}

function realWorldSurfaceLightingProfile(place) {
  const profiles = {
    coast: { sheen: "spray", highlight: "rgba(70,217,255,0.26)", shadow: "rgba(4,14,17,0.24)", material: "salt" },
    city: { sheen: "wet", highlight: "rgba(244,251,248,0.22)", shadow: "rgba(0,0,0,0.3)", material: "asphalt" },
    canyon: { sheen: "dry", highlight: "rgba(255,183,74,0.18)", shadow: "rgba(62,25,10,0.28)", material: "dust" },
    alpine: { sheen: "snow", highlight: "rgba(244,251,248,0.3)", shadow: "rgba(22,43,48,0.25)", material: "ice" },
    harbor: { sheen: "wet", highlight: "rgba(70,217,255,0.24)", shadow: "rgba(0,0,0,0.32)", material: "dock" },
    snow: { sheen: "snow", highlight: "rgba(244,251,248,0.34)", shadow: "rgba(22,43,48,0.28)", material: "ice" },
    airfield: { sheen: "runway", highlight: "rgba(255,209,102,0.22)", shadow: "rgba(0,0,0,0.24)", material: "runway" },
    freight: { sheen: "industrial", highlight: "rgba(255,209,102,0.2)", shadow: "rgba(0,0,0,0.3)", material: "diesel" },
    farm: { sheen: "dry", highlight: "rgba(187,242,74,0.14)", shadow: "rgba(62,42,18,0.26)", material: "dirt" },
    tokyo: { sheen: "night", highlight: "rgba(255,79,216,0.28)", shadow: "rgba(0,0,0,0.36)", material: "wetNeon" },
    desert: { sheen: "heat", highlight: "rgba(255,183,74,0.22)", shadow: "rgba(72,38,14,0.26)", material: "sand" },
    rainforest: { sheen: "wet", highlight: "rgba(54,217,138,0.22)", shadow: "rgba(0,24,12,0.32)", material: "mud" },
    europe: { sheen: "stone", highlight: "rgba(220,232,239,0.22)", shadow: "rgba(0,0,0,0.26)", material: "stone" }
  };
  return profiles[place] || profiles.city;
}

function drawSurfaceCrownAndShoulderLight(w, h, horizon, roadTop, roadBottom, profile, design, theme, pho
```

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```
neMode) {
  const rows = phoneMode ? 7 : 10;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.14 : 0.2;
  for (let i = 0; i < rows; i += 1) {
    const t = 0.1 + i / rows * 0.94;
    const y = horizon + (h - horizon) * t;
    const center = perspectiveRoadCenter(w, t);
    const half = realWorldRoadHalf(roadTop, roadBottom, t) * 0.66;
    const glintW = Math.max(10, half * (0.16 + t * 0.08));
    const glintH = Math.max(2, 4 + t * 14);
    const crown = ctx.createLinearGradient(center - half, y, center + half, y);
    crown.addColorStop(0, profile.shadow);
    crown.addColorStop(0.42, "rgba(255,255,255,0)");
    crown.addColorStop(0.5, profile.highlight);
    crown.addColorStop(0.58, "rgba(255,255,255,0)");
    crown.addColorStop(1, profile.shadow);
    ctx.fillStyle = crown;
    roundRect(center - half, y - glintH * 0.5, half * 2, glintH, Math.max(2, glintH * 0.35));
    ctx.fill();
    if (profile.sheen === "night" || profile.sheen === "wet") {
      ctx.globalCompositeOperation = "screen";
      ctx.fillStyle = profile.highlight;
      roundRect(center - glintW / 2, y - glintH * 0.5, glintW, glintH, glintH * 0.5);
      ctx.fill();
      ctx.globalCompositeOperation = "source-over";
    }
  }
  ctx.restore();
}

function drawSurfaceMaterialStreaks(w, h, horizon, roadTop, roadBottom, profile, design, theme, seed, motion, phoneMode) {
  const count = phoneMode ? 10 : 18;
  ctx.save();
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * 91 + motion * 0.8 + seed * 0.002) % (h - horizon + 180)) - 54);
    if (y < horizon || y > h + 28) continue;
    const t = Math.max(0.05, Math.min(1.06, (y - horizon) / Math.max(1, h - horizon)));
    const lane = -1.55 + seededUnit(seed, i + 30) * 3.1;
    const x = realWorldRoadX(w, lane, t);
    const len = (22 + seededUnit(seed, i + 40) * 86) * (0.38 + t);
    const thickness = (2 + seededUnit(seed, i + 50) * 8) * (0.44 + t * 0.8);
    const style = surfaceMaterialStyle(profile.material, design, theme, seed + i);
    ctx.globalAlpha = phoneMode ? 0.08 + t * 0.045 : 0.12 + t * 0.09;
    ctx.fillStyle = style;
    ctx.save();
    ctx.translate(x, y);
    ctx.rotate((seededUnit(seed, i + 60) - 0.5) * 0.12);
    roundRect(-len / 2, -thickness / 2, len, thickness, Math.max(1.5, thickness * 0.45));
    ctx.fill();
    ctx.restore();
  }
  ctx.restore();
}

function surfaceMaterialStyle(material, design, theme, seed) {
  if (material === "ice") return seededUnit(seed, 1) > 0.5 ? "rgba(244,251,248,0.42)" : "rgba(70,217,255,0.2)";
  if (material === "sand" || material === "dust") return "rgba(255,183,74,0.3)";
  if (material === "dirt" || material === "mud") return material === "mud" ? "rgba(16,50,28,0.36)" : "rgba(117,86,47,0.34)";
  if (material === "runway") return "rgba(255,209,102,0.28)";
  if (material === "diesel") return seededUnit(seed, 2) > 0.55 ? "rgba(0,0,0,0.38)" : "rgba(255,209,102,0.2)";
  if (material === "wetNeon") return seededUnit(seed, 3) > 0.45 ? `${design.accent} || theme[1]44` : "rgba(244,251,248,0.2)";
  if (material === "salt") return "rgba(244,251,248,0.22)";
  if (material === "stone") return "rgba(220,232,239,0.2)";
  return "rgba(0,0,0,0.28)";
}

function drawSurfaceTireRubberAndSeams(w, h, horizon, roadTop, roadBottom, profile, design, theme, seed, motion, phoneMode) {
  const seamCount = phoneMode ? 7 : 12;
  ctx.save();
  ctx.lineCap = "round";
  for (let i = 0; i < seamCount; i += 1) {
    const y = horizon + (((i * 128 + motion * 0.62 + seed * 0.004) % (h - horizon + 220)) - 74);
    if (y < horizon || y > h + 34) continue;
    const t = Math.max(0.05, Math.min(1.05, (y - horizon) / Math.max(1, h - horizon)));
    const lane = i % 3 === 0 ? 0 : i % 2 === 0 ? -0.72 : 0.72;
    const x = realWorldRoadX(w, lane, t);
    const len = (52 + seededUnit(seed, i + 140) * 112) * (0.42 + t * 0.9);
    ctx.globalAlpha = phoneMode ? 0.09 + t * 0.05 : 0.13 + t * 0.08;
    ctx.strokeStyle = profile.sheen === "snow" ? "rgba(22,43,48,0.24)" : "rgba(0,0,0,0.36)";
    ctx.lineWidth = Math.max(1, (1.4 + t * 3.2) * (i % 3 === 0 ? 1.2 : 0.75));
    ctx.beginPath();
    ctx.moveTo(x - len * 0.45, y);
    ctx.quadraticCurveTo(x, y + Math.sin(seed + i) * 4, x + len * 0.
```

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```
48, y + 1 + t * 2);
    ctx.stroke();
    if (i % 4 === 0) {
        drawPerspectivePavementQuad(w, horizon, lane, y + 10 + t * 8, 3 + t * 9, laneWidth() * (0.46 + t * 0.18),
"rgba(244,251,248,0.12)");
    }
}
ctx.restore();
}

function drawSurfaceMovingReflectionPass(w, h, horizon, roadTop, roadBottom, profile, design, theme, seed, motion, phoneMode) {
    const count = phoneMode ? 5 : 9;
    const night = profile.shen === "night" || raceIsNight(selectedRace);
    const wet = profile.shen === "wet" || profile.shen === "spray" || profile.shen === "night" || profile.shen === "snow";
    ctx.save();
    if (wet || night) ctx.globalCompositeOperation = "screen";
    for (let i = 0; i < count; i += 1) {
        const y = horizon + (((i * 176 + motion * 0.5 + seed * 0.005) % (h - horizon + 260)) - 86);
        if (y < horizon || y > h + 46) continue;
        const t = Math.max(0.06, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
        const lane = i % 2 === 0 ? -1.1 : 1.1;
        const x = realWorldRoadX(w, lane + (seededUnit(seed, i + 210) - 0.5) * 0.5, t);
        const width = (22 + seededUnit(seed, i + 220) * 76) * (0.42 + t);
        const height = (5 + seededUnit(seed, i + 230) * 18) * (0.46 + t * 0.72);
        const color = wet ? profile.highlight : surfaceMaterialStyle(profile.material, design, theme, seed + i);
        ctx.globalAlpha = phoneMode ? 0.07 + t * 0.06 : 0.1 + t * 0.09;
        const grad = ctx.createRadialGradient(x, y, 1, x, y, Math.max(width, height));
        grad.addColorStop(0, color);
        grad.addColorStop(1, "rgba(0,0,0,0)");
        ctx.fillStyle = grad;
        ctx.beginPath();
        ctx.ellipse(x, y, width, height, 0, 0, Math.PI * 2);
        ctx.fill();
    }
    ctx.globalCompositeOperation = "source-over";
    if (profile.shen === "heat") {
        ctx.globalAlpha = phoneMode ? 0.08 : 0.12;
        ctx.strokeStyle = "rgba(255,183,74,0.32)";
        ctx.lineWidth = 1.5;
        for (let i = 0; i < 6; i += 1) {
            const y = horizon + h * (0.09 + i * 0.035);
            ctx.beginPath();
            ctx.moveTo(w * 0.28, y);
            ctx.bezierCurveTo(w * 0.38, y - 8, w * 0.56, y + 8, w * 0.72, y - 3);
            ctx.stroke();
        }
    }
    ctx.restore();
}

function drawRealWorldWeatherDisasterPass(w, h, horizon, roadTop, roadBottom, place, route, design, theme, seed) {
    const phoneMode = phoneGraphicsActive();
    const length = Math.max(1, raceLength());
    const progress = Math.max(0, Math.min(1, raceState.distance / length));
    const condition = raceState.roadCondition || routeWeatherCondition(place, progress, raceState.elapsed || 0);
    const motion = (raceState.roadOffset || 0) * (0.62 + Math.min(1.1, Math.max(0, raceState.speed || 0) / 260));
    ctx.save();
    drawWeatherSkyMoodPass(w, h, horizon, place, condition, design, theme, seed, motion, phoneMode);
    drawWeatherRoadConditionSurface(w, h, horizon, roadTop, roadBottom, place, condition, design, theme, seed, motion, phoneMode);
    drawWeatherRailAndHazardPass(w, h, horizon, roadTop, roadBottom, place, route, condition, design, theme, seed, motion, phoneMode);
    ctx.restore();
}

function drawWeatherSkyMoodPass(w, h, horizon, place, condition, design, theme, seed, motion, phoneMode) {
    const sky = condition.sky || "sun";
    ctx.save();
    if (sky === "night") {
        const night = ctx.createLinearGradient(0, 0, 0, horizon + h * 0.18);
        night.addColorStop(0, "rgba(2,7,16,0.34)");
        night.addColorStop(1, "rgba(2,7,16,0)");
        ctx.fillStyle = night;
        ctx.fillRect(0, 0, w, horizon + h * 0.2);
        ctx.globalCompositeOperation = "screen";
        ctx.fillStyle = `${design.light || theme[2] || "#ffd166"}44`;
        for (let i = 0; i < (phoneMode ? 5 : 8); i += 1) {
            const x = w * (0.08 + seededUnit(seed, i + 90) * 0.84);
            const y = horizon * (0.18 + seededUnit(seed, i + 95) * 0.62);
            ctx.beginPath();
            ctx.arc(x, y, 1.2 + seededUnit(seed, i + 96) * 1.8, 0, Math.PI * 2);
            ctx.fill();
        }
    } else if (sky === "sun") {
        const x = w * (0.18 + seededUnit(seed, 102) * 0.18);
        const y = horizon * (0.2 + seededUnit(seed, 103) * 0.22);
        const r = Math.max(24, Math.min(58, w * 0.08));
        ctx.globalCompositeOperation = "screen";
        const sun = ctx.createRadialGradient(x, y, 2, x, y, r * 2.6);
        sun.addColorSt
```


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```
op(0, "rgba(255,245,194,0.68)");
sun.addColorStop(0.36, "rgba(255,209,102,0.28)");
sun.addColorStop(1, "rgba(255,209,102,0)");
ctx.fillStyle = sun;
ctx.beginPath();
ctx.arc(x, y, r * 2.6, 0, Math.PI * 2);
ctx.fill();
} else if (sky === "cloud" || sky === "rain" || sky === "snow" || sky === "storm") {
  const cloudAlpha = sky === "storm" ? 0.34 : sky === "cloud" ? 0.22 : 0.18;
  ctx.fillStyle = "rgba(14,24,31,${cloudAlpha})";
  for (let i = 0; i < (phoneMode ? 5 : 8); i += 1) {
    const x = w * (0.04 + seededUnit(seed, i + 120) * 0.92) + Math.sin(motion * 0.004 + i) * 24;
    const y = horizon * (0.16 + seededUnit(seed, i + 130) * 0.48);
    const sx = w * (0.11 + seededUnit(seed, i + 140) * 0.12);
    const sy = h * (0.018 + seededUnit(seed, i + 150) * 0.026);
    ctx.beginPath();
    ctx.ellipse(x, y, sx, sy, 0, 0, Math.PI * 2);
    ctx.fill();
  }
  if (sky === "storm") {
    ctx.strokeStyle = "rgba(255,245,194,0.28)";
    ctx.lineWidth = phoneMode ? 1.4 : 2;
    const lx = w * (0.64 + seededUnit(seed, 161) * 0.2);
    ctx.beginPath();
    ctx.moveTo(lx, horizon * 0.22);
    ctx.lineTo(lx - w * 0.025, horizon * 0.38);
    ctx.lineTo(lx + w * 0.018, horizon * 0.36);
    ctx.lineTo(lx - w * 0.018, horizon * 0.54);
    ctx.stroke();
  }
}

if (sky === "rain" || sky === "storm") {
  ctx.strokeStyle = sky === "storm" ? "rgba(222,249,255,0.32)" : "rgba(222,249,255,0.22)";
  ctx.lineWidth = phoneMode ? 1 : 1.4;
  const drops = phoneMode ? 26 : 44;
  for (let i = 0; i < drops; i += 1) {
    const x = (seededUnit(seed, i + 180) * w + motion * 0.08) % w;
    const y = horizon * 0.12 + seededUnit(seed, i + 190) * h * 0.74;
    const len = phoneMode ? 7 + seededUnit(seed, i + 200) * 9 : 10 + seededUnit(seed, i + 200) * 16;
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x - len * 0.22, y + len);
    ctx.stroke();
  }
} else if (sky === "snow") {
  ctx.fillStyle = "rgba(244,251,248,0.44)";
  const flakes = phoneMode ? 28 : 48;
  for (let i = 0; i < flakes; i += 1) {
    const x = (seededUnit(seed, i + 210) * w + Math.sin(motion * 0.006 + i) * 18) % w;
    const y = (seededUnit(seed, i + 220) * h + motion * 0.12) % h;
    ctx.beginPath();
    ctx.arc(x, y, 1.1 + seededUnit(seed, i + 230) * 2.2, 0, Math.PI * 2);
    ctx.fill();
  }
}
ctx.restore();
}

function drawWeatherRoadConditionSurface(w, h, horizon, roadTop, roadBottom, place, condition, design, theme, seed, motion, phoneMode) {
  const surface = condition.surface || "dry";
  if (surface === "dry") return;
  const count = phoneMode ? 7 : 12;
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  if (surface === "standingwater" || surface === "wet") {
    ctx.globalCompositeOperation = "screen";
    const wash = ctx.createLinearGradient(0, horizon, 0, h);
    wash.addColorStop(0, "rgba(70,217,255,0)");
    wash.addColorStop(0.66, surface === "standingwater" ? "rgba(103,216,255,0.16)" : "rgba(222,249,255,0.08)");
    wash.addColorStop(1, surface === "standingwater" ? "rgba(103,216,255,0.22)" : "rgba(222,249,255,0.1)");
    ctx.fillStyle = wash;
    ctx.fillRect(0, horizon, w, h - horizon);
    ctx.globalCompositeOperation = "source-over";
  }
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * 151 + motion * 0.82 + seed * 0.004) % (h - horizon + 240)) - 70);
    if (y < horizon + 4 || y > h + 34) continue;
    const t = Math.max(0.07, Math.min(1.06, (y - horizon) / Math.max(1, h - horizon)));
    const lane = -1.45 + seededUnit(seed, i + 260) * 2.9;
    const x = realWorldRoadX(w, lane, t);
    const width = (38 + seededUnit(seed, i + 270) * 102) * (0.38 + t * 0.84);
    const height = (5 + seededUnit(seed, i + 280) * 22) * (0.42 + t * 0.9);
    drawWeatherRoadPatch(surface, x, y, width, height, t, place, design, theme, seed + i * 17);
  }
  if (condition.hazard === "earthquake" || surface === "cracked") {
    drawWeatherQuakeRoadCracks(w, h, horizon, seed, motion, phoneMode);
  }
  ctx.restore();
}

function drawWeatherRoadPatch(surface, x, y, width, height, t, place, design, theme, seed) {
  ctx.save();
  ctx.tra
```

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```
nslate(x, y);
ctx.rotate((seededUnit(seed, 1) - 0.5) * 0.12);
if (surface === "wet" || surface === "standingWater") {
  ctx.globalCompositeOperation = "screen";
  const puddle = ctx.createRadialGradient(0, 0, 1, 0, 0, Math.max(width, height));
  puddle.addColorStop(0, surface === "standingWater" ? "rgba(147,232,255,0.38)" : "rgba(222,249,255,0.2)");
  puddle.addColorStop(0.48, surface === "standingWater" ? "rgba(70,217,255,0.2)" : "rgba(70,217,255,0.1)");
  puddle.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = puddle;
  ctx.beginPath();
  ctx.ellipse(0, 0, width, height, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = "rgba(244,251,248,0.26)";
  ctx.lineWidth = Math.max(1, height * 0.12);
  ctx.beginPath();
  ctx.moveTo(-width * 0.42, -height * 0.12);
  ctx.quadraticCurveTo(0, -height * 0.7, width * 0.44, -height * 0.05);
  ctx.stroke();
} else if (surface === "oilSlick" || surface === "blackIce") {
  ctx.globalCompositeOperation = "screen";
  const gloss = ctx.createRadialGradient(0, 0, 1, 0, 0, Math.max(width, height));
  gloss.addColorStop(0, surface === "blackIce" ? "rgba(8,18,28,0.78)" : "rgba(0,0,0,0.7)");
  gloss.addColorStop(0.58, surface === "blackIce" ? "rgba(70,217,255,0.18)" : `${design.accent || theme[1] || "#46d9ff"}24`);
  gloss.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = gloss;
  ctx.beginPath();
  ctx.ellipse(0, 0, width * 0.82, height, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = surface === "blackIce" ? "rgba(222,249,255,0.36)" : "rgba(255,209,102,0.22)";
  ctx.lineWidth = Math.max(1, height * 0.1);
  ctx.beginPath();
  ctx.moveTo(-width * 0.3, -height * 0.35);
  ctx.lineTo(width * 0.32, -height * 0.18);
  ctx.stroke();
} else if (surface === "snowSlush") {
  ctx.fillStyle = "rgba(244,251,248,0.36)";
  ctx.beginPath();
  ctx.ellipse(0, 0, width * 0.75, height * 0.92, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(70,217,255,0.16)";
  ctx.beginPath();
  ctx.ellipse(width * 0.18, height * 0.1, width * 0.34, height * 0.48, 0, 0, Math.PI * 2);
  ctx.fill();
} else if (surface === "mud" || surface === "dust") {
  ctx.fillStyle = surface === "mud" ? "rgba(72,42,22,0.42)" : "rgba(255,183,74,0.26)";
  ctx.beginPath();
  ctx.ellipse(0, 0, width * 0.68, height * 0.82, 0, 0, Math.PI * 2);
  ctx.fill();
} else if (surface === "cracked") {
  ctx.strokeStyle = "rgba(0,0,0,0.5)";
  ctx.lineWidth = Math.max(1, 1.4 + t * 2.8);
  ctx.beginPath();
  ctx.moveTo(-width * 0.42, 0);
  ctx.lineTo(-width * 0.14, -height * 0.6);
  ctx.lineTo(width * 0.1, height * 0.32);
  ctx.lineTo(width * 0.44, -height * 0.2);
  ctx.stroke();
}
ctx.restore();
}

function drawWeatherQuakeRoadCracks(w, h, horizon, seed, motion, phoneMode) {
  ctx.save();
  ctx.strokeStyle = "rgba(0,0,0,0.56)";
  ctx.lineCap = "round";
  const cracks = phoneMode ? 4 : 7;
  for (let i = 0; i < cracks; i += 1) {
    const y = horizon + (((i * 211 + motion * 0.52 + seed * 0.005) % (h - horizon + 260)) - 82);
    if (y < horizon + 10 || y > h * 0.96) continue;
    const t = Math.max(0.08, Math.min(1.05, (y - horizon) / Math.max(1, h - horizon)));
    const x = realWorldRoadX(w, -0.9 + seededUnit(seed, i + 310) * 1.8, t);
    ctx.lineWidth = Math.max(1.5, 1.8 + t * 4.2);
    ctx.beginPath();
    ctx.moveTo(x - 38 * t, y);
    ctx.lineTo(x - 14 * t, y + 9 * t);
    ctx.lineTo(x + 8 * t, y - 5 * t);
    ctx.lineTo(x + 42 * t, y + 12 * t);
    ctx.stroke();
    ctx.strokeStyle = "rgba(244,251,248,0.12)";
    ctx.beginPath();
    ctx.moveTo(x - 20 * t, y + 5 * t);
    ctx.lineTo(x + 34 * t, y + 16 * t);
    ctx.stroke();
    ctx.strokeStyle = "rgba(0,0,0,0.56)";
  }
  ctx.restore();
}

function drawWeatherRailAndHazardPass(w, h, horizon, roadTop, roadBottom, place, route, condition, design, theme, seed, motion, phoneMode) {
  ctx.save();
  clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 0 : 4);
  if (condition.hazard === "train" || place === "freight" || place === "city" || place === "tokyo" || place === "europe") {
    drawWeatherPassingTrain(w, h, horizon, roadTop, roadBottom, place, design, th
```

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```
eme, seed, motion, phoneMode);
}
if (condition.hazard === "tornado") drawWeatherTornado(w, h, horizon, roadTop, roadBottom, design, theme, seed, motion,
phoneMode);
if (condition.hazard === "tidalSurge" || condition.hazard === "coastalFlood" || condition.hazard === "floodedBridge") {
  drawWeatherCoastalFloodView(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode);
}
if (condition.hazard === "earthquake") drawWeatherEarthquakeRoadsideDamage(w, h, horizon, roadTop, roadBottom, design, theme,
seed, motion, phoneMode);
ctx.restore();
}

function drawWeatherPassingTrain(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode) {
  const side = seededUnit(seed, 351) > 0.48 ? 1 : -1;
  const t = phoneMode ? 0.26 + seededUnit(seed, 352) * 0.16 : 0.22 + seededUnit(seed, 352) * 0.18;
  const y = horizon + (h - horizon) * t + Math.sin(motion * 0.004) * 6;
  const half = realWorldRoadHalf(roadTop, roadBottom, t);
  const baseX = perspectiveRoadCenter(w, t) + side * (half + w * (phoneMode ? 0.17 : 0.2));
  const trainMotion = ((motion * 0.7 + seed * 0.02) % (w * 0.42)) - w * 0.21;
  const x = baseX + side * trainMotion;
  const scale = 0.62 + t * (phoneMode ? 1.0 : 1.15);
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale * side, scale);
  ctx.strokeStyle = "rgba(180,192,188,0.7)";
  ctx.lineWidth = Math.max(2, 3.2 * scale);
  ctx.beginPath();
  ctx.moveTo(-160, 25);
  ctx.lineTo(160, 16);
  ctx.moveTo(-160, 34);
  ctx.lineTo(160, 25);
  ctx.stroke();
  ctx.fillStyle = "rgba(3,8,10,0.34)";
  ctx.beginPath();
  ctx.ellipse(0, 43, 170, 13, -0.03, 0, Math.PI * 2);
  ctx.fill();
  const carColor = place === "tokyo" ? `${design.accent || theme[1] || "#46d9ff"}cc` : place === "freight" ? "rgba(255,183,74,0.82)";
  : "rgba(220,232,239,0.82)";
  for (let i = -2; i <= 2; i += 1) {
    const cx = i * 58;
    ctx.fillStyle = carColor;
    roundRect(cx - 26, -14 + (i % 2) * 2, 52, 31, 5);
    ctx.fill();
    ctx.fillStyle = "rgba(5,8,7,0.72)";
    roundRect(cx - 17, -8, 34, 8, 2);
    ctx.fill();
    ctx.fillStyle = "#050807";
    ctx.beginPath();
    ctx.arc(cx - 15, 19, 5, 0, Math.PI * 2);
    ctx.arc(cx + 15, 19, 5, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.restore();
}

function drawWeatherTornado(w, h, horizon, roadTop, roadBottom, design, theme, seed, motion, phoneMode) {
  const side = seededUnit(seed, 380) > 0.5 ? 1 : -1;
  const x = w * (side > 0 ? 0.78 : 0.22) + Math.sin(motion * 0.003) * 18;
  const baseY = horizon + h * (phoneMode ? 0.22 : 0.2);
  const topY = horizon * 0.36;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.48 : 0.42;
  ctx.strokeStyle = "rgba(60,62,58,0.72)";
  ctx.lineWidth = phoneMode ? 8 : 11;
  ctx.lineCap = "round";
  ctx.beginPath();
  ctx.moveTo(x - side * 32, topY);
  ctx.bezierCurveTo(x + side * 26, topY + 54, x - side * 20, baseY - 42, x + side * 10, baseY);
  ctx.stroke();
  ctx.strokeStyle = "rgba(244,251,248,0.18)";
  ctx.lineWidth = phoneMode ? 3 : 4;
  for (let i = 0; i < 4; i += 1) {
    const y = topY + i * ((baseY - topY) / 4);
    ctx.beginPath();
    ctx.ellipse(x + Math.sin(motion * 0.008 + i) * 12, y, 42 - i * 7, 7, -0.15, 0, Math.PI * 2);
    ctx.stroke();
  }
  ctx.fillStyle = "rgba(117,86,47,0.26)";
  ctx.beginPath();
  ctx.ellipse(x + side * 8, baseY + 10, 72, 14, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.restore();
}

function drawWeatherCoastalFloodView(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode) {
  const side = place === "harbor" || seededUnit(seed, 410) > 0.5 ? 1 : -1;
  ctx.save();
  ctx.globalCompositeOperation = "screen";
  for (let i = 0; i < (phoneMode ? 4 : 6); i += 1) {
    const t = 0.36 + i * 0.11;
    const y = horizon + (h - horizon) * t + Math.sin(motion * 0.01 + i) * 7;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.05 + t * 0.08));
    const width = w * (0.22 + t * 0.22);
    const height = h * (0.018 + t * 0.025);
    const wave = ctx.createRadialGradient(x, y, 1, x, y, width);
    wave.addColorStop(0, "rgba(147,232,255,0.46)");
    wave.addColorStop(0.48, "rgba(70,217,255,0.18)");
    wa
```

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```
ve.addColorStop(1, "rgba(0,0,0,0)");
ctx.fillStyle = wave;
ctx.beginPath();
ctx.ellipse(x, y, width, height, 0.02 * side, 0, Math.PI * 2);
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.38)";
ctx.lineWidth = Math.max(1.4, 2 + t * 4);
ctx.beginPath();
ctx.moveTo(x - side * width * 0.54, y - height * 0.2);
ctx.quadraticCurveTo(x, y - height * 1.3, x + side * width * 0.48, y - height * 0.1);
ctx.stroke();
}
ctx.restore();
}

function drawWeatherEarthquakeRoadsideDamage(w, h, horizon, roadTop, roadBottom, design, theme, seed, motion, phoneMode) {
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.62 : 0.5;
  for (let i = 0; i < (phoneMode ? 4 : 6); i += 1) {
    const side = i % 2 === 0 ? -1 : 1;
    const t = 0.34 + seededUnit(seed, i + 440) * 0.44;
    const y = horizon + (h - horizon) * t;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.04 + t * 0.06));
    const scale = 0.6 + t * 1.0;
    ctx.save();
    ctx.translate(x, y);
    ctx.scale(scale * side, scale);
    ctx.fillStyle = "rgba(72,72,68,0.72)";
    roundRect(-42, -8, 84, 16, 3);
    ctx.fill();
    ctx.strokeStyle = "rgba(0,0,0,0.54)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-30, -8);
    ctx.lineTo(-12, 8);
    ctx.lineTo(8, -6);
    ctx.lineTo(30, 8);
    ctx.stroke();
    ctx.fillStyle = `${design.light || theme[2] || "#ffd166"}66`;
    ctx.fillRect(-38, -18, 14, 7);
    ctx.restore();
  }
  ctx.restore();
}

function drawRealWorldRoadHardware(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion) {
  const phoneMode = phoneGraphicsActive();
  const hardwareCount = phoneMode ? 5 : 14;
  const cityRoad = place === "city" || place === "tokyo" || place === "europe" || place === "harbor";
  for (let i = 0; i < hardwareCount; i += 1) {
    const y = horizon + ((i * 144 + motion * 0.74 + seed * 0.01) % (h - horizon + 240)) - 70;
    if (y < horizon || y > h + 42) continue;
    const t = Math.max(0.04, Math.min(1.04, (y - horizon) / Math.max(1, h - horizon)));
    const lane = cityRoad ? (i % 3) - 1 : -1.55 + seededUnit(seed, i + 120) * 3.1;
    const x = realWorldRoadX(w, lane * 0.55, t);
    const size = (10 + seededUnit(seed, i + 130) * 18) * (0.45 + t);
    const kind = cityRoad ? (i % 3 === 0 ? "manhole" : i % 3 === 1 ? "drain" : "arrow") : (i % 3 === 0 ? "pothole" : i % 3 === 1 ?
"patchedJoint" : "arrow");
    ctx.save();
    ctx.translate(x, y);
    ctx.globalAlpha = phoneMode ? 0.12 + t * 0.08 : 0.18 + t * 0.14;
    if (kind === "manhole") {
      ctx.fillStyle = "rgba(5,8,7,0.62)";
      ctx.beginPath();
      ctx.ellipse(0, 0, size * 0.9, size * 0.34, 0, 0, Math.PI * 2);
      ctx.fill();
      ctx.strokeStyle = "rgba(244,251,248,0.26)";
      ctx.lineWidth = Math.max(1, size * 0.07);
      ctx.beginPath();
      ctx.ellipse(0, 0, size * 0.68, size * 0.25, 0, 0, Math.PI * 2);
      ctx.stroke();
      if (!phoneMode) {
        for (let s = -2; s <= 2; s += 1) {
          ctx.beginPath();
          ctx.moveTo(s * size * 0.16, -size * 0.2);
          ctx.lineTo(s * size * 0.16, size * 0.2);
          ctx.stroke();
        }
      }
    }
    else if (kind === "drain") {
      ctx.fillStyle = "rgba(5,8,7,0.66)";
      roundRect(-size * 1.2, -size * 0.22, size * 2.4, size * 0.44, Math.max(2, size * 0.08));
      ctx.fill();
      ctx.strokeStyle = "rgba(244,251,248,0.22)";
      ctx.lineWidth = Math.max(1, size * 0.05);
      for (let s = phoneMode ? -1 : -3; s <= (phoneMode ? 1 : 3); s += 1) {
        ctx.beginPath();
        ctx.moveTo(s * size * 0.28, -size * 0.19);
        ctx.lineTo(s * size * 0.28, size * 0.19);
        ctx.stroke();
      }
    }
    else if (kind === "pothole") {
      ctx.fillStyle = "rgba(0,0,0,0.48)";
      ctx.beginPath();
      ctx.ellipse(0, 0, size * 1.25, size * 0.34, seededUnit(seed, i + 140) - 0.5, 0, Math.PI * 2);
      ctx.fill();
      ctx.strokeStyle = "rgba(244,251,248,0.11)";
      ctx.lineWidth = Math.max(1, size * 0.04);
      ctx.stroke();
    }
    else if (kind === "patchedJoint") {
      ctx.fillStyle = "rgba(244,251,248,0.18)";
      roundRect(-size * 1.5, -size * 0.18,
```

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```
size * 3, size * 0.36, Math.max(2, size * 0.08));
ctx.fill();
ctx.fillStyle = "rgba(0,0,0,0.22)";
roundRect(-size * 1.32, -size * 0.09, size * 2.64, size * 0.16, 2);
ctx.fill();
} else if (phoneMode) {
  ctx.globalCompositeOperation = "screen";
  ctx.fillStyle = "rgba(222,249,255,0.16)";
  ctx.beginPath();
  ctx.ellipse(0, 0, size * 1.45, size * 0.18, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = "rgba(244,251,248,0.2)";
  ctx.lineWidth = Math.max(1, size * 0.05);
  ctx.beginPath();
  ctx.moveTo(-size * 1.2, 0);
  ctx.lineTo(size * 1.2, size * 0.08);
  ctx.stroke();
} else {
  ctx.fillStyle = "rgba(244,251,248,0.42)";
  ctx.beginPath();
  ctx.moveTo(0, -size * 1.35);
  ctx.lineTo(size * 0.48, -size * 0.22);
  ctx.lineTo(size * 0.18, -size * 0.22);
  ctx.lineTo(size * 0.18, size * 1.12);
  ctx.lineTo(-size * 0.18, size * 1.12);
  ctx.lineTo(-size * 0.18, -size * 0.22);
  ctx.lineTo(-size * 0.48, -size * 0.22);
  ctx.closePath();
  ctx.fill();
}
ctx.restore();
}
}

const shoulderLines = phoneMode ? 2 : 6;
for (let side = -1; side <= 1; side += 2) {
  for (let i = 0; i < shoulderLines; i += 1) {
    const y = horizon + (((i * 164 + motion * 0.8 + side * 53) % (h - horizon + 260)) - 88);
    if (y < horizon || y > h + 30) continue;
    const t = Math.max(0.04, Math.min(1.08, (y - horizon) / Math.max(1, h - horizon)));
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * half * 0.84;
    ctx.globalAlpha = phoneMode ? 0.1 : 0.18;
    ctx.strokeStyle = "rgba(244,251,248,0.58)";
    ctx.lineWidth = Math.max(1, 1.5 + t * 2.5);
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x + side * (18 + t * 34), y + 1 + t * 2);
    ctx.stroke();
  }
}
}

function realWorldPropSet(place) {
  const sets = {
    coast: ["person", "bystanders", "animal", "parked", "palm", "cliff", "fence", "roadSign", "lifeguard", "pier"],
    city: ["building", "person", "bystanders", "animal", "parked", "streetlight", "busStop", "roadSign", "fence", "speedCamera"],
    canyon: ["rock", "cliff", "animal", "person", "roadSign", "fence", "parked", "telephone", "mileMarker", "tunnelMouth"],
    alpine: ["pine", "animal", "mountain", "person", "cabin", "guardrail", "roadSign", "snowbank", "tunnelMouth"],
    harbor: ["crane", "boat", "person", "bystanders", "animal", "warehouse", "streetlight", "parked", "fence", "pier"],
    snow: ["pine", "snowbank", "animal", "person", "cabin", "roadSign", "fence", "mountain", "tunnelMouth"],
    airfield: ["hangar", "beacon", "plane", "person", "bystanders", "fence", "roadSign", "warehouse", "radar", "serviceTruck"],
    freight: ["trailer", "warehouse", "person", "bystanders", "animal", "streetlight", "parked", "roadSign", "fence", "fuelStop"],
    farm: ["barn", "field", "animal", "tractor", "person", "bystanders", "fence", "telephone", "silo", "market"],
    tokyo: ["neon", "building", "person", "bystanders", "animal", "parked", "streetlight", "busStop", "roadSign", "speedCamera"],
    desert: ["dune", "rock", "animal", "person", "roadSign", "fence", "telephone", "parked", "mileMarker", "tunnelMouth"],
    rainforest: ["canopy", "animal", "person", "bystanders", "bridge", "fence", "roadSign", "parked", "market", "waterfall"],
    europe: ["village", "mountain", "animal", "person", "bystanders", "parked", "streetlight", "roadSign", "fence", "market"]
  };
  return sets[place] || sets.city;
}

function drawRealWorldRoadsideLife(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const props = realWorldPropSet(place);
  const count = phoneMode ? 14 : 24;
  const motion = (raceState.roadOffset || 0) * (0.58 + Math.min(1.1, Math.max(0, raceState.speed || 0) / 260));
  ctx.save();
  clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 2 : 8);
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * (phoneMode ? 132 : 112) + motion * (0.58 + seededUnit(seed, i) * 0.2)) % (h - horizon + 320)) - 88);
    if (y <
```

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```
horizon - 8 || y > h + 72) continue;
const t = Math.max(0.04, Math.min(1.1, (y - horizon) / Math.max(1, h - horizon)));
const side = 1 % 2 === 0 ? -1 : 1;
const half = realWorldRoadHalf(roadTop, roadBottom, t);
const projectedX = perspectiveRoadCenter(w, t) + side * (half + w * (0.028 + t * 0.12 + seededUnit(seed, i + 10) * 0.045));
const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, y, side, projectedX, phoneMode, phoneMode ? 6 : 14);
const scale = (0.42 + t * (phoneMode ? 1.0 : 1.2)) * (0.92 + seededUnit(seed, i + 22) * 0.34);
const kind = worldSceneryProp(props[(i + Math.floor(seed % props.length)) % props.length], place, i, phoneMode);
ctx.globalAlpha = Math.min(1, phoneMode ? 0.72 + t * 0.28 : 0.68 + t * 0.28);
drawRealWorldRoadsideProp(kind, x, y, scale, side, place, design, theme, seed + i * 31);
}
drawRealWorldSidewalkBand(w, h, horizon, roadTop, roadBottom, place, design);
ctx.restore();
}

function drawRealWorldSidewalkBand(w, h, horizon, roadTop, roadBottom, place, design) {
  if (place !== "city" && place !== "tokyo" && place !== "europe" && place !== "harbor") return;
  ctx.save();
  ctx.globalAlpha = phoneGraphicsActive() ? 0.12 : 0.2;
  for (let side = -1; side <= 1; side += 2) {
    const grad = ctx.createLinearGradient(0, horizon, 0, h);
    grad.addColorStop(0, "rgba(244,251,248,0)");
    grad.addColorStop(1, place === "tokyo" ? `${design.accent}33` : "rgba(150,160,156,0.35)");
    ctx.fillStyle = grad;
    ctx.beginPath();
    ctx.moveTo(perspectiveRoadCenter(canvas.width, 0) + side * roadTop * 0.92, horizon);
    ctx.lineTo(perspectiveRoadCenter(canvas.width, 0) + side * roadTop * 1.15, horizon);
    ctx.lineTo(perspectiveRoadCenter(canvas.width, 1) + side * roadBottom * 0.9, h + 20);
    ctx.lineTo(perspectiveRoadCenter(canvas.width, 1) + side * roadBottom * 0.68, h + 20);
    ctx.closePath();
    ctx.fill();
  }
  ctx.restore();
}

function drawRealWorldRoadsideProp(kind, x, y, scale, side, place, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.globalAlpha *= Math.max(0.58, Math.min(1, 0.64 + scale * 0.2));
  if (kind === "person") {
    drawRoadsidePeopleCluster(side, accent, light, seed);
  } else if (kind === "bystanders") {
    drawRoadsideBystanderBarrier(side, accent, light, seed);
  } else if (kind === "animal") {
    drawRoadsideAnimalGroup(routeAnimalSpecies(place, seed), side, accent, light, seed);
  } else if (kind === "parked") {
    drawRoadsideParkedVehicle(side, seededUnit(seed, 2) > 0.5 ? accent : light);
  } else if (kind === "building" || kind === "neon" || kind === "village" || kind === "warehouse") {
    drawRoadsideBuilding(kind, side, place, design, theme, seed);
  } else if (kind === "streetlight" || kind === "telephone" || kind === "beacon") {
    drawRoadsidePole(kind, side, accent, light);
  } else if (kind === "fence" || kind === "guardrail") {
    drawRoadsideFence(side, kind === "guardrail" ? "rgba(205,218,214,0.52)" : "rgba(150,117,80,0.58)");
  } else if (kind === "roadSign" || kind === "busStop") {
    drawRoadsideSign(kind, side, selectedRace && selectedRace.sign ? selectedRace.sign : "ROUTE", accent);
  } else if (kind === "palm" || kind === "pine" || kind === "canopy") {
    drawRoadsideTree(kind, side);
  } else if (kind === "crane" || kind === "hangar" || kind === "trailer" || kind === "barn" || kind === "cabin") {
    drawRoadsideBuilding(kind, side, place, design, theme, seed);
  } else if (kind === "boat" || kind === "plane" || kind === "tractor") {
    drawRoadsideMachine(kind, side, accent, light);
  } else if (kind === "billboard" || kind === "speedCamera" || kind === "mileMarker") {
    drawRoadsideServiceSign(kind, side, place, design, theme, seed);
  } else if (kind === "pier" || kind === "container" || kind === "fuelStop" || kind === "silo" || kind === "market" || kind === "radar" || kind === "serviceTruck" || kind === "tunnelMouth" || kind === "waterfall") {
    drawRoadsideWorldFixture(kind, side, place, design, theme, seed);
  } else {
    drawRoa
```

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```
dsideTerrain(kind, side, place, accent, light);
}
ctx.restore();
}

function drawRoadsidePeopleCluster(side, accent, light, seed) {
  const count = seededUnit(seed, 1) > 0.62 ? 3 : 2;
  for (let i = 0; i < count; i += 1) {
    const px = side * (i * 9 - 8);
    const py = (i % 2) * 3;
    const jacket = i % 2 ? accent : light;
    ctx.fillStyle = "rgba(5,8,7,0.38)";
    ctx.beginPath();
    ctx.ellipse(px, py + 18, 7, 2.5, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.42)";
    ctx.lineWidth = 2.2;
    ctx.beginPath();
    ctx.moveTo(px, py - 7);
    ctx.lineTo(px - side * 5, py + 7);
    ctx.moveTo(px, py - 5);
    ctx.lineTo(px + side * 6, py + 3);
    ctx.stroke();
    ctx.fillStyle = jacket;
    roundRect(px - 4, py - 5, 8, 15, 3);
    ctx.fill();
    ctx.fillStyle = "rgba(238,196,150,0.9)";
    ctx.beginPath();
    ctx.arc(px, py - 11, 4, 0, Math.PI * 2);
    ctx.fill();
  }
}

function drawRoadsideBystanderBarrier(side, accent, light, seed) {
  ctx.strokeStyle = "rgba(244,251,248,0.48)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-56 * side, 3);
  ctx.lineTo(62 * side, -3);
  ctx.moveTo(-56 * side, -9);
  ctx.lineTo(62 * side, -15);
  for (let i = -2; i <= 2; i += 1) {
    ctx.moveTo(i * 24 * side, -18);
    ctx.lineTo(i * 24 * side, 12);
  }
  ctx.stroke();
  ctx.save();
  ctx.translate(0, -24);
  drawRoadsidePeopleCluster(side, accent, light, seed);
  ctx.restore();
}

function drawRoadsideAnimalGroup(species, side, accent, light, seed) {
  if (!species) return;
  const count = species === "bird" || species === "shoreBird" ? 3 : species === "sheep" ? 3 : 2;
  for (let i = 0; i < count; i += 1) {
    const x = side * (i * 24 - 18 + seededUnit(seed, i + 6) * 8);
    const y = 8 + (i % 2) * 5;
    const scale = species === "camel" || species === "horse" || species === "cow" ? 0.82 : species === "bird" || species === "shoreBird" ? 0.48 : 0.66;
    drawRoadsideAnimalShape(species, x, y, scale, side, accent, light, seed + i * 13);
  }
}

function drawRoadsideAnimalShape(species, x, y, scale, side, accent, light, seed) {
  drawReferenceAnimalShape(species, x, y, scale, side, accent, light, seed);
}

function drawRoadsideParkedVehicle(side, color) {
  ctx.save();
  ctx.scale(1.15, 0.88);
  const paint = ctx.createLinearGradient(-34, -18, 34, 18);
  paint.addColorStop(0, shade(color, 34));
  paint.addColorStop(0.55, color);
  paint.addColorStop(1, shade(color, -38));
  ctx.fillStyle = "rgba(0,0,0,0.44)";
  ctx.beginPath();
  ctx.ellipse(0, 18, 38, 6, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "#050807";
  roundRect(-34, 2, 13, 16, 4);
  roundRect(21, 2, 13, 16, 4);
  ctx.fill();
  ctx.fillStyle = paint;
  ctx.beginPath();
  ctx.moveTo(-25, -21);
  ctx.lineTo(25, -21);
  ctx.lineTo(37, 2);
  ctx.lineTo(29, 17);
  ctx.lineTo(-29, 17);
  ctx.lineTo(-37, 2);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-16, -15, 32, 12, 4);
  ctx.fill();
  ctx.fillStyle = "rgba(255,51,72,0.76)";
  roundRect(-22, 13, 10, 4, 2);
  roundRect(12, 13, 10, 4, 2);
  ctx.fill();
  ctx.scale(side, 1);
  ctx.restore();
}

function drawRoadsideBuilding(kind, side, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  if (kind === "barn" || kind === "cabin") {
    ctx.fillStyle = kind === "barn" ? "rgba(148,42,34,0.86)" : "rgba(74,54,36,0.88)";
    roundRect(-38, -44, 76, 54, 5);
    ctx.fill();
    ctx.fillStyle = kind === "barn" ? "rgba(110,27,25,0.92)" : "rgba(96,74,54,0.92)";
    ctx.beginPath();
  }
```

```
    ctx.moveTo(-44, -44);
    ctx.lineTo(0, -78);
    ctx.lineTo(44, -44);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = light;
    roundRect(-9, -22, 18, 32, 2);
    ctx.fill();
    return;
}
if (kind === "hangar" || kind === "warehouse" || kind === "trailer") {
  ctx.fillStyle = "rgba(22,30,31,0.9)";
  roundRect(-58, -38, 116, 48, 5);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.18)";
  for (let i = -2; i <= 2; i += 1) ctx.fillRect(i * 20 - 5, -28, 10, 11);
  ctx.strokeStyle = accent;
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(-58, -38);
  ctx.lineTo(0,
```


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```
-64);
  ctx.lineTo(58, -38);
  ctx.stroke();
  return;
}
if (kind === "crane") {
  ctx.strokeStyle = `${light}bb`;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(0, 10);
  ctx.lineTo(0, -92);
  ctx.moveTo(-48 * side, -72);
  ctx.lineTo(58 * side, -84);
  ctx.moveTo(34 * side, -81);
  ctx.lineTo(34 * side, -42);
  ctx.stroke();
  ctx.fillStyle = accent;
  roundRect(27 * side, -42, 14 * side, 16, 2);
  ctx.fill();
  return;
}
const floors = kind === "village" ? 2 : kind === "neon" ? 6 : 5 + Math.floor(seededUnit(seed, 5) * 4);
const bw = kind === "village" ? 58 : 38 + seededUnit(seed, 6) * 22;
const bh = floors * 21;
const wall = ctx.createLinearGradient(-bw / 2, -bh, bw / 2, 8);
wall.addColorStop(0, kind === "neon" || place === "tokyo" ? "rgba(25,12,42,0.94)" : "rgba(17,26,29,0.94)");
wall.addColorStop(1, "rgba(5,8,7,0.86)");
ctx.fillStyle = wall;
roundRect(-bw / 2, -bh, bw, bh + 8, 4);
ctx.fill();
if (kind === "village") {
  ctx.fillStyle = "rgba(146,64,42,0.92)";
  ctx.beginPath();
  ctx.moveTo(-bw * 0.6, -bh);
  ctx.lineTo(0, -bh - 28);
  ctx.lineTo(bw * 0.6, -bh);
  ctx.closePath();
  ctx.fill();
}
ctx.fillStyle = place === "tokyo" || kind === "neon" ? `${accent}99` : `${light}77`;
for (let i = 0; i < floors; i += 1) {
  const wy = -bh + 10 + i * 18;
  ctx.fillRect(-bw * 0.32, wy, bw * 0.18, 5);
  ctx.fillRect(bw * 0.14, wy + (i % 2) * 3, bw * 0.18, 5);
}
if (kind === "neon") {
  ctx.strokeStyle = `${accent}aa`;
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(side * bw * 0.56, -bh + 8);
  ctx.lineTo(side * bw * 0.56, -18);
  ctx.stroke();
}
}

function drawRoadsidePole(kind, side, accent, light) {
  ctx.strokeStyle = kind === "telephone" ? "rgba(100,70,44,0.86)" : "rgba(180,192,188,0.72)";
  ctx.lineWidth = kind === "telephone" ? 4 : 3;
  ctx.beginPath();
  ctx.moveTo(0, 12);
  ctx.lineTo(0, -78);
  if (kind === "telephone") {
    ctx.moveTo(-24 * side, -58);
    ctx.lineTo(24 * side, -58);
  }
  ctx.stroke();
  ctx.globalCompositeOperation = "screen";
  ctx.fillStyle = kind === "beacon" ? `${accent}aa` : `${light}88`;
  ctx.beginPath();
  ctx.ellipse(0, -82, kind === "streetlight" ? 11 : 7, kind === "streetlight" ? 7 : 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.globalCompositeOperation = "source-over";
}

function drawRoadsideFence(side, color) {
  if (phoneGraphicsActive()) {
    ctx.fillStyle = color && color.includes("244") ? "rgba(230,240,236,0.5)" : "rgba(70,112,72,0.45)";
    ctx.beginPath();
    ctx.ellipse(4 * side, -5, 76, 18, -0.05 * side, 0, Math.PI * 2);
    ctx.ellipse(-38 * side, -15, 38, 14, 0.08 * side, 0, Math.PI * 2);
    ctx.ellipse(44 * side, -16, 42, 15, -0.08 * side, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = color;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-68 * side, -18);
    ctx.lineTo(72 * side, -24);
    ctx.moveTo(-68 * side, 0);
    ctx.lineTo(72 * side, -6);
    ctx.stroke();
    return;
  }
  ctx.strokeStyle = color;
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-62 * side, -4);
  ctx.lineTo(66 * side, -10);
  ctx.moveTo(-62 * side, 12);
  ctx.lineTo(66 * side, 8);
  for (let i = -3; i <= 3; i += 1) {
    ctx.moveTo(i * 20 * side, -16);
    ctx.lineTo(i * 20 * side, 18);
  }
  ctx.stroke();
}

function drawRoadsideSign(kind, side, text, accent) {
  ctx.strokeStyle = "rgba(205,218,214,0.58)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(0, 14);
  ctx.lineTo(0, -44);
  ctx.stroke();
}
```

```

ctx.fillStyle = kind === "busStop" ? "rgba(5,8,7,0.76)" : "rgba(12,22,20,0.9)";
roundRect(-38, -74, 76, 34, 5);
ctx.fill();
ctx.strokeStyle = accent;
ctx.lineWidth = 1.5;
ctx.stroke();
ctx.fillStyle = "#f4fbf8";
ctx.font = "900 8px system-ui";
ctx.textAlign = "center";
ctx.textBaseline = "middle";
const label = kind === "busStop" ? "STOP" : String(text || "ROUTE").toUpperCase().slice(0, 9);
ctx.fillText(label, 0, -57, 66);
ctx.scale(side, 1);
}

function drawRoadsideTree(kind, side) {
ctx.strokeStyle = kind === "palm" ? "rgba(132,86,46,0.9)" : "rgba(24,78,52,0.9)";
ctx.lineWidth = 5;
ctx.beginPath();
ctx.moveTo(0, 14);

```

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```
    ctx.lineTo(side * 7, -64);
    ctx.stroke();
    ctx.fillStyle = kind === "pine" ? "rgba(28,88,52,0.88)" : "rgba(54,217,138,0.5)";
    if (kind === "pine") {
      for (let i = 0; i < 3; i += 1) {
        ctx.beginPath();
        ctx.moveTo(side * 4, -76 + i * 22);
        ctx.lineTo(-34, -34 + i * 17);
        ctx.lineTo(38, -32 + i * 17);
        ctx.closePath();
        ctx.fill();
      }
    } else {
      for (let i = 0; i < 5; i += 1) {
        ctx.beginPath();
        ctx.ellipse(side * (5 + i), -68 + i * 3, 38 - i * 4, 12, side * (-0.85 + i * 0.42), 0, Math.PI * 2);
        ctx.fill();
      }
    }
  }
}

function drawRoadsideMachine(kind, side, accent, light) {
  if (kind === "boat") {
    ctx.fillStyle = "rgba(70,217,255,0.34)";
    ctx.beginPath();
    ctx.ellipse(0, 14, 52, 11, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = accent;
    ctx.beginPath();
    ctx.moveTo(-46, -4);
    ctx.quadraticCurveTo(0, -28, 46, -4);
    ctx.lineTo(32, 12);
    ctx.lineTo(-32, 12);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(5,8,7,0.84)";
    roundRect(-12, -19, 24, 12, 3);
    ctx.fill();
  } else if (kind === "plane") {
    ctx.fillStyle = light;
    ctx.beginPath();
    ctx.moveTo(0, -52);
    ctx.lineTo(12, 10);
    ctx.lineTo(-12, 10);
    ctx.closePath();
    ctx.fill();
    ctx.beginPath();
    ctx.moveTo(-62, -14);
    ctx.lineTo(62, -14);
    ctx.lineTo(16, 2);
    ctx.lineTo(-16, 2);
    ctx.closePath();
    ctx.fill();
  } else {
    ctx.fillStyle = accent;
    roundRect(-32, -34, 64, 42, 6);
    ctx.fill();
    ctx.fillStyle = "#050807";
    ctx.beginPath();
    ctx.arc(-26, 12, 15, 0, Math.PI * 2);
    ctx.arc(28, 14, 22, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.scale(side, 1);
}

function drawRoadsideTerrain(kind, side, place, accent, light) {
  if (kind === "mountain" || kind === "cliff" || kind === "rock") {
    const fill = kind === "rock" || place === "canyon" ? "rgba(158,84,48,0.62)" : "rgba(132,160,160,0.46)";
    ctx.fillStyle = fill;
    ctx.beginPath();
    ctx.moveTo(-62, 18);
    ctx.lineTo(-22, -58);
    ctx.lineTo(8, -18);
    ctx.lineTo(38, -78);
    ctx.lineTo(74, 16);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.22)";
    ctx.beginPath();
    ctx.moveTo(26, -52);
    ctx.lineTo(38, -78);
    ctx.lineTo(50, -50);
    ctx.closePath();
    ctx.fill();
  } else if (kind === "dune" || kind === "field" || kind === "snowbank") {
    ctx.fillStyle = kind === "snowbank" ? "rgba(244,251,248,0.62)" : kind === "field" ? "rgba(187,242,74,0.28)" :
    "rgba(255,183,74,0.42)";
    ctx.beginPath();
    ctx.ellipse(0, 8, 72, 22, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = kind === "field" ? "rgba(117,86,47,0.46)" : `${light}55`;
    ctx.lineWidth = 2;
    ctx.beginPath();
    ctx.moveTo(-56, 0);
    ctx.quadraticCurveTo(-12, -10, 58, 7);
    ctx.stroke();
  } else {
    ctx.fillStyle = `${accent}22`;
    ctx.beginPath();
    ctx.ellipse(0, 7, 58, 18, 0, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.scale(side, 1);
}

function drawRoadsideServiceSign(kind, side, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  if (kind === "speedCamera") {
    ctx.strokeStyle = "rgba(205,218,214,0.72)";
    ctx.lineWidth = 3;
  }
```

```

ctx.beginPath();
ctx.moveTo(0, 12);
ctx.lineTo(0, -58);
ctx.stroke();
ctx.fillStyle = "rgba(5,8,7,0.9)";
roundRect(-16, -72, 32, 18, 3);
ctx.fill();
ctx.fillStyle = accent;
ctx.beginPath();
ctx.arc(side * 8, -63, 4, 0, Math.PI * 2);
ctx.fill();
} else if (kind === "mileMarker") {
ctx.fillStyle = "rgba(5,8,7,0.84)";
roundRect(-18, -50, 36, 42, 4);
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.32)";
ctx.lineWidth = 1.5;
ctx.stroke();
ctx.fillStyle = "#f4fbf8";
ctx.font = "900 8px system-ui";
ctx.textAlign = "center";
ctx.fillText("KM", 0, -35);
ctx.fillStyle = light;
ctx.fillText(String(20 + Math.floor(seededUnit(seed, 4) * 79)), 0, -21);
} else {
ctx.strokeStyle = "rgba(180,192,188,0.58)";
ctx.lineWidth = 4;
ctx.beginPath();
ctx.moveTo(-42, 14);
ctx.lineTo(-42, -54);
ctx.moveTo

```

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```
o(42, 14);
  ctx.lineTo(42, -54);
  ctx.stroke();
  const panel = ctx.createLinearGradient(-62, -92, 62, -48);
  panel.addColorStop(0, "rgba(5,8,7,0.92)");
  panel.addColorStop(1, place === "tokyo" ? `${accent}66` : "rgba(20,34,36,0.86)");
  ctx.fillStyle = panel;
  roundRect(-66, -94, 132, 42, 5);
  ctx.fill();
  ctx.strokeStyle = `${accent}aa`;
  ctx.lineWidth = 2;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 9px system-ui";
  ctx.textAlign = "center";
  const words = place === "coast" ? "SCENIC DRIVE" : place === "tokyo" ? "EXPRESS" : place === "city" ? "DOWNTOWN" : "NEXT
SECTOR";
  ctx.fillText(words, 0, -75, 112);
  ctx.fillStyle = `${light}dd`;
  ctx.font = "800 7px system-ui";
  ctx.fillText(routeTurnName(raceState.roadTurn || 0).toUpperCase(), 0, -61, 112);
}
ctx.scale(side, 1);
ctx.restore();
}

function drawRoadsideWorldFixture(kind, side, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  if (kind === "pier" || kind === "bridge") {
    ctx.strokeStyle = kind === "pier" ? "rgba(117,86,47,0.74)" : "rgba(205,218,214,0.5)";
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-76 * side, 4);
    ctx.lineTo(76 * side, -12);
    ctx.moveTo(-76 * side, 16);
    ctx.lineTo(76 * side, 0);
    for (let i = -3; i <= 3; i += 1) {
      ctx.moveTo(i * 22 * side, -16);
      ctx.lineTo(i * 22 * side, 22);
    }
    ctx.stroke();
  } else if (kind === "container") {
    const colors = ["#ff5b6b", "#46d9ff", "#ffd166", "#dce8ef"];
    for (let i = 0; i < 3; i += 1) {
      ctx.fillStyle = colors[(i + Math.floor(seed)) % colors.length];
      roundRect(-58 + i * 38, -34 - (i % 2) * 22, 34, 24, 2);
      ctx.fill();
      ctx.strokeStyle = "rgba(5,8,7,0.5)";
      ctx.lineWidth = 1.5;
      ctx.stroke();
    }
  } else if (kind === "fuelStop" || kind === "market") {
    ctx.fillStyle = "rgba(10,20,20,0.9)";
    roundRect(-52, -38, 104, 48, 4);
    ctx.fill();
    ctx.fillStyle = kind === "fuelStop" ? `${light}cc` : `${accent}aa`;
    roundRect(-42, -56, 84, 18, 3);
    ctx.fill();
    ctx.fillStyle = "#050807";
    ctx.font = "900 8px system-ui";
    ctx.textAlign = "center";
    ctx.fillText(kind === "fuelStop" ? "FUEL" : "MARKET", 0, -44, 70);
    ctx.fillStyle = "rgba(244,251,248,0.18)";
    ctx.fillRect(-36, -24, 18, 22);
    ctx.fillRect(16, -24, 18, 22);
  } else if (kind === "silo") {
    ctx.fillStyle = "rgba(198,205,198,0.62)";
    roundRect(-16, -74, 32, 84, 8);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.36)";
    ctx.beginPath();
    ctx.ellipse(0, -74, 17, 8, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = "rgba(148,42,34,0.8)";
    roundRect(18 * side, -42, 54 * side, 50, 4);
    ctx.fill();
  } else if (kind === "radar") {
    ctx.strokeStyle = "rgba(205,218,214,0.62)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(0, 12);
    ctx.lineTo(0, -58);
    ctx.stroke();
    ctx.strokeStyle = `${accent}99`;
    ctx.lineWidth = 2;
    ctx.beginPath();
    ctx.arc(0, -64, 20, Math.PI * 0.08, Math.PI * 0.92);
    ctx.arc(0, -64, 12, Math.PI * 0.08, Math.PI * 0.92);
    ctx.stroke();
  } else if (kind === "serviceTruck") {
    drawRoadsideParkedVehicle(side, light);
    ctx.fillStyle = "rgba(5,8,7,0.88)";
    roundRect(-20, -28, 40, 18, 3);
    ctx.fill();
    ctx.fillStyle = accent;
    roundRect(-5, -36, 10, 8, 2);
    ctx.fill();
  } else if (kind === "tunnelMouth") {
    ctx.fillStyle = "rgba(5,8,7,0.86)";
    roundRect(-62, -68, 124, 78, 12);
    ctx.fill();
    ctx.fillStyle = "rgba(0,0,0,0.58)";
    ctx.beginPath();
    ctx.ellipse(0, -18, 42, 38, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = `${accent}88`;
    ctx.lineWidth = 3;
    ctx.stroke();
  } else if (kind === "waterfall") {
    ctx.fillStyle = "rgba(18,75,45,0.58)";
    ctx.beginPath();
  }
```

```
ctx.ellipse(0, -18, 72, 40, 0, 0, Math.PI * 2);
ctx.fill();
ctx.fillStyle = "rgba(222,249,255,0.46)";
roundRect(-12, -72, 24, 92, 6);
ctx.fill();
ctx.fillStyle = "rgba(70,217,255,0.24)";
ctx.beginPath();
ctx.ellipse(0, 22, 42, 9, 0, 0, Math.PI * 2);
c
```

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```
tx.fill();
}
ctx.scale(side, 1);
ctx.restore();
}

function drawRealWorldCurveCues(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const turn = raceState.roadTurn || raceState.roadCurve || 0;
  if (Math.abs(turn) < 0.12) return;
  const phoneMode = phoneGraphicsActive();
  const side = turn > 0 ? 1 : -1;
  const count = phoneMode ? 2 : 8;
  const motion = (raceState.roadOffset || 0) * 0.9;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.24 : 0.6;
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * (phoneMode ? 210 : 98) + motion) % (h - horizon + 230)) - 72);
    if (y < horizon || y > h + 30) continue;
    const t = Math.max(0.04, Math.min(1.04, (y - horizon) / Math.max(1, h - horizon)));
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.035 + t * 0.075));
    const scale = 0.42 + t * 0.88;
    ctx.save();
    ctx.translate(x, y);
    ctx.scale(scale, scale);
    ctx.fillStyle = "rgba(5,8,7,0.78)";
    roundRect(-28, -18, 56, 32, 4);
    ctx.fill();
    ctx.strokeStyle = `${design.accent} || theme[1]aa`;
    ctx.lineWidth = 2.2;
    ctx.stroke();
    ctx.strokeStyle = "rgba(244,251,248,0.82)";
    ctx.lineWidth = 4;
    ctx.lineCap = "round";
    for (let c = -1; c <= 1; c += 1) {
      ctx.beginPath();
      ctx.moveTo(-13 * side + c * 10, -8);
      ctx.lineTo(1 * side + c * 10, -1);
      ctx.lineTo(-13 * side + c * 10, 8);
      ctx.stroke();
    }
    ctx.restore();
  }
  ctx.restore();
}

function drawRealWorldHorizonDetails(w, h, horizon, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const offset = (raceState.roadOffset || 0) * 0.013;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.34 : 0.48;
  const base = horizon + h * 0.04;
  if (place === "alpine" || place === "snow" || place === "europe" || place === "canyon" || place === "desert") {
    for (let layer = 0; layer < 2; layer += 1) {
      ctx.fillStyle = layer === 0 ? "rgba(54,72,74,0.36)" : (place === "desert" || place === "canyon" ? "rgba(132,72,38,0.34)" : "rgba(150,174,174,0.3)");
      ctx.beginPath();
      ctx.moveTo(-40, base + layer * h * 0.03);
      for (let i = 0; i <= 8; i += 1) {
        const x = i * w / 7 - ((offset * (layer + 1) * 0.8) % (w / 7));
        const peak = base - h * (0.08 + seededUnit(seed + layer * 19, 1) * 0.16) - layer * h * 0.025;
        ctx.lineTo(x, peak);
        ctx.lineTo(x + w / 14, base + layer * h * 0.035);
      }
      ctx.lineTo(w + 40, base + layer * h * 0.04);
      ctx.closePath();
      ctx.fill();
    }
  }
  if (place === "city" || place === "tokyo") {
    for (let i = 0; i < 18; i += 1) {
      const x = ((i * w * 0.072 - offset * 7) % (w + 140)) - 70;
      const bw = w * (0.028 + seededUnit(seed, i) * 0.032);
      const bh = h * (0.08 + seededUnit(seed, i + 8) * 0.16);
      ctx.fillStyle = place === "tokyo" ? "rgba(20,10,38,0.64)" : "rgba(8,18,21,0.66)";
      roundRect(x, base - bh, bw, bh, 3);
      ctx.fill();
      ctx.fillStyle = i % 2 ? `${design.accent}66` : `${design.light}55`;
      ctx.fillRect(x + bw * 0.24, base - bh + 12, bw * 0.18, Math.max(8, bh * 0.48));
    }
  }
  if (place === "coast" || place === "harbor") {
    const water = ctx.createLinearGradient(0, horizon + h * 0.02, 0, horizon + h * 0.18);
    water.addColorStop(0, "rgba(70,217,255,0.18)");
    water.addColorStop(1, "rgba(4,20,24,0.38)");
    ctx.fillStyle = water;
    ctx.fillRect(0, horizon + h * 0.015, w, h * 0.16);
    ctx.strokeStyle = "rgba(244,251,248,0.2)";
    ctx.lineWidth = 2;
    for (let i = 0; i < 7; i += 1) {
      const y = horizon + h * (0.05 + i * 0.018);
      ctx.beginPath();
      ctx.moveTo(0, y);
      ctx.lineTo(w, y + Math.sin(i + offset) * 2);
      ctx.stroke();
    }
  }
  if (place === "rainforest") {
    ctx.fillStyle = "rgba(18,75,45,0.42)";
    for (let i = 0; i < 22; i += 1) {
      const x = ((i * w * 0.055 - offset * 9) % (w + 100)) - 50;
      const y = base - h * (0.05 + seededUnit(seed, i) * 0.08);
      ctx.beginPath();
      ctx.ellipse(x, y, w * 0.055, h * 0.034, seededUnit(seed, i + 5) - 0.5, 0, Math.PI * 2);
      ctx.fill();
    }
  }
}
```

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```
} else if (place === "farm" || place === "freight" || place === "airfield") {
  ctx.fillStyle = place === "farm" ? "rgba(95,126,54,0.36)" : "rgba(38,48,50,0.34)";
  ctx.fillRect(0, horizon + h * 0.035, w, h * 0.1);
  ctx.strokeStyle = place === "farm" ? "rgba(187,242,74,0.22)" : `${design.light}33`;
  ctx.lineWidth = 2;
  for (let i = 0; i < 10; i += 1) {
    const y = horizon + h * 0.06 + i * h * 0.012;
    ctx.beginPath();
    ctx.moveTo(0, y);
    ctx.lineTo(w, y + Math.sin(i * 1.7 + offset) * 3);
    ctx.stroke();
  }
}
ctx.restore();
}

function drawRealWorldRouteSignaturePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const route = routeWorldInfo(place);
  const length = raceLength();
  const progress = length ? Math.max(0, Math.min(1, raceState.distance / length)) : 0;
  const stage = routeStageInfo(place, progress);
  const motion = (raceState.roadOffset || 0) * 0.02;
  const fixtures = {
    coast: ["pierRails", "beachHouse", "coastalWall"],
    city: ["overpass", "officeGlass", "taxiStand"],
    canyon: ["roadCut", "lookoutRail", "butte"],
    alpine: ["switchbackWall", "stoneTunnel", "snowFence"],
    harbor: ["containerYard", "drawBridge", "dockCrane"],
    snow: ["snowFence", "skiVillage", "stoneTunnel"],
    airfield: ["runwayBeacon", "hangarRow", "radarTower"],
    freight: ["interstateGantry", "truckStop", "containerYard"],
    farm: ["cropRows", "grainElevator", "countyRoad"],
    tokyo: ["expressGantry", "neonBlocks", "railOverpass"],
    desert: ["rallyCamp", "roadCut", "duneRidge"],
    rainforest: ["canopyBridge", "wetMarket", "riverBridge"],
    europe: ["stoneVillage", "alpsTunnel", "switchbackWall"]
  };
  const routeFixtures = fixtures[place] || fixtures.city;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.35 : 0.5;

  for (let i = 0; i < 8; i += 1) {
    const side = i % 2 === 0 ? -1 : 1;
    const y = horizon + h * (0.1 + seededUnit(seed, i) * 0.28);
    const t = Math.max(0.08, Math.min(0.58, (y - horizon) / Math.max(1, h - horizon)));
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.1 + seededUnit(seed, i + 8) * 0.13)) + Math.sin(motion + i) * 14;
    const scale = 0.42 + t * (phoneMode ? 0.62 : 0.84);
    drawRouteSignatureFixture(routeFixtures[(i + stage.index) % routeFixtures.length], x, y, scale, side, place, design, theme, seed
+ i * 41);
  }

  if (!phoneMode || h > 620) {
    const signT = 0.22;
    const signY = horizon + h * 0.12;
    const signX = perspectiveRoadCenter(w, signT);
    const signW = Math.min(w * 0.44, 260);
    ctx.globalAlpha = phoneMode ? 0.32 : 0.48;
    ctx.fillStyle = "rgba(5,8,7,0.64)";
    roundRect(signX - signW / 2, signY - 18, signW, 34, 5);
    ctx.fill();
    ctx.strokeStyle = `${design.accent} || theme[1]77`;
    ctx.lineWidth = 1.5;
    ctx.stroke();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = `900 ${Math.max(8, Math.min(13, w * 0.012))}px system-ui`;
    ctx.textAlign = "center";
    ctx.textBaseline = "middle";
    ctx.fillText(`${route.country} ${stage.label}.toUpperCase(), signX, signY - 2, signW - 18);
    ctx.fillStyle = `${design.light} || theme[2]dd`;
    ctx.font = `800 ${Math.max(7, Math.min(10, w * 0.01))}px system-ui`;
    ctx.fillText(route.cue.toUpperCase(), signX, signY + 11, signW - 18);
  }
  ctx.restore();
}

function drawRouteSignatureFixture(kind, x, y, scale, side, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  if (kind === "overpass" || kind === "expressGantry" || kind === "interstateGantry" || kind === "railOverpass") {
    ctx.strokeStyle = "rgba(180,192,188,0.52)";
    ctx.lineWidth = 5;
    ctx.beginPath();
    ctx.moveTo(-92 * side, -34);
    ctx.lineTo(92 * side, -42);
    ctx.moveTo(-76 * side, -34);
    ctx.lineTo(-76 * side, 24);
    ctx.moveTo(76 * side, -42);
    ctx.lineTo(76 * side, 20);
    ctx.stroke();
    ctx.fillStyle = kind === "expressGantry" || kind === "railOverpass" ? `${accent}44` : "rgba(5,8,7,0.7)";
  }
}
```


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```
roundRect(-58, -66, 116, 24, 3);
ctx.fill();
ctx.fillStyle = "#f4fbf8";
ctx.font = "900 8px system-ui";
ctx.textAlign = "center";
ctx.fillText(kind === "interstateGantry" ? "FREIGHT" : kind === "expressGantry" ? "EXPRESS" : "CITY", 0, -52, 92);
} else if (kind === "containerYard" || kind === "hangarRow" || kind === "truckStop") {
  for (let i = 0; i < 5; i += 1) {
    const px = -70 + i * 34;
    ctx.fillStyle = i % 2 ? `${accent}66` : "rgba(244,251,248,0.3)";
    roundRect(px, -36 - (i % 2) * 14, 30, 24, 2);
    ctx.fill();
  }
  ctx.fillStyle = "rgba(5,8,7,0.76)";
  roundRect(-76, -10, 152, 20, 4);
  ctx.fill();
} else if (kind === "officeGlass" || kind === "neonBlocks" || kind === "stoneVillage" || kind === "skiVillage") {
  for (let i = 0; i < 4; i += 1) {
    const bw = kind === "stoneVillage" || kind === "skiVillage" ? 38 : 28;
    const bh = (kind === "stoneVillage" || kind === "skiVillage" ? 34 : 82) + i * 11;
    const px = -60 + i * 38;
    ctx.fillStyle = kind === "neonBlocks" ? "rgba(20,10,38,0.76)" : kind === "stoneVillage" || kind === "skiVillage" ?
"rgba(76,58,44,0.8)" : "rgba(12,22,26,0.78)";
    roundRect(px, -bh, bw, bh, 3);
    ctx.fill();
    if (kind === "stoneVillage" || kind === "skiVillage") {
      ctx.fillStyle = kind === "skiVillage" ? "rgba(244,251,248,0.52)" : "rgba(146,64,42,0.82)";
      ctx.beginPath();
      ctx.moveTo(px - 4, -bh);
      ctx.lineTo(px + bw / 2, -bh - 20);
      ctx.lineTo(px + bw + 4, -bh);
      ctx.closePath();
      ctx.fill();
    } else {
      ctx.fillStyle = i % 2 ? `${accent}88` : `${light}77`;
      for (let yy = 10; yy < bh - 8; yy += 20) ctx.fillRect(px + 8, -bh + yy, bw - 16, 4);
    }
  }
} else if (kind === "pierRails" || kind === "drawBridge" || kind === "canopyBridge" || kind === "riverBridge") {
  ctx.strokeStyle = kind === "pierRails" ? "rgba(117,86,47,0.74)" : `${accent}66`;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-90 * side, -8);
  ctx.lineTo(90 * side, -20);
  ctx.moveTo(-88 * side, 8);
  ctx.lineTo(88 * side, -4);
  for (let i = -4; i <= 4; i += 1) {
    ctx.moveTo(i * 21 * side, -28);
    ctx.lineTo(i * 21 * side, 16);
  }
  ctx.stroke();
} else if (kind === "roadCut" || kind === "butte" || kind === "duneRidge" || kind === "switchbackWall") {
  ctx.fillStyle = kind === "switchbackWall" ? "rgba(112,132,128,0.42)" : kind === "duneRidge" ? "rgba(255,183,74,0.36)" :
"rgba(158,84,48,0.42)";
  ctx.beginPath();
  ctx.moveTo(-98, 24);
  ctx.lineTo(-48, -52);
  ctx.lineTo(-14, -8);
  ctx.lineTo(32, -66);
  ctx.lineTo(96, 22);
  ctx.closePath();
  ctx.fill();
  ctx.strokeStyle = "rgba(244,251,248,0.16)";
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(-62, 4);
  ctx.quadraticCurveTo(0, -16, 70, 2);
  ctx.stroke();
} else if (kind === "stoneTunnel" || kind === "alpsTunnel") {
  ctx.fillStyle = "rgba(5,8,7,0.84)";
  roundRect(-74, -72, 148, 82, 12);
  ctx.fill();
  ctx.strokeStyle = `${accent}88`;
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.ellipse(0, -18, 50, 42, 0, 0, Math.PI * 2);
  ctx.stroke();
} else if (kind === "runwayBeacon" || kind === "radarTower") {
  ctx.strokeStyle = "rgba(205,218,214,0.62)";
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(0, 18);
  ctx.lineTo(0, -84);
  ctx.stroke();
  ctx.strokeStyle = `${accent}aa`;
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.arc(0, -88, kind === "radarTower" ? 26 : 10, 0, Math.PI * 2);
  ctx.stroke();
  ctx.fillStyle = `${light}aa`;
  ctx.beginPath();
  ctx.arc(0, -88, 5, 0, Math.PI * 2);
  ctx.fill();
} else if (kind === "cropRows" || kind === "countyRoad" || kind === "grainElevator") {
  if (kind === "grainElevator") {
    ctx.fillStyle = "rgba(198,205,198,0.58)";
    roundRect(-18, -84, 36, 92, 8);
    ctx.fill();
    ctx.fillStyle = "rgba(148,42,34,0.78)";
    roundRect(24 * side, -52, 62 * side, 58, 4);
    ctx.fill();
  } else {
    ctx.strokeStyle = "rgba(187,242,74,0.28)";
    ctx.lineWidth = 3;
    for (let i = -4; i <= 4; i += 1) {
      ctx.beginPath();

```

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```
    ctx.moveTo(i * 16, -28);
    ctx.quadraticCurveTo(i * 18 + side * 18, -6, i * 20 + side * 36, 22);
    ctx.stroke();
  }
}
} else if (kind === "rallyCamp" || kind === "wetMarket" || kind === "beachHouse") {
  ctx.fillStyle = "rgba(5,8,7,0.72)";
  roundRect(-56, -38, 112, 46, 5);
  ctx.fill();
  ctx.fillStyle = kind === "beachHouse" ? `${light}88` : `${accent}88`;
  ctx.beginPath();
  ctx.moveTo(-62, -38);
  ctx.lineTo(0, -64);
  ctx.lineTo(62, -38);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.22)";
  ctx.fillRect(-36, -24, 18, 22);
  ctx.fillRect(18, -24, 18, 22);
} else if (kind === "coastalWall" || kind === "lookoutRail" || kind === "snowFence") {
  ctx.strokeStyle = kind === "snowFence" ? "rgba(244,251,248,0.48)" : "rgba(205,218,214,0.42)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-88 * side, -4);
  ctx.lineTo(88 * side, -14);
  ctx.moveTo(-88 * side, 10);
  ctx.lineTo(88 * side, 0);
  for (let i = -4; i <= 4; i += 1) {
    ctx.moveTo(i * 20 * side, -20);
    ctx.lineTo(i * 20 * side, 18);
  }
  ctx.stroke();
} else {
  ctx.fillStyle = `${accent}33`;
  ctx.beginPath();
  ctx.ellipse(0, 0, 70, 18, 0, 0, Math.PI * 2);
  ctx.fill();
}
ctx.restore();
}

function drawRealWorldForegroundDepthPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const alpha = phoneMode ? 0.24 : 0.34;
  const motion = (raceState.roadOffset || 0) * 1.05;
  ctx.save();
  for (let side = -1; side <= 1; side += 2) {
    const shoulder = ctx.createLinearGradient(0, h * 0.58, 0, h);
    shoulder.addColorStop(0, "rgba(0,0,0)");
    shoulder.addColorStop(1, place === "snow" ? "rgba(244,251,248,0.16)" : place === "desert" || place === "canyon" ?
"rgba(132,72,38,0.18)" : "rgba(5,8,7,0.42)");
    ctx.globalAlpha = alpha;
    ctx.fillStyle = shoulder;
    ctx.beginPath();
    ctx.moveTo(perspectiveRoadCenter(w, 0.58) + side * realWorldRoadHalf(roadTop, roadBottom, 0.58), h * 0.58);
    ctx.lineTo(side < 0 ? 0 : w, h * 0.58);
    ctx.lineTo(side < 0 ? 0 : w, h + 10);
    ctx.lineTo(perspectiveRoadCenter(w, 1) + side * realWorldRoadHalf(roadTop, roadBottom, 1) * 0.86, h + 10);
    ctx.closePath();
    ctx.fill();

    const postCount = phoneMode ? 5 : 8;
    for (let i = 0; i < postCount; i += 1) {
      const y = h * 0.62 + (((i * 96 + motion + side * 44) % (h * 0.46 + 120)) - 54);
      if (y < h * 0.58 || y > h + 28) continue;
      const t = Math.max(0.58, Math.min(1.12, (y - horizon) / Math.max(1, h - horizon)));
      const half = realWorldRoadHalf(roadTop, roadBottom, t);
      const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.035 + seededUnit(seed, i + 210) * 0.08));
      const s = 0.5 + t * 0.72;
      ctx.globalAlpha = alpha + t * 0.16;
      if (i % 3 === 0) {
        ctx.fillStyle = "rgba(5,8,7,0.82)";
        roundRect(x - side * 4, y - 24 * s, side * 8, 42 * s, 3);
        ctx.fill();
        ctx.fillStyle = `${design.accent || theme[1]}88`;
        ctx.beginPath();
        ctx.ellipse(x, y - 29 * s, 8 * s, 4 * s, 0, 0, Math.PI * 2);
        ctx.fill();
      } else if (i % 3 === 1) {
        ctx.fillStyle = place === "city" || place === "tokyo" ? "rgba(150,160,156,0.28)" : "rgba(117,86,47,0.3)";
        roundRect(x - side * 18 * s, y - 4 * s, side * 36 * s, 8 * s, 3);
        ctx.fill();
      } else {
        ctx.fillStyle = place === "snow" ? "rgba(244,251,248,0.52)" : place === "desert" ? "rgba(255,183,74,0.32)" :
"rgba(54,217,138,0.24)";
        ctx.beginPath();
        ctx.ellipse(x, y, 18 * s, 6 * s, 0, 0, Math.PI * 2);
        ctx.fill();
      }
    }
  }
  ctx.globalAlpha = 1;
  ctx.restore();
}

function drawRealWorldAmbientLifePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const motion = (raceState.roadOffset || 0) * 0.018;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.28 : 0.42;

  if (place === "city" || place === "tokyo" || place === "freight" || place === "farm" || place === "rainforest" || place ===
"europe") {
    drawRealWorldPowerLines(w, h, horizon, roadTo
```

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```
p, roadBottom, place, design, theme, seed, motion);
}

ctx.globalAlpha = 1;
drawLivingWorldAmbientMotionPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, phoneMode);
ctx.globalAlpha = phoneMode ? 0.36 : 0.52;

const lifeCount = phoneMode ? 8 : 13;
for (let i = 0; i < lifeCount; i += 1) {
  const y = horizon + h * (0.12 + seededUnit(seed, i + 240) * 0.38);
  const t = Math.max(0.08, Math.min(0.68, (y - horizon) / Math.max(1, h - horizon)));
  const side = i % 2 === 0 ? -1 : 1;
  const half = realWorldRoadHalf(roadTop, roadBottom, t);
  const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.08 + seededUnit(seed, i + 260) * 0.16)) + Math.sin(motion + i) *
18;
  const scale = 0.32 + t * (phoneMode ? 0.6 : 0.78);
  const kind = realWorldAmbientKind(place, i, seed);
  drawAmbientLifeShape(kind, x, y, scale, side, place, design, theme, seed + i * 37);
}

ctx.restore();
}

function drawLivingWorldAmbientMotionPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, phoneMode) {
  const route = routeWorldInfo(place);
  const set = livingWorldActivitySet(place);
  const length = Math.max(1, raceLength());
  const spacing = phoneMode ? 500 : 390;
  const lookBehind = phoneMode ? 180 : 260;
  const lookAhead = phoneMode ? 1700 : 2300;
  const first = Math.floor((raceState.distance - lookBehind) / spacing) - 1;
  const last = Math.ceil((raceState.distance + lookAhead) / spacing) + 1;
  const items = [];
  for (let index = first; index <= last; index += 1) {
    const baseDistance = index * spacing + seededUnit(seed, index + 620) * spacing * 0.42;
    if (baseDistance < 0 || baseDistance > length + lookAhead * 0.25) continue;
    const gap = baseDistance - raceState.distance;
    if (gap < -lookBehind || gap > lookAhead) continue;
    const side = seededUnit(seed, index + 630) > 0.5 ? 1 : -1;
    const lane = side * (2.16 + seededUnit(seed, index + 640) * (phoneMode ? 0.34 : 0.54));
    const pos = roadObjectPos(lane, baseDistance);
    if (!pos || pos.y < horizon - 38 || pos.y > h + 96) continue;
    const depth = Math.max(0.04, Math.min(1, pos.depth || (pos.y - horizon) / Math.max(1, h - horizon)));
    const x = roadsideAnchorX(w, h, horizon, roadTop, roadBottom, pos.y, side, pos.x, phoneMode, phoneMode ? 8 : 18);
    const scale = Math.max(0.42, Math.min(phoneMode ? 1.62 : 1.82, pos.scale * (0.98 + depth * 0.86)));
    items.push({
      x,
      y: pos.y,
      scale,
      side,
      depth,
      index,
      kind: set[Math.abs(index + Math.floor(seed % set.length)) % set.length]
    });
  }

  ctx.save();
  ctx.globalAlpha = 1;
  clipOutsideRealWorldRoad(w, h, horizon, roadTop, roadBottom, phoneMode ? 12 : 18);
  items.sort((a, b) => a.y - b.y).forEach((item) => {
    ctx.globalAlpha = Math.min(1, phoneMode ? 0.78 + item.depth * 0.2 : 0.7 + item.depth * 0.24);
    drawGroundLockedDriveByShadow(item.x, item.y, item.scale * 0.72, item.side, theme);
    drawLivingWorldActivityShape(item.kind, item.x, item.y, item.scale, item.side, place, route, design, theme, seed + item.index *
47);
  });
  ctx.restore();
  ctx.globalAlpha = 1;
}

function livingWorldActivitySet(place) {
  const sets = {
    coast: ["beachJoggers", "cyclistLane", "pierPeople", "shoreBirds", "boatWake", "patioCrowd"],
    city: ["sidewalkQueue", "cyclistLane", "joggers", "busStopLife", "patioCrowd", "deliveryVan"],
    canyon: ["hikerLookout", "wildlifeLookout", "tourPullout", "rallyCrew", "birdFlock"],
    alpine: ["hikerLookout", "skiQueue", "wildlifeLookout", "cabinPorch", "snowPlowCrew"],
    harbor: ["dockWorkers", "forkliftSweep", "pierPeople", "boatWake", "shoreBirds"],
    snow: ["skiQueue", "snowPlowCrew", "cabinPorch", "wildlifeLookout", "snowMarkerCrew"],
    airfield: ["runwayCrew", "airportBeacon", "serviceCartMotion", "planeSpotters", "birdFlock"],
    pursuit: ["sidewalkQueue", "policeCheckpoint", "cameraCrew", "deliveryVan", "joggers"],
    freight: ["loadingCrew", "forkliftSweep", "dieselStopLife", "yardWorkers", "serviceTruckMotion"],
    interstate: ["restStopLife", "dieselStopLife", "serviceTruckMotion", "truckScaleCrew", "birdFlock"],
    farm: ["pastureAnimals", "fieldWorker
```

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```
s", "tractorMotion", "farmStand", "roadsideFenceLife"],
monsterpark: ["stadiumCrowd", "pitCrew", "cameraCrew", "marshalBarrier", "serviceTruckMotion"],
military: ["rangeCrew", "militaryPatrol", "watchPostLife", "serviceTruckMotion", "cameraCrew"],
skybase: ["runwayCrew", "airportBeacon", "serviceCartMotion", "planeSpotters", "radarCrew"],
tokyo: ["sidewalkQueue", "cyclistLane", "tramStopLife", "neonCrowd", "deliveryVan"],
desert: ["rallyCrew", "wildlifeLookout", "supportTentLife", "birdFlock", "tourPullout"],
rainforest: ["marketLife", "woodBridgeWalkers", "canopyAnimals", "riverCrew", "hikerLookout"],
europe: ["patioCrowd", "tramStopLife", "cyclistLane", "pastureAnimals", "villageWalkers"]
};
return sets[place] || sets.city;
}

function drawLivingWorldActivityShape(kind, x, y, scale, side, place, route, design, theme, seed) {
  const accent = design.accent || theme[1] || "#46d9ff";
  const light = design.light || theme[2] || "#ffd166";
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  if (kind === "cyclistLane") {
    drawLivingCyclists(side, accent, light, seed);
  } else if (kind === "beachJoggers" || kind === "joggers" || kind === "villageWalkers" || kind === "woodBridgeWalkers") {
    drawLivingJoggers(side, accent, light, seed);
  } else if (kind === "sidewalkQueue" || kind === "patioCrowd" || kind === "neonCrowd" || kind === "stadiumCrowd" || kind === "farmStand" || kind === "marketLife" || kind === "fieldWorkers") {
    drawLivingCrowdMoment(kind, side, accent, light, seed);
  } else if (kind === "busStopLife" || kind === "tramStopLife" || kind === "skiQueue") {
    drawLivingTransitStop(kind, side, accent, light, seed);
  } else if (kind === "pierPeople" || kind === "dockWorkers" || kind === "shoreBirds" || kind === "boatWake" || kind === "riverCrew") {
    drawLivingWaterEdgeMoment(kind, side, place, accent, light, seed);
  } else if (kind === "pastureAnimals" || kind === "wildlifeLookout" || kind === "canopyAnimals" || kind === "roadsideFenceLife") {
    drawLivingAnimalSetback(kind, side, place, accent, light, seed);
  } else if (kind === "airportBeacon" || kind === "runwayCrew" || kind === "serviceCartMotion" || kind === "planeSpotters" || kind === "radarCrew") {
    drawLivingAirfieldMoment(kind, side, route, place, design, theme, seed);
  } else if (kind === "forkliftSweep" || kind === "loadingCrew" || kind === "yardWorkers" || kind === "dieselStopLife" || kind === "serviceTruckMotion" || kind === "restStopLife" || kind === "truckScaleCrew" || kind === "deliveryVan" || kind === "tractorMotion") {
    drawLivingWorksiteMoment(kind, side, route, place, design, theme, seed);
  } else if (kind === "hikerLookout" || kind === "tourPullout" || kind === "rallyCrew" || kind === "supportTentLife" || kind === "snowPlowCrew" || kind === "snowMarkerCrew" || kind === "cabinPorch") {
    drawLivingRouteOutpost(kind, side, route, place, design, theme, seed);
  } else if (kind === "policeCheckpoint" || kind === "cameraCrew" || kind === "pitCrew" || kind === "marshalBarrier" || kind === "rangeCrew" || kind === "militaryPatrol" || kind === "watchPostLife") {
    drawLivingEventCrew(kind, side, route, place, design, theme, seed);
  } else {
    drawReferenceBystanderBarrier(side, accent, light, seed);
  }
  ctx.restore();
}

function drawLivingCyclists(side, accent, light, seed) {
  ctx.save();
  ctx.scale(side, 1);
  ctx.strokeStyle = "rgba(244,251,248,0.56)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-76, 22);
  ctx.lineTo(86, 16);
  ctx.stroke();
  for (let i = 0; i < 2; i += 1) {
    const x = -36 + i * 58 + Math.sin(raceState.elapsed * 2 + seed + i) * 5;
    ctx.strokeStyle = i % 2 ? light : accent;
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.arc(x - 12, 8, 10, 0, Math.PI * 2);
    ctx.arc(x + 14, 8, 10, 0, Math.PI * 2);
    ctx.moveTo(x - 12, 8);
    ctx.lineTo(x + 1, -8);
    ctx.lineTo(x + 14, 8);
    ctx.lineTo(x - 2, 8);
    ctx.lineTo(x + 1, -8);
    ctx.stroke();
    ctx.fillStyle = i % 2 ? light : accent;
    ctx.beginPath();
    ctx.arc(x, -24, 5, 0, Math.PI * 2);
    ctx.fill();
    roundRect(x - 5, -18, 10, 16, 3);
    ctx.fill();
  }
  ctx.resto
```

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```
re();
}

function drawLivingJoggers(side, accent, light, seed) {
  ctx.save();
  ctx.scale(side, 1);
  ctx.strokeStyle = "rgba(244,251,248,0.36)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-82, 19);
  ctx.lineTo(84, 12);
  ctx.stroke();
  const colors = [accent, light, "#f4fbf8", "#ff5b6b"];
  for (let i = 0; i < 4; i += 1) {
    const x = -46 + i * 28 + Math.sin(raceState.elapsed * 2.8 + seed + i) * 3;
    const leg = Math.sin(raceState.elapsed * 5 + i) * 5;
    ctx.strokeStyle = "rgba(5,8,7,0.52)";
    ctx.lineWidth = 2.4;
    ctx.beginPath();
    ctx.moveTo(x, -1);
    ctx.lineTo(x - 6 - leg, 16);
    ctx.moveTo(x, -1);
    ctx.lineTo(x + 6 + leg, 15);
    ctx.stroke();
    ctx.fillStyle = colors[(i + Math.floor(seed)) % colors.length];
    roundRect(x - 5, -20, 10, 18, 4);
    ctx.fill();
    ctx.fillStyle = "rgba(238,196,150,0.9)";
    ctx.beginPath();
    ctx.arc(x, -27, 5, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.restore();
}

function drawLivingCrowdMoment(kind, side, accent, light, seed) {
  if (kind === "stadiumCrowd") {
    ctx.fillStyle = "rgba(5,8,7,0.72)";
    roundRect(-94, -58, 188, 52, 5);
    ctx.fill();
    for (let row = 0; row < 3; row += 1) {
      drawLivingPeopleCluster(0, -36 + row * 13, 0.72, side, "stadium", { accent, light }, [accent, light], seed + row);
    }
    ctx.strokeStyle = `${light}88`;
    ctx.lineWidth = 4;
    ctx.strokeRect(-98, -64, 196, 62);
    return;
  }
  if (kind === "farmStand") {
    ctx.fillStyle = "rgba(84,46,24,0.82)";
    roundRect(-58, -36, 116, 52, 5);
    ctx.fill();
    ctx.fillStyle = light;
    ctx.beginPath();
    ctx.moveTo(-68, -36);
    ctx.lineTo(0, -70);
    ctx.lineTo(68, -36);
    ctx.closePath();
    ctx.fill();
  } else if (kind === "patioCrowd") {
    ctx.fillStyle = "rgba(5,8,7,0.58)";
    roundRect(-70, -30, 140, 32, 4);
    ctx.fill();
    ctx.fillStyle = `${light}88`;
    for (let i = -2; i <= 2; i += 1) ctx.fillRect(i * 22 - 5, -44, 10, 16);
  } else if (kind === "neonCrowd") {
    ctx.fillStyle = `${accent}66`;
    roundRect(-76, -48, 152, 36, 5);
    ctx.fill();
    ctx.fillStyle = "rgba(5,8,7,0.84)";
    roundRect(-66, -40, 132, 20, 4);
    ctx.fill();
  }
  drawReferenceBystanderBarrier(side, accent, light, seed);
}

function drawLivingTransitStop(kind, side, accent, light, seed) {
  ctx.save();
  ctx.scale(side, 1);
  ctx.fillStyle = "rgba(5,8,7,0.72)";
  roundRect(-78, -52, 156, 60, 6);
  ctx.fill();
  ctx.fillStyle = kind === "tramStopLife" ? `${accent}88` : kind === "skiQueue" ? "rgba(244,251,248,0.74)" : `${light}88`;
  roundRect(-88, -66, 176, 14, 4);
  ctx.fill();
  if (kind === "tramStopLife") {
    ctx.strokeStyle = `${light}88`;
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-96, 15);
    ctx.lineTo(96, 8);
    ctx.moveTo(-96, 24);
    ctx.lineTo(96, 17);
    ctx.stroke();
  }
  drawLivingPeopleCluster(0, -8, 0.9, 1, "transit", { accent, light }, [accent, light], seed);
  ctx.restore();
}

function drawLivingWaterEdgeMoment(kind, side, place, accent, light, seed) {
  ctx.save();
  ctx.scale(side, 1);
  ctx.fillStyle = place === "rainforest" ? "rgba(54,217,138,0.2)" : "rgba(70,217,255,0.2)";
  ctx.beginPath();
  ctx.ellipse(0, 20, 118, 18, 0, 0, Math.PI * 2);
  ctx.fill();
  if (kind === "boatWake" || kind === "riverCrew") {
    ctx.strokeStyle = "rgba(244,251,248,0.38)";
    ctx.lineWidth = 3;
    for (let i = 0; i < 3; i += 1) {
      ctx.beginPath();
      ctx.moveTo(-82 + i * 24, 18 + i * 5);
      ctx.quadraticCurveTo(-20, 5 + i * 2, 62, 16 + i * 5);
    }
  }
}
```

```

    ctx.stroke();
  }
  drawWaterSceneBoat(0, -2, 0.5, 1, [accent, light], seed);
} else if (kind === "shoreBirds") {
  drawReferenceAnimalGroup("shoreBird", 1, accent, light, seed);
  drawLivingBirdFlock(0, -58, 0.82, 1, light, seed);
} else {
  ctx.fillStyle = "rgba(102,72,40,0.9)";
  roundRect(-88, -8, 176, phoneGraphicsActive() ? 22 : 16, 5);
  ctx.fill();
  if (!phoneGraphicsActive()) {
    for (let i = -3; i <= 3; i += 1) {
      ctx.fillRect(i * 28 - 3, -8, 6, 42);
    }
  } else {
    ctx.fillStyle = "rgba(244,251,248,0.24)";
    roundRect(-72, -2, 144, 5, 3);
    ctx.fill();
  }
}
drawReferencePeopleLine(1, acc

```

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```
ent, light, seed);
}
ctx.restore();
}

function drawLivingAnimalSetback(kind, side, place, accent, light, seed) {
  ctx.save();
  ctx.scale(side, 1);
  drawReferenceGuardrail(1, place === "farm" || kind === "pastureAnimals" ? "rgba(156,112,66,0.72)" : "rgba(244,251,248,0.42)");
  const species = kind === "canopyAnimals" ? "monkey" : kind === "pastureAnimals" || kind === "roadsideFenceLife" ?
routeAnimalSpecies(place, seed) : routeAnimalSpecies(place, seed + 11);
  ctx.translate(0, -5);
  drawReferenceAnimalGroup(species, 1, accent, light, seed);
  if (kind === "wildlifeLookout") {
    ctx.translate(28, -26);
    drawReferencePeopleLine(1, accent, light, seed + 2);
  }
  ctx.restore();
}

function drawLivingAirfieldMoment(kind, side, route, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  ctx.scale(side, 1);
  if (kind === "airportBeacon" || kind === "radarCrew") {
    ctx.strokeStyle = "rgba(205,218,214,0.7)";
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(0, 18);
    ctx.lineTo(0, -82);
    ctx.stroke();
    ctx.fillStyle = kind === "radarCrew" ? `${accent}88` : `${light}aa`;
    ctx.beginPath();
    ctx.arc(0, -92, 13 + Math.sin(raceState.elapsed * 3 + seed) * 2, 0, Math.PI * 2);
    ctx.fill();
    drawReferencePeopleLine(1, accent, light, seed);
  } else if (kind === "planeSpotters") {
    drawReferenceBystanderBarrier(1, accent, light, seed);
    drawAirTrafficSilhouette(38, -56, 0.34, 1, theme, seed);
  } else {
    drawLivingServiceVehicle("serviceCart", 0, 0, 0.82, 1, route, place, design, theme, seed);
    ctx.translate(48, -16);
    drawReferencePeopleLine(1, accent, light, seed);
  }
  ctx.restore();
}

function drawLivingWorksiteMoment(kind, side, route, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  ctx.scale(side, 1);
  const vehicleKind = kind === "forkliftSweep" ? "cargoForklift" : kind.includes("Truck") || kind.includes("diesel") ||
kind.includes("rest") ? "serviceTruck" : "maintenanceCart";
  drawLivingServiceVehicle(vehicleKind, 0, 0, 0.82, 1, route, place, design, theme, seed);
  if (kind === "forkliftSweep") {
    ctx.fillStyle = `${light}44`;
    ctx.beginPath();
    ctx.moveTo(32, -14);
    ctx.lineTo(102, -38);
    ctx.lineTo(102, 4);
    ctx.closePath();
    ctx.fill();
  }
  ctx.translate(54, -16);
  drawReferencePeopleLine(1, accent, light, seed + 4);
  ctx.strokeStyle = "rgba(244,251,248,0.34)";
  ctx.lineWidth = 3;
  ctx.beginPath();
  ctx.moveTo(-102, 22);
  ctx.lineTo(112, 12);
  ctx.stroke();
  ctx.restore();
}

function drawLivingRouteOutpost(kind, side, route, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  ctx.scale(side, 1);
  if (kind === "snowPlowCrew" || kind === "snowMarkerCrew") {
    drawLivingServiceVehicle(kind === "snowPlowCrew" ? "marshalCart", 0, 0, 0.82, 1, route, place, design, theme,
seed);
  } else if (kind === "supportTentLife" || kind === "rallyCrew") {
    ctx.fillStyle = `${light}88`;
    ctx.beginPath();
    ctx.moveTo(-56, 12);
    ctx.lineTo(0, -62);
    ctx.lineTo(58, 12);
    ctx.closePath();
    ctx.fill();
  } else if (kind === "tourPullout") {
    drawLivingServiceVehicle("tourVan", 0, 0, 0.84, 1, route, place, design, theme, seed);
  } else if (kind === "cabinPorch") {
    ctx.fillStyle = "rgba(84,58,38,0.86)";
    roundRect(-64, -58, 128, 66, 5);
    ctx.fill();
    ctx.fillStyle = `${light}77`;
    roundRect(-38, -34, 22, 22, 3);
    roundRect(18, -34, 22, 22, 3);
    ctx.fill();
  } else {
    drawReferenceGuardrail(1, "rgba(244,251,248,0.5)");
  }
  ctx.translate(48, -18);
  drawReferencePeopleLine(1, accent, light, seed);
  if (kind === "hikerLookout") drawLivingBirdFlock(-34, -68, 0.74, 1, light, seed);
  ctx.restore();
}

function drawLivingEventCrew(kind, side, route, place, design, theme, seed) {
  const accent = design.accent || theme[1];
```

```
const light = design.light || theme[2];
ctx.save();
ctx.scale(side, 1);
if (kind === "militaryPatrol" || kind === "rangeCrew") {
```


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```
drawLivingServiceVehicle("serviceTruck", 0, 0, 0.84, 1, route, place, design, theme, seed);
ctx.fillStyle = "rgba(48,68,42,0.82)";
roundRect(52, -42, 64, 52, 4);
ctx.fill();
} else if (kind === "watchPostLife") {
ctx.fillStyle = "rgba(58,62,54,0.9)";
roundRect(-42, -88, 84, 42, 4);
ctx.fill();
ctx.strokeStyle = "rgba(180,192,188,0.62)";
ctx.lineWidth = 4;
ctx.beginPath();
ctx.moveTo(-34, 12);
ctx.lineTo(-20, -46);
ctx.moveTo(34, 12);
ctx.lineTo(20, -46);
ctx.stroke();
} else if (kind === "pitCrew") {
drawLivingServiceVehicle("pitCart", 0, 0, 0.78, 1, route, place, design, theme, seed);
drawReferenceBystanderBarrier(1, accent, light, seed);
} else if (kind === "cameraCrew") {
drawLivingServiceVehicle("cameraCart", -32, 0, 0.72, 1, route, place, design, theme, seed);
ctx.fillStyle = "rgba(5,8,7,0.78)";
roundRect(32, -46, 38, 24, 4);
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.52)";
ctx.lineWidth = 3;
ctx.beginPath();
ctx.moveTo(50, -20);
ctx.lineTo(34, 16);
ctx.moveTo(50, -20);
ctx.lineTo(66, 16);
ctx.stroke();
} else {
drawReferenceBystanderBarrier(1, accent, light, seed);
}
ctx.translate(58, -18);
drawReferencePeopleLine(1, accent, light, seed + 5);
ctx.restore();
}

function drawLivingBirdFlock(x, y, scale, side, color, seed) {
ctx.save();
ctx.translate(x, y);
ctx.scale(scale * side, scale);
ctx.strokeStyle = color || "rgba(244,251,248,0.7)";
ctx.lineWidth = 2.4;
for (let i = 0; i < 5; i += 1) {
const bx = -44 + i * 22;
const by = Math.sin(seed + i) * 9;
ctx.beginPath();
ctx.moveTo(bx - 8, by);
ctx.quadraticCurveTo(bx, by - 8, bx + 8, by);
ctx.stroke();
}
ctx.restore();
}

function realWorldAmbientKind(place, index, seed) {
const sets = {
coast: ["kites", "cyclists", "surfers", "balconyLights", "joggers", "birdFlock", "boatWake"],
city: ["trafficLights", "storefronts", "busShelter", "crosswalkCrew", "cyclistLane", "sidewalkQueue", "tramStop"],
canyon: ["dustDevils", "lookout", "trailSigns", "tourVan", "wildlifeLookout", "hikers"],
alpine: ["skiLift", "snowFence", "cabinLights", "switchbackPosts", "wildlifeLookout", "joggers"],
harbor: ["dockLights", "cargoForklift", "mooredBoat", "harborWorkers", "boatWake", "birdFlock"],
snow: ["skiLift", "snowMarkers", "cabinLights", "plowBanks", "wildlifeLookout", "sidewalkQueue"],
airfield: ["windsock", "runwayCrew", "serviceCart", "towerLights", "airportBeacon", "forkliftSweep"],
pursuit: ["trafficLights", "storefronts", "crosswalkCrew", "policeQueue", "sidewalkQueue"],
freight: ["weighStation", "yardLights", "loadingCrew", "dieselPump", "forkliftSweep", "sidewalkQueue"],
interstate: ["weighStation", "dieselPump", "serviceCart", "birdFlock", "sidewalkQueue"],
farm: ["windmill", "mailbox", "fieldWorkers", "hayBales", "pastureLine", "birdFlock"],
monsterpark: ["rallyFlags", "sidewalkQueue", "cameraCrew", "pitCrew", "towerLights"],
military: ["towerLights", "serviceCart", "sidewalkQueue", "rangeCrew", "birdFlock"],
skybase: ["windsock", "airportBeacon", "runwayCrew", "serviceCart", "birdFlock"],
tokyo: ["trafficLights", "storefronts", "railLights", "crosswalkCrew", "cyclistLane", "sidewalkQueue", "tramStop"],
desert: ["rallyFlags", "dustDevils", "supportTent", "trailSigns", "wildlifeLookout", "birdFlock"],
rainforest: ["hangingVines", "marketStall", "mistCloud", "woodBridge", "canopyAnimals", "sidewalkQueue"],
europe: ["balconyLights", "stoneWall", "cafeAwning", "switchbackPosts", "cyclistLane", "pastureLine", "tramStop"]
};
const list = sets[place] || sets.city;
return list[(index + Math.floor(seed % list.length)) % list.length];
}

function drawRealWorldPowerLines(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion) {
ctx.save();
ctx.globalAlpha *= 0.58;
for (let side = -1; side <= 1; side += 2) {
ctx.strokeStyle = place === "tokyo" ? `${design.accent}44` : "rgba(180,192,188,0.28)";
ctx.lineWidth = Math.max(1, w * 0.002);
for (let wire = 0; wire < 3; wire += 1) {
ctx.beginPath();

```

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```
for (let step = 0; step <= 8; step += 1) {
  const t = step / 8;
  const y = horizon + h * (0.03 + wire * 0.025 + t * 0.2);
  const roadT = Math.max(0.04, Math.min(0.42, (y - horizon) / Math.max(1, h - horizon)));
  const half = realWorldRoadHalf(roadTop, roadBottom, roadT);
  const sag = Math.sin(t * Math.PI) * h * 0.018;
  const x = perspectiveRoadCenter(w, roadT) + side * (half + w * (0.18 + wire * 0.018)) + Math.sin(motion + step + wire) * 10;
  if (step === 0) ctx.moveTo(x, y + sag);
  else ctx.lineTo(x, y + sag);
}
ctx.stroke();
}
}
ctx.restore();
}

function drawAmbientLifeShape(kind, x, y, scale, side, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  if (kind === "trafficLights") {
    ctx.strokeStyle = "rgba(180,192,188,0.62)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(0, 16);
    ctx.lineTo(0, -80);
    ctx.lineTo(-42 * side, -80);
    ctx.stroke();
    ctx.fillStyle = "rgba(5,8,7,0.92)";
    roundRect(-54 * side, -92, 25 * side, 42, 4);
    ctx.fill();
    ["#ff3348", "#fffd166", "#36d98a"].forEach((color, index) => {
      ctx.fillStyle = color;
      ctx.beginPath();
      ctx.arc(-42 * side, -84 + index * 13, 4, 0, Math.PI * 2);
      ctx.fill();
    });
  } else if (kind === "storefronts" || kind === "cafeAwning" || kind === "marketStall") {
    ctx.fillStyle = "rgba(6,14,14,0.82)";
    roundRect(-58, -42, 116, 54, 4);
    ctx.fill();
    ctx.fillStyle = kind === "cafeAwning" ? `${light}aa` : `${accent}88`;
    for (let i = -3; i <= 3; i += 1) {
      ctx.fillRect(i * 17, -48, 10, 12);
    }
    ctx.fillStyle = "rgba(244,251,248,0.22)";
    ctx.fillRect(-42, -26, 28, 28);
    ctx.fillRect(16, -26, 28, 28);
  } else if (kind === "crosswalkCrew" || kind === "harborWorkers" || kind === "runwayCrew" || kind === "fieldWorkers" || kind === "loadingCrew") {
    drawRoadsidePeopleCluster(side, accent, light, seed);
    ctx.fillStyle = `${light}88`;
    ctx.beginPath();
    ctx.moveTo(side * 28, -14);
    ctx.lineTo(side * 42, -2);
    ctx.lineTo(side * 28, 10);
    ctx.closePath();
    ctx.fill();
  } else if (kind === "windsock") {
    ctx.strokeStyle = "rgba(205,218,214,0.7)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(0, 18);
    ctx.lineTo(0, -72);
    ctx.stroke();
    ctx.fillStyle = "rgba(255,91,107,0.78)";
    ctx.beginPath();
    ctx.moveTo(0, -72);
    ctx.lineTo(54 * side, -64);
    ctx.lineTo(44 * side, -48);
    ctx.lineTo(0, -54);
    ctx.closePath();
    ctx.fill();
  } else if (kind === "windmill") {
    ctx.strokeStyle = "rgba(205,218,214,0.58)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(0, 18);
    ctx.lineTo(0, -76);
    ctx.stroke();
    ctx.strokeStyle = "rgba(244,251,248,0.5)";
    ctx.beginPath();
    for (let a = 0; a < 4; a += 1) {
      const ang = a * Math.PI * 0.5 + raceState.elapsed * 0.4;
      ctx.moveTo(0, -76);
      ctx.lineTo(Math.cos(ang) * 34, -76 + Math.sin(ang) * 34);
    }
    ctx.stroke();
  } else if (kind === "kites" || kind === "surfers") {
    ctx.strokeStyle = `${accent}77`;
    ctx.lineWidth = 2;
    ctx.beginPath();
    ctx.moveTo(0, 10);
    ctx.quadraticCurveTo(side * 26, -30, side * 56, -66);
    ctx.stroke();
    ctx.fillStyle = kind === "kites" ? `${light}99` : `${accent}66`;
    ctx.beginPath();
    ctx.moveTo(side * 56, -78);
    ctx.lineTo(side * 76, -64);
    ctx.lineTo(side * 56, -50);
    ctx.lineTo(side * 36, -64);
    ctx.closePath();
    ctx.fill();
  } else if (kind === "skiLift") {
    ctx.strokeStyle = "rgba(244,251,248,0.34)";
    ctx.lineWidth = 2;
    ctx.beginPath();
    ctx.moveTo(-80 * side, -78);
    ctx.lineTo(80 * side, -92);
    ctx.stroke();
    ctx.fillStyle = "rgba(5,8,7,0.74)";
```

```
    roundRect(18 * side, -80, 28 * side, 18, 3);
    ctx.fill();
  } else if (kind === "dustDevils" || kind === "mistCloud") {
    ctx.fillStyle = kind === "dustDevils" ? "rgba(255,183,74,0.18)" : "rgba(222,249,255,0.18)";
    for (let i = 0; i < 5; i += 1) {
      ctx.beginPath();
      ctx.ellipse(side * i * 7, -i * 9
```

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```
, 22 - i * 2, 0 + i, i * 0.2, 0, Math.PI * 2);
    ctx.fill();
  }
} else if (kind === "cyclists" || kind === "cyclistLane") {
  drawLivingCyclists(side, accent, light, seed);
} else if (kind === "joggers" || kind === "hikers") {
  drawLivingJoggers(side, accent, light, seed);
} else if (kind === "sidewalkQueue" || kind === "policeQueue" || kind === "cameraCrew" || kind === "pitCrew" || kind === "rangeCrew" || kind === "tramStop") {
  if (kind === "tramStop") drawLivingTransitStop("tramStopLife", side, accent, light, seed);
  else drawReferenceBystanderBarrier(side, accent, light, seed);
} else if (kind === "birdFlock") {
  drawLivingBirdFlock(0, -28, 1, side, light, seed);
} else if (kind === "boatWake") {
  drawLivingWaterEdgeMoment("boatWake", side, place, accent, light, seed);
} else if (kind === "pastureLine" || kind === "wildlifeLookout" || kind === "canopyAnimals") {
  drawLivingAnimalSetback(kind === "pastureLine" ? "pastureAnimals" : kind, side, place, accent, light, seed);
} else if (kind === "airportBeacon" || kind === "forkliftSweep") {
  if (kind === "airportBeacon") drawLivingAirfieldMoment(kind, side, routeWorldInfo(place), place, design, theme, seed);
  else drawLivingWorksiteMoment(kind, side, routeWorldInfo(place), place, design, theme, seed);
} else {
  ctx.fillStyle = `${accent}22`;
  ctx.beginPath();
  ctx.ellipse(0, 4, 56, 18, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = `${light}66`;
  roundRect(-26, -22, 52, 24, 4);
  ctx.fill();
}
ctx.scale(side, 1);
ctx.restore();
}

function drawRealWorldPavementStoryPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const motion = (raceState.roadOffset || 0) * (0.78 + Math.min(1.2, Math.max(0, raceState.speed || 0) / 230));
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  drawRealWorldRoutePaintGlyphs(w, h, horizon, place, design, theme, seed, motion);
  drawRealWorldSurfaceEffects(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode);
  ctx.restore();
}

function drawRealWorldRoutePaintGlyphs(w, h, horizon, place, design, theme, seed, motion) {
  const labels = {
    coast: ["COAST", "VIEW", "PCH"],
    city: ["LANE", "ONLY", "SLOW"],
    canyon: ["CURVE", "MESA", "WATCH"],
    alpine: ["PASS", "ICE", "LOW"],
    harbor: ["PIER", "DOCK", "BRG"],
    snow: ["SNOW", "ICE", "PASS"],
    airfield: ["RWY", "TAXI", "04"],
    freight: ["LOAD", "TRUCK", "EXIT"],
    farm: ["COUNTY", "GRAVEL", "RALLY"],
    tokyo: ["EXP", "TUNNEL", "NIGHT"],
    desert: ["RALLY", "DUNE", "CHECK"],
    rainforest: ["RAIN", "BRIDGE", "SLOW"],
    europe: ["ALPS", "VILLAGE", "PASS"]
  };
  const set = labels[place] || labels.city;
  const count = phoneGraphicsActive() ? 3 : 5;
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * 226 + motion * 0.38 + seed * 0.008) % (h - horizon + 300)) - 86);
    if (y < horizon || y > h * 0.95) continue;
    const t = Math.max(0.08, Math.min(0.98, (y - horizon) / Math.max(1, h - horizon)));
    const x = realWorldRoadX(w, i % 2 ? -0.76 : 0.76, t);
    ctx.save();
    ctx.translate(x, y);
    ctx.scale(0.72 + t * 0.76, 0.34 + t * 0.38);
    ctx.globalAlpha = phoneGraphicsActive() ? 0.11 + t * 0.05 : 0.16 + t * 0.08;
    ctx.fillStyle = place === "airfield" || place === "freight" || place === "farm" ? "rgba(255,209,102,0.76)" : "rgba(244,251,248,0.72)";
    ctx.font = "900 18px system-ui";
    ctx.textAlign = "center";
    ctx.textBaseline = "middle";
    ctx.fillText(set[(i + Math.floor(seed)) % set.length], 0, 0, 92);
    ctx.restore();
  }
}

function drawRealWorldSurfaceEffects(w, h, horizon, roadTop, roadBottom, place, design, theme, seed, motion, phoneMode) {
  const wet = ["coast", "city", "harbor", "tokyo", "rainforest", "europe"].includes(place);
  const dusty = ["canyon", "desert", "farm", "freight"].includes(place);
  const snowy = ["snow", "alpine"].includes(place);
  const count = phoneMode ? 12 : 22;
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * 76 + motion * (wet ? 1.02 : 0.72) + seed * 0.003) % (h - horizon + 160)) - 52
```

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```
);
  if (y < horizon || y > h + 20) continue;
  const t = Math.max(0.04, Math.min(1.06, (y - horizon) / Math.max(1, h - horizon)));
  const lane = -1.9 + seededUnit(seed, i + 320) * 3.8;
  const x = realWorldRoadX(w, lane, t);
  const rw = (12 + seededUnit(seed, i + 330) * 64) * (0.45 + t);
  const rh = (2 + seededUnit(seed, i + 340) * 7) * (0.5 + t * 0.6);
  ctx.save();
  if (wet) {
    ctx.globalCompositeOperation = "screen";
    ctx.globalAlpha = phoneMode ? 0.07 + t * 0.05 : 0.11 + t * 0.09;
    ctx.fillStyle = seededUnit(seed, i + 350) > 0.5 ? `${design.accent}55` : "rgba(222,249,255,0.45)";
    ctx.beginPath();
    ctx.ellipse(x, y, rw, rh, 0, 0, Math.PI * 2);
    ctx.fill();
  } else if (dusty) {
    ctx.globalAlpha = phoneMode ? 0.06 + t * 0.04 : 0.1 + t * 0.08;
    ctx.fillStyle = place === "farm" ? "rgba(187,242,74,0.18)" : "rgba(255,183,74,0.22)";
    ctx.beginPath();
    ctx.ellipse(x, y, rw * 0.78, rh * 1.8, 0, 0, Math.PI * 2);
    ctx.fill();
  } else if (snowy) {
    ctx.globalAlpha = phoneMode ? 0.08 + t * 0.05 : 0.12 + t * 0.09;
    ctx.fillStyle = "rgba(244,251,248,0.45)";
    ctx.beginPath();
    ctx.ellipse(x, y, rw * 0.62, rh * 1.6, 0, 0, Math.PI * 2);
    ctx.fill();
  } else {
    ctx.globalAlpha = phoneMode ? 0.06 : 0.1;
    ctx.fillStyle = "rgba(244,251,248,0.18)";
    roundRect(x - rw * 0.4, y - rh, rw * 0.8, rh * 1.7, Math.max(2, rh));
    ctx.fill();
  }
  ctx.restore();
}

if (place === "airfield") {
  for (let i = 0; i < 5; i += 1) {
    const y = horizon + (((i * 188 + motion * 0.5) % (h - horizon + 280)) - 92);
    if (y < horizon || y > h * 0.92) continue;
    const t = Math.max(0.08, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
    drawPerspectivePavementQuad(w, horizon, 0, y, 8 + t * 16, laneWidth() * 4.8, "rgba(244,251,248,0.11)");
  }
}

function drawRealWorldNearSideInfrastructurePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const motion = (raceState.roadOffset || 0) * 1.14;
  ctx.save();
  const count = phoneMode ? 7 : 11;
  for (let i = 0; i < count; i += 1) {
    const y = h * 0.64 + (((i * 118 + motion + seed * 0.004) % (h * 0.48 + 140)) - 52);
    if (y < h * 0.6 || y > h + 40) continue;
    const t = Math.max(0.6, Math.min(1.12, (y - horizon) / Math.max(1, h - horizon)));
    const side = i % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.055 + seededUnit(seed, i + 410) * 0.11));
    const scale = 0.58 + t * 0.66;
    const kind = ["barrel", "hydrant", "bench", "bollard", "roadCone", "lightPool"][i + Math.floor(seed) % 6];
    drawNearSideInfrastructure(kind, x, y, scale, side, place, design, theme, seed + i * 53);
  }
  ctx.restore();
}

function drawNearSideInfrastructure(kind, x, y, scale, side, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.globalAlpha = phoneGraphicsActive() ? 0.34 : 0.48;
  if (kind === "barrel" || kind === "roadCone") {
    ctx.fillStyle = kind === "roadCone" ? "rgba(255,91,107,0.82)" : `${light}99`;
    ctx.beginPath();
    ctx.moveTo(-12, 14);
    ctx.lineTo(-6, -24);
    ctx.lineTo(6, -24);
    ctx.lineTo(12, 14);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.72)";
    ctx.fillRect(-8, -5, 16, 4);
    ctx.fillRect(-10, 6, 20, 4);
  } else if (kind === "hydrant") {
    ctx.fillStyle = place === "city" || place === "tokyo" ? "rgba(255,91,107,0.76)" : `${accent}66`;
    roundRect(-9, -24, 18, 34, 5);
    ctx.fill();
    ctx.fillRect(-18, -12, 36, 8);
  } else if (kind === "bench") {
    ctx.fillStyle = "rgba(117,86,47,0.62)";
    roundRect(-34, -18, 68, 10, 3);
    roundRect(-34, -3, 68, 8, 3);
    ctx.fill();
    ctx.strokeStyle = "rgba(5,8,7,0.58)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-24, 5);
    ctx.lineTo(-24, 22);
    ctx.moveTo(24, 5);
    ctx.lineTo(24, 22);
    ctx.stroke();
  } else if (kind ===
```

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```
"bollard") {
  ctx.fillStyle = "rgba(205,218,214,0.58)";
  roundRect(-6, -30, 12, 44, 4);
  ctx.fill();
  ctx.fillStyle = `${accent}88`;
  ctx.fillRect(-6, -12, 12, 6);
} else {
  ctx.globalCompositeOperation = "screen";
  ctx.fillStyle = place === "snow" ? "rgba(244,251,248,0.24)" : `${accent}22`;
  ctx.beginPath();
  ctx.ellipse(0, 10, 46, 11, 0, 0, Math.PI * 2);
  ctx.fill();
}
ctx.fillStyle = "rgba(0,0,0,0.32)";
ctx.beginPath();
ctx.ellipse(0, 20, 28, 5, 0, 0, Math.PI * 2);
ctx.fill();
ctx.scale(side, 1);
ctx.restore();
}

function drawRealWorldDistrictMotionPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const motion = (raceState.roadOffset || 0) * 0.026;
  const count = phoneMode ? 7 : 12;
  ctx.save();
  ctx.globalAlpha = phoneMode ? 0.42 : 0.34;
  for (let i = 0; i < count; i += 1) {
    const side = i % 2 === 0 ? -1 : 1;
    const y = horizon + h * (0.11 + seededUnit(seed, i + 520) * 0.34);
    const t = Math.max(0.06, Math.min(0.58, (y - horizon) / Math.max(1, h - horizon)));
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half + w * (0.16 + seededUnit(seed, i + 530) * 0.2)) + Math.sin(motion + i * 0.7) * 28;
    const scale = 0.32 + t * 0.78;
    const kind = realWorldDistrictKind(place, i, seed);
    drawDistrictMotionShape(kind, x, y, scale, side, place, design, theme, seed + i * 61);
  }
  ctx.restore();
}

function realWorldDistrictKind(place, index, seed) {
  const sets = {
    coast: ["sideRoadCar", "boatWake", "pedBridge", "coastCyclist"],
    city: ["sideRoadCar", "metroTrain", "pedBridge", "deliveryVan"],
    canyon: ["tourBus", "dustTruck", "lookoutCar", "rangerGate"],
    alpine: ["snowPlow", "skiliftCar", "stoneWall", "tourBus"],
    harbor: ["forklift", "smallBoat", "railCrane", "sideRoadTruck"],
    snow: ["snowPlow", "skiliftCar", "tourBus", "snowGroomer"],
    airfield: ["taxiPlane", "serviceCart", "runwayLightRow", "radarSweep"],
    freight: ["sideRoadTruck", "yardForklift", "railCar", "weighBooth"],
    farm: ["farmTruck", "tractorRow", "mailRoute", "fieldIrrigation"],
    tokyo: ["metroTrain", "sideRoadCar", "neonBridge", "deliveryVan"],
    desert: ["rallySupport", "dustTruck", "rangerGate", "camelSign"],
    rainforest: ["riverBoat", "woodBridge", "marketCart", "mistBank"],
    europe: ["tourBus", "stoneWall", "tramLine", "villageCar"]
  };
  const list = sets[place] || sets.city;
  return list[(index + Math.floor(seed % list.length)) % list.length];
}

function drawDistrictMotionShape(kind, x, y, scale, side, place, design, theme, seed) {
  const accent = design.accent || theme[1];
  const light = design.light || theme[2];
  const phoneMode = phoneGraphicsActive();
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  if (kind === "sideRoadCar" || kind === "deliveryVan" || kind === "tourBus" || kind === "farmTruck" || kind === "sideRoadTruck" || kind === "lookoutCar" || kind === "dustTruck" || kind === "snowPlow" || kind === "villageCar") {
    const longBody = kind === "tourBus" || kind === "deliveryVan" || kind === "sideRoadTruck" || kind === "snowPlow";
    const paint = kind === "snowPlow" ? "#dce8ef" : kind === "farmTruck" ? "#36d98a" : kind === "tourBus" ? "#ffd166" : accent;
    ctx.fillStyle = "rgba(0,0,0,0.34)";
    ctx.beginPath();
    ctx.ellipse(0, 11, longBody ? 54 : 36, 6, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = paint;
    roundRect(longBody ? -50 : -34, -16, longBody ? 100 : 68, 28, 5);
    ctx.fill();
    ctx.fillStyle = "rgba(5,8,7,0.82)";
    ctx.fillRect(longBody ? -36 : -20, -10, longBody ? 28 : 18, 10);
    ctx.fillRect(longBody ? 8 : 7, -10, longBody ? 26 : 18, 10);
    ctx.fillStyle = "#050807";
    ctx.beginPath();
    ctx.arc(longBody ? -34 : -22, 13, 7, 0, Math.PI * 2);
    ctx.arc(longBody ? 34 : 22, 13, 7, 0, Math.PI * 2);
    ctx.fill();
    if (kind === "snowPlow") {
      ctx.fillStyle = "rgba(255,209,102,0.76)";
      ctx.beginPath();
      ctx.moveTo(50 * side, -6);
      ctx.lineTo(74 * side, 6);
      ctx.lineTo(50 * side, 18);
      ctx.closePath();
      ctx.fi
```

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```
ll();
}
} else if (kind === "metroTrain" || kind === "tramLine" || kind === "railCar") {
  ctx.strokeStyle = phoneMode ? "rgba(180,192,188,0.74)" : "rgba(180,192,188,0.5)";
  ctx.lineWidth = phoneMode ? 4 : 3;
  ctx.beginPath();
  ctx.moveTo(-92 * side, 16);
  ctx.lineTo(92 * side, 8);
  ctx.moveTo(-92 * side, 28);
  ctx.lineTo(92 * side, 20);
  ctx.stroke();
  ctx.fillStyle = kind === "railCar" ? "rgba(255,209,102,0.78)" : `${accent}86`;
  const cars = phoneMode ? 3 : 5;
  for (let i = -Math.floor(cars / 2); i <= Math.floor(cars / 2); i += 1) {
    const carW = phoneMode ? 42 : 26;
    roundRect(i * (phoneMode ? 47 : 30) - carW / 2, -18 + (i % 2) * 2, carW, 22, 3);
    ctx.fill();
    if (phoneMode) {
      ctx.fillStyle = "rgba(5,8,7,0.58)";
      roundRect(i * 47 - 14, -12, 28, 6, 2);
      ctx.fill();
      ctx.fillStyle = kind === "railCar" ? "rgba(255,209,102,0.78)" : `${accent}86`;
    }
  }
} else if (kind === "smallBoat" || kind === "riverBoat" || kind === "boatWake") {
  ctx.fillStyle = "rgba(70,217,255,0.18)";
  ctx.beginPath();
  ctx.ellipse(0, 16, 74, 13, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = kind === "riverBoat" ? "#36d98a" : accent;
  ctx.beginPath();
  ctx.moveTo(-58, -4);
  ctx.quadraticCurveTo(0, -28, 58, -4);
  ctx.lineTo(40, 12);
  ctx.lineTo(-40, 12);
  ctx.closePath();
  ctx.fill();
  ctx.strokeStyle = "rgba(222,249,255,0.35)";
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(-74, 20);
  ctx.quadraticCurveTo(-18, 32, 54, 20);
  ctx.stroke();
} else if (kind === "taxiPlane") {
  ctx.fillStyle = "rgba(244,251,248,0.62)";
  ctx.beginPath();
  ctx.moveTo(0, -44);
  ctx.lineTo(15, 16);
  ctx.lineTo(-15, 16);
  ctx.closePath();
  ctx.fill();
  ctx.beginPath();
  ctx.moveTo(-70, -10);
  ctx.lineTo(70, -10);
  ctx.lineTo(18, 10);
  ctx.lineTo(-18, 10);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = accent;
  ctx.fillRect(-6, -40, 12, 52);
} else if (kind === "forklift" || kind === "yardForklift" || kind === "serviceCart" || kind === "marketCart") {
  ctx.fillStyle = light;
  roundRect(-24, -18, 48, 28, 4);
  ctx.fill();
  ctx.strokeStyle = "rgba(5,8,7,0.62)";
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(28 * side, -28);
  ctx.lineTo(28 * side, 14);
  ctx.moveTo(28 * side, 8);
  ctx.lineTo(56 * side, 8);
  ctx.stroke();
  ctx.fillStyle = "#050807";
  ctx.beginPath();
  ctx.arc(-18, 13, 7, 0, Math.PI * 2);
  ctx.arc(20, 13, 7, 0, Math.PI * 2);
  ctx.fill();
} else if (kind === "pedBridge" || kind === "neonBridge" || kind === "woodBridge") {
  ctx.strokeStyle = kind === "woodBridge" ? "rgba(117,86,47,0.58)" : `${accent}55`;
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(-96 * side, -14);
  ctx.lineTo(96 * side, -22);
  ctx.moveTo(-80 * side, 8);
  ctx.lineTo(80 * side, 0);
  if (phoneMode) {
    ctx.lineWidth = 7;
    ctx.moveTo(-92 * side, -2);
    ctx.lineTo(92 * side, -8);
  } else {
    for (let i = -4; i <= 4; i += 1) {
      ctx.moveTo(i * 22 * side, -30);
      ctx.lineTo(i * 22 * side, 12);
    }
  }
  ctx.stroke();
} else if (kind === "runwayLightRow" || kind === "radarSweep") {
  ctx.globalCompositeOperation = "screen";
  ctx.fillStyle = `${light}66`;
  for (let i = -4; i <= 4; i += 1) {
    ctx.beginPath();
    ctx.arc(i * 18, 0, 4, 0, Math.PI * 2);
    ctx.fill();
  }
  if (kind === "radarSweep") {
    ctx.strokeStyle = `${accent}88`;
    ctx.lineWidth = 2;
    ctx.beginPath();
    ctx.arc(0, -28, 32, -0.3, 1.1);
    ctx.stroke();
  }
} else {
  ctx.fillStyle = `${accent}22`;
}
```

```
    ctx.beginPath();
    ctx.ellipse(0, 0, 68, 18, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = `${light}44`;
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(-56, -8);
    ctx.quadraticCurveTo(0, -26, 58, -4);
    ctx.stroke();
  }
  ctx.scale(side, 1);
  ctx.restore();
}

function drawRealWorldRoadRulePass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const motion = (raceState.roadOf
```


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```
fset || 0) * (0.7 + Math.min(1.1, Math.max(0, raceState.speed || 0) / 240));
ctx.save();
clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
const ruleCount = phoneMode ? 4 : 7;
for (let i = 0; i < ruleCount; i += 1) {
  const y = horizon + ((i * 194 + motion * 0.56 + seed * 0.006) % (h - horizon + 290)) - 94;
  if (y < horizon || y > h * 0.96) continue;
  const t = Math.max(0.08, Math.min(1.02, (y - horizon) / Math.max(1, h - horizon)));
  const type = realWorldRoadRuleKind(place, i, seed);
  drawRoadRuleMark(type, w, horizon, y, t, place, design, theme, seed + i * 71);
}
ctx.restore();
}

function realWorldRoadRuleKind(place, index, seed) {
  const sets = {
    coast: ["merge", "bikeLane", "bridgeJoint", "edgeHash"],
    city: ["turnOnly", "busLane", "stopBar", "crosswalk"],
    canyon: ["edgeHash", "slowCurve", "bridgeJoint", "gravelEdge"],
    alpine: ["iceWarning", "snowEdge", "bridgeJoint", "slowCurve"],
    harbor: ["dockLane", "bridgeJoint", "merge", "stopBar"],
    snow: ["iceWarning", "snowEdge", "slowCurve", "bridgeJoint"],
    airfield: ["runwayBars", "taxiLine", "holdShort", "beaconDots"],
    freight: ["truckLane", "weighStation", "merge", "bridgeJoint"],
    farm: ["gravelEdge", "countyLine", "tractorCross", "edgeHash"],
    tokyo: ["turnOnly", "busLane", "tunnelBands", "merge"],
    desert: ["rallyMarks", "gravelEdge", "slowCurve", "edgeHash"],
    rainforest: ["wetBridge", "slowCurve", "bridgeJoint", "edgeHash"],
    europe: ["villageCrossing", "slowCurve", "bikeLane", "bridgeJoint"]
  };
  const list = sets[place] || sets.city;
  return list[(index + Math.floor(seed % list.length)) % list.length];
}

function drawRoadRuleMark(type, w, horizon, y, t, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const alpha = phoneMode ? 0.08 + t * 0.035 : 0.18 + t * 0.08;
  ctx.save();
  ctx.globalAlpha = alpha;
  const lane = type === "busLane" || type === "bikeLane" || type === "truckLane" ? -1.15 : type === "merge" ? 1.1 : 0;
  const x = realWorldRoadX(w, lane, t);
  if (phoneMode && type === "merge") {
    drawPerspectivePavementQuad(w, horizon, 0.9, y, 4 + t * 5, laneWidth() * 0.78, "rgba(244,251,248,0.34)");
    ctx.restore();
    return;
  }
  if (type === "bridgeJoint" || type === "tunnelBands" || type === "runwayBars") {
    const stripes = phoneMode ? (type === "runwayBars" ? 4 : 2) : (type === "runwayBars" ? 6 : 4);
    for (let i = 0; i < stripes; i += 1) {
      drawPerspectivePavementQuad(w, horizon, 0, y + i * (5 + t * 6), 3 + t * 5, laneWidth() * (type === "runwayBars" ? 4.8 : 4.1),
        type === "tunnelBands" ? `${design.accent}55` : "rgba(244,251,248,0.5)");
    }
  }
  } else if (type === "edgeHash" || type === "gravelEdge" || type === "snowEdge") {
    for (let side = -1; side <= 1; side += 2) {
      const hashCount = phoneMode ? 2 : 6;
      for (let i = 0; i < hashCount; i += 1) {
        const yy = y + i * (10 + t * 8);
        const tx = realWorldRoadX(w, side * 1.84, t);
        ctx.strokeStyle = type === "snowEdge" ? "rgba(244,251,248,0.72)" : type === "gravelEdge" ? "rgba(255,183,74,0.5)" :
"rgba(244,251,248,0.44)";
        ctx.lineWidth = 2 + t * 3;
        ctx.beginPath();
        ctx.moveTo(tx, yy);
        ctx.lineTo(tx + side * (16 + t * 30), yy + 2);
        ctx.stroke();
      }
    }
  }
  } else if (type === "stopBar" || type === "holdShort" || type === "wetBridge" || type === "villageCrossing" || type ===
"crosswalk") {
    const bars = phoneMode ? (type === "crosswalk" || type === "villageCrossing" ? 3 : 1) : (type === "crosswalk" || type ===
"villageCrossing" ? 6 : 2);
    for (let i = 0; i < bars; i += 1) {
      drawPerspectivePavementQuad(w, horizon, 0, y + i * (11 + t * 8), 6 + t * 8, laneWidth() * 4.2, type === "holdShort" ?
"rgba(255,209,102,0.58)" : "rgba(244,251,248,0.58)");
    }
  }
  } else {
    ctx.translate(x, y);
    ctx.scale(0.7 + t * 0.9, 0.36 + t * 0.42);
    ctx.fillStyle = phoneMode ? "rgba(244,251,248,0.54)" : type === "busLane" || type === "bikeLane" ? `${design.accent}66` : type
=== "iceWarning" ? "rgba(244,251,248,0.68)" : type === "truckLane" || type === "weighStation" ? "rgba(255,209,102,0.7)" :
"rgba(244,251,248,0.

```

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```
62)";
  if (type === "merge") {
    ctx.beginPath();
    ctx.moveTo(-36, 32);
    ctx.lineTo(-4, -32);
    ctx.lineTo(8, -22);
    ctx.lineTo(-8, -4);
    ctx.lineTo(34, -28);
    ctx.lineTo(40, -16);
    ctx.lineTo(-4, 22);
    ctx.closePath();
    ctx.fill();
  } else {
    const text = type === "turnOnly" ? "ONLY" : type === "busLane" ? "BUS" : type === "bikelane" ? "BIKE" : type === "truckLane" ?
    "TRUCK" : type === "weighStation" ? "WEIGH" : type === "tractorCross" ? "XING" : type === "taxiLine" ? "TAXI" : type ===
    "iceWarning" ? "ICE" : type === "rallyMarks" ? "RALLY" : "SLOW";
    ctx.font = "900 18px system-ui";
    ctx.textAlign = "center";
    ctx.textBaseline = "middle";
    ctx.fillText(text, 0, 0, 88);
  }
}
ctx.restore();
}

function drawRealWorldLightShadowPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const motion = (raceState.roadOffset || 0) * 0.42;
  const count = phoneMode ? 4 : 7;
  ctx.save();
  clipRealWorldRoad(w, h, horizon, roadTop, roadBottom);
  for (let i = 0; i < count; i += 1) {
    const y = horizon + (((i * 232 + motion + seed * 0.002) % (h - horizon + 340)) - 110);
    if (y < horizon || y > h + 40) continue;
    const t = Math.max(0.06, Math.min(1.06, (y - horizon) / Math.max(1, h - horizon)));
    const shadeKind = realWorldShadowKind(place, i, seed);
    const alpha = phoneMode ? 0.06 + t * 0.04 : 0.1 + t * 0.07;
    ctx.globalAlpha = alpha;
    if (shadeKind === "underpass") {
      drawPerspectivePavementQuad(w, horizon, 0, y, 24 + t * 70, laneWidth() * 5, "rgba(0,0,0,0.82)");
      drawPerspectivePavementQuad(w, horizon, 0, y + 20 + t * 52, 6 + t * 14, laneWidth() * 5, `${design.accent}33`);
    } else if (shadeKind === "treeShadow" || shadeKind === "wireShadow") {
      ctx.strokeStyle = shadeKind === "wireShadow" ? "rgba(0,0,0,0.62)" : "rgba(0,0,0,0.46)";
      ctx.lineWidth = Math.max(1, 2 + t * 4);
      for (let s = -2; s <= 2; s += 1) {
        const x1 = realWorldRoadX(w, s * 0.72, t);
        ctx.beginPath();
        ctx.moveTo(x1 - 80 * t, y + s * 6);
        ctx.quadraticCurveTo(x1, y + 14 + s * 4, x1 + 90 * t, y + 4 + s * 6);
        ctx.stroke();
      }
    } else {
      ctx.globalCompositeOperation = "screen";
      const glow = ctx.createRadialGradient(realWorldRoadX(w, i % 2 ? -1.2 : 1.2, t), y, 3, realWorldRoadX(w, i % 2 ? -1.2 : 1.2,
t), y, 120 * t);
      glow.addColorStop(0, ` ${theme[2] || "#ffd166"}55`);
      glow.addColorStop(1, "rgba(0,0,0,0)");
      ctx.fillStyle = glow;
      ctx.fillRect(0, y - 80 * t, w, 160 * t);
      ctx.globalCompositeOperation = "source-over";
    }
  }
  ctx.restore();
}

function realWorldShadowKind(place, index, seed) {
  const sets = {
    city: ["underpass", "wireShadow", "lightPool"],
    tokyo: ["underpass", "lightPool", "wireShadow"],
    rainforest: ["treeShadow", "treeShadow", "wetLight"],
    alpine: ["treeShadow", "underpass", "snowLight"],
    snow: ["treeShadow", "snowLight", "underpass"],
    coast: ["wireShadow", "lightPool", "treeShadow"],
    harbor: ["underpass", "lightPool", "wireShadow"],
    freight: ["underpass", "wireShadow", "yardLight"],
    airfield: ["lightPool", "wireShadow", "runwayGlow"]
  };
  const list = sets[place] || ["treeShadow", "wireShadow", "lightPool"];
  return list[(index + Math.floor(seed % list.length)) % list.length];
}

function drawRealWorldEdgeMaterialPass(w, h, horizon, roadTop, roadBottom, place, design, theme, seed) {
  const phoneMode = phoneGraphicsActive();
  const motion = (raceState.roadOffset || 0) * 1.2;
  const count = phoneMode ? 12 : 20;
  ctx.save();
  for (let i = 0; i < count; i += 1) {
    const y = h * 0.56 + (((i * 74 + motion + seed * 0.005) % (h * 0.52 + 120)) - 44);
    if (y < h * 0.52 || y > h + 35) continue;
    const t = Math.max(0.54, Math.min(1.12, (y - horizon) / Math.max(1, h - horizon)));
    const side = i % 2 === 0 ? -1 : 1;
    const half = realWorldRoadHalf(roadTop, roadBottom, t);
    const x = perspectiveRoadCenter(w, t) + side * (half * 0.92 + seededUnit(seed, i + 630) * w * 0.12);
    const size = (8 + seededUnit(seed, i + 640) * 24) * (0.
```

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```
6 + t * 0.8);
ctx.save();
ctx.globalAlpha = phoneMode ? 0.12 + t * 0.08 : 0.18 + t * 0.11;
if (place === "snow" || place === "alpine") {
  ctx.fillStyle = "rgba(244,251,248,0.58)";
  ctx.beginPath();
  ctx.ellipse(x, y, size * 1.7, size * 0.38, 0, 0, Math.PI * 2);
  ctx.fill();
} else if (place === "desert" || place === "canyon" || place === "farm") {
  ctx.fillStyle = place === "farm" ? "rgba(117,86,47,0.28)" : "rgba(255,183,74,0.28)";
  ctx.beginPath();
  ctx.ellipse(x, y, size * 1.8, size * 0.34, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = "rgba(255,209,102,0.2)";
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(x - size, y);
  ctx.lineTo(x + size * 1.2, y + Math.sin(i) * 4);
  ctx.stroke();
} else if (place === "rainforest" || place === "city" || place === "tokyo" || place === "europe") {
  ctx.fillStyle = place === "rainforest" ? "rgba(54,217,138,0.22)" : "rgba(244,251,248,0.14)";
  roundRect(x - size * 1.2, y - size * 0.12, size * 2.4, size * 0.25, Math.max(2, size * 0.08));
  ctx.fill();
  ctx.fillStyle = place === "rainforest" ? "rgba(36,110,62,0.34)" : "rgba(0,0,0,0.26)";
  ctx.beginPath();
  ctx.ellipse(x + side * size * 0.4, y + size * 0.08, size * 0.55, size * 0.16, 0, 0, Math.PI * 2);
  ctx.fill();
} else {
  ctx.fillStyle = "rgba(244,251,248,0.18)";
  ctx.beginPath();
  ctx.ellipse(x, y, size, size * 0.24, 0, 0, Math.PI * 2);
  ctx.fill();
}
ctx.restore();
}
ctx.restore();
}

function drawGenAiHeroBackdrop(w, h, horizon, design, theme, place, route, seed) {
  ctx.save();
  const phoneMode = phoneGraphicsActive();
  const alpha = phoneMode ? 0.58 : 0.72;
  ctx.globalAlpha = alpha;
  const farBase = horizon + h * 0.025;
  if (place === "city" || place === "tokyo") {
    for (let i = 0; i < 18; i += 1) {
      const bw = w * (0.025 + seededUnit(seed, i) * 0.035);
      const bh = h * (0.16 + seededUnit(seed, i + 11) * 0.22);
      const x = ((i * w * 0.078 - (raceState.roadOffset || 0) * 0.018) % (w + 140)) - 70;
      const tower = ctx.createLinearGradient(x, farBase - bh, x + bw, farBase);
      tower.addColorStop(0, place === "tokyo" ? "rgba(255,79,216,0.18)" : "rgba(70,217,255,0.12)");
      tower.addColorStop(0.32, "rgba(15,25,28,0.72)");
      tower.addColorStop(1, "rgba(5,8,7,0.92)");
      ctx.fillStyle = tower;
      roundRect(x, farBase - bh, bw, bh, 3);
      ctx.fill();
      ctx.fillStyle = i % 2 ? `${design.accent}66` : `${design.light}55`;
      for (let yy = 12; yy < bh - 8; yy += 22) ctx.fillRect(x + bw * 0.24, farBase - bh + yy, bw * 0.52, 3);
    }
  } else if (place === "coast" || place === "harbor") {
    const water = ctx.createLinearGradient(0, horizon - h * 0.01, 0, horizon + h * 0.19);
    water.addColorStop(0, "rgba(70,217,255,0.22)");
    water.addColorStop(1, "rgba(4,22,26,0.54)");
    ctx.fillStyle = water;
    ctx.fillRect(0, horizon - h * 0.01, w, h * 0.22);
    for (let i = 0; i < 7; i += 1) {
      const x = ((i * w * 0.18 - (raceState.roadOffset || 0) * 0.026) % (w + 160)) - 80;
      ctx.fillStyle = place === "harbor" ? "rgba(10,42,46,0.78)" : "rgba(35,54,45,0.72)";
      ctx.beginPath();
      ctx.ellipse(x, horizon + h * 0.085, w * 0.08, h * 0.028, 0, 0, Math.PI * 2);
      ctx.fill();
      if (place === "harbor") {
        ctx.strokeStyle = `${design.light}88`;
        ctx.lineWidth = 3;
        ctx.beginPath();
        ctx.moveTo(x, horizon + h * 0.07);
        ctx.lineTo(x + w * 0.035, horizon - h * 0.09);
        ctx.lineTo(x + w * 0.09, horizon - h * 0.085);
        ctx.stroke();
      }
    }
  }
} else if (place === "canyon" || place === "desert") {
  for (let i = 0; i < 8; i += 1) {
    const x = ((i * w * 0.16 - (raceState.roadOffset || 0) * 0.014) % (w + 180)) - 90;
    const ridgeH = h * (0.09 + seededUnit(seed, i + 3) * 0.12);
    ctx.fillStyle = place === "desert" ? "rgba(255,183,74,0.24)" : "rgba(255,91,107,0.22)";
    ctx.beginPath();
    ctx.moveTo(x - w * 0.08, farBase + h * 0.03);
    ctx.quadraticCurveTo(x, farBase - ridgeH, x + w * 0.12, farBase + h * 0.035);
    ctx.closePath();
    ctx.fill
  }
}
```

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```
();  
}  
} else if (place === "alpine" || place === "snow" || place === "europe") {  
  for (let i = 0; i < 8; i += 1) {  
    const x = i * w * 0.2 - ((raceState.roadOffset || 0) * 0.01 % (w * 0.2));  
    const peak = h * (0.18 + seededUnit(seed, i) * 0.13);  
    ctx.fillStyle = "rgba(130,160,160,0.36)";  
    ctx.beginPath();  
    ctx.moveTo(x - w * 0.16, farBase + h * 0.05);  
    ctx.lineTo(x, farBase - peak);  
    ctx.lineTo(x + w * 0.18, farBase + h * 0.05);  
    ctx.closePath();  
    ctx.fill();  
    ctx.fillStyle = "rgba(244,251,248,0.42)";  
    ctx.beginPath();  
    ctx.moveTo(x - w * 0.035, farBase - peak * 0.58);  
    ctx.lineTo(x, farBase - peak);  
    ctx.lineTo(x + w * 0.045, farBase - peak * 0.56);  
    ctx.closePath();  
    ctx.fill();  
  }  
} else if (place === "farm" || place === "freight") {  
  for (let i = 0; i < 8; i += 1) {  
    const x = ((i * w * 0.15 - (raceState.roadOffset || 0) * 0.02) % (w + 140)) - 70;  
    ctx.fillStyle = place === "farm" ? "rgba(95,70,42,0.7)" : "rgba(30,34,30,0.78)";  
    roundRect(x, farBase - h * 0.08, w * 0.075, h * 0.08, 4);  
    ctx.fill();  
    ctx.fillStyle = place === "farm" ? "rgba(160,40,31,0.72)" : `${design.light}66`;  
    ctx.beginPath();  
    ctx.moveTo(x - w * 0.01, farBase - h * 0.08);  
    ctx.lineTo(x + w * 0.037, farBase - h * 0.13);  
    ctx.lineTo(x + w * 0.085, farBase - h * 0.08);  
    ctx.closePath();  
    ctx.fill();  
  }  
} else if (place === "rainforest") {  
  for (let i = 0; i < 16; i += 1) {  
    const x = ((i * w * 0.08 - (raceState.roadOffset || 0) * 0.025) % (w + 120)) - 60;  
    const treeH = h * (0.13 + seededUnit(seed, i) * 0.12);  
    ctx.strokeStyle = "rgba(24,78,52,0.72)";  
    ctx.lineWidth = 4;  
    ctx.beginPath();  
    ctx.moveTo(x, farBase + h * 0.07);  
    ctx.lineTo(x, farBase - treeH);  
    ctx.stroke();  
    ctx.fillStyle = "rgba(54,217,138,0.32)";  
    ctx.beginPath();  
    ctx.ellipse(x, farBase - treeH, w * 0.045, h * 0.035, -0.3, 0, Math.PI * 2);  
    ctx.fill();  
  }  
}  
ctx.globalAlpha = alpha * 0.82;  
ctx.fillStyle = "rgba(5,8,7,0.74)";  
roundRect(w * 0.39, horizon - h * 0.055, w * 0.22, h * 0.045, 6);  
ctx.fill();  
ctx.strokeStyle = `${design.accent}77`;  
ctx.lineWidth = 1.4;  
ctx.stroke();  
ctx.fillStyle = "#f4fbf8";  
ctx.font = `900 ${Math.max(8, Math.min(13, w * 0.013))}px system-ui`;  
ctx.textAlign = "center";  
ctx.fillText(route.scene.toUpperCase(), w * 0.5, horizon - h * 0.026);  
ctx.restore();  
}  
  
function drawGenAiSurfaceDetails(w, h, theme, horizon, roadTop, roadBottom, design, seed) {  
  ctx.save();  
  const phoneMode = phoneGraphicsActive();  
  if (phoneCleanRoadActive()) {  
    ctx.restore();  
    return;  
  }  
  const laneX = (lane, t) => perspectiveRoadCenter(w, t) + lane * laneWidth() * (0.34 + t * 1.05);  
  ctx.beginPath();  
  ctx.moveTo(perspectiveRoadCenter(w, 0) - roadTop, horizon);  
  ctx.lineTo(perspectiveRoadCenter(w, 0) + roadTop, horizon);  
  ctx.lineTo(perspectiveRoadCenter(w, 1) + roadBottom * 0.63, h + 10);  
  ctx.lineTo(perspectiveRoadCenter(w, 1) - roadBottom * 0.63, h + 10);  
  ctx.closePath();  
  ctx.clip();  
  const tint = ctx.createLinearGradient(0, horizon, 0, h);  
  tint.addColorStop(0, `${design.surface}11`);  
  tint.addColorStop(0.5, `${design.surface}42`);  
  tint.addColorStop(1, "rgba(0,0,0,0.42)");  
  ctx.fillStyle = tint;  
  ctx.fillRect(0, horizon, w, h - horizon);  
  const speed = Math.max(0, raceState.speed || 0);  
  const motion = (raceState.roadOffset || 0) * (1.2 + Math.min(1.4, speed / 180));  
  const grimeCount = phoneMode ? 0 : 42;  
  for (let i = 0; i < grimeCount; i += 1) {  
    const y = horizon + (((i * 47 + motion * (0.54 + seededUnit(seed, i) * 0.28)) % (h - horizon + 150)) - 42);  
    if (y < horizon || y > h + 20) continue;  
    const t = Math.max(0, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));  
    const x = laneX(-2 + seededUnit(seed, i + 20) * 4, t);  
    const len = phoneMode ? 5 + t * 20 : 10 + t * 70;  
    ctx.globalAlpha = phoneMode ? 0.035 + t * 0.045 : 0.08 + t * 0.18;  
    ctx.strokeStyle = phoneMode ? "rgba(244,251,248,0.22)" : (seededUnit(seed, i + 30) > 0.54 ? "rgba(244,251,248,0.34)" :  
    "rgba(0,0,0,0.78)");  
    ctx
```

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```
x.lineWidth = phoneMode ? 0.7 + t * 1.1 : 1 + t * 3.5;
ctx.beginPath();
ctx.moveTo(x, y);
ctx.lineTo(x + Math.sin(i) * len * 0.18, y + len * 0.16);
ctx.stroke();
}
const racingLine = ctx.createLinearGradient(0, horizon, 0, h);
racingLine.addColorStop(0, "rgba(255,255,255,0)");
racingLine.addColorStop(0.42, `${design.accent}28`);
racingLine.addColorStop(1, "rgba(255,255,255,0)");
ctx.strokeStyle = racingLine;
ctx.lineWidth = Math.max(4, w * 0.006);
ctx.globalAlpha = phoneMode ? 0.18 : 0.42;
ctx.beginPath();
for (let step = 0; step <= 18; step += 1) {
  const t = step / 18;
  const y = horizon + (h - horizon) * t;
  const lane = Math.sin((raceState.distance || 0) * 0.0009 + t * 3.5) * 0.72;
  const x = laneX(lane, t);
  if (step === 0) ctx.moveTo(x, y);
  else ctx.lineTo(x, y);
}
ctx.stroke();
ctx.globalAlpha = 1;
ctx.restore();
}

function drawGenAiTrackProp(prop, x, y, scale, side, design, theme, route, seed) {
  ctx.save();
  ctx.translate(x, y);
  ctx.scale(scale, scale);
  ctx.globalAlpha = Math.max(0.25, Math.min(0.9, 0.34 + scale * 0.28));
  const accent = design.accent;
  const light = design.light;
  if (prop === "tower" || prop === "neon" || prop === "village") {
    const bw = 24 + seededUnit(seed, 1) * 26;
    const bh = 70 + seededUnit(seed, 2) * 90;
    ctx.fillStyle = prop === "neon" ? "rgba(14,10,30,0.82)" : "rgba(12,18,20,0.8)";
    roundRect(-bw / 2, -bh, bw, bh, 3);
    ctx.fill();
    ctx.fillStyle = prop === "neon" ? `${accent}aa` : `${light}77`;
    for (let i = 0; i < 5; i += 1) ctx.fillRect(-bw * 0.32, -bh + 10 + i * 15, bw * 0.64, 4);
  } else if (prop === "cliff" || prop === "rock" || prop === "dune" || prop === "snowcap") {
    ctx.fillStyle = prop === "snowcap" ? "rgba(220,232,239,0.48)" : prop === "dune" ? "rgba(255,183,74,0.38)" :
"rgba(132,72,38,0.52)";
    ctx.beginPath();
    ctx.moveTo(-54, 12);
    ctx.quadraticCurveTo(-10, -42 - seededUnit(seed, 3) * 26, 58, 14);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.25)";
    ctx.fillRect(-18, -8, 38, 5);
  } else if (prop === "crowd" || prop === "camera") {
    ctx.fillStyle = "rgba(5,8,7,0.74)";
    roundRect(-45, -22, 90, 22, 4);
    ctx.fill();
    for (let i = 0; i < 9; i += 1) {
      ctx.fillStyle = i % 3 === 0 ? accent : i % 3 === 1 ? light : "#f4fbf8";
      ctx.beginPath();
      ctx.arc(-34 + i * 8, -26 - (i % 2) * 3, 3.2, 0, Math.PI * 2);
      ctx.fill();
    }
    if (prop === "camera") {
      ctx.strokeStyle = "rgba(244,251,248,0.42)";
      ctx.lineWidth = 2;
      ctx.beginPath();
      ctx.moveTo(30, -20);
      ctx.lineTo(42, -42);
      ctx.stroke();
      ctx.fillStyle = "rgba(5,8,7,0.9)";
      roundRect(35, -48, 20, 12, 3);
      ctx.fill();
    }
  } else if (prop === "barrier" || prop === "guard" || prop === "brake") {
    const text = prop === "brake" ? "100" : route.country.toUpperCase();
    ctx.fillStyle = "rgba(5,8,7,0.76)";
    roundRect(-42, -30, 84, 24, 4);
    ctx.fill();
    ctx.strokeStyle = accent;
    ctx.lineWidth = 2;
    ctx.stroke();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = "900 12px system-ui";
    ctx.textAlign = "center";
    ctx.fillText(text, 0, -14);
  } else if (prop === "crane" || prop === "gantry" || prop === "hangar" || prop === "tunnel" || prop === "bridge") {
    ctx.strokeStyle = `${light}88`;
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.moveTo(-55, -8);
    ctx.lineTo(-55, -72);
    ctx.lineTo(52, -72);
    ctx.lineTo(52, -8);
    ctx.stroke();
    ctx.fillStyle = "rgba(5,8,7,0.62)";
    roundRect(-35, -58, 70, 18, 3);
    ctx.fill();
  } else {
    ctx.fillStyle = `${accent}66`;
    ctx.beginPath();
    ctx.ellipse(0, -14, 34, 16, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.beginPath();
    ctx.moveTo(0, -8);
    ctx.lineTo(side * 6, 32);
    ctx.stroke();
  }
  ctx.restore();
}

function drawGenAiCrowdAndCameras(w, h, horizon, roadTop, roadBottom, design) {
  const phoneMode = phoneGraphicsActive();
```

```
const bands = phoneMode ? 2 : 3;
ctx.save();
for (let side = -1; side <= 1; side += 2) {
  for (let band = 0; band < bands; band += 1) {
```

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```
const t = 0.18 + band * 0.18;
const y = horizon + (h - horizon) * t + Math.sin(raceState.elapsed + band) * 2;
const center = perspectiveRoadCenter(w, t);
const roadHalf = roadTop * 0.92 + (roadBottom * 0.58 - roadTop * 0.92) * t;
const x = center + side * (roadHalf + w * (0.05 + band * 0.025));
const standW = w * (0.12 + band * 0.03);
const standH = h * (0.025 + band * 0.012);
ctx.globalAlpha = 0.18 + band * 0.09;
ctx.fillStyle = "rgba(5,8,7,0.78)";
roundRect(x - standW / 2, y - standH, standW, standH, 4);
ctx.fill();
for (let i = 0; i < 16; i += 1) {
  ctx.fillStyle = i % 4 === 0 ? design.accent : i % 4 === 1 ? design.light : "rgba(244,251,248,0.76)";
  ctx.fillRect(x - standW * 0.42 + i * standW * 0.052, y - standH - 3 - (i % 2) * 2, 2 + band, 5 + band);
}
}
}
ctx.restore();
}

function drawDriftStylePass(w, h, theme) {
  if (!raceState.active || raceState.resetTimer > 0) return;
  if (phoneCleanRoadActive()) return;
  const speed = Math.abs(raceState.speed || 0);
  const drift = Math.max(0, Math.min(1, ((raceState.slip || 0) - 0.14) * 1.95 + Math.abs(raceState.lateralVelocity || 0) * 0.07));
  if (drift <= 0.03 || speed < 30) return;
  const vehicle = selectedVehicle();
  const carX = w / 2 + raceState.lane * laneWidth();
  const baseY = cameraMode === "cockpit" ? h * 0.78 : cameraMode === "hood" ? h * 0.91 : h * 0.87;
  const steer = raceState.steerAngle || 0;
  const slide = raceState.lateralVelocity || 0;
  ctx.save();
  ctx.globalAlpha = Math.min(0.86, 0.26 + drift * 0.58);
  ctx.strokeStyle = "rgba(1,2,2,0.72)";
  ctx.lineWidth = Math.max(2, w * 0.005 + drift * 4);
  ctx.lineCap = "round";
  for (let side = -1; side <= 1; side += 2) {
    const tireX = carX + side * laneWidth() * 0.2;
    ctx.beginPath();
    ctx.moveTo(tireX, baseY + h * 0.02);
    ctx.bezierCurveTo(tireX - slide * 18 - steer * 30, baseY + h * 0.08, tireX - slide * 34 - steer * 55, baseY + h * 0.2, tireX -
slide * 50 - steer * 78, h + 22);
    ctx.stroke();
  }
  ctx.globalCompositeOperation = "screen";
  const smoke = ctx.createRadialGradient(carX, baseY + h * 0.02, 4, carX - slide * 35, baseY + h * 0.04, w * (0.12 + drift * 0.14));
  const smokeColor = vehicle.type === "snowmobile" ? "rgba(244,251,248," : selectedRace.place === "desert" || selectedRace.place ===
"canyon" ? "rgba(255,183,74," : "rgba(190,205,205,";
  smoke.addColorStop(0, `${smokeColor}${0.12 + drift * 0.18}`);
  smoke.addColorStop(0.5, `${smokeColor}${0.06 + drift * 0.1}`);
  smoke.addColorStop(1, `${smokeColor}0`);
  ctx.fillStyle = smoke;
  ctx.fillRect(0, h * 0.48, w, h * 0.52);
  if (drift > 0.35 && cameraMode !== "cockpit") {
    ctx.fillStyle = theme[1];
    ctx.font = `900 ${Math.max(13, Math.min(20, w * 0.018))}px system-ui`;
    ctx.textAlign = "center";
    ctx.globalAlpha = Math.min(0.62, drift);
    ctx.fillText("DRIFT", carX + Math.sign(slide || steer || 1) * laneWidth() * 0.55, baseY - h * 0.18);
  }
  ctx.restore();
}

function drawWebGLFlatPavementPass(w, h, theme) {
  if (phoneGraphicsActive()) {
    const floorY = cameraMode === "cockpit" ? h * 0.52 : cameraMode === "hood" ? h * 0.56 : h * 0.58;
    const speed = Math.max(0, Math.abs(raceState.speed || 0));
    const offset = raceState.roadOffset || 0;
    ctx.save();
    const shade = ctx.createLinearGradient(0, floorY, 0, h);
    shade.addColorStop(0, "rgba(0,0,0)");
    shade.addColorStop(0.35, "rgba(5,7,0.42)");
    shade.addColorStop(1, "rgba(0,0,0,0.72)");
    ctx.fillStyle = shade;
    ctx.fillRect(0, floorY, w, h - floorY);
    if (phoneCleanRoadActive()) {
      ctx.restore();
      return;
    }
  }
  for (let i = 0; i < 16; i += 1) {
    const y = floorY + (((i * 34 + offset * (0.34 + speed / 760)) % (h - floorY + 60)) - 20);
    if (y < floorY || y > h) continue;
    const t = (y - floorY) / Math.max(1, h - floorY);
    ctx.globalAlpha = 0.08 + t * 0.16;
    ctx.strokeStyle = i % 2 ? "rgba(244,251,248,0.12)" : "rgba(2,3,3,0.48)";
    ctx.lineWidth = 1 + t * 3;
    ctx.beginPath();
    ctx.moveTo(w * 0.18, y);
    ctx.lineTo(w * 0.82, y + 1);
    ctx.stroke();
  }
  ctx.restore()
}
```

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```
);
    return;
}
const floorY = cameraMode === "cockpit" ? h * 0.36 : cameraMode === "hood" ? h * 0.39 : h * 0.42;
const turn = raceState.roadTurn || raceState.roadCurve || 0;
const speed = Math.max(0, Math.abs(raceState.speed || 0));
const offset = raceState.roadOffset || 0;
ctx.save();

const roadTop = w * (cameraMode === "cockpit" ? 0.16 : cameraMode === "hood" ? 0.19 : 0.18);
const roadBottom = w * (cameraMode === "cockpit" ? 0.96 : cameraMode === "hood" ? 1.04 : 1.1);
const roadCenter = (t) => w * 0.5 + turn * (1 - t) * (1 - t) * w * 0.12;
ctx.beginPath();
ctx.moveTo(roadCenter(0) - roadTop, floorY);
ctx.lineTo(roadCenter(0) + roadTop, floorY);
ctx.lineTo(roadCenter(1) + roadBottom * 0.55, h + 10);
ctx.lineTo(roadCenter(1) - roadBottom * 0.55, h + 10);
ctx.closePath();
const asphalt = ctx.createLinearGradient(0, floorY, 0, h);
asphalt.addColorStop(0, "rgba(19,23,23,0.99)");
asphalt.addColorStop(0.42, "rgba(11,14,14,1)");
asphalt.addColorStop(1, "rgba(2,3,3,1)");
ctx.fillStyle = asphalt;
ctx.fill();
ctx.clip();

for (let i = 0; i < 20; i += 1) {
    const y = floorY + (((i * 32 + offset * (0.42 + speed / 650)) % (h - floorY + 72)) - 24);
    if (y < floorY || y > h) continue;
    const t = (y - floorY) / Math.max(1, h - floorY);
    ctx.globalAlpha = 0.08 + t * 0.2;
    ctx.strokeStyle = i % 2 ? "rgba(244,251,248,0.14)" : "rgba(6,8,8,0.42)";
    ctx.lineWidth = 1 + t * 3.5;
    ctx.beginPath();
    const center = roadCenter(t);
    const half = w * (0.24 + t * 0.42);
    ctx.moveTo(center - half, y);
    ctx.lineTo(center + half, y + 1);
    ctx.stroke();
}

for (let side = -1; side <= 1; side += 2) {
    for (let i = 0; i < 8; i += 1) {
        const y = floorY + (((i * 58 + offset * 0.42) % (h - floorY + 80)) - 16);
        if (y < floorY || y > h) continue;
        const t = (y - floorY) / Math.max(1, h - floorY);
        const center = roadCenter(t);
        const roadHalf = w * (0.25 + t * 0.4);
        ctx.globalAlpha = 0.1 + t * 0.18;
        ctx.fillStyle = side < 0 ? "rgba(255,91,107,0.22)" : "rgba(70,217,255,0.2)";
        roundRect(center + side * roadHalf - side * (14 + t * 24), y, 14 + t * 32, 2 + t * 4, 2);
        ctx.fill();
    }
}
ctx.restore();
}

function drawScenery(w, h, theme) {
    const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
    ctx.save();
    if (place === "coast") drawCoastalScenery(w, h, theme);
    if (place === "city") drawMetroScenery(w, h, theme);
    if (place === "canyon") drawCanyonScenery(w, h, theme);
    if (place === "alpine") drawAlpineScenery(w, h, theme);
    if (place === "harbor") drawHarborScenery(w, h, theme);
    if (place === "snow") drawSnowScenery(w, h, theme);
    if (place === "airfield") drawAirfieldScenery(w, h, theme);
    if (place === "freight") drawFreightScenery(w, h, theme);
    if (place === "farm") drawFarmScenery(w, h, theme);
    if (place === "tokyo") drawTokyoScenery(w, h, theme);
    if (place === "desert") drawWorldDesertScenery(w, h, theme);
    if (place === "rainforest") drawRainforestScenery(w, h, theme);
    if (place === "europe") drawEuropeanScenery(w, h, theme);
    drawAtmosphericDepth(w, h, theme);
    drawRoadSigns(w, h, theme);
    ctx.restore();
}

function drawRouteLightingPass(w, h, theme) {
    const night = raceIsNight(selectedRace);
    const turn = raceState.roadTurn || raceState.roadCurve || 0;
    ctx.save();
    if (!night) {
        const sunX = w * (0.18 + (Math.sin((raceState.distance || 0) * 0.00004) + 1) * 0.08);
        const sunY = h * 0.16;
        const glow = ctx.createRadialGradient(sunX, sunY, 4, sunX, sunY, w * 0.34);
        glow.addColorStop(0, "rgba(255,209,102,0.26)");
        glow.addColorStop(1, "rgba(255,209,102,0)");
        ctx.fillStyle = glow;
        ctx.fillRect(0, 0, w, h * 0.5);
        ctx.restore();
        return;
    }
    const skyShade = ctx.createLinearGradient(0, 0, 0, h);
    skyShade.addColorStop(0, "rgba(0,0,0,0.34)");
    skyShade.addColorStop(0.48, "rgba(0,0,0,0.08)");
    skyShade.addColorStop(1, "rgba(0,0,0,0.26)");
    ctx.fillStyle = skyShade;
    ctx.fillRect(0, 0, w, h);
    ctx.globalCompositeOperation = "screen";
    const horizon = phoneGraphicsActive() ? h * 0.34 : h * 0.44;
    for (let i = 0; i < 11; i +=
```


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```
1) {
  const depth = (i + 1) / 12;
  const y = horizon + Math.pow(depth, 1.7) * h * 0.52;
  const center = perspectiveRoadCenter(w, depth) + turn * (1 - depth) * w * 0.05;
  const half = w * (0.18 + depth * 0.43);
  for (let side = -1; side <= 1; side += 2) {
    const x = center + side * half;
    const lamp = ctx.createRadialGradient(x, y, 2, x, y, w * (0.025 + depth * 0.04));
    lamp.addColorStop(0, "rgba(255,229,148,0.34)");
    lamp.addColorStop(1, "rgba(255,229,148,0)");
    ctx.fillStyle = lamp;
    ctx.fillRect(x - w * 0.09, y - h * 0.08, w * 0.18, h * 0.16);
    ctx.strokeStyle = "rgba(244,251,248,0.14)";
    ctx.lineWidth = Math.max(1, 2 * depth);
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x, y + h * (0.04 + depth * 0.04));
    ctx.stroke();
  }
}
ctx.restore();
}

function drawAtmosphericDepth(w, h, theme) {
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const haze = ctx.createLinearGradient(0, h * 0.16, 0, h * 0.55);
  haze.addColorStop(0, "rgba(244,251,248,0.06)");
  haze.addColorStop(0.65, "rgba(70,217,255,0.05)");
  haze.addColorStop(1, "rgba(5,8,7,0)");
  ctx.fillStyle = haze;
  ctx.fillRect(0, h * 0.12, w, h * 0.44);

  if (place === "coast") {
    for (let i = 0; i < 10; i += 1) {
      const x = ((i * 131 + raceState.roadOffset * 0.03) % (w + 120)) - 60;
      ctx.strokeStyle = "rgba(244,251,248,0.22)";
      ctx.lineWidth = 3;
      ctx.beginPath();
      ctx.moveTo(x, h * 0.31);
      ctx.lineTo(x - 20, h * 0.42);
      ctx.stroke();
      ctx.fillStyle = "rgba(187,242,74,0.18)";
      ctx.beginPath();
      ctx.ellipse(x - 24, h * 0.28, 22, 7, -0.35, 0, Math.PI * 2);
      ctx.ellipse(x + 3, h * 0.27, 20, 6, 0.35, 0, Math.PI * 2);
      ctx.fill();
    }
  }

  if (place === "city") {
    ctx.strokeStyle = "rgba(255,255,255,0.16)";
    ctx.lineWidth = 4;
    for (let i = 0; i < 5; i += 1) {
      const x = w * (0.12 + i * 0.19);
      ctx.beginPath();
      ctx.moveTo(x, h * 0.34);
      ctx.lineTo(x + Math.sin(i) * 36, h * 0.2);
      ctx.stroke();
    }
  }

  if (place === "alpine") {
    ctx.fillStyle = "rgba(244,251,248,0.22)";
    for (let i = 0; i < 8; i += 1) {
      const x = ((i * 151 + raceState.roadOffset * 0.02) % (w + 160)) - 80;
      ctx.beginPath();
      ctx.moveTo(x, h * 0.31);
      ctx.lineTo(x + 38, h * 0.18);
      ctx.lineTo(x + 76, h * 0.31);
      ctx.closePath();
      ctx.fill();
    }
  }

  if (place === "harbor") {
    ctx.strokeStyle = "rgba(244,251,248,0.24)";
    ctx.lineWidth = 3;
    for (let i = 0; i < 9; i += 1) {
      const x = ((i * 137 + raceState.roadOffset * 0.05) % (w + 150)) - 75;
      ctx.beginPath();
      ctx.moveTo(x, h * 0.24);
      ctx.lineTo(x + 22, h * 0.36);
      ctx.stroke();
    }
  }

  if (place === "snow") {
    ctx.fillStyle = "rgba(244,251,248,0.42)";
    for (let i = 0; i < 70; i += 1) {
      const x = (i * 59 + raceState.roadOffset * 0.06) % w;
      const y = h * 0.12 + ((i * 37 + raceState.elapsed * 28) % (h * 0.42));
      ctx.fillRect(x, y, 2, 2);
    }
  }

  if (place === "airfield") {
    ctx.strokeStyle = "rgba(255,209,102,0.22)";
    ctx.lineWidth = 2;
    for (let i = 0; i < 13; i += 1) {
      const x = ((i * 91 + raceState.roadOffset * 0.08) % (w + 120)) - 60;
      ctx.beginPath();
      ctx.moveTo(x, h * 0.36);
      ctx.lineTo(x + 34, h * 0.36);
      ctx.stroke();
    }
  }

  if (place === "tokyo") {
    ctx.strokeStyle = "rgba(255,79,216,0.22)";
    ctx.lineWidth = 3;
    for (let i = 0; i < 12; i += 1) {
      const x = ((i * 91 + raceState.roadOffset * 0.11) % (w + 120)) - 60;
      ctx.beginPath();

```

```

        ctx.moveTo(x, h * 0.19);
        ctx.lineTo(x + 18, h * 0.43);
        ctx.stroke();
    }
}

if (place === "farm" || place === "freight") {
    ctx.strokeStyle = "rgba(255,209,102,0.16)";
    ctx.lineWidth = 2;
    for (let i = 0; i < 16; i += 1) {
        const y = h * 0.38 + i * 12;
        ctx.beginPath();
        ctx.moveTo(0, y);
        ctx.lineTo(w, y + Math.sin(i) * 10);
        ctx.stroke();
    }
}

if (place === "rainforest") {
    ctx.fillStyle = "rgba(54,217,138,0.16)";
    for (let i = 0; i < 70; i += 1) {
        c

```

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```
    onst x = ((i * 47 + raceState.roadOffset * 0.05) % (w + 80)) - 40;
    const y = h * 0.18 + (i % 18) * 16;
    ctx.beginPath();
    ctx.ellipse(x, y, 16 + (i % 5) * 3, 6, Math.sin(i), 0, Math.PI * 2);
    ctx.fill();
  }
}

function drawMetroScenery(w, h, theme) {
  const lake = ctx.createLinearGradient(0, h * 0.33, 0, h * 0.56);
  lake.addColorStop(0, "rgba(38,84,96,0.38)");
  lake.addColorStop(1, "rgba(5,8,7,0.16)");
  ctx.fillStyle = lake;
  ctx.fillRect(0, h * 0.34, w, h * 0.18);
  ctx.strokeStyle = "rgba(244,251,248,0.12)";
  for (let i = 0; i < 7; i += 1) {
    const y = h * 0.39 + i * 18;
    ctx.beginPath();
    ctx.moveTo(0, y);
    ctx.bezierCurveTo(w * 0.3, y - 8, w * 0.62, y + 10, w, y - 3);
    ctx.stroke();
  }
  for (let i = 0; i < 42; i += 1) {
    const x = ((i * 93 + raceState.roadOffset * 0.08) % (w + 180)) - 90;
    const bh = 92 + ((i * 47) % 230);
    const tower = ctx.createLinearGradient(x, h * 0.34 - bh, x + 70, h * 0.34);
    tower.addColorStop(0, i % 3 === 0 ? "rgba(70,217,255,0.26)" : "rgba(255,255,255,0.12)");
    tower.addColorStop(1, "rgba(5,8,7,0.4)");
    ctx.fillStyle = tower;
    ctx.fillRect(x, h * 0.34 - bh, 42 + (i % 5) * 11, bh);
    ctx.fillStyle = theme[i % 2 + 1];
    ctx.globalAlpha = 0.58;
    for (let y = 18; y < bh - 12; y += 22) ctx.fillRect(x + 9 + (y % 3) * 8, h * 0.34 - bh + y, 7, 8);
    ctx.globalAlpha = 1;
  }
  ctx.strokeStyle = "rgba(255,255,255,0.14)";
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(w * 0.07, h * 0.34);
  ctx.lineTo(w * 0.93, h * 0.34);
  ctx.stroke();
}

function drawCoastalScenery(w, h, theme) {
  const sun = ctx.createRadialGradient(w * 0.16, h * 0.16, 8, w * 0.16, h * 0.16, 130);
  sun.addColorStop(0, "rgba(255,209,102,0.58)");
  sun.addColorStop(1, "rgba(255,209,102,0)");
  ctx.fillStyle = sun;
  ctx.fillRect(0, 0, w, h * 0.4);
  const water = ctx.createLinearGradient(0, h * 0.32, 0, h * 0.58);
  water.addColorStop(0, "rgba(70,217,255,0.2)");
  water.addColorStop(1, "rgba(5,8,7,0.1)");
  ctx.fillStyle = water;
  ctx.fillRect(0, h * 0.32, w, h * 0.22);
  ctx.strokeStyle = "rgba(244,251,248,0.18)";
  ctx.lineWidth = 2;
  for (let i = 0; i < 8; i += 1) {
    const y = h * 0.37 + i * 18;
    ctx.beginPath();
    ctx.moveTo(0, y);
    ctx.bezierCurveTo(w * 0.25, y + 14, w * 0.5, y - 12, w, y + 8);
    ctx.stroke();
  }
  ctx.strokeStyle = theme[1];
  ctx.lineWidth = 5;
  ctx.beginPath();
  ctx.moveTo(w * 0.08, h * 0.32);
  ctx.quadraticCurveTo(w * 0.5, h * 0.12, w * 0.92, h * 0.32);
  ctx.stroke();
  for (let i = 0; i < 9; i += 1) {
    const x = w * 0.12 + i * w * 0.095;
    ctx.strokeStyle = "rgba(244,251,248,0.3)";
    ctx.lineWidth = 3;
    ctx.beginPath();
    ctx.moveTo(x, h * 0.18);
    ctx.lineTo(x, h * 0.34);
    ctx.stroke();
  }
}

function drawCanyonScenery(w, h, theme) {
  const layers = [
    { y: 0.35, color: "rgba(255,91,107,0.28)", scale: 1 },
    { y: 0.42, color: "rgba(255,209,102,0.22)", scale: 0.74 },
    { y: 0.49, color: "rgba(111,54,30,0.45)", scale: 0.55 }
  ];
  layers.forEach((layer, layerIndex) => {
    ctx.fillStyle = layer.color;
    ctx.beginPath();
    ctx.moveTo(0, h * layer.y);
    for (let i = 0; i <= 10; i += 1) {
      const x = i * w * 0.1;
      const y = h * layer.y - (((i * 37 + layerIndex * 19) % 90) + 20) * layer.scale;
      ctx.lineTo(x, y);
    }
    ctx.lineTo(w, h * 0.56);
    ctx.lineTo(0, h * 0.56);
    ctx.closePath();
    ctx.fill();
  });
  ctx.fillStyle = "rgba(255,209,102,0.32)";
  ctx.beginPath();
  ctx.arc(w * 0.82, h * 0.16, 42, 0, Math.PI * 2);
  ctx.fill();
}

function drawAlpineScenery(w, h, theme) {
  const fog = ctx.createLinearGradient(0, h * 0.16, 0, h * 0.56);
  fog.addColorStop(0, "rgba(185,216,224,0.16)");
  fog.addColorStop(1, "rgba(185,216,224,0)");
  ctx.fillStyle = fog;
  ctx.fillRect(0, h * 0.14, w, h * 0.44);
}
```

```
ctx.fillStyle = "rgba(244,251,248,0.16)";
ctx.beginPath();
ctx.moveTo(0, h * 0.38);
ctx.lineTo(w * 0.18, h * 0.16);
ctx.lineTo(w * 0.34, h * 0.38);
ctx.lineTo(w * 0.54, h * 0.12);
ctx.lineTo(w * 0.78, h * 0.38);
ctx.lineTo(w, h * 0.2);
ctx.lineTo(w, h * 0.52);
ctx.lineTo(0, h * 0.52);
ctx.closePath();
ctx.fill();
ctx.fillStyle =
```

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```
"rgba(187,242,74,0.1)";
for (let i = 0; i < 38; i += 1) {
  const x = ((i * 71 + raceState.roadOffset * 0.04) % (w + 80)) - 40;
  const y = h * 0.36 + (i % 5) * 20;
  ctx.beginPath();
  ctx.moveTo(x, y - 38);
  ctx.lineTo(x - 16, y);
  ctx.lineTo(x + 16, y);
  ctx.closePath();
  ctx.fill();
}
}

function drawHarborScenery(w, h, theme) {
  const water = ctx.createLinearGradient(0, h * 0.28, 0, h * 0.62);
  water.addColorStop(0, "rgba(70,217,255,0.28)");
  water.addColorStop(1, "rgba(4,24,32,0.48)");
  ctx.fillStyle = water;
  ctx.fillRect(0, h * 0.3, w, h * 0.28);
  ctx.strokeStyle = "rgba(244,251,248,0.18)";
  for (let i = 0; i < 10; i += 1) {
    const y = h * 0.34 + i * 18;
    ctx.beginPath();
    ctx.moveTo(0, y);
    ctx.bezierCurveTo(w * 0.28, y + 12, w * 0.56, y - 10, w, y + 6);
    ctx.stroke();
  }
  for (let i = 0; i < 16; i += 1) {
    const x = ((i * 113 + raceState.roadOffset * 0.05) % (w + 160)) - 80;
    const y = h * (0.3 + (i % 3) * 0.055);
    ctx.fillStyle = i % 2 ? "rgba(255,209,102,0.28)" : "rgba(70,217,255,0.24)";
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x + 36, y + 8);
    ctx.lineTo(x + 24, y + 18);
    ctx.lineTo(x - 18, y + 14);
    ctx.closePath();
    ctx.fill();
  }
  ctx.fillStyle = "rgba(5,8,7,0.42)";
  for (let i = 0; i < 12; i += 1) {
    const x = ((i * 97 + raceState.roadOffset * 0.03) % (w + 100)) - 50;
    ctx.fillRect(x, h * 0.22, 28 + (i % 4) * 18, h * 0.12);
  }
}

function drawSnowScenery(w, h, theme) {
  const skyGlow = ctx.createLinearGradient(0, h * 0.1, 0, h * 0.54);
  skyGlow.addColorStop(0, "rgba(244,251,248,0.2)");
  skyGlow.addColorStop(1, "rgba(118,168,188,0)");
  ctx.fillStyle = skyGlow;
  ctx.fillRect(0, h * 0.1, w, h * 0.48);
  ctx.fillStyle = "rgba(244,251,248,0.28)";
  ctx.beginPath();
  ctx.moveTo(0, h * 0.42);
  ctx.lineTo(w * 0.14, h * 0.18);
  ctx.lineTo(w * 0.3, h * 0.42);
  ctx.lineTo(w * 0.48, h * 0.14);
  ctx.lineTo(w * 0.72, h * 0.42);
  ctx.lineTo(w, h * 0.2);
  ctx.lineTo(w, h * 0.54);
  ctx.lineTo(0, h * 0.54);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(10,35,33,0.48)";
  for (let i = 0; i < 44; i += 1) {
    const x = ((i * 67 + raceState.roadOffset * 0.05) % (w + 90)) - 45;
    const y = h * 0.36 + (i % 4) * 24;
    ctx.beginPath();
    ctx.moveTo(x, y - 32);
    ctx.lineTo(x - 13, y);
    ctx.lineTo(x + 13, y);
    ctx.closePath();
    ctx.fill();
  }
}

function drawAirfieldScenery(w, h, theme) {
  const desert = ctx.createLinearGradient(0, h * 0.32, 0, h * 0.58);
  desert.addColorStop(0, "rgba(255,209,102,0.18)");
  desert.addColorStop(1, "rgba(77,55,35,0.34)");
  ctx.fillStyle = desert;
  ctx.fillRect(0, h * 0.32, w, h * 0.26);
  ctx.fillStyle = "rgba(5,8,7,0.5)";
  for (let i = 0; i < 9; i += 1) {
    const x = ((i * 151 + raceState.roadOffset * 0.04) % (w + 180)) - 90;
    const y = h * 0.31 + (i % 2) * 24;
    ctx.fillRect(x, y, 104, 42);
    ctx.beginPath();
    ctx.moveTo(x + 10, y);
    ctx.lineTo(x + 52, y - 28);
    ctx.lineTo(x + 94, y);
    ctx.closePath();
    ctx.fill();
  }
  ctx.strokeStyle = "rgba(244,251,248,0.32)";
  ctx.lineWidth = 3;
  for (let i = 0; i < 5; i += 1) {
    const x = ((i * 197 + raceState.roadOffset * 0.08) % (w + 220)) - 110;
    ctx.beginPath();
    ctx.moveTo(x, h * 0.25);
    ctx.lineTo(x + 80, h * 0.29);
    ctx.lineTo(x + 36, h * 0.32);
    ctx.stroke();
  }
}

function drawFreightScenery(w, h, theme) {
  const plains = ctx.createLinearGradient(0, h * 0.34, 0, h * 0.58);
  plains.addColorStop(0, "rgba(255,209,102,0.22)");
  plains.addColorStop(1, "rgba(57,66,31,0.28)");
  ctx.fillStyle = plains;
  ctx.fillRect(0, h * 0.34, w, h * 0.24);
```

```
ctx.strokeStyle = "rgba(244,251,248,0.18)";
ctx.lineWidth = 3;
for (let i = 0; i < 10; i += 1) {
  const x = ((i * 146 + raceState.roadOffset * 0.08) % (w + 180)) - 90;
  ctx.beginPath();
  ctx.moveTo(x, h * 0.3);
  ctx.lineTo(x + 32, h * 0.46);
  ctx.stroke();
}
for (let i = 0; i < 8; i += 1) {
  const x = ((i * 181 + raceState.roadOffset * 0.045) % (w + 220)) - 110;
  ctx.fillStyle = "rgba(5,8,7,0.48)";
  roundRect(x, h * 0.28, 120, 34, 4);
  ctx.fill();
  ctx.fillStyle = theme[1];
  ctx.globalAlpha
```

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```
    pha = 0.45;
    ctx.fillRect(x + 14, h * 0.3, 28, 6);
    ctx.fillRect(x + 58, h * 0.3, 42, 6);
    ctx.globalAlpha = 1;
  }
}

function drawFarmScenery(w, h, theme) {
  const fields = ctx.createLinearGradient(0, h * 0.32, 0, h * 0.6);
  fields.addColorStop(0, "rgba(187,242,74,0.2)");
  fields.addColorStop(1, "rgba(56,88,31,0.42)");
  ctx.fillStyle = fields;
  ctx.fillRect(0, h * 0.32, w, h * 0.28);
  ctx.strokeStyle = "rgba(255,209,102,0.22)";
  ctx.lineWidth = 2;
  for (let i = 0; i < 20; i += 1) {
    const y = h * 0.36 + i * 13;
    ctx.beginPath();
    ctx.moveTo(0, y);
    ctx.bezierCurveTo(w * 0.25, y + 8, w * 0.62, y - 7, w, y + 4);
    ctx.stroke();
  }
  for (let i = 0; i < 9; i += 1) {
    const x = ((i * 173 + raceState.roadOffset * 0.045) % (w + 200)) - 100;
    const y = h * 0.32 + (i % 3) * 26;
    ctx.fillStyle = i % 2 ? "rgba(160,40,31,0.5)" : "rgba(244,251,248,0.38)";
    ctx.fillRect(x, y, 58, 38);
    ctx.fillStyle = "rgba(117,86,47,0.72)";
    ctx.beginPath();
    ctx.moveTo(x - 5, y);
    ctx.lineTo(x + 29, y - 24);
    ctx.lineTo(x + 63, y);
    ctx.closePath();
    ctx.fill();
  }
}

function drawTokyoScenery(w, h, theme) {
  const glow = ctx.createLinearGradient(0, h * 0.08, 0, h * 0.56);
  glow.addColorStop(0, "rgba(255,79,216,0.12)");
  glow.addColorStop(0.56, "rgba(70,217,255,0.1)");
  glow.addColorStop(1, "rgba(5,8,7,0)");
  ctx.fillStyle = glow;
  ctx.fillRect(0, h * 0.08, w, h * 0.5);
  for (let i = 0; i < 44; i += 1) {
    const x = ((i * 83 + raceState.roadOffset * 0.12) % (w + 180)) - 90;
    const bh = 100 + ((i * 41) % 260);
    ctx.fillStyle = i % 2 ? "rgba(7,12,24,0.74)" : "rgba(16,10,28,0.78)";
    ctx.fillRect(x, h * 0.36 - bh, 36 + (i % 4) * 16, bh);
    ctx.fillStyle = i % 3 ? theme[1] : theme[2];
    ctx.globalAlpha = 0.6;
    for (let y = 18; y < bh - 12; y += 24) ctx.fillRect(x + 8, h * 0.36 - bh + y, 10 + (i % 3) * 7, 7);
    ctx.globalAlpha = 1;
  }
  ctx.strokeStyle = "rgba(244,251,248,0.22)";
  ctx.lineWidth = 5;
  ctx.beginPath();
  ctx.moveTo(0, h * 0.37);
  ctx.quadraticCurveTo(w * 0.5, h * 0.28, w, h * 0.37);
  ctx.stroke();
}

function drawWorldDesertScenery(w, h, theme) {
  const sand = ctx.createLinearGradient(0, h * 0.3, 0, h * 0.6);
  sand.addColorStop(0, "rgba(255,183,74,0.24)");
  sand.addColorStop(1, "rgba(92,55,27,0.42)");
  ctx.fillStyle = sand;
  ctx.fillRect(0, h * 0.3, w, h * 0.3);
  for (let layer = 0; layer < 3; layer += 1) {
    ctx.fillStyle = "rgba(255,183,74,${0.12 + layer * 0.08})";
    ctx.beginPath();
    ctx.moveTo(0, h * (0.42 + layer * 0.04));
    for (let i = 0; i <= 8; i += 1) {
      const x = i * w * 0.125;
      const y = h * (0.42 + layer * 0.04) - Math.sin(i * 0.8 + layer + raceState.roadOffset * 0.001) * 34;
      ctx.lineTo(x, y);
    }
    ctx.lineTo(w, h * 0.62);
    ctx.lineTo(0, h * 0.62);
    ctx.closePath();
    ctx.fill();
  }
  ctx.fillStyle = "rgba(244,251,248,0.18)";
  ctx.beginPath();
  ctx.arc(w * 0.78, h * 0.16, 48, 0, Math.PI * 2);
  ctx.fill();
}

function drawRainforestScenery(w, h, theme) {
  const canopy = ctx.createLinearGradient(0, h * 0.2, 0, h * 0.6);
  canopy.addColorStop(0, "rgba(54,217,138,0.16)");
  canopy.addColorStop(1, "rgba(10,54,35,0.5)");
  ctx.fillStyle = canopy;
  ctx.fillRect(0, h * 0.24, w, h * 0.36);
  ctx.strokeStyle = "rgba(54,217,138,0.22)";
  ctx.lineWidth = 5;
  for (let i = 0; i < 36; i += 1) {
    const x = ((i * 71 + raceState.roadOffset * 0.04) % (w + 100)) - 50;
    const y = h * 0.34 + (i % 5) * 24;
    ctx.beginPath();
    ctx.moveTo(x, y + 50);
    ctx.lineTo(x + Math.sin(i) * 16, y - 45);
    ctx.stroke();
    ctx.fillStyle = i % 2 ? "rgba(54,217,138,0.32)" : "rgba(187,242,74,0.18)";
    ctx.beginPath();
    ctx.ellipse(x - 12, y - 34, 30, 10, -0.4, 0, Math.PI * 2);
    ctx.ellipse(x + 16, y - 28, 28, 9, 0.45, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.strokeStyle = "rgba(244,251,248,0.14)";
  ctx.lineWidth = 2;
  for (let i = 0; i < 18; i += 1) {
    const x = (i * 53 + raceState.elapsed * 40) % w;
```

```
    ctx.beginPath();  
    ctx.moveTo(x, h * 0.16);  
    ctx.lineTo(x - 18, h * 0.58);  
    ctx.stroke();  
  }  
}  
  
function drawEuropeanScenery(w, h, theme) {  
  const valley = ctx.createLinearGradi
```


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```
ent(0, h * 0.3, 0, h * 0.58);
valley.addColorStop(0, "rgba(220,232,239,0.18)");
valley.addColorStop(1, "rgba(24,58,52,0.32)");
ctx.fillStyle = valley;
ctx.fillRect(0, h * 0.3, w, h * 0.28);
ctx.fillStyle = "rgba(220,232,239,0.36)";
ctx.beginPath();
ctx.moveTo(0, h * 0.4);
ctx.lineTo(w * 0.12, h * 0.13);
ctx.lineTo(w * 0.28, h * 0.4);
ctx.lineTo(w * 0.48, h * 0.1);
ctx.lineTo(w * 0.72, h * 0.4);
ctx.lineTo(w * 0.9, h * 0.18);
ctx.lineTo(w, h * 0.4);
ctx.lineTo(w, h * 0.56);
ctx.lineTo(0, h * 0.56);
ctx.closePath();
ctx.fill();
for (let i = 0; i < 12; i += 1) {
  const x = (i * 137 + raceState.roadOffset * 0.035) % (w + 160) - 80;
  const y = h * 0.33 + (i % 3) * 28;
  ctx.fillStyle = "rgba(244,251,248,0.44)";
  ctx.fillRect(x, y, 44, 30);
  ctx.fillStyle = "rgba(160,40,31,0.58)";
  ctx.beginPath();
  ctx.moveTo(x - 4, y);
  ctx.lineTo(x + 22, y - 18);
  ctx.lineTo(x + 48, y);
  ctx.closePath();
  ctx.fill();
}
}

function drawRoadSigns(w, h, theme) {
  const signText = selectedRace && selectedRace.sign ? selectedRace.sign : "Race Route";
  const x = w * 0.75;
  const y = h * 0.25 + Math.sin(raceState.roadOffset * 0.004) * 8;
  ctx.strokeStyle = "rgba(244,251,248,0.28)";
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(x, y + 28);
  ctx.lineTo(x, y + 94);
  ctx.stroke();
  ctx.fillStyle = "rgba(5,8,7,0.76)";
  roundRect(x - 82, y - 18, 164, 46, 6);
  ctx.fill();
  ctx.strokeStyle = theme[1];
  ctx.lineWidth = 2;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 14px system-ui";
  ctx.textAlign = "center";
  ctx.fillText(signText, x, y + 10);
}

function drawRoad(w, h, theme) {
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const horizon = cameraMode === "cockpit" ? h * 0.28 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const roadTop = cameraMode === "cockpit" ? w * 0.12 : cameraMode === "hood" ? w * 0.15 : w * 0.14;
  const roadBottom = cameraMode === "cockpit" ? w * 1.05 : cameraMode === "hood" ? w * 0.95 : w * 1.02;
  const phoneMode = phoneGraphicsActive();
  const roadCenter = (t) => w * 0.5 - cameraLaneOffset(0.24 + t * 0.86) * laneWidth() * (0.18 + t * 0.95);
  const topCenter = roadCenter(0);
  const bottomCenter = roadCenter(1);
  const glare = ctx.createRadialGradient(bottomCenter, h * 0.62, w * 0.08, bottomCenter, h * 0.74, w * 0.75);
  glare.addColorStop(0, "rgba(244,251,248,0.08)");
  glare.addColorStop(1, "rgba(244,251,248,0)");
  ctx.fillStyle = glare;
  ctx.fillRect(0, horizon, w, h - horizon);
  ctx.fillStyle = "rgba(0,0,0,0.34)";
  ctx.beginPath();
  ctx.moveTo(topCenter - roadTop, horizon);
  ctx.lineTo(topCenter + roadTop, horizon);
  ctx.lineTo(bottomCenter + roadBottom, h);
  ctx.lineTo(bottomCenter - roadBottom, h);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "#222826";
  const asphalt = ctx.createLinearGradient(0, horizon, 0, h);
  if (place === "snow") {
    asphalt.addColorStop(0, "#596967");
    asphalt.addColorStop(0.45, "#394845");
    asphalt.addColorStop(1, "#1a2422");
  } else if (place === "harbor") {
    asphalt.addColorStop(0, "#28434a");
    asphalt.addColorStop(0.45, "#182d33");
    asphalt.addColorStop(1, "#0b1619");
  } else if (place === "airfield") {
    asphalt.addColorStop(0, "#3d3f3b");
    asphalt.addColorStop(0.45, "#282a27");
    asphalt.addColorStop(1, "#151613");
  } else if (place === "freight") {
    asphalt.addColorStop(0, "#393c34");
    asphalt.addColorStop(0.45, "#252820");
    asphalt.addColorStop(1, "#131610");
  } else if (place === "farm") {
    asphalt.addColorStop(0, "#5d5542");
    asphalt.addColorStop(0.45, "#3d3526");
    asphalt.addColorStop(1, "#211b13");
  } else if (place === "tokyo") {
    asphalt.addColorStop(0, "#252335");
    asphalt.addColorStop(0.45, "#171722");
    asphalt.addColorStop(1, "#0b0a12");
  } else if (place === "desert") {
    asphalt.addColorStop(0, "#6b5234");
    asphalt.addColorStop(0.45, "#473622");
    asphalt.addColorStop(1, "#21170e");
  } else if (place === "rainforest") {
    asphalt.addColorStop(0, "#314037");
    asphalt.addColorStop(0.45, "#1e2b24");
    asphalt.addColorStop(1, "#101812");
  } else if (p
```

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```
lace === "europe") {
  asphalt.addColorStop(0, "#3d4649");
  asphalt.addColorStop(0.45, "#283234");
  asphalt.addColorStop(1, "#141c1d");
} else {
  asphalt.addColorStop(0, "#2e3330");
  asphalt.addColorStop(0.45, "#1f2422");
  asphalt.addColorStop(1, "#111615");
}
ctx.fillStyle = asphalt;
ctx.beginPath();
ctx.moveTo(topCenter - roadTop * 0.86, horizon);
ctx.lineTo(topCenter + roadTop * 0.86, horizon);
ctx.lineTo(bottomCenter + roadBottom * 0.7, h);
ctx.lineTo(bottomCenter - roadBottom * 0.7, h);
ctx.closePath();
ctx.fill();
ctx.save();
ctx.globalAlpha = phoneMode ? 0.88 : 0.17;
for (let i = 0; i < (phoneMode ? 44 : 130); i += 1) {
  const y = ((i * 23 + raceState.roadOffset * 0.9) % (h + 40)) - 20;
  if (y < horizon) continue;
  const t = (y - horizon) / (h - horizon);
  const x = roadCenter(t) + (Math.sin(i * 13.7) * roadBottom * 0.55 * t);
  ctx.fillStyle = i % 3 ? "#f4fbf8" : "rgba(180,192,188,0.72)";
  ctx.fillRect(x, y, 1 + t * 3, 1 + t * 5);
}
ctx.restore();
if (!phoneMode) {
  ctx.globalAlpha = 0.16;
  ctx.strokeStyle = "#f4fbf8";
  ctx.lineWidth = 1;
  for (let i = 0; i < 26; i += 1) {
    const y = ((i * 42 + raceState.roadOffset * 0.6) % (h + 80)) - 40;
    const t = Math.max(0, (y - horizon) / (h - horizon));
    const center = roadCenter(t);
    ctx.beginPath();
    ctx.moveTo(center - roadBottom * 0.65 * t, y);
    ctx.lineTo(center + roadBottom * 0.65 * t, y + 4);
    ctx.stroke();
  }
}
ctx.globalAlpha = 1;
const baseRoadX = (lane, t) => roadCenter(t) + lane * laneWidth() * (0.3 + t * 1.05);
const basePaintSegment = (lane, y, length, width, style) => {
  const y1 = Math.max(horizon, y);
  const y2 = Math.min(h + 80, y + length);
  const t1 = Math.max(0, Math.min(1.08, (y1 - horizon) / (h - horizon)));
  const t2 = Math.max(0, Math.min(1.12, (y2 - horizon) / (h - horizon)));
  const x1 = baseRoadX(lane, t1);
  const x2 = baseRoadX(lane, t2);
  const w1 = width * (0.24 + t1 * 0.58);
  const w2 = width * (0.24 + t2 * 0.98);
  ctx.fillStyle = style;
  ctx.beginPath();
  ctx.moveTo(x1 - w1 / 2, y1);
  ctx.lineTo(x1 + w1 / 2, y1);
  ctx.lineTo(x2 + w2 / 2, y2);
  ctx.lineTo(x2 - w2 / 2, y2);
  ctx.closePath();
  ctx.fill();
};
const paintedLanes = phoneMode ? [-0.5, 0.5] : [-1.5, -0.5, 0.5, 1.5];
paintedLanes.forEach((lane) => {
  for (let i = 0; i < (phoneMode ? 10 : 14); i += 1) {
    const y = ((i * 116 + raceState.roadOffset * 1.42) % (h + 190)) - 95;
    if (y < horizon) continue;
    const t = Math.max(0, (y - horizon) / (h - horizon));
    const dashH = 20 + t * (phoneMode ? 54 : 72);
    const x = baseRoadX(lane, t);
    const dash = ctx.createLinearGradient(x, y, x, y + dashH);
    dash.addColorStop(0, phoneMode ? "rgba(244,251,248,0.04)" : "rgba(244,251,248,0.08)");
    dash.addColorStop(0.35, phoneMode ? "rgba(244,251,248,0.44)" : "rgba(244,251,248,0.64)");
    dash.addColorStop(1, phoneMode ? "rgba(244,251,248,0.08)" : "rgba(244,251,248,0.12)");
    basePaintSegment(lane, y, dashH, 7 + t * (phoneMode ? 5 : 8), dash);
  }
});
if (!phoneMode) {
  ctx.strokeStyle = "rgba(244,251,248,0.72)";
  ctx.lineWidth = 2;
  ctx.globalAlpha = 0.42;
  for (let lane = -2.5; lane <= 2.5; lane += 1) {
    ctx.beginPath();
    ctx.moveTo(topCenter + lane * laneWidth() * 0.28, horizon);
    ctx.lineTo(bottomCenter + lane * laneWidth() * 1.28, h);
    ctx.stroke();
  }
}
ctx.strokeStyle = phoneMode ? "rgba(244,251,248,0.32)" : "rgba(244,251,248,0.58)";
ctx.lineWidth = phoneMode ? 3 : 4;
ctx.globalAlpha = phoneMode ? 0.26 : 0.7;
ctx.beginPath();
ctx.moveTo(topCenter - roadTop * 0.9, horizon);
ctx.lineTo(bottomCenter - roadBottom * 0.7, h);
ctx.moveTo(topCenter + roadTop * 0.9, horizon);
ctx.lineTo(bottomCenter + roadBottom * 0.7, h);
ctx.stroke();
ctx.globalAlpha = 1;
drawRoadsideDetails(w, h, theme, horizon, roadBottom);
drawGuardrails(w, h, theme, horizon, roadTop, roadBottom);
drawHeadlightBeams(w, h, theme, horizon);
}

function drawGuardrails(w, h, theme, horizon, roadTop, roadBottom) {
  ctx.save();
  const phoneMode = ph
```

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```
oneGraphicsActive();
const topCenter = perspectiveRoadCenter(w, 0.04);
const bottomCenter = perspectiveRoadCenter(w, 1);
for (let side = -1; side <= 1; side += 2) {
  ctx.strokeStyle = "rgba(205,218,214,0.38)";
  ctx.lineWidth = 5;
  ctx.beginPath();
  ctx.moveTo(topCenter + side * roadTop * 1.08, horizon + 12);
  ctx.lineTo(bottomCenter + side * roadBottom * 0.82, h);
  ctx.stroke();
  ctx.strokeStyle = "rgba(5,8,7,0.78)";
  ctx.lineWidth = 2;
  ctx.beginPath();
  ctx.moveTo(topCenter + side * roadTop * 1.04, horizon + 26);
  ctx.lineTo(bottomCenter + side * roadBottom * 0.78, h);
  ctx.stroke();
  if (!phoneMode) {
    for (let i = 0; i < 18; i += 1) {
      const y = ((i * 74 + raceState.roadOffset * 1.2) % (h + 120)) - 60;
      if (y < horizon) continue;
      const t = (y - horizon) / (h - horizon);
      const x = perspectiveRoadCenter(w, t) + side * (roadTop * 1.08 + (roadBottom * 0.82 - roadTop * 1.08) * t);
      ctx.fillStyle = i % 2 ? "rgba(255,255,255,0.34)" : "rgba(180,192,188,0.34)";
      roundRect(x - side * 4, y, side * 10, 22 + t * 34, 3);
      ctx.fill();
    }
  }
}
ctx.restore();
}

function drawRoadsideDetails(w, h, theme, horizon, roadBottom) {
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const phoneMode = phoneGraphicsActive();
  for (let i = 0; i < (phoneMode ? 9 : 22); i += 1) {
    const y = ((i * 92 + raceState.roadOffset * 1.08) % (h + 160)) - 80;
    if (y < horizon) continue;
    const t = (y - horizon) / (h - horizon);
    const side = i % 2 === 0 ? -1 : 1;
    const x = perspectiveRoadCenter(w, t) + side * roadBottom * (0.28 + t * 0.64);
    const size = 10 + t * 28;
    if (place === "canyon") {
      const rock = ctx.createLinearGradient(x - size, y - size, x + size, y + size);
      rock.addColorStop(0, "rgba(255,150,83,0.46)");
      rock.addColorStop(1, "rgba(82,42,27,0.56)");
      ctx.fillStyle = rock;
      ctx.beginPath();
      ctx.moveTo(x, y - size);
      ctx.lineTo(x - size * 0.7, y + size);
      ctx.lineTo(x + size * 0.8, y + size * 0.7);
      ctx.closePath();
      ctx.fill();
    } else if (place === "alpine") {
      ctx.fillStyle = "rgba(23,45,35,0.72)";
      ctx.beginPath();
      ctx.moveTo(x, y - size * 1.2);
      ctx.lineTo(x - size * 0.7, y + size);
      ctx.lineTo(x + size * 0.7, y + size);
      ctx.closePath();
      ctx.fill();
      ctx.strokeStyle = "rgba(244,251,248,0.2)";
      ctx.beginPath();
      ctx.moveTo(x, y - size * 0.9);
      ctx.lineTo(x, y + size * 0.8);
      ctx.stroke();
    } else if (place === "harbor") {
      ctx.fillStyle = "rgba(117,86,47,0.72)";
      ctx.fillRect(x - size * 0.2, y - size * 0.35, size * 0.4, size * 1.6);
      ctx.strokeStyle = "rgba(244,251,248,0.2)";
      ctx.lineWidth = Math.max(2, size * 0.08);
      ctx.beginPath();
      ctx.moveTo(x - size * 0.7, y + size * 0.12);
      ctx.lineTo(x + size * 0.7, y + size * 0.12);
      ctx.stroke();
    } else if (place === "snow") {
      ctx.fillStyle = "rgba(244,251,248,0.76)";
      ctx.beginPath();
      ctx.ellipse(x, y + size * 0.72, size * 0.8, size * 0.22, 0, 0, Math.PI * 2);
      ctx.fill();
      ctx.fillStyle = "rgba(10,35,33,0.72)";
      ctx.beginPath();
      ctx.moveTo(x, y - size);
      ctx.lineTo(x - size * 0.62, y + size * 0.6);
      ctx.lineTo(x + size * 0.62, y + size * 0.6);
      ctx.closePath();
      ctx.fill();
    } else if (place === "airfield") {
      ctx.fillStyle = i % 3 === 0 ? "rgba(255,209,102,0.9)" : "rgba(244,251,248,0.5)";
      ctx.beginPath();
      ctx.arc(x, y, size * 0.2, 0, Math.PI * 2);
      ctx.fill();
      ctx.strokeStyle = "rgba(244,251,248,0.22)";
      ctx.lineWidth = 2;
      ctx.beginPath();
      ctx.moveTo(x, y);
      ctx.lineTo(x, y + size * 1.4);
      ctx.stroke();
    } else if (place === "freight") {
      ctx.fillStyle = "rgba(244,251,248,0.24)";
      roundRect(x - size * 0.7, y - size * 0.45, size * 1.4, size * 0.54, 4);
      ctx.fill();
      ctx.fillStyle = "rgba(5,8,7,0.76)";
      ctx.fillRect(x - size * 0.52, y - size * 0.3, size * 0.32, size * 0.12);
      ctx.fillRect(x + size * 0.16, y - size * 0.3, si
```

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```
ze * 0.20, size * 0.12);
} else if (place === "farm") {
  ctx.fillStyle = i % 3 === 0 ? "rgba(187,242,74,0.58)" : "rgba(255,209,102,0.46)";
  ctx.beginPath();
  ctx.moveTo(x, y - size * 0.85);
  ctx.lineTo(x - size * 0.36, y + size * 0.2);
  ctx.lineTo(x + size * 0.36, y + size * 0.2);
  ctx.closePath();
  ctx.fill();
  ctx.strokeStyle = "rgba(117,86,47,0.62)";
  ctx.lineWidth = Math.max(2, size * 0.08);
  ctx.beginPath();
  ctx.moveTo(x, y + size * 0.2);
  ctx.lineTo(x, y + size);
  ctx.stroke();
} else if (place === "tokyo") {
  ctx.fillStyle = i % 2 ? "rgba(255,79,216,0.66)" : "rgba(70,217,255,0.58)";
  roundRect(x - size * 0.26, y - size * 1.2, size * 0.52, size * 1.5, 4);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  ctx.fillRect(x - size * 0.13, y - size * 1.05, size * 0.26, size * 0.96);
} else if (place === "desert") {
  ctx.fillStyle = "rgba(255,183,74,0.52)";
  ctx.beginPath();
  ctx.moveTo(x, y - size * 0.15);
  ctx.lineTo(x - size, y + size * 0.68);
  ctx.lineTo(x + size, y + size * 0.68);
  ctx.closePath();
  ctx.fill();
} else if (place === "rainforest") {
  ctx.strokeStyle = "rgba(38,78,46,0.82)";
  ctx.lineWidth = Math.max(3, size * 0.12);
  ctx.beginPath();
  ctx.moveTo(x, y - size * 0.4);
  ctx.lineTo(x, y + size * 1.3);
  ctx.stroke();
  ctx.fillStyle = "rgba(54,217,138,0.58)";
  ctx.beginPath();
  ctx.ellipse(x - size * 0.32, y - size * 0.46, size * 0.74, size * 0.24, -0.5, 0, Math.PI * 2);
  ctx.ellipse(x + size * 0.32, y - size * 0.36, size * 0.7, size * 0.22, 0.5, 0, Math.PI * 2);
  ctx.fill();
} else if (place === "europe") {
  ctx.fillStyle = "rgba(244,251,248,0.68)";
  ctx.beginPath();
  ctx.moveTo(x, y - size);
  ctx.lineTo(x - size * 0.7, y + size * 0.7);
  ctx.lineTo(x + size * 0.7, y + size * 0.7);
  ctx.closePath();
  ctx.fill();
} else {
  if (phoneMode) {
    ctx.fillStyle = place === "coast" ? "rgba(38,118,76,0.44)" : "rgba(44,62,58,0.46)";
    ctx.beginPath();
    ctx.ellipse(x, y + size * 0.2, size * 1.05, size * 0.32, -0.08 * side, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = place === "coast" ? "rgba(70,217,255,0.18)" : "rgba(0,0,0,0.26)";
    ctx.beginPath();
    ctx.ellipse(x + side * size * 0.55, y + size * 0.4, size * 0.9, size * 0.18, -0.08 * side, 0, Math.PI * 2);
    ctx.fill();
    continue;
  }
  ctx.strokeStyle = "rgba(244,251,248,0.28)";
  ctx.lineWidth = Math.max(2, size * 0.08);
  ctx.beginPath();
  ctx.moveTo(x, y - size);
  ctx.lineTo(x, y + size * 1.5);
  ctx.stroke();
  ctx.fillStyle = theme[1];
  ctx.globalAlpha = 0.65;
  ctx.beginPath();
  ctx.arc(x, y - size, size * 0.22, 0, Math.PI * 2);
  ctx.fill();
  ctx.globalAlpha = 1;
  if (place === "city" && i % 5 === 0) {
    const signW = size * 2.1;
    const signX = side < 0 ? x - signW : x;
    ctx.fillStyle = "rgba(5,8,7,0.82)";
    roundRect(signX, y - size * 1.2, signW, size * 0.78, 4);
    ctx.fill();
    ctx.strokeStyle = theme[1];
    ctx.lineWidth = 1.5;
    ctx.stroke();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = `900 ${Math.max(8, size * 0.22)}px system-ui`;
    ctx.textAlign = "center";
    ctx.fillText("DOWNTOWN", signX + signW / 2, y - size * 0.73);
  }
}
}
}

function drawHeadlightBeams(w, h, theme, horizon) {
  const boostInput = input.boost || input.gamepadBoost;
  const alpha = cameraMode === "cockpit" ? 0.18 : 0.11;
  const beam = ctx.createLinearGradient(0, horizon, 0, h);
  beam.addColorStop(0, "rgba(255,255,255,0)");
  beam.addColorStop(0.62, `rgba(255,248,214,${alpha})`);
  beam.addColorStop(1, `rgba(70,217,255,${boostInput ? 0.22 : 0.08})`);
  ctx.fillStyle = beam;
  ctx.beginPath();
  ctx.moveTo(w * 0.43, h * 0.7);
  ctx.lineTo(w * 0.17, h);
  ctx.lineTo(w * 0.83, h);
  ctx.lineTo(w * 0.57, h * 0.7);
  ctx.closePath();
  ctx.fill();
  if (!boostInput) return;
  ctx.strokeStyle = theme[1];
  ctx.lineWidth = 3;
  ctx.globalAlpha = 0.45;
```

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```
for (let i = 0; i < 8; i += 1) {
  const x = w * (0.18 + i * 0.09);
  ctx.beginPath();
  ctx.moveTo(w * 0.5, h * 0.52);
  ctx.lineTo(x, h);
  ctx.stroke();
}
ctx.globalAlpha = 1;
}

function objectPos(lane, y) {
  const h = canvas.height;
  const t = Math.max(0, y / h);
  const turn = raceState.roadTurn || raceState.roadCurve || 0;
  const phoneCurve = phoneGraphicsActive() ? 0.29 : 0.19;
  const curveShift = turn * (1 - Math.min(1, t)) * (1 - Math.min(1, t) * 0.24) * canvas.width * phoneCurve;
  const laneSpread = laneWidth() * (0.42 + t * 0.72);
  return {
    x: canvas.width / 2 + curveShift + (lane - cameraLaneOffset(t)) * laneSpread,
    y,
    scale: 0.52 + t * 0.55
  };
}

function phoneRoadObjectPos(lane, distance) {
  const w = canvas.width;
  const h = canvas.height;
  const gap = Number(distance) - raceState.distance;
  const horizon = cameraMode === "cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const nearY = cameraMode === "cockpit" ? h * 0.83 : cameraMode === "hood" ? h * 0.9 : h * 0.87;
  const farGap = 1680;
  let depth;
  if (gap >= 0) {
    const raw = 1 - Math.max(0, Math.min(1, gap / farGap));
    depth = 0.12 + Math.pow(raw, 0.62) * 0.86;
  } else {
    depth = 1 + Math.min(0.32, Math.abs(gap) / 520);
  }
  const clampedDepth = Math.max(0.09, Math.min(1.22, depth));
  const y = gap >= 0
    ? horizon + (nearY - horizon) * clampedDepth
    : nearY + Math.min(h * 0.16, Math.abs(gap) * 0.16);
  const turn = raceState.roadTurn || raceState.roadCurve || 0;
  const farPull = (1 - Math.min(1, clampedDepth)) * (1 - Math.min(1, clampedDepth));
  const curveShift = turn * farPull * w * 0.28;
  const laneSpread = laneWidth() * (0.54 + Math.min(1.08, clampedDepth) * 1.02);
  return {
    x: w / 2 + curveShift + (lane - cameraLaneOffset(clampedDepth)) * laneSpread,
    y,
    scale: Math.max(0.34, Math.min(1.26, 0.32 + clampedDepth * 0.84)),
    depth: clampedDepth
  };
}

function roadSpawnDistance(minRatio = 0.34, maxRatio = 0.44) {
  const h = Math.max(1, canvas.height || window.innerHeight || 720);
  const ratio = minRatio + Math.random() * Math.max(0.01, maxRatio - minRatio);
  return raceState.distance + (h * 0.68 - h * ratio) / 0.11;
}

function screenYFromRoadDistance(distance) {
  return canvas.height * 0.68 - (distance - raceState.distance) * 0.11;
}

function roadDistanceFromScreenY(y) {
  const h = Math.max(1, canvas.height || window.innerHeight || 720);
  const clampedY = Math.max(h * 0.34, Math.min(h * 0.98, Number(y) || h * 0.34));
  return raceState.distance + (h * 0.68 - clampedY) / 0.11;
}

function ensureRoadDistance(object, fallbackY = -100) {
  if (!object) return raceState.distance;
  const distance = Number(object.distance);
  if (Number.isFinite(distance)) return distance;
  object.distance = roadDistanceFromScreenY(Number.isFinite(Number(object.y)) ? Number(object.y) : fallbackY);
  return object.distance;
}

function roadObjectY(object) {
  const y = screenYFromRoadDistance(ensureRoadDistance(object));
  object.y = y;
  return y;
}

function roadObjectPos(lane, distance) {
  if (phoneGraphicsActive()) return phoneRoadObjectPos(lane, distance);
  if (useWebGLRenderer()) return visibleRoadObjectPos(lane, distance);
  return objectPos(lane, screenYFromRoadDistance(distance));
}

function roadContactSink(scale = 1, vehicleType = "car") {
  if (isAirVehicleType(vehicleType) || isWaterVehicleType(vehicleType)) return 0;
  const h = canvas.height || 720;
  const heavy = ["semi", "truck", "monster", "tank", "tractor"].includes(vehicleType);
  return Math.max(12, Math.min(h * 0.12, h * (heavy ? 0.058 : 0.046) + scale * (heavy ? 24 : 20)));
}

function visibleRoadObjectPos(lane, distance) {
  const w = canvas.width;
  const h = canvas.height;
  if (webglRenderer && typeof webglRenderer.projectRoadPoint === "function") {
    const projected = webglRenderer.projectRoadPoint(lane, distance);
    if (projected && Number.isFinite(projected.x) && Number.isFinite(projected.y)) {
      const clampedY = Math.max(h * 0.34, Math.min(h * 1.06, projected.y));
      const t = Math.max(0, Math.min(1, (clampedY - h * 0.34) / Math.max(1, h * 0.7)));
      const turn = raceStat
```

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```
e.roadTurn || raceState.roadCurve || 0;
const curveShift = turn * (1 - t) * (1 - t * 0.22) * w * 0.17;
const laneSpread = laneWidth() * (0.32 + t * 1.08);
return {
  x: w / 2 + curveShift + (lane - cameraLaneOffset(t)) * laneSpread,
  y: clampedY,
  scale: Math.max(0.34, Math.min(1.24, projected.scale || (0.36 + t * 0.78))),
  depth: t
};
}
}
const rawGap = Number(distance) - raceState.distance;
const gap = Math.max(-140, rawGap);
const horizon = cameraMode === "cockpit" ? h * 0.44 : cameraMode === "hood" ? h * 0.48 : h * 0.5;
const nearY = cameraMode === "cockpit" ? h * 0.84 : cameraMode === "hood" ? h * 0.9 : h * 0.88;
const depth = gap >= 0
  ? Math.max(0.18, Math.min(1, 1 / (1 + gap / 88)))
  : Math.min(1.28, 1 + Math.abs(gap) / 360);
const y = gap >= 0
  ? horizon + (nearY - horizon) * depth
  : nearY + Math.min(h * 0.2, Math.abs(gap) * 0.18);
const turn = raceState.roadTurn || raceState.roadCurve || 0;
const phoneCurve = phoneGraphicsActive() ? 0.27 : 0.17;
const curveShift = turn * (1 - depth) * (1 - depth * 0.22) * w * phoneCurve;
const laneSpread = laneWidth() * (0.32 + depth * 1.08);
return {
  x: w / 2 + curveShift + (lane - cameraLaneOffset(depth)) * laneSpread,
  y,
  scale: Math.min(1.34, 0.36 + depth * 0.78),
  depth
};
}

function isVehicleScreenYVisible(y) {
  return y >= canvas.height * 0.44 && y <= canvas.height * 1.08;
}

function drawObjects() {
  const draws = [];
  const phoneMode = phoneGraphicsActive();
  const phoneDrawnVehicles = [];
  const canDrawPhoneVehicle = (gap, lane, priority = false) => {
    if (!phoneMode) return true;
    if (gap > 1180) return false;
    if (gap < -130) return false;
    if (!priority && gap < 95 && Math.abs(lane - raceState.lane) < 0.95) return false;
    const conflict = phoneDrawnVehicles.some((item) => Math.abs(item.gap - gap) < 250 && Math.abs(item.lane - lane) < 1.05);
    if (conflict && !priority) return false;
    phoneDrawnVehicles.push({ gap, lane });
    return true;
  };
  raceState.opponents.forEach((opponent) => {
    const gap = opponent.distance - raceState.distance;
    if (gap < (phoneMode ? -180 : -55)) return;
    if (phoneMode && gap > 1250) return;
    if (!phoneMode && gap > 3600) return;
    if (!canDrawPhoneVehicle(gap, opponent.lane, true)) return;
    const p = roadObjectPos(opponent.lane, opponent.distance);
    if (!isVehicleScreenYVisible(p.y)) return;
    const vehicle = vehicleById(opponent.vehicleId);
    const groundY = p.y + roadContactSink(p.scale, vehicle.type);
    const label = opponent.name;
    draws.push({
      y: groundY,
      draw: () => {
        if (phoneMode) drawPhoneOpponentGapCue(p.x, groundY, p.scale, gap, opponent.lane, opponent.damage || 0, opponent.wrecked);
        drawTrafficRearCar(p.x, groundY, 70 * p.scale, 112 * p.scale, opponent.color, false, vehicle.type, label, opponent.damage || 0, opponent.wrecked, opponent.spin || 0);
      }
    });
  });
  raceState.coinsOnRoad.forEach((coin) => {
    coin.lane = safePickupLane(coin.lane);
    const p = roadObjectPos(coin.lane, ensureRoadDistance(coin));
    if (!isVehicleScreenYVisible(p.y)) return;
    draws.push({
      y: p.y,
      draw: () => drawRouteMarker(p.x, p.y, (coin.r * 2.8) * p.scale, coin.pulse)
    });
  });
  raceState.rivals.forEach((rival) => {
    const distance = ensureRoadDistance(rival);
    const gap = distance - raceState.distance;
    const priority = gap > -30 && gap < 260 && Math.abs(rival.lane - raceState.lane) < 1.08;
    if (!canDrawPhoneVehicle(gap, rival.lane, priority)) return;
    const p = roadObjectPos(rival.lane, distance);
    if (!isVehicleScreenYVisible(p.y)) return;
    const groundY = p.y + roadContactSink(p.scale, rival.type || "car");
    draws.push({
      y: groundY,
      draw: () => drawTrafficRearCar(p.x, groundY, rival.w * p.scale * 1.1, rival.h * p.scale * 0.82, rival.color, false, rival.type || "car", "", rival.damage || 0, rival.wrecked, rival.spin || 0)
    });
  });
  raceState.police.forEach((unit) => {
    const distance = ensureRoadDistance(unit);
    const gap = distance - raceState.distance;
    const priority
```

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```
= gap > -30 && gap < 290 && Math.abs(unit.lane - raceState.lane) < 1.12;
if (!canDrawPhoneVehicle(gap, unit.lane, priority)) return;
const p = roadObjectPos(unit.lane, distance);
if (!isVehicleScreenYVisible(p.y)) return;
const unitType = unit.type || "car";
const groundY = p.y + roadContactSink(p.scale, unitType);
draws.push({
  y: groundY,
  draw: () => drawTrafficRearCar(p.x, groundY, unit.w * p.scale * 1.16, unit.h * p.scale * 0.84, "#f4fbf8", true, unitType,
unit.label || "", unit.damage || 0, unit.wrecked, unit.spin || 0)
});
});
(raceState.oncoming || []).forEach((unit) => {
  const distance = ensureRoadDistance(unit);
  const gap = distance - raceState.distance;
  const priority = gap > -30 && gap < 320 && Math.abs(unit.lane - raceState.lane) < 1.12;
  if (!canDrawPhoneVehicle(gap, unit.lane, priority)) return;
  const p = roadObjectPos(unit.lane, distance);
  if (!isVehicleScreenYVisible(p.y)) return;
  const groundY = p.y + roadContactSink(p.scale, unit.type || "car");
  draws.push({
    y: groundY,
    draw: () => drawTrafficRearCar(p.x, groundY, unit.w * p.scale * 1.08, unit.h * p.scale * 0.82, unit.color || "#ffd166", false,
unit.type || "car", "ONCOMING", unit.damage || 0, unit.wrecked, unit.spin || Math.PI)
  });
});
(raceState.civilians || []).forEach((person) => {
  const distance = ensureRoadDistance(person);
  const p = roadObjectPos(person.lane, distance);
  if (!isVehicleScreenYVisible(p.y)) return;
  draws.push({
    y: p.y + 8,
    draw: () => drawCivilianMarker(p.x, p.y, p.scale, person)
  });
});
(raceState.routeFeatures || []).forEach((feature) => {
  const distance = ensureRoadDistance(feature);
  const gap = distance - raceState.distance;
  if (gap < -120 || gap > 1800) return;
  const p = roadObjectPos(feature.lane, distance);
  if (!isVehicleScreenYVisible(p.y)) return;
  draws.push({
    y: p.y + 16,
    draw: () => drawRouteFeatureMarker(p.x, p.y, p.scale, feature)
  });
});
draws.sort((a, b) => a.y - b.y).forEach((item) => item.draw());
}

function drawRouteFeatureMarker(x, y, scale, feature) {
  const s = Math.max(0.36, Math.min(1.22, scale || 1));
  const size = routeFeatureSize(feature);
  const w = size.w * s;
  const h = size.h * s;
  const accent = feature.type === "hide" ? "#46d9ff" : "#bbf24a";
  ctx.save();
  ctx.translate(x, y + 6 * s);
  ctx.globalAlpha = 0.92;
  const glow = ctx.createRadialGradient(0, 8 * s, 4, 0, 8 * s, w * 0.86);
  glow.addColorStop(0, feature.type === "hide" ? "rgba(70,217,255,0.22)" : "rgba(187,242,74,0.22)");
  glow.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = glow;
  ctx.fillRect(-w, -h * 0.6, w * 2, h * 1.4);
  ctx.fillStyle = "rgba(0,0,0,0.4)";
  ctx.beginPath();
  ctx.ellipse(0, h * 0.42, w * 0.62, h * 0.12, 0, 0, Math.PI * 2);
  ctx.fill();
  if (feature.type === "shortcut") {
    ctx.fillStyle = "rgba(12,18,14,0.86)";
    ctx.beginPath();
    ctx.moveTo(-w * 0.58, h * 0.34);
    ctx.lineTo(w * 0.58, h * 0.18);
    ctx.lineTo(w * 0.48, h * 0.42);
    ctx.lineTo(-w * 0.62, h * 0.56);
    ctx.closePath();
    ctx.fill();
    ctx.strokeStyle = accent;
    ctx.lineWidth = Math.max(1.5, 3 * s);
    ctx.stroke();
    ctx.fillStyle = accent;
    roundRect(-w * 0.42, -h * 0.26, w * 0.84, h * 0.26, 5 * s);
    ctx.fill();
  } else if (feature.icon === "pole") {
    ctx.strokeStyle = "#d8c08b";
    ctx.lineWidth = Math.max(2, 5 * s);
    ctx.beginPath();
    ctx.moveTo(0, -h * 0.62);
    ctx.lineTo(0, h * 0.42);
    ctx.stroke();
    ctx.strokeStyle = "rgba(244,251,248,0.58)";
    ctx.lineWidth = Math.max(1, 2 * s);
    ctx.beginPath();
    ctx.moveTo(-w * 0.46, -h * 0.45);
    ctx.lineTo(w * 0.46, -h * 0.45);
    ctx.moveTo(-w * 0.5, -h * 0.38);
    ctx.lineTo(w * 0.5, -h * 0.38);
    ctx.stroke();
  } else {
    ctx.fillStyle = feature.icon === "cave" ? "rgba(13,10,8,0.94)" : feature.icon === "mountain" ? "rgba(36,45,42,0.94)" :
"rgba(13,21,21,0.94)";
    if (feature.icon === "cave" || feature.icon === "mountain") {
      ctx.beginPath();
      ctx.moveTo(-w * 0.6, h * 0.36);
      ctx.lineTo(-w * 0.18, -h * 0.48);
      ctx.lineTo(w * 0.18, -h * 0

```

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```
.36);
    ctx.lineTo(w * 0.6, h * 0.36);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(0,0,0,0.72)";
    ctx.beginPath();
    ctx.ellipse(0, h * 0.18, w * 0.24, h * 0.2, 0, 0, Math.PI * 2);
    ctx.fill();
  } else {
    roundRect(-w * 0.48, -h * 0.52, w * 0.96, h * 0.92, 5 * s);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.16)";
    for (let i = 0; i < 3; i += 1) {
      roundRect(-w * 0.34, -h * 0.34 + i * h * 0.22, w * 0.68, h * 0.06, 2 * s);
      ctx.fill();
    }
  }
  ctx.strokeStyle = accent;
  ctx.lineWidth = Math.max(1.4, 2.4 * s);
  ctx.stroke();
}
ctx.fillStyle = "rgba(5,8,7,0.82)";
roundRect(-w * 0.54, -h * 0.74, w * 1.08, h * 0.22, 5 * s);
ctx.fill();
ctx.strokeStyle = accent;
ctx.lineWidth = Math.max(1, 1.5 * s);
ctx.stroke();
ctx.fillStyle = "#f4fbf8";
ctx.font = `900 ${Math.max(7, Math.min(12, 9 * s))}px system-ui`;
ctx.textAlign = "center";
ctx.textBaseline = "middle";
ctx.fillText(feature.type === "hide" ? "HIDE" : "CUT", 0, -h * 0.63);
ctx.restore();
}

function drawCivilianMarker(x, y, scale, person) {
  const s = Math.max(0.42, Math.min(1.18, scale || 1));
  const step = Math.sin(person.step || 0) * 3 * s;
  if (person && person.kind === "animal") {
    ctx.save();
    ctx.translate(x, y + 12 * s);
    ctx.globalAlpha = person.crossing ? 0.94 : 0.76;
    drawReferenceAnimalShape(person.species || "dog", 0, 0, Math.max(0.72, s * 0.9), person.side || 1, "#bbf24a", "#ffd166",
    person.step || 0);
    if (person.crossing) {
      ctx.strokeStyle = "rgba(255,209,102,0.76)";
      ctx.lineWidth = Math.max(1.4, 2.4 * s);
      ctx.beginPath();
      ctx.arc(0, 0, 25 * s, 0, Math.PI * 2);
      ctx.stroke();
    }
    ctx.restore();
    return;
  }
  ctx.save();
  ctx.translate(x, y + 8 * s);
  ctx.globalAlpha = person.crossing ? 0.92 : 0.72;
  ctx.fillStyle = "rgba(0,0,0,0.42)";
  ctx.beginPath();
  ctx.ellipse(0, 13 * s, 12 * s, 4 * s, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = person.crossing ? "#ffd166" : "rgba(244,251,248,0.72)";
  ctx.lineWidth = Math.max(1.4, 2.4 * s);
  ctx.lineCap = "round";
  ctx.beginPath();
  ctx.moveTo(0, -5 * s);
  ctx.lineTo(0, 8 * s);
  ctx.moveTo(-5 * s, 0);
  ctx.lineTo(5 * s, 2 * s);
  ctx.moveTo(0, 8 * s);
  ctx.lineTo(-5 * s, 15 * s + step);
  ctx.moveTo(0, 8 * s);
  ctx.lineTo(6 * s, 15 * s - step);
  ctx.stroke();
  ctx.fillStyle = "rgba(244,251,248,0.9)";
  ctx.beginPath();
  ctx.arc(0, -10 * s, 4.2 * s, 0, Math.PI * 2);
  ctx.fill();
  if (person.crossing) {
    ctx.fillStyle = "rgba(255,209,102,0.16)";
    ctx.beginPath();
    ctx.arc(0, 1 * s, 24 * s, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.restore();
}

function drawPhoneOpponentGapCue(x, y, scale, gap, lane, damage = 0, wrecked = false) {
  if (!phoneGraphicsActive()) return;
  const ahead = gap >= 0;
  const label = ahead ? `${Math.round(Math.max(0, gap))}m` : "PASS";
  const laneName = lane < -0.7 ? "L" : lane > 0.7 ? "R" : "MID";
  const badgeW = Math.max(40, Math.min(62, 38 + scale * 24));
  const badgeH = Math.max(13, Math.min(18, 12 + scale * 6));
  const badgeY = y - Math.max(34, 52 * scale);
  ctx.save();
  ctx.globalAlpha = wrecked ? 0.54 : 0.78;
  ctx.fillStyle = damage > 60 ? "rgba(255,91,107,0.78)" : "rgba(5,8,7,0.76)";
  roundRect(x - badgeW / 2, badgeY - badgeH / 2, badgeW, badgeH, 5);
  ctx.fill();
  ctx.strokeStyle = ahead ? "rgba(70,217,255,0.88)" : "rgba(187,242,74,0.88)";
  ctx.lineWidth = 1.4;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `900 ${Math.max(8, Math.min(11, 8 + scale * 4))}px Inter, system-ui, sans-serif`;
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(label, x, badgeY - badgeH * 0.08);
  ctx.fillStyle = "rgba(244,251,248,0.72)";
  ctx.font = `800 ${Math.max(6, Math.min(8, 6 + scale * 2))}px Inter, system-ui, sans-serif`;
  ctx.fillText(laneName, x, badgeY + badgeH * 0.34);
  ctx.restore();
}
```



```
function drawRouteMarker(x, y, size, pulse) {  
  ctx.save();  
  ctx.translate(x, y);  
  const glow = 0.18 + Math.sin(pulse) * 0.05;  
  const phoneMode = phoneGraphicsActive();  
  ctx.globalAlpha = 0.45;  
  ctx.fillStyle = phoneMode ? `rgba(70,217,255,${glow})`  
}
```

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```
)` : `rgba(255,209,102,${glow})`;
  ctx.beginPath();
  ctx.ellipse(0, size * 0.25, size * 1.1, size * 0.28, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.globalAlpha = 1;
  ctx.fillStyle = phoneMode ? "rgba(70,217,255,0.82)" : "rgba(255,209,102,0.82)";
  roundRect(-size * 0.46, -size * 0.1, size * 0.92, size * 0.22, size * 0.11);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.72)";
  ctx.beginPath();
  ctx.arc(0, size * 0.01, Math.max(2, size * 0.13), 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = phoneMode ? "rgba(70,217,255,0.45)" : "rgba(255,209,102,0.45)";
  ctx.lineWidth = Math.max(1, size * 0.045);
  ctx.beginPath();
  ctx.ellipse(0, size * 0.02, size * 0.68, size * 0.26, 0, 0, Math.PI * 2);
  ctx.stroke();
  ctx.restore();
}

function drawRaceStandings(w, h) {
  const pos = playerPosition();
  const vehicle = selectedVehicle();
  ctx.save();
  const panelW = Math.min(178, w * 0.34);
  const panelH = 66;
  const x = w - panelW - 14;
  const y = cameraMode === "chase" ? h * 0.13 : 14;
  ctx.fillStyle = "rgba(5,8,7,0.68)";
  roundRect(x, y, panelW, panelH, 7);
  ctx.fill();
  ctx.strokeStyle = "rgba(244,251,248,0.16)";
  ctx.stroke();
  ctx.fillStyle = pos === 1 ? "#ffd166" : "#f4fbf8";
  ctx.font = "900 18px system-ui";
  ctx.textAlign = "left";
  const fieldSize = raceRankings().length;
  ctx.fillText(`P${pos}/${fieldSize}`, x + 12, y + 23);
  ctx.fillStyle = "rgba(244,251,248,0.72)";
  ctx.font = "800 11px system-ui";
  const leader = raceRankings()[0];
  ctx.fillText(`${leader.name} leads`, x + 12, y + 39);
  ctx.fillStyle = "rgba(187,242,74,0.9)";
  ctx.fillText(`${raceState.teamScore || 0} > 0 ? `Crew ${Math.round(raceState.teamScore)}` : `${raceState.overtakes || 0} overtakes`,
x + 12, y + 54);
  ctx.fillStyle = vehicle.color;
  roundRect(x + panelW - 48, y + 15, 32, 18, 5);
  ctx.fill();
  ctx.restore();
}

function drawTrafficLabel(w, h, label) {
  if (!label) return;
  ctx.fillStyle = "rgba(5,8,7,0.72)";
  roundRect(-w * 0.36, -h * 0.76, w * 0.72, h * 0.16, 4);
  ctx.fill();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `${Math.max(7, w * 0.12)}px system-ui`;
  ctx.textAlign = "center";
  ctx.fillText(label, 0, -h * 0.64);
}

function drawVehicleGroundContact(w, h, vehicleType = "car", speedFactor = 0.5) {
  const speed = Math.max(0.18, Math.min(1.15, Math.abs(speedFactor)));
  const air = isAirVehicleType(vehicleType);
  const water = isWaterVehicleType(vehicleType);
  const snow = vehicleType === "snowmobile";
  ctx.save();
  const contactY = h * (air ? 0.58 : water ? 0.76 : 0.92);
  const contact = ctx.createRadialGradient(0, contactY - h * 0.05, w * 0.08, 0, contactY, w * 0.82);
  contact.addColorStop(0, air ? "rgba(0,0,0,0.34)" : "rgba(0,0,0,0.82)");
  contact.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = contact;
  ctx.beginPath();
  ctx.ellipse(0, contactY, w * 0.78, h * (air ? 0.1 : 0.2), 0, 0, Math.PI * 2);
  ctx.fill();

  if (!air && !water) {
    ctx.fillStyle = snow ? "rgba(244,251,248,0.54)" : "rgba(0,0,0,0.96)";
    ctx.beginPath();
    ctx.ellipse(-w * 0.34, h * 0.58, w * 0.14, h * 0.052, 0, 0, Math.PI * 2);
    ctx.ellipse(w * 0.34, h * 0.58, w * 0.14, h * 0.052, 0, 0, Math.PI * 2);
    ctx.ellipse(-w * 0.35, h * 0.91, w * 0.19, h * 0.082, 0, 0, Math.PI * 2);
    ctx.ellipse(w * 0.35, h * 0.91, w * 0.19, h * 0.082, 0, 0, Math.PI * 2);
    ctx.fill();
  }

  ctx.globalAlpha = (air ? 0.14 : 0.34) + speed * 0.18;
  ctx.strokeStyle = water ? "rgba(70,217,255,0.62)" : snow ? "rgba(244,251,248,0.7)" : "rgba(3,5,5,0.82)";
  ctx.lineWidth = Math.max(1.2, w * 0.018);
  ctx.lineCap = "round";
  ctx.beginPath();
  ctx.moveTo(-w * 0.32, contactY - h * 0.05);
  ctx.lineTo(-w * (0.4 + speed * 0.24), contactY + h * 0.24);
  ctx.moveTo(w * 0.32, contactY - h * 0.05);
  ctx.lineTo(w * (0.4 + speed * 0.24), contactY + h * 0.24);
  ctx.stroke();
  ctx.restore();
}

function spriteContactRatio(vehicleType = "car") {
  if (isAirVehicleType(vehicleType)) return vehicleType === "drone" ? 0.6 : vehicleType === "bomber" ? 0.7 : 0.68;
  if (isWaterVehicleType(vehicleType)) return 0.78;
  if (vehicleType === "snowmobile") return 0.94;
  return 0.955;
}

function spriteContactLift(h, vehicleType = "car") {
  return h
}
```

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```
* spriteContactRatio(vehicleType);
}

function drawRoadContactShadow(w, h, vehicleType = "car", intensity = 1) {
  const air = isAirVehicleType(vehicleType);
  const water = isWaterVehicleType(vehicleType);
  const snow = vehicleType === "snowmobile";
  ctx.save();
  ctx.globalAlpha = Math.max(0.28, Math.min(1, intensity)) * (air ? 0.36 : 0.96);
  const shadow = ctx.createRadialGradient(0, -h * 0.025, w * 0.08, 0, 0, w * 0.78);
  shadow.addColorStop(0, water ? "rgba(10,70,82,0.74)" : "rgba(0,0,0,0.92)");
  shadow.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = shadow;
  ctx.beginPath();
  ctx.ellipse(0, 0, w * (air ? 0.45 : 0.76), h * (air ? 0.065 : 0.13), 0, 0, Math.PI * 2);
  ctx.fill();

  if (!air && !water) {
    ctx.globalAlpha = Math.max(0.52, Math.min(1, intensity));
    ctx.fillStyle = snow ? "rgba(244,251,248,0.78)" : "rgba(0,0,0,0.98)";
    const tireW = w * 0.19;
    const tireH = h * 0.075;
    ctx.beginPath();
    ctx.ellipse(-w * 0.36, -h * 0.01, tireW, tireH, 0, 0, Math.PI * 2);
    ctx.ellipse(w * 0.36, -h * 0.01, tireW, tireH, 0, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.restore();
}

function drawVehicleRoadLock(w, h, vehicleType = "car", speedFactor = 0.5) {
  const air = isAirVehicleType(vehicleType);
  const water = isWaterVehicleType(vehicleType);
  const snow = vehicleType === "snowmobile";
  const speed = Math.max(0.18, Math.min(1.15, Math.abs(speedFactor)));
  ctx.save();
  ctx.globalAlpha = air ? 0.3 : 0.94;
  ctx.fillStyle = water ? "rgba(70,217,255,0.56)" : snow ? "rgba(244,251,248,0.7)" : "rgba(0,0,0,0.98)";
  if (air || water) {
    ctx.beginPath();
    ctx.ellipse(0, h * 0.78, w * 0.42, h * 0.06, 0, 0, Math.PI * 2);
    ctx.fill();
  } else {
    roundRect(-w * 0.52, h * 0.55, w * 0.18, h * 0.5, Math.max(4, w * 0.04));
    roundRect(w * 0.34, h * 0.55, w * 0.18, h * 0.5, Math.max(4, w * 0.04));
    ctx.fill();
    ctx.fillStyle = snow ? "rgba(244,251,248,0.78)" : "rgba(0,0,0,1)";
    ctx.beginPath();
    ctx.ellipse(-w * 0.38, h * 0.74, w * 0.12, h * 0.046, 0, 0, Math.PI * 2);
    ctx.ellipse(w * 0.38, h * 0.74, w * 0.12, h * 0.046, 0, 0, Math.PI * 2);
    ctx.ellipse(-w * 0.36, h * 0.98, w * 0.18, h * 0.072, 0, 0, Math.PI * 2);
    ctx.ellipse(w * 0.36, h * 0.98, w * 0.18, h * 0.072, 0, 0, Math.PI * 2);
    ctx.fill();
  }

  ctx.globalAlpha = air ? 0.18 : 0.54;
  ctx.strokeStyle = water ? "rgba(70,217,255,0.65)" : snow ? "rgba(244,251,248,0.72)" : "rgba(1,2,2,0.72)";
  ctx.lineWidth = Math.max(1.4, w * 0.02);
  ctx.lineCap = "round";
  ctx.beginPath();
  const trailY = air || water ? h * 0.78 : h * 1.0;
  ctx.moveTo(-w * 0.28, trailY);
  ctx.lineTo(-w * (0.32 + speed * 0.2), trailY + h * 0.2);
  ctx.moveTo(w * 0.28, trailY);
  ctx.lineTo(w * (0.32 + speed * 0.2), trailY + h * 0.2);
  ctx.stroke();
  ctx.restore();
}

function drawGroundPinnedVehicleContact(w, h, vehicleType = "car", speedFactor = 0.5) {
  const air = isAirVehicleType(vehicleType);
  const water = isWaterVehicleType(vehicleType);
  const snow = vehicleType === "snowmobile";
  const speed = Math.max(0.18, Math.min(1.15, Math.abs(speedFactor)));
  ctx.save();
  ctx.globalAlpha = air ? 0.34 : 0.96;
  const shadow = ctx.createRadialGradient(0, -h * 0.04, w * 0.08, 0, 0, w * 0.82);
  shadow.addColorStop(0, water ? "rgba(10,70,82,0.76)" : "rgba(0,0,0,0.9)");
  shadow.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = shadow;
  ctx.beginPath();
  ctx.ellipse(0, 0, w * (air ? 0.48 : 0.82), h * (air ? 0.07 : 0.13), 0, 0, Math.PI * 2);
  ctx.fill();

  if (!air && !water) {
    ctx.globalAlpha = 1;
    ctx.fillStyle = snow ? "rgba(244,251,248,0.82)" : "rgba(0,0,0,0.98)";
    ctx.beginPath();
    ctx.ellipse(-w * 0.36, -h * 0.006, w * 0.17, h * 0.052, 0, 0, Math.PI * 2);
    ctx.ellipse(w * 0.36, -h * 0.006, w * 0.17, h * 0.052, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.globalAlpha = 0.34 + speed * 0.16;
    ctx.strokeStyle = snow ? "rgba(244,251,248,0.72)" : "rgba(1,2,2,0.74)";
    ctx.lineWidth = Math.max(1.3, w * 0.018);
    ctx.lineCap = "round";
    ctx.beginPath();
    ctx.moveTo(-w * 0.3, 0);
    ctx.lineTo(-w * (0.36 + speed * 0.24), h * 0.18);
    ctx.moveTo(w * 0.3, 0);
    ctx.lineTo(w * (0.36 + speed * 0.24), h * 0.18);
  }
}
```

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```
0.18);
    ctx.stroke();
  }
  ctx.restore();
}

function drawVisibleTireRoadLock(w, h, vehicleType = "car") {
  const air = isAirVehicleType(vehicleType);
  const water = isWaterVehicleType(vehicleType);
  if (air || water) return;
  const wide = ["semi", "truck", "monster", "tank", "tractor"].includes(vehicleType);
  const snow = vehicleType === "snowmobile";
  const tireSpread = wide ? 0.42 : 0.36;
  const tireW = w * (wide ? 0.18 : 0.15);
  const tireH = h * (wide ? 0.055 : 0.045);
  ctx.save();
  ctx.globalAlpha = snow ? 0.68 : 0.95;
  ctx.fillStyle = snow ? "rgba(244,251,248,0.72)" : "rgba(0,0,0,0.98)";
  roundRect(-w * (wide ? 0.64 : 0.58), -h * 0.035, w * (wide ? 1.28 : 1.16), h * 0.072, Math.max(2, w * 0.025));
  ctx.fill();
  ctx.globalAlpha = 0.42;
  ctx.strokeStyle = snow ? "rgba(244,251,248,0.82)" : "rgba(244,251,248,0.16)";
  ctx.lineWidth = Math.max(1, w * 0.008);
  ctx.beginPath();
  ctx.moveTo(-w * 0.48, -h * 0.002);
  ctx.lineTo(w * 0.48, -h * 0.002);
  ctx.stroke();
  ctx.globalAlpha = snow ? 0.5 : 0.82;
  ctx.strokeStyle = snow ? "rgba(244,251,248,0.78)" : "rgba(0,0,0,0.78)";
  ctx.lineWidth = Math.max(3, w * 0.034);
  ctx.beginPath();
  ctx.moveTo(-w * tireSpread, tireH * 0.24);
  ctx.lineTo(-w * tireSpread, h * 0.26);
  ctx.moveTo(w * tireSpread, tireH * 0.24);
  ctx.lineTo(w * tireSpread, h * 0.26);
  ctx.stroke();
  ctx.globalAlpha = 1;
  ctx.fillStyle = snow ? "rgba(244,251,248,0.86)" : "rgba(0,0,0,0.98)";
  ctx.beginPath();
  ctx.ellipse(-w * tireSpread, -h * 0.006, tireW * 1.14, tireH * 1.28, 0, 0, Math.PI * 2);
  ctx.ellipse(w * tireSpread, -h * 0.006, tireW * 1.14, tireH * 1.28, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.globalAlpha = 0.58;
  ctx.strokeStyle = snow ? "rgba(244,251,248,0.72)" : "rgba(1,2,2,0.76)";
  ctx.lineWidth = Math.max(1.4, w * 0.02);
  ctx.lineCap = "round";
  ctx.beginPath();
  ctx.moveTo(-w * tireSpread, tireH * 0.35);
  ctx.lineTo(-w * (tireSpread + 0.18), h * 0.16);
  ctx.moveTo(w * tireSpread, tireH * 0.35);
  ctx.lineTo(w * (tireSpread + 0.18), h * 0.16);
  ctx.stroke();
  ctx.restore();
}

function drawPhoneAssetVehicleSprite(w, h, color, vehicleType = "car", police = false, damage = 0, contactAnchored = false) {
  const assets = window.VelocityPhoneAssets;
  if (!assets || !assets.ready || typeof assets.getVehicleSprite !== "function") return false;
  const type = police ? "car" : vehicleType;
  const sprite = assets.getVehicleSprite(type, color, { police, damage });
  if (!sprite) return false;
  const wide = ["semi", "truck", "monster", "tank", "tractor"].includes(type);
  const floating = isWaterVehicleType(type) || isAirVehicleType(type);
  const snow = type === "snowmobile";
  const spriteW = w * (contactAnchored ? (type === "semi" ? 1.86 : wide ? 1.78 : snow ? 1.68 : 1.72) : (type === "semi" ? 1.72 : wide ? 1.58 : 1.5));
  const spriteH = h * (contactAnchored ? (type === "semi" ? 1.18 : wide ? 1.12 : snow ? 1.0 : 1.06) : (type === "semi" ? 1.66 : floating || snow ? 1.48 : 1.56));
  const contactRatio = spriteContactRatio(type);
  const speedFactor = Math.abs(raceState.speed || 0) / 220;
  ctx.save();
  ctx.imageSmoothingEnabled = true;
  if (contactAnchored) {
    drawGroundPinnedVehicleContact(w, h, type, speedFactor);
    ctx.drawImage(sprite, -spriteW / 2, -spriteH * contactRatio, spriteW, spriteH);
    drawGroundPinnedVehicleContact(w * 0.72, h * 0.72, type, speedFactor);
    drawVisibleTireRoadLock(w, h, type);
  } else {
    drawVehicleGroundContact(w, h, type, speedFactor);
    ctx.drawImage(sprite, -spriteW / 2, -spriteH * 0.3, spriteW, spriteH);
    drawVehicleRoadLock(w, h, type, speedFactor);
  }
  ctx.restore();
  return true;
}

function drawPhoneAssetTexturePass(w, h, theme) {
  const assets = window.VelocityPhoneAssets;
  if (!assets || !assets.ready || typeof assets.getRoadTexture !== "function") return;
  if (phoneCleanRoadActive()) return;
  const texture = assets.getRoadTexture(selectedRace ? selectedRace.place : "city", theme);
  const pattern = ctx.createPattern(texture, "repeat");
  if (!pattern) return;
  const phoneMode = assets.isPhoneViewport && assets.isPhoneViewport();
  const clipPhoneRoad = () => {
    const horizon = cameraMode ===
```

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```
"cockpit" ? h * 0.27 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
const roadTop = cameraMode === "cockpit" ? w * 0.11 : cameraMode === "hood" ? w * 0.14 : w * 0.13;
const roadBottom = cameraMode === "cockpit" ? w * 1.02 : cameraMode === "hood" ? w * 0.98 : w * 1.08;
const turn = raceState.roadTurn || raceState.roadCurve || 0;
const roadCenter = (t) => {
  const clamped = Math.max(0, Math.min(1.08, t));
  const farPull = (1 - clamped) * (1 - clamped);
  return w * 0.5 + turn * farPull * w * 0.26;
};
ctx.beginPath();
ctx.moveTo(roadCenter(0) - roadTop, horizon);
ctx.lineTo(roadCenter(0) + roadTop, horizon);
ctx.lineTo(roadCenter(1) + roadBottom * 0.62, h + 8);
ctx.lineTo(roadCenter(1) - roadBottom * 0.62, h + 8);
ctx.closePath();
ctx.clip();
};
ctx.save();
if (phoneMode) clipPhoneRoad();
ctx.globalAlpha = phoneMode ? 0.08 : 0.13;
const tile = texture.width || 256;
ctx.translate(-((raceState.roadOffset * 0.28) % tile), -((raceState.roadOffset * 0.08) % tile));
ctx.fillStyle = pattern;
ctx.fillRect(-256, h * 0.3, w + 512, h * 0.78 + 256);
ctx.restore();

ctx.save();
if (phoneMode) clipPhoneRoad();
const bloom = ctx.createRadialGradient(w * 0.5, h * 0.72, w * 0.05, w * 0.5, h * 0.78, w * 0.58);
bloom.addColorStop(0, "rgba(244,251,248,0.07)");
bloom.addColorStop(0.52, "rgba(244,251,248,0.035)");
bloom.addColorStop(1, "rgba(0,0,0,0)");
ctx.fillStyle = bloom;
ctx.fillRect(0, h * 0.34, w, h * 0.66);
ctx.restore();
}

function drawRoadWeightPass(w, h, theme) {
  const horizon = cameraMode === "cockpit" ? h * 0.28 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const roadTop = cameraMode === "cockpit" ? w * 0.12 : cameraMode === "hood" ? w * 0.15 : w * 0.14;
  const roadBottom = cameraMode === "cockpit" ? w * 1.05 : cameraMode === "hood" ? w * 0.95 : w * 1.02;
  const roadCenter = (t) => w * 0.5 - cameraLaneOffset(0.24 + t * 0.86) * laneWidth() * (0.18 + t * 0.95);
  const topCenter = roadCenter(0);
  const bottomCenter = roadCenter(1);
  ctx.save();
  ctx.beginPath();
  ctx.moveTo(topCenter - roadTop * 0.88, horizon);
  ctx.lineTo(topCenter + roadTop * 0.88, horizon);
  ctx.lineTo(bottomCenter + roadBottom * 0.7, h);
  ctx.lineTo(bottomCenter - roadBottom * 0.7, h);
  ctx.closePath();
  ctx.clip();

  const asphaltWeight = ctx.createLinearGradient(0, horizon, 0, h);
  asphaltWeight.addColorStop(0, "rgba(0,0,0,0)");
  asphaltWeight.addColorStop(0.55, "rgba(0,0,0,0.2)");
  asphaltWeight.addColorStop(1, "rgba(0,0,0,0.64)");
  ctx.fillStyle = asphaltWeight;
  ctx.fillRect(0, horizon, w, h - horizon);

  ctx.globalAlpha = 0.32;
  for (let lane = -1.5; lane <= 1.5; lane += 1) {
    for (let i = 0; i < 7; i += 1) {
      const y = horizon + (((i * 92 + raceState.roadOffset * 0.72) % (h - horizon + 110)) - 30);
      if (y < horizon || y > h) continue;
      const t = Math.max(0, Math.min(1, (y - horizon) / Math.max(1, h - horizon)));
      const x = roadCenter(t) + lane * laneWidth() * (0.28 + t * 1.1);
      ctx.fillStyle = "rgba(0,0,0,0.56)";
      roundRect(x - (4 + t * 7), y, 8 + t * 14, 2 + t * 5, 2);
      ctx.fill();
    }
  }
  ctx.restore();
}

function drawRoadMotionPass(w, h, theme) {
  const speed = Math.max(0, Math.abs(raceState.speed || 0));
  if (speed < 3 && raceState.active) return;
  const horizon = cameraMode === "cockpit" ? h * 0.28 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const roadTop = cameraMode === "cockpit" ? w * 0.12 : cameraMode === "hood" ? w * 0.15 : w * 0.14;
  const roadBottom = cameraMode === "cockpit" ? w * 1.05 : cameraMode === "hood" ? w * 0.95 : w * 1.02;
  const motion = raceState.roadOffset * (1.35 + Math.min(1.9, speed / 130));
  const loop = (value, span) => ((value % span) + span) % span;
  const roadCenter = (t) => w * 0.5 - cameraLaneOffset(0.24 + t * 0.86) * laneWidth() * (0.18 + t * 0.95);
  const roadX = (lane, t) => roadCenter(t) + lane * laneWidth() * (0.28 + t * 1.18);
  const topCenter = roadCenter(0);
  const bottomCenter = roadCenter(1);
  const paintSegment = (lane, y, length, width, style, alpha = 1) => {
    const y1 = Math.max(horizon, y);
    const y2 = Math.min(h + 80, y + length);
  }
```

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```
ngth);
    if (y2 <= horizon) return;
    const t1 = Math.max(0, Math.min(1.08, (y1 - horizon) / (h - horizon)));
    const t2 = Math.max(0, Math.min(1.12, (y2 - horizon) / (h - horizon)));
    const x1 = roadX(lane, t1);
    const x2 = roadX(lane, t2);
    const w1 = width * (0.22 + t1 * 0.72);
    const w2 = width * (0.22 + t2 * 1.15);
    ctx.globalAlpha = alpha;
    ctx.fillStyle = style;
    ctx.beginPath();
    ctx.moveTo(x1 - w1 / 2, y1);
    ctx.lineTo(x1 + w1 / 2, y1);
    ctx.lineTo(x2 + w2 / 2, y2);
    ctx.lineTo(x2 - w2 / 2, y2);
    ctx.closePath();
    ctx.fill();
  };

  ctx.save();
  ctx.beginPath();
  ctx.moveTo(topCenter - roadTop * 0.88, horizon);
  ctx.lineTo(topCenter + roadTop * 0.88, horizon);
  ctx.lineTo(bottomCenter + roadBottom * 0.7, h);
  ctx.lineTo(bottomCenter - roadBottom * 0.7, h);
  ctx.closePath();
  ctx.clip();

  const bandAlpha = Math.min(0.42, 0.12 + speed / 520);
  for (let i = 0; i < 30; i += 1) {
    const span = h - horizon + 180;
    const y = horizon - 90 + loop(i * 58 + motion * 1.18, span);
    if (y < horizon || y > h + 30) continue;
    const t = Math.max(0, Math.min(1, (y - horizon) / (h - horizon)));
    const roadHalf = roadTop * 0.62 + (roadBottom * 0.63 - roadTop * 0.62) * t;
    const center = roadCenter(t);
    ctx.globalAlpha = bandAlpha * (0.3 + t * 0.9);
    ctx.strokeStyle = i % 2 ? "rgba(0,0,0,0.78)" : "rgba(244,251,248,0.18)";
    ctx.lineWidth = 1 + t * 7;
    ctx.beginPath();
    ctx.moveTo(center - roadHalf, y);
    ctx.lineTo(center + roadHalf, y + 3 + t * 14);
    ctx.stroke();
  }

  const dashAlpha = Math.min(0.9, 0.4 + speed / 310);
  for (let lane = -1.5; lane <= 1.5; lane += 1) {
    for (let i = 0; i < 11; i += 1) {
      const span = h - horizon + 260;
      const y = horizon - 130 + loop(i * 128 + motion * 1.72, span);
      if (y < horizon || y > h + 70) continue;
      const t = Math.max(0, Math.min(1, (y - horizon) / (h - horizon)));
      const dashH = 14 + t * 42;
      const x = roadX(lane, t);
      const dash = ctx.createLinearGradient(x, y, x, y + dashH);
      dash.addColorStop(0, "rgba(244,251,248,0)");
      dash.addColorStop(0.28, "rgba(244,251,248,0.78)");
      dash.addColorStop(1, "rgba(244,251,248,0.08)");
      paintSegment(lane, y, dashH, 8 + t * 7, dash, dashAlpha * (0.32 + t * 0.5));
    }
  }

  ctx.globalAlpha = Math.min(0.5, speed / 360);
  ctx.strokeStyle = "rgba(244,251,248,0.5)";
  ctx.lineWidth = 2;
  for (let i = 0; i < 26; i += 1) {
    const span = h - horizon + 220;
    const y = horizon - 90 + loop(i * 46 + motion * 2.1, span);
    if (y < h * 0.5) continue;
    const t = Math.max(0, Math.min(1, (y - horizon) / (h - horizon)));
    const side = i % 2 ? -1 : 1;
    const x = roadCenter(t) + side * (roadTop * 1.08 + (roadBottom * 0.82 - roadTop * 1.08) * t);
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x + side * w * (0.04 + t * 0.1), y + h * (0.04 + t * 0.12));
    ctx.stroke();
  }

  ctx.restore();
  ctx.globalAlpha = 1;
}

function drawPhoneUltraGraphicsPass(w, h, theme) {
  if (!phoneGraphicsActive()) return;
  if (phoneCleanRoadActive()) return;
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  const horizon = cameraMode === "cockpit" ? h * 0.28 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
  const roadTop = cameraMode === "cockpit" ? w * 0.12 : cameraMode === "hood" ? w * 0.15 : w * 0.14;
  const roadBottom = cameraMode === "cockpit" ? w * 1.05 : cameraMode === "hood" ? w * 0.95 : w * 1.02;
  const speed = Math.max(0, raceState.speed || 0);

  ctx.save();
  ctx.beginPath();
  ctx.moveTo(w / 2 - roadTop * 0.88, horizon);
  ctx.lineTo(w / 2 + roadTop * 0.88, horizon);
  ctx.lineTo(w / 2 + roadBottom * 0.7, h);
  ctx.lineTo(w / 2 - roadBottom * 0.7, h);
  ctx.closePath();
  ctx.clip();

  const rubber = ctx.createLinearGradient(0, horizon, 0, h);
  rubber.addColorStop(0, "rgba(0,0,0,0)");
  rubber.addColorStop(0.42, "rgba(0,0,0,0.18)");
  rubber.addColorStop(1, "rgba(0,0,0,0.46)");
  ctx.fillStyle = rubber;
  ctx.fillRect(0, horizon, w, h - horizon);

  ctx.globalAlpha = 0.38;
  ctx.strokeStyle = "rgba(1,2,2,0.78)";
  ctx.lineWidth = Math.max(2, w * 0.004);
  for (le
```

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```
t lane = -1; lane <= 1; lane += 2) {
  const tireBase = laneWidth() * (0.18 + lane * 0.03);
  ctx.beginPath();
  ctx.moveTo(w * 0.5 + lane * roadTop * 0.22, horizon + h * 0.04);
  ctx.bezierCurveTo(
    w * 0.5 + lane * tireBase,
    h * 0.55,
    w * 0.5 + lane * tireBase * 1.85 - (raceState.steerAngle || 0) * 18,
    h * 0.78,
    w * 0.5 + lane * tireBase * 2.4 - (raceState.lateralVelocity || 0) * 22,
    h
  );
  ctx.stroke();
}

ctx.globalAlpha = place === "tokyo" || place === "city" || place === "harbor" || place === "rainforest" ? 0.36 : 0.18;
for (let i = 0; i < 18; i += 1) {
  const y = ((i * 67 + raceState.roadOffset * 1.1) % (h + 130)) - 40;
  if (y < horizon) continue;
  const t = Math.max(0, (y - horizon) / (h - horizon));
  const spread = w * (0.1 + t * 0.44);
  const x = w * 0.5 + Math.sin(i * 2.11) * w * 0.08 * t;
  const reflection = ctx.createLinearGradient(x - spread, y, x + spread, y);
  reflection.addColorStop(0, "rgba(255,255,255,0)");
  reflection.addColorStop(0.46, "rgba(244,251,248,0.22)");
  reflection.addColorStop(0.52, "rgba(255,255,255,0.22)");
  reflection.addColorStop(1, "rgba(255,255,255,0)");
  ctx.strokeStyle = reflection;
  ctx.lineWidth = 1 + t * 7;
  ctx.beginPath();
  ctx.moveTo(x - spread, y);
  ctx.quadraticCurveTo(x, y + 5 + t * 18, x + spread, y + 2);
  ctx.stroke();
}

ctx.restore();

ctx.save();
const foreground = ctx.createLinearGradient(0, h * 0.64, 0, h);
foreground.addColorStop(0, "rgba(0,0,0,0)");
foreground.addColorStop(0.52, speed > 70 ? "rgba(0,0,0,0.22)" : "rgba(0,0,0,0.14)");
foreground.addColorStop(1, "rgba(0,0,0,0.48)");
ctx.fillStyle = foreground;
ctx.fillRect(0, h * 0.58, w, h * 0.42);

if (speed > 45) {
  ctx.globalAlpha = Math.min(0.34, speed / 620);
  ctx.strokeStyle = "rgba(244,251,248,0.42)";
  ctx.lineWidth = 3;
  for (let i = 0; i < 18; i += 1) {
    const side = i % 2 ? -1 : 1;
    const x = side < 0 ? (i * 59 + raceState.roadOffset * 0.8) % (w * 0.28) : w - ((i * 59 + raceState.roadOffset * 0.8) % (w *
0.28));
    const y = h * (0.46 + (i % 9) * 0.06);
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x + side * w * 0.12, y + h * 0.18);
    ctx.stroke();
  }
  ctx.restore();
}

function drawPhoneFilmicPost(w, h, theme) {
  if (!phoneGraphicsActive()) return;
  const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
  ctx.save();
  ctx.globalCompositeOperation = "screen";
  const bloom = ctx.createRadialGradient(w * 0.5, h * 0.54, w * 0.05, w * 0.5, h * 0.58, w * 0.7);
  bloom.addColorStop(0, "rgba(255,255,255,0.06)");
  bloom.addColorStop(0.42, "rgba(244,251,248,0.06)");
  bloom.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = bloom;
  ctx.fillRect(0, 0, w, h);
  ctx.restore();

  ctx.save();
  ctx.globalAlpha = 0.12;
  ctx.fillStyle = place === "desert" || place === "canyon" ? "rgba(255,183,74,0.22)"
    : place === "snow" || place === "europe" ? "rgba(220,232,239,0.22)"
    : "rgba(70,217,255,0.16)";
  ctx.fillRect(0, 0, w, h);
  ctx.globalAlpha = 0.08;
  ctx.fillStyle = "rgba(255,255,255,0.72)";
  for (let i = 0; i < 90; i += 1) {
    const x = (i * 71 + Math.floor(raceState.elapsed * 13) * 17) % w;
    const y = (i * 43 + Math.floor(raceState.elapsed * 11) * 23) % h;
    ctx.fillRect(x, y, 1, 1);
  }
  ctx.restore();
}

function drawPhoneLaneRadar(w, h, theme) {
  if (!phoneGraphicsActive() || !raceState.active) return;
  const panelW = Math.min(260, w * 0.31);
  const panelH = 34;
  const x = w * 0.5 - panelW / 2;
  const y = 12;
  const startX = x + panelW * 0.16;
  const span = panelW * 0.72;
  const midY = y + panelH * 0.54;
  ctx.save();
  ctx.globalAlpha = 0.86;
  ctx.fillStyle = "rgba(5,8,7,0.58)";
  roundRect(x, y, panelW, panelH, 7);
  ctx.fill();
  ctx.strokeStyle = "rgba(244,251,248,0.16)";
  ctx.lineWidth = 1;
  ctx.stroke();
  for (let lane = -2; lane <= 2; lane += 1) {
    const rowY = midY + lane * panelH * 0.085;
    ctx.globalAlpha = lane === 0 ? 0.45 : 0.24;
    ctx.strokeStyle = lane === 0 ? "rgba(244,251,248,0.58)" : "rgba(244,251,248,0.32)";
```

```
ctx.beginPath();  
ctx.moveTo(startX, row
```


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```
Y);
    ctx.lineTo(startX + span, rowY);
    ctx.stroke();
}
ctx.globalAlpha = 1;
ctx.fillStyle = theme[1] || "#46d9ff";
ctx.beginPath();
ctx.arc(startX, midY, 4.6, 0, Math.PI * 2);
ctx.fill();
raceState.opponents.forEach((opponent) => {
    const gap = opponent.distance - raceState.distance;
    if (gap < -220 || gap > 1680 || opponent.finished) return;
    const px = startX + Math.max(0, Math.min(1, (gap + 220) / 1900)) * span;
    const py = midY + Math.max(-2.2, Math.min(2.2, opponent.lane)) * panelH * 0.085;
    ctx.globalAlpha = opponent.wrecked ? 0.45 : 0.9;
    ctx.fillStyle = opponent.wrecked ? "#ff5b6b" : (opponent.color || theme[2] || "#ffd166");
    ctx.beginPath();
    ctx.arc(px, py, 3.8, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(5,8,7,0.7)";
    ctx.lineWidth = 1.2;
    ctx.stroke();
});
(raceState.oncoming || []).forEach((unit) => {
    const gap = ensureRoadDistance(unit) - raceState.distance;
    if (gap < -220 || gap > 1680 || unit.wrecked) return;
    const px = startX + Math.max(0, Math.min(1, (gap + 220) / 1900)) * span;
    const py = midY + Math.max(-2.2, Math.min(2.2, unit.lane)) * panelH * 0.085;
    ctx.globalAlpha = 0.9;
    ctx.fillStyle = "#ff5b6b";
    ctx.beginPath();
    ctx.arc(px, py, 3.6, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(5,8,7,0.7)";
    ctx.lineWidth = 1.2;
    ctx.stroke();
});
ctx.restore();
}

function drawTrafficRearCar(x, y, width, height, color, police = false, vehicleType = "car", label = "", damage = 0, wrecked = false, spin = 0) {
    ctx.save();
    ctx.translate(x, y);
    const t = Math.max(0.3, Math.min(1.25, y / canvas.height));
    const w = width * (0.76 + t * 0.28);
    const h = height * (0.72 + t * 0.12);

    drawRoadContactShadow(w, h, vehicleType, 1);

    if (wrecked || damage > 45) ctx.rotate(Math.max(-0.22, Math.min(0.22, (spin || 0) * 0.18)));

    if (drawPhoneAssetVehicleSprite(w, h, color, vehicleType, police, damage, true)) {
        ctx.save();
        ctx.translate(0, -spriteContactLift(h, vehicleType));
        drawOpponentRaceIdentity(w, h, label, police, vehicleType, damage, wrecked);
        drawTrafficLabel(w, h, label);
        drawTrafficDamageBadge(w, h, damage, wrecked);
        ctx.restore();
        ctx.restore();
        return;
    }

    ctx.translate(0, -h * 0.46);

    if (!police && vehicleType !== "car") {
        drawSpecialVehicleSilhouette(w, h, color, vehicleType, false);
        drawOpponentRaceIdentity(w, h, label, false, vehicleType, damage, wrecked);
        drawVehicleDamageMarks(w, h, damage);
        drawTrafficDamageBadge(w, h, damage, wrecked);
        drawTrafficLabel(w, h, label);
        ctx.restore();
        return;
    }

    ctx.fillStyle = "#050807";
    roundRect(-w * 0.56, -h * 0.03, w * 0.16, h * 0.36, 5);
    ctx.fill();
    roundRect(w * 0.4, -h * 0.03, w * 0.16, h * 0.36, 5);
    ctx.fill();
    drawWheelRim(-w * 0.49, h * 0.08, w * 0.075, h * 0.15);
    drawWheelRim(w * 0.49, h * 0.08, w * 0.075, h * 0.15);
    drawWheelRim(-w * 0.48, h * 0.3, w * 0.065, h * 0.105, "rgba(255,209,102,0.24)");
    drawWheelRim(w * 0.48, h * 0.3, w * 0.065, h * 0.105, "rgba(255,209,102,0.24)");

    const body = ctx.createLinearGradient(-w * 0.48, -h * 0.48, w * 0.48, h * 0.44);
    body.addColorStop(0, police ? "ffffff" : shade(color, 38));
    body.addColorStop(0.48, police ? "#dce2e1" : color);
    body.addColorStop(1, police ? "#9da9a7" : shade(color, -44));
    ctx.fillStyle = body;
    ctx.beginPath();
    ctx.moveTo(-w * 0.34, -h * 0.5);
    ctx.lineTo(w * 0.34, -h * 0.5);
    ctx.lineTo(w * 0.52, -h * 0.12);
    ctx.lineTo(w * 0.42, h * 0.42);
    ctx.lineTo(-w * 0.42, h * 0.42);
    ctx.lineTo(-w * 0.52, -h * 0.12);
    ctx.closePath();
    ctx.fill();
    ctx.strokeStyle = "rgba(255,255,255,0.22)";
    ctx.lineWidth = Math.max(1, w * 0.016);
    ctx.stroke();
    drawVehicleTrimDetails(w, h, color, police);
    drawRealWorldVehicleSkinDetails(w, h, color, vehicleType, damage, police);
    drawOpponentRaceIdentity(w, h, label, police, vehicleType, damage, wrecked);

    ctx.fillStyle = "rgba(5,8,7,0.86)";
    roundRect(-w * 0.28, -h * 0.35, w * 0.56, h * 0.2, 7);
    ctx.fill();
    ctx.fillStyle = "rgba(222,249,255,0.22)";
    roundRect(-w * 0.22, -h * 0.31, w * 0.24, h * 0.035, 3);
    ctx.fill();
    c
```

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```
tx.fillStyle = "rgba(5,8,7,0.92)";
roundRect(-w * 0.34, h * 0.02, w * 0.68, h * 0.21, 8);
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.18)";
ctx.lineWidth = Math.max(1, w * 0.012);
ctx.beginPath();
ctx.moveTo(-w * 0.26, h * 0.06);
ctx.lineTo(w * 0.26, h * 0.06);
ctx.moveTo(0, h * 0.02);
ctx.lineTo(0, h * 0.23);
ctx.stroke();

if (police) {
  ctx.fillStyle = "#111817";
  ctx.fillRect(-w * 0.38, -h * 0.08, w * 0.76, h * 0.09);
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `${Math.max(7, w * 0.13)}px system-ui`;
  ctx.textAlign = "center";
  ctx.fillText("POLICE", 0, h * 0.17);
  const flash = Math.sin(raceState.elapsed * 18) > 0;
  ctx.fillStyle = flash ? "#ff3348" : "#46d9ff";
  roundRect(-w * 0.18, -h * 0.22, w * 0.36, h * 0.06, 3);
  ctx.fill();
}

drawTrafficLabel(w, h, label);

ctx.fillStyle = "#ff3348";
roundRect(-w * 0.35, h * 0.3, w * 0.18, h * 0.06, 3);
ctx.fill();
roundRect(w * 0.17, h * 0.3, w * 0.18, h * 0.06, 3);
ctx.fill();
ctx.fillStyle = "rgba(244,251,248,0.58)";
roundRect(-w * 0.08, h * 0.32, w * 0.16, h * 0.035, 3);
ctx.fill();
ctx.fillStyle = "rgba(5,8,7,0.9)";
roundRect(-w * 0.17, h * 0.43, w * 0.12, h * 0.035, 3);
ctx.fill();
roundRect(w * 0.05, h * 0.43, w * 0.12, h * 0.035, 3);
ctx.fill();
drawVehicleDamageMarks(w, h, damage);
drawTrafficDamageBadge(w, h, damage, wrecked);
ctx.restore();
}

function drawOpponentRaceIdentity(w, h, label = "", police = false, vehicleType = "car", damage = 0, wrecked = false) {
  if (!label && !police) return;
  const seed = hashText(`${label} | "${vehicleType}"`);
  const number = String((seed % 89) + 10);
  const stripe = seededUnit(seed, 2) > 0.5 ? 1 : -1;
  const accent = police ? "#46d9ff" : (seededUnit(seed, 3) > 0.5 ? "#ffd166" : "#46d9ff");
  ctx.save();
  if (phoneCleanRoadActive()) {
    ctx.globalAlpha = wrecked ? 0.44 : 0.76;
    ctx.fillStyle = police ? "rgba(5,8,7,0.82)" : "rgba(5,8,7,0.7)";
    roundRect(-w * 0.17, h * 0.07, w * 0.34, h * 0.12, 4);
    ctx.fill();
    ctx.strokeStyle = police ? "#ff3348" : accent;
    ctx.lineWidth = Math.max(1, w * 0.011);
    ctx.stroke();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = `900 ${Math.max(8, w * 0.14)}px system-ui`;
    ctx.textAlign = "center";
    ctx.fillText(police ? "HEAT" : number, 0, h * 0.17);
    if (damage > 55) {
      ctx.globalCompositeOperation = "screen";
      ctx.globalAlpha = Math.min(0.48, damage / 170);
      ctx.fillStyle = "#ff5b6b";
      ctx.beginPath();
      ctx.ellipse(0, h * 0.3, w * 0.16, h * 0.036, 0, 0, Math.PI * 2);
      ctx.fill();
    }
    ctx.restore();
    return;
  }
  ctx.globalAlpha = wrecked ? 0.48 : 0.82;
  ctx.strokeStyle = accent;
  ctx.lineWidth = Math.max(2, w * 0.028);
  ctx.beginPath();
  ctx.moveTo(-w * 0.32 * stripe, -h * 0.42);
  ctx.lineTo(w * 0.34 * stripe, h * 0.36);
  ctx.stroke();
  ctx.globalAlpha = wrecked ? 0.36 : 0.68;
  ctx.strokeStyle = "rgba(244,251,248,0.58)";
  ctx.lineWidth = Math.max(1, w * 0.012);
  ctx.beginPath();
  ctx.moveTo(-w * 0.2 * stripe, -h * 0.46);
  ctx.lineTo(w * 0.44 * stripe, h * 0.24);
  ctx.stroke();
  ctx.globalAlpha = 0.86;
  ctx.fillStyle = police ? "rgba(5,8,7,0.78)" : "rgba(5,8,7,0.66)";
  roundRect(-w * 0.16, h * 0.08, w * 0.32, h * 0.13, 4);
  ctx.fill();
  ctx.strokeStyle = police ? "#ff3348" : accent;
  ctx.lineWidth = Math.max(1, w * 0.01);
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `900 ${Math.max(8, w * 0.15)}px system-ui`;
  ctx.textAlign = "center";
  ctx.fillText(police ? "HEAT" : number, 0, h * 0.18);
  if (label) {
    ctx.globalAlpha = 0.7;
    ctx.fillStyle = "rgba(222,249,255,0.32)";
    ctx.beginPath();
    ctx.arc(-w * 0.07, -h * 0.23, w * 0.045, 0, Math.PI * 2);
    ctx.arc(w * 0.07, -h * 0.23, w * 0.045, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(5,8,7,0.72)";
    ctx.lineWidth = Math.max(1, w * 0.012);
    ctx.beginPath();
    ctx.moveTo(-w * 0.1, -h * 0.18);
    ctx.lineTo(w * 0.1, -h * 0.18);
    ctx.stroke();
  }
}
```

```
}  
if (damage > 55) {  
  ctx.globalCompositeOperation = "screen";  
  ctx.globalAlpha = Math.min(0.55, damage / 150);  
  ctx.fillStyle = "#ff5b6b";  
  ctx.beginPath();
```

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```
        ctx.ellipse(stripe * w * 0.24, h * 0.3, w * 0.12, h * 0.035, 0, 0, Math.PI * 2);
        ctx.fill();
    }
    ctx.restore();
}

function drawTrafficDamageBadge(w, h, damage, wrecked) {
    if (damage < 12 && !wrecked) return;
    ctx.save();
    const barWidth = w * 0.58;
    const pct = Math.min(1, damage / 100);
    ctx.fillStyle = "rgba(5,8,7,0.74)";
    roundRect(-barWidth / 2, -h * 0.88, barWidth, h * 0.055, 3);
    ctx.fill();
    ctx.fillStyle = wrecked ? "#ff5b6b" : damage > 58 ? "#ffd166" : "#bbf24a";
    roundRect(-barWidth / 2, -h * 0.88, barWidth * pct, h * 0.055, 3);
    ctx.fill();
    if (wrecked) {
        ctx.fillStyle = "rgba(255,91,107,0.72)";
        ctx.beginPath();
        ctx.moveTo(-w * 0.16, -h * 0.72);
        ctx.lineTo(0, -h * 0.56);
        ctx.lineTo(w * 0.16, -h * 0.72);
        ctx.closePath();
        ctx.fill();
    }
    ctx.restore();
}

function drawWheelRim(x, y, rx, ry, accent = "rgba(70,217,255,0.26)") {
    ctx.save();
    ctx.fillStyle = "rgba(4,6,6,0.96)";
    ctx.beginPath();
    ctx.ellipse(x, y, rx, ry, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.lineWidth = Math.max(1, rx * 0.18);
    ctx.stroke();
    ctx.strokeStyle = accent;
    ctx.lineWidth = Math.max(1, rx * 0.12);
    ctx.beginPath();
    ctx.moveTo(x - rx * 0.58, y);
    ctx.lineTo(x + rx * 0.58, y);
    ctx.moveTo(x, y - ry * 0.55);
    ctx.lineTo(x, y + ry * 0.55);
    ctx.moveTo(x - rx * 0.38, y - ry * 0.38);
    ctx.lineTo(x + rx * 0.38, y + ry * 0.38);
    ctx.moveTo(x + rx * 0.38, y - ry * 0.38);
    ctx.lineTo(x - rx * 0.38, y + ry * 0.38);
    ctx.stroke();
    ctx.restore();
}

function drawVehicleTrimDetails(w, h, color, police = false) {
    ctx.save();
    const trim = police ? "#111817" : shade(color, -46);
    ctx.fillStyle = trim;
    roundRect(-w * 0.65, -h * 0.24, w * 0.14, h * 0.06, 3);
    ctx.fill();
    roundRect(w * 0.51, -h * 0.24, w * 0.14, h * 0.06, 3);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.22)";
    ctx.lineWidth = Math.max(1, w * 0.011);
    ctx.beginPath();
    ctx.moveTo(-w * 0.33, -h * 0.08);
    ctx.lineTo(-w * 0.42, h * 0.38);
    ctx.moveTo(w * 0.33, -h * 0.08);
    ctx.lineTo(w * 0.42, h * 0.38);
    ctx.moveTo(-w * 0.22, -h * 0.36);
    ctx.lineTo(w * 0.22, -h * 0.36);
    ctx.stroke();
    ctx.fillStyle = "rgba(5,8,7,0.86)";
    roundRect(-w * 0.23, h * 0.38, w * 0.46, h * 0.065, 4);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.74)";
    roundRect(-w * 0.075, h * 0.392, w * 0.15, h * 0.032, 3);
    ctx.fill();
    ctx.fillStyle = "rgba(255,248,214,0.8)";
    roundRect(-w * 0.37, -h * 0.46, w * 0.17, h * 0.055, 3);
    ctx.fill();
    roundRect(w * 0.2, -h * 0.46, w * 0.17, h * 0.055, 3);
    ctx.fill();
    ctx.fillStyle = "rgba(5,8,7,0.84)";
    roundRect(-w * 0.34, h * 0.25, w * 0.68, h * 0.055, 3);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.18)";
    roundRect(-w * 0.22, -h * 0.31, w * 0.44, h * 0.045, 3);
    ctx.fill();
    if (raceIsNight(selectedRace)) {
        ctx.save();
        ctx.globalCompositeOperation = "screen";
        ctx.fillStyle = police ? "rgba(70,217,255,0.35)" : "rgba(255,248,214,0.28)";
        ctx.beginPath();
        ctx.ellipse(-w * 0.285, -h * 0.48, w * 0.18, h * 0.04, -0.08, 0, Math.PI * 2);
        ctx.ellipse(w * 0.285, -h * 0.48, w * 0.18, h * 0.04, 0.08, 0, Math.PI * 2);
        ctx.fill();
        ctx.fillStyle = "rgba(255,51,72,0.34)";
        ctx.beginPath();
        ctx.ellipse(-w * 0.3, h * 0.42, w * 0.14, h * 0.035, 0, 0, Math.PI * 2);
        ctx.ellipse(w * 0.3, h * 0.42, w * 0.14, h * 0.035, 0, 0, Math.PI * 2);
        ctx.fill();
        ctx.restore();
    }
    ctx.restore();
}

function drawRealWorldVehicleSkinDetails(w, h, color, vehicleType = "car", damage = 0, police = false) {
    ctx.save();
    const air = isAirVehicleType(vehicleType);
    const water = isWaterVehicleType(vehicleType);
    const accent = police ? "#46d9ff" : shade(color, -34);
    ctx.globalAlpha = phoneGraphicsActive() ? 0.58 : 0.74;
    ctx.strokeStyle = "rgba(244,251,248,0.24)";
```

```
ctx.lineWidth = Math.max(1, w * 0.012);
ctx.beginPath();
ctx.moveTo(-w * 0.38, -h * 0.08);
ctx.quadraticCurveTo(-w * 0.46, h * 0.12, -w * 0.34, h * 0.36);
ctx.moveTo(w * 0.38, -h * 0.08);
ctx.quadraticCurveTo(w * 0.46, h * 0.12, w * 0.34, h * 0.36);
ctx.moveTo(-w * 0.2, -h * 0.4);
ctx.lineTo(w * 0.2, -h * 0.4);
```

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```
ctx.stroke();

ctx.fillStyle = "rgba(5,8,7,0.84)";
roundRect(-w * 0.18, h * 0.34, w * 0.36, h * 0.058, Math.max(2, w * 0.018));
ctx.fill();
ctx.fillStyle = "rgba(244,251,248,0.72)";
roundRect(-w * 0.065, h * 0.352, w * 0.13, h * 0.03, Math.max(2, w * 0.012));
ctx.fill();

if (!air && !water) {
  ctx.fillStyle = "rgba(0,0,0,0.72)";
  roundRect(-w * 0.52, h * 0.5, w * 0.17, h * 0.05, Math.max(2, w * 0.018));
  roundRect(w * 0.35, h * 0.5, w * 0.17, h * 0.05, Math.max(2, w * 0.018));
  ctx.fill();
  ctx.strokeStyle = "rgba(255,255,255,0.18)";
  ctx.lineWidth = Math.max(1, w * 0.009);
  ctx.beginPath();
  ctx.moveTo(-w * 0.46, h * 0.09);
  ctx.lineTo(-w * 0.46, h * 0.38);
  ctx.moveTo(w * 0.46, h * 0.09);
  ctx.lineTo(w * 0.46, h * 0.38);
  ctx.stroke();
}

if (vehicleType === "semi" || vehicleType === "truck" || vehicleType === "monster" || vehicleType === "tractor") {
  ctx.fillStyle = "rgba(5,8,7,0.34)";
  for (let i = -2; i <= 2; i += 1) roundRect(i * w * 0.13 - w * 0.035, -h * 0.02, w * 0.07, h * 0.34, 2);
  ctx.fill();
  ctx.strokeStyle = `${accent}77`;
  ctx.lineWidth = Math.max(1, w * 0.01);
  ctx.beginPath();
  ctx.moveTo(-w * 0.38, -h * 0.28);
  ctx.lineTo(w * 0.38, -h * 0.28);
  ctx.stroke();
} else if (vehicleType === "tank") {
  ctx.fillStyle = "rgba(5,8,7,0.34)";
  for (let i = 0; i < 8; i += 1) {
    const px = -w * 0.32 + i * w * 0.09;
    ctx.beginPath();
    ctx.arc(px, h * 0.18, Math.max(1.5, w * 0.017), 0, Math.PI * 2);
    ctx.fill();
  }
} else if (vehicleType === "f1" || vehicleType === "prototype") {
  ctx.strokeStyle = `${accent}88`;
  ctx.lineWidth = Math.max(1.5, w * 0.016);
  ctx.beginPath();
  ctx.moveTo(-w * 0.33, h * 0.33);
  ctx.lineTo(w * 0.33, h * 0.33);
  ctx.moveTo(0, -h * 0.5);
  ctx.lineTo(0, h * 0.38);
  ctx.stroke();
} else if (water) {
  ctx.strokeStyle = "rgba(70,217,255,0.62)";
  ctx.lineWidth = Math.max(1.4, w * 0.016);
  ctx.beginPath();
  ctx.moveTo(-w * 0.32, h * 0.22);
  ctx.quadraticCurveTo(0, h * 0.35, w * 0.32, h * 0.22);
  ctx.stroke();
} else if (air) {
  ctx.fillStyle = "rgba(222,249,255,0.34)";
  for (let i = -2; i <= 2; i += 1) {
    ctx.beginPath();
    ctx.ellipse(i * w * 0.08, -h * 0.14, w * 0.025, h * 0.025, 0, 0, Math.PI * 2);
    ctx.fill();
  }
}

drawRealisticRearVehicleSignature(w, h, color, vehicleType, police, air, water);

if (damage > 24) {
  ctx.globalAlpha = Math.min(0.45, damage / 180);
  ctx.strokeStyle = "rgba(255,91,107,0.82)";
  ctx.lineWidth = Math.max(1.5, w * 0.014);
  ctx.beginPath();
  ctx.moveTo(-w * 0.24, -h * 0.18);
  ctx.lineTo(-w * 0.02, h * 0.08);
  ctx.moveTo(w * 0.18, -h * 0.02);
  ctx.lineTo(w * 0.32, h * 0.18);
  ctx.stroke();
}
ctx.restore();
}

function drawRealisticRearVehicleSignature(w, h, color, vehicleType = "car", police = false, air = false, water = false) {
  if (air || water) return;
  const heavy = ["semi", "truck", "monster", "tractor"].includes(vehicleType);
  const tank = vehicleType === "tank";
  const f1 = vehicleType === "f1" || vehicleType === "prototype";
  const led = police ? "#46d9ff" : heavy ? "#ff5b6b" : "#ff3348";
  ctx.save();
  ctx.globalAlpha = phoneGraphicsActive() ? 0.86 : 0.78;
  if (tank) {
    ctx.fillStyle = "rgba(4,6,4,0.68)";
    roundRect(-w * 0.34, h * 0.28, w * 0.68, h * 0.1, Math.max(3, w * 0.018));
    ctx.fill();
    ctx.fillStyle = "rgba(187,242,74,0.46)";
    for (let i = -3; i <= 3; i += 1) {
      ctx.beginPath();
      ctx.arc(i * w * 0.08, h * 0.33, Math.max(2, w * 0.014), 0, Math.PI * 2);
      ctx.fill();
    }
  }
  ctx.restore();
  return;
}

ctx.fillStyle = "rgba(3,5,5,0.88)";
if (heavy) {
  roundRect(-w * 0.36, h * 0.18, w * 0.72, h * 0.16, Math.max(4, w * 0.026));
  ctx.fill();
} else {
  ctx.beginPath();
  ctx.moveTo(-w * 0.36, h * 0.2);
  ctx.lineTo(w * 0.36, h * 0.2);
}
```

```
    ctx.lineTo(w * 0.28, h * 0.42);
    ctx.lineTo(-w * 0.28, h * 0.42);
    ctx.closePath();
    ctx.fill();
}
ctx.fillStyle = led;
ctx.shadowColor = led;
ctx.shadowBlur = Math.max(3, w * 0.025);
if (heavy) {
    roundRect(-w * 0.32, h *
```

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```
0.19, w * 0.11, h * 0.055, 4);
    roundRect(w * 0.21, h * 0.19, w * 0.11, h * 0.055, 4);
    ctx.fill();
  } else {
    roundRect(-w * 0.32, h * 0.21, w * 0.2, h * 0.04, Math.max(3, w * 0.02));
    roundRect(w * 0.12, h * 0.21, w * 0.2, h * 0.04, Math.max(3, w * 0.02));
    ctx.fill();
    ctx.fillStyle = police ? "#ff3348" : "rgba(255,91,107,0.86)";
    roundRect(-w * 0.26, h * 0.27, w * 0.12, h * 0.018, 3);
    roundRect(w * 0.14, h * 0.27, w * 0.12, h * 0.018, 3);
    ctx.fill();
  }
  ctx.shadowBlur = 0;
  ctx.fillStyle = "rgba(244,251,248,0.76)";
  roundRect(-w * 0.08, h * (f1 ? 0.29 : 0.32), w * 0.16, h * 0.035, 4);
  ctx.fill();
  if (!heavy) {
    ctx.fillStyle = "rgba(1,2,2,0.96)";
    for (let i = -2; i <= 2; i += 1) {
      roundRect(i * w * 0.045 - w * 0.012, h * 0.39, w * 0.024, h * 0.09, 3);
      ctx.fill();
    }
    ctx.fillStyle = "rgba(0,0,0,0.86)";
    ctx.beginPath();
    ctx.ellipse(-w * 0.22, h * 0.43, w * 0.07, h * 0.025, 0, 0, Math.PI * 2);
    ctx.ellipse(w * 0.22, h * 0.43, w * 0.07, h * 0.025, 0, 0, Math.PI * 2);
    ctx.fill();
  }
  ctx.strokeStyle = "rgba(255,255,255,0.18)";
  ctx.lineWidth = Math.max(1, w * 0.01);
  ctx.beginPath();
  ctx.moveTo(-w * 0.36, -h * 0.04);
  ctx.bezierCurveTo(-w * 0.48, h * 0.12, -w * 0.43, h * 0.32, -w * 0.28, h * 0.46);
  ctx.moveTo(w * 0.36, -h * 0.04);
  ctx.bezierCurveTo(w * 0.48, h * 0.12, w * 0.43, h * 0.32, w * 0.28, h * 0.46);
  ctx.stroke();
  ctx.restore();
}

function drawLicensePlate(w, h, plate) {
  const text = sanitizePlate(plate || "RACER");
  ctx.save();
  ctx.fillStyle = "rgba(244,251,248,0.88)";
  roundRect(-w * 0.16, h * 0.36, w * 0.32, h * 0.058, 3);
  ctx.fill();
  ctx.fillStyle = "#06100e";
  ctx.font = `900 ${Math.max(6, Math.min(13, w * 0.075))}px system-ui, sans-serif`;
  ctx.textAlign = "center";
  ctx.textBaseline = "middle";
  ctx.fillText(text, 0, h * 0.389, w * 0.28);
  ctx.restore();
}

function drawSpecialVehicleSilhouette(w, h, color, vehicleType, player) {
  const paint = ctx.createLinearGradient(-w * 0.5, -h * 0.48, w * 0.48, h * 0.48);
  paint.addColorStop(0, shade(color, 42));
  paint.addColorStop(0.5, color);
  paint.addColorStop(1, shade(color, -48));
  ctx.fillStyle = paint;

  if (vehicleType === "f1" || vehicleType === "prototype") {
    ctx.fillStyle = "#050807";
    roundRect(-w * 0.62, h * 0.2, w * 0.18, h * 0.2, 6);
    ctx.fill();
    roundRect(w * 0.44, h * 0.2, w * 0.18, h * 0.2, 6);
    ctx.fill();
    ctx.fillStyle = paint;
    ctx.beginPath();
    ctx.moveTo(0, -h * 0.58);
    ctx.lineTo(w * 0.36, -h * 0.2);
    ctx.lineTo(w * 0.26, h * 0.45);
    ctx.lineTo(-w * 0.26, h * 0.45);
    ctx.lineTo(-w * 0.36, -h * 0.2);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = "rgba(5,8,7,0.82)";
    ctx.beginPath();
    ctx.ellipse(0, -h * 0.1, w * 0.18, h * 0.16, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = shade(color, -35);
    roundRect(-w * 0.48, h * 0.36, w * 0.96, h * 0.08, 3);
    ctx.fill();
  } else if (vehicleType === "policecar") {
    ctx.fillStyle = "#050807";
    roundRect(-w * 0.58, -h * 0.2, w * 0.16, h * 0.54, 7);
    ctx.fill();
    roundRect(w * 0.42, -h * 0.2, w * 0.16, h * 0.54, 7);
    ctx.fill();
    ctx.fillStyle = paint;
    roundRect(-w * 0.46, -h * 0.5, w * 0.92, h * 0.86, 10);
    ctx.fill();
    ctx.fillStyle = "rgba(5,8,7,0.82)";
    roundRect(-w * 0.24, -h * 0.36, w * 0.48, h * 0.22, 5);
    ctx.fill();
    ctx.fillStyle = "#ff3348";
    roundRect(-w * 0.24, -h * 0.58, w * 0.22, h * 0.055, 3);
    ctx.fill();
    ctx.fillStyle = "#46d9ff";
    roundRect(w * 0.02, -h * 0.58, w * 0.22, h * 0.055, 3);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.86)";
    ctx.font = `900 ${Math.max(7, w * 0.1)}px system-ui`;
    ctx.textAlign = "center";
    ctx.textBaseline = "middle";
    ctx.fillText("POLICE", 0, h * 0.14, w * 0.62);
  } else if (vehicleType === "semi") {
    ctx.fillStyle = "#050807";
    roundRect(-w * 0.72, -h * 0.02, w * 0.18, h * 0.5, 7);
    ctx.fill();
    roundRect(w * 0.54, -h * 0.02, w * 0.18, h * 0.5, 7);
    ctx.fill();
    roundRect(-w * 0.72, -h * 0.42, w * 0.16, h * 0.22, 6);
    ctx.fill();
  }
}
```



```
roundRect(w * 0.56, -h * 0.42, w * 0.16, h * 0.22,
```

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```
6);
ctx.fill();
ctx.fillStyle = paint;
roundRect(-w * 0.42, -h * 0.58, w * 0.84, h * 0.38, 6);
ctx.fill();
ctx.fillStyle = shade(color, -28);
roundRect(-w * 0.5, -h * 0.16, w * 1.0, h * 0.68, 4);
ctx.fill();
ctx.fillStyle = "rgba(5,8,7,0.82)";
roundRect(-w * 0.27, -h * 0.5, w * 0.54, h * 0.18, 4);
ctx.fill();
ctx.fillStyle = "rgba(244,251,248,0.48)";
ctx.fillRect(-w * 0.34, -h * 0.1, w * 0.68, h * 0.08);
ctx.fillRect(-w * 0.34, h * 0.08, w * 0.68, h * 0.08);
} else if (vehicleType === "tractor") {
  ctx.fillStyle = "#050807";
  roundRect(-w * 0.64, h * 0.12, w * 0.26, h * 0.36, 9);
  ctx.fill();
  roundRect(w * 0.38, h * 0.12, w * 0.26, h * 0.36, 9);
  ctx.fill();
  roundRect(-w * 0.48, -h * 0.42, w * 0.14, h * 0.2, 5);
  ctx.fill();
  roundRect(w * 0.34, -h * 0.42, w * 0.14, h * 0.2, 5);
  ctx.fill();
  ctx.fillStyle = paint;
  roundRect(-w * 0.3, -h * 0.48, w * 0.6, h * 0.36, 8);
  ctx.fill();
  ctx.fillStyle = shade(color, -28);
  roundRect(-w * 0.42, -h * 0.06, w * 0.84, h * 0.38, 8);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * 0.2, -h * 0.38, w * 0.4, h * 0.16, 4);
  ctx.fill();
  ctx.strokeStyle = "rgba(255,209,102,0.6)";
  ctx.lineWidth = Math.max(2, w * 0.035);
  ctx.beginPath();
  ctx.moveTo(-w * 0.44, h * 0.38);
  ctx.lineTo(w * 0.44, h * 0.38);
  ctx.stroke();
} else if (vehicleType === "truck" || vehicleType === "monster" || vehicleType === "tank") {
  const tire = vehicleType === "monster" ? 0.24 : 0.17;
  ctx.fillStyle = "#050807";
  roundRect(-w * 0.67, -h * 0.08, w * tire, h * 0.5, 7);
  ctx.fill();
  roundRect(w * (0.67 - tire), -h * 0.08, w * tire, h * 0.5, 7);
  ctx.fill();
  ctx.fillStyle = paint;
  roundRect(-w * 0.48, -h * 0.48, w * 0.96, h * 0.82, vehicleType === "tank" ? 3 : 7);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * 0.27, -h * 0.32, w * 0.54, h * 0.22, 5);
  ctx.fill();
  if (vehicleType === "tank") {
    ctx.fillStyle = shade(color, -25);
    roundRect(-w * 0.22, -h * 0.18, w * 0.44, h * 0.3, 4);
    ctx.fill();
    ctx.fillRect(-w * 0.04, -h * 0.66, w * 0.08, h * 0.5);
  }
} else if (vehicleType === "boat" || vehicleType === "snowmobile") {
  ctx.beginPath();
  ctx.moveTo(0, -h * 0.58);
  ctx.quadraticCurveTo(w * 0.5, -h * 0.05, w * 0.24, h * 0.48);
  ctx.lineTo(-w * 0.24, h * 0.48);
  ctx.quadraticCurveTo(-w * 0.5, -h * 0.05, 0, -h * 0.58);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * 0.16, -h * 0.12, w * 0.32, h * 0.24, 5);
  ctx.fill();
  ctx.strokeStyle = vehicleType === "boat" ? "rgba(70,217,255,0.58)" : "rgba(244,251,248,0.58)";
  ctx.lineWidth = Math.max(2, w * 0.04);
  ctx.beginPath();
  ctx.moveTo(-w * 0.44, h * 0.36);
  ctx.lineTo(-w * 0.68, h * 0.5);
  ctx.moveTo(w * 0.44, h * 0.36);
  ctx.lineTo(w * 0.68, h * 0.5);
  ctx.stroke();
} else if (vehicleType === "drone") {
  ctx.fillStyle = "rgba(244,251,248,0.66)";
  for (let sx = -1; sx <= 1; sx += 2) {
    for (let sy = -1; sy <= 1; sy += 2) {
      ctx.beginPath();
      ctx.ellipse(sx * w * 0.36, sy * h * 0.25, w * 0.22, h * 0.055, 0, 0, Math.PI * 2);
      ctx.fill();
    }
  }
  ctx.strokeStyle = shade(color, -28);
  ctx.lineWidth = Math.max(2, w * 0.045);
  ctx.beginPath();
  ctx.moveTo(-w * 0.34, -h * 0.25);
  ctx.lineTo(w * 0.34, h * 0.25);
  ctx.moveTo(w * 0.34, -h * 0.25);
  ctx.lineTo(-w * 0.34, h * 0.25);
  ctx.stroke();
  ctx.fillStyle = paint;
  roundRect(-w * 0.22, -h * 0.22, w * 0.44, h * 0.44, 10);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * 0.12, -h * 0.08, w * 0.24, h * 0.16, 4);
  ctx.fill();
} else if (vehicleType === "helicopter") {
  ctx.fillStyle = "rgba(244,251,248,0.68)";
  roundRect(-w * 0.62, -h * 0.64, w * 1.24, h * 0.05, 2);
  ctx.fill();
  ctx.fillStyle = paint;
  roundRect(-w * 0.28, -h * 0.36, w * 0.56, h * 0.56, 14);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * 0.17, -h * 0.24, w * 0.34, h * 0.18, 5);
  ctx.fill();
}
```

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```
ctx.strokeStyle = shade(color, -30);
ctx.lineWidth = Math.max(2, w * 0.05);
ctx.beginPath();
ctx.moveTo(0, h * 0.18);
ctx.lineTo(0, h * 0.56);
ctx.stroke();
} else if (vehicleType === "airplane" || vehicleType === "fighterjet" || vehicleType === "bomber") {
  const bomber = vehicleType === "bomber";
  const fighter = vehicleType === "fighterjet";
  ctx.fillStyle = paint;
  ctx.beginPath();
  ctx.moveTo(0, -h * (fighter ? 0.7 : 0.62));
  ctx.lineTo(w * (bomber ? 0.28 : 0.2), h * 0.48);
  ctx.lineTo(-w * (bomber ? 0.28 : 0.2), h * 0.48);
  ctx.closePath();
  ctx.fill();
  ctx.beginPath();
  ctx.moveTo(-w * (bomber ? 0.82 : fighter ? 0.72 : 0.68), h * (fighter ? -0.02 : 0.02));
  ctx.lineTo(w * (bomber ? 0.82 : fighter ? 0.72 : 0.68), h * (fighter ? -0.02 : 0.02));
  ctx.lineTo(w * (bomber ? 0.34 : 0.22), h * 0.22);
  ctx.lineTo(-w * (bomber ? 0.34 : 0.22), h * 0.22);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * (bomber ? 0.16 : 0.1), -h * 0.34, w * (bomber ? 0.32 : 0.2), h * 0.26, 4);
  ctx.fill();
  if (fighter) {
    ctx.fillStyle = shade(color, -30);
    ctx.beginPath();
    ctx.moveTo(-w * 0.22, h * 0.42);
    ctx.lineTo(-w * 0.44, h * 0.66);
    ctx.lineTo(-w * 0.12, h * 0.52);
    ctx.closePath();
    ctx.fill();
    ctx.beginPath();
    ctx.moveTo(w * 0.22, h * 0.42);
    ctx.lineTo(w * 0.44, h * 0.66);
    ctx.lineTo(w * 0.12, h * 0.52);
    ctx.closePath();
    ctx.fill();
  }
}
}

const groundVehicle = !isWaterVehicleType(vehicleType) && !isAirVehicleType(vehicleType) && vehicleType !== "snowmobile";
if (groundVehicle) {
  const rimAccent = vehicleType === "tractor" ? "rgba(255,209,102,0.38)" : "rgba(70,217,255,0.28)";
  const frontY = vehicleType === "f1" || vehicleType === "prototype" ? h * 0.33 : h * 0.18;
  const rearY = vehicleType === "f1" || vehicleType === "prototype" ? h * 0.43 : h * 0.38;
  drawWheelRim(-w * 0.55, frontY, w * 0.075, h * 0.12, rimAccent);
  drawWheelRim(w * 0.55, frontY, w * 0.075, h * 0.12, rimAccent);
  drawWheelRim(-w * 0.5, rearY, w * 0.07, h * 0.105, rimAccent);
  drawWheelRim(w * 0.5, rearY, w * 0.07, h * 0.105, rimAccent);
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * 0.26, h * 0.31, w * 0.52, h * 0.075, 3);
  ctx.fill();
  ctx.fillStyle = "rgba(255,248,214,0.86)";
  roundRect(-w * 0.34, -h * 0.5, w * 0.18, h * 0.055, 3);
  ctx.fill();
  roundRect(w * 0.16, -h * 0.5, w * 0.18, h * 0.055, 3);
  ctx.fill();
  ctx.strokeStyle = "rgba(244,251,248,0.2)";
  ctx.lineWidth = Math.max(1, w * 0.012);
  ctx.beginPath();
  ctx.moveTo(-w * 0.32, -h * 0.08);
  ctx.lineTo(-w * 0.36, h * 0.3);
  ctx.moveTo(w * 0.32, -h * 0.08);
  ctx.lineTo(w * 0.36, h * 0.3);
  ctx.stroke();
} else if (vehicleType === "boat" || vehicleType === "snowmobile") {
  ctx.fillStyle = "rgba(244,251,248,0.18)";
  roundRect(-w * 0.18, -h * 0.4, w * 0.36, h * 0.055, 3);
  ctx.fill();
  ctx.fillStyle = vehicleType === "boat" ? "rgba(70,217,255,0.44)" : "rgba(244,251,248,0.44)";
  roundRect(-w * 0.34, h * 0.35, w * 0.68, h * 0.045, 3);
  ctx.fill();
} else if (vehicleType === "drone") {
  ctx.fillStyle = "rgba(244,251,248,0.18)";
  ctx.beginPath();
  ctx.ellipse(0, 0, w * 0.5, h * 0.08, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(70,217,255,0.42)";
  roundRect(-w * 0.18, h * 0.22, w * 0.36, h * 0.045, 3);
  ctx.fill();
} else if (vehicleType === "helicopter") {
  ctx.fillStyle = "rgba(244,251,248,0.22)";
  ctx.beginPath();
  ctx.ellipse(0, -h * 0.62, w * 0.66, h * 0.055, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.fillStyle = "rgba(5,8,7,0.82)";
  roundRect(-w * 0.1, h * 0.48, w * 0.2, h * 0.08, 3);
  ctx.fill();
} else if (vehicleType === "airplane" || vehicleType === "fighterjet" || vehicleType === "bomber") {
  ctx.fillStyle = "rgba(255,248,214,0.82)";
  roundRect(-w * 0.08, -h * 0.54, w * 0.16, h * 0.05, 3);
  ctx.fill();
  ctx.fillStyle = "rgba(70,217,255,0.36)";
  roundRect(-w * 0.48, h * 0.08, w * 0.96, h * 0.035, 3);
  ctx.fill();
}

ctx.strokeStyle = "rgba(
```

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```
255,255,255,0.24)";
ctx.lineWidth = Math.max(1, w * 0.016);
ctx.stroke();
drawRealWorldVehicleSkinDetails(w, h, color, vehicleType, player ? raceState.damage || 0 : 0, false);
if (player) drawVehicleDamageMarks(w, h, raceState.damage || 0);
if (player) {
  ctx.fillStyle = "rgba(255,209,102,0.55)";
  ctx.beginPath();
  ctx.ellipse(0, h * 0.52, w * 0.24, h * 0.05, 0, 0, Math.PI * 2);
  ctx.fill();
}
}

function drawDemoPursuitTraffic(w, h) {
  const t = performance.now() / 1000;
  const demo = [
    { lane: -1.15 + Math.sin(t * 0.7) * 0.15, y: h * 0.43, color: "#ff5b6b", police: false, s: 0.58 },
    { lane: 1.05 + Math.sin(t * 0.9) * 0.12, y: h * 0.5, color: "#f4fbf8", police: true, s: 0.68 },
    { lane: 0.08 + Math.sin(t * 1.1) * 0.18, y: h * 0.61, color: "#46d9ff", police: false, s: 0.86 }
  ];
  demo.forEach((car) => {
    const p = objectPos(car.lane, car.y);
    drawTrafficRearCar(p.x, p.y, 78 * car.s * p.scale, 118 * car.s * p.scale, car.color, car.police);
  });
  ctx.save();
  ctx.globalAlpha = 0.18 + Math.max(0, Math.sin(t * 12)) * 0.1;
  let wash = ctx.createRadialGradient(w * 0.18, h * 0.48, 10, w * 0.18, h * 0.48, w * 0.55);
  wash.addColorStop(0, "rgba(255,51,72,0.44)");
  wash.addColorStop(1, "rgba(255,51,72,0)");
  ctx.fillStyle = wash;
  ctx.fillRect(0, 0, w, h);
  wash = ctx.createRadialGradient(w * 0.82, h * 0.48, 10, w * 0.82, h * 0.48, w * 0.55);
  wash.addColorStop(0, "rgba(70,217,255,0.4)");
  wash.addColorStop(1, "rgba(70,217,255,0)");
  ctx.fillStyle = wash;
  ctx.fillRect(0, 0, w, h);
  ctx.restore();
}

function drawCar(w, h) {
  if (cameraMode === "cockpit") return;
  const vehicle = selectedVehicle();
  const color = selectedVehicleColor(vehicle);
  if (phoneGraphicsActive() && cameraMode === "hood") {
    drawPhoneDriverHood(w, h, Object.assign({}, vehicle, { color }));
    return;
  }
  const carLaneFollowDepth = useWebGLRenderer() && cameraMode === "chase" ? 0.42 : 1;
  let x = w / 2 + (raceState.lane - cameraLaneOffset(carLaneFollowDepth)) * laneWidth();
  let y = cameraMode === "hood" ? h * 0.99 : useWebGLRenderer() ? h * 0.84 : h * 0.86;
  let projectionScale = 1;
  if (useWebGLRenderer() && cameraMode === "chase" && webglRenderer.typeof webglRenderer.projectWorldRoadPoint === "function") {
    const anchor = webglRenderer.projectWorldRoadPoint(raceState.lane, 5.2);
    if (anchor && Number.isFinite(anchor.x) && Number.isFinite(anchor.y)) {
      y = Math.max(h * 0.78, Math.min(h * 0.96, anchor.y + h * 0.045));
      projectionScale = Math.max(0.92, Math.min(1.12, anchor.scale || 1));
    }
  }
  const sizeBoost = vehicle.type === "semi" ? 1.28 : vehicle.type === "monster" || vehicle.type === "tank" ? 1.18 : vehicle.type === "tractor" ? 1.08 : vehicle.type === "f1" || vehicle.type === "snowmobile" ? 0.9 : 1;
  const phoneChaseScale = phoneGraphicsActive() && cameraMode === "chase" ? 0.66 : 1;
  const carWidth = (cameraMode === "hood" ? 220 : 118) * sizeBoost * projectionScale * phoneChaseScale;
  const carHeight = (cameraMode === "hood" ? 190 : 178) * sizeBoost * projectionScale * phoneChaseScale;
  if (cameraMode === "chase") {
    const phoneLift = phoneGraphicsActive() ? h * 0.065 : 0;
    drawPlayerChaseCar(x, y + phoneLift + 18 + roadContactSink(projectionScale, vehicle.type), carWidth * 1.26, carHeight * 1.02, color, vehicle.type);
  } else {
    drawVehicle(x, y, carWidth, carHeight, color, true, false, vehicle.type);
  }
  if ((input.boost || input.gamepadBoost) && raceState.active) {
    ctx.fillStyle = "rgba(255,209,102,0.82)";
    ctx.beginPath();
    ctx.moveTo(x - carWidth * 0.28, y + carHeight * 0.44);
    ctx.lineTo(x, y + carHeight * 0.95 + Math.random() * 34);
    ctx.lineTo(x + carWidth * 0.28, y + carHeight * 0.44);
    ctx.closePath();
    ctx.fill();
  }
}

function drawPhoneDriverHood(w, h, vehicle) {
  const hoodY = h * 0.84;
  const color = vehicle.color || "#46d9ff";
  ctx.save();
  const hoodShift = Math.max(-w * 0.035, Math.min(w * 0.035, (raceState.steerAngle || 0) * w * 0.018 + (raceState.lateralVelocity || 0) * w * 0.012));
  ctx.translate(hoodShift, 0);
  const shadow = ctx.createLinearGradient(0, h * 0.68, 0, h);
  shadow.addColorStop(
```

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```
0, "rgba(0,0,0,0)");
shadow.addColorStop(0.62, "rgba(0,0,0,0.62)");
shadow.addColorStop(1, "rgba(0,0,0,0.92)");
ctx.fillStyle = shadow;
ctx.fillRect(0, h * 0.62, w, h * 0.38);

const hood = ctx.createLinearGradient(w * 0.28, hoodY, w * 0.72, h);
hood.addColorStop(0, shade(color, -24));
hood.addColorStop(0.45, shade(color, 42));
hood.addColorStop(1, shade(color, -52));
ctx.fillStyle = hood;
ctx.beginPath();
ctx.moveTo(w * 0.18, h + 18);
ctx.quadraticCurveTo(w * 0.28, hoodY + 4, w * 0.42, hoodY - h * 0.015);
ctx.lineTo(w * 0.58, hoodY - h * 0.015);
ctx.quadraticCurveTo(w * 0.72, hoodY + 4, w * 0.82, h + 18);
ctx.closePath();
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.24)";
ctx.lineWidth = Math.max(2, w * 0.004);
ctx.stroke();

ctx.globalAlpha = 0.72;
ctx.strokeStyle = "rgba(5,8,7,0.58)";
ctx.lineWidth = Math.max(3, w * 0.006);
ctx.beginPath();
ctx.moveTo(w * 0.5, hoodY);
ctx.lineTo(w * 0.5, h);
ctx.stroke();
ctx.globalAlpha = 1;
ctx.fillStyle = "rgba(5,8,7,0.82)";
roundRect(w * 0.36, hoodY + h * 0.08, w * 0.28, h * 0.045, 5);
ctx.fill();
ctx.fillStyle = "rgba(244,251,248,0.2)";
roundRect(w * 0.375, hoodY + h * 0.095, w * 0.25, h * 0.012, 4);
ctx.fill();
if ((input.boost || input.gamepadBoost) && raceState.active) {
  ctx.globalCompositeOperation = "screen";
  ctx.fillStyle = "rgba(255,209,102,0.28)";
  ctx.beginPath();
  ctx.moveTo(w * 0.42, h * 0.97);
  ctx.lineTo(w * 0.5, h * 0.82);
  ctx.lineTo(w * 0.58, h * 0.97);
  ctx.closePath();
  ctx.fill();
}
ctx.restore();
}

function drawPlayerChaseCar(x, y, width, height, color, vehicleType = "car") {
  ctx.save();
  ctx.translate(x, y);
  const bodyYaw = Math.max(-0.12, Math.min(0.12, (raceState.steerAngle || 0) * 0.055 + (raceState.lateralVelocity || 0) * 0.018));
  ctx.rotate(bodyYaw);

  drawRoadContactShadow(width, height, vehicleType, 1);

  if (drawPhoneAssetVehicleSprite(width, height, color, vehicleType, false, raceState.damage || 0, true)) {
    ctx.save();
    ctx.translate(0, -height * 0.18);
    drawLicensePlate(width * 0.78, height * 0.78, driverPlate(activeProfile));
    ctx.restore();
    ctx.restore();
    return;
  }

  ctx.translate(0, -height * 0.58);

  if (vehicleType !== "car") {
    drawSpecialVehicleSilhouette(width, height, color, vehicleType, true);
    drawLicensePlate(width * 0.82, height * 0.82, driverPlate(activeProfile));
    ctx.restore();
    return;
  }

  const paint = ctx.createLinearGradient(-width * 0.5, -height * 0.52, width * 0.42, height * 0.5);
  paint.addColorStop(0, shade(color, 56));
  paint.addColorStop(0.22, color);
  paint.addColorStop(0.58, shade(color, -22));
  paint.addColorStop(1, shade(color, -56));

  ctx.fillStyle = "#050807";
  roundRect(-width * 0.56, -height * 0.18, width * 0.16, height * 0.5, 8);
  ctx.fill();
  roundRect(width * 0.4, -height * 0.18, width * 0.16, height * 0.5, 8);
  ctx.fill();
  drawWheelRim(-width * 0.49, -height * 0.01, width * 0.08, height * 0.16);
  drawWheelRim(width * 0.49, -height * 0.01, width * 0.08, height * 0.16);
  drawWheelRim(-width * 0.48, height * 0.27, width * 0.07, height * 0.12, "rgba(255,209,102,0.24)");
  drawWheelRim(width * 0.48, height * 0.27, width * 0.07, height * 0.12, "rgba(255,209,102,0.24)");

  ctx.fillStyle = paint;
  ctx.beginPath();
  ctx.moveTo(-width * 0.34, -height * 0.52);
  ctx.lineTo(width * 0.34, -height * 0.52);
  ctx.quadraticCurveTo(width * 0.54, -height * 0.38, width * 0.5, -height * 0.08);
  ctx.lineTo(width * 0.43, height * 0.36);
  ctx.quadraticCurveTo(width * 0.31, height * 0.55, 0, height * 0.58);
  ctx.quadraticCurveTo(-width * 0.31, height * 0.55, -width * 0.43, height * 0.36);
  ctx.lineTo(-width * 0.5, -height * 0.08);
  ctx.quadraticCurveTo(-width * 0.54, -height * 0.38, -width * 0.34, -height * 0.52);
  ctx.closePath();
  ctx.fill();
  ctx.strokeStyle = "rgba(255,255,255,0.28)";
  ctx.lineWidth = Math.max(2, width * 0.018);
  ctx.stroke();
  drawVehicleTrimDetails(width, height, color, false);
  drawRealWorldVehicleSkinDetails(width, height, color, vehicleType, raceState.damage || 0, false);
  drawOpponentRaceIdentity(width, height, activeProfile ? acti
```

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```
veProfile.name : "Driver", false, vehicleType, raceState.damage || 0, false);

const windshield = ctx.createLinearGradient(0, -height * 0.46, 0, height * 0.08);
windshield.addColorStop(0, "rgba(222,249,255,0.72)");
windshield.addColorStop(0.38, "rgba(58,88,94,0.82)");
windshield.addColorStop(1, "rgba(3,6,7,0.96)");
ctx.fillStyle = windshield;
ctx.beginPath();
ctx.moveTo(-width * 0.28, -height * 0.39);
ctx.lineTo(width * 0.28, -height * 0.39);
ctx.lineTo(width * 0.34, -height * 0.08);
ctx.quadraticCurveTo(0, height * 0.03, -width * 0.34, -height * 0.08);
ctx.closePath();
ctx.fill();

ctx.fillStyle = "rgba(5,8,7,0.9)";
roundRect(-width * 0.32, height * 0.12, width * 0.64, height * 0.27, 10);
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.18)";
ctx.stroke();

ctx.strokeStyle = "rgba(5,8,7,0.52)";
ctx.lineWidth = Math.max(2, width * 0.024);
ctx.beginPath();
ctx.moveTo(0, -height * 0.49);
ctx.lineTo(0, height * 0.55);
ctx.moveTo(-width * 0.42, -height * 0.05);
ctx.lineTo(width * 0.42, -height * 0.05);
ctx.moveTo(-width * 0.38, height * 0.36);
ctx.lineTo(width * 0.38, height * 0.36);
ctx.stroke();

ctx.fillStyle = "rgba(255,255,255,0.18)";
ctx.beginPath();
ctx.moveTo(-width * 0.31, -height * 0.48);
ctx.lineTo(-width * 0.08, -height * 0.43);
ctx.lineTo(-width * 0.23, -height * 0.3);
ctx.closePath();
ctx.moveTo(width * 0.31, -height * 0.48);
ctx.lineTo(width * 0.08, -height * 0.43);
ctx.lineTo(width * 0.23, -height * 0.3);
ctx.closePath();
ctx.fill();

ctx.fillStyle = "#ff3348";
roundRect(-width * 0.36, height * 0.42, width * 0.22, height * 0.055, 4);
ctx.fill();
roundRect(width * 0.14, height * 0.42, width * 0.22, height * 0.055, 4);
ctx.fill();
ctx.globalAlpha = 0.28;
ctx.fillStyle = "#ff3348";
ctx.beginPath();
ctx.ellipse(-width * 0.25, height * 0.44, width * 0.32, height * 0.08, 0, 0, Math.PI * 2);
ctx.ellipse(width * 0.25, height * 0.44, width * 0.32, height * 0.08, 0, 0, Math.PI * 2);
ctx.fill();
ctx.globalAlpha = 1;

ctx.fillStyle = "rgba(5,8,7,0.9)";
roundRect(-width * 0.19, height * 0.49, width * 0.38, height * 0.05, 4);
ctx.fill();
ctx.fillStyle = "rgba(244,251,248,0.64)";
roundRect(-width * 0.08, height * 0.5, width * 0.16, height * 0.03, 3);
ctx.fill();

drawVehicleDamageMarks(width, height, raceState.damage || 0);
ctx.restore();
}

function drawVehicleDamageMarks(width, height, damage) {
  if (!damage || damage < 14) return;
  ctx.save();
  const dents = Math.min(8, Math.floor(damage / 12));
  ctx.strokeStyle = "rgba(5,8,7,0.72)";
  ctx.lineWidth = Math.max(1.5, width * 0.018);
  ctx.lineCap = "round";
  for (let i = 0; i < dents; i += 1) {
    const side = i % 2 ? -1 : 1;
    const x = side * width * (0.12 + (i % 3) * 0.085);
    const y = -height * 0.25 + (i % 5) * height * 0.14;
    ctx.beginPath();
    ctx.moveTo(x, y);
    ctx.lineTo(x + side * width * 0.16, y + height * (0.04 + (i % 2) * 0.035));
    ctx.stroke();
  }
  ctx.globalAlpha = Math.min(0.45, damage / 160);
  ctx.fillStyle = "rgba(0,0,0,0.7)";
  for (let i = 0; i < dents; i += 1) {
    const x = ((i * 41) % 100 - 50) / 100 * width * 0.62;
    const y = ((i * 29) % 100 - 48) / 100 * height * 0.72;
    ctx.beginPath();
    ctx.ellipse(x, y, width * 0.035, height * 0.025, i * 0.4, 0, Math.PI * 2);
    ctx.fill();
  }
  if (damage > 58) {
    ctx.globalAlpha = Math.min(0.62, damage / 140);
    ctx.fillStyle = "rgba(255,91,107,0.42)";
    roundRect(-width * 0.3, -height * 0.48, width * 0.18, height * 0.055, 3);
    ctx.fill();
  }
  ctx.restore();
}

function drawCameraOverlay(w, h, theme) {
  if (cameraMode === "chase") {
    ctx.save();
    const mirror = ctx.createLinearGradient(0, 0, 0, h * 0.18);
    mirror.addColorStop(0, "rgba(3,6,6,0.78)");
    mirror.addColorStop(1, "rgba(3,6,6,0)");
    ctx.fillStyle = mirror;
    ctx.fillRect(0, 0, w, h * 0.18);
    ctx.fillStyle = "rgba(5,8,7,0.58)";
    roundRect(w * 0.39, h * 0.055, w * 0.22, h * 0.055, 6);
    ctx.fill();
  }
}
```

```
ctx.strokeStyle = "rgba(244,251,248,0.18)";  
ctx.stroke();  
ctx.fillStyle = theme[1];  
ctx.font = "900 12px system-ui";  
ctx.textAlign = "center";  
ctx.fi
```

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```
llText(raceState.chaseActive ? "PURSUIT CAMERA" : "CHASE CAMERA", w * 0.5, h * 0.09);
ctx.restore();
return;
}
if (cameraMode === "hood") {
  const hood = ctx.createLinearGradient(0, h * 0.72, 0, h);
  hood.addColorStop(0, "rgba(70,217,255,0.05)");
  hood.addColorStop(0.58, "#101817");
  hood.addColorStop(1, "#050807");
  ctx.fillStyle = hood;
  ctx.beginPath();
  ctx.moveTo(w * 0.14, h);
  ctx.quadraticCurveTo(w * 0.5, h * 0.68, w * 0.86, h);
  ctx.closePath();
  ctx.fill();
  ctx.strokeStyle = theme[1];
  ctx.lineWidth = 4;
  ctx.globalAlpha = 0.8;
  ctx.beginPath();
  ctx.moveTo(w * 0.36, h * 0.91);
  ctx.lineTo(w * 0.64, h * 0.91);
  ctx.stroke();
  ctx.globalAlpha = 1;
  return;
}
if (cameraMode !== "cockpit") return;
ctx.save();
ctx.fillStyle = "rgba(2,5,5,0.64)";
ctx.fillRect(0, 0, w, h * 0.1);
const dash = ctx.createLinearGradient(0, h * 0.74, 0, h);
dash.addColorStop(0, "rgba(6,10,10,0.52)");
dash.addColorStop(0.42, "#0a0f0e");
dash.addColorStop(1, "#020403");
ctx.fillStyle = dash;
ctx.fillRect(0, h * 0.75, w, h * 0.25);
ctx.fillStyle = "#080d0c";
ctx.beginPath();
ctx.moveTo(0, h);
ctx.lineTo(w * 0.2, h * 0.78);
ctx.lineTo(w * 0.8, h * 0.78);
ctx.lineTo(w, h);
ctx.closePath();
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.28)";
ctx.lineWidth = 10;
roundRect(w * 0.08, h * 0.12, w * 0.84, h * 0.58, 18);
ctx.stroke();
ctx.strokeStyle = "rgba(244,251,248,0.16)";
ctx.lineWidth = 2;
ctx.beginPath();
ctx.moveTo(w * 0.08, h * 0.68);
ctx.quadraticCurveTo(w * 0.5, h * 0.76, w * 0.92, h * 0.68);
ctx.stroke();
ctx.strokeStyle = theme[1];
ctx.lineWidth = 3;
ctx.globalAlpha = 0.65;
ctx.beginPath();
ctx.moveTo(w * 0.5, h * 0.14);
ctx.lineTo(w * 0.5 + raceState.lane * 22, h * 0.68);
ctx.stroke();
ctx.globalAlpha = 1;
ctx.fillStyle = "#111817";
ctx.beginPath();
ctx.arc(w * 0.5, h * 0.9, Math.min(w, h) * 0.13, Math.PI, 0);
ctx.fill();
ctx.strokeStyle = "#46d9ff";
ctx.lineWidth = 8;
ctx.beginPath();
ctx.arc(w * 0.5, h * 0.9, Math.min(w, h) * 0.1, Math.PI * 1.06, Math.PI * 1.94);
ctx.stroke();
ctx.fillStyle = "#bbf24a";
ctx.font = "900 18px system-ui";
ctx.textAlign = "center";
ctx.fillText(`${Math.round(raceState.speed)} MPH`, w * 0.5, h * 0.9);
ctx.fillStyle = "rgba(5,8,7,0.74)";
roundRect(w * 0.11, h * 0.8, w * 0.22, h * 0.1, 8);
ctx.fill();
ctx.fillStyle = raceState.chaseActive ? "#ff5b6b" : "#a9bbb5";
ctx.font = "900 14px system-ui";
ctx.textAlign = "left";
ctx.fillText(raceState.chaseActive ? "PURSUIT ACTIVE" : "STREET CLEAR", w * 0.13, h * 0.835);
ctx.fillStyle = "#f4fbf8";
ctx.font = "800 12px system-ui";
ctx.fillText(`HEAT ${Math.round(raceState.heat)}%`, w * 0.13, h * 0.865);
ctx.fillStyle = "rgba(5,8,7,0.74)";
roundRect(w * 0.67, h * 0.8, w * 0.22, h * 0.1, 8);
ctx.fill();
ctx.fillStyle = theme[1];
ctx.font = "900 14px system-ui";
ctx.textAlign = "right";
ctx.fillText(selectedRace ? selectedRace.mood.toUpperCase() : "STREET RACE", w * 0.87, h * 0.835);
ctx.fillStyle = "#f4fbf8";
ctx.font = "800 12px system-ui";
ctx.fillText(`${Math.round((raceState.distance / raceLength()) * 100)}% ROUTE`, w * 0.87, h * 0.865);
if (raceState.chaseActive) {
  ctx.globalAlpha = 0.18 + Math.max(0, Math.sin(raceState.elapsed * 16)) * 0.14;
  ctx.fillStyle = "#ff3348";
  ctx.fillRect(0, 0, w * 0.5, h);
  ctx.fillStyle = "#46d9ff";
  ctx.fillRect(w * 0.5, 0, w * 0.5, h);
  ctx.globalAlpha = 1;
}
ctx.restore();
}

function drawDamageOverlay(w, h, theme) {
  if (!raceState.active) return;
  const damage = Math.max(0, raceState.damage || 0);
  const resetting = raceState.resetTimer > 0;
  const damageAlert = (raceState.damageAlertTimer || 0) > 0;
  const hazardWarning = (raceState.hazardWarningTimer || 0) > 0;
  if (damage < 10 && !resetting && !damageAlert && !hazardWarning) return;
  ctx.save();
```



```
const danger = Math.min(0.34, damage / 280 + (resetting ? 0.18 : 0));
const edge = ctx.createRadialGradient(w * 0.5, h * 0.52, h * 0.28, w * 0.5, h * 0.52, h * 0.9);
edge.addColorStop(0, "rgba(0,0,0,0)");
edge.addColorStop(0.68, "rgba(0,0,
```

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```
0,0)");
edge.addColorStop(1, `rgba(255,51,72,${danger})`);
ctx.fillStyle = edge;
ctx.fillRect(0, 0, w, h);

if (hazardWarning || damageAlert) {
  const label = hazardWarning ? raceState.hazardWarningLabel : raceState.damageAlertLabel;
  const panelW = Math.min(phoneGraphicsActive() ? 330 : 420, w - 28);
  const panelH = phoneGraphicsActive() ? 34 : 40;
  const x = (w - panelW) / 2;
  const y = phoneGraphicsActive() ? h * 0.2 : h * 0.16;
  ctx.globalAlpha = hazardWarning ? 0.9 : 0.82;
  ctx.fillStyle = hazardWarning ? "rgba(255,209,102,0.16)" : "rgba(255,91,107,0.18)";
  roundRect(x, y, panelW, panelH, 7);
  ctx.fill();
  ctx.strokeStyle = hazardWarning ? "rgba(255,209,102,0.82)" : "rgba(255,91,107,0.84)";
  ctx.lineWidth = 2;
  ctx.stroke();
  ctx.globalAlpha = 1;
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `900 ${phoneGraphicsActive() ? 13 : 15}px system-ui`;
  ctx.textAlign = "center";
  ctx.fillText(label || "WATCH TRAFFIC", w / 2, y + panelH * 0.64);
}

if (cameraMode === "cockpit" || damage > 48) {
  ctx.strokeStyle = `rgba(244,251,248,${Math.min(0.28, damage / 240)})`;
  ctx.lineWidth = Math.max(1, w * 0.002);
  const cracks = Math.floor(Math.min(8, damage / 12));
  for (let i = 0; i < cracks; i += 1) {
    const anchorX = w * (0.2 + ((i * 0.17) % 0.6));
    const anchorY = h * (0.18 + ((i * 0.13) % 0.38));
    ctx.beginPath();
    ctx.moveTo(anchorX, anchorY);
    ctx.lineTo(anchorX + Math.sin(i * 1.9) * w * 0.08, anchorY + h * (0.05 + (i % 3) * 0.025));
    ctx.lineTo(anchorX + Math.cos(i * 1.4) * w * 0.1, anchorY + h * (0.1 + (i % 2) * 0.03));
    ctx.stroke();
  }
}

if (resetting) {
  ctx.fillStyle = "rgba(5,8,7,0.78)";
  roundRect(w * 0.5 - 132, h * 0.36, 264, 78, 8);
  ctx.fill();
  ctx.strokeStyle = theme[2];
  ctx.lineWidth = 2;
  ctx.stroke();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 20px system-ui";
  ctx.textAlign = "center";
  ctx.fillText(`RESET ${Math.ceil(raceState.resetTimer)}`, w * 0.5, h * 0.36 + 32);
  ctx.fillStyle = "#a9bbb5";
  ctx.font = "800 12px system-ui";
  ctx.fillText(raceState.resetReason || "Vehicle recovery", w * 0.5, h * 0.36 + 55);
}
ctx.restore();
}

function drawRaceGoalIntro(w, h, theme) {
  if (!raceState.active || raceState.goalIntroTimer <= 0) return;
  const phoneMode = phoneGraphicsActive();
  const total = 5.8;
  const fadeIn = Math.min(1, (total - raceState.goalIntroTimer) / 0.45);
  const fadeOut = Math.min(1, raceState.goalIntroTimer / 0.85);
  const alpha = Math.max(0, Math.min(1, fadeIn, fadeOut));
  if (alpha <= 0) return;
  const route = routeWorldInfo(selectedRace && selectedRace.place ? selectedRace.place : "city");
  const panelW = Math.min(phoneMode ? 390 : 560, w - 28);
  const panelH = phoneMode ? 92 : 108;
  const x = (w - panelW) / 2;
  const y = Math.max(phoneMode ? 48 : 72, h * (phoneMode ? 0.15 : 0.18));
  ctx.save();
  ctx.globalAlpha = alpha;
  ctx.fillStyle = "rgba(5,8,7,0.8)";
  roundRect(x, y, panelW, panelH, 8);
  ctx.fill();
  ctx.strokeStyle = `${theme[1]}aa`;
  ctx.lineWidth = 2;
  ctx.stroke();
  ctx.fillStyle = theme[1];
  ctx.font = `900 ${phoneMode ? 12 : 14}px system-ui`;
  ctx.textAlign = "center";
  ctx.fillText("RACE GOAL", w / 2, y + (phoneMode ? 22 : 26));
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `900 ${phoneMode ? 18 : 24}px system-ui`;
  const goal = selectedRace ? selectedRace.target : "Finish the race";
  ctx.fillText(goal, w / 2, y + (phoneMode ? 48 : 58));
  ctx.fillStyle = "#a9bbb5";
  ctx.font = `800 ${phoneMode ? 11 : 13}px system-ui`;
  ctx.fillText(`${route.country}: ${route.scene} | ${activeScenario().name} | Hideouts + shortcut branches`, w / 2, y + (phoneMode ? 72 : 84));
  ctx.restore();
}

function drawCinematicGrade(w, h, theme) {
  const speedLines = Math.max(0, (raceState.speed - 190) / 150);
  if (speedLines > 0) {
    ctx.save();
    ctx.globalAlpha = Math.min(0.16, speedLines * 0.14);
    const edgeRush = ctx.createLinearGradient(0, h * 0.42, 0, h);
    edgeRush.addColorStop(0, "rgba(255,255,255,0)");
    edgeRush.addColorStop(0.7, "rgba(244,251,248,0.13)");
    edgeRush.addColorStop(1, "rgba(255,255,255,0)");
  }
}
```

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```
;
    ctx.fillStyle = edgeRush;
    ctx.fillRect(0, h * 0.5, w, h * 0.5);
    ctx.restore();
}
const slipGlow = Math.min(0.24, (raceState.slip || 0) * 0.18 + (raceState.brakeHeat || 0) * 0.08);
if (slipGlow > 0.01 && raceState.active) {
    ctx.save();
    const gripWash = ctx.createLinearGradient(0, h * 0.55, 0, h);
    gripWash.addColorStop(0, "rgba(255,255,255,0)");
    gripWash.addColorStop(1, "rgba(244,251,248,${slipGlow})");
    ctx.fillStyle = gripWash;
    ctx.fillRect(0, h * 0.55, w, h * 0.45);
    ctx.restore();
}
const vignette = ctx.createRadialGradient(w * 0.5, h * 0.52, h * 0.22, w * 0.5, h * 0.52, h * 0.9);
vignette.addColorStop(0, "rgba(0,0,0,0)");
vignette.addColorStop(1, "rgba(0,0,0,0.58)");
ctx.fillStyle = vignette;
ctx.fillRect(0, 0, w, h);
}

function drawRealisticDrivingPass(w, h, theme) {
    const place = selectedRace && selectedRace.place ? selectedRace.place : "city";
    const speed = Math.max(0, raceState.speed);
    const horizon = cameraMode === "cockpit" ? h * 0.28 : cameraMode === "hood" ? h * 0.31 : h * 0.34;
    const webGLActive = useWebGLRenderer();
    ctx.save();

    const depthFog = ctx.createLinearGradient(0, horizon - h * 0.08, 0, horizon + h * 0.18);
    depthFog.addColorStop(0, "rgba(244,251,248,0.02)");
    depthFog.addColorStop(0.52, place === "desert" || place === "canyon" ? "rgba(205,178,142,0.08)" : "rgba(210,230,235,0.08)");
    depthFog.addColorStop(1, "rgba(5,8,7,0)");
    ctx.fillStyle = depthFog;
    ctx.fillRect(0, Math.max(0, horizon - h * 0.12), w, h * 0.36);
    if (webGLActive) {
        ctx.restore();
        return;
    }
    if (phoneCleanRoadActive()) {
        ctx.restore();
        return;
    }
}

const wetPlaces = ["city", "tokyo", "coast", "harbor", "rainforest"];
if (wetPlaces.includes(place)) {
    ctx.globalAlpha = place === "tokyo" ? 0.32 : 0.2;
    for (let i = 0; i < 16; i += 1) {
        const y = ((i * 47 + raceState.roadOffset * 0.72) % (h * 0.62)) + horizon;
        const t = Math.min(1, Math.max(0, (y - horizon) / (h - horizon)));
        const ribbonW = w * (0.12 + t * 0.5);
        const x = w * 0.5 + Math.sin(i * 1.7 + raceState.elapsed) * w * 0.04;
        const shine = ctx.createLinearGradient(x - ribbonW, y, x + ribbonW, y);
        shine.addColorStop(0, "rgba(244,251,248,0)");
        shine.addColorStop(0.5, "rgba(244,251,248,0.26)");
        shine.addColorStop(1, "rgba(244,251,248,0)");
        ctx.strokeStyle = shine;
        ctx.lineWidth = 1 + t * 5;
        ctx.beginPath();
        ctx.moveTo(x - ribbonW, y);
        ctx.quadraticCurveTo(x, y + 5 + t * 12, x + ribbonW, y + 2);
        ctx.stroke();
    }
    ctx.globalAlpha = 1;
}

if (raceState.active && speed > 35) {
    const carX = w / 2 + raceState.lane * laneWidth();
    const baseY = cameraMode === "cockpit" ? h * 0.76 : cameraMode === "hood" ? h * 0.9 : h * 0.86;
    const slipAlpha = Math.min(0.32, 0.05 + (raceState.slip || 0) * 0.22);
    ctx.strokeStyle = "rgba(8,10,10,${slipAlpha})";
    ctx.lineWidth = Math.max(2, w * 0.006);
    for (let side = -1; side <= 1; side += 2) {
        const tireX = carX + side * laneWidth() * 0.22;
        ctx.beginPath();
        ctx.moveTo(tireX, baseY);
        ctx.bezierCurveTo(
            tireX - raceState.lateralVelocity * 15,
            baseY + h * 0.06,
            tireX - raceState.steerAngle * 24,
            h,
            tireX - raceState.steerAngle * 48,
            h + 20
        );
        ctx.stroke();
    }
}

if (speed > 105) {
    const blur = Math.min(0.18, (speed - 105) / 900);
    ctx.globalAlpha = blur;
    ctx.strokeStyle = "rgba(244,251,248,0.7)";
    ctx.lineWidth = 2;
    for (let i = 0; i < 34; i += 1) {
        const side = i % 2 ? -1 : 1;
        const y = h * (0.36 + (i % 17) * 0.04);
        const x0 = w * 0.5 + side * w * (0.18 + (i % 5) * 0.05);
        ctx.beginPath();
        ctx.moveTo(x0, y);
        ctx.lineTo(x0 + side * w * 0.16, y + h * 0.018);
        ctx.stroke();
    }
    ctx.globalAlpha = 1;
}

if (["city", "tokyo", "airfield", "freight"].includes(place) || cameraMode === "cockpit") {
    const cone = ctx.createRadialGradient(w * 0.5, h * 0.78, w * 0.04, w * 0.5, h * 0.7, w * 0.42);
    cone.addColorStop(0, "rgba(255,246,202,0.16)");
    cone.addColorStop(1, "rgba(255,246,202,0)");
}
```

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```
    ctx.fillStyle = cone;
    ctx.fillRect(0, horizon, w, h - horizon);
}

if (place === "desert" || place === "canyon" || place === "freight") {
    ctx.globalAlpha = Math.min(0.28, 0.08 + speed / 980);
    ctx.fillStyle = "rgba(255,183,74,0.22)";
    for (let i = 0; i < 26; i += 1) {
        const x = (i * 79 + raceState.roadOffset * 0.11) % w;
        const y = h * 0.38 + ((i * 31 + raceState.elapsed * 55) % (h * 0.5));
        ctx.beginPath();
        ctx.ellipse(x, y, 18 + (i % 4) * 8, 3 + (i % 3), 0.15, 0, Math.PI * 2);
        ctx.fill();
    }
    ctx.globalAlpha = 1;
}

if (place === "snow") {
    ctx.globalAlpha = 0.34;
    ctx.fillStyle = "rgba(244,251,248,0.75)";
    for (let i = 0; i < 46; i += 1) {
        const x = (i * 61 + raceState.elapsed * 34) % w;
        const y = (i * 43 + raceState.elapsed * 52 + raceState.roadOffset * 0.05) % h;
        ctx.fillRect(x, y, 2, 2);
    }
    ctx.globalAlpha = 1;
}

if (place === "coast" || place === "desert") {
    const sunX = place === "coast" ? w * 0.18 : w * 0.78;
    const sunY = h * 0.16;
    const flare = ctx.createRadialGradient(sunX, sunY, 0, sunX, sunY, w * 0.28);
    flare.addColorStop(0, "rgba(255,226,150,0.35)");
    flare.addColorStop(1, "rgba(255,226,150,0)");
    ctx.fillStyle = flare;
    ctx.fillRect(0, 0, w, h * 0.48);
}

ctx.restore();
}

function drawVehicle(x, y, width, height, color, player, police = false, vehicleType = "car") {
    ctx.save();
    ctx.translate(x, y);
    drawRoadContactShadow(width, height, vehicleType, 1);

    if (drawPhoneAssetVehicleSprite(width, height, color, vehicleType, police, player ? raceState.damage || 0 : 0, true)) {
        if (player) {
            ctx.save();
            ctx.translate(0, -height * 0.18);
            drawLicensePlate(width * 0.78, height * 0.78, driverPlate(activeProfile));
            ctx.restore();
        }
        ctx.restore();
        return;
    }

    ctx.translate(0, -height * 0.58);

    if (!police && vehicleType !== "car") {
        drawSpecialVehicleSilhouette(width, height, color, vehicleType, player);
        if (player) drawLicensePlate(width * 0.82, height * 0.82, driverPlate(activeProfile));
        ctx.restore();
        return;
    }

    const tireW = width * 0.17;
    const tireH = height * 0.3;
    ctx.fillStyle = "#050807";
    roundRect(-width * 0.56, -height * 0.29, tireW, tireH, 5);
    ctx.fill();
    roundRect(width * 0.39, -height * 0.29, tireW, tireH, 5);
    ctx.fill();
    roundRect(-width * 0.56, height * 0.14, tireW, tireH, 5);
    ctx.fill();
    roundRect(width * 0.39, height * 0.14, tireW, tireH, 5);
    ctx.fill();
    drawWheelRim(-width * 0.475, -height * 0.14, width * 0.07, height * 0.12);
    drawWheelRim(width * 0.475, -height * 0.14, width * 0.07, height * 0.12);
    drawWheelRim(-width * 0.475, height * 0.26, width * 0.07, height * 0.12, "rgba(255,209,102,0.24)");
    drawWheelRim(width * 0.475, height * 0.26, width * 0.07, height * 0.12, "rgba(255,209,102,0.24)");

    const body = ctx.createLinearGradient(-width * 0.48, -height * 0.52, width * 0.48, height * 0.52);
    body.addColorStop(0, shade(color, 42));
    body.addColorStop(0.2, color);
    body.addColorStop(0.54, shade(color, -18));
    body.addColorStop(1, shade(color, -42));
    ctx.fillStyle = police ? "#f0f2f0" : body;
    ctx.beginPath();
    ctx.moveTo(-width * 0.26, -height * 0.54);
    ctx.lineTo(width * 0.26, -height * 0.54);
    ctx.quadraticCurveTo(width * 0.51, -height * 0.48, width * 0.5, -height * 0.2);
    ctx.lineTo(width * 0.45, height * 0.38);
    ctx.quadraticCurveTo(width * 0.28, height * 0.57, 0, height * 0.58);
    ctx.quadraticCurveTo(-width * 0.28, height * 0.57, -width * 0.45, height * 0.38);
    ctx.lineTo(-width * 0.5, -height * 0.2);
    ctx.quadraticCurveTo(-width * 0.51, -height * 0.48, -width * 0.26, -height * 0.54);
    ctx.closePath();
    ctx.fill();
    ctx.strokeStyle = "rgba(255,255,255,0.28)";
    ctx.lineWidth = Math.max(1, width * 0.018);
    ctx.stroke();
    drawVehicleTrimDetails(width, height, color, police);
    drawRealWorldVehicleSkinDetails(width, height, color, vehicleType, player ? raceState.damage || 0 : 0, police);
    if (player) drawLicensePlate(width, height, driverPlate(activeProfile));

    ctx.fillStyle = police ? "#111817" : shade(color, -36);
    ctx.globalAlpha = 0.64;
    ctx.begi
```

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```
nPath();
ctx.moveTo(-width * 0.38, -height * 0.24);
ctx.lineTo(-width * 0.22, -height * 0.48);
ctx.lineTo(width * 0.22, -height * 0.48);
ctx.lineTo(width * 0.38, -height * 0.24);
ctx.closePath();
ctx.fill();
ctx.globalAlpha = 1;

if (police) {
  ctx.fillStyle = "#111817";
  ctx.fillRect(-width * 0.34, -height * 0.16, width * 0.68, height * 0.11);
  ctx.fillStyle = "#f4fbf8";
  ctx.font = `${Math.max(8, width * 0.14)}px system-ui`;
  ctx.textAlign = "center";
  ctx.fillText("POLICE", 0, height * 0.24);
}

const glass = ctx.createLinearGradient(0, -height * 0.42, 0, height * 0.24);
glass.addColorStop(0, "rgba(220,244,255,0.58)");
glass.addColorStop(0.45, "rgba(43,66,70,0.84)");
glass.addColorStop(1, "rgba(7,11,12,0.92)");
ctx.fillStyle = glass;
roundRect(-width * 0.31, -height * 0.39, width * 0.62, height * 0.21, 8);
ctx.fill();
roundRect(-width * 0.29, -height * 0.11, width * 0.58, height * 0.32, 10);
ctx.fill();
ctx.strokeStyle = "rgba(244,251,248,0.28)";
ctx.lineWidth = Math.max(1, width * 0.015);
ctx.stroke();
ctx.fillStyle = player ? "#dce8ef" : "#e6d69b";
ctx.beginPath();
ctx.arc(0, -height * 0.02, width * 0.12, 0, Math.PI * 2);
ctx.fill();
ctx.fillStyle = "#050807";
roundRect(-width * 0.09, -height * 0.04, width * 0.18, height * 0.045, 2);
ctx.fill();
ctx.strokeStyle = "rgba(255,255,255,0.18)";
ctx.beginPath();
ctx.arc(0, -height * 0.02, width * 0.12, Math.PI * 1.04, Math.PI * 1.96);
ctx.stroke();

ctx.fillStyle = "rgba(255,248,214,0.94)";
ctx.beginPath();
ctx.moveTo(-width * 0.36, -height * 0.49);
ctx.lineTo(-width * 0.11, -height * 0.45);
ctx.lineTo(-width * 0.24, -height * 0.36);
ctx.closePath();
ctx.moveTo(width * 0.36, -height * 0.49);
ctx.lineTo(width * 0.11, -height * 0.45);
ctx.lineTo(width * 0.24, -height * 0.36);
ctx.closePath();
ctx.fill();
ctx.globalAlpha = 0.18;
ctx.fillStyle = "fff8d6";
ctx.beginPath();
ctx.ellipse(-width * 0.25, -height * 0.42, width * 0.28, height * 0.1, -0.28, 0, Math.PI * 2);
ctx.ellipse(width * 0.25, -height * 0.42, width * 0.28, height * 0.1, 0.28, 0, Math.PI * 2);
ctx.fill();
ctx.globalAlpha = 1;

ctx.fillStyle = "fff3348";
roundRect(-width * 0.32, height * 0.42, width * 0.17, height * 0.055, 3);
ctx.fill();
roundRect(width * 0.15, height * 0.42, width * 0.17, height * 0.055, 3);
ctx.fill();
if (police) {
  const flash = Math.sin(raceState.elapsed * 18) > 0;
  ctx.fillStyle = flash ? "fff3348" : "#46d9ff";
  roundRect(-width * 0.21, -height * 0.25, width * 0.42, height * 0.055, 3);
  ctx.fill();
  ctx.globalAlpha = 0.35;
  ctx.fillStyle = flash ? "fff3348" : "#46d9ff";
  ctx.beginPath();
  ctx.arc(0, -height * 0.25, width * 0.45, 0, Math.PI * 2);
  ctx.fill();
  ctx.globalAlpha = 1;
}
ctx.strokeStyle = "rgba(5,8,7,0.5)";
ctx.lineWidth = Math.max(1.5, width * 0.02);
ctx.beginPath();
ctx.moveTo(0, -height * 0.51);
ctx.lineTo(0, height * 0.5);
ctx.moveTo(-width * 0.41, -height * 0.16);
ctx.lineTo(width * 0.41, -height * 0.16);
ctx.moveTo(-width * 0.37, height * 0.28);
ctx.lineTo(width * 0.37, height * 0.28);
ctx.stroke();
ctx.fillStyle = "rgba(244,251,248,0.22)";
ctx.fillRect(-width * 0.34, height * 0.19, width * 0.68, height * 0.026);

ctx.strokeStyle = "rgba(255,255,255,0.22)";
ctx.lineWidth = Math.max(1, width * 0.012);
ctx.beginPath();
ctx.moveTo(-width * 0.43, -height * 0.13);
ctx.quadraticCurveTo(-width * 0.56, height * 0.02, -width * 0.42, height * 0.33);
ctx.moveTo(width * 0.43, -height * 0.13);
ctx.quadraticCurveTo(width * 0.56, height * 0.02, width * 0.42, height * 0.33);
ctx.stroke();
if (player) drawVehicleDamageMarks(width, height, raceState.damage || 0);
ctx.restore();
}

function shade(hex, amount) {
  const raw = hex.replace("#", "");
  const num = parseInt(raw.length === 3 ? raw.split("").map((c) => c + c).join("") : raw, 16);
  const r = Math.max(0, Math.min(255, (num >> 16) + amount));
  const g = Math.max(0, Math.min(255, ((num >> 8) & 255) + amount));
  const b = Math.max(0, Math.min(255, (num & 255) + amount));
  return `rgb(${r}, ${g}, ${b})`;
}

function roundRect(x, y, wi
```

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```
dth, height, radius) {
  ctx.beginPath();
  ctx.moveTo(x + radius, y);
  ctx.lineTo(x + width - radius, y);
  ctx.quadraticCurveTo(x + width, y, x + width, y + radius);
  ctx.lineTo(x + width, y + height - radius);
  ctx.quadraticCurveTo(x + width, y + height, x + width - radius, y + height);
  ctx.lineTo(x + radius, y + height);
  ctx.quadraticCurveTo(x, y + height, x, y + height - radius);
  ctx.lineTo(x, y + radius);
  ctx.quadraticCurveTo(x, y, x + radius, y);
  ctx.closePath();
}

function drawParticles() {
  raceState.particles.forEach((p) => {
    const fade = Math.max(0, p.life / (p.maxLife || 0.7));
    ctx.globalAlpha = fade;
    ctx.fillStyle = p.color;
    ctx.beginPath();
    ctx.arc(p.x, p.y, (p.radius || 4) * (1.1 - fade * 0.35), 0, Math.PI * 2);
    ctx.fill();
    ctx.globalAlpha = 1;
  });
}

function drawAttract(w, h) {
  ctx.save();
  ctx.textAlign = "center";
  ctx.fillStyle = "rgba(5,8,7,0.38)";
  roundRect(w / 2 - 330, h * 0.36 - 54, 660, 118, 8);
  ctx.fill();
  ctx.fillStyle = "#f4fbf8";
  ctx.font = "900 34px system-ui";
  ctx.fillText("Press Play. Hit The Street.", w / 2, h * 0.36);
  ctx.fillStyle = "#a9bbb5";
  ctx.font = "700 16px system-ui";
  ctx.fillText("Performance cars, police heat, and cinematic routes built for replay.", w / 2, h * 0.36 + 34);
  ctx.restore();
}

function drawPause(w, h) {
  ctx.fillStyle = "rgba(0,0,0,0.56)";
  ctx.fillRect(0, 0, w, h);
  ctx.fillStyle = "#f4fbf8";
  ctx.textAlign = "center";
  ctx.font = "900 48px system-ui";
  ctx.fillText("Paused", w / 2, h / 2);
}

function syncViewportSize() {
  const viewport = window.visualViewport;
  const width = Math.max(320, Math.floor(viewport ? viewport.width : window.innerWidth));
  const height = Math.max(240, Math.floor(viewport ? viewport.height : window.innerHeight));
  document.documentElement.style.setProperty("--vw", `${width}px`);
  document.documentElement.style.setProperty("--vh", `${height}px`);
}

function fitCanvas() {
  syncViewportSize();
  const rect = canvas.getBoundingClientRect();
  const racingViewport = document.body.classList.contains("race-live");
  const minWidth = racingViewport ? 320 : 640;
  const minHeight = racingViewport ? 240 : 420;
  const width = Math.max(minWidth, Math.floor(rect.width));
  const height = Math.max(minHeight, Math.floor(rect.height));
  canvas.width = width;
  canvas.height = height;
  if (glCanvas) {
    glCanvas.width = width;
    glCanvas.height = height;
  }
  if (webglRenderer) webglRenderer.resize(width, height);
}

function bindEvents() {
  $$("#.age-card").forEach((btn) => {
    btn.addEventListener("click", () => {
      selectedAge = btn.dataset.age;
      $$("#.age-card").forEach((item) => item.classList.toggle("selected", item === btn));
    });
  });
  $("#quickPlay").addEventListener("click", () => enterProfile(true));
  $("#startProfile").addEventListener("click", () => enterProfile(false));
  $("#installApp").addEventListener("click", installApp);
  $("#resetProfiles").addEventListener("click", () => {
    if (!confirm("Clear all local Velocity Vault profiles on this device?")) return;
    profiles = {};
    saveProfiles();
    clearSavedRace();
    renderSavedRaceButton();
    showToast("Local profiles cleared.");
  });
  $("#logoutBtn").addEventListener("click", () => {
    activeProfile = null;
    raceState.active = false;
    $("#modeChip").textContent = "Select Profile";
    $("#missionChip").textContent = "Mission Ready";
    updateHud();
    showView("login");
  });
  $$(".tab").forEach((btn) => {
    btn.addEventListener("click", () => {
      tab = btn.dataset.tab;
      $$(".tab").forEach((item) => item.classList.toggle("active", item === btn));
      $$(".tab-panel").forEach((panel) => panel.classList.toggle("active", panel.id === `${tab}Tab`));
    });
  });
  $("#launchRace").addEventListener("click", launchRace);
  $("#resumeRace").addEventListener("click", resumeSavedRace);
  $("#saveRace").addEventListener("click", saveRace);
  $("#resetCar").addEventListener("click", manualResetVehicle);
  $("#endRace").addEventListener("click", () => endRace(true));
}
```

```
$("#toGarage").addEventListener("click", () => {  
  showView("garage");  
  renderHub(  

```

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```
});
});
$("#nextRace").addEventListener("click", () => {
  const idx = races.findIndex((race) => race.id === selectedRace.id);
  selectedRace = races[Math.min(races.length - 1, idx + 1)];
  showView("garage");
  renderHub();
});
$("#pauseBtn").addEventListener("click", () => {
  input.paused = !input.paused;
  $("#pauseBtn").textContent = input.paused ? "Play" : "Pause";
});
$("#refreshDirector").addEventListener("click", () => {
  renderDirector();
  renderRaces();
  renderUpgrades();
  showToast("AI tuning refreshed for this driver.");
});
$$(".camera-btn").forEach((btn) => {
  btn.addEventListener("click", () => setCameraMode(btn.dataset.camera));
});
$$(".renderer-btn").forEach((btn) => {
  btn.addEventListener("click", () => setRendererMode(btn.dataset.renderer));
});
bindHold($("#leftBtn"), "left");
bindHold($("#rightBtn"), "right");
bindHold($("#gasBtn"), "gas");
bindHold($("#brakeBtn"), "brake");
bindHold($("#boostBtn"), "boost");
$$(".mobile-control").forEach((button) => bindMobileControl(button));
bindDriveStickPointerControl();
bindFloatingPhoneJoystick();
const mobileLayer = $(".mobile-drive-controls");
mobileLayer.addEventListener("touchstart", handleMobileTouchStart, { passive: false });
mobileLayer.addEventListener("touchmove", handleMobileTouchMove, { passive: false });
mobileLayer.addEventListener("touchend", handleMobileTouchEnd, { passive: false });
mobileLayer.addEventListener("touchcancel", handleMobileTouchEnd, { passive: false });
window.addEventListener("keydown", (event) => setKey(event, true));
window.addEventListener("keyup", (event) => setKey(event, false));
const resizeGame = () => {
  syncViewportSize();
  fitCanvas();
};
window.addEventListener("resize", resizeGame);
window.addEventListener("orientationchange", () => setTimeout(resizeGame, 180));
if (window.visualViewport) {
  window.visualViewport.addEventListener("resize", resizeGame);
  window.visualViewport.addEventListener("scroll", resizeGame);
}
window.addEventListener("gamepadconnected", (event) => setController(event.gamepad));
window.addEventListener("gamepaddisconnected", () => clearController());
window.addEventListener("pagehide", persistCurrentProgress);
document.addEventListener("visibilitychange", () => {
  if (document.visibilityState === "hidden") persistCurrentProgress();
});
window.addEventListener("beforeinstallprompt", (event) => {
  event.preventDefault();
  deferredInstallPrompt = event;
  showToast("App install is available.");
});
window.addEventListener("appinstalled", () => {
  deferredInstallPrompt = null;
  showToast("Velocity Vault installed.");
});
});

function persistCurrentProgress() {
  if (raceState.active) {
    saveRace(true);
  } else {
    saveProfiles();
  }
}

function bindHold(button, key) {
  const start = (event) => {
    event.preventDefault();
    input[key] = true;
    if (button.setPointerCapture && event.pointerId !== undefined) {
      try {
        button.setPointerCapture(event.pointerId);
      } catch {
        // Some browsers reject capture after synthetic pointer events.
      }
    }
  };
  const stop = (event) => {
    if (event) event.preventDefault();
    input[key] = false;
    if (button.releasePointerCapture && event && event.pointerId !== undefined) {
      try {
        button.releasePointerCapture(event.pointerId);
      } catch {
        // Capture may already be released by the browser.
      }
    }
  };
  button.addEventListener("pointerdown", start);
  button.addEventListener("pointerup", stop);
  button.addEventListener("pointercancel", stop);
  button.addEventListener("lostpointercapture", stop);
  button.addEventListener("contextmenu", (event) => event.preventDefault());
}

function bindMobileControl(button) {
  const stickButton = button.closest("&& button.closest(\".drive-stick\") ? button.closest(\".drive-stick\") : null;
  if (stickButton) {
    bindDriveStickButtonFallback(button, stickButton);
    return;
  }
  const control = button.dataset.control;
  const start = (event) => {
    if (event.pointerType === "touch" && mobileTouchState.usingTouc
```


game.js continued 161/162

```
hEvents) return;
    event.preventDefault();
    startAudio();
    if (audioSystem && audioSystem.ctx.state === "suspended") audioSystem.ctx.resume();
    if (control === "size") {
        setTouchControlSize(touchControlSize === "mini" ? "full" : "mini");
        return;
    }
    if (control === "mode") {
        setTouchDriveMode(touchDriveMode === "toggle" ? "hold" : "toggle");
        return;
    }
    if (touchDriveMode === "toggle") {
        toggleDriveInput(control);
        updateRaceUi();
        return;
    }
    setDriveInput(control, true);
    updateRaceUi();
    if (button.setPointerCapture && event.pointerId !== undefined) {
        try {
            button.setPointerCapture(event.pointerId);
        } catch {
            // Some browsers reject capture after synthetic pointer events.
        }
    }
};
const stop = (event) => {
    if (event && event.pointerType === "touch" && mobileTouchState.usingTouchEvents) return;
    if (event) event.preventDefault();
    if (touchDriveMode === "toggle" || control === "mode" || control === "size") return;
    setDriveInput(control, false);
    updateRaceUi();
    if (button.releasePointerCapture && event && event.pointerId !== undefined) {
        try {
            button.releasePointerCapture(event.pointerId);
        } catch {
            // Capture may already be released by the browser.
        }
    }
};
button.addEventListener("pointerdown", start);
button.addEventListener("pointerup", stop);
button.addEventListener("pointercancel", stop);
button.addEventListener("lostpointercapture", stop);
button.addEventListener("contextmenu", (event) => event.preventDefault());
}

function directStickControl(control) {
    if (control === "gas") return "stick:gas";
    if (control === "brake") return "stick:brake";
    if (control === "left") return "stick:left:steer=-1";
    if (control === "right") return "stick:right:steer=1";
    return "stick:neutral";
}

function bindDriveStickButtonFallback(button, stick) {
    const pointerKey = `drive-stick-button-${button.dataset.control} || "neutral"`;
    const readControl = (event) => driveStickControlAt(stick, event.clientX, event.clientY) ||
    directStickControl(button.dataset.control);
    const apply = (event) => {
        if (event.pointerType === "touch" && mobileTouchState.usingTouchEvents) return;
        if (useFloatingStickControls()) return;
        if (touchControlSize !== "mini") return;
        event.preventDefault();
        startAudio();
        if (audioSystem && audioSystem.ctx.state === "suspended") audioSystem.ctx.resume();
        const control = readControl(event);
        if (touchDriveMode === "toggle") mobileTouchState.latchedStickControl = control.includes("neutral") ? "" : control;
        mobileTouchState.active.set(pointerKey, control);
        applyMobileTouchSnapshot();
        if (button.setPointerCapture && event.pointerId !== undefined) {
            try {
                button.setPointerCapture(event.pointerId);
            } catch {
                // Some browsers reject capture after synthetic pointer events.
            }
        }
    };
    const stop = (event) => {
        if (event && event.pointerType === "touch" && mobileTouchState.usingTouchEvents) return;
        if (event) event.preventDefault();
        if (touchDriveMode !== "toggle") mobileTouchState.active.delete(pointerKey);
        applyMobileTouchSnapshot();
        if (button.releasePointerCapture && event && event.pointerId !== undefined) {
            try {
                button.releasePointerCapture(event.pointerId);
            } catch {
                // Capture may already be released by the browser.
            }
        }
    };
    button.addEventListener("pointerdown", apply);
    button.addEventListener("pointermove", apply);
    button.addEventListener("pointerup", stop);
    button.addEventListener("pointercancel", stop);
    button.addEventListener("lostpointercapture", stop);
    button.addEventListener("contextmenu", (event) => event.preventDefault());
}

function setController(gamepad) {
    controllerState.connected = true;
    controllerState.name = gamepad.id || "Gamepad";
    controllerState.index = gamepad.index;
    $("#controllerChip").textContent = "Controller: On";
    $("#controllerName").textContent = controllerState.name;
    showToast("Controller connected.");
}

function clearController() {
    contr
```

game.js continued 162/162

```
ollerState.connected = false;
controllerState.name = "";
controllerState.index = null;
input.gamepadSteer = 0;
input.gamepadGas = false;
input.gamepadBrake = false;
input.gamepadBoost = false;
input.gamepadPauseHeld = false;
input.gamepadResetHeld = false;
$("#controllerChip").textContent = "Controller: Off";
$("#controllerName").textContent = "Bluetooth controller ready";
showToast("Controller disconnected.");
}

function pollGamepad() {
  if (!navigator.getGamepads) return;
  const pads = navigator.getGamepads();
  const pad = controllerState.index !== null ? pads[controllerState.index] : Array.from(pads).find(Boolean);
  if (!pad) {
    if (controllerState.connected) clearController();
    return;
  }
  if (!controllerState.connected) setController(pad);
  const stick = Math.abs(pad.axes[0] || 0) > 0.18 ? pad.axes[0] : 0;
  const dpadLeft = pad.buttons[14] && pad.buttons[14].pressed;
  const dpadRight = pad.buttons[15] && pad.buttons[15].pressed;
  input.gamepadSteer = dpadLeft ? -1 : dpadRight ? 1 : stick;
  input.gamepadGas = Boolean((pad.buttons[7] && pad.buttons[7].value > 0.18) || (pad.buttons[0] && pad.buttons[0].pressed));
  input.gamepadBrake = Boolean((pad.buttons[6] && pad.buttons[6].value > 0.18) || (pad.buttons[2] && pad.buttons[2].pressed));
  input.gamepadBoost = Boolean((pad.buttons[1] && pad.buttons[1].pressed) || (pad.buttons[5] && pad.buttons[5].pressed));
  const pausePressed = Boolean((pad.buttons[9].pressed) || (pad.buttons[8] && pad.buttons[8].pressed));
  if (pausePressed && !input.gamepadPauseHeld) {
    input.paused = !input.paused;
    $("#pauseBtn").textContent = input.paused ? "Play" : "Pause";
  }
  input.gamepadPauseHeld = pausePressed;
  const resetPressed = Boolean(pad.buttons[3] && pad.buttons[3].pressed);
  if (resetPressed && !input.gamepadResetHeld) manualResetVehicle();
  input.gamepadResetHeld = resetPressed;
}

function setKey(event, down) {
  if (event.key === "ArrowLeft" || event.key.toLowerCase() === "a") input.left = down;
  if (event.key === "ArrowRight" || event.key.toLowerCase() === "d") input.right = down;
  if (event.key === "ArrowUp" || event.key.toLowerCase() === "w") input.gas = down;
  if (event.key === "ArrowDown" || event.key.toLowerCase() === "s") input.brake = down;
  if (event.key === " ") input.boost = down;
  if (event.key.toLowerCase() === "p" && down) {
    input.paused = !input.paused;
    $("#pauseBtn").textContent = input.paused ? "Play" : "Pause";
  }
  if (event.key.toLowerCase() === "r" && down) manualResetVehicle();
}

bindEvents();
syncViewportSize();
fitCanvas();
updateHud();
applyPhoneGraphicsDefaults();
setCameraMode(cameraMode, true);
setTouchDriveMode(touchDriveMode, true);
setTouchControlSize(touchControlSize, true);
initWebGLRenderer();
setRendererMode(rendererMode, true);
registerOfflineApp();
drawFrame();
```

phone-asset-pipeline.js

```
"use strict";

(function () {
  const spriteCache = new Map();
  const textureCache = new Map();

  function makeCanvas(width, height) {
    const canvas = document.createElement("canvas");
    canvas.width = width;
    canvas.height = height;
    return canvas;
  }

  function hexToRgb(hex) {
    if (!hex || typeof hex !== "string") return [70, 217, 255];
    const clean = hex.replace("#", "");
    const full = clean.length === 3 ? clean.split("").map((c) => c + c).join("") : clean;
    const value = parseInt(full, 16);
    return [(value >> 16) & 255, (value >> 8) & 255, value & 255];
  }

  function shade(hex, amount) {
    const [r, g, b] = hexToRgb(hex);
    return `rgb(${Math.max(0, Math.min(255, r + amount))}, ${Math.max(0, Math.min(255, g + amount))}, ${Math.max(0, Math.min(255, b + amount))})`;
  }

  function rgba(hex, alpha) {
    const [r, g, b] = hexToRgb(hex);
    return `rgba(${r}, ${g}, ${b}, ${alpha})`;
  }

  function isAirType(type) {
    return ["airplane", "helicopter", "fighterjet", "bomber", "drone"].includes(type);
  }

  function isWaterType(type) {
    return type === "boat";
  }

  function roundRect(ctx, x, y, width, height, radius) {
    const r = Math.min(radius, Math.abs(width) / 2, Math.abs(height) / 2);
    ctx.beginPath();
    ctx.moveTo(x + r, y);
    ctx.lineTo(x + width - r, y);
    ctx.quadraticCurveTo(x + width, y, x + width, y + r);
    ctx.lineTo(x + width, y + height - r);
    ctx.quadraticCurveTo(x + width, y + height, x + width - r, y + height);
    ctx.lineTo(x + r, y + height);
    ctx.quadraticCurveTo(x, y + height, x, y + height - r);
    ctx.lineTo(x, y + r);
    ctx.quadraticCurveTo(x, y, x + r, y);
    ctx.closePath();
  }

  function pathVehicleBody(ctx, w, h, type) {
    const cx = w / 2;
    const top = h * 0.16;
    const bottom = h * 0.86;
    const narrow = type === "f1" || type === "prototype";
    const wide = type === "truck" || type === "monster" || type === "tank" || type === "semi";
    const upper = narrow ? w * 0.21 : wide ? w * 0.31 : w * 0.24;
    const shoulder = narrow ? w * 0.34 : wide ? w * 0.43 : w * 0.38;
    const lower = narrow ? w * 0.27 : wide ? w * 0.4 : w * 0.34;
    ctx.beginPath();
    ctx.moveTo(cx - upper, top);
    ctx.lineTo(cx + upper, top);
    ctx.quadraticCurveTo(cx + shoulder, h * 0.22, cx + shoulder, h * 0.42);
    ctx.lineTo(cx + lower, bottom);
    ctx.quadraticCurveTo(cx + lower * 0.72, h * 0.93, cx, h * 0.94);
    ctx.quadraticCurveTo(cx - lower * 0.72, h * 0.93, cx - lower, bottom);
    ctx.lineTo(cx - shoulder, h * 0.42);
    ctx.quadraticCurveTo(cx - shoulder, h * 0.22, cx - upper, top);
    ctx.closePath();
  }

  function drawWheel(ctx, x, y, rx, ry, accent) {
    ctx.save();
    ctx.fillStyle = "rgba(3,5,5,0.96)";
    ctx.beginPath();
    ctx.ellipse(x, y, rx, ry, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.28)";
    ctx.lineWidth = Math.max(2, rx * 0.14);
    ctx.stroke();
    ctx.strokeStyle = accent;
    ctx.lineWidth = Math.max(1, rx * 0.1);
    ctx.beginPath();
    ctx.moveTo(x - rx * 0.62, y);
    ctx.lineTo(x + rx * 0.62, y);
    ctx.moveTo(x, y - ry * 0.62);
    ctx.lineTo(x, y + ry * 0.62);
    ctx.moveTo(x - rx * 0.42, y - ry * 0.42);
    ctx.lineTo(x + rx * 0.42, y + ry * 0.42);
    ctx.moveTo(x + rx * 0.42, y - ry * 0.42);
    ctx.lineTo(x - rx * 0.42, y + ry * 0.42);
    ctx.stroke();
    ctx.restore();
  }

  function drawPaintGrain(ctx, w, h, color) {
    ctx.save();
    ctx.globalAlpha = 0.15;
    for (let i = 0; i < 95; i += 1) {
      const x = (i * 37) % w;
      const y = (i * 61) % h;
      ctx.strokeStyle = i % 2 ? "rgba(255,255,255,0.28)" : rgba(color, 0.35);
      ctx.lineWidth = 1;
      ctx.beginPath();
      ctx.moveTo(x, y);
      ctx.lineTo(x + 18 + (i % 5) * 4, y + 3);
      ctx.stroke();
    }
  }
})
```

```
    }  
    ctx.restore();  
  }  
  
function drawDamage(ctx, w, h, damage) {  
  if (!damage || damage < 12) return;  
  ctx.save();  
  const count = Math.min(9, Math.floor(damage / 12));  
  ctx.strokeStyle = "rgba(3,5,5,0.72)";  
  ctx.lineWidth = Math.max(2, w * 0.012);  
  ctx.lineCap = "round";  
  for (let i = 0; i < count; i += 1) {  
    const side = i % 2 ? -1 : 1;  
  }  
}
```

phone-asset-pipeline.js continued 2/9

```
const x = w * 0.5 + side * w * (0.08 + (i % 3) * 0.055);
const y = h * (0.28 + (i % 5) * 0.1);
ctx.beginPath();
ctx.moveTo(x, y);
ctx.lineTo(x + side * w * 0.11, y + h * (0.02 + (i % 2) * 0.025));
ctx.stroke();
}
if (damage > 58) {
  ctx.fillStyle = "rgba(255,91,107,0.46)";
  roundRect(ctx, w * 0.38, h * 0.18, w * 0.2, h * 0.045, 5);
  ctx.fill();
}
ctx.restore();
}

function drawBodyRealismDetails(ctx, w, h, type, color, police) {
  const paint = police ? "#f2f5f2" : color;
  ctx.save();
  pathVehicleBody(ctx, w, h, type === "semi" ? "truck" : type);
  ctx.clip();

  const sideShade = ctx.createLinearGradient(w * 0.12, 0, w * 0.88, 0);
  sideShade.addColorStop(0, "rgba(0,0,0,0.48)");
  sideShade.addColorStop(0.22, "rgba(255,255,255,0.08)");
  sideShade.addColorStop(0.5, "rgba(255,255,255,0.18)");
  sideShade.addColorStop(0.78, "rgba(255,255,255,0.06)");
  sideShade.addColorStop(1, "rgba(0,0,0,0.46)");
  ctx.fillStyle = sideShade;
  ctx.fillRect(0, 0, w, h);

  ctx.globalAlpha = 0.42;
  ctx.strokeStyle = "rgba(255,255,255,0.7)";
  ctx.lineWidth = Math.max(2, w * 0.012);
  ctx.beginPath();
  ctx.moveTo(w * 0.36, h * 0.18);
  ctx.bezierCurveTo(w * 0.42, h * 0.34, w * 0.33, h * 0.56, w * 0.28, h * 0.82);
  ctx.moveTo(w * 0.64, h * 0.18);
  ctx.bezierCurveTo(w * 0.58, h * 0.34, w * 0.67, h * 0.56, w * 0.72, h * 0.82);
  ctx.stroke();

  ctx.globalAlpha = 0.22;
  ctx.strokeStyle = rgba(paint, 0.8);
  ctx.lineWidth = Math.max(1, w * 0.006);
  for (let i = 0; i < 18; i += 1) {
    const y = h * (0.18 + i * 0.038);
    ctx.beginPath();
    ctx.moveTo(w * 0.22, y);
    ctx.lineTo(w * 0.78, y + h * 0.012);
    ctx.stroke();
  }
  ctx.restore();

  ctx.save();
  ctx.strokeStyle = "rgba(3,5,5,0.68)";
  ctx.lineWidth = Math.max(2, w * 0.012);
  ctx.beginPath();
  ctx.moveTo(w * 0.26, h * 0.34);
  ctx.lineTo(w * 0.2, h * 0.78);
  ctx.moveTo(w * 0.74, h * 0.34);
  ctx.lineTo(w * 0.8, h * 0.78);
  ctx.moveTo(w * 0.37, h * 0.7);
  ctx.lineTo(w * 0.63, h * 0.7);
  ctx.stroke();
  ctx.fillStyle = "rgba(2,4,4,0.88)";
  roundRect(ctx, w * 0.24, h * 0.17, w * 0.1, h * 0.04, 5);
  ctx.fill();
  roundRect(ctx, w * 0.66, h * 0.17, w * 0.1, h * 0.04, 5);
  ctx.fill();
  ctx.restore();
}

function drawAirWaterRealismDetails(ctx, w, h, type) {
  ctx.save();
  ctx.globalAlpha = 0.38;
  ctx.strokeStyle = "rgba(255,255,255,0.78)";
  ctx.lineWidth = Math.max(2, w * 0.01);
  if (type === "airplane" || type === "fighterjet" || type === "bomber") {
    ctx.beginPath();
    ctx.moveTo(w * 0.5, h * 0.12);
    ctx.lineTo(w * 0.5, h * 0.86);
    ctx.moveTo(w * 0.18, h * 0.52);
    ctx.lineTo(w * 0.82, h * 0.52);
    ctx.stroke();
  } else if (type === "drone") {
    ctx.beginPath();
    ctx.moveTo(w * 0.24, h * 0.28);
    ctx.lineTo(w * 0.76, h * 0.72);
    ctx.moveTo(w * 0.76, h * 0.28);
    ctx.lineTo(w * 0.24, h * 0.72);
    ctx.moveTo(w * 0.5, h * 0.24);
    ctx.lineTo(w * 0.5, h * 0.78);
    ctx.stroke();
  } else if (type === "helicopter") {
    ctx.beginPath();
    ctx.moveTo(w * 0.5, h * 0.18);
    ctx.lineTo(w * 0.5, h * 0.84);
    ctx.moveTo(w * 0.28, h * 0.38);
    ctx.lineTo(w * 0.72, h * 0.38);
    ctx.stroke();
  } else {
    ctx.beginPath();
    ctx.moveTo(w * 0.5, h * 0.12);
    ctx.bezierCurveTo(w * 0.38, h * 0.38, w * 0.34, h * 0.66, w * 0.42, h * 0.86);
    ctx.moveTo(w * 0.5, h * 0.12);
    ctx.bezierCurveTo(w * 0.62, h * 0.38, w * 0.66, h * 0.66, w * 0.58, h * 0.86);
    ctx.stroke();
  }
  ctx.restore();
}

function drawSpriteContact(ctx, w, h, type) {
  const air = isAirType(type);
  const water = isWaterType(type);
```

```
const snow = type === "snowmobile";
ctx.save();
const baseY = h * (air ? 0.68 : water ? 0.78 : 0.84);
const contact = ctx.createRadialGradient(w * 0.5, baseY - h * 0.03, w * 0.06, w * 0.5, baseY, w * 0.48);
contact.addColorStop(0, air ? "rgba(0,0,0,0.22)" : "rgba(0,0,0,0.62)");
contact.addColorStop(1, "rgba(0,0,0,0)");
ctx.fillStyle = contact;
ctx.beginPa
```

phone-asset-pipeline.js continued 3/9

```
th();
ctx.ellipse(w * 0.5, baseY, w * 0.42, h * (air ? 0.045 : 0.075), 0, 0, Math.PI * 2);
ctx.fill();

if (!air && !water) {
  ctx.fillStyle = snow ? "rgba(244,251,248,0.42)" : "rgba(3,5,5,0.82)";
  ctx.beginPath();
  ctx.ellipse(w * 0.29, h * 0.4, w * 0.09, h * 0.025, 0, 0, Math.PI * 2);
  ctx.ellipse(w * 0.71, h * 0.4, w * 0.09, h * 0.025, 0, 0, Math.PI * 2);
  ctx.ellipse(w * 0.29, h * 0.8, w * 0.11, h * 0.034, 0, 0, Math.PI * 2);
  ctx.ellipse(w * 0.71, h * 0.8, w * 0.11, h * 0.034, 0, 0, Math.PI * 2);
  ctx.fill();
}

ctx.strokeStyle = water ? "rgba(70,217,255,0.48)" : snow ? "rgba(244,251,248,0.52)" : "rgba(3,5,5,0.5)";
ctx.lineWidth = Math.max(2, w * 0.012);
ctx.lineCap = "round";
ctx.beginPath();
ctx.moveTo(w * 0.3, baseY - h * 0.035);
ctx.lineTo(w * 0.18, baseY + h * 0.075);
ctx.moveTo(w * 0.7, baseY - h * 0.035);
ctx.lineTo(w * 0.82, baseY + h * 0.075);
ctx.stroke();
ctx.restore();
}

function drawCarLike(ctx, w, h, type, color, police, damage) {
  const paint = police ? "#f2f5f2" : color;
  const accent = police ? "rgba(70,217,255,0.45)" : rgba(color, 0.45);
  const wide = type === "truck" || type === "monster" || type === "tank";
  const narrow = type === "f1" || type === "prototype";
  const semi = type === "semi";
  const tractor = type === "tractor";

  const wheelRX = w * (tractor ? 0.09 : narrow ? 0.075 : wide ? 0.095 : 0.078);
  const wheelRY = h * (tractor ? 0.13 : narrow ? 0.08 : wide ? 0.115 : 0.105);
  const wheelX = w * (wide ? 0.2 : 0.18);
  drawWheel(ctx, w * 0.5 - wheelX, h * 0.38, wheelRX, wheelRY, accent);
  drawWheel(ctx, w * 0.5 + wheelX, h * 0.38, wheelRX, wheelRY, accent);
  drawWheel(ctx, w * 0.5 - wheelX, h * 0.76, wheelRX, wheelRY, "rgba(255,209,102,0.34)");
  drawWheel(ctx, w * 0.5 + wheelX, h * 0.76, wheelRX, wheelRY, "rgba(255,209,102,0.34)");

  if (semi) {
    const trailer = ctx.createLinearGradient(w * 0.18, h * 0.36, w * 0.82, h * 0.88);
    trailer.addColorStop(0, shade(paint, 32));
    trailer.addColorStop(0.55, shade(paint, -12));
    trailer.addColorStop(1, shade(paint, -42));
    ctx.fillStyle = trailer;
    roundRect(ctx, w * 0.18, h * 0.36, w * 0.64, h * 0.5, 14);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.22)";
    for (let i = 0; i < 3; i += 1) {
      roundRect(ctx, w * 0.24, h * (0.46 + i * 0.11), w * 0.52, h * 0.025, 4);
      ctx.fill();
    }
  }

  const body = ctx.createLinearGradient(w * 0.18, h * 0.14, w * 0.82, h * 0.9);
  body.addColorStop(0, shade(paint, 52));
  body.addColorStop(0.3, paint);
  body.addColorStop(0.68, shade(paint, -26));
  body.addColorStop(1, shade(paint, -58));
  ctx.fillStyle = body;
  pathVehicleBody(ctx, w, h, semi ? "truck" : type);
  ctx.fill();
  ctx.strokeStyle = "rgba(255,255,255,0.28)";
  ctx.lineWidth = Math.max(2, w * 0.01);
  ctx.stroke();
  drawBodyRealismDetails(ctx, w, h, semi ? "semi" : type, color, police);

  if (type === "tank") {
    ctx.fillStyle = shade(paint, -30);
    roundRect(ctx, w * 0.35, h * 0.42, w * 0.3, h * 0.2, 8);
    ctx.fill();
    roundRect(ctx, w * 0.47, h * 0.12, w * 0.06, h * 0.35, 4);
    ctx.fill();
  }

  if (type === "f1" || type === "prototype") {
    ctx.fillStyle = shade(paint, -42);
    roundRect(ctx, w * 0.18, h * 0.8, w * 0.64, h * 0.05, 5);
    ctx.fill();
    ctx.fillStyle = "rgba(4,6,6,0.9)";
    ctx.beginPath();
    ctx.ellipse(w * 0.5, h * 0.42, w * 0.11, h * 0.085, 0, 0, Math.PI * 2);
    ctx.fill();
  }

  if (tractor) {
    ctx.fillStyle = shade(paint, -30);
    roundRect(ctx, w * 0.28, h * 0.48, w * 0.44, h * 0.26, 12);
    ctx.fill();
    ctx.strokeStyle = "rgba(255,209,102,0.55)";
    ctx.lineWidth = Math.max(3, w * 0.018);
    ctx.beginPath();
    ctx.moveTo(w * 0.27, h * 0.79);
    ctx.lineTo(w * 0.73, h * 0.79);
    ctx.stroke();
  }

  const glass = ctx.createLinearGradient(0, h * 0.18, 0, h * 0.58);
  glass.addColorStop(0, "rgba(222,249,255,0.78)");
  glass.addColorStop(0.38, "rgba(53,83,90,0.78)");
  glass.addColorStop(1, "rgba(3,6,7,0.96)");
```

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```
    });
    ctx.fillStyle = glass;
    roundRect(ctx, w * 0.32, h * 0.22, w * 0.36, h * 0.15, 10);
    ctx.fill();
    if (!narrow) {
        roundRect(ctx, w * 0.28, h * 0.46, w * 0.44, h * 0.2, 12);
        ctx.fill();
    }

    ctx.strokeStyle = "rgba(244,251,248,0.24)";
    ctx.lineWidth = Math.max(2, w * 0.008);
    ctx.beginPath();
    ctx.moveTo(w * 0.5, h * 0.17);
    ctx.lineTo(w * 0.5, h * 0.86);
    ctx.moveTo(w * 0.31, h * 0.43);
    ctx.lineTo(w * 0.69, h * 0.43);
    ctx.stroke();

    ctx.fillStyle = shade(paint, -48);
    roundRect(ctx, w * 0.18, h * 0.38, w * 0.085, h * 0.035, 5);
    ctx.fill();
    roundRect(ctx, w * 0.735, h * 0.38, w * 0.085, h * 0.035, 5);
    ctx.fill();

    ctx.fillStyle = "rgba(255,248,214,0.95)";
    roundRect(ctx, w * 0.32, h * 0.18, w * 0.12, h * 0.035, 5);
    ctx.fill();
    roundRect(ctx, w * 0.56, h * 0.18, w * 0.12, h * 0.035, 5);
    ctx.fill();
    ctx.fillStyle = police ? "#46d9ff" : "#ff3348";
    roundRect(ctx, w * 0.31, h * 0.78, w * 0.15, h * 0.038, 5);
    ctx.fill();
    ctx.fillStyle = "#ff3348";
    roundRect(ctx, w * 0.54, h * 0.78, w * 0.15, h * 0.038, 5);
    ctx.fill();

    if (police) {
        ctx.fillStyle = "rgba(5,8,7,0.86)";
        roundRect(ctx, w * 0.3, h * 0.42, w * 0.4, h * 0.045, 5);
        ctx.fill();
        ctx.fillStyle = "#f4fbf8";
        ctx.font = `900 ${Math.max(12, w * 0.055)}px system-ui`;
        ctx.textAlign = "center";
        ctx.fillText("POLICE", w * 0.5, h * 0.62);
        ctx.fillStyle = "#ff3348";
        roundRect(ctx, w * 0.4, h * 0.32, w * 0.08, h * 0.032, 4);
        ctx.fill();
        ctx.fillStyle = "#46d9ff";
        roundRect(ctx, w * 0.52, h * 0.32, w * 0.08, h * 0.032, 4);
        ctx.fill();
    }

    ctx.fillStyle = "rgba(244,251,248,0.7)";
    roundRect(ctx, w * 0.42, h * 0.82, w * 0.16, h * 0.03, 5);
    ctx.fill();
    drawPaintGrain(ctx, w, h, paint);
    drawDamage(ctx, w, h, damage);
}

function drawAirOrWater(ctx, w, h, type, color, damage) {
    const paint = ctx.createLinearGradient(w * 0.26, h * 0.12, w * 0.74, h * 0.9);
    paint.addColorStop(0, shade(color, 48));
    paint.addColorStop(0.55, color);
    paint.addColorStop(1, shade(color, -44));
    ctx.fillStyle = paint;
    ctx.beginPath();
    if (type === "airplane" || type === "fighterjet" || type === "bomber") {
        const bomber = type === "bomber";
        const fighter = type === "fighterjet";
        ctx.moveTo(w * 0.5, h * 0.08);
        ctx.lineTo(w * (bomber ? 0.68 : 0.62), h * 0.86);
        ctx.lineTo(w * 0.5, h * 0.92);
        ctx.lineTo(w * (bomber ? 0.32 : 0.38), h * 0.86);
        ctx.closePath();
        ctx.fill();
        ctx.beginPath();
        ctx.moveTo(w * (bomber ? 0.06 : fighter ? 0.1 : 0.12), h * (fighter ? 0.48 : 0.52));
        ctx.lineTo(w * (bomber ? 0.94 : fighter ? 0.9 : 0.88), h * (fighter ? 0.48 : 0.52));
        ctx.lineTo(w * (bomber ? 0.68 : 0.62), h * 0.63);
        ctx.lineTo(w * (bomber ? 0.32 : 0.38), h * 0.63);
        ctx.closePath();
        ctx.fill();
        if (fighter) {
            ctx.fillStyle = shade(color, -34);
            ctx.beginPath();
            ctx.moveTo(w * 0.38, h * 0.78);
            ctx.lineTo(w * 0.22, h * 0.94);
            ctx.lineTo(w * 0.47, h * 0.86);
            ctx.closePath();
            ctx.fill();
            ctx.beginPath();
            ctx.moveTo(w * 0.62, h * 0.78);
            ctx.lineTo(w * 0.78, h * 0.94);
            ctx.lineTo(w * 0.53, h * 0.86);
            ctx.closePath();
            ctx.fill();
        }
    } else if (type === "drone") {
        ctx.fillStyle = "rgba(244,251,248,0.68)";
        for (let x = 0.23; x <= 0.77; x += 0.54) {
            for (let y = 0.28; y <= 0.72; y += 0.44) {
                ctx.beginPath();
                ctx.ellipse(w * x, h * y, w * 0.16, h * 0.04, 0, 0, Math.PI * 2);
                ctx.fill();
            }
        }
        ctx.strokeStyle = shade(color, -30);
        ctx.lineWidth = Math.max(4, w * 0.035);
        ctx.beginPath();
        ctx.moveTo(w * 0.24, h * 0.28);
        ctx.lineTo(w * 0.76, h * 0.72);
        ctx.moveTo(w * 0.76, h * 0.28);
        ctx.lineTo(w * 0.24, h * 0.72);
    }
}
```



```
ctx.stroke();
ctx.fillStyle = paint;
roundRect(ctx, w * 0.34, h * 0.36, w * 0.32, h * 0.28, 18);
ctx.fill();
} else if (type === "helicopter") {
  roun
```

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```
dRect(ctx, w * 0.3, h * 0.26, w * 0.4, h * 0.42, 24);
    ctx.fill();
    ctx.fillStyle = "rgba(244,251,248,0.72)";
    roundRect(ctx, w * 0.13, h * 0.13, w * 0.74, h * 0.035, 5);
    ctx.fill();
    ctx.fillStyle = shade(color, -34);
    roundRect(ctx, w * 0.47, h * 0.64, w * 0.06, h * 0.25, 5);
    ctx.fill();
  } else {
    ctx.moveTo(w * 0.5, h * 0.1);
    ctx.quadraticCurveTo(w * 0.82, h * 0.5, w * 0.62, h * 0.88);
    ctx.lineTo(w * 0.38, h * 0.88);
    ctx.quadraticCurveTo(w * 0.18, h * 0.5, w * 0.5, h * 0.1);
    ctx.closePath();
    ctx.fill();
  }
  ctx.fillStyle = "rgba(3,6,7,0.86)";
  roundRect(ctx, w * 0.39, h * 0.35, w * 0.22, h * 0.18, 10);
  ctx.fill();
  ctx.fillStyle = "rgba(255,248,214,0.86)";
  roundRect(ctx, w * 0.43, h * 0.18, w * 0.14, h * 0.035, 5);
  ctx.fill();
  drawAirWaterRealismDetails(ctx, w, h, type);
  drawPaintGrain(ctx, w, h, color);
  drawDamage(ctx, w, h, damage);
}

function drawRearWheel(ctx, x, groundY, rx, ry, accent, heavy = false) {
  ctx.save();
  const cy = groundY - ry;
  ctx.fillStyle = "rgba(2,4,4,0.98)";
  ctx.beginPath();
  ctx.ellipse(x, cy, rx, ry, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.strokeStyle = "rgba(244,251,248,0.22)";
  ctx.lineWidth = Math.max(2, rx * 0.13);
  ctx.stroke();
  ctx.globalAlpha = 0.42;
  ctx.strokeStyle = accent;
  ctx.lineWidth = Math.max(1, rx * 0.12);
  for (let i = -1; i <= 1; i += 1) {
    ctx.beginPath();
    ctx.moveTo(x - rx * 0.62, cy + i * ry * 0.26);
    ctx.lineTo(x + rx * 0.62, cy + i * ry * 0.26);
    ctx.stroke();
  }
  ctx.globalAlpha = heavy ? 0.52 : 0.34;
  ctx.fillStyle = "rgba(0,0,0,0.92)";
  ctx.beginPath();
  ctx.ellipse(x, groundY + ry * 0.08, rx * 1.12, ry * 0.18, 0, 0, Math.PI * 2);
  ctx.fill();
  ctx.restore();
}

function drawModernRearSignature(ctx, w, h, type, color, policeLivery, groundY) {
  const heavy = ["semi", "truck", "monster", "tractor"].includes(type);
  const tank = type === "tank";
  const f1 = type === "f1" || type === "prototype";
  const snow = type === "snowmobile";
  if (tank) {
    ctx.save();
    ctx.fillStyle = "rgba(2,4,4,0.82)";
    roundRect(ctx, w * 0.18, groundY - h * 0.105, w * 0.64, h * 0.07, 8);
    ctx.fill();
    ctx.fillStyle = "rgba(187,242,74,0.42)";
    for (let i = 0; i < 7; i += 1) {
      ctx.beginPath();
      ctx.arc(w * (0.26 + i * 0.08), groundY - h * 0.07, w * 0.014, 0, Math.PI * 2);
      ctx.fill();
    }
    ctx.restore();
    return;
  }
  ctx.save();
  ctx.globalCompositeOperation = "source-over";
  const led = policeLivery ? "#46d9ff" : type === "semi" || heavy ? "#ff5b6b" : "#ff3348";
  const plateY = groundY - h * (heavy ? 0.18 : f1 ? 0.15 : 0.17);
  const tailY = groundY - h * (heavy ? 0.23 : f1 ? 0.19 : 0.25);
  ctx.fillStyle = "rgba(3,5,5,0.94)";
  if (f1) {
    roundRect(ctx, w * 0.18, groundY - h * 0.16, w * 0.64, h * 0.06, 8);
  } else if (heavy) {
    roundRect(ctx, w * 0.17, groundY - h * 0.25, w * 0.66, h * 0.14, 10);
  } else {
    ctx.beginPath();
    ctx.moveTo(w * 0.22, groundY - h * 0.25);
    ctx.lineTo(w * 0.78, groundY - h * 0.25);
    ctx.lineTo(w * 0.68, groundY - h * 0.1);
    ctx.lineTo(w * 0.32, groundY - h * 0.1);
    ctx.closePath();
  }
  ctx.fill();

  ctx.fillStyle = led;
  ctx.shadowColor = led;
  ctx.shadowBlur = Math.max(4, w * 0.025);
  if (heavy) {
    roundRect(ctx, w * 0.22, tailY, w * 0.08, h * 0.052, 5);
    ctx.fill();
    roundRect(ctx, w * 0.7, tailY, w * 0.08, h * 0.052, 5);
    ctx.fill();
    ctx.shadowBlur = 0;
    ctx.fillStyle = "rgba(244,251,248,0.44)";
    for (let i = 0; i < 4; i += 1) {
      roundRect(ctx, w * 0.27, groundY - h * (0.38 - i * 0.06), w * 0.46, h * 0.012, 3);
      ctx.fill();
    }
  } else {
    roundRect(ctx, w * 0.24, tailY, w * 0.19, h * 0.035, 7);
    ctx.fill();
    roundRect(ctx, w * 0.57, tailY, w * 0.19, h * 0.035, 7);
    ctx.fill();
    ctx.fillStyle = policeLivery ? "#ff3348" : "rgba(255,91,107,0.92)";
  }
}
```

```
roundRect(ctx, w * 0.28, tailY + h * 0.042, w * 0.12, h * 0.016, 5);
```

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```
roundRect(ctx, w * 0.6, tailY + h * 0.042, w * 0.12, h * 0.016, 5);
    ctx.fill();
  }
  ctx.shadowBlur = 0;

  ctx.fillStyle = "rgba(244,251,248,0.86)";
  roundRect(ctx, w * 0.42, plateY, w * 0.16, h * 0.038, 5);
  ctx.fill();
  ctx.fillStyle = "rgba(3,5,5,0.78)";
  roundRect(ctx, w * 0.36, groundY - h * 0.09, w * 0.28, h * 0.024, 5);
  ctx.fill();

  if (!heavy && !snow) {
    ctx.fillStyle = "rgba(3,5,5,0.92)";
    for (let i = -2; i <= 2; i += 1) {
      roundRect(ctx, w * 0.5 + i * w * 0.045 - w * 0.012, groundY - h * 0.085, w * 0.024, h * 0.07, 4);
      ctx.fill();
    }
    ctx.strokeStyle = "rgba(255,255,255,0.22)";
    ctx.lineWidth = Math.max(1.5, w * 0.008);
    ctx.beginPath();
    ctx.moveTo(w * 0.24, h * 0.32);
    ctx.bezierCurveTo(w * 0.16, h * 0.5, w * 0.2, h * 0.74, w * 0.32, groundY - h * 0.13);
    ctx.moveTo(w * 0.76, h * 0.32);
    ctx.bezierCurveTo(w * 0.84, h * 0.5, w * 0.8, h * 0.74, w * 0.68, groundY - h * 0.13);
    ctx.stroke();
  } else if (heavy) {
    ctx.fillStyle = "rgba(3,5,5,0.86)";
    roundRect(ctx, w * 0.19, groundY - h * 0.045, w * 0.18, h * 0.05, 5);
    roundRect(ctx, w * 0.63, groundY - h * 0.045, w * 0.18, h * 0.05, 5);
    ctx.fill();
  }
  ctx.restore();
}

function drawRearGroundVehicle(ctx, w, h, type, color, police, damage) {
  const pursuit = type === "policecar";
  const policeLivery = police || pursuit;
  const paint = policeLivery ? color || "#f2f5f2" : color;
  const accent = policeLivery ? "rgba(70,217,255,0.56)" : "rgba(color, 0.58)";
  const semi = type === "semi";
  const tractor = type === "tractor";
  const tank = type === "tank";
  const monster = type === "monster";
  const truck = type === "truck";
  const f1 = type === "f1" || type === "prototype";
  const snow = type === "snowmobile";
  const wide = semi || tractor || tank || monster || truck || pursuit;
  const groundY = h * (snow ? 0.94 : 0.955);

  ctx.save();
  const contact = ctx.createRadialGradient(w * 0.5, groundY - h * 0.025, w * 0.07, w * 0.5, groundY, w * 0.55);
  contact.addColorStop(0, snow ? "rgba(244,251,248,0.42)" : "rgba(0,0,0,0.72)");
  contact.addColorStop(1, "rgba(0,0,0,0)");
  ctx.fillStyle = contact;
  ctx.beginPath();
  ctx.ellipse(w * 0.5, groundY, w * (wide ? 0.52 : 0.44), h * 0.06, 0, 0, Math.PI * 2);
  ctx.fill();

  if (snow) {
    ctx.strokeStyle = "rgba(244,251,248,0.82)";
    ctx.lineWidth = Math.max(7, w * 0.036);
    ctx.lineCap = "round";
    ctx.beginPath();
    ctx.moveTo(w * 0.26, groundY);
    ctx.lineTo(w * 0.43, groundY - h * 0.02);
    ctx.moveTo(w * 0.57, groundY - h * 0.02);
    ctx.lineTo(w * 0.74, groundY);
    ctx.stroke();
  } else if (tank) {
    ctx.fillStyle = "rgba(1,2,2,0.98)";
    roundRect(ctx, w * 0.11, h * 0.44, w * 0.22, groundY - h * 0.44, 14);
    ctx.fill();
    roundRect(ctx, w * 0.67, h * 0.44, w * 0.22, groundY - h * 0.44, 14);
    ctx.fill();
    ctx.strokeStyle = "rgba(244,251,248,0.16)";
    ctx.lineWidth = Math.max(2, w * 0.012);
    for (let i = 0; i < 7; i += 1) {
      const y = h * 0.49 + i * h * 0.052;
      ctx.beginPath();
      ctx.moveTo(w * 0.14, y);
      ctx.lineTo(w * 0.3, y);
      ctx.moveTo(w * 0.7, y);
      ctx.lineTo(w * 0.86, y);
      ctx.stroke();
    }
  } else {
    const rx = w * (monster ? 0.105 : semi || tractor ? 0.088 : f1 ? 0.074 : 0.082);
    const ry = h * (monster ? 0.155 : semi || tractor ? 0.14 : f1 ? 0.09 : 0.125);
    const left = w * (wide ? 0.2 : 0.24);
    const right = w - left;
    drawRearWheel(ctx, left, groundY, rx, ry, accent, wide);
    drawRearWheel(ctx, right, groundY, rx, ry, accent, wide);
    if (semi || tractor || truck) {
      drawRearWheel(ctx, left + w * 0.08, groundY - h * 0.01, rx * 0.78, ry * 0.8, "rgba(255,209,102,0.28)", true);
      drawRearWheel(ctx, right - w * 0.08, groundY - h * 0.01, rx * 0.78, ry * 0.8, "rgba(255,209,102,0.28)", true);
    }
  }
}

const body = ctx.createLinearGradient(w * 0.18, h * 0.14, w * 0.82, groundY);
body.addColorStop(0, shade(paint, 54));
body.addColorStop(0,
```

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```
34, paint);
body.addColorStop(0.7, shade(paint, -30));
body.addColorStop(1, shade(paint, -62));
ctx.fillStyle = body;

if (f1) {
  ctx.fillStyle = shade(paint, -38);
  roundRect(ctx, w * 0.12, h * 0.73, w * 0.76, h * 0.075, 9);
  ctx.fill();
  ctx.fillStyle = body;
  ctx.beginPath();
  ctx.moveTo(w * 0.5, h * 0.28);
  ctx.lineTo(w * 0.68, h * 0.5);
  ctx.lineTo(w * 0.61, h * 0.82);
  ctx.lineTo(w * 0.39, h * 0.82);
  ctx.lineTo(w * 0.32, h * 0.5);
  ctx.closePath();
  ctx.fill();
  ctx.fillStyle = "rgba(4,6,6,0.92)";
  ctx.beginPath();
  ctx.ellipse(w * 0.5, h * 0.53, w * 0.12, h * 0.09, 0, 0, Math.PI * 2);
  ctx.fill();
} else if (semi) {
  roundRect(ctx, w * 0.19, h * 0.33, w * 0.62, h * 0.43, 18);
  ctx.fill();
  ctx.fillStyle = shade(paint, 18);
  roundRect(ctx, w * 0.24, h * 0.17, w * 0.52, h * 0.26, 16);
  ctx.fill();
  ctx.fillStyle = "rgba(4,7,8,0.9)";
  roundRect(ctx, w * 0.31, h * 0.22, w * 0.38, h * 0.12, 8);
  ctx.fill();
  ctx.fillStyle = "rgba(244,251,248,0.28)";
  for (let i = 0; i < 3; i += 1) {
    roundRect(ctx, w * 0.27, h * (0.48 + i * 0.1), w * 0.46, h * 0.026, 4);
    ctx.fill();
  }
} else if (tractor) {
  roundRect(ctx, w * 0.3, h * 0.25, w * 0.4, h * 0.24, 14);
  ctx.fill();
  ctx.fillStyle = shade(paint, -28);
  roundRect(ctx, w * 0.23, h * 0.48, w * 0.54, h * 0.26, 12);
  ctx.fill();
  ctx.strokeStyle = "rgba(255,209,102,0.58)";
  ctx.lineWidth = Math.max(4, w * 0.022);
  ctx.beginPath();
  ctx.moveTo(w * 0.23, h * 0.77);
  ctx.lineTo(w * 0.77, h * 0.77);
  ctx.stroke();
} else if (tank) {
  roundRect(ctx, w * 0.24, h * 0.32, w * 0.52, h * 0.46, 9);
  ctx.fill();
  ctx.fillStyle = shade(paint, -34);
  roundRect(ctx, w * 0.34, h * 0.38, w * 0.32, h * 0.2, 8);
  ctx.fill();
  ctx.fillRect(w * 0.47, h * 0.18, w * 0.06, h * 0.26);
} else {
  const topHalf = w * (truck || monster ? 0.24 : 0.2);
  const shoulder = w * (truck || monster ? 0.38 : 0.34);
  const lower = w * (truck || monster ? 0.43 : 0.37);
  const top = h * (snow ? 0.3 : 0.18);
  const belly = groundY - h * 0.055;
  ctx.beginPath();
  ctx.moveTo(w * 0.5 - topHalf, top);
  ctx.lineTo(w * 0.5 + topHalf, top);
  ctx.quadraticCurveTo(w * 0.5 + shoulder, h * 0.26, w * 0.5 + shoulder, h * 0.48);
  ctx.lineTo(w * 0.5 + lower, belly);
  ctx.quadraticCurveTo(w * 0.5 + lower * 0.7, groundY - h * 0.015, w * 0.5, groundY - h * 0.012);
  ctx.quadraticCurveTo(w * 0.5 - lower * 0.7, groundY - h * 0.015, w * 0.5 - lower, belly);
  ctx.lineTo(w * 0.5 - shoulder, h * 0.48);
  ctx.quadraticCurveTo(w * 0.5 - shoulder, h * 0.26, w * 0.5 - topHalf, top);
  ctx.closePath();
  ctx.fill();
}

ctx.strokeStyle = "rgba(255,255,255,0.28)";
ctx.lineWidth = Math.max(2, w * 0.012);
ctx.stroke();

if (snow) {
  ctx.strokeStyle = "rgba(244,251,248,0.92)";
  ctx.lineWidth = Math.max(7, w * 0.034);
  ctx.lineCap = "round";
  ctx.beginPath();
  ctx.moveTo(w * 0.24, groundY);
  ctx.lineTo(w * 0.43, groundY - h * 0.01);
  ctx.moveTo(w * 0.57, groundY - h * 0.01);
  ctx.lineTo(w * 0.76, groundY);
  ctx.stroke();
} else if (tank) {
  ctx.fillStyle = "rgba(1,2,2,1)";
  roundRect(ctx, w * 0.11, groundY - h * 0.12, w * 0.23, h * 0.13, 10);
  ctx.fill();
  roundRect(ctx, w * 0.66, groundY - h * 0.12, w * 0.23, h * 0.13, 10);
  ctx.fill();
} else {
  const rx = w * (monster ? 0.108 : semi || tractor ? 0.092 : f1 ? 0.078 : 0.086);
  const ry = h * (monster ? 0.16 : semi || tractor ? 0.14 : f1 ? 0.092 : 0.128);
  const left = w * (wide ? 0.2 : 0.24);
  const right = w - left;
  drawRearWheel(ctx, left, groundY, rx, ry, accent, true);
  drawRearWheel(ctx, right, groundY, rx, ry, accent, true);
}

ctx.fillStyle = "rgba(5,8,7,0.9)";
roundRect(ctx, w * 0.31, h * (semi ? 0.46 : 0.32), w * 0.38, h * 0.13, 10);
ctx.fill();
if (!f1 && !semi && !tractor && !tank) {
  roundRect(ctx, w * 0.2
```

phone-asset-pipeline.js continued 8/9

```
7, h * 0.55, w * 0.46, h * 0.19, 12);
    ctx.fill();
}
ctx.fillStyle = "rgba(222,249,255,0.24)";
roundRect(ctx, w * 0.34, h * (semi ? 0.49 : 0.35), w * 0.14, h * 0.026, 4);
ctx.fill();
roundRect(ctx, w * 0.52, h * (semi ? 0.49 : 0.35), w * 0.14, h * 0.026, 4);
ctx.fill();

if (policeLivery) {
    ctx.fillStyle = "rgba(5,8,7,0.88)";
    roundRect(ctx, w * 0.29, h * 0.53, w * 0.42, h * 0.052, 5);
    ctx.fill();
    ctx.fillStyle = "#f4fbf8";
    ctx.font = '900 ${Math.max(16, w * 0.07)}px system-ui';
    ctx.textAlign = "center";
    ctx.fillText("POLICE", w * 0.5, h * 0.65);
    ctx.fillStyle = "#ff3348";
    roundRect(ctx, w * 0.4, h * 0.27, w * 0.08, h * 0.034, 4);
    ctx.fill();
    ctx.fillStyle = "#46d9ff";
    roundRect(ctx, w * 0.52, h * 0.27, w * 0.08, h * 0.034, 4);
    ctx.fill();
}

ctx.fillStyle = "#ff3348";
roundRect(ctx, w * 0.28, groundY - h * 0.12, w * 0.17, h * 0.048, 5);
ctx.fill();
roundRect(ctx, w * 0.55, groundY - h * 0.12, w * 0.17, h * 0.048, 5);
ctx.fill();
ctx.fillStyle = "rgba(244,251,248,0.68)";
roundRect(ctx, w * 0.43, groundY - h * 0.085, w * 0.14, h * 0.028, 4);
ctx.fill();

ctx.strokeStyle = "rgba(5,8,7,0.46)";
ctx.lineWidth = Math.max(2, w * 0.016);
ctx.beginPath();
ctx.moveTo(w * 0.5, h * 0.2);
ctx.lineTo(w * 0.5, groundY - h * 0.08);
ctx.stroke();

drawModernRearSignature(ctx, w, h, type, paint, policeLivery, groundY);
drawPaintGrain(ctx, w, h, paint);
drawDamage(ctx, w, h, damage);
ctx.restore();
}

function getVehicleSprite(type = "car", color = "#46d9ff", options = {}) {
    const police = Boolean(options.police);
    const damage = Math.max(0, Math.min(100, Number(options.damage) || 0));
    const damageBucket = Math.floor(damage / 14);
    const key = `${type}|${color}|${police ? 1 : 0}|${damageBucket}`;
    if (spriteCache.has(key)) return spriteCache.get(key);

    const canvas = makeCanvas(384, 552);
    const ctx = canvas.getContext("2d");
    ctx.imageSmoothingEnabled = true;

    if (isWaterType(type) || isAirType(type)) {
        const shadow = ctx.createRadialGradient(192, 422, 26, 192, 422, 176);
        shadow.addColorStop(0, "rgba(0,0,0,0.52)");
        shadow.addColorStop(1, "rgba(0,0,0,0)");
        ctx.fillStyle = shadow;
        ctx.beginPath();
        ctx.ellipse(192, 422, 168, 56, 0, 0, Math.PI * 2);
        ctx.fill();
        drawSpriteContact(ctx, 384, 552, type);
        drawAirOrWater(ctx, 384, 552, type, police ? "#f4fbf8" : color, damage);
    } else {
        drawRearGroundVehicle(ctx, 384, 552, type, color, police, damage);
    }

    spriteCache.set(key, canvas);
    return canvas;
}

function getRoadTexture(place = "city", theme = ["#09100f", "#46d9ff", "#ffd166"]) {
    const key = `${place}|${theme.join("|")}`;
    if (textureCache.has(key)) return textureCache.get(key);
    const canvas = makeCanvas(512, 512);
    const ctx = canvas.getContext("2d");
    ctx.fillStyle = place === "snow" ? "rgba(205,222,224,0.14)" : place === "desert" || place === "canyon" ? "rgba(156,92,42,0.13)" : "rgba(244,251,248,0.08)";
    ctx.fillRect(0, 0, 512, 512);
    for (let i = 0; i < 420; i += 1) {
        const x = (i * 47) % 512;
        const y = (i * 83) % 512;
        const len = 8 + (i % 9) * 7;
        ctx.globalAlpha = 0.08 + (i % 5) * 0.018;
        ctx.strokeStyle = i % 3 ? "rgba(255,255,255,0.55)" : rgba(theme[1], 0.75);
        ctx.lineWidth = 1 + (i % 3);
        ctx.beginPath();
        ctx.moveTo(x, y);
        ctx.lineTo(x + len, y + (i % 2 ? 2 : -2));
        ctx.stroke();
    }
    ctx.globalAlpha = 0.18;
    ctx.strokeStyle = place === "snow" ? "rgba(244,251,248,0.9)" : "rgba(0,0,0,0.85)";
    ctx.lineCap = "round";
    for (let i = 0; i < 54; i += 1) {
        const x = (i * 109) % 512;
        const y = (i * 67) % 512;
        ctx.lineWidth = 1 + (i % 3);
        ctx.beginPath();
        ctx.moveTo(x, y);
        ctx.bezierCurveTo(x + 24, y + 9, x - 18, y + 31, x + 38, y + 45);
        ctx.stroke();
    }
    ctx.globalAlpha = place === "tokyo" || place === "city" || place === "harbor" || place === "rainforest" ? 0.22 : 0.1;
    for (let i = 0; i < 18; i += 1) {
```

phone-asset-pipeline.js continued 9/9

```
const x = (i * 131) % 512;
const y = (i * 191) % 512;
const glow = ctx.createRadialGradient(x, y, 4, x, y, 92);
glow.addColorStop(0, rgba(theme[1], 0.9));
glow.addColorStop(1, "rgba(0,0,0,0)");
ctx.fillStyle = glow;
ctx.fillRect(x - 96, y - 96, 192, 192);
}
ctx.globalAlpha = 1;
textureCache.set(key, canvas);
return canvas;
}

function isPhoneViewport() {
  return window.matchMedia && (window.matchMedia("(pointer: coarse)").matches || Math.min(window.innerWidth, window.innerHeight) <
720);
}

window.VelocityPhoneAssets = {
  getVehicleSprite,
  getRoadTexture,
  isPhoneViewport,
  ready: true
};
})();
```

webgl-pipeline.js

```
"use strict";

(function () {
  function hexToRgba(hex, alpha = 1) {
    if (!hex || typeof hex !== "string") return [1, 1, 1, alpha];
    const clean = hex.replace("#", "");
    const value = parseInt(clean.length === 3 ? clean.split("").map((c) => c + c).join("") : clean, 16);
    return [(value >> 16) & 255] / 255, ((value >> 8) & 255) / 255, (value & 255) / 255, alpha];
  }

  function shade(color, factor) {
    return [
      Math.max(0, Math.min(1, color[0] * factor)),
      Math.max(0, Math.min(1, color[1] * factor)),
      Math.max(0, Math.min(1, color[2] * factor)),
      color[3]
    ];
  }

  function perspective(fovy, aspect, near, far) {
    const f = 1 / Math.tan(fovy / 2);
    const nf = 1 / (near - far);
    return [
      f / aspect, 0, 0, 0,
      0, f, 0, 0,
      0, 0, (far + near) * nf, -1,
      0, 0, (2 * far * near) * nf, 0
    ];
  }

  function normalize(v) {
    const length = Math.hypot(v[0], v[1], v[2]) || 1;
    return [v[0] / length, v[1] / length, v[2] / length];
  }

  function cross(a, b) {
    return [
      a[1] * b[2] - a[2] * b[1],
      a[2] * b[0] - a[0] * b[2],
      a[0] * b[1] - a[1] * b[0]
    ];
  }

  function dot(a, b) {
    return a[0] * b[0] + a[1] * b[1] + a[2] * b[2];
  }

  function lookAt(eye, center, up) {
    const z = normalize([eye[0] - center[0], eye[1] - center[1], eye[2] - center[2]]);
    const x = normalize(cross(up, z));
    const y = cross(z, x);
    return [
      x[0], y[0], z[0], 0,
      x[1], y[1], z[1], 0,
      x[2], y[2], z[2], 0,
      -dot(x, eye), -dot(y, eye), -dot(z, eye), 1
    ];
  }

  function multiply(a, b) {
    const out = new Array(16).fill(0);
    for (let col = 0; col < 4; col += 1) {
      for (let row = 0; row < 4; row += 1) {
        out[col * 4 + row] =
          a[0 * 4 + row] * b[col * 4 + 0] +
          a[1 * 4 + row] * b[col * 4 + 1] +
          a[2 * 4 + row] * b[col * 4 + 2] +
          a[3 * 4 + row] * b[col * 4 + 3];
      }
    }
    return out;
  }

  function compile(gl, type, source) {
    const shader = gl.createShader(type);
    gl.shaderSource(shader, source);
    gl.compileShader(shader);
    if (!gl.getShaderParameter(shader, gl.COMPILE_STATUS)) {
      throw new Error(gl.getShaderInfoLog(shader) || "WebGL shader compile failed");
    }
    return shader;
  }

  function linkProgram(gl) {
    const vertex = compile(gl, gl.VERTEX_SHADER, `
      attribute vec3 aPosition;
      attribute vec4 aColor;
      uniform mat4 uMvp;
      varying vec4 vColor;
      void main() {
        gl_Position = uMvp * vec4(aPosition, 1.0);
        vColor = aColor;
      }
    `);
    const fragment = compile(gl, gl.FRAGMENT_SHADER, `
      precision mediump float;
      varying vec4 vColor;
      void main() {
        gl_FragColor = vColor;
      }
    `);
    const program = gl.createProgram();
    gl.attachShader(program, vertex);
    gl.attachShader(program, fragment);
    gl.linkProgram(program);
    if (!gl.getProgramParameter(program, gl.LINK_STATUS)) {
      throw new Error(gl.getProgramInfoLog(program) || "WebGL program link failed");
    }
    return program;
  }

  function VelocityWebGLPipeline(canvas) {

```



```

const gl = canvas.getContext("webgl", {
  alpha: false,
  antialias: true,
  depth: true,
  powerPreference: "high-performance"
});
if (!gl) throw new Error("WebGL is not available in this browser");

const program = linkProgram(gl);
const buffer = gl.createBuffer();
const aPosition = gl.getAttribLocation(program, "aPosition");
const aColor = gl.getAttribLocation(program, "aColor");
const uMvp = gl.getUniformLocation(program, "uMvp");
let vertices = [];
let lastProjection = null;

function visuallLaneValue(data) {
  const state = data && data.raceState ? data.raceState : {};
  const visuallLane = Number(state.visuallLane);
  if (Number.isFinite(visuallLane)) return visuallLane;
  return Number(state.lane) || 0;
}

function laneToWorldX(lane) {
  return Number(lane) * 2.08;
}

function cameraProjection(data) {
  const laneX = laneToWorldX(visuallLaneValue(data));
  const shake = data.raceState.cameraShake || 0;
  const curve =

```

webgl-pipeline.js continued 2/12

```
roadCurveValue(data);
const jitter = shake > 0 ? (Math.random() - 0.5) * shake * 0.012 : 0;
const phoneFrame = data.width <= 940 || data.height <= 540;
let eye = phoneFrame ? [laneX * 0.08 - curve * 0.78 + jitter, 1.62, -15.8] : [laneX * 0.06 - curve * 0.9 + jitter, 2.1,
-14.6];
let target = phoneFrame ? [laneX * 0.015 + curve * 3.45, 1.24, 62] : [laneX * 0.015 + curve * 3.4, 1.96, 50];
if (data.cameraMode === "hood") {
  eye = phoneFrame ? [laneX * 0.3 - curve * 0.56 + jitter, 1.18, -7.8] : [laneX * 0.34 - curve * 0.62 + jitter, 1.42, -7.4];
  target = phoneFrame ? [laneX * 0.08 + curve * 3.65, 1.02, 64] : [laneX * 0.1 + curve * 3.6, 1.38, 52];
} else if (data.cameraMode === "cockpit") {
  eye = [laneX * 0.2 - curve * 0.5 + jitter, 1.58, -3.8];
  target = [laneX * 0.05 + curve * 3.8, 1.52, 56];
}
const view = lookAt(eye, target, [0, 1, 0]);
const fov = phoneFrame && data.cameraMode !== "cockpit" ? 2.28 : data.cameraMode === "cockpit" ? 2.25 : 2.55;
const proj = perspective(Math.PI / fov, data.width / Math.max(1, data.height), 0.1, 220);
return {
  mvp: multiply(proj, view),
  width: data.width,
  height: data.height,
  raceDistance: data.raceState.distance,
  roadTurn: curve,
  roadOffset: data.raceState.roadOffset || 0,
  cameraMode: data.cameraMode
};
}

function projectPoint(projection, x, y, z) {
  const m = projection.mvp;
  const clipX = m[0] * x + m[4] * y + m[8] * z + m[12];
  const clipY = m[1] * x + m[5] * y + m[9] * z + m[13];
  const clipW = m[3] * x + m[7] * y + m[11] * z + m[15];
  if (!Number.isFinite(clipW) || clipW <= 0.02) return null;
  const ndcX = clipX / clipW;
  const ndcY = clipY / clipW;
  if (!Number.isFinite(ndcX) || !Number.isFinite(ndcY)) return null;
  const screenY = (0.5 - ndcY * 0.5) * projection.height;
  const t = Math.max(0, Math.min(1.16, screenY / Math.max(1, projection.height)));
  return {
    x: (ndcX * 0.5 + 0.5) * projection.width,
    y: screenY,
    scale: Math.max(0.34, Math.min(1.24, 0.42 + t * 0.78)),
    clipW,
    z
  };
};

function screenYFromRoadDistance(data, distance) {
  return data.height * 0.68 - (distance - data.raceState.distance) * 0.11;
}

function zFromRoadScreenY(data, y) {
  const t = Math.max(0.32, Math.min(1.02, y / Math.max(1, data.height)));
  return 116 - t * 108;
}

function zFromRoadDistance(data, distance) {
  return zFromRoadScreenY(data, screenYFromRoadDistance(data, distance));
}

function addVertex(point, color) {
  vertices.push(point[0], point[1], point[2], color[0], color[1], color[2], color[3]);
}

function tri(a, b, c, color) {
  addVertex(a, color);
  addVertex(b, color);
  addVertex(c, color);
}

function quad(a, b, c, d, color) {
  tri(a, b, c, color);
  tri(a, c, d, color);
}

function box(x, y, z, w, h, d, color) {
  const x0 = x - w / 2;
  const x1 = x + w / 2;
  const y0 = y - h / 2;
  const y1 = y + h / 2;
  const z0 = z - d / 2;
  const z1 = z + d / 2;
  quad([x0, y1, z0], [x1, y1, z0], [x1, y1, z1], [x0, y1, z1], shade(color, 1.14));
  quad([x0, y0, z1], [x1, y0, z1], [x1, y0, z0], [x0, y0, z0], shade(color, 0.55));
  quad([x0, y0, z0], [x0, y1, z0], [x0, y1, z1], [x0, y0, z1], shade(color, 0.76));
  quad([x1, y0, z1], [x1, y1, z1], [x1, y1, z0], [x1, y0, z0], shade(color, 0.88));
  quad([x0, y0, z1], [x0, y1, z1], [x1, y1, z1], [x1, y0, z1], shade(color, 1.02));
  quad([x1, y0, z0], [x1, y1, z0], [x0, y1, z0], [x0, y0, z0], shade(color, 0.64));
}

function taperedBox(x, y, z, bottomW, topW, h, d, color) {
  const y0 = y - h / 2;
  const y1 = y + h / 2;
  const z0 = z - d / 2;
  const z1 = z + d / 2;
  const b0 = x - bottomW / 2;
  const b1 = x + bottomW / 2;
  const t0 = x - topW / 2;
  const t1 = x + topW / 2;
  quad([t0, y1, z0], [t1, y1, z0], [t1, y1, z1], [t0, y1, z1], shade(color, 1.16));
  quad([b0, y0, z1], [b1, y0, z1], [b1,
```

webgl-pipeline.js continued 3/12

```
y0, z0], [b0, y0, z0], shade(color, 0.55));
quad([b0, y0, z0], [t0, y1, z0], [t0, y1, z1], [b0, y0, z1], shade(color, 0.78));
quad([b1, y0, z1], [t1, y1, z1], [t1, y1, z0], [b1, y0, z0], shade(color, 0.9));
quad([b0, y0, z1], [t0, y1, z1], [t1, y1, z1], [b1, y0, z1], shade(color, 1.02));
quad([b1, y0, z0], [t1, y1, z0], [t0, y1, z0], [b0, y0, z0], shade(color, 0.66));
}

function lowMound(x, z, w, h, d, color) {
  taperedBox(x, h * 0.3, z, w, w * 0.72, h * 0.6, d, shade(color, 0.82));
  taperedBox(x, h * 0.78, z - d * 0.05, w * 0.72, w * 0.42, h * 0.44, d * 0.76, color);
  taperedBox(x, h * 1.08, z - d * 0.1, w * 0.42, w * 0.18, h * 0.24, d * 0.48, shade(color, 1.18));
}

function signPanel(x, z, textColor, baseColor) {
  box(x, 2.15, z, 3.2, 1.25, 0.14, baseColor);
  box(x, 2.22, z - 0.08, 2.35, 0.14, 0.16, textColor);
  box(x - 0.9, 1.7, z + 0.05, 0.12, 1.0, 0.12, [0.42, 0.44, 0.4, 1]);
  box(x + 0.9, 1.7, z + 0.05, 0.12, 1.0, 0.12, [0.42, 0.44, 0.4, 1]);
}

function person(x, z, shirtColor, scale = 1) {
  box(x, 0.18 * scale, z, 0.12 * scale, 0.36 * scale, 0.08 * scale, [0.04, 0.045, 0.04, 1]);
  box(x, 0.62 * scale, z, 0.28 * scale, 0.52 * scale, 0.16 * scale, shirtColor);
  box(x, 1.02 * scale, z, 0.22 * scale, 0.22 * scale, 0.18 * scale, [0.72, 0.52, 0.38, 1]);
  box(x - 0.24 * scale, 0.68 * scale, z, 0.08 * scale, 0.42 * scale, 0.08 * scale, shade(shirtColor, 0.75));
  box(x + 0.24 * scale, 0.68 * scale, z, 0.08 * scale, 0.42 * scale, 0.08 * scale, shade(shirtColor, 0.75));
}

function spectatorCluster(x, z, accent, accent2) {
  for (let j = 0; j < 5; j += 1) {
    const row = Math.floor(j / 3);
    const px = x + (j % 3 - 1) * 0.55;
    const pz = z + row * 0.34;
    person(px, pz, j % 2 ? accent : accent2, 0.74);
  }
  box(x, 0.28, z + 0.85, 2.4, 0.14, 0.14, [0.78, 0.82, 0.78, 1]);
}

function animalGroup(x, z, species = "dog", side = 1) {
  const count = species === "birds" || species === "sheep" ? 3 : 2;
  for (let j = 0; j < count; j += 1) {
    const px = x + side * (j * 0.75 - 0.45);
    const pz = z + (j % 2) * 0.34;
    const large = species === "cow" || species === "horse" || species === "camel";
    const body = species === "cow" || species === "sheep" || species === "birds" ? [0.84, 0.84, 0.76, 1]
      : species === "deer" || species === "horse" || species === "dog" ? [0.48, 0.28, 0.16, 1]
      : species === "camel" ? [0.76, 0.52, 0.28, 1]
      : [0.25, 0.34, 0.2, 1];
    if (species === "birds") {
      box(px, 0.24, pz, 0.32, 0.12, 0.18, body);
      box(px + side * 0.18, 0.34, pz, 0.12, 0.12, 0.12, body);
      box(px - side * 0.08, 0.16, pz, 0.04, 0.24, 0.04, [0.08, 0.08, 0.06, 1]);
      box(px + side * 0.08, 0.16, pz, 0.04, 0.24, 0.04, [0.08, 0.08, 0.06, 1]);
    } else {
      box(px, large ? 0.42 : 0.3, pz, large ? 0.92 : 0.62, large ? 0.42 : 0.3, large ? 0.36 : 0.26, body);
      box(px + side * (large ? 0.52 : 0.36), large ? 0.54 : 0.4, pz - 0.04, large ? 0.24 : 0.18, large ? 0.26 : 0.2, large ?
0.22 : 0.16, body);
      box(px - side * 0.28, 0.12, pz + 0.14, 0.08, 0.24, 0.08, [0.06, 0.045, 0.035, 1]);
      box(px + side * 0.22, 0.12, pz + 0.14, 0.08, 0.24, 0.08, [0.06, 0.045, 0.035, 1]);
      if (species === "deer") {
        box(px + side * 0.6, 0.78, pz - 0.08, 0.05, 0.34, 0.05, [0.9, 0.76, 0.48, 1]);
        box(px + side * 0.42, 0.78, pz - 0.08, 0.05, 0.3, 0.05, [0.9, 0.76, 0.48, 1]);
      }
    }
  }
}

function livingCrowdLine(x, z, side, accent, accent2, kind = "crowd") {
  const rail = kind === "farm" ? [0.58, 0.42, 0.22, 1] : [0.76, 0.8, 0.76, 1];
  box(x, 0.25, z + 0.72, 3.1, 0.12, 0.14, rail);
  box(x, 0.42, z + 0.72, 0.1, 0.46, 0.16, rail);
  box(x - 1.25, 0.42, z + 0.72, 0.1, 0.46, 0.16, rail);
  box(x + 1.25, 0.42, z + 0.72, 0.1, 0.46, 0.16, rail);
  const count = kind === "stadium" ? 7 : 4;
  for (let j = 0; j < count; j += 1) {
    const px = x + (j - (count - 1) / 2) * 0.46;
    const
```

webgl-pipeline.js continued 4/12

```
pz = z + (j % 2) * 0.22;
person(px, pz, j % 2 ? accent : accent2, kind === "stadium" ? 0.68 : 0.62);
}
if (kind === "stadium") box(x, 1.1, z + 1.16, 3.7, 1.2, 0.22, [0.08, 0.09, 0.08, 1]);
}

function cyclistPair(x, z, side, accent, accent2) {
  for (let j = 0; j < 2; j += 1) {
    const px = x + side * (j * 0.95 - 0.45);
    const paint = j % 2 ? accent : accent2;
    box(px - side * 0.22, 0.18, z, 0.24, 0.12, 0.08, [0.02, 0.025, 0.02, 1]);
    box(px + side * 0.22, 0.18, z, 0.24, 0.12, 0.08, [0.02, 0.025, 0.02, 1]);
    box(px, 0.34, z, 0.62, 0.08, 0.08, paint);
    person(px, z - 0.06, paint, 0.48);
  }
  box(x, 0.06, z + 0.48, 2.1, 0.04, 0.12, [0.78, 0.82, 0.78, 1]);
}

function serviceCrew(x, z, side, accent, accent2, heavy = false) {
  const body = heavy ? [0.74, 0.56, 0.14, 1] : accent;
  box(x, 0.34, z, heavy ? 1.4 : 1.0, 0.58, heavy ? 0.9 : 0.68, body);
  box(x + side * 0.34, 0.74, z - 0.05, 0.36, 0.42, 0.42, shade(body, 0.76));
  box(x - 0.34, 0.08, z + 0.38, 0.24, 0.1, 0.18, [0.02, 0.025, 0.02, 1]);
  box(x + 0.34, 0.08, z + 0.38, 0.24, 0.1, 0.18, [0.02, 0.025, 0.02, 1]);
  person(x + side * 1.05, z + 0.2, accent2, 0.6);
  person(x + side * 1.48, z + 0.42, accent, 0.58);
  box(x + side * 1.4, 0.28, z + 0.75, 1.4, 0.08, 0.12, [0.78, 0.82, 0.78, 1]);
}

function pastureLine(x, z, side, species, accent, accent2) {
  box(x, 0.32, z + 0.72, 4.2, 0.12, 0.12, [0.58, 0.42, 0.22, 1]);
  for (let j = -2; j <= 2; j += 1) box(x + j * 0.82, 0.42, z + 0.72, 0.08, 0.62, 0.1, [0.58, 0.42, 0.22, 1]);
  animalGroup(x - side * 0.8, z - 0.2, species, side);
  if (species === "sheep") animalGroup(x + side * 0.95, z + 0.35, "sheep", side);
  person(x + side * 1.9, z + 0.2, accent2, 0.58);
}

function waterEdgeLife(x, z, side, accent, accent2) {
  box(x, 0.08, z, 3.8, 0.12, 1.7, [0.1, 0.36, 0.42, 1]);
  box(x, 0.24, z + 0.84, 3.6, 0.12, 0.14, [0.52, 0.34, 0.16, 1]);
  for (let j = -1; j <= 1; j += 1) person(x + j * 0.56, z + 0.36, j % 2 ? accent : accent2, 0.58);
  animalGroup(x + side * 1.6, z - 0.2, "birds", side);
}

function streetLight(x, z, accent) {
  const side = x < 0 ? 1 : -1;
  box(x, 1.35, z, 0.1, 2.7, 0.1, [0.38, 0.42, 0.39, 1]);
  box(x + side * 0.55, 2.62, z, 1.1, 0.08, 0.08, [0.38, 0.42, 0.39, 1]);
  box(x + side * 1.05, 2.52, z, 0.34, 0.12, 0.32, accent);
  box(x + side * 1.05, 2.43, z, 0.6, 0.03, 0.56, shade(accent, 1.35));
}

function roadsideTree(x, z, color, kind = "tree") {
  box(x, 0.74, z, 0.22, 1.45, 0.22, [0.34, 0.2, 0.1, 1]);
  if (kind === "palm") {
    taperedBox(x - 0.45, 1.82, z, 1.6, 0.25, 0.22, 1.0, color);
    taperedBox(x + 0.45, 1.86, z, 1.6, 0.25, 0.22, 1.0, color);
    taperedBox(x, 2.02, z - 0.45, 1.4, 0.2, 0.2, 1.15, color);
  } else {
    taperedBox(x, 1.85, z, 1.8, 0.7, 1.3, 1.6, color);
    taperedBox(x + 0.35, 2.45, z - 0.1, 1.3, 0.45, 1.0, 1.25, shade(color, 1.18));
  }
}

function barrierRun(x, z, accent) {
  box(x, 0.58, z, 0.14, 0.75, 0.16, [0.62, 0.66, 0.62, 1]);
  box(x, 0.86, z, 0.16, 0.12, 3.2, [0.78, 0.82, 0.78, 1]);
  box(x, 0.68, z, 0.18, 0.08, 3.0, shade(accent, 0.9));
}

function isAirType(type) {
  return ["airplane", "helicopter", "fighterjet", "bomber", "drone"].includes(type);
}

function groundContactPad(type, x, z, scale, paint) {
  const air = isAirType(type);
  const water = type === "boat";
  const wide = type === "semi" || type === "monster" || type === "tank" || type === "truck";
  const shadow = water ? [0.02, 0.18, 0.22, 1] : [0.006, 0.008, 0.007, 1];
  const dust = water ? [0.1, 0.5, 0.62, 1] : shade(paint, 0.24);
  const padW = (air ? 1.65 : wide ? 2.45 : 1.95) * scale;
  const padD = (air ? 2.1 : wide ? 3.2 : 2.75) * scale;
  box(x, 0.012 * scale, z, padW, 0.024 * scale, padD, shadow);
  box(x, 0.018 * scale, z + 0.42 * scale, padW * 0.72, 0.014 * scale, padD * 0.32, shade(shadow, 1.4));
  if (air || water) {
    box(x, 0.045 * scale, z + 0.88 * scale, padW
```

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```
* 0.58, 0.018 * scale, 0.36 * scale, dust);
    return;
  }
  const tireX = wide ? 0.86 : 0.64;
  const rearZ = wide ? 1.0 : 0.78;
  const frontZ = wide ? -0.95 : -0.82;
  box(x - tireX * scale, 0.044 * scale, z + rearZ * scale, 0.42 * scale, 0.03 * scale, 0.34 * scale, shadow);
  box(x + tireX * scale, 0.044 * scale, z + rearZ * scale, 0.42 * scale, 0.03 * scale, 0.34 * scale, shadow);
  box(x - tireX * scale, 0.044 * scale, z + frontZ * scale, 0.36 * scale, 0.03 * scale, 0.3 * scale, shadow);
  box(x + tireX * scale, 0.044 * scale, z + frontZ * scale, 0.36 * scale, 0.03 * scale, 0.3 * scale, shadow);
  box(x - tireX * scale, 0.72 * scale, 0.038 * scale, z + (rearZ + 0.54) * scale, 0.13 * scale, 0.025 * scale, 0.9 * scale,
shade(shadow, 1.25));
  box(x + tireX * scale, 0.72 * scale, 0.038 * scale, z + (rearZ + 0.54) * scale, 0.13 * scale, 0.025 * scale, 0.9 * scale,
shade(shadow, 1.25));
}

function wrapZ(index, spacing, offset, speed = 0.07) {
  const total = spacing * 22;
  let z = index * spacing - ((offset * speed * 2.35) % total);
  while (z < 0) z += total;
  return z + 8;
}

function roadCurveValue(data) {
  const state = data && data.raceState ? data.raceState : {};
  return Math.max(-1.34, Math.min(1.34, Number(state.roadTurn || state.roadCurve || 0)));
}

function roadWorldXFor(turn, offset, x, z) {
  const depth = Math.max(0, Math.min(1.12, Number(z) / 150));
  const sweep = turn * depth * depth * 8.4;
  const surfaceWave = Math.sin((Number(z) || 0) * 0.035 + (Number(offset) || 0) * 0.0028) * Math.abs(turn) * depth * 0.72;
  return x + sweep + surfaceWave;
}

function roadWorldX(data, x, z) {
  return roadWorldXFor(roadCurveValue(data), data.raceState.roadOffset || 0, x, z);
}

function addRoad(data) {
  const place = data.selectedRace.place;
  const theme = data.selectedRace.theme || ["#09100f", "#46d9ff", "#ffd166"];
  const accent = hexToRgba(theme[1], 1);
  const accent2 = hexToRgba(theme[2], 1);
  const turn = roadCurveValue(data);
  const phoneFrame = data.width <= 940 || data.height <= 540;
  const roadColors = {
    snow: [0.34, 0.42, 0.41, 1],
    harbor: [0.12, 0.25, 0.3, 1],
    airfield: [0.24, 0.24, 0.22, 1],
    farm: [0.31, 0.25, 0.16, 1],
    desert: [0.42, 0.31, 0.18, 1],
    tokyo: [0.12, 0.11, 0.18, 1],
    rainforest: [0.12, 0.18, 0.14, 1],
    europe: [0.18, 0.24, 0.25, 1]
  };
  const groundColors = {
    coast: [0.05, 0.2, 0.2, 1],
    city: [0.06, 0.07, 0.08, 1],
    canyon: [0.28, 0.13, 0.08, 1],
    alpine: [0.05, 0.14, 0.12, 1],
    harbor: [0.03, 0.16, 0.2, 1],
    snow: [0.6, 0.68, 0.66, 1],
    airfield: [0.24, 0.18, 0.12, 1],
    freight: [0.2, 0.22, 0.13, 1],
    farm: [0.12, 0.28, 0.12, 1],
    tokyo: [0.04, 0.03, 0.08, 1],
    desert: [0.48, 0.32, 0.16, 1],
    rainforest: [0.03, 0.18, 0.1, 1],
    europe: [0.07, 0.18, 0.16, 1]
  };
  let roadColor = roadColors[place] || [0.14, 0.16, 0.15, 1];
  let groundColor = groundColors[place] || [0.04, 0.08, 0.06, 1];
  let roadMark = [0.78, 0.82, 0.78, 1];
  let roadMarkDim = [0.48, 0.52, 0.5, 1];
  if (phoneFrame) {
    roadColor = shade(roadColor, place === "snow" ? 0.56 : 0.38);
    groundColor = shade(groundColor, 0.34);
    roadMark = [0.34, 0.37, 0.36, 1];
    roadMarkDim = [0.18, 0.2, 0.2, 1];
  }
  quad([-70, -0.05, 0], [70, -0.05, 0], [70, -0.05, 170], [-70, -0.05, 170], groundColor);
  quad([roadWorldX(data, -5.8, 0), 0, 0], [roadWorldX(data, 5.8, 0), 0, 0], [roadWorldX(data, 7.4, 170), 0, 170],
[roadWorldX(data, -7.4, 170), 0, 170], roadColor);
  for (let segment = 0; segment < 12; segment += 1) {
    const z0 = segment * 14.2;
    const z1 = (segment + 1) * 14.2;
    const t0 = z0 / 170;
    const t1 = z1 / 170;
    const half0 = 5.8 + t0 * 1.6;
    const half1 = 5.8 + t1 * 1.6;
    const tone = phoneFrame ? (segment % 2 ? 0.82 : 0.92) : (segment % 2 ? 0.98 : 1.06);
    quad(
      [roadWorldX(data, -half
```

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```
f0, z0), 0.006, z0],
    [roadWorldX(data, half0, z0), 0.006, z0],
    [roadWorldX(data, half1, z1), 0.006, z1],
    [roadWorldX(data, -half1, z1), 0.006, z1],
    shade(roadColor, tone)
  );
}
quad([roadWorldX(data, -8.7, 0), -0.02, 0], [roadWorldX(data, -5.9, 0), -0.02, 0], [roadWorldX(data, -7.7, 170), -0.02, 170],
[roadWorldX(data, -10.2, 170), -0.02, 170], shade(groundColor, 0.72));
quad([roadWorldX(data, 5.9, 0), -0.02, 0], [roadWorldX(data, 8.7, 0), -0.02, 0], [roadWorldX(data, 10.2, 170), -0.02, 170],
[roadWorldX(data, 7.7, 170), -0.02, 170], shade(groundColor, 0.72));
for (let i = 0; i < 42; i += 1) {
  const z = wrapZ(i, 4.1, data.raceState.roadOffset, 0.16);
  const x = ((i * 1.37) % 10.6) - 5.3;
  const patch = phoneFrame ? (i % 3 ? shade(roadColor, 0.72) : shade(roadColor, 0.98)) : (i % 3 ? shade(roadColor, 0.78) :
shade(roadColor, 1.18));
  box(roadWorldX(data, x, z), 0.026, z, 0.38 + (i % 4) * 0.18, 0.014, 1.1 + (i % 5) * 0.28, patch);
}
for (let i = 0; i < 36; i += 1) {
  const z = wrapZ(i, 5.1, data.raceState.roadOffset, 0.16);
  const color = i % 2 ? roadMarkDim : shade(roadMarkDim, 0.72);
  box(roadWorldX(data, -6.25, z), 0.046, z, 0.16, 0.016, 1.05, color);
  box(roadWorldX(data, 6.25, z), 0.046, z, 0.16, 0.016, 1.05, color);
}
for (let i = 0; i < 34; i += 1) {
  const z = wrapZ(i, 5.6, data.raceState.roadOffset, 0.14);
  box(roadWorldX(data, -5.45, z), 0.04, z, 0.14, 0.018, 0.24, roadMark);
  box(roadWorldX(data, 5.45, z), 0.04, z, 0.14, 0.018, 0.24, roadMark);
}
for (let lane = -1; lane <= 1; lane += 1) {
  for (let i = 0; i < 24; i += 1) {
    const z = wrapZ(i, 7.2, data.raceState.roadOffset, 0.1);
    box(roadWorldX(data, lane * 2.08, z), 0.034, z, 0.1, 0.016, 1.55, [0.86, 0.9, 0.86, 1]);
  }
}
for (let i = 0; i < 28; i += 1) {
  const z = wrapZ(i, 6.5, data.raceState.roadOffset, 0.13);
  const side = i % 2 ? -1 : 1;
  barrierRun(roadWorldX(data, side * 8.4, z), z, i % 3 ? roadMark : roadMarkDim);
  box(roadWorldX(data, side * 6.55, z + 1.9), 0.08, z + 1.9, 0.45, 0.05, 1.0, i % 2 ? [0.85, 0.86, 0.82, 1] : [0.72, 0.08,
0.06, 1]);
  box(roadWorldX(data, side * 6.55, z - 1.9), 0.08, z - 1.9, 0.45, 0.05, 1.0, i % 2 ? [0.72, 0.08, 0.06, 1] : [0.85, 0.86,
0.82, 1]);
}
if (Math.abs(turn) > 0.22) {
  const side = turn > 0 ? 1 : -1;
  for (let i = 0; i < 12; i += 1) {
    const z = wrapZ(i, 9.2, data.raceState.roadOffset, 0.14);
    box(roadWorldX(data, side * 7.25, z), 0.75, z, 0.18, 1.0, 0.12, roadMarkDim);
    box(roadWorldX(data, side * 7.05, z - 0.18), 1.32, z - 0.18, 1.2, 0.34, 0.12, i % 2 ? accent : accent2);
    box(roadWorldX(data, side * 7.05, z - 0.34), 1.32, z - 0.34, 0.48, 0.08, 0.14, [0.9, 0.94, 0.88, 1]);
  }
}
if (place === "city" || place === "tokyo") {
  for (let i = 0; i < 10; i += 1) {
    const z = wrapZ(i, 10, data.raceState.roadOffset, 0.09);
    for (let stripe = -2; stripe <= 2; stripe += 1) {
      box(roadWorldX(data, stripe * 1.15, z), 0.034, z, 0.72, 0.018, 1.2, [0.85, 0.86, 0.82, 1]);
    }
  }
}
if ([ "city", "tokyo", "coast", "harbor", "rainforest" ].includes(place)) {
  for (let i = 0; i < 22; i += 1) {
    const z = wrapZ(i, 7.8, data.raceState.roadOffset, 0.12);
    box(roadWorldX(data, Math.sin(i * 1.7) * 1.6, z), 0.032, z, 1.0 + (i % 4) * 0.42, 0.012, 0.16, i % 2 ? roadMarkDim :
roadMark);
  }
}
if ([ "city", "tokyo", "europe", "freight" ].includes(place)) {
  for (let i = 0; i < 8; i += 1) {
    const z = wrapZ(i, 22, data.raceState.roadOffset, 0.08);
    box(-6.8, 3.1, z, 0.22, 6.2, 0.22, [0.34, 0.36, 0.34, 1]);
    box(6.8, 3.1, z, 0.22, 6.2, 0.22, [0.34, 0.36, 0.34, 1]);
    box(0, 6.12, z, 13.8, 0.22, 0.3, [0.34, 0.36, 0.34, 1]);
    box(-2.8, 5.72, z - 0.18, 2.4, 0.3, 0.16, i % 2 ? accent : [0.08, 0.1, 0.1, 1]);
    box(2.8, 5.72, z - 0.18, 2.4, 0.3, 0.16, i % 2
```

webgl-pipeline.js continued 7/12

```
? [0.08, 0.1, 0.1, 1] : accent);
    }
}

function addScenery(data) {
    const place = data.selectedRace.place;
    const theme = data.selectedRace.theme || ["#09100f", "#46d9ff", "#ffd166"];
    const accent = hexToRgba(theme[1], 1);
    const accent2 = hexToRgba(theme[2], 1);
    const offset = data.raceState.roadOffset;
    const phoneFrame = data.width <= 940 || data.height <= 540;
    const count = phoneFrame ? 12 : 18;
    for (let i = 0; i < count; i += 1) {
        const z = wrapZ(i, 12.6, offset, 0.038);
        const side = i % 2 ? -1 : 1;
        const x = roadWorldX(data, side * (phoneFrame ? 9.8 + (i % 4) * 2.2 : 11.2 + (i % 5) * 3.2), z);
        if (place === "city" || place === "tokyo") {
            const height = 5 + (i % 7) * 2.2;
            taperedBox(x, height / 2, z, 2.6 + (i % 3), 2.1 + (i % 2), height, 2.8, place === "tokyo" ? [0.08, 0.06, 0.16, 1] : [0.08,
0.1, 0.11, 1]);
            box(x, height * 0.55, z - 1.42, 1.7, 0.14, 0.08, i % 2 ? accent : accent2);
            box(x, height * 0.32, z - 1.45, 0.24, 0.1, 0.1, i % 2 ? accent2 : accent);
            box(x, height * 0.72, z - 1.45, 0.24, 0.1, 0.1, i % 2 ? accent : accent2);
            if (place === "tokyo" && i % 5 === 0) {
                box(side * 6.6, 3.1, z + 1.2, 0.2, 4.5, 0.18, accent2);
                box(side * 6.95, 3.1, z + 1.2, 0.2, 4.5, 0.18, accent);
            }
            if (i % 4 === 0) streetLight(side * 7.55, z + 1.1, place === "tokyo" ? accent2 : accent);
            if (i % 14 === 0) signPanel(x - side * 2.8, z + 1.5, place === "tokyo" ? accent2 : accent, [0.04, 0.05, 0.06, 1]);
            if (i % 9 === 0) spectatorCluster(side * 8.9, z + 2.4, accent, accent2);
            if (i % 13 === 0) animalGroup(side * 9.8, z + 2.9, "dog", side);
            if (i % 5 === 0) livingCrowdLine(roadWorldX(data, side * 9.6, z + 1.7), z + 1.7, side, accent, accent2, place === "tokyo"
? "stadium" : "crowd");
            if (i % 6 === 0) cyclistPair(roadWorldX(data, side * 8.25, z + 0.7), z + 0.7, side, accent, accent2);
        } else if (place === "farm") {
            box(x, 0.8, z, 2.2, 1.6, 2.4, i % 2 ? [0.5, 0.1, 0.08, 1] : [0.72, 0.66, 0.38, 1]);
            taperedBox(x, 1.95, z - 1.25, 2.45, 1.3, 0.6, 2.55, [0.44, 0.22, 0.12, 1]);
            box(side * 8.5, 0.35, z + 3, 0.12, 0.7, 4, [0.72, 0.56, 0.32, 1]);
            if (i % 4 === 0) roadsideTree(side * 10.8, z + 1.5, [0.12, 0.44, 0.14, 1]);
            if (i % 5 === 0) box(x + side * 3.4, 0.65, z - 1, 2.2, 1.3, 1.8, [0.86, 0.72, 0.34, 1]);
            if (i % 9 === 0) spectatorCluster(side * 9.4, z + 2, accent, accent2);
            if (i % 6 === 0) animalGroup(side * 11.6, z + 2.8, i % 12 === 0 ? "cow" : "sheep", side);
            if (i % 4 === 0) pastureLine(roadWorldX(data, side * 12.2, z + 2.8), z + 2.8, side, i % 8 === 0 ? "cow" : "sheep", accent,
accent2);
            if (i % 7 === 0) serviceCrew(roadWorldX(data, side * 9.6, z - 1.2), z - 1.2, side, accent, accent2, true);
        } else if (place === "freight" || place === "interstate") {
            box(x, 0.9, z, 4.8, 1.8, 2.2, i % 2 ? [0.22, 0.25, 0.24, 1] : [0.68, 0.7, 0.72, 1]);
            box(x + side * 3.2, 1.4, z + 1.4, 1.1, 2.8, 1.1, [0.12, 0.14, 0.14, 1]);
            if (i % 14 === 0) signPanel(x - side * 4.1, z + 1.8, accent, [0.04, 0.07, 0.06, 1]);
            if (i % 7 === 0) box(side * 9.4, 0.9, z - 2.4, 3.6, 1.8, 1.9, [0.52, 0.08, 0.06, 1]);
            if (i % 8 === 0) streetLight(side * 7.7, z + 2.1, accent2);
            if (i % 5 === 0) serviceCrew(roadWorldX(data, side * 10.6, z + 1.4), z + 1.4, side, accent, accent2, i % 10 === 0);
            if (place === "interstate" && i % 6 === 0) livingCrowdLine(roadWorldX(data, side * 11.4, z - 1.2), z - 1.2, side, accent,
accent2, "crowd");
        } else if (place === "desert" || place === "canyon") {
            lowMound(x, z, 4.2 + (i % 3) * 1.4, 1.2 + (i % 5) * 0.28, 3.3, place === "canyon" ? [0.55, 0.22, 0.12, 1] : [0.74, 0.48,
0.22, 1]);
            if (i % 6 === 0) lowMound(side * 18.5, z + 2.7, 7.5, 2.6, 5.2, place === "canyon" ? [0.64, 0.25, 0.13, 1] : [0.78, 0.5,
0.22, 1]);
            if (i % 16 === 0) box(side * 10.2, 0.6, z + 1.2, 2.2, 1.2, 1.2, [0.12, 0.08, 0.04, 1]);
        }
    }
}
```

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```
if (i % 11 === 0) animalGroup(side * 10.8, z + 2.4, place === "desert" ? "camel" : "deer", side);
if (i % 9 === 0) livingCrowdLine(roadWorldX(data, side * 10.4, z + 1.8), z + 1.8, side, accent, accent2, "crowd");
} else if (place === "rainforest") {
  taperedBox(x, 2.0, z, 0.7, 0.42, 4, 0.7, [0.12, 0.22, 0.12, 1]);
  taperedBox(x, 4.3, z, 3.6, 2.2, 1.8, 2.5, i % 2 ? [0.1, 0.42, 0.22, 1] : [0.18, 0.52, 0.18, 1]);
  if (i % 4 === 0) roadsideTree(side * 8.6, z + 1.8, [0.08, 0.38, 0.18, 1]);
  if (i % 11 === 0) spectatorCluster(side * 8.7, z + 1.4, accent, accent2);
  if (i % 9 === 0) animalGroup(side * 10.5, z + 2.4, "monkey", side);
  if (i % 7 === 0) livingCrowdLine(roadWorldX(data, side * 9.8, z + 1.2), z + 1.2, side, accent, accent2, "crowd");
} else if (place === "snow" || place === "alpine" || place === "europe") {
  lowMound(x, z, 5.8 + (i % 4) * 1.3, 2.3 + (i % 5) * 0.5, 3.8, [0.74, 0.82, 0.84, 1]);
  if (i % 3 === 0) box(x + side * 2.5, 1.2, z + 2.2, 1.8, 2.4, 1.7, [0.08, 0.2, 0.16, 1]);
  if (i % 5 === 0) roadsideTree(side * 9.2, z + 1.5, [0.08, 0.24, 0.18, 1]);
  if (place === "europe" && i % 8 === 0) spectatorCluster(side * 8.8, z + 2.2, accent, accent2);
  if (i % 10 === 0) animalGroup(side * 10.4, z + 2.5, place === "europe" ? "sheep" : "deer", side);
  if (place === "europe" && i % 5 === 0) cyclistPair(roadWorldX(data, side * 8.4, z + 0.8), z + 0.8, side, accent, accent2);
  if (i % 6 === 0) livingCrowdLine(roadWorldX(data, side * 9.7, z + 1.8), z + 1.8, side, accent, accent2, place === "europe"
? "crowd" : "farm");
} else if (place === "harbor" || place === "coast") {
  box(x, 0.22, z, 5.5, 0.25, 6.5, [0.04, 0.22, 0.28, 1]);
  box(x + side * 1.4, 0.9, z, 0.28, 1.8, 0.28, [0.58, 0.44, 0.26, 1]);
  if (i % 6 === 0) {
    box(side * 14.2, 2.1, z, 0.24, 4.2, 0.24, [0.72, 0.48, 0.2, 1]);
    box(side * 12.4, 4.0, z - 0.6, 3.8, 0.22, 0.24, [0.72, 0.48, 0.2, 1]);
    taperedBox(side * 15.4, 0.5, z + 1.8, 2.4, 1.5, 0.72, 3.2, accent2);
  }
  if (i % 4 === 0) roadsideTree(side * 10.4, z + 0.9, [0.18, 0.48, 0.2, 1], "palm");
  if (i % 5 === 0) taperedBox(x - side * 2.8, 0.65, z + 1.6, 2.2, 1.1, 0.55, 3.1, accent2);
  if (i % 9 === 0) spectatorCluster(side * 9.1, z + 2.8, accent, accent2);
  if (i % 10 === 0) animalGroup(side * 9.8, z + 2.2, "birds", side);
  if (i % 5 === 0) waterEdgeLife(roadWorldX(data, side * 10.8, z + 1.6), z + 1.6, side, accent, accent2);
  if (place === "coast" && i % 7 === 0) cyclistPair(roadWorldX(data, side * 8.3, z + 0.6), z + 0.6, side, accent, accent2);
} else if (place === "airfield") {
  box(x, 1.1, z, 4.4, 2.2, 3.4, [0.18, 0.18, 0.16, 1]);
  taperedBox(x, 2.35, z - 1.8, 4.8, 2.4, 0.6, 3.6, [0.24, 0.24, 0.22, 1]);
  if (i % 6 === 0) box(side * 8.8, 0.35, z + 2, 1.8, 0.7, 2.6, accent);
  if (i % 8 === 0) streetLight(side * 7.6, z + 1.7, accent2);
  if (i % 5 === 0) serviceCrew(roadWorldX(data, side * 9.6, z + 1.4), z + 1.4, side, accent, accent2, false);
  if (i % 9 === 0) livingCrowdLine(roadWorldX(data, side * 10.8, z + 2.1), z + 2.1, side, accent, accent2, "crowd");
} else if (place === "monsterpark" || place === "military" || place === "skybase" || place === "pursuit") {
  const baseColor = place === "military" ? [0.16, 0.2, 0.12, 1] : place === "monsterpark" ? [0.32, 0.17, 0.08, 1] : [0.12,
0.14, 0.16, 1];
  box(x, 0.78, z, 4.6, 1.55, 2.4, baseColor);
  if (place === "monsterpark" && i % 4 === 0) livingCrowdLine(roadWorldX(data, side * 9.8, z + 1.6), z + 1.6, side, accent,
accent2, "stadium");
  if (place === "military" && i % 5 === 0) serviceCrew(roadWorldX(data, side * 10.2, z + 1.4), z + 1.4, side, accent,
accent2, true);
  if (place === "skybase" && i % 5 === 0) serviceCrew(roadWorldX(data, side * 9.8, z + 1.4), z + 1.4, side, accent, accent2,
false);
  if (place === "pursuit" && i % 6 === 0) livingCrowdLine(roadWorldX(data,
```


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```
side * 9.6, z + 1.8), z + 1.8, side, accent, accent2, "crowd");
    }
}

function addVehicle(type, x, z, colorHex, scale = 1, police = false, damage = 0) {
    const paint = police ? [0.9, 0.92, 0.9, 1] : hexToRgba(colorHex, 1);
    const dark = [0.01, 0.015, 0.014, 1];
    const red = [1, 0.08, 0.12, 1];
    const blue = [0.1, 0.55, 1, 1];
    const glass = [0.52, 0.72, 0.78, 1];
    const chrome = [0.76, 0.82, 0.78, 1];
    groundContactPad(type, x, z, scale, paint);
    if (type === "semi") {
        taperedBox(x, 0.8 * scale, z + 0.95 * scale, 1.75 * scale, 1.25 * scale, 1.2 * scale, 1.6 * scale, paint);
        taperedBox(x, 0.95 * scale, z - 1.25 * scale, 2.05 * scale, 1.78 * scale, 1.6 * scale, 3.2 * scale, shade(paint, 0.78));
        box(x, 1.14 * scale, z + 1.45 * scale, 0.95 * scale, 0.16 * scale, 0.08 * scale, [0.72, 0.86, 0.9, 1]);
    } else if (type === "tractor") {
        taperedBox(x, 0.7 * scale, z, 1.45 * scale, 0.95 * scale, 0.8 * scale, 1.55 * scale, paint);
        taperedBox(x, 1.25 * scale, z - 0.2 * scale, 1.0 * scale, 0.72 * scale, 0.9 * scale, 0.8 * scale, shade(paint, 0.9));
        box(x - 0.75 * scale, 0.45 * scale, z + 0.2 * scale, 0.35 * scale, 0.65 * scale, 0.75 * scale, dark);
        box(x + 0.75 * scale, 0.45 * scale, z + 0.2 * scale, 0.35 * scale, 0.65 * scale, 0.75 * scale, dark);
    } else if (type === "fi" || type === "prototype") {
        taperedBox(x, 0.36 * scale, z, 0.9 * scale, 0.36 * scale, 0.34 * scale, 2.5 * scale, paint);
        taperedBox(x, 0.54 * scale, z - 0.2 * scale, 0.58 * scale, 0.32 * scale, 0.32 * scale, 0.55 * scale, dark);
        box(x, 0.28 * scale, z + 1.0 * scale, 1.5 * scale, 0.12 * scale, 0.25 * scale, shade(paint, 0.7));
    } else if (type === "truck" || type === "monster" || type === "tank") {
        taperedBox(x, 0.75 * scale, z, 1.85 * scale, 1.35 * scale, type === "tank" ? 0.75 * scale : 1.0 * scale, 2.1 * scale,
paint);
        taperedBox(x, 1.25 * scale, z - 0.25 * scale, 1.1 * scale, 0.78 * scale, 0.65 * scale, 0.9 * scale, shade(paint, 0.82));
        if (type === "tank") box(x, 1.55 * scale, z - 1.15 * scale, 0.22 * scale, 0.18 * scale, 1.8 * scale, shade(paint, 0.7));
    } else if (type === "boat" || type === "snowmobile") {
        taperedBox(x, 0.42 * scale, z, 1.45 * scale, 0.7 * scale, 0.36 * scale, 2.3 * scale, paint);
        taperedBox(x, 0.68 * scale, z - 0.25 * scale, 0.62 * scale, 0.35 * scale, 0.42 * scale, 0.75 * scale, dark);
    } else if (type === "airplane" || type === "fighterjet" || type === "bomber") {
        const bomber = type === "bomber";
        const fighter = type === "fighterjet";
        taperedBox(x, 0.7 * scale, z, (bomber ? 0.92 : fighter ? 0.58 : 0.68) * scale, 0.38 * scale, 0.45 * scale, (bomber ? 3.3 :
2.9) * scale, paint);
        box(x, 0.66 * scale, z, (bomber ? 3.2 : fighter ? 2.8 : 2.55) * scale, 0.13 * scale, 0.42 * scale, shade(paint, 0.88));
        if (fighter) {
            box(x - 0.52 * scale, 0.72 * scale, z + 1.1 * scale, 0.34 * scale, 0.08 * scale, 0.58 * scale, shade(paint, 0.72));
            box(x + 0.52 * scale, 0.72 * scale, z + 1.1 * scale, 0.34 * scale, 0.08 * scale, 0.58 * scale, shade(paint, 0.72));
        }
    } else if (type === "helicopter") {
        taperedBox(x, 0.95 * scale, z, 1.2 * scale, 0.78 * scale, 0.65 * scale, 1.45 * scale, paint);
        box(x, 1.45 * scale, z, 2.3 * scale, 0.08 * scale, 0.1 * scale, [0.9, 0.94, 0.94, 1]);
    } else if (type === "drone") {
        taperedBox(x, 0.72 * scale, z, 0.74 * scale, 0.3 * scale, 0.48 * scale, 0.74 * scale, paint);
        box(x - 0.72 * scale, 0.8 * scale, z - 0.62 * scale, 0.52 * scale, 0.04 * scale, 0.16 * scale, [0.9, 0.94, 0.94, 1]);
        box(x + 0.72 * scale, 0.8 * scale, z - 0.62 * scale, 0.52 * scale, 0.04 * scale, 0.16 * scale, [0.9, 0.94, 0.94, 1]);
        box(x - 0.72 * scale, 0.8 * scale, z + 0.62 * scale, 0.52 * scale, 0.04 * scale, 0.16 * scale, [0.9, 0.94, 0.94, 1]);
        box(x + 0.72 * scale, 0.8 * scale, z + 0.62 * scale, 0.52 * scale, 0.04 * scale, 0.16 * scale, [0.9, 0.94, 0.94, 1]);
    } else {
        taperedBox(x, 0.55 * scale, z, 1.46 * scale,
```

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```
0.9 * scale, 0.55 * scale, 2.1 * scale, paint));
taperedBox(x, 0.95 * scale, z - 0.25 * scale, 0.88 * scale, 0.55 * scale, 0.45 * scale, 0.82 * scale, dark);
box(x, 0.78 * scale, z + 0.8 * scale, 0.7 * scale, 0.08 * scale, 0.08 * scale, [0.86, 0.93, 1, 1]);
}
const groundVehicle = !isAirType(type);
if (groundVehicle) {
  box(x - 0.42 * scale, 0.54 * scale, z + 1.04 * scale, 0.28 * scale, 0.08 * scale, 0.08 * scale, [1, 0.92, 0.62, 1]);
  box(x + 0.42 * scale, 0.54 * scale, z + 1.04 * scale, 0.28 * scale, 0.08 * scale, 0.08 * scale, [1, 0.92, 0.62, 1]);
  box(x, 0.46 * scale, z + 1.08 * scale, 0.58 * scale, 0.06 * scale, 0.08 * scale, shade(dark, 1.5));
  box(x - 0.34 * scale, 0.58 * scale, z - 1.2 * scale, 0.34 * scale, 0.07 * scale, 0.08 * scale, police ? blue : red);
  box(x + 0.34 * scale, 0.58 * scale, z - 1.2 * scale, 0.34 * scale, 0.07 * scale, 0.08 * scale, police ? red : red);
  box(x, 0.42 * scale, z - 1.24 * scale, (type === "semi" || type === "truck" ? 1.38 : 0.82) * scale, 0.06 * scale, 0.08 *
scale, dark);
  if (type !== "semi" && type !== "truck" && type !== "tractor" && type !== "tank") {
    box(x - 0.22 * scale, 0.34 * scale, z - 1.32 * scale, 0.14 * scale, 0.08 * scale, 0.09 * scale, chrome);
    box(x + 0.22 * scale, 0.34 * scale, z - 1.32 * scale, 0.14 * scale, 0.08 * scale, 0.09 * scale, chrome);
    for (let fin = -2; fin <= 2; fin += 1) {
      box(x + fin * 0.1 * scale, 0.34 * scale, z - 1.2 * scale, 0.025 * scale, 0.14 * scale, 0.22 * scale, dark);
    }
  }
  box(x - 0.82 * scale, 0.66 * scale, z - 0.16 * scale, 0.08 * scale, 0.36 * scale, 0.72 * scale, shade(paint, 0.72));
  box(x + 0.82 * scale, 0.66 * scale, z - 0.16 * scale, 0.08 * scale, 0.36 * scale, 0.72 * scale, shade(paint, 0.84));
  box(x - 0.96 * scale, 0.88 * scale, z + 0.34 * scale, 0.18 * scale, 0.1 * scale, 0.2 * scale, shade(paint, 0.64));
  box(x + 0.96 * scale, 0.88 * scale, z + 0.34 * scale, 0.18 * scale, 0.1 * scale, 0.2 * scale, shade(paint, 0.72));
  box(x, 1.02 * scale, z + 0.38 * scale, 0.72 * scale, 0.08 * scale, 0.36 * scale, glass);
  box(x, 0.94 * scale, z - 0.66 * scale, 0.86 * scale, 0.08 * scale, 0.28 * scale, shade(glass, 0.6));
  box(x, 0.52 * scale, z - 1.22 * scale, 0.36 * scale, 0.055 * scale, 0.07 * scale, chrome);
  box(x, 0.42 * scale, z + 1.12 * scale, 0.78 * scale, 0.055 * scale, 0.08 * scale, dark);
  box(x - 0.46 * scale, 0.82 * scale, z - 0.22 * scale, 0.05 * scale, 0.52 * scale, 1.42 * scale, chrome);
  box(x + 0.46 * scale, 0.82 * scale, z - 0.22 * scale, 0.05 * scale, 0.52 * scale, 1.42 * scale, chrome);
  if (type === "f1" || type === "prototype" || type === "car") {
    box(x, 0.8 * scale, z - 1.16 * scale, 1.28 * scale, 0.1 * scale, 0.16 * scale, shade(paint, 0.65));
  }
  if (type === "semi" || type === "truck") {
    box(x, 1.25 * scale, z - 0.86 * scale, 1.55 * scale, 0.06 * scale, 0.12 * scale, chrome);
    box(x, 0.72 * scale, z - 1.84 * scale, 1.45 * scale, 0.08 * scale, 0.1 * scale, shade(paint, 1.18));
  }
  if (type === "policecar") {
    box(x - 0.2 * scale, 1.19 * scale, z - 0.18 * scale, 0.28 * scale, 0.08 * scale, 0.16 * scale, red);
    box(x + 0.2 * scale, 1.19 * scale, z - 0.18 * scale, 0.28 * scale, 0.08 * scale, 0.16 * scale, blue);
    box(x, 0.73 * scale, z - 1.0 * scale, 0.72 * scale, 0.055 * scale, 0.08 * scale, [0.94, 0.96, 0.96, 1]);
  }
} else {
  box(x - 0.32 * scale, 0.78 * scale, z + 1.12 * scale, 0.18 * scale, 0.08 * scale, 0.08 * scale, [1, 0.92, 0.62, 1]);
  box(x + 0.32 * scale, 0.78 * scale, z + 1.12 * scale, 0.18 * scale, 0.08 * scale, 0.08 * scale, [1, 0.92, 0.62, 1]);
}
if (damage > 18) {
  const dent = [0.02, 0.022, 0.02, 1];
  box(x - 0.28 * scale, 1.04 * scale, z - 0.34 * scale, 0.48 * scale, 0.035 * scale, 0.16 * scale, dent);
  box(x + 0.34 * scale, 0.86 * scale, z + 0.18 * scale, 0.42 * scale, 0.035 * scale, 0.14 * scale, dent);
  if (damage > 52) box(x - 0.1 * sca
```

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```
le, 0.72 * scale, z + 0.9 * scale, 0.36 * scale, 0.06 * scale, 0.1 * scale, [1, 0.18, 0.16, 1]);
    if (damage > 76) box(x + 0.18 * scale, 1.2 * scale, z - 0.86 * scale, 0.28 * scale, 0.08 * scale, 0.18 * scale, [0.05, 0.05,
0.045, 1]);
    }
    box(x - 0.64 * scale, 0.17 * scale, z + 0.72 * scale, 0.24 * scale, 0.34 * scale, 0.5 * scale, dark);
    box(x + 0.64 * scale, 0.17 * scale, z + 0.72 * scale, 0.24 * scale, 0.34 * scale, 0.5 * scale, dark);
    box(x - 0.64 * scale, 0.17 * scale, z - 0.72 * scale, 0.24 * scale, 0.34 * scale, 0.5 * scale, dark);
    box(x + 0.64 * scale, 0.17 * scale, z - 0.72 * scale, 0.24 * scale, 0.34 * scale, 0.5 * scale, dark);
    box(x - 0.64 * scale, 0.2 * scale, z + 0.72 * scale, 0.1 * scale, 0.055 * scale, 0.2 * scale, chrome);
    box(x + 0.64 * scale, 0.2 * scale, z + 0.72 * scale, 0.1 * scale, 0.055 * scale, 0.2 * scale, chrome);
    box(x - 0.64 * scale, 0.2 * scale, z - 0.72 * scale, 0.1 * scale, 0.055 * scale, 0.2 * scale, chrome);
    box(x + 0.64 * scale, 0.2 * scale, z - 0.72 * scale, 0.1 * scale, 0.055 * scale, 0.2 * scale, chrome);
    box(x - 0.32 * scale, 0.62 * scale, z - 1.15 * scale, 0.36 * scale, 0.08 * scale, 0.08 * scale, police ? blue : red);
    box(x + 0.32 * scale, 0.62 * scale, z - 1.15 * scale, 0.36 * scale, 0.08 * scale, 0.08 * scale, police ? red : red);
}

function addMovingObjects(data) {
    const defs = data.vehicleDefs || [];
    const getVehicle = (id) => defs.find((vehicle) => vehicle.id === id) || defs[0] || { type: "car", color: "#46d9ff" };
    const screenYFromRoadDistance = (distance) => data.height * 0.68 - (distance - data.raceState.distance) * 0.11;
    const distanceFromScreenY = (y) => {
        const clampedY = Math.max(data.height * 0.34, Math.min(data.height * 0.98, Number(y) || data.height * 0.34));
        return data.raceState.distance + (data.height * 0.68 - clampedY) / 0.11;
    };
    const zFromRoadDistance = (distance) => {
        const y = screenYFromRoadDistance(distance);
        const t = Math.max(0.32, Math.min(1.02, y / Math.max(1, data.height)));
        return 116 - t * 108;
    };
    const projectRoadObject = (object) => {
        const distance = Number.isFinite(Number(object.distance)) ? Number(object.distance) : distanceFromScreenY(object.y);
        const y = screenYFromRoadDistance(distance);
        if (y < data.height * 0.32 || y > data.height * 1.08) return null;
        return { distance, z: zFromRoadDistance(distance) };
    };
    data.raceState.opponents.forEach((opponent) => {
        const projected = projectRoadObject(opponent);
        if (!projected) return;
        const z = projected.z;
        if (z < 4 || z > 132) return;
        const vehicle = getVehicle(opponent.vehicleId);
        addVehicle(vehicle.type, roadWorldX(data, laneToWorldX(opponent.lane), z), z, opponent.color || vehicle.color,
opponent.wrecked ? 0.9 : 0.95, false, opponent.damage || 0);
    });
    data.raceState.rivals.forEach((rival) => {
        const projected = projectRoadObject(rival);
        if (!projected) return;
        const z = projected.z;
        if (z < 4 || z > 132) return;
        addVehicle(rival.type || "car", roadWorldX(data, laneToWorldX(rival.lane), z), z, rival.color, rival.type === "semi" ? 1.08
: 0.9, false, rival.damage || 0);
    });
    data.raceState.police.forEach((unit) => {
        const projected = projectRoadObject(unit);
        if (!projected) return;
        const z = projected.z;
        if (z < 4 || z > 132) return;
        addVehicle("car", roadWorldX(data, laneToWorldX(unit.lane), z), z, "#f4fbf8", 0.98, true, unit.damage || 0);
    });
    data.raceState.coinsOnRoad.forEach((coin) => {
        const projected = projectRoadObject(coin);
        if (!projected) return;
        const z = projected.z;
        if (z < 4 || z > 132) return;
        box(roadWorldX(data, laneToWorldX(coin.lane), z), 0.52, z, 0.52, 0.72, 0.12, [1, 0.77, 0.26, 1]);
    });
    if (data.cameraMode === "chase") {
        addVehicle(data.selectedVehicle.type, roadWorldX(data, laneToWorldX(data.raceState.lane), 5.2), 5.2,
data.selectedVehicle.color, data
```

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```
.selectedVehicle.type === "semi" ? 1.2 : 1.05, false, data.raceState.damage || 0);
    } else if (data.cameraMode === "hood") {
        const hoodX = roadWorldX(data, laneToWorldX(data.raceState.lane), 1.2);
        box(hoodX, 0.22, 1.2, 2.2, 0.28, 2.3, hexToRgba(data.selectedVehicle.color, 1));
        if ((data.raceState.damage || 0) > 25) box(hoodX - 0.5, 0.39, 1.68, 0.5, 0.04, 0.16, [0.02, 0.022, 0.02, 1]);
    }
}

function render(data) {
    vertices = [];
    const theme = data.selectedRace.theme || ["#09100f", "#46d9ff", "#ffd166"];
    const clear = hexToRgba(theme[0], 1);
    const phoneFrame = data.width <= 940 || data.height <= 540;
    gl.viewport(0, 0, gl.canvas.width, gl.canvas.height);
    gl.clearColor(clear[0] * 0.72, clear[1] * 0.72, clear[2] * 0.72, 1);
    gl.clear(gl.COLOR_BUFFER_BIT | gl.DEPTH_BUFFER_BIT);
    gl.enable(gl.DEPTH_TEST);
    gl.disable(gl.CULL_FACE);

    const projection = cameraProjection(data);
    const mvp = projection.mvp;
    lastProjection = {
        ...projection,
        data
    };

    addRoad(data);
    addScenery(data);

    gl.useProgram(program);
    gl.uniformMatrix4fv(uMvp, false, new Float32Array(mvp));
    gl.bindBuffer(gl.ARRAY_BUFFER, buffer);
    gl.bufferData(gl.ARRAY_BUFFER, new Float32Array(vertices), gl.DYNAMIC_DRAW);
    gl.enableVertexAttribArray(aPosition);
    gl.vertexAttribPointer(aPosition, 3, gl.FLOAT, false, 28, 0);
    gl.enableVertexAttribArray(aColor);
    gl.vertexAttribPointer(aColor, 4, gl.FLOAT, false, 28, 12);
    gl.drawArrays(gl.TRIANGLES, 0, vertices.length / 7);
}

function resize(width, height) {
    if (canvas.width !== width) canvas.width = width;
    if (canvas.height !== height) canvas.height = height;
}

function projectRoadPoint(lane, distance) {
    if (!lastProjection || !lastProjection.data) return null;
    const data = lastProjection.data;
    const z = zFromRoadDistance(data, Number(distance));
    return projectPoint(lastProjection, roadWorldXFor(lastProjection.roadTurn || 0, lastProjection.roadOffset || 0,
        laneToWorldX(lane), z), 0, z);
}

function projectWorldRoadPoint(lane, z) {
    if (!lastProjection) return null;
    return projectPoint(lastProjection, roadWorldXFor(lastProjection.roadTurn || 0, lastProjection.roadOffset || 0,
        laneToWorldX(lane), z), 0, z);
}

return {
    ready: true,
    render,
    resize,
    projectRoadPoint,
    projectWorldRoadPoint
};
}

window.VelocityWebGLPipeline = VelocityWebGLPipeline;
})();
```

manifest.json

```
{
  "name": "Velocity Vault",
  "short_name": "Velocity",
  "description": "A standalone racing game with local profiles, upgrades, missions, controller support, and offline play.",
  "id": "/",
  "start_url": "./index.html",
  "scope": "./",
  "display": "standalone",
  "orientation": "any",
  "background_color": "#09100f",
  "theme_color": "#09100f",
  "categories": ["games", "entertainment"],
  "icons": [
    {
      "src": "app-icon.svg",
      "sizes": "any",
      "type": "image/svg+xml",
      "purpose": "any maskable"
    }
  ],
  "shortcuts": [
    {
      "name": "Race Hub",
      "short_name": "Hub",
      "description": "Open Velocity Vault at the driver gate.",
      "url": "./index.html"
    }
  ]
}
```

sw.js

```
"use strict";

const cacheName = "velocity-vault-v92-weather-disaster-trains";
const appShell = [
  "./",
  "./index.html",
  "./styles.css",
  "./phone-asset-pipeline.js",
  "./webgl-pipeline.js",
  "./game.js",
  "./manifest.json",
  "./app-icon.svg"
];

self.addEventListener("install", (event) => {
  event.waitUntil(
    caches.open(cacheName)
      .then((cache) => cache.addAll(appShell))
      .then(() => self.skipWaiting())
  );
});

self.addEventListener("activate", (event) => {
  event.waitUntil(
    caches.keys()
      .then((keys) => Promise.all(keys.filter((key) => key !== cacheName).map((key) => caches.delete(key))))
      .then(() => self.clients.claim())
  );
});

self.addEventListener("fetch", (event) => {
  if (event.request.method !== "GET") return;
  event.respondWith(
    caches.match(event.request).then((cached) => {
      if (cached) return cached;
      return fetch(event.request).then((response) => {
        const copy = response.clone();
        caches.open(cacheName).then((cache) => cache.put(event.request, copy));
        return response;
      }).catch(() => caches.match("./index.html"));
    })
  );
});
});
```

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Velocity Vault

A standalone browser racing game built with local-only profiles, age-band selection, upgrades, missions, and canvas racing.

Play

Open `index.html` in a browser.

Use `Play` to launch straight into a race, or `Enter Garage` to review vehicles, upgrades, missions, and AI tuning first.

Controls:

- `Left/Right` or `A/D`: steer
- `Up` or `W`: accelerate
- `Down` or `S`: brake, then reverse when stopped
- `Space`: boost while accelerating
- `P`: pause
- `R`: reset the vehicle after a crash or bad position
- On touch screens, use the on-screen controls. Phone Mini mode now uses a floating analog joystick: touch the driving screen, drag left/right to change lanes, and the car keeps accelerating while you steer. Pull the stick down only when you want brake/reverse. Full controls are still available from the `Size` button.
- Landscape scrolling stays available in menus and the garage; the full-screen no-scroll layout only turns on while a race is live.
- Bluetooth/gamepad controllers: pair the controller with the device, press any controller button in the browser, then use left stick or D-pad to steer, RT / A to accelerate, LT / X to brake or reverse, B / RB to boost, Y to reset, and Start / Select to pause

Driver views:

- `Full Car`: third-person chase camera with the full performance car visible.
- `Half Car`: low hood-style view with the front of the car filling the lower screen.
- `Windshield`: cockpit-style view looking out through the windshield with dashboard framing.
- `WebGL 3D`: hardware-accelerated 3D racing pipeline with a classic 2D fallback button for older browsers.

How To Play

1. Choose `Play` to start immediately, or `Enter Garage` to pick a vehicle, route, upgrades, and missions.
2. Read the goal card at the start of each race. Goals can ask you to collect route markers, keep focus high, dodge rivals, or reach a target score.
3. Hold gas to build speed. On phones, use Mini mode by touching the driving screen; the floating joystick appears under your thumb and keeps gas active while you drag left or right.
4. Steer smoothly to pass opponents. Watch the lane radar, distance badges, and opponent names to see where rivals are sitting on the road.
5. Pull the phone joystick down, press `Down/S`, or use controller LT to brake. Once stopped, brake becomes reverse.
6. Use boost only while accelerating. Boost helps pass rivals, escape police heat, and recover speed after corners.
7. Crashes do not end the race. Heavy damage starts a reset countdown, then the car is stabilized so you can continue.
8. Save a race from the live race controls, then use `Resume Saved Race` in the Race Hub to continue that driver profile later.

Real-world style visuals:

- Race routes use real-world inspired locations across the country and world: Pacific Coast Highway, Chicago lakefront skyline, Sedona red rock canyon, Rocky Mountain pass, Miami harbor, Aspen snowfields, Nevada airfield, Texas freight interstate, Iowa farm roads, Tokyo expressway, Sahara desert, Amazon rainforest, and Swiss Alps.
- Scenery includes skyline buildings, bridge cables, water, canyon walls, mountain silhouettes, street lamps, wet asphalt glare, roadside route signs, farmland, big-rig freight stops, neon towers, desert dunes, rainforest canopy, alpine villages, and pass-specific colors.
- The v57 local GenAI scene director adds original route-specific racing scenes at runtime: trackside crowds, camera crews, brake boards, barriers, gantries, cranes, hangars, barns, cliffs, dunes, snowbanks, neon towers, tunnel frames, and cleaner phone-safe route atmosphere.
- Drifting now has stronger style cues with tire smoke, dust or snow wash by surface, darker skid arcs, racing-line marks, and a `DRIFT` callout during bigger slides.
- Vehicles are original generic classes with shaded 3D-style silhouettes: street supercars, F1 open-wheel racers, grand prix prototypes, performance trucks, semi truck racers, racing tractors, monster trucks, armored tanks, snowmobiles, race boats, helicopters, and sport airplanes. No real automaker, military, or aircraft trademarks or licensed models are used.
- Opponents now get generated race identity overlays with

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invented race numbers, livery stripes, driver/helmet silhouettes in glass areas, damage glow, and clearer name/position cues so rivals feel more like named racers instead of anonymous traffic.

- Engine, tire, boost, shield, and magnet upgrades apply across the whole roster, so semis and tractors can be upgraded into faster and more agile racing builds too.
- A local WebGL game-engine-style renderer (``webgl-pipeline.js``) draws the 3D road, route scenery, opponents, traffic, and vehicle models. It uses browser WebGL directly so the app stays standalone and secure without CDN scripts.
- A phone-focused local asset pipeline (``phone-asset-pipeline.js``) now generates cached high-resolution vehicle sprites, road texture maps, paint grain, glass, rim detail, light clusters, damage marks, and bloom overlays directly on the device. This gives the installed phone app richer assets without downloading licensed models or relying on remote servers.
- Phone graphics now add an extra cinematic realism pass with denser asphalt texture, rubber tire lanes, wet/oil reflections, stronger foreground depth, subtle film grain, route-tinted lighting, and higher-resolution vehicle paint/glass highlights. The harsh angular road marks were softened so the road reads like asphalt instead of spikes. On phones, the app uses the clean mobile canvas renderer plus the lower hood camera so the first race stays readable and avoids stacked road layers; the camera buttons still let players switch back to full-car or windshield views.
- Phone graphics now add a console-style chase pass inspired by modern open-world and street-racing presentation: lower camera framing, darker asphalt, softer horizon scenery, wet reflection ribbons, peripheral speed blur, headlight wash, and cleaner road markings so the phone view feels less like a flat arcade lane game.
- Phone clean-road mode now removes the remaining spike-like road artifacts on phones by disabling texture scratches, racing-line glow, tire/skid curves, wet streak ribbons, diagonal speed marks, extra contrast dashes, and diagonal snow/rain strokes while keeping flat asphalt, shoulders, and grounded lane dashes.
- The track now has stronger sweeping bends and route-country presentation for the phone/browser app. The same turn model drives the WebGL camera, road mesh, lane projection, traffic projection, road-edge chevrons, and phone overlay so the course feels like it is bending through cities, highways, coast roads, mountains, and world routes instead of staying on a flat straight strip.
- Road paint is now ground-locked with short dash plates and low pavement bands instead of long rising guide lines, so lane markings stay visually attached to the road while vehicles remain planted on the pavement.
- WebGL phone mode no longer draws the older transparent canvas road sheet over the 3D road. The actual WebGL pavement stays visible, while the overlay is limited to sky, scenery, route signs, and low flat pavement bands so nothing reads like a see-through ramp.
- Tokyo/city atmosphere markers now render as low horizontal neon glints instead of tall diagonal guide strokes, removing the last see-through marker lines that appeared to angle upward over the road.
- Phone WebGL mode now covers the pale see-through road sheet with an opaque, low asphalt surface while keeping WebGL projection for the planted vehicles. Mini-toggle mode also latches drive-stick direction until neutral is tapped, and the opponent grid is spread across longer gaps and separate lanes for clearer passing.
- Phone WebGL mode now removes the remaining mid-screen transparent road wash and moves the opaque asphalt lock higher under the horizon, so the race view no longer shows a see-through angled ramp. The compact 4-way stick now owns all mini-stick input by itself, and the rival grid starts much farther apart across separate lanes so you can chase, pass, and be passed instead of running into a bunched pack.
- Phone WebGL mode now removes the angled pavement trapezoid entirely on small screens and uses only horizontal low-ground shading, so the extra invisible road plane is no longer drawn over the real WebGL road. Phone scenery also avoids sharp f

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loading triangle shapes, and the compact stick now has direct button fallback so pressing the up arrow always accelerates even on browsers that do not deliver the parent stick drag event.

- Phone racing now uses one clean mobile road stack: the old desktop road/scenery layer is skipped on phones, asphalt is opaque instead of see-through, texture is clipped inside the road, yellow chevron markers are replaced with flat road pips, and nearby opponents show distance/lane badges plus a compact lane radar so passing and collision avoidance are easier to read.
- Phone racing now uses a stronger v51 readability override: the mobile road gets high-contrast shoulders and brighter lane paint, opponents are forced into larger race gaps, and phone-only projection spreads far vehicles vertically so they no longer visually collapse into one cluster at the horizon.
- Phone racing now has a v52 passability fix: the AI field actively leaves an open passing lane near the player, random traffic is capped on small screens, overlapped cars are not drawn under the hood, hood view shows only the lower vehicle nose instead of a full top-down car, and phone scenery has a dedicated skyline/roadside backdrop.
- Phone racing now has a v58 fair-damage pass: hidden or unreadable traffic can no longer damage the player, close hazards are forced visible, road-edge drift warns and slows instead of silently destroying the vehicle, and the HUD names the cause whenever damage happens.
- Phone racing now has a v59 lane-feel pass: hood/cockpit/phone cameras follow the player lane, so lane stripes, markers, traffic, and rivals slide across the screen during steering instead of feeling like the car is only changing an invisible number.
- Online/browser racing now has a v62 lane-feel pass: desktop canvas and WebGL chase views use the same smoothed lane camera anchor as phone play, while the WebGL overlay keeps the player car and passable traffic moving visibly across the lane during right/left steering.
- The WebGL route pass now uses smoother world landmarks, neon signs, mounds, hangars, barns, freight stops, streetlights, trees, barriers, spectators, weather overlays, speed streaks, and detailed vehicle bodies instead of the first sharp spike scenery and plain box vehicles.
- The WebGL route pass also adds overhead highway gantries, neon vertical signs, harbor cranes, extra wet-road reflection strips, bridge-style structures, route curve camera targets, and route-specific foreground details so the world feels less empty and more like a real drive.
- Roads now include darker asphalt patches, reflective studs, shoulders, rumble strips, center paint, city crosswalk-style stripes, signs, spectator groups, guardrail details, and route-specific landmarks so each place reads more like a real location.
- Vehicles now have extra WebGL body details such as glass, headlights, grilles, side panels, mirrors, plates, rim highlights, rear wings on fast cars, larger stance cues for trucks, visible damage panels, and route-appropriate traffic types.
- WebGL mode now layers the high-detail canvas vehicle renderer over the 3D road, so player cars, opponents, police, and traffic keep readable wheels, glass, mirrors, trim, labels, brake lights, and damage marks instead of looking like simple boxes.
- Phone WebGL mode now prefers the generated asset sprites over the older simple procedural silhouettes, giving the app a more model-like vehicle read while staying light enough for mobile browsers.
- Race opponents now line up against the player, change pace during the route, show names on the track, show visible damage bars, take collision damage, limp when wrecked, and affect finish position, rewards, and wins.
- Opponent pack racing now uses a longer staggered grid, side-lane passing behavior, draft/catch-up pressure, and overtake tracking. You can pass opponents, they can draft back and pass you, and the phone HUD/finish screen show overtake progress.
- The starting grid now splits the pack with three opponents ahead and two behind, wider side-lane behavior, less early traffic clutter, and close-pass collision tuning so the player can actually overtake named rivals and climb positio

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n instead of staring at one grouped wall of vehicles.

- The v63 progression pass adds vehicle ownership, reputation unlocks, higher vehicle/upgrade pricing, per-vehicle saved builds, paint cycling, player license plates based on callsigns, and route rules so boats stay on water, aircraft race sky routes, tanks use military-style routes with a boost-button cannon, tractors stay on farm rallies, semis run freight, and monster trucks get off-road/arena-style routes.
- The v63 race pass tightens player hitboxes so hidden traffic should not damage the car from far ahead, spreads online/browser opponents much farther apart, applies the same passing-lane behavior outside phone mode, makes route markers harder to collect, lowers police frequency outside true pursuit conditions, adds civilian penalties, adds selected-route oncoming traffic, and adds stronger day/night lighting with street lamps and vehicle light glow.
- The v64 scenario pass adds more vehicle models and classes, including rally, drift, buggy, heavy tank, hydroplane, patrol boat, attack helicopter, and jet plane builds. It also adds local ghost/crew multiplayer-style race scenarios, longer hot-pursuit escalation, route hideouts such as garages, caves, hangars, barns, mountain tunnels, and marina cover, plus shortcut branches that open different road, water, sky, farm, freight, military, and world-route decisions during races.
- The v65 close-pass pass widens side-contact detection and makes nearby rivals steer apart faster, so passing reads as side-by-side racing or a scrape instead of the player car visually floating over another vehicle.
- The v66 visibility pass tightens desktop/WebGL hitboxes to a short road-distance window around the player car, adds route-stage scenery that changes during each race, and draws an actual finish gate/checkered road line so the route has a visible end instead of feeling endless.
- The v67 pickup-lane pass keeps coins and route markers inside safe painted lanes instead of letting outer pickups project onto the shoulder or off the road during curves.
- The v68 real-world detail pass adds shared browser, WebGL, and phone scenery layers with asphalt patches, raised reflectors, rumble strips, crosswalk/route paint, curve boards, sidewalks, roadside people, parked vehicles, telephone poles, fences, buildings, barns, cranes, hangars, water, hills, mountains, and location-specific road life. Vehicle bodies also get extra panel seams, plates, tire contact cues, light trim, grilles, utility details, and damage marks while staying original and unlicensed.
- The v69 real-world expansion adds route-signature environment pieces and more pavement hardware: overpasses, interstate gantries, expressway rails, piers, tunnel mouths, grain elevators, fuel stops, markets, speed cameras, mile markers, containers, radar towers, runway beacons, road drains, manholes, potholes, lane arrows, patched joints, shoulder wear, and foreground posts/curbs so phone and browser races feel more like moving through actual places.
- The v70 real-world texture pass adds ambient local life and surface storytelling: traffic lights, storefronts, cafe awnings, power lines, crosswalk crews, windsocks, windmills, ski lifts, kites, mist/dust clouds, route-specific road words, puddle reflections, road spray, snow/dust edges, cones, barrels, hydrants, benches, bollards, and extra near-road light pools while keeping the phone road flat and readable.
- The v71 place-detail pass adds place identity, district motion, and road-rule texture: route-specific horizon landmarks, destination boards, side-road cars and trucks, metro/tram/rail silhouettes, tiny boats, planes, service carts, pedestrian bridges, bridge expansion joints, underpass shadows, crosswalk/stop/hold-short markings, merge/bus/bike/truck lane paint, edge hash marks, snow/sand/leaves/wet buildup, and moving light pools so routes feel more like active real places without cluttering the driving line.
- The v72 route-atmosphere pass adds believable world edges around the race surface: place-specific shoulder terrain, water/wave/dock edges, city curbs, neon sidewalk glow, farm rows, desert sand, canyon rock, snow banks, runw

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ay and freight-yard service lines, micro landmarks such as kiosks, signs, cones, hay bales, guard posts, lamps, and subtle route weather like rain, snow, dust, sea spray, leaves, and night glow while keeping the main driving lane visible.

- The v73 route-sector pass makes scenery change more clearly as each race progresses: each sector gets unique horizon set pieces, sector boards, edge markings, tunnel lamps, bridge cables, checkpoint crews, city storefronts, airfield lights, freight-yard signs, farm gates, coastal props, and stage-specific roadside details so the route reads less like a repeating backdrop and more like driving through changing real places.
- The v74 living-world pass adds active roadside life outside the racing line: parked and service vehicles, route crews, spectators, workers, tents, kiosks, food stands, traffic signals, marshal cars, rescue trucks, fuel trucks, forklifts, farm pickups, snow plows, airfield carts, freight-yard activity, and place-specific crowd clusters so each route feels occupied without hiding the road or touch controls.
- The v75 surface-lighting pass gives the road more physical material feel: subtle road crown, moving wet/gloss reflections, rubber and seam marks, route-specific pavement streaks, coastal spray sheen, snow/ice shine, desert heat shimmer, farm dirt, freight diesel marks, runway glints, and neon night reflections while preserving a flat, readable phone driving lane.
- The v76 depth-infrastructure pass adds more real-world scale behind the race: parallax city blocks, mountains, dunes, water spans, fields, ports, rail yards, sound walls, guardrails, snow fences, stone walls, power lines, tram wires, lift cables, gantries, bridge cables, beacons, and far-side traffic silhouettes so the route feels deeper without putting clutter in the driving line.
- The v81 ground-visibility pass tightens player hit detection to the visible car position, reduces ahead-of-car side-scrrape damage, brightens the readable road surface, and adds grounded real-world props such as storefronts, streetlights, guardrails, parked vehicles, crowds, route signs, barns, fences, containers, hangars, palms, rocks, pines, and barriers that sit on the roadside instead of floating in the haze.
- The v82 drive-by world pass spaces larger, brighter roadside landmarks by real route distance, sorts them far-to-near, and changes the landmark set by race sector so you drive past grounded billboards, bridges, tunnel portals, service stations, docks, traffic lights, homes, grandstands, towers, barns, shops, crowds, and route signs instead of seeing one repeated backdrop.
- The v83 surface-view pass keeps scenery outside the drivable road, paints the road back over shoulder props so details no longer cut through the lane, and adds dedicated water and air views: boat races now get a water channel with shorelines, docks, buoys, and other boats, while airplane and helicopter races get clouds, flight gates, distant terrain, and aircraft silhouettes instead of road scenery.
- The v84 specialty-world pass makes races declare appropriate vehicle types, adds dedicated semi, monster truck, and tank events, adds more upgradeable variants in those categories, and gives each specialty route its own scenery: convoy lanes, weigh stations, diesel stops, dirt arena berms, crush cars, grandstands, bunkers, range targets, watch towers, and concrete proving-ground details. Profiles also silently auto-save race state every few seconds and on app/page hide so local coins, owned vehicles, per-vehicle upgrades, paint, plates, reputation, and live race progress are remembered after reopening.
- The v85 all-vehicle upgrade pass expands the garage and race rules so police cars, offshore boats, racing helicopters, fighter jets, bombers, drones, sport planes, tanks, semis, tractors, monster trucks, and standard road cars all use the same saved per-vehicle upgrades, colors, plates, ownership, reputation unlocks, and race compatibility system. It also adds dedicated police interceptor and sky-command routes with pursuit hideouts, runway/skybase traffic, air gates, lightbar visuals, and type-specific phone/WebGL sprites.
- The v86 world-spacing

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pass spreads scenery farther down the route, removes most repeated sign-like props from phone mode, turns police roadblocks into lower physical barricades, limits stacked stage boards, and swaps excess labels for grounded buildings, parked cars, lights, fences, people, barriers, and service vehicles so races feel more like driving through a real place instead of a cluster of signs.

- The v87 reference-world scene pass uses real driving-photo/video cues such as long guardrails, shorelines, sidewalks, parked vehicles, storefront rows, utility poles, fields, cliffs, mountains, service yards, and set-back buildings. It spaces these as drive-by scene blocks by route distance, reduces repeated signs in both phone and browser views, and gives the WebGL route wider roadside spacing so the world reads more like passing through actual places.
- The v88 living-world safety pass adds more bystanders and route-appropriate animals to the scenery: sidewalk spectators behind barriers, beach and dock crowds, farm pasture animals, mountain and desert wildlife, rainforest animals, and city dogs or birds. Some people or animals can become clear avoidable crossing hazards with on-screen warnings, reputation/coin penalties, and slowdown instead of unexplained damage.
- The v89 ambient living-world pass adds more visible drive-by life that stays set back from the road: cyclists, joggers, sidewalk queues, patio crowds, dock workers, pier people, pasture animals, wildlife lookouts, airfield crews, airport beacons, freight forklifts, diesel-stop activity, stadium crowds, military/range crews, and shoreline/airfield motion. The same idea now feeds the WebGL route so browser and phone views feel more like passing through occupied places instead of empty scenery.
- The v90 visible-detail pass makes the real-world build easier to see on phones: larger foreground landmarks, brighter roadside people/animals/crews, closer shoulder placement, stronger sidewalk/ground pads, more visible buildings/fences/lights/signs, and a late foreground pass drawn after the road so details are no longer washed into the background. The WebGL route also keeps scenery active on shorter landscape frames with a lighter object count.
- The v91 road-drama and vehicle-detail pass removes the remaining phone grid look from guardrails, docks, road seams, and turn arrows; adds cleaner grounded asphalt and softer curve cues; adds more visible route moments for tunnels, city underpasses, hills, mountain drives, and sweeping turns; and gives canvas, phone, and WebGL vehicles more realistic rear signatures with LED tail lights, diffusers, exhausts, plates, heavy-vehicle panels, and tank tread detail.
- The v92 weather-disaster pass adds route-specific sun, clouds, rain, snow, night, storm, tornado, quake, flood, tidal-surge, and train moments. Wet roads now show clear blue glossy sheen and puddles, oil and black ice show dark shiny patches, snow routes add slush, quake routes add cracks and shake, and the handling model lightly reduces grip/speed on slick or damaged surfaces. Phone mode also removes the remaining oversized side-road rails and phone grid/arrow clutter while making distant scenery more opaque.
- Opponents can now make wheel-to-wheel side contact when the player catches them, causing both vehicles to slow, push sideways, smoke, spin, lose focus, and keep damaged state instead of phasing through.
- Traffic uses rear-view performance silhouettes instead of overhead arcade sprites, with realistic road dashes and route markers replacing blocky lane pickups. Traffic and police no longer disappear on contact; damaged vehicles remain on the road until they move out of view.
- Driving now uses more realistic throttle, brake, reverse, steering inertia, lateral velocity, grip, road-edge slowdown, tire slip, corrected steering direction across touch, keyboard, and gamepad input, vehicle body yaw, damage-based acceleration loss, and damage-based handling loss instead of automatic forward motion or lane-snapping movement.
- Collision damage now appears in the HUD, finish stats, damage bar, car scuffs, smoke, cracked windshield overlays, impact flashes, and reset countdowns. At critica

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- damage the vehicle enters a timed reset, and the manual `Reset Car` button or `R` key can recover from a bad crash position.
- Police pursuit mode adds heat, sirens, flashing light bars, interceptor units, contact penalties, escape scoring, and cockpit pursuit alerts.
- Sound is generated locally with Web Audio: engine tone, boost feel, pickup chimes, collision hits, and police siren sweeps. No external audio files or network calls are used.
- Driving effects include tire smoke, launch haze, boost exhaust, brake/slip wash, speed streaks, road reflections, depth haze, headlight wash, dust/snow/weather layers, camera shake, off-road drag, and car body lean during hard steering.
- Phone landscape racing moves touch controls to a transparent driver-pad overlay so steering, center, gas, brake/reverse, boost, pause, and touch mode switching stay usable without covering the driving screen.
- Touch controls use a multi-touch tracker plus pointer fallback so gas, brake, boost, and steering can stay active together until finger-up/cancel even if both thumbs are down at once.
- Phone mode has a separate in-game control layer fixed to the visible app window in both portrait and landscape. The default Mini layout is now a floating one-thumb joystick that appears under your touch, keeps gas active while you steer, releases cleanly when you lift your thumb, and leaves more of the driving screen open. Tapping `Full` expands the larger separate controls.
- The Mini floating joystick sends analog steering into the driving physics, keeps throttle active while steering sideways, supports brake/reverse only after a deliberate downward pull, and visually follows the player's touch instead of staying locked to a fixed D-pad position.
- Phone collision detection now uses a smaller distance-and-lane overlap check that matches the phone road projection, so traffic should only count as a crash when it is actually in your lane and close to your vehicle. Phone impacts also do less damage and push vehicles apart instead of repeatedly resetting the player.
- Race starts now show a clear goal card on the driving screen with the current challenge, route, longer-race pacing, and crash-recovery status.
- Crashes and total focus loss no longer end the race. Critical damage now starts a reset countdown, stabilizes the vehicle, and lets the player continue the same run.
- Generated vehicle sprites now draw with stronger tire contact shadows, dark tire contact patches, short road streaks, and lower pavement anchoring so cars, trucks, semis, tractors, and opponents read as driving on the road instead of floating above it.
- Phone and browser race views now generate rear/chase-view road vehicle sprites with the wheel bottoms used as the tire-contact anchor, so cars, trucks, semis, tractors, tanks, and snowmobiles sit on the pavement instead of reading like overhead stickers.
- Vehicle tire anchors are now pushed to the bottom of the generated sprite, and every grounded vehicle draws a hard pavement clamp directly across the tire line so there is no transparent gap that can read as floating.
- The player chase camera now places the car lower on the road plane and draws stronger vertical tire contact streaks, so the vehicle reads as planted on the foreground pavement instead of hovering over it.
- Grounded vehicle sprites now use the true generated wheel line as their contact anchor, then sink that anchor into the road by vehicle weight. This makes tires overlap the asphalt instead of hovering just above it.
- Race-view vehicles now draw wider and shorter, with neutral road markings replacing the yellow accent streaks that cluttered the screen. The chase camera favors a planted rear-view car shape instead of a tall angled sprite.
- Browser and phone vehicles now get heavier road grounding with lowered sprite anchoring, black tire sidewall pins, hard road-contact pads, and WebGL ground shadows under every moving vehicle.
- Traffic, police, opponents, and route markers now use road-distance projection instead of raw screen-height movement, so vehicles enter from the road horizon and stay locked to the pavement instead of falling down from the sky.
- Vehicle drawing

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now uses the tire/contact line as the screen anchor, so generated sprites grow upward from the pavement instead of hovering around a center point.

- Road motion now has an explicit speed-linked pass with moving asphalt bands, lane dashes, shoulder streaks, texture drift, and faster WebGL track markers so the pavement visibly moves when the player accelerates.
- Road paint now uses flat perspective quads and lower WebGL ground markers, while vehicle sprites use a lower tire-contact anchor and stronger contact shadows so tires and road lines sit on the pavement instead of floating above it.
- WebGL mode now projects every canvas vehicle sprite from the WebGL road plane, then pins the generated vehicle art to a tire-contact origin with visible tire pads and contact shadows. This keeps opponents, police, traffic, and the player car visually attached to the same pavement perspective as the 3D road.
- WebGL chase, hood, and cockpit cameras now use a flatter forward-looking road angle, plus a stronger foreground asphalt pass with horizontal pavement bands. The near road reads as flat ground under the tires instead of a ramp climbing up the screen.
- Vehicle sprites in WebGL mode now use a visible-road projection instead of the steep camera projection, with stronger spacing by distance and far-to-near draw ordering. This keeps cars from clustering high in the scene or appearing to climb into the sky while the moving pavement stays underneath them.
- Opponent, police, route marker, and traffic visibility now use the same visible road projection as their final draw position, keeping the far pack on the actual road horizon instead of letting older screen-height math place them above the road.
- WebGL mode now keeps road paint in the 3D ground layer and draws the high-detail vehicle sprites on top, preventing 2D road streaks or duplicate low-poly vehicles from appearing to float in front of the race.
- The Vehicles garage now paints real generated preview images for every vehicle class, including cars, F1, prototypes, trucks, semis, tractors, monster trucks, tanks, snowmobiles, boats, helicopters, and airplanes.
- 'Save Race' stores the current run on the device, and 'Resume Saved Race' appears in the Race Hub for the same driver profile.
- New and existing local profiles now get a starter garage budget and early reputation so vehicle switching, upgrades, freight racing, farm racing, and Tokyo-style routes are available quickly instead of feeling locked.
- Race routes now run more than seven times longer than the original short sprint format and cannot finish before the opening minute-plus race window, so events land closer to a real 1-3 minute phone race instead of a quick arcade burst.

Visual direction references:

- Need for Speed-style street-racing energy: heat, police pressure, risk/reward pursuit decisions, expressive speed effects.
- Forza Horizon-style scenic variety: coast roads, city streets, canyon routes, and mountain passes.
- Tokyo night-highway feel: neon signage, wet road reflections, tunnel/highway rhythm, and rival traffic tension.
- Sim/cockpit cues: windshield framing, steering wheel, dashboard speed readout, heat alert, and road-focused forward camera.

Graphics note: this is a practical standalone browser build with both WebGL 3D rendering and high-detail canvas fallback. It is not using licensed console-game assets, but the rendering pipeline is now structured so deeper 3D models, tracks, lighting, and physics can be added over time.

Install As An App

Velocity Vault is now a Progressive Web App. To play it everywhere without this computer being on, put this folder on an HTTPS web host, open the hosted link once on the phone or computer, then install it:

- Android / Chrome / Edge: tap 'Install / Add to Phone' or use the browser menu and choose install.
- iPhone / iPad Safari: tap Share, then 'Add to Home Screen'.
- Windows / Edge / Chrome: click 'Install / Add to Phone' or use the browser install icon.

After installation, the service worker caches the game files for offline play. The app can then launch from the phone or computer home screen without your PC being awake.

Good hosting options include Ape

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x Vault Command Cloud, GitHub Pages, Netlify, Cloudflare Pages, Vercel, or any HTTPS web host. The temporary local download link from this PC is useful for transfer, but it cannot keep the phone app available after this computer turns off.

Host On Apex Vault Command Cloud

Velocity Vault is prepared for Apex Vault Command Cloud as a secure static PWA deployment at ``https://apexvault.duckdns.org/``. Upload ``VelocityVault-ApexVaultCommandCloud-StaticBundle.zip``, or connect the GitHub repository and use the repository root as the static deploy root.

Recommended Apex settings:

- App type: static web app, static site, or PWA.
- Entry file: ``index.html``.
- HTTPS: on.
- Service workers/offline app support: on.
- Unknown route fallback: ``index.html``.
- Required backend/runtime: none.
- Environment variables: none.
- Security headers: use ``_headers`` or ``apex-vault-command-cloud.json``.
- ZIP shape: ``index.html`` must be at the top level of the ZIP, not inside another folder.
- Apex-safe manifest: the upload bundle uses ``manifest.json`` instead of ``webmanifest`` because Apex blocks ``webmanifest`` assets.
- Apex-safe asset list: the upload bundle excludes ``md``, ``pdf``, ``webmanifest``, extensionless, and underscore-prefixed files because Apex blocks some non-app asset types.

The Apex package keeps the game local-only: no third-party scripts, no CDN files, no server database, and no remote assets. Driver profiles, saved races, coins, reputation, upgrades, custom vehicle builds, and PIN hashes remain stored in each device browser.

When the local game changes, Apex Vault Command Cloud does not update automatically from this folder. Rebuild and reupload ``VelocityVault-ApexVaultCommandCloud-StaticBundle.zip`` after each change unless the Apex host is later connected to an automatic GitHub/deploy hook.

After publishing, open the secure Apex link on the phone, then use the browser's ``Install``, ``Add to Phone``, or ``Add to Home Screen`` option.

Put It On This Computer

You can use Velocity Vault on this computer in three ways:

- Open ``index.html`` directly from this folder.
- Use the ``Velocity Vault`` desktop shortcut if one has been created.
- After publishing to GitHub Pages, open the secure hosted link in Edge or Chrome and click ``Install / Add to Phone``. That creates a real app-style launcher in Windows.

The installed browser app is the cleanest computer version because it opens in its own window and caches files for offline play.

Send It To Kids Or Testers

The best way to send the game to kids' phones is to publish the ``publish`` folder to GitHub Pages and send them the secure link:

``https://YOUR-GITHUB-USERNAME.github.io/velocity-vault/``

They can open the link on their phone and install it:

- iPhone / iPad: open in Safari, tap Share, tap ``Add to Home Screen``.
- Android: open in Chrome or Edge, tap ``Install / Add to Phone`` or ``Add to Home screen``.

Once installed, the game appears like an app icon and can play offline after the first load. Each phone keeps its own local driver profiles and PINs; profiles do not sync between devices.

You can also send ``VelocityVault-Phone-Download.zip``, but most phones do not run zipped web apps as cleanly as a hosted secure link. The GitHub Pages link is the smoother option.

Local AI Director

Velocity Vault includes a local agentic tuning director in the Garage AI tab. It reviews only the current on-device profile stats and rotates daily bonus events, reward multipliers, rival pacing, coin density, and upgrade recommendations.

This keeps the game feeling fresh without sending player data to any server. Because the game is fully standalone, it does not fetch online trend updates; new content can be added by editing the race, mission, event, and upgrade lists in ``game.js``.

Security Notes

- No network calls.
- No third-party scripts, fonts, or assets.
- Strict Content Security Policy in ``index.html``.
- Offline app caching uses ``sw.js`` and same-origin files only.
- WebGL rendering is local-only and does not load shaders, models, scripts, or textures from the network.
- Phone graphics assets are generated locally at runtime and cached in memory; no third-party model, texture, or image

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files are fetched.

- Profiles and optional PIN hashes stay in `localStorage` on this device.
- Controller support uses the browser Gamepad API only.
- The PIN is only for casual local separation between players; it is not a replacement for operating-system account security.

APEX-VAULT-COMMAND-CLOUD.md

Apex Vault Command Cloud Hosting

Velocity Vault is ready to host as a secure static Progressive Web App on Apex Vault Command Cloud.

Target host: `https://apexvault.duckdns.org/`

What To Upload

Use `VelocityVault-ApexVaultCommandCloud-StaticBundle.zip`.

The ZIP contains `index.html` at the top level, plus only the playable game files Apex should accept: `.html`, `.css`, `.js`, `.json`, and `.svg`. It does not include `.md`, `.pdf`, `.webmanifest`, extensionless, or underscore-prefixed files because the Apex upload gate blocks some non-app asset types.

It does not need a database, server runtime, paid API, or external asset host.

Apex Setup

1. Create a new static web app, static site, or PWA deployment in Apex Vault Command Cloud.
2. Upload the ZIP contents, or connect the GitHub repository and point the deploy root at the repository root.
3. Set the entry file to `index.html`.
4. Turn on HTTPS.
5. Turn on service workers or PWA/offline support if Apex asks for that setting.
6. Set unknown route fallback to `index.html`.
7. Apply the security headers from `\_headers` or `apex-vault-command-cloud.json`.
8. Publish the site and open the secure browser link on a phone or computer.
9. Install it from the browser menu with `Install`, `Add to Phone`, or `Add to Home Screen`.

Required Files

- `index.html`
- `styles.css`
- `game.js`
- `phone-asset-pipeline.js`
- `webgl-pipeline.js`
- `manifest.json`
- `sw.js`
- `app-icon.svg`

The local project still keeps the README, Apex notes, source PDF, and deployment manifest outside the Apex upload ZIP.

Security Profile

- Local-only game code.
- No third-party scripts.
- No CDN dependencies.
- No remote images, fonts, shaders, models, or audio.
- No backend database needed.
- Driver profiles, coins, upgrades, vehicle builds, saved races, and optional PIN hashes stay in the browser's local storage on each device.
- HTTPS is required for phone installation and reliable service-worker caching.

After Publishing

Send the secure Apex link to testers. Each phone can install the game like an app, then play offline after the first full load.

When the local files change, rebuild and reupload `VelocityVault-ApexVaultCommandCloud-StaticBundle.zip` for Apex to show the new version. The hosted copy will not update by itself from this folder unless Apex is later connected to a GitHub/deploy hook.

If Apex gives you a custom domain option, point the domain to this deployment and keep HTTPS enabled.

apex-vault-command-cloud.json

```
{
  "appName": "Velocity Vault",
  "deploymentName": "velocity-vault",
  "version": "v80-apex-safe-static-assets",
  "siteUrl": "https://apexvault.duckdns.org/",
  "description": "Standalone secure PWA racing game prepared for Apex Vault Command Cloud static hosting.",
  "runtime": "static-pwa",
  "entry": "index.html",
  "rootDirectory": ".",
  "requiredHttps": true,
  "installablePwa": true,
  "offlineSupport": true,
  "serviceWorker": "sw.js",
  "manifest": "manifest.json",
  "assets": [
    "index.html",
    "styles.css",
    "game.js",
    "phone-asset-pipeline.js",
    "webgl-pipeline.js",
    "manifest.json",
    "sw.js",
    "app-icon.svg"
  ],
  "routes": [
    {
      "path": "/",
      "serve": "index.html"
    },
    {
      "path": "/index.html",
      "serve": "index.html"
    },
    {
      "path": "/*",
      "serve": "index.html",
      "fallbackOnly": true
    }
  ],
  "mimeTypes": {
    ".html": "text/html; charset=utf-8",
    ".css": "text/css; charset=utf-8",
    ".js": "text/javascript; charset=utf-8",
    ".json": "application/json; charset=utf-8",
    ".svg": "image/svg+xml"
  },
  "securityHeaders": {
    "Content-Security-Policy": "default-src 'self'; script-src 'self'; style-src 'self'; img-src 'self' data:; connect-src 'self'; manifest-src 'self'; worker-src 'self'; object-src 'none'; base-uri 'none'; form-action 'none'; frame-ancestors 'none'; upgrade-insecure-requests",
    "Referrer-Policy": "no-referrer",
    "X-Content-Type-Options": "nosniff",
    "X-Frame-Options": "DENY",
    "Permissions-Policy": "camera=(), microphone=(), geolocation=(), payment=(), usb=(), xr-spatial-tracking=(), fullscreen=(self), gamepad=(self)",
    "Cross-Origin-Opener-Policy": "same-origin",
    "Cross-Origin-Resource-Policy": "same-origin"
  },
  "cacheRules": [
    {
      "path": "/index.html",
      "cacheControl": "no-cache"
    },
    {
      "path": "/sw.js",
      "cacheControl": "no-cache"
    },
    {
      "path": "/manifest.json",
      "cacheControl": "public, max-age=3600"
    },
    {
      "path": "/*.js",
      "cacheControl": "public, max-age=31536000"
    },
    {
      "path": "/*.css",
      "cacheControl": "public, max-age=31536000"
    },
    {
      "path": "/*.svg",
      "cacheControl": "public, max-age=31536000"
    }
  ],
  "environmentVariables": [],
  "networkPolicy": {
    "outboundCallsRequired": false,
    "thirdPartyScriptsRequired": false,
    "databaseRequired": false
  },
  "uploadChecklist": [
    "Upload VelocityVault-ApexVaultCommandCloud-StaticBundle.zip to https://apexvault.duckdns.org/ as a static web/PWA app.",
    "The upload ZIP intentionally excludes .md, .pdf, .webmanifest, extensionless, and underscore-prefixed files because the Apex upload gate blocks some non-app asset types.",
    "Set index.html as the entry file and the deployment root as the uploaded folder root.",
    "Require HTTPS so phone installation and service-worker offline play work correctly.",
    "Enable SPA fallback to index.html for unknown paths.",
    "Apply the equivalent security headers in this manifest if Apex exposes a header settings screen.",
    "After deployment, open the secure URL on a phone and use Add to Home Screen or Install."
  ]
}
```